Nuclear Structure and Decay Data for A=35 Isobars*

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Abstract: This work presents a comprehensive and critical evaluation of experimental nuclear spectroscopic data from reactions and decays for all 11 known nuclides with mass number 35 (Ne, Na, Mg, Al, Si, P, S, Cl, Ar, K, Ca). Recommended values are produced for level energies, spins and parities, half-lives, and radiation properties including energies, branching ratios, and multipolarities of γ rays, as well as characteristics of β radiation decays, based on a rigorous assessment of all available experimental data. Discrepancies among existing results are carefully addressed. This work supersedes earlier full evaluations of A=35 published by 2011Ch48, 1990En08 (also 1998En04 update) and 1978En02.

^{35}Cl remains the most extensively studied nuclide, investigated through 28 different reactions and decays, followed by ^{35}S, ^{35}P, and ^{35}Ar. For ^{35}Na, only two excited states have been observed, and its β^--decay scheme has not yet been established. For ^{35}Mg, no γ rays have been placed with certainty, and no decay dataset exists on ^{35}Mg$\rightarrow$^{35}Al. ^{35}Ar resonances of astrophysical interest have been studied via transfer reactions and ^{35}K ε+β^+ decays, but their properties are yet to be determined. For ^{35}K, many excited states discovered via ^{35}Ca ε+β^+ decay are unresolved. For ^{35}Ca, no γ rays have been observed and only one excited state has been identified. Notably, ^{35}Ca is the only nuclide in this mass chain that decays by ε+β^+-delayed two protons, although the emission mechanism remains unclear.

Cutoff Date: Literature available up to September 30, 2025 has been consulted, with the help of the primary bibliographic source, the NSR database (2014Pr09) available at Brookhaven National Laboratory web page: www.nndc.bnl.gov/nsr/.

General Policies and Organization of Material: See the April 2026 issue of the *Nuclear Data Sheets* or https://www.nndc.bnl.gov/nds/docs/NDSPolicies.pdf.

General Comments: The statistical analysis of γ-ray data and deduced level schemes is carried out through a suite of current versions of computer codes available at the IAEA-NDS web page of ENSDF codes (https://www-nds.iaea.org/public/ensdf_pgm) such as GLSC, Java-RULER, BrIccMixing, and ConsistencyCheck. The direct feedings to excited states in β^- and ε decays have generally been computed from I(γ+ce) intensity balances at each level; the associated log *ft* values are calculated using the BetaShape code. Theoretical total conversion coefficients are calculated using version v2.3e of BrIcc code (2008Ki07) for frozen-orbit option with an implicit uncertainty of 1.4% when not stated. Q values and particle separation energies have been adopted from 2021Wa16 (AME2020). Static magnetic dipole moments are taken from 2019StZV (long-lived states) and 2020StZV (short-lived states) evaluations, and static electric quadrupole moments from 2021StZZ evaluation, when available in these publications. Nuclear rms charge radii have been adopted from 2013An02 evaluation. In cases where weighted averaging procedures have been used, the assigned uncertainty in the result is generally not lower than the lowest uncertainty in the data points used in the procedure.

Acknowledgements: Earlier evaluations (2011Ch48, 1998En04 update, 1990En08, 1978En02, 1973EnVA, 1967En05, 1962En05, 1957En01, 1954En19), and compilations available in the XUNDL database provided valuable resources for the present work. The evaluators are grateful to the reviewers at different stages of this work, and to the personnel at the National Nuclear Data Center (NNDC) at Brookhaven National Laboratory for facilitating this work.

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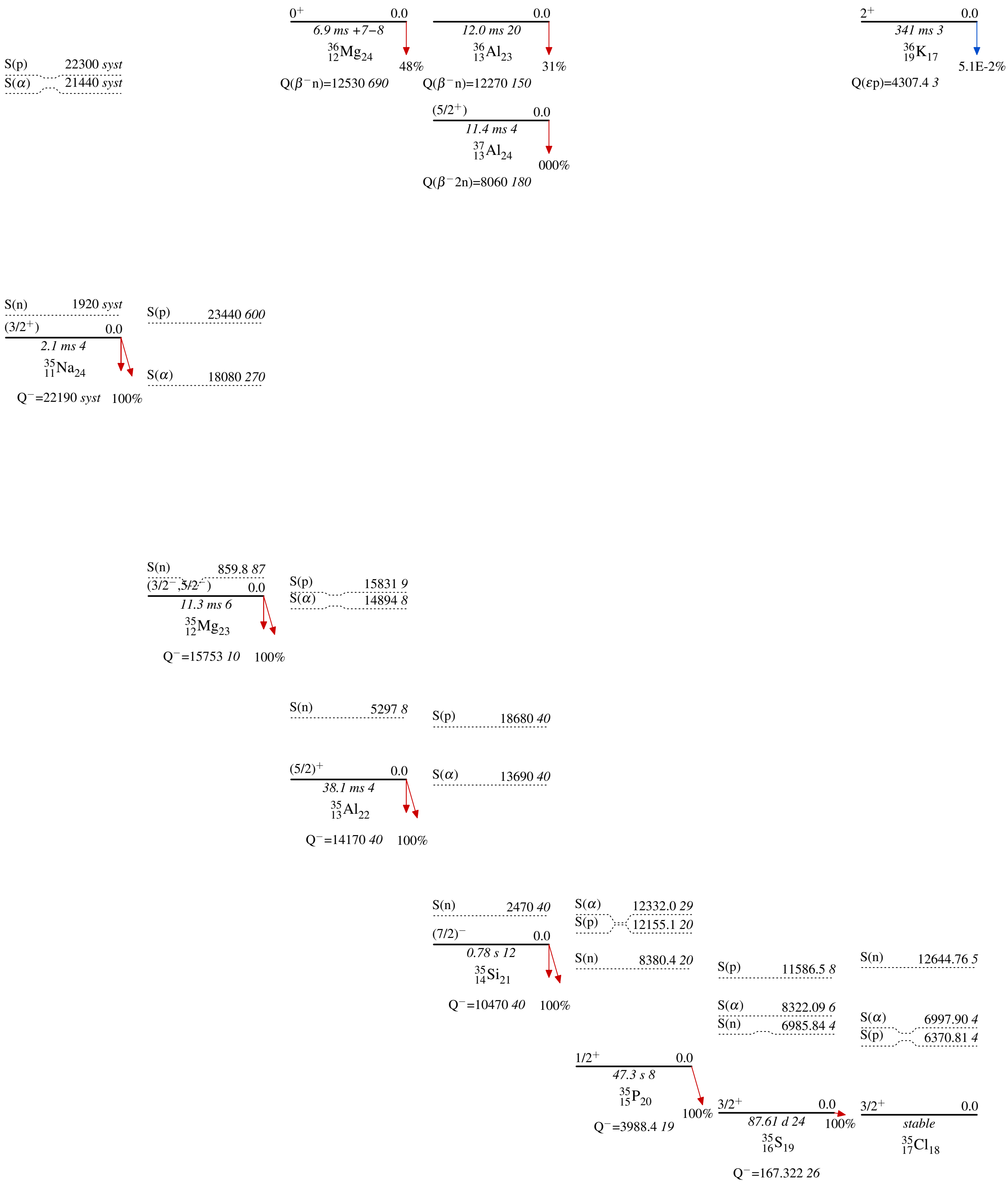
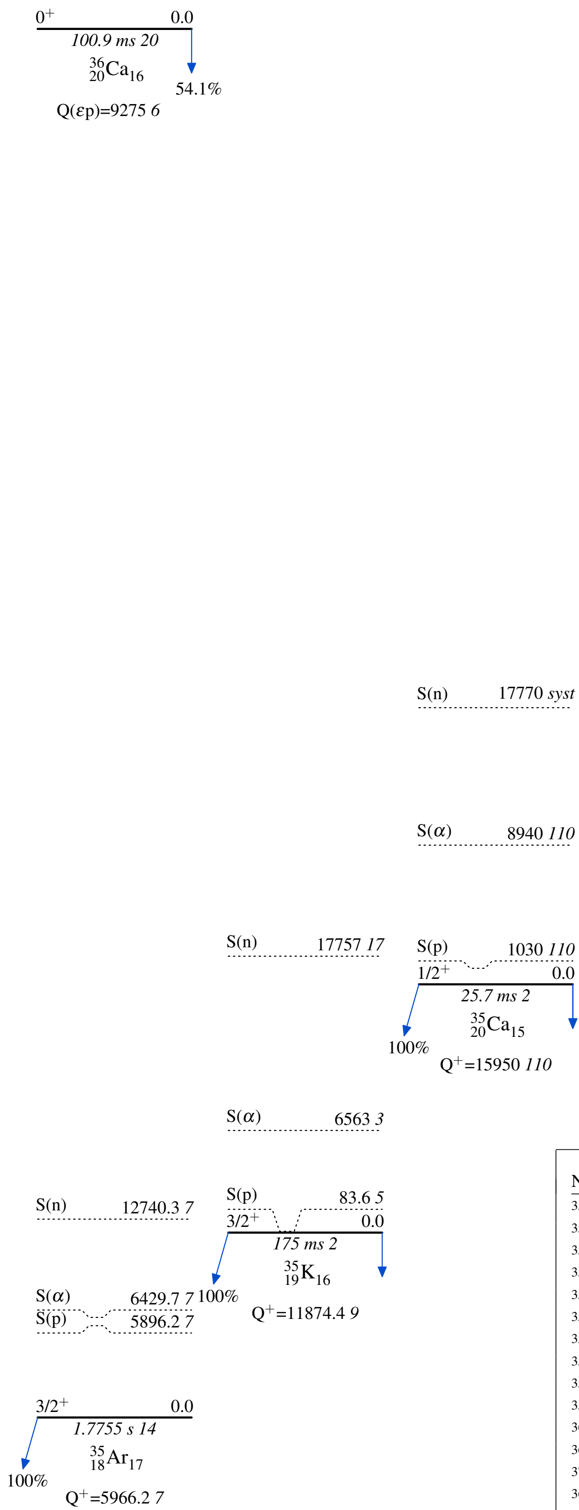
Index for A=35

Skeleton Scheme for A=35



Skeleton Scheme for A=35 (continued)



Ground-State and Isomeric-Level Properties

Nuclide	Level	Jπ	$T_{1/2}$	Decay Mode
^{35}Na	0.0	(3/2$^+$)	2.1 ms *4*	%β^-=100; %β^-n>0; %β^-2n=?...
^{35}Mg	0.0	(3/2$^-$,5/2$^-$)	11.3 ms *6*	%β^-=100; %β^-n=52 *46*; %β^-2n=?
^{35}Al	0.0	(5/2)$^+$	38.1 ms *4*	%β^-=100; %β^-n=37 *4*; %β^-2n=?
^{35}Si	0.0	(7/2)$^-$	0.78 s *12*	%β^-=100; %β^-n<5
^{35}P	0.0	1/2$^+$	47.3 s *8*	%β^-=100
^{35}S	0.0	3/2$^+$	87.61 d *24*	%β^-=100
^{35}Cl	0.0	3/2$^+$	stable	
^{35}Ar	0.0	3/2$^+$	1.7755 s *14*	%ε+%β^+=100
^{35}K	0.0	3/2$^+$	175 ms *2*	%ε+%β^+=100; %εp=0.37 *15*
^{35}Ca	0.0	1/2$^+$	25.7 ms *2*	%ε+%β^+=100; %εp=95.8 *3*; %ε2p=4.2 *3*
^{36}Mg	0.0	0$^+$	6.9 ms *+7−8*	%β^-n=48 *12*
^{36}Al	0.0		12.0 ms *20*	%β^-n<31
^{37}Al	0.0	(5/2$^+$)	11.4 ms *4*	%β^-2n≥000
^{36}K	0.0	2$^+$	341 ms *3*	%εp=5.1×10^{-2}
^{36}Ca	0.0	0$^+$	100.9 ms *20*	%εp=54.1 *12*

Adopted Levels:not observed

Q(β^-)=24430 *calc*; S(n)=−2370 *calc*; S(p)=29480 *calc* 2019Mo01
S(2n)=−470 (2019Mo01, FRDM).

2019Ah07: ^{9}Be(^{48}Ca,X) using a 345-MeV/nucleon, 450-pnA ^{48}Ca beam provided by the cascade operation of the RIBF accelerator complex at RIKEN and impinging on a 20-mm-thick beryllium target. Projectile fragments were separated and identified using ΔE-tof-Bρ by the large-acceptance two-stage separator BigRIPS. Tof was measured using two thin plastic scintillators placed at the intermediate and final foci of the second stage of BigRIPS. Bρ was measured from position measurement at the intermediate focus using the plastic scintillator. ΔE was measured using a stack of six silicon detectors installed at the final focus. Optimum settings of Bρ were tuned to transmit ^{33}F for 14 hours and ^{36}Ne+^{39}Na for 7.8 hours. The Be target was irradiated with 1.4×10^{17} and 7.8×10^{16} ^{48}Ca ions, respectively. Measured Z vs A/Z particle-identification plot. No ^{35}Ne events were observed in either setting. Under ^{33}F setting, the expected ^{35}Ne yields obtained from LISE++ are 177 *53* using the production σ=37.8 fb from EPAX 2.15 systematics and 69 *17* using the production σ=14.8 fb *36* from Q_g systematics.

2022Ah02: Same experimental setup as 2019Ah07 with 540-pnA ^{48}Ca beam. Optimum settings of Bρ were tuned to transmit ^{39}Na for 46.1 hours and ^{36}Ne for 25.3 hours. Measured Z vs A/Z particle-identification plot. No ^{35}Ne events were observed in either setting. Also see 2025Ku15.

2023Ra22: Relativistic Hartree-Bogoliubov model with nuclear energy density functional (EDF) calculations of binding energies.

2020Mi15: VS-IMSRG ab initio calculations of ground-state energies and S(2n).

^{35}Ne Levels

E(level)	Comments
0?	%n=?; %2n=? Evaluators estimate the probability of not observing ^{35}Ne events by chance is 2.6×10^{-23} using the lowest expected yield of 52 events (2019Ah07) and Poisson probability distributions. ^{35}Ne is determined to be unbound at a confidence level of 1-2.6×10^{-23}. The heaviest bound neon isotope is ^{34}Ne. J$^{\pi}$: 5/2$^-$ calculated projection of the odd-neutron angular momentum along the symmetry axis and parity of the wave function (2019Mo01). $T_{1/2}$: 2.7 ms calculated with respect to Gamow-Teller QRPA transitions and phenomenological first-forbidden contributions (2019Mo01).

Adopted Levels, Gammas

Q(β^-)=22190 *syst*; S(n)=1920 *syst*; S(p)=22300 *syst*; Q(α)=−21440 *syst* 2021Wa16
ΔQ(β^-)=720, ΔS(n)=300, ΔS(p)=840, ΔQ(α)=860 (syst,2021Wa16).
S(2n)=2090 *810*, Q(β^-n)=21440 *670* (syst,2021Wa16).
Isotope discovery (2012Th10): Ir(p,X)^{35}Na at CERN (1983La12).
^{35}Na production:
2025Ku15,2022Ah02,2019Ah07: ^{9}Be(^{48}Ca,X) at RIKEN.
2002LuZT: Ta(^{48}Ca,X) at GANIL.
^{35}Na decay measurements:
2022Cr03: ^{9}Be(^{48}Ca,X) at FRIB. Measured $T_{1/2}$.
2013StZY: ^{9}Be(^{48}Ca,X) at RIKEN. Measured $T_{1/2}$ and β^--delayed γ rays.
1983La12,1984La03: Ir(p,X) at CERN. Measured $T_{1/2}$ and β^--delayed neutrons.
^{35}Na mass measurements: None.
Theoretical calculations (binding energies, deformation, quadrupole moments, radii, levels, J, π, mass, $T_{1/2}$, etc.): 2024Lu06, 2023Zh15, 2023Ba27, 2022Ot01, 2020Ch21, 2020Ts03, 2013Li39, 2013Sh05, 2009Ly01, 2004Ge02, 2004Lu10, 2002Sa08, 1997Mo25, 1991Pa19, 1991Pa21, 1989Ly01, 1987SaZQ, 1985Ly02, 1975Ca27.

^{35}Na Levels

Cross Reference (XREF) Flags

A ^{9}Be(^{48}Ca,^{35}Na)
B C(^{36}Mg,^{35}Naγ)

E(level)	J$^{\pi}$†	$T_{1/2}$	XREF	Comments
0.0‡	(3/2$^+$)	2.1 ms *4*	AB	%β^-=100; %β^-n>0; %β^-2n=?; %β^-3n=?; %β^-4n=? ^{35}Na β^--delayed neutrons have been observed by 1983La12. Experimental %β^-n values are unknown. Theoretical %β^-0n=1.4, %β^-1n=73.5, %β^-2n=20.1, %β^-3n=4.8 (2021Mi17). Theoretical %β^-0n=14.0, %β^-1n=73.0, %β^-2n=10.0, %β^-3n=3.0 (2019Mo01). $T_{1/2}$: weighted average of 2.4 ms *3* (stat) *2* (syst) (2022Cr03, implant-β correlation), 2.4 ms *3* (stat) *6* (syst) (2013StZY, implant-β correlation), and 1.5 ms *5* (1983La12,1984La03, decay curve of βn-coin).
373‡ *5*	(5/2$^+$)		B	
1014‡ *17*	(7/2$^+$)		B	

† From Monte-Carlo shell-model calculations using the SPDF-M effective interaction (2014Do05).
‡ Band(A): K^{π}=(3/2$^+$) rotational band predicted by the shell model (2014Do05).

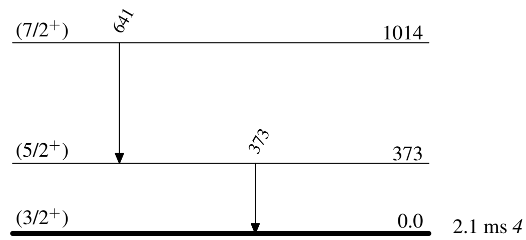
γ(^{35}Na)

E_i(level)	J^{π}_i	E_{γ}†	E_f	J^{π}_f
373	(5/2$^+$)	373 *5*	0.0	(3/2$^+$)
1014	(7/2$^+$)	641 *16*	373	(5/2$^+$)

† From C(^{36}Mg,^{35}Naγ).

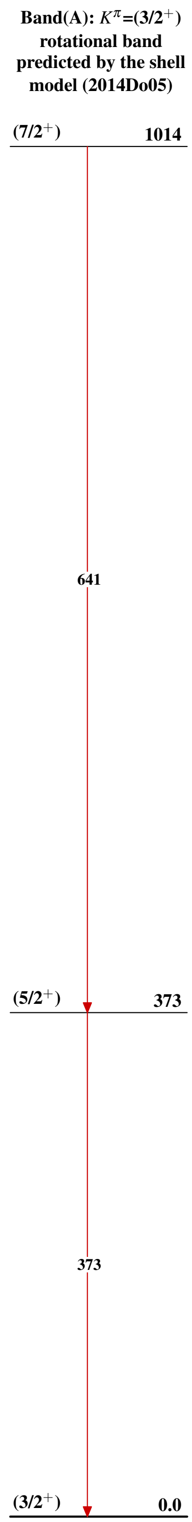
Adopted Levels, Gammas

Level Scheme



$^{35}_{11}Na_{24}$

Adopted Levels, Gammas



$^{35}_{11}Na_{24}$

^{9}Be(^{48}Ca,^{35}Na) 2013StZY,2022Cr03

2013StZY: ^{35}Na was produced via the projectile fragmentation of a 345-MeV/nucleon, 70-pnA $^{48}Ca^{20+}$ primary beam from the linear accelerator RILAC and the four cyclotrons RRC, fRC, IRC, and SRC at RIKEN impinging on a 15-mm-thick ^{9}Be target. The secondary cocktail beam was selected by the BigRIPS separator and the zero-degree spectrometer (ZDS) using the Bρ-ΔE-ToF method, and implanted into the Cylindrical Active Implantation Target for Exotic Nuclei (CAITEN) consisting of a segmented movable hollow-cylindrical-shaped plastic scintillator and a stationary ring of 24 position-sensitive photomultiplier tubes (PSPMTs) arranged on a ring inside the scintillator at the height of the beam line. To reduce background buildup, the scintillator barrel was rotated quickly and axially-moved slowly in vertical direction, resulting in a helix-shaped motion. β particles were detected by the CAITEN and γ rays were detected using three HPGe clover detectors. Measured Eγ, $\beta\gamma$-coin, and implant-β correlation, and deduced the $T_{1/2}$ of ^{35}Na. Comparisons with QRPA and shell-model calculations.

2022Cr03: ^{35}Na was produced via the projectile fragmentation of a 172.3-MeV/nucleon, 120-pnA $^{48}Ca^{20+}$ primary beam from the FRIB linac impinging on an 8.89-mm-thick ^{9}Be target. The secondary cocktail beam centered around ^{42}Si was selected by the ARIS separator and implanted into a 5-mm-thick YSO segmented scintillator sandwiched between two plastic scintillator veto detectors. Surrounding the implantation array were 11 HPGe clover detectors and 15 fast-timing $LaBr_3$ detectors, and the VANDLE array of 88 neutron detectors. Two Si PIN detectors and a plastic scintillator were placed 1.5 m upstream in the last diagnostic detector box of the beamline. Ions were identified event-by-event by energy loss in the upstream PIN detector (Z determination) and by time-of-flight between a plastic timing scintillator at the start of stage 3 of ARIS and the timing scintillator in the diagnostic detector box (A/Q determination) over a flight path of 33.5 m. Measured implant-β correlation and deduced $T_{1/2}$ of ^{35}Na. Comparisons with QRPA and shell-model calculations.

^{35}Na Levels

E(level)	$T_{1/2}$	Comments
0.0	2.1 ms *4*	$T_{1/2}$: From the Adopted Levels of ^{35}Na. Values from this dataset: 2.4 ms *3* (stat) *2* (syst) (2022Cr03, implant-β correlation); 2.4 ms *3* (stat) *6* (syst) (2013StZY, implant-β correlation).

C(^{36}Mg,^{35}Naγ) 2014Do05

2014Do05: A 345-MeV/nucleon ^{48}Ca primary beam was accelerated by the superconducting ring cyclotron (SRC) at RIKEN. A 236-MeV/nucleon ^{36}Mg secondary beam was produced by the fragmentation of ^{48}Ca on a Be target and separated by the BigRIPS separator using the Bρ-ΔE-Bρ method. The secondary reaction targets were 2.54 g/cm^2 thick carbon and 2.13 g/cm^2 thick CH_2 polyethylene. The secondary reaction products were identified by the magnetic spectrometer ZeroDegree using the Bρ-ΔE-Bρ method. γ rays in coincidence with ^{35}Na were detected using the DALI2 array of 186 large NaI(Tl) detectors. Measured Eγ with Doppler correction, Iγ, and $\gamma\gamma$-coin. Deduced levels, J, π, and bands. Compared with shell-model calculations using the SPDF-M effective interaction.

^{35}Na Levels

E(level)[†]	J$^{\pi}$[‡]
0[#]	(3/2$^+$)
373[#] *5*	(5/2$^+$)
1014[#] *17*	(7/2$^+$)

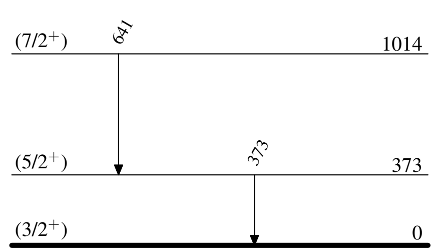
[†] From a least-squares fit to γ-ray energies.
[‡] From shell-model calculations.
[#] Band(A): K^{π}=(3/2$^+$) rotational band predicted by the shell model.

γ(^{35}Na)

E$_\gamma$	E$_i$(level)	J$^{\pi}_i$	E$_f$	J$^{\pi}_f$
373 *5*	373	(5/2$^+$)	0	(3/2$^+$)
641 *16*	1014	(7/2$^+$)	373	(5/2$^+$)

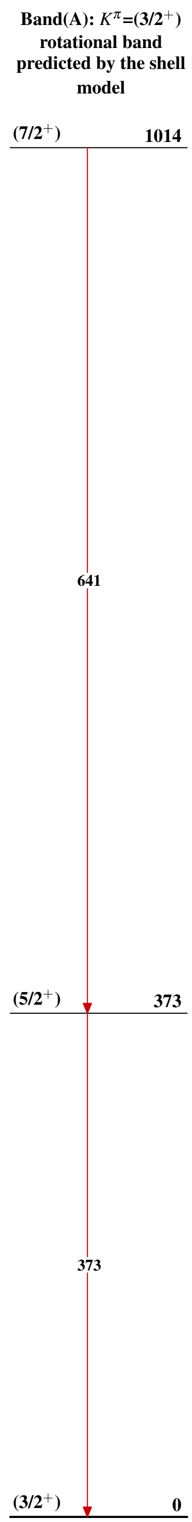
C(^{36}Mg,^{35}Naγ) 2014Do05

Level Scheme



$^{35}_{11}Na_{24}$

C(^{36}Mg,^{35}Naγ) 2014Do05



$^{35}_{11}Na_{24}$

Adopted Levels, Gammas

$Q(\beta^-)$=15753 *10*; S(n)=859.8 *87*; S(p)=2.344×10^4 *60*; $Q(\alpha)$=-1.808×10^4 *27*

$Q(\beta^-)$,S(n),S(p),$Q(\alpha)$: Deduced by the evaluators using mass excesses of 15529.5 *71* for ^{35}Mg measured by 2025Ly01, and 8318 *5* for ^{34}Mg: a weighted average of 8323 *7* (2019As04) and 8315 *5* (2025Ly01); −224 *7* for ^{35}Al, 31680 *600* for ^{34}Na, and 31180 *270* for ^{31}Ne from 2021Wa16. Values from 2021Wa16: $Q(\beta^-)$=15860 *270*, S(n)=750 *270*, S(p)=23330 *660*, $Q(\alpha)$=−17970 *380*.

S(2n)=5576.0 *76*, $Q(\beta^-n)$=10455.8 *74*, from mass excesses of 15529.5 *71* for ^{35}Mg measured by 2025Ly01; 4962.9 *27* for ^{33}Mg and −2997.6 *21* for ^{34}Al from 2021Wa16. Values from 2021Wa16: S(2n)=5470 *270*, $Q(\beta^-n)$=10570 *270*.

S(2p)=45070 *660* (syst) (2021Wa16).

Isotope discovery (2012Th10): Ta(^{48}Ca,X) projectile fragmentation at GANIL (1989Gu03,1991Or01).

^{35}Mg production:

2012Kw02: ^{35}Mg produced by ^{nat}Ni(^{40}Ar,X) at E(^{40}Ar)=140 MeV/nucleon at NSCL. Measured fragmentation cross sections, parallel momentum transfers, and widths. Compared with empirical formula EPAX, and predictions from internuclear cascade and deep inelastic models using Monte Carlo ISABEL-GEMINI and DIT-GEMINI codes.

2011FuZZ: ^{35}Mg produced by ^{9}Be(^{48}Ca,X) fragmentation at E(^{48}Ca)=345 MeV/nucleon at RIKEN. Measured thick target fragmentation, deduced production cross sections, and compared with EPAX-2.15 systematics.

2007Ts09: Analyzed fragmentation σ of n-rich Na, Mg isotopes (including ^{35}Mg) from ^{9}Be,^{181}Ta(^{48}Ca,X). Used systematics based on average binding energy to extrapolate towards drip line, predicting σ for ^{40}Mg and discussing ^{39}Na. Compared different extrapolation models.

^{35}Mg decay measurements:

2013StZY: ^{9}Be(^{48}Ca,X) at RIKEN. Measured $T_{1/2}$.

1999YoZW: ^{9}Be(^{48}Ca,X) and ^{181}Ta(^{48}Ca,X) at RIKEN. Measured $T_{1/2}$ and $\%\beta^-$n.

^{35}Mg radius measurements:

2011Ka01: ^{35}Mg produced by ^{9}Be(^{48}Ca,X) fragmentation at GSI. Measured interaction cross sections with C and CH_2 targets at 900 MeV/nucleon. Deduced rms matter radii.

2006Kh08: ^{35}Mg produced by ^{181}Ta(^{48}Ca,X) fragmentation at E(^{48}Ca)=60.3 MeV/nucleon at GANIL. Measured energy-integrated reaction cross sections at 30-65 MeV/nucleon using a silicon telescope as both active target and detector. Deduced reduced strong absorption radii, isospin dependence, and possible halo structure or large deformation.

^{35}Mg mass measurements: 2025Ly01, 2007Ju03, 2001Sa72, 2000Sa21, 1991Or01.

Theoretical calculations (binding energies, deformation, quadrupole moments, radii, levels, J, π, mass, $T_{1/2}$, etc): 2023Ra22, 2021Ka07, 2020Mi15, 2016Ba59, 2016Sa46, 2016Sh05, 2015Sh21, 2014Ga13, 2014Wa14, 2013Ch31, 2013Li39, 2013Sh05, 2012Fo27, 2012Ho19, 2007Ha53, 2006Zh19, 2005Ch71, 2004Kh16, 1996Re10, 1991Pa19, 1991Pa21.

The level scheme is based on that of 2011Ga15 and 2017Mo26.

^{35}Mg Levels

Cross Reference (XREF) Flags

A	^{35}Na β^- decay (2.1 ms)
B	^{9}Be(^{38}Si,^{35}Mgγ)
C	C(^{36}Mg,^{35}Mgγ),(^{37}Al,^{35}Mgγ)

E(level)†	J^π	$T_{1/2}$	XREF	Comments
0	(3/2$^-$,5/2$^-$)	11.3 ms *6*	BC	$\%\beta^-$=100; $\%\beta^-$n=52 *46*; $\%\beta^-$2n=? $\%\beta^-$n: from 1995ReZZ,2008ReZZ. Other: 52 *11* (1999YoZW, preliminary). Theoretical $\%\beta^-$0n=29, $\%\beta^-$1n=66, $\%\beta^-$2n=5 (2021Mi17). Theoretical $\%\beta^-$0n=65, $\%\beta^-$1n=32, $\%\beta^-$2n=3 (2019Mo01). J^π: 3/2$^-$ from shell−model calculations with the SDPF−M and SDPF−M+2$p_{1/2}$ interactions (2017Mo26). Nearly degenerate 30-keV 3/2$^-$ and 5/2$^-$ g.s. from Monte Carlo shell-model calculations with the SDPF-M interaction (2011Ga15), and 3/2$^-$ g.s. from shell-model calculations with the SDPF-U interaction (2011Ga15). 3/2$^-$ from projection of the odd-neutron angular momentum along the symmetry axis and parity of the wave function (2019Mo01). Others: 3/2$^+$ from antisymmetrized molecular dynamics (AMD) calculations with the Gogny D1S force (2017Mo26).

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued)

^{35}Mg Levels (continued)

E(level)†	J^{π}	XREF	Comments
			$T_{1/2}$: 11.3 ms *5* (stat) *4* (syst) (2013StZY, implant-β correlation). Other: 72 ms *43* (2008ReZZ,1995ReZZ) and ≈9 ms (1999YoZW, implant-β correlation, preliminary). Reduced strong absorption radius r_0^2=1.64 fm^2 *15* from the energy-integrated σ of Si(^{35}Mg,X) (2006Kh08). The rms matter radius R^{m}_{rms}=3.40 fm *24* from the interaction σ of C,CH_2(^{35}Mg,X) (2011Ka01).
0+x		BC	E(level): x≤80 keV (2011Ga15 detection threshold); x≤200 keV (2017Mo26 detection threshold). J^{π}: Shell−model calculations with the SDPF−M+2$p_{1/2}$ interaction predict a 1/2$^-$ level at 141 keV (2017Mo26). Shell-model calculations with the SDPF-M interaction predict a 5/2$^-$ level at 84 keV (2017Mo26). Monte Carlo shell-model calculations with the SDPF-M interaction predict a 3/2$^-$ level at 30 keV (2011Ga15).
0+y?		C	E(level): y≤200 keV (2017Mo26 detection threshold). 2017Mo26 suggested a low-lying level from the observed 42(1)% neutron knockout from the l=3 orbit in ^{36}Mg in the inclusive parallel momentum distribution.
206+x *8*		C	J^{π}: 2017Mo26 states that based on the observed weak γ-ray intensity, this level is not the 1/2$^-$ level at 141 keV predicted by shell−model calculations with the SDPF−M+2$p_{1/2}$ interaction.
445+x *5*	(3/2$^+$,5/2$^+$)	BC	J^{π}: L(^{36}Mg,^{35}Mg)=(2) from 0$^+$. 3/2$^+$ from shell−model calculations with the SDPF−M+2$p_{1/2}$ interaction (2017Mo26).
619+x *7*	(1/2$^-$,3/2$^-$)	BC	J^{π}: L(^{36}Mg,^{35}Mg)=(1) from 0$^+$. 3/2$^-$ from shell−model calculations with the SDPF−M+2$p_{1/2}$ interaction (2017Mo26).
670+x *8*		BC	

† From Eγ data in (^{38}Si,^{35}Mgγ) and (^{36}Mg,^{35}Mgγ),(^{37}Al,^{35}Mgγ).

γ(^{35}Mg)

E_i(level)	J^{π}_i	E_{γ}	I_{γ}	E_f	Comments
206+x		206 *8*	100	0+x	E_{γ}: From 2017Mo26.
445+x	(3/2$^+$,5/2$^+$)	445 *5*	100	0+x	E_{γ}: weighted average of 443 *7* (2017Mo26) and 446 *5* (2011Ga15).
619+x	(1/2$^-$,3/2$^-$)	619 *7*	100	0+x	E_{γ}: weighted average of 616 *8* (2017Mo26) and 621 *7* (2011Ga15).
670+x		670† *8*	100	0+x	E_{γ}: From 2011Ga15, as this γ is not resolved from the 616γ in 2017Mo26, but its presence is indicated in the fit of the spectrum. 2017Mo26 states that the origin of the 670γ remains vague.

† Placement of transition in the level scheme is uncertain.

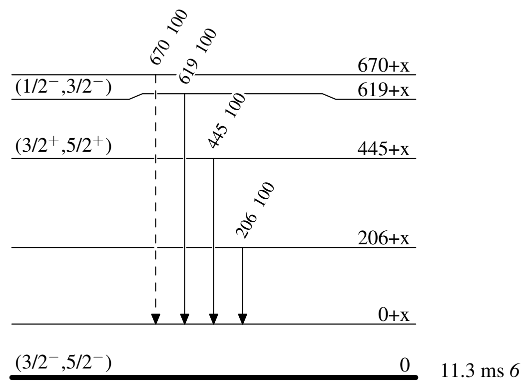
Adopted Levels, Gammas

Legend

Level Scheme

Intensities: Relative photon branching from each level

γ Decay (Uncertain)



$^{35}_{12}Mg_{23}$

^{35}Na β^- decay (2.1 ms) 2013StZY,2022Cr03,1983La12

Parent: ^{35}Na: E=0; J$^\pi$=(3/2$^+$); $T_{1/2}$=2.1 ms *4*; Q(β^-)=22190 *syst*; %β^- decay=100
^{35}Na-J$^\pi$,$T_{1/2}$: From the Adopted Levels of ^{35}Na.
^{35}Na-Q(β^-): 22190 *720* (syst,2021Wa16).
^{35}Na-%β^- decay: %β^-=100, %β^-n>0.

2013StZY: ^{35}Na was produced via the projectile fragmentation of a 345-MeV/nucleon, 70-pnA $^{48}Ca^{20+}$ primary beam from the linear accelerator RILAC and the four cyclotrons RRC, fRC, IRC, and SRC at RIKEN impinging on a 15-mm-thick ^{9}Be target. The secondary cocktail beam was selected by the BigRIPS separator and the zero-degree spectrometer (ZDS) using the Bρ-ΔE-ToF method, and implanted into the Cylindrical Active Implantation Target for Exotic Nuclei (CAITEN) consisting of a segmented movable hollow-cylindrical-shaped plastic scintillator and a stationary ring of 24 position-sensitive photomultiplier tubes (PSPMTs) arranged on a ring inside the scintillator at the height of the beam line. To reduce background buildup, the scintillator barrel was rotated quickly and axially-moved slowly in vertical direction, resulting in a helix-shaped motion. β particles were detected by the CAITEN and γ rays were detected using three HPGe clover detectors. Measured Eγ, $\beta\gamma$-coin, and implant-β correlation, and deduced the $T_{1/2}$ of ^{35}Na. Comparisons with QRPA and shell-model calculations.

2022Cr03: ^{35}Na was produced via the projectile fragmentation of a 172.3-MeV/nucleon, 120-pnA $^{48}Ca^{20+}$ primary beam from the FRIB linac impinging on an 8.89-mm-thick ^{9}Be target. The secondary cocktail beam centered around ^{42}Si was selected by the ARIS separator and implanted into a 5-mm-thick YSO segmented scintillator sandwiched between two plastic scintillator veto detectors. Surrounding the implantation array were 11 HPGe clover detectors and 15 fast-timing $LaBr_3$ detectors, and the VANDLE array of 88 neutron detectors. Two Si PIN detectors and a plastic scintillator were placed 1.5 m upstream in the last diagnostic detector box of the beamline. Ions were identified event by event by energy loss in the upstream PIN detector (Z determination) and time of flight between a plastic timing scintillator at the start of stage 3 of ARIS and the timing scintillator in the diagnostic detector box (A/Q determination) over a flight path of 33.5 m. Measured implant-β correlation and deduced $T_{1/2}$ of ^{35}Na. Comparisons with QRPA and shell-model calculations.

1983La12,1984La03: ^{35}Na was produced via the fragmentation of a 30 g/cm^2 iridium target by 10-GeV protons from the CERN synchrotron. The recoiling nuclear fragments are thermalized in heated graphite. The alkali elements selectively diffuse out of the target, are surface ionized, and analyzed by the online mass spectrometer, then focused at the bottom of a well-shaped NE213 liquid scintillator neutron detector. β particles were detected using a well-shaped thin plastic scintillator. Measured β and neutron counts. Deduced the $T_{1/2}$ of ^{35}Na. P_n values were deduced for other isotopes but not for ^{35}Na.

^{35}Na β^--delayed neutrons have been observed by 1983La12. Experimental %β^-n values are unknown.

γ(^{35}Mg)

Eγ	Comments
^{x}661	2013StZY observed this γ ray in the background-subtracted γ-ray spectrum in a time window of 0-10 ms after the implantation of ^{35}Na but did not observe it in later time windows. 2013StZY proposed that it could either be the deexcitation of the first 2$^+$ to 0$^+$ g.s. in ^{34}Mg or the deexcitation of an excited state in ^{35}Mg. Based on theoretical calculations (2019Mo01,2021Mi17), the β^-1n should be the dominant branch, implying that this γ ray is more likely from ^{34}Mg.

^{x} γ ray not placed in level scheme.

^{9}Be(^{38}Si,^{35}Mgγ) 2011Ga15

^{38}Si→2p+1n+^{35}Mg.

2011Ga15: A ^{38}Si secondary beam was produced via the projectile fragmentation of a 140-MeV/nucleon ^{48}Ca primary beam impinging on a ^{9}Be target and selected by the A1900 separator at NSCL, MSU. E(^{38}Si)=83 MeV/nucleon in the middle of the 376(4)-mg/cm^2 ^{9}Be secondary target. The reaction leading to ^{35}Mg from ^{38}Si is likely dominated by the two-proton knockout into the continuum of ^{36}Mg and subsequent neutron emission. The reaction residues were selected and identified by the S800 spectrograph using the ΔE-ToF method. The γ rays in coincidence with reaction residues were detected using a 32-fold segmented high-purity germanium detector array (SeGA) at 90° and 37°. Measured Doppler-corrected Eγ(>80 keV) and deduced levels. Results were compared with Monte Carlo shell-model calculations using the SDPF-M effective interaction that allows for unrestricted mixing of neutron particle-hole configurations across the N = 20 gap, and with conventional shell-model calculations with the SDPF-U effective interaction that does not include neutron intruder configurations in the model space.

^{35}Mg Levels

E(level)†	J$^\pi$	Comments
0	(5/2$^-$)	J$^\pi$: From shell-model calculations with the SDPF-M interaction.
0+x		
0+y		
446+x *5*		
621+y *7*		
670+y *8*		

† From measured Eγ. x≤80 keV and y≤80 keV (detection threshold). Shell-model calculations with the SDPF-M interaction predicts a 3/2$^-$ level at 30 keV.

γ(^{35}Mg)

2011Ga15 states that the γ-ray transitions they observed were not in coincidence and depopulate excited states at ≈500 and ≈700 keV to the ground state or the alleged near-degenerate excited state below 100-keV excitation energy. 2017Mo26 states that the difference of the γ-ray emission between 1n-knockout and 1p1n-removal reaction channels suggests that the 621 and 670-keV γ rays were emitted from two different excited states. Another possible placement of the 621 and 670-keV γ rays: 2011Ga15 states that it might be possible that the two transitions originate from the same level and that their energy difference of 49(11) keV corresponds to the energy spacing of the alleged (5/2$^-$,3/2$^-$) doublet near the ground state. In Adopted Gammas, all γ rays in this dataset are placed to feed the same final levels based on suggestions by 2017Mo26.

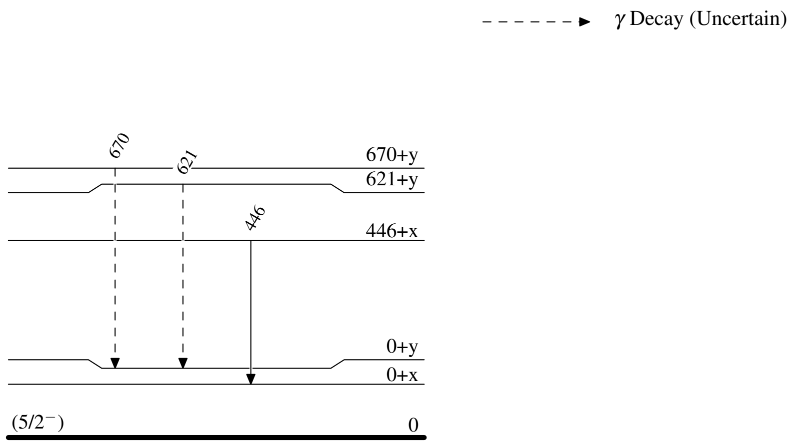
E$_\gamma$	E$_i$(level)	E$_f$	Comments
446 *5*	446+x	0+x	E$_\gamma$: strong γ transition observed in 2011Ga15; could correspond to a transition to the g.s. or the shell-model predicted low-lying excited state from any of the shell-model predicted levels at 350 keV with J^π=7/2$^-$; 360 keV with J^π=3/2$^+$, 400 keV with J^π=1/2$^-$, or 560 keV with J^π=7/2$^-$.
621† *7*	621+y	0+y	
670† *8*	670+y	0+y	

† Placement of transition in the level scheme is uncertain.

$^{9}Be(^{38}Si,^{35}Mg\gamma)$ 2011Ga15

Level Scheme

Legend



$^{35}_{12}Mg_{23}$

C(^{36}Mg,^{35}Mgγ),(^{37}Al,^{35}Mgγ) 2017Mo26

^{36}Mg→1n+^{35}Mg from J^π=0$^+$ ^{36}Mg ground state.
^{37}Al→1p1n+^{35}Mg from J^π=(5/2$^+$) ^{37}Al ground state.
2017Mo26: A secondary beam composed of ^{36}Mg and ^{37}Al was produced via the projectile fragmentation of a 345-MeV/nucleon ^{48}Ca primary beam impinging on a ^{9}Be target and selected by the BigRIPS separator at RIKEN. E(^{36}Mg)=235 MeV/nucleon and E(^{37}Al)=246 MeV/nucleon in front of the 2.54 g/cm^2 carbon secondary target. The reactions leading to ^{35}Mg from ^{36}Mg and ^{37}Al are likely 1n-knockout and 1p1n-removal reactions, respectively. The reaction residues were selected and identified by the Zero Degree spectrometer using the Bρ-ΔE-ToF method. The γ rays in coincidence with ^{35}Mg residues were detected using the DALI2 array of 186 NaI(Tl) crystals at 20°−150°. Measured Eγ(>200 keV), Iγ, (^{35}Mg)γ-coin, the inclusive one-neutron knockout cross section and exclusive γ-ray emission cross sections, and parallel momentum distributions of ^{35}Mg in coincidence with γ rays. Deduced levels, L-transfers, J, and π. Compared with shell-model calculations using the SDPF-M interaction in the sd shell with $\nu 1f_{7/2}$ and $\nu 2p_{3/2}$ orbits, and the SDPF-M interaction in a model space up to $\nu 2p_{1/2}$, and antisymmetrized molecular dynamics (AMD) model calculations using the Gogny D1S force.

^{35}Mg Levels

E(level)†	J$^\pi$‡	L‡	Comments
0			J$^\pi$: 3/2$^-$ from shell−model calculations with the SDPF−M and SDPF−M+2p$_{1/2}$ interactions; 3/2$^+$ from AMD with the Gogny D1S force (2017Mo26).
0+x			E(level): <200 keV; the detection threshold in 2017Mo26. The original placement in 2017Mo26 of all γ rays is to the ground state, which cannot be confirmed with existing data.
0+y?	(5/2$^-$,7/2$^-$)		E(level): <200 keV; the detection threshold. 2017Mo26 suggested a low-lying level from the observed 42(1)% neutron knockout from the l=3 orbit in the inclusive parallel momentum distribution.
206+x *8*			J$^\pi$: γ-ray intensity is too low to be assigned to the 1/2$^-$ level at 141 keV from shell-model calculations with the SDPF−M+2p$_{1/2}$ interaction (2017Mo26).
443+x *7*	(3/2$^+$,5/2$^+$)	(2)	
616+x *8*	(1/2$^-$,3/2$^-$)	(1)	
670+x *8*			

† From Eγ data. 2017Mo26 suggested that all the four observed γ rays were emitted independently and fed either the ground state or a low-lying excited state below 200 keV. But the suggested experimental level scheme in FIG.4 of 2017Mo26 (with no γ rays shown) indicates all four γ rays feeding the ground state, which is not adopted by the evaluators since it cannot be confirmed with existing data.
‡ 2017Mo26 deduced L by comparing the measured and eikonal-calculated parallel momentum distributions of ^{35}Mg residues. J^π options are deduced accordingly.

γ(^{35}Mg)

2017Mo26 states that no clear $\gamma\gamma$ coincidence was observed.
It is possible that the final levels of 206γ, 443γ, 616γ, and 670γ are not the same level from authors′ comparisons with shell-model calculations with different interactions.

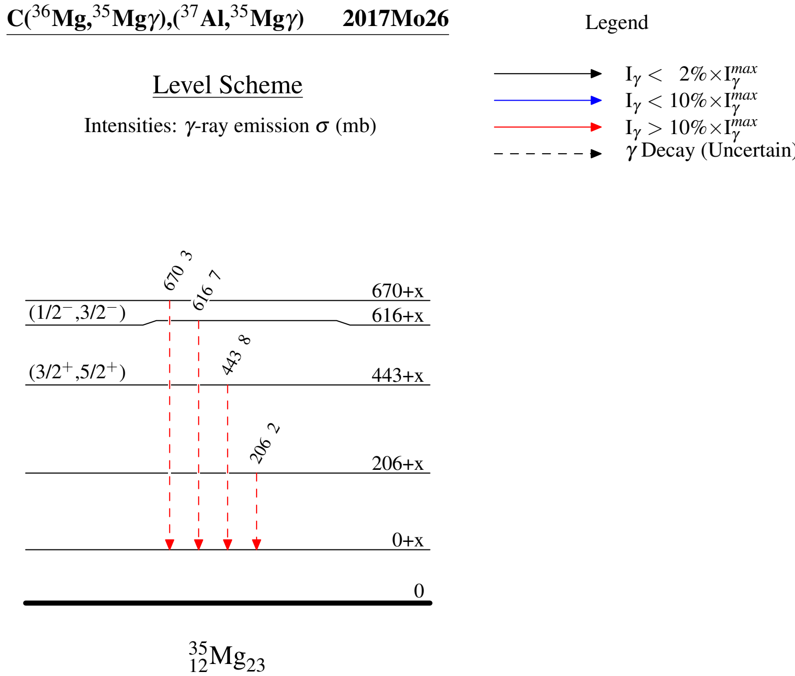
E$_\gamma$	I$_\gamma$†	E$_i$(level)	J$^\pi_i$	E$_f$	Comments
206‡ *8*	2 *1*	206+x		0+x	
443‡ *7*	8 *1*	443+x	(3/2$^+$,5/2$^+$)	0+x	2017Mo26 assigned this γ as the 3/2$^+$ at 788 keV to 3/2$^-$ g.s. transition in shell−model calculations with the SDPF−M+2p$_{1/2}$ interaction (2017Mo26).
616‡ *8*	7 *1*	616+x	(1/2$^-$,3/2$^-$)	0+x	2017Mo26 assigned this γ as the 3/2$^-$ at 664 keV to 1/2$^-$ at 141 keV transition in shell−model calculations with the SDPF−M+2p$_{1/2}$ interaction (2017Mo26).
670‡ *8*	3 *1*	670+x		0+x	E$_\gamma$: From 2011Ga15, as this γ is not resolved from the 616γ in 2017Mo26, but its presence is indicated in the fit of the spectrum. 2017Mo26 states that the origin of the 670γ remains vague.

Continued on next page (footnotes at end of table)

C(^{36}Mg,^{35}Mgγ),(^{37}Al,^{35}Mgγ) 2017Mo26 (continued)

γ(^{35}Mg) (continued)

† γ-ray emission σ (mb).
‡ Placement of transition in the level scheme is uncertain.



Adopted Levels, Gammas

Q(β^-)=14170 *40*; S(n)=5297 *8*; S(p)=15831 *9*; Q(α)=−14894 *8*   2021Wa16

S(p),Q(α): Deduced by evaluators using mass excesses of 8318 *5* for ^{34}Mg: a weighted average of 8323 *7* (2019As04) and 8315 *5* (2025Ly01), and 12245.3 *26* for ^{31}Na measured by 2025Ly01; −224 *7* for ^{35}Al from 2021Wa16. Values from 2021Wa16: S(p)=15836 *10*, Q(α)=−14895 *16*.

S(2p)=38400.5 *85*, Q(ε)=−15753 *10*, from mass excesses of 23598.8 *43* for ^{33}Na and 15529.5 *71* for ^{35}Mg measured by 2025Ly01; −224 *7* for ^{35}Al from 2021Wa16. Values from 2021Wa16: S(2p)=38580 *450*, Q(ε)=−15860 *270*.

S(2n)=7869 *10*, Q(β^-n)=11697 *7* (2021Wa16).

Isotope discovery (2012Th10): C(^{40}Ar,X) projectile fragmentation at Berkeley (1979Sy01).

^{35}Al production:

2015Mo17: ^{9}Be(^{40}Ar,X) at E(^{40}Ar)=95 MeV/nucleon at RIKEN. Measured angular distributions and transverse momentum distributions of fragments. Deduced formulation for the width of transverse momentum distribution as a function of fragment velocity.

2012Kw02: E(^{40}Ar)=140 MeV/nucleon impinged on ^{9}Be, natural nickel, and ^{181}Ta targets at NSCL. Measured fragmentation cross sections, parallel momentum transfers, and widths. Compared with empirical formula EPAX, and predictions from internuclear cascade and deep inelastic models using Monte Carlo ISABEL-GEMINI and DIT-GEMINI codes.

2012Zh06: ^{9}Be(^{40}Ar,X) at E(^{40}Ar)=57 MeV/nucleon at HIRFL. Measured momentum distributions and production cross sections of fragments. Observed competition between projectile fragmentation and other mechanisms. Compared with EPAX, abrasion-ablation, and HIPSE models. Studied target dependence of fragment cross sections.

2007No13: ^{9}Be(^{40}Ar,X) at E(^{40}Ar)=100 MeV/nucleon at RIKEN. Measured fragment momentum distributions and production cross sections.

^{35}Al decay measurements:

2017Ha23: ^{9}Be(^{40}Ar,X) at HIRFL. Measured $T_{1/2}$.

2005Ti11,2006AnZW: (^{36}S,X) at GANIL. Measured $T_{1/2}$, β^--delayed γ and neutron spectroscopy.

2001Nu01,2002Nu02: U(p,X) at CERN. Measured $T_{1/2}$, β^--delayed γ and neutron spectroscopy.

1999YoZW: ^{9}Be(^{48}Ca,X) and ^{181}Ta(^{48}Ca,X) at RIKEN. Measured $T_{1/2}$ and %β^-n.

1995ReZZ,2008ReZZ: ^{232}Th(p,X) at LAMPF. Measured $T_{1/2}$ and %β^-n, and average E_n.

1989Le16,1989MuZU: ^{181}Ta(^{48}Ca,X) at GANIL. Measured $T_{1/2}$ and %β^-n.

1988Mu08,1988MuZY,1988BaYZ: ^{181}Ta(^{86}Kr,X) at GANIL. $T_{1/2}$ and %β^-n.

^{35}Al radius measurement:

2006Kh08: ^{35}Al produced by ^{181}Ta(^{48}Ca,X) fragmentation at E(^{48}Ca)=60.3 MeV/nucleon at GANIL. Measured energy-integrated reaction cross sections at 30-65 MeV/nucleon using a silicon telescope as both active target and detector. Deduced reduced strong absorption radii, isospin dependence, and possible halo structure or large deformation.

^{35}Al knockout-reaction measurements:

2012No05: ^{33,34,35,36}Al produced by Be(^{48}Ca,X) fragmentation at E(^{48}Ca)=900 MeV/nucleon at GSI. Measured 1n removal cross sections and longitudinal momentum distributions of the residues. Deduced single-particle occupancies in the ground states of ^{33,34,35}Al. σ(^{35}Al→^{34}Al)=75 mb *4* and σ(^{36}Al→^{35}Al)=95 mb *5*.

2010Ro23: ^{35}Al produced by ^{9}Be(^{40}Ar,X) fragmentation at E(^{40}Ar)=700 MeV/nucleon at GSI. Measured 1n knockout cross sections and longitudinal momentum distributions of the residues. σ(^{35}Al→^{34}Al)=65 mb *18*.

^{35}Al in-beam γ spectroscopy:

2006FuZX: ^{35}Al produced by Be,C(^{40}Ar,X) and observed one 760.1-keV *21* γ ray from He(^{35}Al,^{35}Alγ) without placing it in the level scheme.

^{35}Al mass measurements: 2017Ga20, 2007Ju03, 1991Or01, 1991Zh24, 1987Gi05.

Theoretical calculations (binding energies, deformation, quadrupole moments, radii, levels, J, π, mass, $T_{1/2}$, etc): 2016Sa46, 2014Ca21, 2013Li39, 2013Sh05, 2011Ki12, 2009Yo05, 2004Kh16, 1994Po05.

^{35}Al Levels

Cross Reference (XREF) Flags

A	^{36}Mg β^-n decay (6.9 ms)	D	Pb(^{35}Al,^{34}Alnγ)
B	^{9}Be(^{36}Si,^{35}Alγ)	E	Coulomb excitation
C	He(^{35}Al,^{35}Al′γ)		

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued)

^{35}Al Levels (continued)

E(level)†	J^π	$T_{1/2}$	XREF	Comments
0	(5/2)$^+$	38.1 ms *4*	AB DE	%β^-=100; %β^-n=37 *4*; %β^-2n=? J^π: From shell-model calculations and L(^{36}Si,^{35}Al)=2 from 0$^+$. $T_{1/2}$: weighted average of 38.4 ms *3* (2017Ha23, implant-β correlation), 36.8 ms *5* (2005Ti11,2006AnZW, implant-β correlation), 38.6 ms *4* (2001Nu01,2002Nu02, three β-counting rates and one γ-counting rate), 30 ms *4* (1995ReZZ,2008ReZZ, implant-β/n correlation), 170 ms *+90−50* (1989Le16,1989MuZU, β-decay curve), 30 ms *10* (1988DuZT, beam pulsations), and 130 ms *+100−50* (1988Mu08,1988BaYZ, β counting rate). %β^-n: Unweighted average of 41 *2* (2005Ti11,2006AnZW), 41 *13* (2001Nu01,2002Nu02), 26 *4* (1995ReZZ,2008ReZZ), and 40 *10* (1989Le16,1989MuZU). Others: 43 *9* (1999YoZW, preliminary), 87 *+37−25* (1988Mu08,1988BaYZ) with a non-physical upper bound. Note that the original value reported in 2005Ti11 is 38 *2* (2006AnZW report 38 *3*) and the evaluators have added intensities of γ rays deexciting the 4379 (0.9% *2*) and 4255 (2.0% *8*) levels in ^{34}Si not included in their reported %β^-n value, as the authors state that these levels are fed by either neutrons with energy below their detection threshold or by weak transitions that cannot be identified in the neutron spectrum. %β^-2n: No evidence was found for any A=33 activity and 2n emission was thus found negligible (2001Nu01). Reduced strong absorption radius r_0^2=1.188 fm^2 *14* from the energy-integrated σ of Si(^{35}Al,X) (2006Kh08). Major configurations for J^π=5/2$^+$ of ^{35}Al g.s. from Pb(^{35}Al,^{34}Alnγ): (g.s., 4$^-$ in ^{34}Al)$\otimes\nu p_{3/2}$, S=0.36 *9* (2017Ch36); (46 keV, 1$^+$ in ^{34}Al)$\otimes\nu d_{3/2}$, S=1.47 *22* (2017Ch36); (1.4 MeV, 2$^+$ in ^{34}Al)$\otimes\nu s_{1/2}$, S=0.16 *1* (2021Bh12); (2.5 MeV, 3$^-$ in ^{34}Al)$\otimes\nu p_{3/2}$, S=1.48 *18* (2021Bh12).
802 *3*			AB	
1007 *4*			B E	XREF: E(1020) B(E2)↑=0.0142 *52* from Coulomb excitation assuming E2 excitation mode (1999Ib01). Others: B(E1)↑≤0.00020 *9*, B(E2)↑≤0.0125 *56*, B(M1)↑≤0.0024 *11*.
1866 *4*			B	
1975 *4*	3/2$^+$,5/2$^+$‡		B	
2734 *7*			B	
3245 *5*	3/2$^+$,5/2$^+$‡		B	
4275? *9*	3/2$^+$,5/2$^+$‡		B	XREF: B(?) Level exists assuming that the 4275γ feeds the ground state of ^{35}Al (2014St18).

† From a least-squares fit to γ-ray energies.
‡ From L(^{36}Si,^{35}Al)=2 of parallel momentum distributions of ^{35}Al residues (2014St18).

γ(^{35}Al)

E_i(level)	J^π_i	E_γ†	I_γ†	E_f	J^π_f	Comments
802		802 *4*	100	0	(5/2)$^+$	
1007		1006 *6*	100	0	(5/2)$^+$	E_γ: weighted average of 1003 *4* from (^{36}Si,^{35}Alγ) and 1020 *9* from Coulomb excitation.
1866		859 *4*	100 *8*	1007		
		1064 *4*	22 *6*	802		
1975	3/2$^+$,5/2$^+$	968 *4*	59 *4*	1007		
		1174 *5*	37 *4*	802		
		1972 *6*	100 *7*	0	(5/2)$^+$	

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued)

$\gamma(^{35}Al)$ (continued)

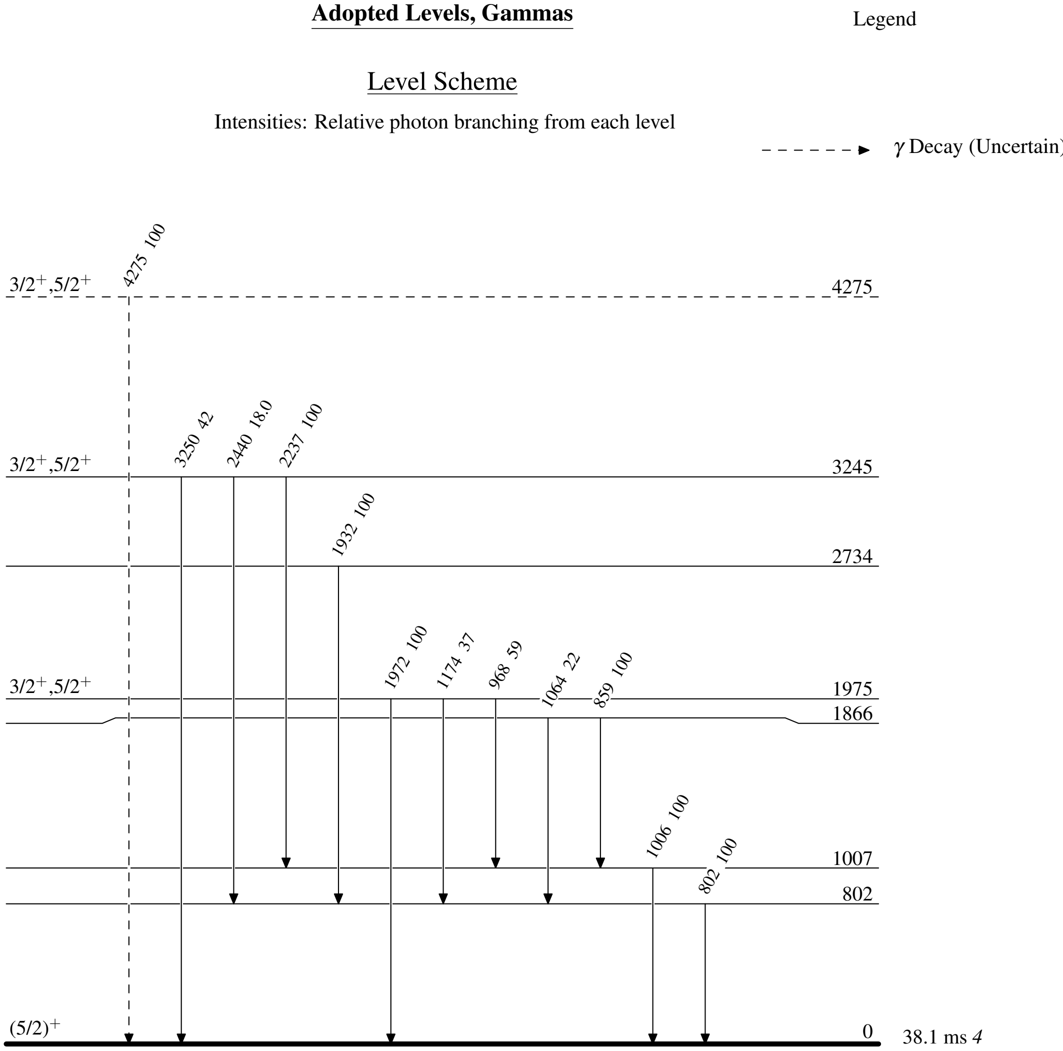
E_i(level)	J^π_i	E_γ†	I_γ†	E_f	J^π_f	Comments
2734		1932 *6*	100	802		
3245	$3/2^+,5/2^+$	2237 *6*	100 *8*	1007		
		2440 *7*	18.0 *26*	802		
		3250 *8*	42 *5*	0	$(5/2)^+$	
4275?	$3/2^+,5/2^+$	4275‡ *9*	100	0	$(5/2)^+$	2014St18 states that the 4275γ transition could populate the ground state or one of the lowest two excited states. In the absence of any observed $\gamma\gamma$ coincidences, they tentatively place the 4275-keV γ transition feeding the ground state.

† From $^9Be(^{36}Si,^{35}Al\gamma)$, unless otherwise noted.
‡ Placement of transition in the level scheme is uncertain.

Adopted Levels, Gammas

Level Scheme

Intensities: Relative photon branching from each level



$^{35}_{13}Al_{22}$

^{36}Mg β^-n decay (6.9 ms) 2023Lu07

Parent: ^{36}Mg: E=0; $J^\pi=0^+$; $T_{1/2}$=6.9 ms *+7−8*; Q(β^-n)=1.253×10^4 *69*; %β^-n decay=48 *12*

^{36}Mg-$T_{1/2}$: Weighted average of 3.9 ms *13* (2004Gr20,2003Gr22, implant-β correlation), 7.6 ms *+5−8* (2013StZY, implant-β correlation, original $T_{1/2}$=7.6 ms *1* (stat) *+5−8* (syst)), 7.2 ms *12* (2022Cr03, implant-β correlation, original $T_{1/2}$=7.2 ms *1* (stat) *12* (syst)), and 6.8 ms *10* (2023Lu07, implant-$\beta\gamma$ correlation). Other: ≈5 ms (1999YoZW, implant-β correlation, preliminary).

^{36}Mg-Q(β^-n): From 2021Wa16.

^{36}Mg-%β^-n decay: From 1999YoZW (preliminary; not adopted in Adopted Levels of ^{36}Mg).

2023Lu07: Exp 1: ^{36}Mg and ^{36}Al were produced via the projectile fragmentation of a 140-MeV/nucleon, 80-pnA ^{48}Ca primary beam from the NSCL cyclotrons impinging on a 642-mg/cm^2-thick ^{9}Be target. The secondary cocktail beam centered around ^{33}Na was selected by the A1900 separator and implanted into a $CeBr_3$ scintillator sandwiched between two plastic scintillator veto detectors. Surrounding the implantation array were the SeGA array of 16 segmented Ge detectors and 15 $LaBr_3$ detectors. Exp 2: ^{36}Mg and ^{36}Al were produced via the projectile fragmentation of a 172.3-MeV/nucleon, 120-pnA ^{48}Ca primary beam from the FRIB linac impinging on an 8.89-mm-thick ^{9}Be target. The secondary cocktail beam centered around ^{42}Si was selected by the ARIS separator and implanted into a 5-mm-thick YSO segmented scintillator sandwiched between two plastic scintillator veto detectors. Surrounding the implantation array were 11 HPGe clover detectors and 15 fast-timing $LaBr_3$ detectors, and the VANDLE array of 88 neutron detectors. Measured Eγ, Iγ, $\beta\gamma$-coin, $\gamma\gamma$-coin, implant-$\beta\gamma$ correlation. Deduced $T_{1/2}$ of ^{36}Mg g.s., ^{36}Al g.s. and a ^{36}Al isomer. Comparisons with FSU shell-model calculations.

2013StZY: ^{36}Mg was produced via the projectile fragmentation of a 345-MeV/nucleon, 70-pnA $^{48}Ca^{20+}$ primary beam from the linear accelerator RILAC and the four cyclotrons RRC, fRC, IRC, and SRC at RIKEN impinging on an 15-mm-thick ^{9}Be target. The secondary cocktail beam was selected by the BigRIPS separator and the zero-degree spectrometer (ZDS) using the Bρ-ΔE-ToF method, and implanted into the Cylindrical Active Implantation Target for Exotic Nuclei (CAITEN) consisting of a segmented movable hollow-cylindrical-shaped plastic scintillator and a stationary ring of 24 position-sensitive photomultiplier tubes (PSPMTs) arranged on a ring inside the scintillator at the height of the beam line. To reduce background buildup, the scintillator barrel was rotated quickly and axially-moved slowly in vertical direction, resulting in a helix-shaped motion. β particles were detected by the CAITEN and γ rays were detected using three HPGe clover detectors. Measured Eγ, $\beta\gamma$-coin, and implant-β correlation. Deduced parent $T_{1/2}$. Comparisons with QRPA and shell-model calculations.

^{35}Al Levels

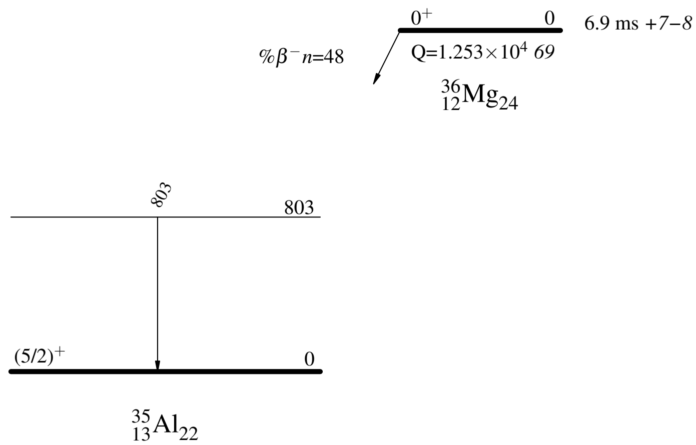
E(level)	J^π	Comments
0	(5/2)$^+$	J^π: From the Adopted Levels.
803		

γ(^{35}Al)

E$_\gamma$	E$_i$(level)	E$_f$	J^π_f	Comments
803	803	0	(5/2)$^+$	E$_\gamma$: From 2023Lu07. 2023Lu07 observed the 803γ within the 100-ms window following the arrival of a ^{36}Mg ion in both NSCL and FRIB experiments. 2013StZY observed a 804-keV γ ray that was not in coincidence with other γ rays. 2013StZY states that the 804γ is a prompt line that predominantly appears in the time window 0-10 ms after an implantation of ^{36}Mg, so it must originate from a deexcitation in one of the daughter nuclei ^{34,35,36}Al.

^{36}Mg β^-n decay (6.9 ms) 2023Lu07

Decay Scheme



He(^{35}Al,^{35}Al$'\gamma$) 2006FuZX

2006FuZX: ^{35}Al was produced via the projectile fragmentation of a 63-MeV/nucleon ^{40}Ar primary beam impinging on 1-mm-thick C and Be targets and selected by the RIKEN Projectile-Fragment Separator (RIPS). The secondary cocktail beam was directed onto a 150-mg/cm^2 liquid He target and the outgoing particles were identified by energy loss, kinetic energy, and time-of-flight measured using a Si detector, a NaI calorimeter, and a couple of plastic scintillators. The positions and directions of incident and outgoing particles were measured using PPACs. Only outgoing particles were identified so that the excited states could be populated by any allowed reactions. Deexcited γ rays emitted from outgoing particle moving at v/c≈0.3 were detected by the Gamma-Ray detector Array with Position and Energy sensitivity (GRAPE) consisting of segmented Ge detectors surrounding the liquid He target.

γ(^{35}Al)

E_γ †
^{x}760.1 *21*

† From 2006FuZX.
^{x} γ ray not placed in level scheme.

^{9}Be(^{36}Si,^{35}Alγ) 2014St18

^{36}Si→p+^{35}Al from J^{π}=0^{+} ^{36}Si ground state.

2014St18: A ^{36}Si secondary beam was produced via the projectile fragmentation of a 140-MeV/nucleon ^{48}Ca primary beam impinging on a ^{9}Be target at NSCL, MSU and was selected by the A1900 separator. The states of ^{35}Al and ^{35}Si were populated by the one-proton/neutron knockout reactions, respectively, from the ^{36}Si beam at a midtarget energy of 97.7 MeV/nucleon *5* on a 287-mg/cm^{2} ^{9}Be secondary target. Knockout residues were identified from their energy loss measured by an ionization chamber at the focal plane of the S800 spectrometer and from their ToF measured between two scintillators at the object position and at the focal plane of the S800 spectrometer. The position and angle of the residues were measured using two cathode-readout drift chambers. Prompt γ rays from the deexcitation of the residues were detected by the GRETINA Ge array. Measured Doppler-corrected Eγ, Iγ, (^{35}Al)γ-coin, $\gamma\gamma$-coin, the parallel momentum distributions of populated states in ^{35}Al residues. Deduced levels, J, π, L-transfers, inclusive and exclusive knockout cross section for producing ^{35}Al from ^{36}Si. Calculations using eikonal model and shell model calculations with SDPF-U and SDPF-MU interactions.

^{35}Al Levels

Total knockout σ=22 mb *1* for producing ^{35}Al from ^{36}Si.

E(level)†	J$^{\pi}$‡	L‡	Comments
0	(5/2)$^{+}$	2	J$^{\pi}$: interpreted as the 1d$_{5/2}$ proton removal from ^{36}Si (2014St18). Partial knockout σ=13 mb *2*.
801 *3*			Partial knockout σ=1.0 mb *7*.
1005 *3*			Partial knockout σ=0.8 mb *9*.
1865 *4*			Partial knockout σ=1.0 mb *2*.
1973 *4*	3/2^{+},5/2^{+}	2	Partial knockout σ=3.2 mb *5*.
2733 *7*			Partial knockout σ=0.5 mb *1*.
3244 *5*	3/2^{+},5/2^{+}	2	Partial knockout σ=2.6 mb *3*.
4275? *9*	3/2^{+},5/2^{+}	2	Partial knockout σ=0.5 mb *1*.

† From a least-squares fit to γ-ray energies.

‡ 2014St18 deduced L by comparing the measured and eikonal-calculated parallel momentum distributions of residuals. J^{π} options are deduced accordingly.

γ(^{35}Al)

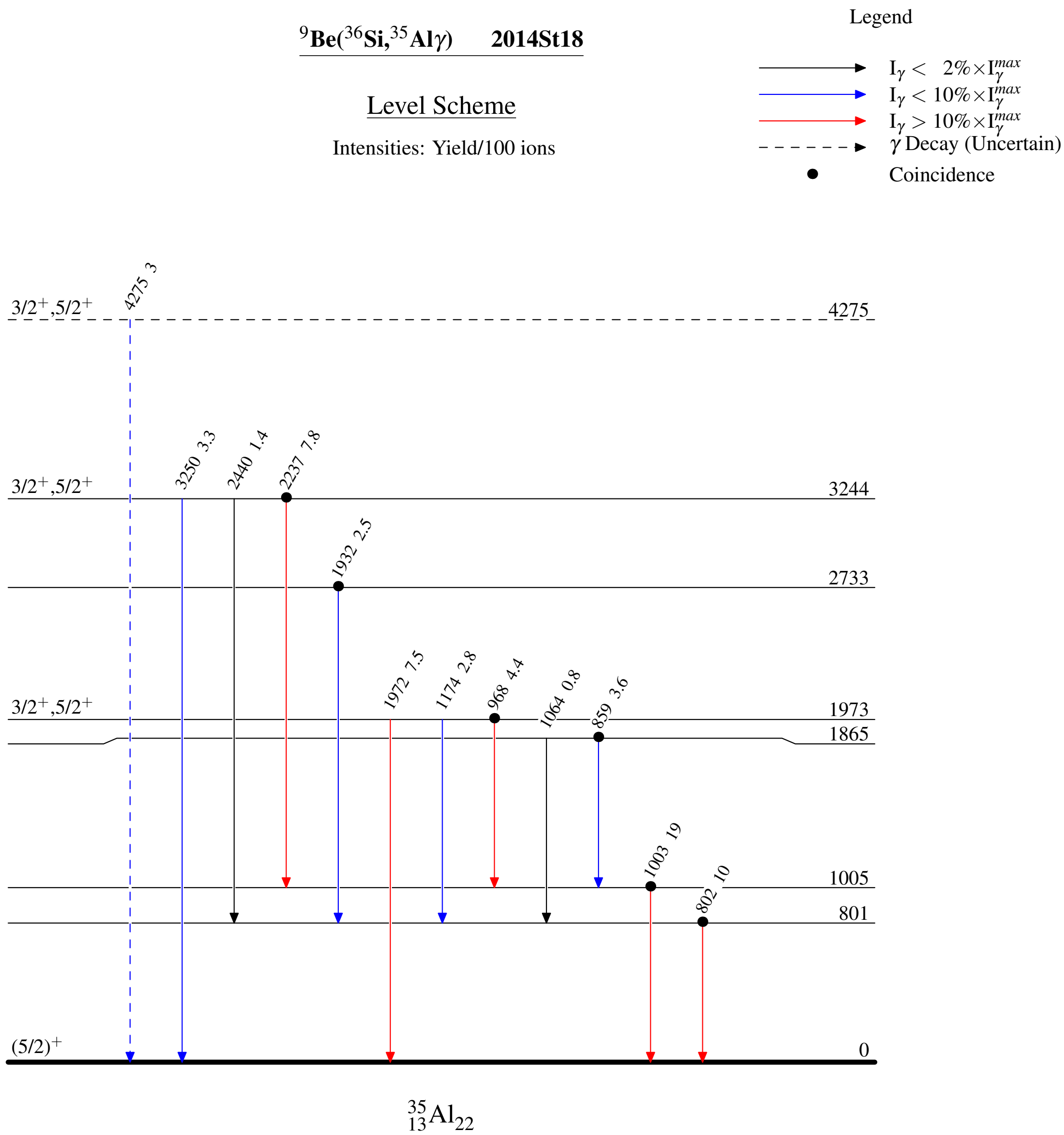
Eγ	Iγ	E$_i$(level)	J$^{\pi}_i$	E$_f$	J$^{\pi}_f$	Comments
802 *4*	10 *1*	801		0	(5/2)$^{+}$	
859 *4*	3.6 *3*	1865		1005		
968 *4*	4.4 *3*	1973	3/2^{+},5/2^{+}	1005		
1003 *4*	19 *1*	1005		0	(5/2)$^{+}$	
1064 *4*	0.8 *2*	1865		801		
1174 *5*	2.8 *3*	1973	3/2^{+},5/2^{+}	801		
^{x}1473 *5*	1.1 *2*					
1932 *6*	2.5 *3*	2733		801		
1972 *6*	7.5 *5*	1973	3/2^{+},5/2^{+}	0	(5/2)$^{+}$	
2237 *6*	7.8 *6*	3244	3/2^{+},5/2^{+}	1005		
2440 *7*	1.4 *2*	3244	3/2^{+},5/2^{+}	801		
^{x}3060 *8*	1.6 *4*					
3250 *8*	3.3 *4*	3244	3/2^{+},5/2^{+}	0	(5/2)$^{+}$	
4275† *9*	3 *1*	4275?	3/2^{+},5/2^{+}	0	(5/2)$^{+}$	2014St18 states that the 4275γ transition could populate the ground state or one of the lowest two excited states. In the absence of any observed $\gamma\gamma$ coincidences, they tentatively place the 4275-keV γ transition feeding the ground state.

Continued on next page (footnotes at end of table)

^{9}Be(^{36}Si,^{35}Alγ) 2014St18 (continued)

γ(^{35}Al) (continued)

† Placement of transition in the level scheme is uncertain.
^{x} γ ray not placed in level scheme.



Pb(^{35}Al,^{34}Alnγ) 2017Ch36,2021Bh12

Coulomb dissociation of ^{35}Al on Pb target.

2017Ch36,2014ChZZ: ^{35}Al was produced via the projectile fragmentation of a 531-MeV/nucleon ^{40}Ar primary beam from the Heavy Ion Synchrotron (SIS18) at GSI. The secondary cocktail beam was separated by the FRS separator and impinged on a 2 g/cm^2 Pb target and a 0.93 g/cm^2 C target. Projectiles and reaction fragments were detected using 8 DSSDs, separated by a large-area dipole magnet (ALADIN) and tracked using two large scintillator fiber detectors (GFIs). Neutrons from the excited projectiles were detected using the high-efficiency Large Area Neutron Detector (LAND). γ rays from the deexcited projectile and projectile-like fragments were detected using a spherical 4π Crystal Ball detector array of 162 NaI(Tl) crystals. Measured E(fragment), E_n, Eγ, Coulomb dissociation cross sections. Deduced relative populations of ^{34}Al, ^{35}Al g.s. configuration. Comparisons with shell-model calculations with the SDPF-M interaction. The measured inclusive differential CD cross section (integrated up to 5.0 MeV relative energy) for ^{35}Al→^{34}Al+n using a Pb target is 78 mb *13*.

2021Bh12: A further analysis of the data from 2017Ch36. The ^{35}Al(γ,n)^{34}Al photoabsorption cross section was obtained from fitting the direct breakup model to the measured differential Coulomb dissociation cross section of ^{35}Al breaking up into ^{34}Al core excited states. The ^{34}Al(n,γ)^{35}Al neutron capture cross sections were obtained from the photoabsorption cross sections using the detailed balance theorem.

^{35}Al Levels

E(level)	J$^\pi$	Comments
0	(5/2$^+$,3/2$^+$,1/2$^+$)	J$^\pi$: From comparisons of measured differential Coulomb dissociation cross section of ^{35}Al breaking up into ^{34}Al in its g.s. and/or 46-keV isomer with theoretical calculations from the direct breakup model using the plane-wave approximation assuming the valence neutron at different orbitals. 2017Ch36 states that the differential CD cross section of ^{35}Al→^{34}Al+n has been interpreted in the light of a direct breakup model, and it suggests that the possible ground-state spin and parity of ^{35}Al could be, tentatively, 1/2$^+$ or 3/2$^+$ or 5/2$^+$. Major configurations and spectroscopic factor for J^π=5/2$^+$ of ^{35}Al g.s.: (g.s., 4$^-$ in ^{34}Al)⊗$\nu p_{3/2}$, S=0.36 *9* (2017Ch36); (46 keV, 1$^+$ in ^{34}Al)⊗$\nu d_{3/2}$, S=1.47 *22* (2017Ch36); (1.4 MeV, 2$^+$ in ^{34}Al)⊗$\nu s_{1/2}$, S=0.16 *1* (2021Bh12); (2.5 MeV, 3$^-$ in ^{34}Al)⊗$\nu p_{3/2}$, S=1.48 *18* (2021Bh12). Other configurations for J^π=1/2$^+$,3/2$^+$ of ^{35}Al g.s.: (g.s., 4$^-$ in ^{34}Al)⊗$\nu f_{7/2}$, S=1.03 *43* and (46 keV, 1$^+$ in ^{34}Al)⊗$\nu s_{1/2}$, S=0.62 *7*; (46 keV, 1$^+$ in ^{34}Al)⊗$\nu s_{1/2}$, S=0.72 *8*; (46 keV, 1$^+$ in ^{34}Al)⊗$\nu s_{1/2}$, S=0.45 *7* and (46 keV,1$^+$ in ^{34}Al)⊗$\nu d_{5/2}$, S=0.94 *22*.

Coulomb excitation 1999Ib01,2000PrZX

1999Ib01: $^{197}Au(^{35}Al,^{35}Al')$ Nuclei of interest were produced via the projectile fragmentation of a 70-MeV/nucleon, ^{48}Ca primary beam from the NSCL K1200 cyclotron impinging on a 285-mg/cm^2-thick ^9Be target. The secondary cocktail beam was selected by the A1200 separator and impinged on a 532 mg/cm^2 ^{197}Au target. The position and direction of each incident beam particle were measured using two upstream parallel-plate avalanche counters (PPAC). Scattered beam particles $\theta_{lab}<3.8°$ were detected using a downstream position-sensitive plastic phoswich detector in coincidence with γ rays detected using an array of 38 cylindrical NaI(Tl) detectors centered around the ^{197}Au target. Measured Doppler-corrected Eγ, Iγ, and excitation cross sections. Deduced levels and E2 transition probabilities for ^{35}Al, ^{37}Si, ^{39}P, ^{41}S, ^{43}S, and ^{45}Cl.

2000PrZX: $^{197}Au(^{35}Al,^{35}Al')$ The same experimental setup as 1999Ib01 with an 80-MeV/nucleon ^{48}Ca primary beam. Scattered beam particles $\theta_{c.m.}<3.3°$ were detected.

^{35}Al Levels

E(level)	J$^\pi$	Comments
0	(5/2)$^+$	J$^\pi$: From the Adopted Levels.
1020 *9*		E(level): From measured Eγ. B(E2)↑=0.0142 *52* from 1999Ib01, assuming E2 excitation mode. Others: B(E1)↑≤0.00020 *9*, B(E2)↑≤0.0125 *56*, B(M1)↑≤0.0024 *11* (2000PrZX). Measured σ=30 mb *14* (2000PrZX).

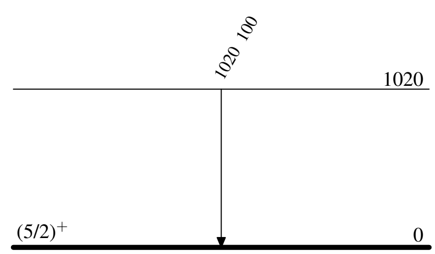
$\gamma(^{35}Al)$

E$_i$(level)	Eγ	Iγ	E$_f$	J$^\pi_f$	Comments
1020	1020 *9*	100	0	(5/2)$^+$	Eγ: Weighted average of 1006 *19* (1999Ib01) and 1023 *9* (2000PrZX). 2000PrZX reports Eγ=1023 *8* in Table 4.9 and Eγ=1023 *9* in Table 4.10.

Coulomb excitation 1999Ib01,2000PrZX

Level Scheme

Intensities: % photon branching from each level



$^{35}_{13}Al_{22}$

Adopted Levels, Gammas

Q(β^-)=10470 *40*; S(n)=2470 *40*; S(p)=18680 *40*; Q(α)=−13690 *40* 2021Wa16
S(2n)=10020 *40*, S(2p)=33930 *40*, Q(β^-n)=2090 *40* (2021Wa16).
Isotope discovery (2012Th10): ^{232}Th(^{40}Ar,X) at Dubna (1971Ar32).
^{35}Si production:
2015Mo17: ^{9}Be(^{40}Ar,X) at E(^{40}Ar)=95 MeV/nucleon at RIKEN. Measured angular distributions and transverse momentum distributions of fragments. Deduced formulation for the width of transverse momentum distribution as a function of fragment velocity.
2012Kw02: E(^{40}Ar)=140 MeV/nucleon impinged on ^{9}Be, natural nickel, and ^{181}Ta targets at NSCL. Measured fragmentation cross sections, parallel momentum transfers, and widths. Compared with empirical formula EPAX, and predictions from internuclear cascade and deep inelastic models using Monte Carlo ISABEL-GEMINI and DIT-GEMINI codes.
2012Zh06: ^{9}Be,^{181}Ta(^{40}Ar,X) at E(^{40}Ar)=57 MeV/nucleon at HIRFL. Measured momentum distributions and production cross sections of fragments. Observed competition between projectile fragmentation and other mechanisms. Compared with EPAX, abrasion-ablation, and HIPSE models. Studied target dependence of fragment cross sections.
2007No13: ^{9}Be(^{40}Ar,X) at E(^{40}Ar)=100 MeV/nucleon at RIKEN. Measured fragment momentum distributions and production cross sections.
2006Ro34: ^{2}H(^{42}S,X) at E(^{42}S)=99.8 MeV/nucleon at NSCL. Measured production cross sections.
1997Fo01: ^{208}Pb(^{37}Cl,X) at E(^{37}Cl)=230 MeV at Legnaro. Measured yields.
^{35}Si decay measurements:
1986Du07,1986HuZW,1987DuZU,1988DuZS,1988DuZT: ^{9}Be(^{40}Ar,X) at GANIL. Measured $T_{1/2}$ and β^--delayed γ rays.
2007Ne14: Polarized ^{35}Si from ^{9}Be(^{36}S,X) 1n pickup at GANIL. ^{35}Si g.s. magnetic moment and *g*-factor using β-NMR.
^{35}Si radius measurements:
2006Kh08: ^{35}Si produced by ^{181}Ta(^{48}Ca,X) fragmentation at E(^{48}Ca)=60.3 MeV/nucleon at GANIL. Measured energy-integrated reaction cross sections at 30-65 MeV/nucleon using a silicon telescope as both active target and detector. Deduced reduced strong absorption radii, isospin dependence, and possible halo structure or large deformation.
1999Ai02: Si(^{35}Si,X) at NSCL. Measured energy-integrated reaction cross sections at E=38-80 MeV/ nucleon. Deduced strong absorption radii.
^{35}Si mass measurements: 1986Fi06, 1986Sm05, 1984Ma49.
Theoretical calculations (binding energies, deformation, quadrupole moments, radii, levels, J^π, etc.): 2011Ka03, 2009No01, 2008Wi11, 2007Ch82, 2004Kh16, 1999Du05, 1994Mo37, 1994Po05, 1987Wa10, 1986Wo02.

^{35}Si Levels

Cross Reference (XREF) Flags

A	^{35}Al β^- decay (38.1 ms)	D	^{2}H(^{34}Si,^{35}Siγ)
B	^{36}Al β^-n decay (12.0 ms)	E	^{9}Be(^{36}Si,^{35}Siγ)
C	^{37}Al β^-2n decay (11.4 ms)	F	^{208}Pb(^{36}S,^{35}Siγ)

E(level)†	J^π	$T_{1/2}$	XREF	Comments
0	(7/2)$^-$	0.78 s *12*	ABCDEF	%β^-=100; %β^-n<5 (1995ReZZ,2008ReZZ) μ=(−)1.639 *4* (2007Ne14,2019StZV) μ: From original (−)1.638 *4* using β-NMR (2007Ne14) with corrections by 2019StZV. J^π: L=3 from 0$^+$ in ^{2}H(^{34}Si,^{35}Si) and ^{9}Be(^{36}Si,^{35}Si) and $\nu f_{7/2}$ configuration from shell model. $T_{1/2}$: From $\beta\gamma$(t) (1988DuZS,1988DuZT). Other: 0.87 s *17* by the same group (1986Du07). Reduced strong absorption radius r_0^2=1.261 fm^2 *35* from the energy-integrated σ of Si(^{35}Si,X) (2006Kh08) and r_0^2=1.258 fm^2 *92* from the energy-integrated σ of Si(^{35}Si,X) (1999Ai02).
909.95 *23*	(3/2)$^-$	55 ps *14*	ABCDEF	J^π: L=1 from 0$^+$ in ^{2}H(^{34}Si,^{35}Si) and ^{9}Be(^{36}Si,^{35}Si) and $\nu p_{3/2}$ configuration from shell model. $T_{1/2}$: From analysis of broadened line shapes in ^{9}Be(^{36}Si,^{35}Siγ).

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Adopted Levels, Gammas (continued)

^{35}Si Levels (continued)

E(level)†	J^{π}	$T_{1/2}$	XREF	Comments
973.88 *18*	(3/2$^+$)	5.9 ns *6*	A EF	J^{π}: $\nu d_{3/2}$ configuration from shell model and 715γ from 1689, 1/2$^+$. $T_{1/2}$: From $\beta\gamma$(t) in ^{35}Al β^- decay.
1689.4 *28*	1/2$^+$		E	J^{π}: L=0 from 0$^+$ in ^{9}Be(^{36}Si,^{35}Si).
1970 *6*			E	
2044 *5*	(1/2)$^-$		DE	J^{π}: L=1 from 0$^+$ in ^{2}H(^{34}Si,^{35}Si) and $\nu p_{1/2}$ configuration from shell model.
2168.2 *4*	(5/2$^+$)		A E	J^{π}: From shell model and its possible isobaric analog state in ^{35}P with L=2 from 0$^+$ in ^{1}H(^{34}Si,p) from R-matrix analysis in 2012Im01. L=2,3 from 0$^+$ in ^{9}Be(^{36}Si,^{35}Si).
2275 *6*			E	
2377 *7*			E	
3140	(3/2$^+$,5/2$^+$,7/2$^+$)‡		A	
3450	(3/2$^+$,5/2$^+$,7/2$^+$)‡		A	
3611? *8*			E	XREF: E(?) Level exists assuming that the 3611γ feeds the ground state of ^{35}Si (2014St18).
3770	(3/2$^+$,5/2$^+$,7/2$^+$)‡		A	
5190	(3/2$^+$,5/2$^+$,7/2$^+$)‡		A	
≈5500	(5/2)$^-$		D	E(level): A broad level at ≈5500 deduced from E_p in ^{2}H(^{34}Si,^{35}Si). J^{π}: L=3 from 0$^+$ in ^{2}H(^{34}Si,^{35}Si) and $\nu f_{5/2}$ configuration from shell model.
5760	(3/2$^+$,5/2$^+$,7/2$^+$)‡		A	
6330	(3/2$^+$,5/2$^+$,7/2$^+$)‡		A	
7360	(3/2$^+$,5/2$^+$,7/2$^+$)‡		A	
7690	(3/2$^+$,5/2$^+$,7/2$^+$)‡		A	

† From a least-squares fit to γ-ray energies for levels connected with γ transitions; from ^{35}Al β^--delayed neutron decays for other levels, unless otherwise noted.

‡ Possible allowed β^- feeding from (5/2)$^+$ parent in ^{35}Al.

γ(^{35}Si)

E_i(level)	J^{π}_i	E_{γ}†	I_{γ}†	E_f	J^{π}_f	Mult.	$\alpha$$^{\#}$	Comments
909.95	(3/2)$^-$	910.11 *30*	100	0	(7/2)$^-$	[E2]	4.13×10^{-5} *6*	B(E2)(W.u.)=2.4 *+8−5* E_{γ}: others: 910 *3* from ^{2}H(^{34}Si,^{35}Siγ), 908 *4* from ^{9}Be(^{36}Si,^{35}Siγ), and 910 from ^{208}Pb(^{36}S,^{35}Siγ).
973.88	(3/2$^+$)	64.1 *3*	100	909.95	(3/2)$^-$	[E1]	0.0368 *8*	B(E1)(W.u.)=3.52×10^{-4} *+41−34*
		973.78 *20*	11.8 *24*	0	(7/2)$^-$	[M2]	5.05×10^{-5} *7*	B(M2)(W.u.)=0.057 *+13−12* E_{γ}: other: 974 from ^{208}Pb(^{36}S,^{35}Siγ).
1689.4	1/2$^+$	715‡ *4*	14.6‡ *16*	973.88	(3/2$^+$)			
		780‡ *4*	100‡ *8*	909.95	(3/2)$^-$			
1970		1970‡ *6*	100	0	(7/2)$^-$			
2044	(1/2)$^-$	1134‡ *5*	100	909.95	(3/2)$^-$			E_{γ}: Other: 1134 *6* from ^{2}H(^{34}Si,^{35}Siγ).
2168.2	(5/2$^+$)	1194.2 *4*	35 *8*	973.88	(3/2$^+$)			

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Adopted Levels, Gammas (continued)

$\gamma(^{35}Si)$ (continued)

E_i(level)	E_γ†	I_γ†	E_f	J^π_f	Comments
2168.2	2168.2 *6*	100 *20*	0	(7/2)$^-$	E_γ: Other: 2164 *6* from $^9Be(^{36}Si,^{35}Si\gamma)$.
2275	2275‡ *6*	100	0	(7/2)$^-$	
2377	2377‡ *7*	100	0	(7/2)$^-$	
3611?	3611‡@ *8*	100	0	(7/2)$^-$	2014St18 observed this surprising γ transition at 3611 keV, which must depopulate a state that is unbound to neutron emission by at least 1.1 MeV, considering S(n)=2470 *40* (2021Wa16). 2014St18 proposes two possible explanations involving hindered neutron emission or multistep process.

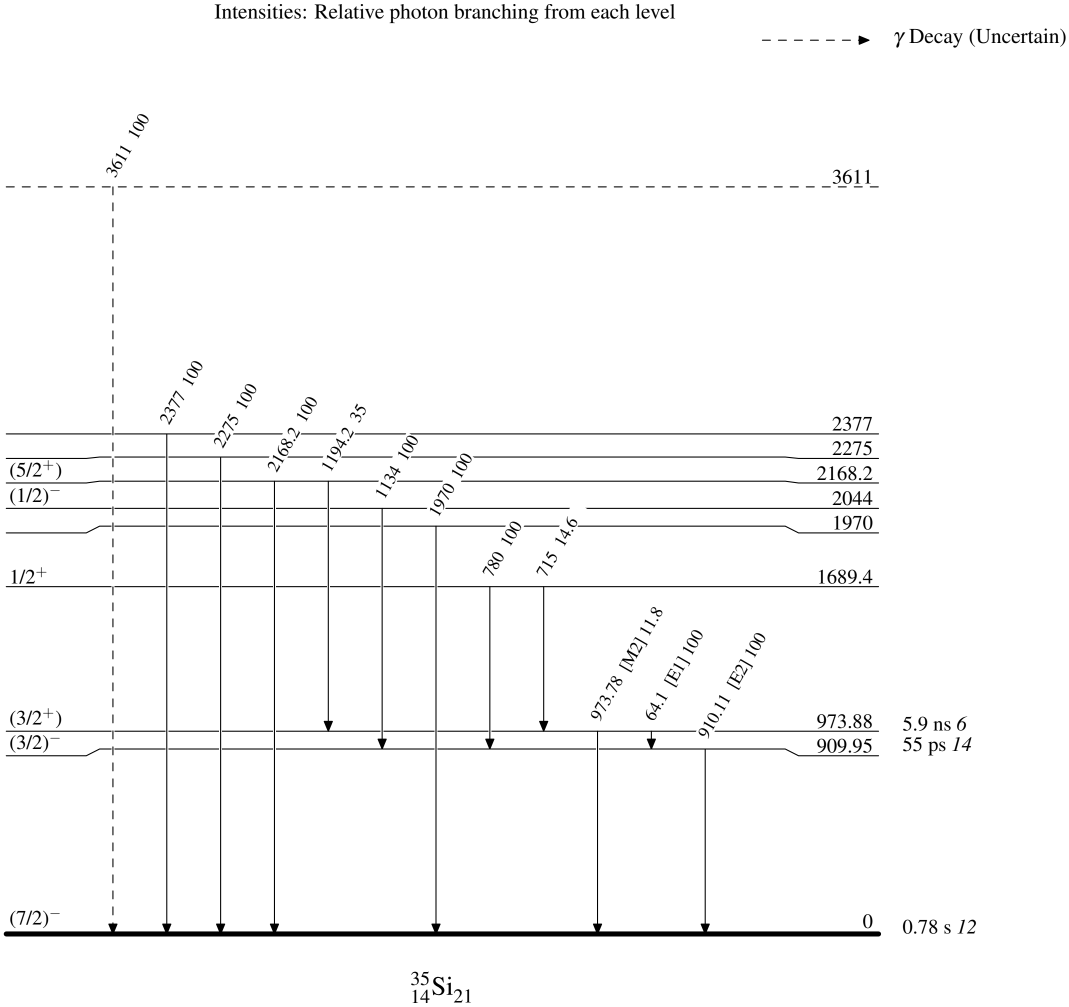
† From ^{35}Al β^- decay, unless otherwise noted.
‡ From $^9Be(^{36}Si,^{35}Si\gamma)$.
# Total theoretical internal conversion coefficients, calculated using the BrIcc code (2008Ki07) with "Frozen Orbitals" approximation based on γ-ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.
@ Placement of transition in the level scheme is uncertain.

Adopted Levels, Gammas

Legend

Level Scheme

Intensities: Relative photon branching from each level



^{35}Al β^- decay (38.1 ms) 2005Ti11,2001Nu01

Parent: ^{35}Al: E=0; J$^\pi$=(5/2)$^+$; $T_{1/2}$=38.1 ms *4*; Q(β^-)=14170 *40*; %β^- decay=100

^{35}Al-J$^\pi$,$T_{1/2}$: From the Adopted Levels of ^{35}Al.

^{35}Al-Q(β^-): From 2021Wa16.

2005Ti11,2006AnZW: A ^{35}Al secondary beam at ≈2 pps was produced via the fragmentation of a 78-MeV/nucleon ^{36}S primary beam and selected by the LISE3 spectrometer at GANIL. A total of 3.46×10^5 ^{35}Al ions were continuously implanted into an NE102A plastic scintillator also for detecting β. The implantation detector was sandwiched between two silicon detectors for monitoring beam and for veto, respectively. Neutrons were detected using the TONNERRE array consisting of 19 plastic scintillator modules. γ rays were detected using two EXOGAM clover modules and a LEPS detector. Measured Eγ, Iγ, E_n, I_n, $\beta\gamma$-coin, βn-coin, and βnγ-coin. Deduced the decay scheme consisting of ^{35}Si and ^{34}Si levels, ^{35}Al $T_{1/2}$, decay branching ratios, log *ft*, B(GT), and β-delayed neutron emission probability. Comparisons with shell-model calculations.

2001Nu01,2002Nu02: Exp 1: A ^{35}Al secondary beam at 8 pps was produced via the fragmentation of a UC target with 1.4 GeV protons at ISOLDE, CERN with subsequent surface-ionization and mass separation. ^{35}Al ions were collected onto a moving tape. β particles were detected using a thin cylindrical plastic scintillator, γ rays were detected using two Ge detectors, and neutrons were detected using eight low-threshold plastic scintillators. Measured Eγ, Iγ, E_n, I_n, $\beta\gamma$-coin, $\gamma\gamma$-coin, βn-coin. Deduced the decay scheme consisting of ^{35}Si and ^{34}Si levels, ^{35}Al $T_{1/2}$, decay branching ratios, log *ft*, and β-delayed neutron emission probability. Comparisons with shell-model calculations. Exp 2: A lifetime measurement for the 974-keV level in ^{35}Si used a thin plastic scintillator for detecting β and a BaF_2 detector for detecting γ.

Other experimental studies on ^{35}Al $T_{1/2}$ and β-delayed neutron emission probability: 2017Ha23, 1999YoZW, 1995ReZZ/2008ReZZ, 1989MuZU, 1989Le16, 1988MuZY, 1988Mu08, 1988DuZT, 1988BaYZ, 1987DuZU, 1987BaZI.

Theoretical studies involving ^{35}Al decay: 2018Yo06, 2013Li39.

^{35}Si Levels

E(n) under comments are deduced from $E(n)_{c.m.}$=E(level)(^{35}Si)−S(n)(^{35}Si)-E(level)(^{34}Si), with E(level)(^{35}Si) reported in 2005Ti11 based on the neutron time-of-flight spectrum. 2005Ti11 used E(level)(^{34}Si)=3326 for E(n1). 2005Ti11 used S(n)(^{35}Si)=2474 *43*, consistent with S(n)(^{35}Si)=2470 *40* from 2021Wa16.

E(level)†	J$^\pi$‡	$T_{1/2}$‡	Comments
0	(7/2)$^-$	0.78 s *12*	
909.95 *23*	(3/2)$^-$	55 ps *14*	2001Nu01 reports a %Iβ^-<0.9 from 2001Nu01 without considering the conversion electron of the transition from 974 level to 910 level. The net feeding is deduced to be −1.9 *9* from the γ+ce intensity balance.
973.88 *18*	(3/2$^+$)	5.9 ns *6*	$T_{1/2}$: lifetime=8.5 ns *9* from $\beta\gamma$(t) in 2001Nu01, also adopted in the Adopted Levels.
2168.16 *36*	(5/2$^+$)		
3140	(3/2$^+$,5/2$^+$,7/2$^+$)		Deduced $E(n0)_{c.m.}$=666.
3450	(3/2$^+$,5/2$^+$,7/2$^+$)		Deduced $E(n0)_{c.m.}$=976.
3770	(3/2$^+$,5/2$^+$,7/2$^+$)		Deduced $E(n0)_{c.m.}$=1296.
5190	(3/2$^+$,5/2$^+$,7/2$^+$)		Deduced $E(n0)_{c.m.}$=2716.
5760	(3/2$^+$,5/2$^+$,7/2$^+$)		Deduced $E(n0)_{c.m.}$=3286.
6330	(3/2$^+$,5/2$^+$,7/2$^+$)		Deduced $E(n0)_{c.m.}$=3856; $E(n1)_{c.m.}$=530.
7360	(3/2$^+$,5/2$^+$,7/2$^+$)		Deduced $E(n1)_{c.m.}$=1560.
7690	(3/2$^+$,5/2$^+$,7/2$^+$)		Deduced $E(n1)_{c.m.}$=1890.

† From a least-squares fit to γ-ray energies for levels connected with γ transitions. For unbound levels, from 2005Ti11 based on the neutron time-of-flight spectrum.

‡ From the Adopted Levels of ^{35}Si.

^{35}Al β^- decay (38.1 ms) 2005Ti11,2001Nu01 (continued)

β^- radiations

The decay scheme is incomplete due to possible unobserved levels in a large gap of about 6.5 MeV between Q-value=14.17 MeV and the highest populated level at E=7.69 MeV.

E(decay)	E(level)	Iβ^- †#	Log *ft* ‡	Comments
(6.48×10^3 *4*)	7690	2.7 *2*	4.5	
(6.81×10^3 *4*)	7360	2.6 *2*	4.6	
(7.84×10^3 *4*)	6330	6.8 *3*	4.5	
(8.41×10^3 *4*)	5760	4.5 *2*	4.8	
(8.98×10^3 *4*)	5190	8.9 *3*	4.6	
(1.040×10^4 *4*)	3770	3.2 *2*	5.4	
(1.072×10^4 *4*)	3450	6.0 *3*	5.2	
(1.103×10^4 *4*)	3140	3.3 *2*	5.5	
(1.200×10^4 *4*)	2168.16	9.5	5.2	Iβ^-: 9.2 *19* from 2001Nu01; 6.7 *9* from 2005Ti11 with only one branch 2168γ observed.
(1.320×10^4 *4*)	973.88	52	4.7	Iβ^-: 48 *9* from 2001Nu01; 50 *3* from 2005Ti11.
(1.417×10^4 *4*)	0	3.0 *10*	6.1 *2*	Iβ^-: from 2001Nu01, which states that the β branch to the ^{35}Si ground state was evaluated by comparing the total γ intensity due to the deexcitation of excited states of ^{35}Si with the decay of ^{35}Si activity and assuming no direct ^{35}Si production.

† For the two bound excited states, Iβ^- is from the γ+ce intensity balance at each state. Quoted Iβ^- values without uncertainties are considered upper limits due to the incomplete decay scheme, and the associated log *ft* values are considered lower limits. For unbound excited states (>2470), Iβ^- is from the absolute Iβ^- in 2005Ti11 based on neutron intensities.
‡ For unbound levels, 2005Ti11 did not report uncertainties for neutron energies or level energies. The evaluators estimate the uncertainties of log *ft* are <0.1, assuming a 200-keV uncertainty in level energies.
# Absolute intensity per 100 decays.

γ(^{35}Si)

Iγ normalization: 0.47 *4* from Σ%I(γ+ce to g.s.)=60 *4*, deduced from 100−%β^-n-%Iβ^-(g.s.), where %β^-n=37 *4* from the Adopted Levels of ^{35}Al and %Iβ^-(g.s.)=3.0 *10* from 2001Nu01. The deduced normalization factor should be considered an upper limit due to potential missing γ transitions from unobserved levels in the gap to the ground state. Other: 0.45 from 2001Nu01, based on %β^-n=41 *13*.

E$_\gamma$ †	I$_\gamma$ †‡	E$_i$(level)	J$^\pi_i$	E$_f$	J$^\pi_f$	Mult.	α #	Comments
64.1 *3*	100	973.88	(3/2$^+$)	909.95	(3/2)$^-$	[E1]	0.0368 *8*	%Iγ=47
910.11 *30*	99.7 *19*	909.95	(3/2)$^-$	0	(7/2)$^-$	[E2]	4.13×10^{-5} *6*	%Iγ=47
973.78 *20*	11.8 *24*	973.88	(3/2$^+$)	0	(7/2)$^-$	[M2]	5.05×10^{-5} *7*	%Iγ=5.6
^{x}1130.4 *4*	3.2 *9*							%Iγ=1.5 2014St18 suggests this γ is likely the 1134γ from 2042 to 908 transition observed in the ^{36}Si→n+^{35}Si knockout reaction.
1194.2 *4*	5.3 *12*	2168.16	(5/2$^+$)	973.88	(3/2$^+$)			%Iγ=2.5
2168.2 *6*	15 *3*	2168.16	(5/2$^+$)	0	(7/2)$^-$			%Iγ=7.1
^{x}5629 *3*	2.4 *12*							%Iγ=1.1

† From 2001Nu01.
‡ For absolute intensity per 100 decays, multiply by 0.47.

Continued on next page (footnotes at end of table)

^{35}Al β^- decay (38.1 ms) 2005Ti11,2001Nu01 (continued)

$\gamma(^{35}Si)$ (continued)

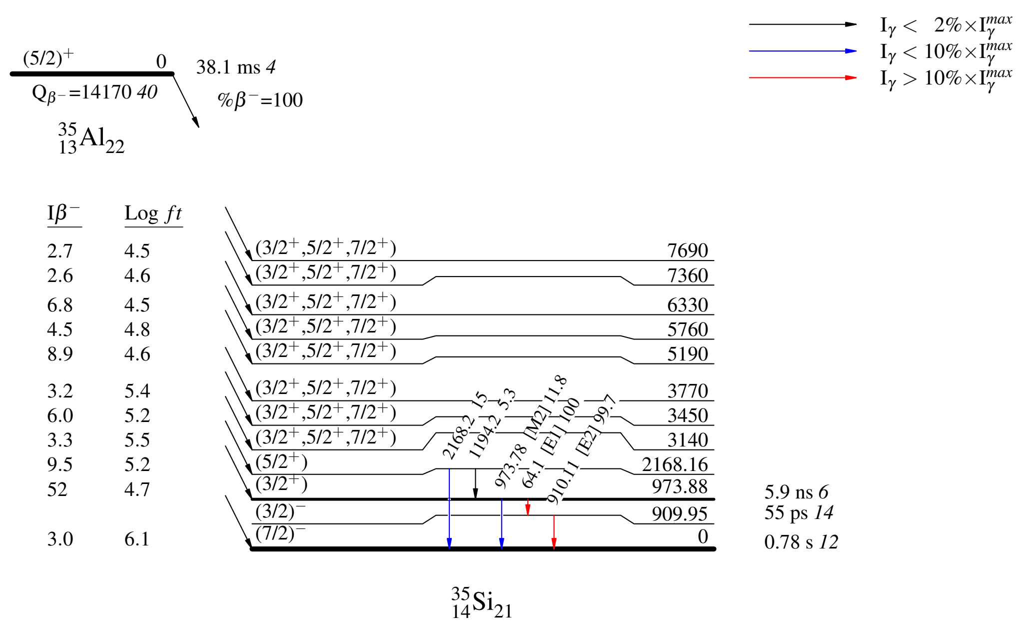
[#] Total theoretical internal conversion coefficients, calculated using the BrIcc code (2008Ki07) with "Frozen Orbitals" approximation based on γ-ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

[x] γ ray not placed in level scheme.

^{35}Al β^- decay (38.1 ms) 2005Ti11,2001Nu01

Decay Scheme

Intensities: Relative I_γ



^{36}Al β^-n decay (12.0 ms) 2023Lu07

Parent: ^{36}Al: E=0; $T_{1/2}$=12.0 ms *20*; Q(β^-n)=1.227×10^4 *15*; %β^-n decay<31

^{36}Al-$T_{1/2}$: From implant-$\beta\gamma$ correlation (2023Lu07). Other: 14.7 ms *10* (2023Lu07, implant-β correlation), 94 ms *37* (1995ReZZ/2008ReZZ, implant-β correlation), ≈15 ms (1999YoZW, implant-β correlation, preliminary). 2023Lu07 adopted 12.0 ms *20* instead of 14.7 ms *10* based on the cleanest determination with the γ-ray gating.

^{36}Al-Q(β^-n): From 2021Wa16.

^{36}Al-%β^-n decay: From 1995ReZZ/2008ReZZ.

2023Lu07: Exp 1: ^{36}Mg and ^{36}Al were produced via the projectile fragmentation of a 140-MeV/nucleon, 80-pnA ^{48}Ca primary beam from the NSCL cyclotrons impinging on a 642-mg/cm^2-thick ^{9}Be target. The secondary cocktail beam centered around ^{33}Na was selected by the A1900 separator and implanted into a $CeBr_3$ scintillator sandwiched between two plastic scintillator veto detectors. Surrounding the implantation array were the SeGA array of 16 segmented Ge detectors and 15 $LaBr_3$ detectors. Exp 2: ^{36}Mg and ^{36}Al were produced via the projectile fragmentation of a 172.3-MeV/nucleon, 120-pnA ^{48}Ca primary beam from the FRIB linac impinging on an 8.89-mm-thick ^{9}Be target. The secondary cocktail beam centered around ^{42}Si was selected by the ARIS separator and implanted into a 5-mm-thick YSO segmented scintillator sandwiched between two plastic scintillator veto detectors. Surrounding the implantation array were 11 HPGe clover detectors and 15 fast-timing $LaBr_3$ detectors, and the VANDLE array of 88 neutron detectors. Both the NSCL and FRIB experiments in 2023Lu07 measured Eγ, Iγ, $\beta\gamma$-coin, $\gamma\gamma$-coin, implant-$\beta\gamma$ correlation and deduced $T_{1/2}$ of ^{36}Mg g.s., ^{36}Al g.s. and a ^{36}Al isomer. Comparisons with FSU shell-model calculations.

^{35}Si Levels

E(level)	$J^{\pi\dagger}$	$T_{1/2}$†
0	(7/2)$^-$	0.78 s *12*
910	(3/2)$^-$	55 ps *14*

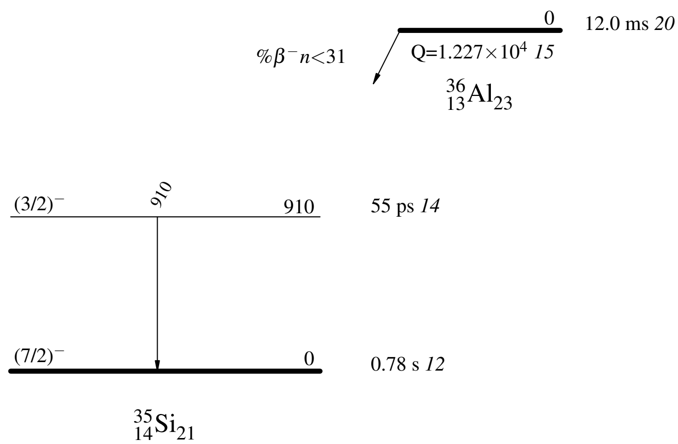
† From the Adopted Levels of ^{35}Si.

γ(^{35}Si)

E_γ	E_i(level)	J^π_i	E_f	J^π_f	Comments
910	910	(3/2)$^-$	0	(7/2)$^-$	E_γ: From 2023Lu07. 2023Lu07 observed the 910γ within the 100-ms window following the arrival of a ^{36}Al ion in both NSCL and FRIB experiments. 2023Lu07 also observed the 910γ within the 100-ms window following the arrival of a ^{36}Mg ion in both NSCL and FRIB experiments. 2013StZY observed the 910γ in ^{36}Mg decay with its maximum intensity in the 20-30 ms time window after the implantation of ^{36}Mg, indicating its origin from the granddaughter generation.

^{36}Al β^{-}n decay (12.0 ms) 2023Lu07

Decay Scheme



^{37}Al β^-2n decay (11.4 ms) 2013StZY

Parent: ^{37}Al: E=0; J^π=(5/2$^+$); $T_{1/2}$=11.4 ms *4*; Q(β^-2n)=8.06×10^3 *18*; %β^-2n decay≥000

^{37}Al-J^π: From shell-model predictions (2013StZY).

^{37}Al-$T_{1/2}$: Weighted average of 10.7 ms *13* (2004Gr20,2003Gr22, implant-β correlation), 11.5 ms *4* (2013StZY, implant-$\beta\gamma$ correlation). Other: 11.8 ms *1* (stat) *+22−34* (syst) (2013StZY, implant-β correlation).

^{37}Al-Q(β^-2n): Deduced from mass excesses in 2021Wa16.

^{37}Al-%β^-2n decay: %β^-2n≥1 *1* (2013StZY).

2013StZY: ^{37}Al was produced via the projectile fragmentation of a 345-MeV/nucleon, 70-pnA $^{48}Ca^{20+}$ primary beam from the linear accelerator RILAC and the four cyclotrons RRC, fRC, IRC, and SRC at RIKEN impinging on an 15-mm-thick ^{9}Be target. The secondary cocktail beam was selected by the BigRIPS separator and the zero-degree spectrometer (ZDS) using the Bρ-ΔE-ToF method, and implanted into the Cylindrical Active Implantation Target for Exotic Nuclei (CAITEN) consisting of a segmented movable hollow-cylindrical-shaped plastic scintillator and a stationary ring of 24 position-sensitive photomultiplier tubes (PSPMTs) arranged on a ring inside the scintillator at the height of the beam line. To reduce background buildup, the scintillator barrel was rotated quickly and axially-moved slowly in vertical direction, resulting in a helix-shaped motion. β particles were detected by the CAITEN and γ rays were detected using three HPGe clover detectors. Measured Eγ, $\beta\gamma$-coin, and implant-β correlation, and deduced $T_{1/2}$. Comparisons with QRPA and shell-model calculations.

^{35}Si Levels

E(level)	J^π †	$T_{1/2}$ †
0	(7/2)$^-$	0.78 s *12*
910	(3/2)$^-$	55 ps *14*

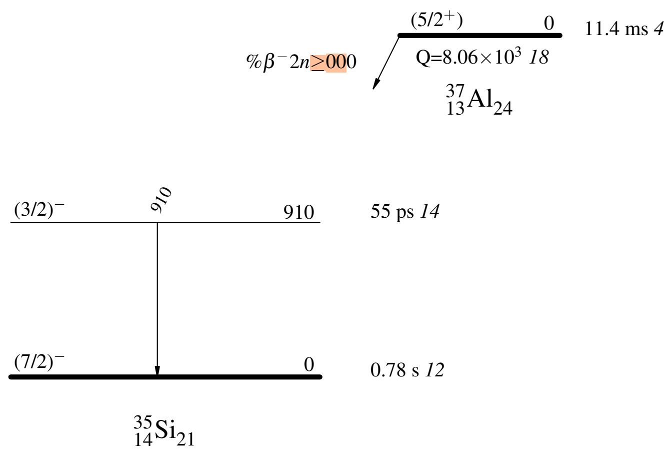
† From the Adopted Levels of ^{35}Si.

γ(^{35}Si)

E_γ	E_i(level)	J^π_i	E_f	J^π_f	Comments
910	910	(3/2)$^-$	0	(7/2)$^-$	I_γ: 2013StZY reports 2 *3* relative to Iγ=100 for the 156γ in ^{37}Si from ^{37}Al decay.

^{37}Al β^-2n decay (11.4 ms) 2013StZY

Decay Scheme



^{2}H(^{34}Si,^{35}Siγ) 2014Bu01

^{34}Si(d,p)^{35}Si from J^{π}=0^{+} ^{34}Si g.s. in inverse kinematics.

2014Bu01: A 20.5-MeV/nucleon, 1.1×10^{5} pps, and 95% pure ^{34}Si secondary beam was produced via the fragmentation of a 55-MeV/nucleon $^{36}S^{16+}$ primary beam impinging on a 1075-μm-thick Be target, separated by the LISE3 spectrometer at GANIL, and incident on a 2.6- mg/cm^{2} *1* CD_2 secondary target. Incoming beam ions were tracked using two position-sensitive multiwire proportional chambers upstream of the target. Outgoing ions were identified using an ionization chamber (IC) downstream of target for energy loss and a plastic scintillator behind the IC for TOF measurements. Protons from the (d,p) reaction were detected using four modules of the MUST2 array placed 10 cm from the target covering polar angles ranging from 105° to 150° with respect to the beam direction and a 16-strip annular Si detector at a distance of 11.3 cm covering polar angles from 156° to 168°. γ rays were detected using four segmented Ge detectors from the EXOGAM array perpendicular to the beam axis at a mean distance of 5 cm, and 9 cm downstream from the target with efficiency ε=3.8% *2* at 1 MeV. Measured $\sigma(E_p,\theta)$, Doppler-corrected Eγ, Iγ, (^{34}Si)p-coin. Deduced levels, J, π, L-transfer and spectroscopic factors. Comparisons with shell-model calculations.

2007GeZX: 30-AMeV ^{34}Si beam on 30-mg/cm^{2} CD_2 secondary target at GANIL. Heavy ions produced in reactions were identified by the VAMOS spectrometer. γ rays were detected using the EXOGAM germanium clover array. Measured Doppler-corrected Eγ, Iγ, $\gamma\gamma$-coin, and (^{35}Si)γ-coin. Deduced levels, J, π. Compared with shell-model calculations.

^{35}Si Levels

E(level)†	J$^{\pi}$‡	L#	S#	Comments
0	7/2$^-$	3	0.56 *6*	Interpreted as the 1$f_{7/2}$ neutron on top of the ^{34}Si core.
910 *3*	3/2$^-$	1	0.69 *10*	Interpreted as the 2$p_{3/2}$ neutron on top of the ^{34}Si core. E(level): Other: 906 *32* from measured E_p. 2014Bu01 deduced that a contamination of the proton spectrum at E(level)=906 *32* due to transfer to the 3/2$^+$ level at 970 keV is less than 30% of the 3/2$^-$ component with a confidence limit of 3σ.
2044 *7*	1/2$^-$	1	0.73 *10*	Interpreted as the 2$p_{1/2}$ neutron on top of the ^{34}Si core. E(level): Other: 2060 *50* from measured E_p. J$^{\pi}$: 2014Bu01 states that J^{π} is likely to be 1/2$^-$ as its large spectroscopic factor value excludes another large L=1, 3/2$^-$ component.
≈5500	5/2$^-$	3	0.32 *3*	Interpreted as the 1$f_{5/2}$ neutron on top of the ^{34}Si core. S: 2014Bu01 reports S=0.32 *4* in Fig. 2, S=0.32 *2* in text on page 3, S=0.32 *3* in text on page 4, and a full error bar ≈0.05 in Fig. 3.

† From a least-squares fit to γ-ray energies, except for a broad level observed at ≈5500 keV from E_p. Another broad structure is observed at 3330 keV *120* from E_p and likely corresponds to the elastic deuteron break-up process, the cross section of which was estimated to be 0.1 mb/MeV (2014Bu01).

‡ As given in 2014Bu01 based on L-transfers and shell-model predictions.

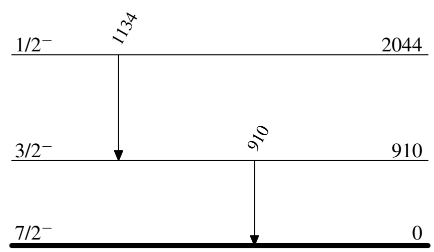
From TWOFNR-ADWA analysis of measured proton angular distributions (2014Bu01). Additional uncertainties of ≈15% in spectroscopic factors due to global potentials in the ADWA calculation are not included.

γ(^{35}Si)

E$_\gamma$	E$_i$(level)	J$^{\pi}_{i}$	E$_f$	J$^{\pi}_{f}$	Comments
910 *3*	910	3/2$^-$	0	7/2$^-$	E$_\gamma$: Other: 905 *1* in 2007GeZX, with uncertainty likely statistical only considering it is from a γ-ray spectrum with a bin size of 4 keV.
1134 *6*	2044	1/2$^-$	910	3/2$^-$	E$_\gamma$: Other: 1133 *3* in 2007GeZX, with uncertainty likely statistical only considering it is a very weak γ ray seen in a γ-ray spectrum with a bin size of 4 keV.

$^{2}H(^{34}Si,^{35}Si\gamma)$ 2014Bu01

Level Scheme



$^{35}_{14}Si_{21}$

^{9}Be(^{36}Si,^{35}Siγ) 2014St18,2015St06

^{36}Si→n+^{35}Si from $J^{\pi}=0^{+}$ ^{36}Si ground state.

2014St18: A ^{36}Si secondary beam was produced via the projectile fragmentation of a 140-MeV/nucleon ^{48}Ca primary beam impinging on a ^{9}Be target at NSCL, MSU and was selected by the A1900 separator. The states of ^{35}Al and ^{35}Si were populated by the one-proton/neutron knockout reactions, respectively, from the ^{36}Si beam at a midtarget energy of 97.7 MeV/nucleon *5* on a 287-mg/cm^{2} ^{9}Be secondary target. Knockout residues were identified from their energy loss measured by an ionization chamber at the focal plane of the S800 spectrometer and from their ToF measured between two scintillators at the object position and at the focal plane of the S800 spectrometer. The position and angle of the residues were measured using two cathode-readout drift chambers. Prompt γ rays from the deexcitation of the residues were detected by the GRETINA Ge array. Measured Doppler-corrected Eγ, Iγ, (^{35}Si)γ-coin, $\gamma\gamma$-coin, the parallel momentum distributions of populated states in ^{35}Si residues. Deduced levels, J, π, L-transfers, inclusive and exclusive knockout cross section for producing ^{35}Si from ^{36}Si. Calculations using eikonal reaction model and shell model spectroscopic strengths with SDPF-U and SDPF-MU interactions.

2015St06: Further theoretical discussions on the data from 2014St18.

^{35}Si Levels

Total knockout σ=81 mb *2* for producing ^{35}Si from ^{36}Si.

E(level)†	J$^{\pi}$‡	T$_{1/2}$	L#	Comments
0	(7/2^{-})		3	Partial knockout σ=52 mb *4*, including an unresolved population of the 3/2^{+} isomer at 973 keV (2014St18). Partial knockout σ=23 mb *6*, resolved by the fitting parallel momentum distribution subtracting all events that decay by prompt γ emission with a linear combination of the theoretical distributions for neutron removal from the f$_{7/2}$ and d$_{3/2}$ orbits together via the χ^{2} minimization method (2015St06).
908 *4*	(3/2^{-})	55 ps *14*	1	T$_{1/2}$: lifetime=80 ps *20* extracted from fitting Geant4 simulation to broadened line shapes via the maximum likelihood method (2014St18). Partial knockout σ=8 mb *3*.
973 *7*	(3/2^{+})			Partial knockout σ=29 mb *6*, resolved by 2015St06.
1688 *6*	1/2^{+}		0	Partial knockout σ=13 mb *1*.
1970 *6*				Partial knockout σ=1.1 mb *2*.
2042 *7*	(1/2^{-})		0,1	Partial knockout σ=1.3 mb *2*.
2164 *6*	(5/2^{+})		2,3	Partial knockout σ=1.1 mb *2*.
2275 *6*				Partial knockout σ=1.6 mb *2*.
2377 *7*				Partial knockout σ=2.1 mb *2*. L: 2014St18 showed L=0,1,2,3 fits to the measured parallel momentum distribution but did not specify L. L=0 seems unlikely.
3611? *8*				2014St18 observed a surprising γ transition at 3611 keV, which must depopulate a state that is unbound to neutron emission by at least 1.1 MeV, considering S(n)=2470 *40* (2021Wa16). One possible explanation is this ^{35}Si level is populated by 1n knockout from the deeply-bound 1d$_{5/2}$ orbit in ^{36}Si and followed by the emission of the other 1d$_{5/2}$ neutron, resulting in a 2p-2h configuration dominated by two holes in the deeply bound neutron 1d$_{5/2}$ orbit. Such a configuration should have a small overlap with the ground state in ^{34}Si that has a large contribution of 0p-0h configurations, leading to a significantly hindered neutron decay. The γ decay, on the other hand, would be a fast 5/2^{+} → 7/2^{-} g.s. E1 decay and could potentially compete with the neutron emission. A second possible explanation could be that this state is weakly populated in a more complicated multistep reaction pathway. The 3611-keV γ results predominantly from events with residue momenta in the tail of the momentum distribution. The low-momentum tail is associated with more inelastic reactions, which would be consistent with a multistep process. Partial knockout σ=0.8 mb *2*.

† From a least-squares fit to γ-ray energies.

Continued on next page (footnotes at end of table)

^{9}Be(^{36}Si,^{35}Siγ) 2014St18,2015St06 (continued)

^{35}Si Levels (continued)

‡ As given in 2014St18 based on L-transfers and shell-model predictions.
# 2014St18 deduced L by comparing the measured and eikonal-calculated parallel momentum distributions of residuals.

γ(^{35}Si)

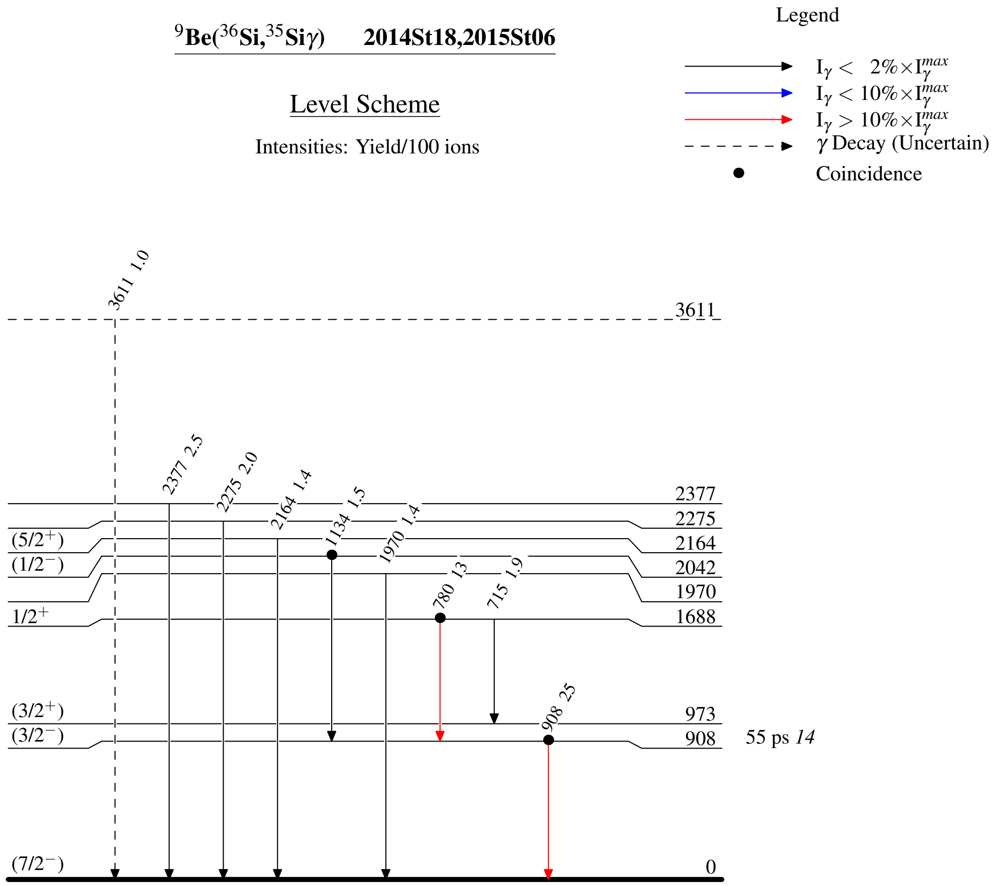
E_γ †	I_γ †	E_i(level)	J_i^π	E_f	J_f^π	Comments
715 *4*	1.9 *2*	1688	1/2$^+$	973	(3/2$^+$)	
780 *4*	13 *1*	1688	1/2$^+$	908	(3/2$^-$)	
908 *4*	25 *2*	908	(3/2$^-$)	0	(7/2$^-$)	B(E2)↓=0.0017 *+4−5* (2014St18)
1134 *5*	1.5 *2*	2042	(1/2$^-$)	908	(3/2$^-$)	
1970 *6*	1.4 *2*	1970		0	(7/2$^-$)	
2164 *6*	1.4 *2*	2164	(5/2$^+$)	0	(7/2$^-$)	
2275 *6*	2.0 *3*	2275		0	(7/2$^-$)	
2377 *7*	2.5 *3*	2377		0	(7/2$^-$)	
3611‡ *8*	1.0 *2*	3611?		0	(7/2$^-$)	

† From 2014St18, unless otherwise noted.
‡ Placement of transition in the level scheme is uncertain.

^{9}Be(^{36}Si,^{35}Siγ) 2014St18,2015St06

Level Scheme

Intensities: Yield/100 ions



$^{35}_{14}Si_{21}$

^{208}Pb(^{36}S,^{35}Siγ) 2010WaZT

2010WaZT,2010Wa20: A 215-MeV $^{36}S^{9+}$ beam was produced from the XTU-Tandem ALPI-superconducting linear accelerator complex at the INFN Legnaro National Laboratory, Italy. The target was 300 μg/cm^2 99.7% enriched ^{208}Pb with 20 μg/cm^2 carbon backing. Projectile-like fragments produced in binary grazing reactions were separated and identified by the PRISMA spectrometer. γ rays were detected using the CLARA array of 22 escape-suppressed Ge clover detectors covering the azimuthal angles from 98° to 180°. Measured Eγ with Doppler corrections, Iγ, and (^{35}Si)γ-coin. Deduced levels. Compared with shell-model calculations.

^{35}Si Levels

E(level)†	J^{π}‡
0	(7/2$^-$)
910	(3/2$^-$)
974	(3/2$^+$)

† From Eγ data.
‡ As given in 2010WaZT, based on shell-model calculations.

γ(^{35}Si)

E$_\gamma$†	I$_\gamma$†	E$_i$(level)	J^{π}_i	E$_f$	J^{π}_f
910	100 *25*	910	(3/2$^-$)	0	(7/2$^-$)
974‡	21 *10*	974	(3/2$^+$)	0	(7/2$^-$)

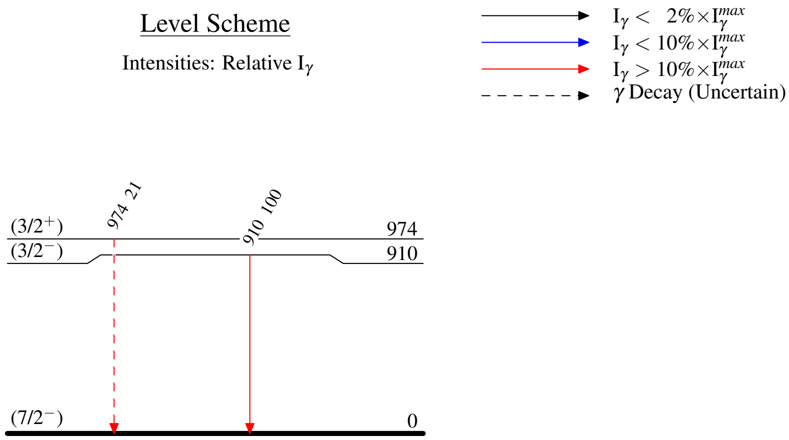
† From 2010WaZT. The author report ΔEγ=1 keV, but it is very likely statistical only since 910γ and 974γ are weak with poor statistics in the γ-ray spectrum, and thus is not adopted by the evaluators.
‡ Placement of transition in the level scheme is uncertain.

$^{208}Pb(^{36}S,^{35}Si\gamma)$ 2010WaZT

Level Scheme

Intensities: Relative I_γ

Legend



$^{35}_{14}Si_{21}$

Adopted Levels, Gammas

Q(β^-)=3988.4 *19*; S(n)=8380.4 *20*; S(p)=12155.1 *20*; Q(α)=−12332.0 *29* 2021Wa16
S(2n)=14663.1 *22*, S(2p)=30938 *7* (2021Wa16).
Isotope discovery (2012Th10): ^{232}Th(^{40}Ar,X) at Dubna (1971Ar32) and ^{37}Cl(γ,2p)^{35}P at Mainz (1971Gr53).
^{35}P production:
2012Kw02: E(^{40}Ar)=140 MeV/nucleon impinged on ^{9}Be and natural nickel targets at NSCL. Measured fragmentation cross sections, parallel momentum transfers, and widths. Compared with empirical formula EPAX, and predictions from internuclear cascade and deep inelastic models using Monte Carlo ISABEL-GEMINI and DIT-GEMINI codes.
2012Zh06: ^{9}Be,^{181}Ta(^{40}Ar,X) at E(^{40}Ar)=57 MeV/nucleon at HIRFL. Measured momentum distributions and production cross sections of fragments. Observed competition between projectile fragmentation and other mechanisms. Compared with EPAX, abrasion-ablation, and HIPSE models. Studied target dependence of fragment cross sections.
2007No13: ^{9}Be,^{181}Ta(^{40}Ar,X) and (^{40}Ar,X) at E(^{40}Ar)=100 MeV/nucleon at RIKEN. Measured fragment momentum distributions and production cross sections.
1997Vo03: ^{56}Fe(p,X) at E_p=800 MeV at LANL. Measured γ radiation. Deduced production cross sections.
^{35}P decay measurements:
1986Wa22: ^{35}P produced via ^{36}S(t,α)^{35}P at BNL. Measured β-delayed γ rays.
1972Ap01: ^{35}P produced via ^{37}Cl(t,αp)^{35}P at LANL. Measured $T_{1/2}$ and β-delayed γ rays.
1972Go31: ^{35}P produced via ^{18}O(^{19}F,2p)^{35}P and ^{36}S(t,α)^{35}P at BNL. Measured $T_{1/2}$ and β-delayed γ rays.
1971Gr53: ^{35}P produced via ^{37}Cl(γ,2p)^{35}P at Mainz. Measured $T_{1/2}$ and γ rays.
^{35}P radius measurement:
1999Ai02: Si(^{35}P,X) at NSCL. Measured energy-integrated reaction cross sections at E=38-80 MeV/ nucleon. Deduced strong absorption radii.
^{35}P mass measurements:
^{34}S(^{18}O,^{17}F) and ^{37}Cl(^{11}B,^{13}N) (1988Or01), ^{36}S(^{6}Li,^{7}Be) (1985Dr06), ^{36}S(d,3He) (1985Kh04), ^{36}S(^{14}C,^{15}N) (1984Ma49).
Theoretical calculations (binding energies, dipole moments, quadrupole moments, radii, levels, J, π, etc.): 2012BoZT, 2009No01, 2004Kh16, 2003Sm02, 1999Du05, 1988Wa04, 1987Wa10, 1986Wo02,1983Wi08, 1975JeZX.

^{35}P Levels

Cross Reference (XREF) Flags

A	^{35}Si β^- decay (0.78 s)	E	^{9}Be(^{36}S,^{35}Pγ)	I	^{37}Cl(^{11}B,^{13}N)
B	^{1}H(^{34}Si,p):resonances	F	^{34}S(^{18}O,^{17}F)	J	^{160}Gd(^{37}Cl,Xγ)
C	^{2}H(^{34}Si,^{35}Pγ)	G	^{36}S(pol d,^{3}He)	K	^{208}Pb(^{36}S,^{35}Pγ)
D	^{2}H(^{36}S,^{3}He)	H	^{36}S(d,^{3}He)		

E(level)[†]	J$^\pi$	$T_{1/2}$ or Γ[#]	XREF	Comments
0	1/2$^+$	47.3 s *8*	A CDEFGHIJK	%β^-=100 J$^\pi$: L(pol d,^{3}He)=0 from 0$^+$ and analyzing power. $T_{1/2}$: weighted average of 45 s *2* (1971Gr53, γ(t)), 47.7 s *29* (1972Go31, γ(t)), 48.2 s *17* (1972Go31, γ(t)), and 47.4 s *8* (1972Ap01, γ(t)).
2386.9 *11*	3/2$^+$	<0.69 ps	A CDEFG I K	XREF: F(2420) J$^\pi$: L(pol d,^{3}He)=2 from 0$^+$ and L-1/2 transfer from analyzing power.
3860.4 *11*	5/2$^+$	<0.69 ps	A CDE GHIJK	J$^\pi$: L(pol d,^{3}He)=2 from 0$^+$ and L+1/2 transfer from analyzing power.
4101.7 *11*	(7/2$^-$)[‡]	>69 ps	A C E JK	
4250 *20*			I	
4382.0 *12*	(5/2$^-$)		A E K	XREF: A(?) J$^\pi$: possibly allowed β^- feeding from 7/2$^-$ parent; 1994.9γ to 3/2$^+$.
4494.1 *12*	(7/2$^-$)[‡]	2.29 ps *49*	A C E G JK	XREF: G(4474) J$^\pi$: L=(3) in ^{9}Be(^{36}S,^{35}P) from 0$^+$.
4666.2 *16*	5/2$^+$		DE GHI	XREF: I(4640) J$^\pi$: L(pol d,^{3}He)=2 from 0$^+$ and L+1/2 transfer from analyzing power.
4767.0 *13*	(9/2$^-$)[‡]		E K	

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued)

^{35}P Levels (continued)

E(level)†	Jπ	$T_{1/2}$ or Γ#	XREF	Comments
4869.6 *12*	(5/2⁻,7/2⁻)		A C K	Jπ: possibly allowed β^- feeding from 7/2⁻ parent; 1009.7γ to 5/2⁺.
4962.8 *12*	(9/2⁻)‡		A E K	XREF: A(?)
5010 *20*			I	1988Or01 observed a 5010 *20* level in (^{11}B,^{13}N) and a 5070 *40* level in (^{18}O,^{17}F) but did not assign them as the same level.
5090.2 *13*	(11/2⁻)‡		EF K	XREF: F(5070)
5199.3 *16*	5/2⁺		DE GHI	XREF: I(5220) Jπ: L(pol d,^{3}He)=2 from 0⁺ and L+1/2 transfer from analyzing power.
5487.9 *13*			K	
5561.0 *13*	(5/2⁻)		A K	Jπ: possibly allowed β^- feeding from 7/2⁻ parent; 3173.5γ to 3/2⁺.
5709.5 *23*	(1/2⁻)		DE	Jπ: L=(1) in ^{9}Be(^{36}S,^{35}P) from 0⁺; interpreted as the deeply bound $1p_{1/2}$ proton removal from 0⁺ in ^{9}Be(^{36}S,^{35}P); 5709γ to 1/2⁺.
5.86×10³ *5*			F I	XREF: F(5890)I(5840) E(level): weighted average of 5890 *70* from (^{18}O,^{17}F) and 5840 *50* from (^{11}B,^{13}N).
6222.7 *13*	(7/2⁻,9/2,11/2⁻)		K	Jπ: 1132γ to (11/2⁻) and 1729γ to (7/2⁻).
6440 *60*			F	
7050 *60*			F	
7440 *60*			Fg	XREF: g(7520)
7526.9 *23*	(1/2⁻)		E g	XREF: g(7520) Jπ: L=(1) in ^{9}Be(^{36}S,^{35}P) from 0⁺; interpreted as the deeply bound $1p_{1/2}$ proton removal from 0⁺ in ^{9}Be(^{36}S,^{35}P); 7526γ to 1/2⁺.
7590 *20*			I	
7920 *60*			F	
8390 *40*			I	
8.60×10³ *10*			F	
9290 *50*			F	
14938 *24*	1/2⁺	<12.7 keV	B	Jπ: L=0 in ^{1}H(^{34}Si,p).
15161 *3*	5/2⁻,7/2⁻	<4.4 keV	B	Jπ: L=3 in ^{1}H(^{34}Si,p).
15306 *24*	3/2⁺,5/2⁺	<30.4 keV	B	Jπ: L=2 in ^{1}H(^{34}Si,p).
15964 *18*	3/2⁺,5/2⁺	84 keV *25*	B	Jπ: L=2 in ^{1}H(^{34}Si,p).
16145 *36*	1/2⁻,3/2⁻	0.35 MeV *9*	B	Jπ: L=1 in ^{1}H(^{34}Si,p).
16605 *44*	1/2⁺	0.22 MeV *15*	B	Jπ: L=0 in ^{1}H(^{34}Si,p).
17254 *12*	3/2⁺,5/2⁺	<11.6 keV	B	Jπ: L=2 in ^{1}H(^{34}Si,p).
17355 *15*	1/2⁻,3/2⁻	32 keV *22*	B	Jπ: L=1 in ^{1}H(^{34}Si,p).

† From a least-squares fit to γ-ray energies for levels connected with γ transitions, from particle-transfer reactions for other levels, or from proton elastic scattering for resonances.

‡ Comparisons with shell-model calculations (2019Gr08).

$T_{1/2}$ from the differential recoil-distance method (2019Gr08) in ^{208}Pb(^{36}S,^{35}Pγ) and widths from the R-matrix analysis of (^{34}Si,p) for resonances, unless otherwise noted.

Adopted Levels, Gammas (continued)

$\gamma(^{35}P)$

E_i(level)	J^π_i	E_γ	I_γ	E_f	J^π_f	Mult.	δ	α#	Comments
2386.9	$3/2^+$	2386.3 *6*	100	0	$1/2^+$	[M1,E2]		0.00046 *5*	E_γ: weighted average of 2386.4 *6* from ^{35}Si β^- decay, 2386 *2* from $^9Be(^{36}S,^{35}P\gamma)$, and 2386 *1* from $^{208}Pb(^{36}S,^{35}P\gamma)$. B(M1)(W.u.)>0.0023 if M1, B(E2)(W.u.)>1.6 if E2.
3860.4	$5/2^+$	1473.5 *5*	15.6 *14*	2386.9	$3/2^+$	[M1,E2]		8.3×10^{-5} *13*	E_γ: weighted average of 1473.4 *5* from ^{35}Si β^- decay, 1473 *2* from $^9Be(^{36}S,^{35}P\gamma)$, and 1474 *1* from $^{208}Pb(^{36}S,^{35}P\gamma)$. I_γ: weighted average of 14.1 *33* from ^{35}Si β^- decay and 15.9 *14* from $^{208}Pb(^{36}S,^{35}P\gamma)$. B(M1)(W.u.)>0.0012 if M1, B(E2)(W.u.)>2.1 if E2.
		3860.2 *10*	100.0 *32*	0	$1/2^+$	[E2]		1.12×10^{-3} *2*	B(E2)(W.u.)>0.12 E_γ: weighted average of 3859.5 *10* from ^{35}Si β^- decay, 3860 *2* from $^9Be(^{36}S,^{35}P\gamma)$, and 3861 *1* from $^{208}Pb(^{36}S,^{35}P\gamma)$. I_γ: From $^{208}Pb(^{36}S,^{35}P\gamma)$. Other: 100 *7* from ^{35}Si β^- decay.
4101.7	$(7/2^-)$	241.3 *5*	100† *7*	3860.4	$5/2^+$	[E1]		0.000665 *10*	B(E1)(W.u.)<4.4×10^{-4} E_γ: weighted average of 241.4 *3* from ^{35}Si β^- decay, 237 *2* from $^9Be(^{36}S,^{35}P\gamma)$, and 241 *1* from $^{208}Pb(^{36}S,^{35}P\gamma)$. I_γ: other: 100 *4* from ^{35}Si β^- decay.
		1714.8 *6*	6.6† *17*	2386.9	$3/2^+$	[M2]		7.93×10^{-5} *11*	B(M2)(W.u.)<0.16 E_γ: weighted average of 1714.7 *6* from ^{35}Si β^- decay and 1715 *1* from $^{208}Pb(^{36}S,^{35}P\gamma)$. I_γ: other: 22 *5* from ^{35}Si β^- decay.
		4101.4 *10*	54† *8*	0	$1/2^+$	[E3]		0.000924 *13*	B(E3)(W.u.)<4.8 E_γ: weighted average of 4100.8 *10* from ^{35}Si β^- decay and 4102 *1* from $^{208}Pb(^{36}S,^{35}P\gamma)$. I_γ: other: 135 *8* from ^{35}Si β^- decay.
4382.0	$(5/2^-)$	1994.9 *6*	100	2386.9	$3/2^+$				E_γ: weighted average of 1994.8 *6* from ^{35}Si β^- decay, 1995 *2* from $^9Be(^{36}S,^{35}P\gamma)$, and 1995 *1* from $^{208}Pb(^{36}S,^{35}P\gamma)$. Placement by 1988DuZS, 2008Wi09, and 2016Mu03. 1988DuZT and 1987Wa10 placed this γ as the 6096 to 4101 transition. 1988Or01 placed this γ as the 6488 to 4493 transition.
4494.1	$(7/2^-)$	392.3 *3*	100 *5*	4101.7	$(7/2^-)$	[M1+E2]	<0.22	0.000199 *12*	B(M1)(W.u.)=0.117 *+42−29* E_γ: weighted average of 392.3 *3* from ^{35}Si β^- decay, 391

$^{35}_{15}P_{20}$-3

Adopted Levels, Gammas (continued)

$\gamma(^{35}P)$ (continued)

E_i(level)	J^{π}_i	E_γ	I_γ	E_f	J^{π}_f	Mult.	α#	Comments
								2 from ^{9}Be(^{36}S,^{35}Pγ), and 392 *1* from ^{208}Pb(^{36}S,^{35}Pγ). I_γ: From ^{35}Si β^- decay. Other: 100 *17* from ^{208}Pb(^{36}S,^{35}Pγ). δ: deduced by evaluators from RUL=100 for B(E2)(W.u.).
4494.1	(7/2$^-$)	633.6 *5*	34 *5*	3860.4	5/2$^+$	[E1]	4.64×10^{-5} *7*	B(E1)(W.u.)=2.8×10^{-4} *+8−6* E_γ: weighted average of 633.7 *5* from ^{35}Si β^- decay, 634 *2* from ^{9}Be(^{36}S,^{35}Pγ), and 633 *1* from ^{208}Pb(^{36}S,^{35}Pγ). I_γ: weighted average of 38 *5* from ^{35}Si β^- decay and 27 *7* from ^{208}Pb(^{36}S,^{35}Pγ).
4666.2	5/2$^+$	804‡ *2*		3860.4	5/2$^+$			
		2279‡ *2*		2386.9	3/2$^+$			
		4668‡ *2*		0	1/2$^+$			
4767.0	(9/2$^-$)	273 *1*	40.0† *25*	4494.1	(7/2$^-$)			E_γ: weighted average of 274 *2* from ^{9}Be(^{36}S,^{35}Pγ) and 273 *1* from ^{208}Pb(^{36}S,^{35}Pγ).
		664 *1*	100† *47*	4101.7	(7/2$^-$)			E_γ: weighted average of 666 *2* from ^{9}Be(^{36}S,^{35}Pγ) and 664 *1* from ^{208}Pb(^{36}S,^{35}Pγ).
4869.6	(5/2$^-$,7/2$^-$)	374† *1*	60† *20*	4494.1	(7/2$^-$)			
		487† *1*	<40†	4382.0	(5/2$^-$)			
		767.9 *4*	100† *20*	4101.7	(7/2$^-$)			E_γ: weighted average of 768.0 *4* from ^{35}Si β^- decay and 767 *1* from ^{208}Pb(^{36}S,^{35}Pγ). I_γ: other: 100 *18* from ^{35}Si β^- decay.
		1009.7 *5*	<20†	3860.4	5/2$^+$			E_γ: weighted average of 1009.9 *5* from ^{35}Si β^- decay and 1009 *1* from ^{208}Pb(^{36}S,^{35}Pγ). I_γ: other: 152 *32* from ^{35}Si β^- decay.
4962.8	(9/2$^-$)	468.9 *4*	100† *8*	4494.1	(7/2$^-$)			E_γ: weighted average of 468.9 *4* from ^{35}Si β^- decay, 469 *2* from ^{9}Be(^{36}S,^{35}Pγ), and 468 *2* from ^{208}Pb(^{36}S,^{35}Pγ).
		859† *3*	66† *9*	4101.7	(7/2$^-$)			
5090.2	(11/2$^-$)	128 *1*	50† *25*	4962.8	(9/2$^-$)			E_γ: weighted average of 127 *2* from ^{9}Be(^{36}S,^{35}Pγ) and 128 *1* from ^{208}Pb(^{36}S,^{35}Pγ).
		322 *1*	100† *35*	4767.0	(9/2$^-$)			E_γ: weighted average of 321 *2* from ^{9}Be(^{36}S,^{35}Pγ) and 322 *1* from ^{208}Pb(^{36}S,^{35}Pγ).
5199.3	5/2$^+$	1337‡ *2*		3860.4	5/2$^+$			
		2811‡ *2*		2386.9	3/2$^+$			
		5202‡ *2*		0	1/2$^+$			
5487.9		993† *1*	100† *20*	4494.1	(7/2$^-$)			
		1387† *1*	60† *20*	4101.7	(7/2$^-$)			

$^{35}_{15}P_{20}$-4

Adopted Levels, Gammas (continued)

$\gamma(^{35}P)$ (continued)

E_i(level)	J^π_i	E_γ	I_γ	E_f	J^π_f	Comments
5561.0	$(5/2^-)$	1459.4 *7*	34 *12*	4101.7	$(7/2^-)$	E_γ: weighted average of 1459.7 *5* from ^{35}Si β^- decay and 1458 *1* from ^{208}Pb(^{36}S,^{35}Pγ). I_γ: From ^{35}Si β^- decay.
		3173.5 *10*	100 *17*	2386.9	$3/2^+$	3173.5γ is not observed in ^{208}Pb(^{36}S,^{35}Pγ), but the weaker 1459γ deexciting the same level is observed in ^{208}Pb(^{36}S,^{35}Pγ). Further experiments are needed to resolve the discrepancy. E_γ,I_γ: From ^{35}Si β^- decay.
5709.5	$(1/2^-)$	5709‡ *2*		0	$1/2^+$	
6222.7	$(7/2^-,9/2,11/2^-)$	1132† *1*	<25†	5090.2	$(11/2^-)$	
		1260† *1*	100† *25*	4962.8	$(9/2^-)$	
		1729† *1*	100† *25*	4494.1	$(7/2^-)$	
7526.9	$(1/2^-)$	7526 *2*		0	$1/2^+$	

† From ^{208}Pb(^{36}S,^{35}Pγ).

‡ From ^{9}Be(^{36}S,^{35}Pγ).

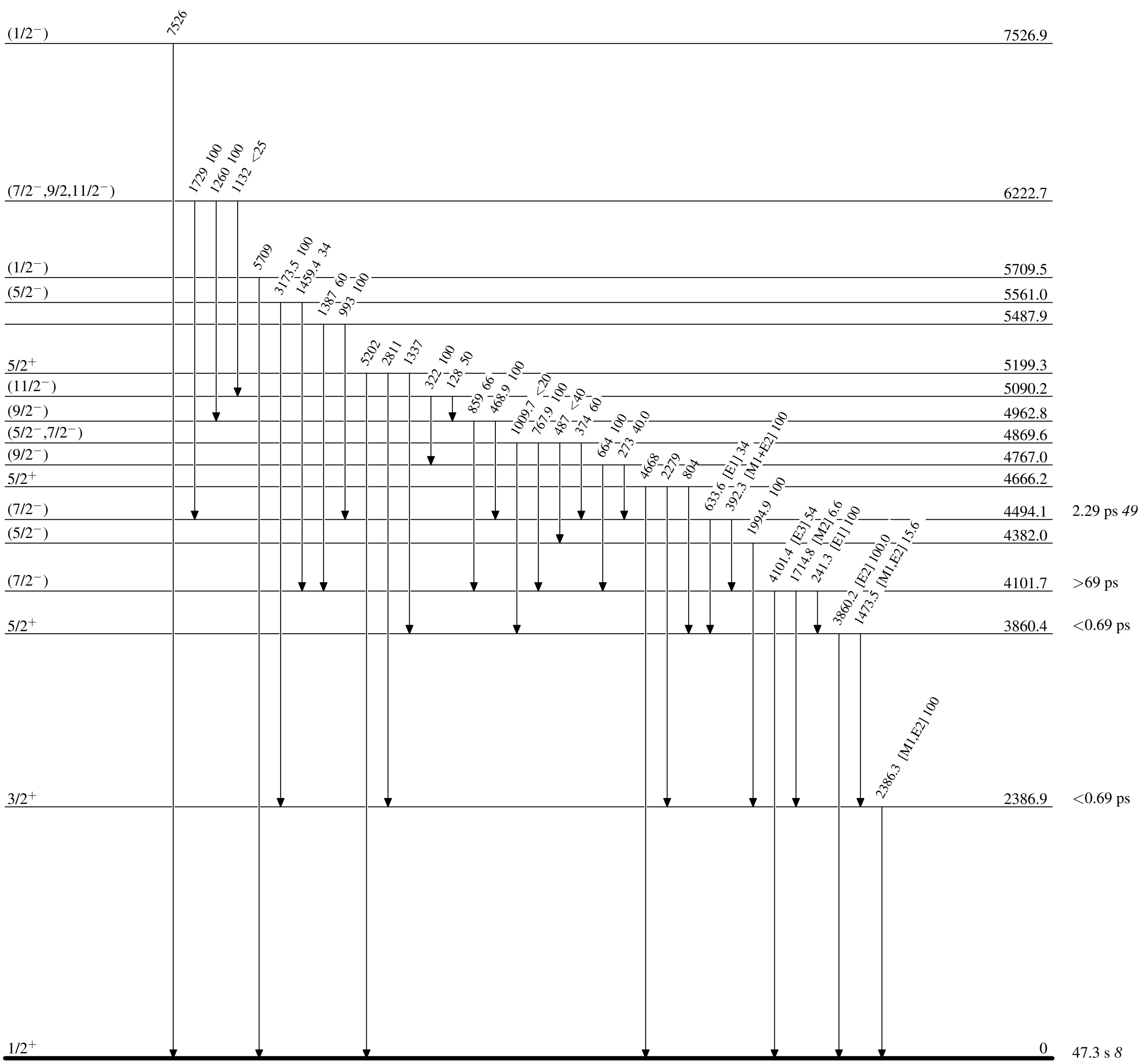
Total theoretical internal conversion coefficients, calculated using the BrIcc code (2008Ki07) with "Frozen Orbitals" approximation based on γ-ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

$^{35}_{15}P_{20}$-5

Adopted Levels, Gammas

Level Scheme

Intensities: Relative photon branching from each level



$^{35}_{15}P_{20}$

^{35}Si β^- decay (0.78 s) 1988DuZS,1986Du07,1988DuZT

Parent: ^{35}Si: E=0; J$^\pi$=(7/2)$^-$; $T_{1/2}$=0.78 s *12*; Q(β^-)=10470 *40*; %β^- decay=100

^{35}Si-J$^\pi$,$T_{1/2}$: From the Adopted Levels of ^{35}Si.

^{35}Si-Q(β^-): From 2021Wa16.

1988DuZS,1986Du07,1988DuZT,1987DuZU: ^{35}Si was produced by the fragmentation of an ^{40}Ar beam of 2x10^{11} particles/s at 60 MeV/nucleon on a 190 mg/cm^2 Be target at GANIL. Decay observed with a 1 mm thick plastic scintillator and a 174 cm^3 intrinsic Ge detector with 1.2% absolute efficiency at 1.33 MeV. Measured $\beta\gamma$(t), Eγ, Iγ. Deduced levels, J, π, parent $T_{1/2}$.

2007Ne14: measured ^{35}Si ground state magnetic moment and *g*-factor using the β-NMR method.

1987Wa10: shell-model calculations for ^{35}Si β^- decay scheme, ^{35}P levels, decay branching ratios, log *ft*, and Gamow-Teller transition strengths.

The decay scheme is considered incomplete due to a large gap of about 4.9 MeV between the highest observed level at E=5561 and Q(β^-) value=10470 *40* (2021Wa16). There may be missing transitions from unobserved levels in the gap.

^{35}P Levels

E(level)†	J$^\pi$‡	$T_{1/2}$‡
0	1/2$^+$	47.3 s *8*
2386.5 *5*	3/2$^+$	<0.69 ps
3859.7 *5*	5/2$^+$	<0.69 ps
4101.2 *5*	(7/2$^-$)	>69 ps
4381.3? *8*	(5/2$^-$)	
4493.5 *6*	(7/2$^-$)	2.29 ps *49*
4869.4 *6*	(5/2$^-$,7/2$^-$)	
4962.4? *7*	(9/2$^-$)	
5560.7 *7*	(5/2$^-$)	

† From a least-squares fit to γ-ray energies.

‡ From the Adopted Levels.

β^- radiations

E(decay)	E(level)	Iβ^-†‡	Log *ft*†	Comments
(4.91×10^3 *4*)	5560.7	12.7	4.6	Iβ=12.4 *18* and log *ft*=4.6 from 1988DuZS.
(5.51×10^3 *4*)	4962.4?	5.1	5.2	Placed based on the 4959 to 4493 transition observed in ^{208}Pb(^{36}S,Xγ) (2008Wi09) and adopted 468.9Iγ from 1988DuZS to deduce its Iβ.
(5.60×10^3 *4*)	4869.4	10.8	4.9	Iβ=10.8 *16* and log *ft*=4.9 from 1988DuZS.
(5.98×10^3 *4*)	4493.5	16.6	4.9	Iβ=21.4 *11* and log *ft*=4.8 from 1988DuZS. 468.9Iγ feeding this level is deducted from its Iβ.
(6.09×10^3 *4*)	4381.3?	9.7	5.1	Iβ=9.4 *13* and log *ft*=5.1 from 1988DuZS.
(6.37×10^3 *4*)	4101.2	46.1	4.6	Iβ=45.9 *31* and log *ft*=4.5 from 1988DuZS.
(8.08×10^3 *4*)	2386.5	1.9	8.6^{1u}	

† β-feeding from γ-ray intensity balance at each level. Quoted Iβ^- values are considered upper limits due to the incomplete decay scheme, and the associated log *ft* values are considered lower limits.

‡ Absolute intensity per 100 decays.

$\gamma(^{35}P)$

Iγ normalization: From Σ%I(γ to g.s.)=100. The deduced normalization factor of 0.27 should be considered an upper limit due to potential missing γ transitions from unobserved levels in the gap to the ground state.

E_γ[‡]	I_γ[‡@]	E_i(level)	J^π_i	E_f	J^π_f	Mult.[†]	α[&]	Comments
241.4 3	100 4	4101.2	(7/2$^-$)	3859.7	5/2$^+$	[E1]	0.000664 10	%Iγ=27
392.3 3	58.2 28	4493.5	(7/2$^-$)	4101.2	(7/2$^-$)	[M1+E2]	4.4×10^{-4} 25	%Iγ=16
468.9[a] 4	18.7 25	4962.4?	(9/2$^-$)	4493.5	(7/2$^-$)			%Iγ=5.0 Unplaced γ ray in 1988DuZS. The placement is based on a 466γ observed in ^{208}Pb(^{36}S,Xγ) (2008Wi09).
633.7 5	21.9 28	4493.5	(7/2$^-$)	3859.7	5/2$^+$	[E1]	4.64×10^{-5} 7	%Iγ=5.9
768.0 4	15.9 29	4869.4	(5/2$^-$,7/2$^-$)	4101.2	(7/2$^-$)			%Iγ=4.3
1009.9 5	24 5	4869.4	(5/2$^-$,7/2$^-$)	3859.7	5/2$^+$			%Iγ=6.5
1459.7 5	12 4	5560.7	(5/2$^-$)	4101.2	(7/2$^-$)			%Iγ=3.2
1473.4 5	17 4	3859.7	5/2$^+$	2386.5	3/2$^+$	[M1,E2]	8.3×10^{-5} 13	%Iγ=4.6
1714.7 6	22 5	4101.2	(7/2$^-$)	2386.5	3/2$^+$	[M2]	7.93×10^{-5} 11	%Iγ=5.9
1994.8[a] 6	36 6	4381.3?	(5/2$^-$)	2386.5	3/2$^+$			%Iγ=9.7 Placement from 1988DuZS, consistent with the 1995γ 4381 to 2386 transition observed in ^{208}Pb(^{36}S,Xγ) (2008Wi09) and the 1995γ 4382 to 2386 transition observed in ^{9}Be(^{36}S,^{35}Pγ) (2016Mu03). 1988DuZT and 1987Wa10 placed this γ as the 6096 to 4101 transition. 1988Or01 placed this γ as the 6488 to 4493 transition.
2386.4 6	117 7	2386.5	3/2$^+$	0	1/2$^+$	[M1,E2]	0.00046 5	%Iγ=32
3173.5 10	35 6	5560.7	(5/2$^-$)	2386.5	3/2$^+$			%Iγ=9.5 1988Or01 suggested that this γ ray cannot be placed into the decay scheme of ^{35}Si, and it could be a transition to the 4101 or 4493 levels based on intensity balances.
^{x}3349.1[#] 10	46[#] 6							%Iγ=12
^{x}3590.0[#] 11	60[#] 7							%Iγ=16
3859.5 10	121 8	3859.7	5/2$^+$	0	1/2$^+$	[E2]	1.12×10^{-3} 2	%Iγ=33
4100.8 10	135 8	4101.2	(7/2$^-$)	0	1/2$^+$	[E3]	0.000924 13	%Iγ=36

† From the Adopted Levels.

‡ From 1988DuZS, unless otherwise noted.

From 1986Du07. 1988Or01 tentatively suggested that these γ rays deexcite a level at 7450, but this suggestion has not been experimentally confirmed.

@ For absolute intensity per 100 decays, multiply by 0.27.

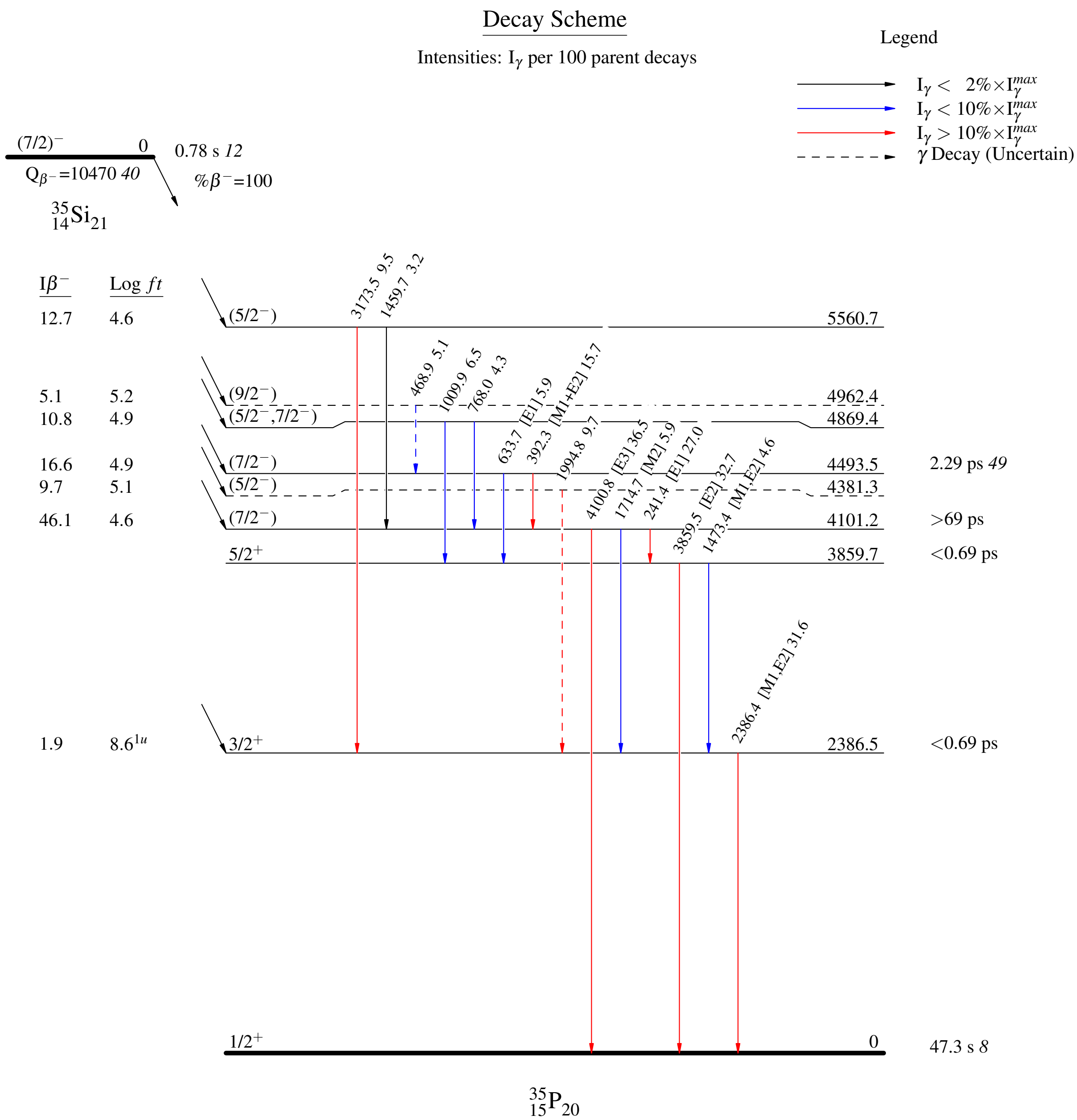
& Total theoretical internal conversion coefficients, calculated using the BrIcc code (2008Ki07) with "Frozen Orbitals" approximation based on γ-ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

$\gamma(^{35}$P) (continued)

[a] Placement of transition in the level scheme is uncertain.
[x] γ ray not placed in level scheme.

$^{35}_{15}$P$_{20}$-9

^{35}Si β^- decay (0.78 s) 1988DuZS,1986Du07,1988DuZT



$^{1}H(^{34}Si,p)$:resonances **2012Im01**

$J^{\pi}=0^{+}$ for ^{34}Si ground state.

2012Im01: A ^{34}Si beam at 7×10^{4} pps and a purity of 97% was produced by the projectile fragmentation of a 63-MeV/nucleon ^{40}Ar primary beam and separated by the RIPS separator at RIKEN. The secondary target was a 10.9-mg/cm^{2} *5* polyethylene film. An incident energy of 4.4 MeV/nucleon *12* for the ^{34}Si beam was determined by the timing difference between a plastic scintillator and two PPACs placed upstream of the target. The PPACs also record the positions and angles of the projectiles incident upon the target. Outgoing particles were detected and identified by a three-layer ΔE-E telescope consisting of 0.5-mm DSSD, 1.5-mm silicon, and 1.5-mm silicon detectors mounted at 0° with an E_{lab} resolution σ=130 keV. Measured excitation functions of proton elastic scattering on ^{34}Si for θ_{lab} < 10° using thick target inverse kinematics. Deduced E_R, L-transfer, Γ_p, and Γ from R-matrix analysis for 8 resonances in the highly excited states in ^{35}P, which are isobaric analog states of ^{35}Si states.

^{35}P Levels

All data are from 2012Im01.

E(level)†	Γ	L	S‡	Comments
14938 *24*	<12.7 keV	0		Γ: from original value of 4.6 keV *81* in 2012Im01. E_R=2783 *24*, Γ_p=4.6 keV *28*.
15161 *3*	<4.4 keV	3	0.63 *16*	Γ: from original value of 1.6 keV *28* in 2012Im01. E_R=3006 *2*, Γ_p=1.6 keV *4*. IAR of the 7/2^{-} g.s. of ^{35}Si.
15306 *24*	<30.4 keV	2	0.19 *15*	Γ: from original value of 10.4 keV *200* in 2012Im01. E_R=3151 *24*, Γ_p=3.3 keV *27*.
15964 *18*	84 keV *25*	2	0.79 *20*	E_R=3809 *18*, Γ_p=26.7 keV *69* in 2012Im01.
16145 *36*	354 keV *87*	1	1.37 *32*	E_R=3990 *36*, Γ_p=185 keV *43*.
16605 *44*	0.22 MeV *15*	0	0.45 *28*	E_R=4450 *44*, Γ_p=58.4 keV *370*.
17254 *12*	<11.6 keV	2	0.04 *1*	Γ: from original value of 3.8 keV *78* in 2012Im01. E_R=5099 *12*, Γ_p=3.8 keV *9*.
17355 *15*	32 keV *22*	1	0.12 *7*	E_R=5200 *15*, Γ_p=20.9 keV *120*.

† Excitation energies are deduced by evaluators from $E_R+S_p(^{35}P)$, where $S_p(^{35}P)$=12155.1 *20* (2021Wa16). E_R given in 2012Im01 are in the center-of-mass system.

‡ Spectroscopic factors are derived from Γ_p using the formula from 1968Th07 as described in 2012Im01.

^{2}H(^{34}Si,^{35}Pγ) 2007GeZX

^{34}Si(d,n)^{35}P from J^{π}=0^{+} ^{34}Si g.s. in inverse kinematics.

2007GeZX: 30-AMeV ^{34}Si beam on 30-mg/cm^2 CD_2 secondary target at GANIL. Heavy ions produced in reactions were identified by the VAMOS spectrometer. γ rays were detected using the EXOGAM germanium clover array. Measured Doppler-corrected Eγ, Iγ, $\gamma\gamma$-coin, and (^{35}P)γ-coin. Deduced levels, J, π. Compared with shell-model calculations.

^{35}P Levels

E(level)†
0
2386.5 *8*
3859.9 *8*
4100.9 *13*
4492.9 *16*
4868.9 *13*

† From a least-squares fit to γ-ray energies, assuming ΔEγ=1 keV since uncertainties were not given in 2007GeZX.

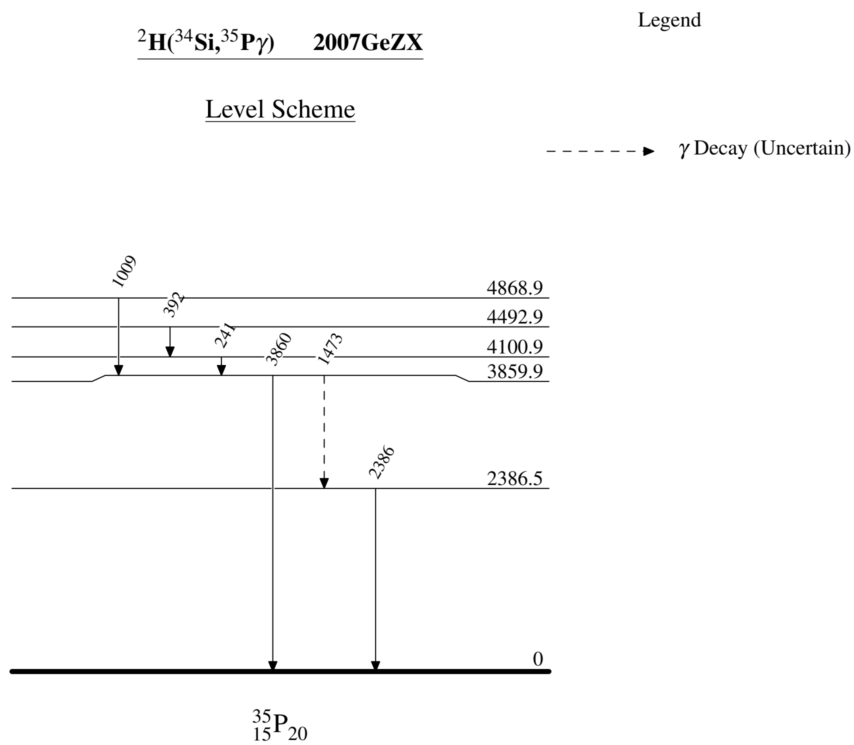
γ(^{35}P)

E_γ†	E_i(level)	E_f
241	4100.9	3859.9
392	4492.9	4100.9
1009	4868.9	3859.9
1473‡	3859.9	2386.5
2386	2386.5	0
3860	3859.9	0

† From 2007GeZX.
‡ Placement of transition in the level scheme is uncertain.

$^{2}H(^{34}Si,^{35}P\gamma)$ 2007GeZX

Level Scheme



^{2}H(^{36}S,^{3}He) 2020Sa44

2020Sa44: E(^{36}S)=15.3 MeV/nucleon was provided by the Argonne Tandem Linear Accelerator System. Targets were 81, 127, 529-μg/cm^2 deuterated plastic. The reaction was studied in inverse kinematics with the HELical Orbit Spectrometer (HELIOS). The ^{3}He ions spiralled in a 2.85-T magnetic field and were detected by a position-sensitive silicon array. The heavy-ion products were measured with a 65-μm thick silicon detector installed between the target and the silicon array. Measured σ(E(^{3}He),θ). Deduced relative spectroscopic factors for 6 levels from the PTOLEMY-DWBA analysis of the angular distributions. A potential 1/2$^+$ neutron 2p2h excitation bandhead in 2.5-3.6 MeV was investigated, and an upper limit for the transfer reaction cross section to populate such an intruder configuration was deduced.

^{35}P Levels

2020Sa44 states that the measurement of the beam current was not made with sufficient accuracy to calculate absolute spectroscopic factors. The C^{2}S values are relative spectroscopic factors normalized such that the ground-state value is 2.

E(level)†	J$^\pi$‡	C^{2}S#
0 *1*	1/2$^+$	2.0
2388 *13*	3/2$^+$	0.33 *9*
3860 *2*	5/2$^+$	2.9 *10*
4666 *9*	5/2$^+$	0.71 *34*
5202 *8*	5/2$^+$	1.1 *6*
5706 *38*	(1/2$^-$)	0.23 *5*

† From 2020Sa44.
‡ Assumed by 2020Sa44 for deducing C^{2}S, consistent with the Adopted Levels.
# Uncertainties are dominated by systematic components.

^{9}Be(^{36}S,^{35}Pγ) 2016Mu03

J^π=0$^+$ for ^{36}S ground state.

2016Mu03: A secondary beam of ^{36}S was produced via the projectile fragmentation of a 140-MeV/nucleon ^{48}Ca primary beam impinging on an 846-mg/cm^2 ^{9}Be target at the coupled cyclotron facility at NSCL, MSU. The ^{36}S nuclei were selected using the A1900 separator with an intensity of 8.1×10^5 pps and a purity of 89.7%. The excited states of ^{35}P were populated by the one-proton knockout reaction from the ^{36}S beam at 88 MeV/nucleon (midtarget energy) on a 100-mg/cm^2 ^{9}Be secondary target located at the reaction target position of the S800 spectrometer. The projectile-like residues were identified from their energy loss measured by an ionization chamber located at the focal plane of the S800 spectrometer and from their ToF measured between two scintillators situated at the object position and at the focal plane of the S800 spectrometer. Prompt γ rays from the deexcitation of the ^{35}P residues were detected by seven modules of the GRETINA Ge array. Measured Eγ, Iγ, (^{35}P)γ-coin, $\gamma\gamma$-coin, the inclusive knockout cross section for producing ^{35}P from ^{36}S, the fractional populations and parallel momentum distributions of 8 populated states in ^{35}P residues. Deduced levels, J, π, L-transfers, partial knockout cross sections and spectroscopic factors. Calculated single-particle cross sections (σ_{sp}) for proton removal and parallel momentum distributions using the eikonal model. This work is also part of the thesis 2015MuZY.

^{35}P Levels

b_f values under comments are population fraction in the knock-out reaction, σ_f^{sp} values are for calculated single-particle cross sections, and R_s values are quenching factors from a systematics study in 2014To14. The partial cross sections to each level can be deduced from $b_f\times\sigma_{inc}^{exp}$.

The inclusive cross section σ_{inc}^{exp} for producing ^{35}P from ^{36}S is measured to be 51 mb *1*. 95.3% is attributed to a direct knockout process from the p-sd shell, i.e., $\sigma_{inc,KO}^{exp}$=0.953×σ_{inc}^{exp}. The remaining 4.7% is possibly attributed to a charge-exchange or a two-step process: ^{36}S core excitation by inelastic scattering, followed by a proton removal. This is proposed based on their shifted parallel momentum distributions, their decay modes to negative parity states, their high excitation energy (around 4.7 MeV), and the fact that they were not observed in the ^{36}S(d,^{3}He) reaction study (1985Kh04).

E(level)†	J$^\pi$@	L#	C^{2}S&	Comments
0.0	1/2$^+$	0	2.2 *7*	b_f=31.4% *72*, σ_f^{sp}=13.5 mb, R_s=0.55 *11*.
2388 *1*	3/2$^+$	2	0.7 *3*	b_f=7.8% *36*, σ_f^{sp}=10.2 mb, R_s=0.52 *10*.
3862 *1*	5/2$^+$	2	2.7 *8*	b_f=27.3% *58*, σ_f^{sp}=10.6 mb, R_s=0.49 *10*.
4101 *2*	(7/2$^-$)‡			b_f=−0.7% *7*.
4383 *3*	(5/2$^-$)‡			b_f=2.0% *4*.
4494 *2*	(7/2$^-$)	(3)	0.30 *7*	L: 3 reported in 2016Mu03. The parentheses are added by evaluators based on the statement in 2016Mu03 that the parallel momentum distribution has limited statistics and seems more consistent with L=3. b_f=2.9% *5*, σ_f^{sp}=10.7 mb, R_s=0.48 *10*.
4667 *2*	5/2$^+$	2	1.0 *2*	b_f=9.5% *10*, σ_f^{sp}=10.3 mb, R_s=0.47 *9*.
4768 *2*	(9/2$^-$)‡			b_f=0.9% *3*.
4962 *3*	(9/2$^-$)‡			b_f=0.3% *2*.
5089 *3*	(11/2$^-$)‡			b_f=1.5% *2*.
5200 *2*	5/2$^+$	2	1.5 *3*	b_f=14.5% *14*, σ_f^{sp}=10.2 mb, R_s=0.47 *9*.
5710 *2*	(1/2$^-$)	(1)	0.21 *16*	b_f=1.4% *10*, σ_f^{sp}=10.8 mb, R_s=0.47 *9*. J$^\pi$: Interpreted as the deeply bound 1p$_{1/2}$ proton removal. L: 1 reported in 2016Mu03. The parentheses are added by evaluators based on the parallel momentum distribution (Fig. 3) in 2016Mu03.
7527 *2*	(1/2$^-$)	(1)	0.20 *6*	b_f=1.2% *3*, σ_f^{sp}=10.2 mb, R_s=0.44 *9*. J$^\pi$: Interpreted as the deeply bound 1p$_{1/2}$ proton removal. L: 1 reported in 2016Mu03. The parentheses are added by evaluators based on the parallel momentum distribution (Fig. 3) in 2016Mu03.

† From a least-squares fit to γ-ray energies.

‡ Populated by a two-step reaction mechanism. 2016Mu03 states that they are likely to have J≥5/2 and negative parities.

Continued on next page (footnotes at end of table)

^{9}Be(^{36}S,^{35}Pγ) 2016Mu03 (continued)

^{35}P Levels (continued)

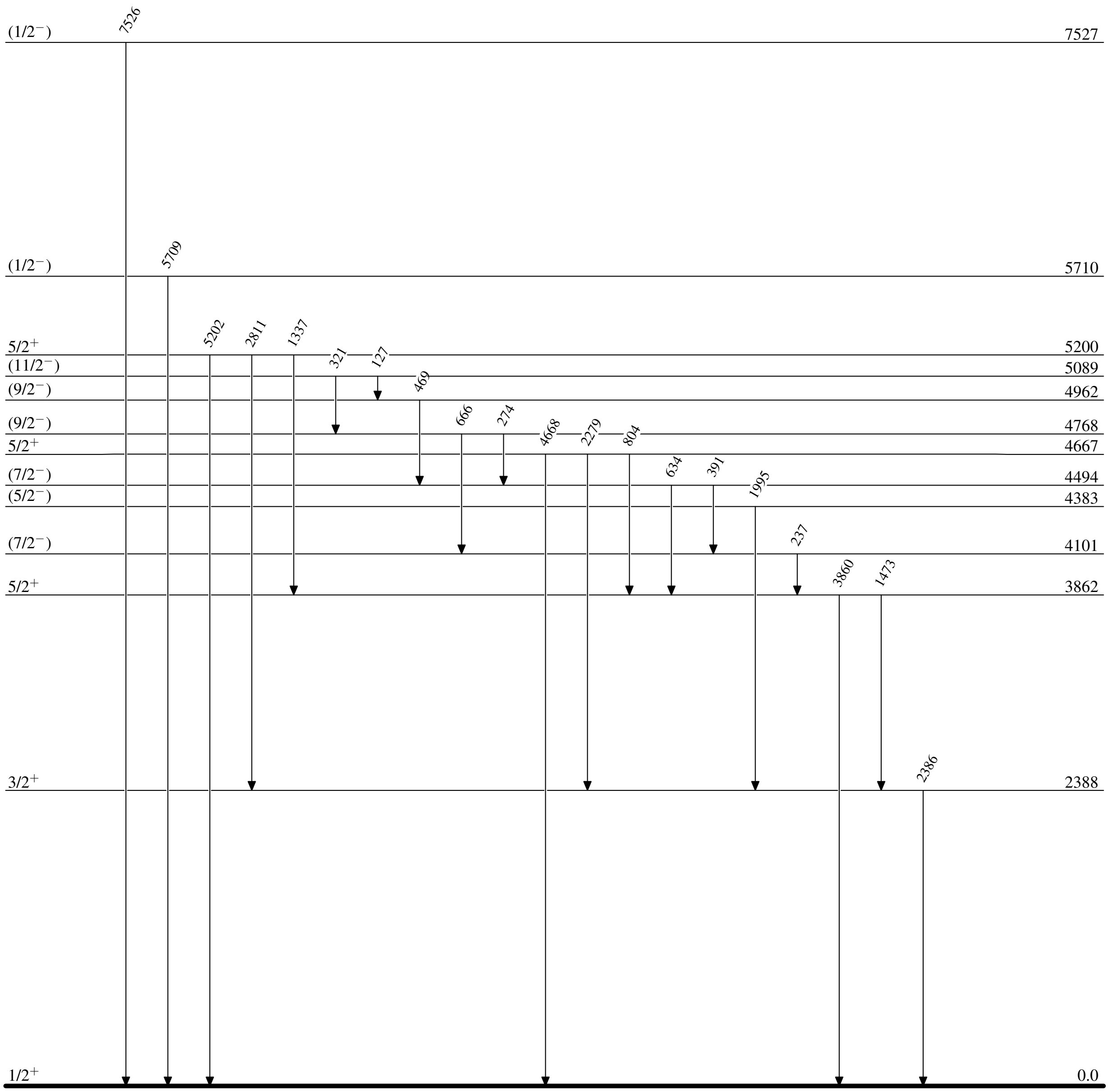
2016Mu03 deduced L by comparing the measured and eikonal-calculated parallel momentum distributions of residuals.
@ From the Adopted Levels, also assumed by 2016Mu03 for deducing C^2S.
& The spectroscopic factor $C^2S=b_f\sigma_{inc}^{exp}/R_s\sigma_f^{sp}$, where b_f is the population fraction, σ_{inc}^{exp} is the measured total inclusive cross section, R_s is the quenching factor from a systematics study 2014To14, and σ_f^{sp} is the eikonal model calculated single-particle cross section.

γ(^{35}P)

E_γ	E_i(level)	J_i^π	E_f	J_f^π
127 2	5089	(11/2$^-$)	4962	(9/2$^-$)
237 2	4101	(7/2$^-$)	3862	5/2$^+$
274 2	4768	(9/2$^-$)	4494	(7/2$^-$)
321 2	5089	(11/2$^-$)	4768	(9/2$^-$)
391 2	4494	(7/2$^-$)	4101	(7/2$^-$)
469 2	4962	(9/2$^-$)	4494	(7/2$^-$)
634 2	4494	(7/2$^-$)	3862	5/2$^+$
666 2	4768	(9/2$^-$)	4101	(7/2$^-$)
804 2	4667	5/2$^+$	3862	5/2$^+$
1337 2	5200	5/2$^+$	3862	5/2$^+$
1473 2	3862	5/2$^+$	2388	3/2$^+$
1995 2	4383	(5/2$^-$)	2388	3/2$^+$
2279 2	4667	5/2$^+$	2388	3/2$^+$
2386 2	2388	3/2$^+$	0.0	1/2$^+$
2811 2	5200	5/2$^+$	2388	3/2$^+$
3860 2	3862	5/2$^+$	0.0	1/2$^+$
4668 2	4667	5/2$^+$	0.0	1/2$^+$
5202 2	5200	5/2$^+$	0.0	1/2$^+$
5709 2	5710	(1/2$^-$)	0.0	1/2$^+$
7526 2	7527	(1/2$^-$)	0.0	1/2$^+$

^{9}Be(^{36}S,^{35}Pγ) 2016Mu03

Level Scheme



$^{35}_{15}P_{20}$

$^{34}S(^{18}O,^{17}F)$ 1988Or01

1988Or01: A 108-MeV $^{18}O^{7+}$ beam was produced from the ANU 14UD Pelletron accelerator. Targets were enriched Ag_2S. Reaction products were momentum-analyzed with an Enge split-pole spectrometer (FWHM=250 keV at 5.75-10.25°) and detected with a multi-element gas-filled detector at the focal plane. Measured $\sigma(E(^{17}F))$. Deduced levels and ground state mass excess (−24.87 MeV *4*). Comparisons with shell-model calculations. Proposed a decay scheme of ^{35}Si for the γ-ray transitions observed but not placed in a decay scheme by 1986Du07.

^{35}P Levels

E(level)†
0
2420 *40*
5070 *40*
5890 *70*
6440 *60*
7050 *60*
7440 *60*
7920 *60*
8.60×10^3 *10*
9290 *50*

† From 1988Or01.

^{36}S(d,^{3}He) 1984Th08

$J^{\pi}=0^{+}$ for ^{36}S ground state.

1984Th08: A 30-MeV deuteron beam was produced from the BNL Double MP tandem facility. Targets were 20.1 and 23.0 $\mu g/cm^2$ sulphur with ^{36}S enriched to 81.1%. Reaction products were momentum-analyzed with the BNL Q3D magnetic spectrometer (FWHM≈32 keV) and detected with a multi-wire proportional counter backed by a topping plastic scintillator. Measured $\sigma(E(^3He),\theta)$. Deduced levels, J, π, L-transfers, spectroscopic factors from the finite-range DWUCK4-DWBA analysis of the angular distributions.

^{35}P Levels

Spectroscopic factor $C^2S=(2j+1)\times\sigma(\theta)_{exp}/\sigma(\theta)_{DWBA}/N$, where j denotes the total angular momentum of the transferred nucleon. N=29.5 is a normalization factor adopted by 1984Th08. The discrepancy from the original N=2.95 in 1966Ba54 likely arises from different units of cross sections.

E(level)[†]	L[‡]	C^2S[‡]
0	0	2.3 *12*
3864 *10*	2	1.45,1.10[#]
4664 *10*	2	0.53,0.41[#]
5202 *10*	2	0.40,0.30[#]

[†] From 1984Th08.
[‡] From DWBA analysis of measured $\sigma(\theta)$. The uncertainty of C^2S is estimated to be 50% by 1984Th08.
[#] Quoted values are for j=L−1/2 ($1d_{3/2}$) and L+1/2 ($1d_{5/2}$), respectively.

^{36}S(pol d,^{3}He) 1985Kh04

$J^{\pi}=0^{+}$ for ^{36}S ground state.

1985Kh04: 52-MeV unpolarized and vector-polarized deuteron beams of 100 nA were produced from the Karlsruhe cyclotron. The target was a 1 mg/cm^{2} ^{208}Pb sulfide with 81.1% enriched ^{36}S on ^{12}C backing. Reaction products were detected using two 300-μm and 1500-μm thick ΔE-E surface-barrier detector telescopes (FWHM≈90 keV). Measured angular distributions of differential cross sections $\sigma(E(^{3}He),\theta)$ and angular distributions of analyzing powers $iT_{11}(E(^{3}He),\theta)$. Deduced mass excess, levels, J, π, L-transfers, and spectroscopic factors from standard local, zero-range JULIE-DWBA analysis of $\sigma(E(^{3}He),\theta)$ and $iT_{11}(E(^{3}He),\theta)$.

^{35}P Levels

Spectroscopic factor $C^{2}S=\sigma(\theta)_{exp}/\sigma(\theta)_{DWBA}/N$, where N=2.95 is a normalization factor adopted by 1985Kh04 from 1974Ma34, originally from 1966Ba54.

E(level)†	J^{π}	L‡	$C^{2}S$‡	Comments
0	1/2^{+}	0	1.63	
2386 *6*	3/2^{+}#	2	0.31#	
3857 *2*	5/2^{+}@	2	2.91@	
4474 *21*			<0.2	$1d_{5/2}$ proton transfer assumed in DWBA calculations, but 2016Mu03 proposed $1f_{7/2}$ (L=3) based on the measured parallel momentum distribution.
4665 *3*	5/2^{+}@	2	1.06@	
5189 *13*	5/2^{+}@	2	1.38@	
7520 *30*			<0.4	

† From 1985Kh04.
‡ From DWBA analysis of measured $\sigma(\theta)$.
# L-1/2 transfer from analyzing power measurements; $1d_{3/2}$ proton transfer assumed in DWBA calculations.
@ L+1/2 transfer from analyzing power measurements; $1d_{5/2}$ proton transfer assumed in DWBA calculations.

^{37}Cl(^{11}B,^{13}N) 1988Or01

1988Or01: An 81-MeV $^{11}B^{5+}$ beam was produced from the ANU 14UD Pelletron accelerator. Targets were enriched $BaCl_2$. Reaction products were momentum-analyzed with an Enge split-pole spectrometer (FWHM=200 keV at 5.75-10.25°) and detected with a multi-element gas-filled detector at the focal plane. Measured σ(E(^{13}N)). Deduced levels. Comparisons with shell-model and DWBA calculations. Proposed a decay scheme of ^{35}Si based on the γ transitions observed by 1986Du07.

^{35}P Levels

E(level)†	Comments
0	dσ/dΩ=120 μb/sr *30*.
2389 *4*	dσ/dΩ=200 μb/sr *50*.
3860 *10*	dσ/dΩ=35 μb/sr *9*.
4250 *20*	
4640 *20*	dσ/dΩ=30 μb/sr *8*.
5010 *20*	
5220 *40*	dσ/dΩ≈10 μb/sr *3*.
5840 *50*	
7590 *20*	dσ/dΩ=45 μb/sr *11*.
8390 *40*	

† From 1988Or01.

^{160}Gd(^{37}Cl,Xγ) 1994Fo04

Possibly $^{160}Gd+^{37}Cl \rightarrow ^{160}Dy+^{35}P+2n$.

1994Fo04: A 167-MeV ^{37}Cl beam was produced from the Argonne Tandem Linac Accelerator System (ATLAS). Targets were 1 mg/cm^2 98.1% enriched ^{160}Gd backed by 15 mg/cm^2 gold. γ rays were detected using the Argonne-Notre Dame BGO γ-ray facility consisting of 12 Compton-suppressed Ge detectors and a 50-element bismuth germanate (BGO) array. Measured Eγ, Iγ, $\gamma\gamma$-coin. Deduced levels.

^{35}P Levels

E(level)†
0
3860
4101
4493

† From Eγ data.

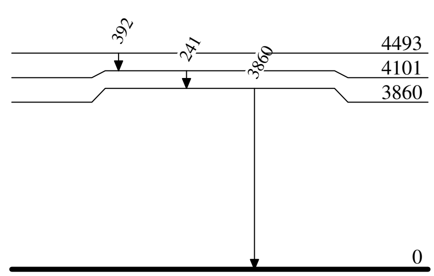
γ(^{35}P)

E$_\gamma$†	E$_i$(level)	E$_f$
241	4101	3860
392	4493	4101
3860	3860	0

† From 1994Fo04.

^{160}Gd(^{37}Cl,Xγ) 1994Fo04

Level Scheme



$^{35}_{15}P_{20}$

Libby: A general comment (not necessary to apply to dataset. 2008Wi09 and 2015Ch56 use different beam energies, which would result in different populations of excited states. You can average branching ratios, but I wouldn't average intensities relative to a single transitions.

^{208}Pb(^{36}S,^{35}Pγ) 2008Wi09,2015Ch56,2019Gr08

A general comment (not necessary to apply to dataset. 2008Wi09 and 2015Ch56 use different

2008Wi09,2010WiZZ: A 230-MeV ^{36}S beam was produced by the Argonne Tandem Linac Accelerator System (ATLAS) with an intensity of 1.5 pnA on a 0.5 mg/cm^2 ^{208}Pb target and an intensity of 0.3 pnA on a 44 mg/cm^2 ^{208}Pb target. In the thin-target run, binary transfer products were detected using a heavy-ion parallel-plate avalanche counter (PPAC) array (CHICO) (Time resolution ≈0.7ns). The polar angle covered was 12°−85° with respect to the beam. γ rays were detected by Gammasphere consisting of 101 HPGe detectors with FWHM=2-10 keV at Eγ=1 MeV. Event-by-event Doppler shift correction was applied. In the thick-target run, binary transfer products were stopped in the target. γ rays were detected by Gammasphere consisting of 95 HPGe detectors with FWHM=2-3 keV at Eγ=1 MeV. Measured Eγ, Iγ, $\gamma\gamma$-coin. Deduced levels. Comparisons with shell-model calculations. Branching-ratio limits were reported for predicted transitions to the 2$\hbar\omega$ bandheads in ^{35}P and ^{34}Si. An e-mail reply from Mathis Wiedeking in April, 2010 (2010WiZZ) provides relative γ-ray intensities, supplementing 2008Wi09.

2015Ch56: A 215-MeV ^{36}S beam was produced using the combination of XTU tandem Van de Graaff accelerator and ALPI superconducting linear accelerator at the INFN Legnaro National Laboratory. The target was 300-μg/cm^2 99.7% enriched ^{208}Pb on a 20 μg/cm^2 carbon backing. Projectile-like fragments produced in multinucleon binary grazing reactions were separated and identified by the PRISMA spectrometer. γ rays were detected using the CLARA array of 22 EUROBALL escape-suppressed HPGe clover detectors. Doppler corrections of γ-ray energies were performed event by event. Measured Eγ, Iγ, (^{35}P)γ-coin, and $\gamma\gamma$-coin. 2015Ch56 also revisited the $\gamma\gamma\gamma$-coin of ^{36}S+^{176}Yb deep-inelastic data by J. Ollier Ph.D. thesis, University of Paisley (2004) to strengthen the evidence for γ-ray placements (see Ref. [39] in 2015Ch56). Deduced levels, J, π. Comparisons with shell-model calculations.

2019Gr08: A 225-MeV ^{36}S beam was provided by Tandem-ALPI accelerator complex at the INFN Legnaro National Laboratory. The target was 1 mg/cm^2 99.7% enriched ^{208}Pb with 1 mg/cm^2 Nb backing and mounted onto the Cologne differential plunger. Projectile-like fragments produced in binary grazing reactions were separated and identified by the PRISMA spectrometer. γ rays were detected using the AGATA demonstrator array of five triple cluster modules of 36-fold segmented Ge crystals covering backward angles from 135° to 175°. Doppler corrections of γ-ray energies were performed event by event. Measured Eγ, (^{35}P)γ-coin, and level lifetimes using the differential recoil-distance method (DRDM). Comparison with shell-model calculations.

^{35}P Levels

E(level)†	J$^\pi$‡	T$_{1/2}$#	Comments
0	1/2$^+$		
2386.7 *7*	3/2$^+$	<0.69 ps	T$_{1/2}$: estimated mean lifetime τ<1 ps (2019Gr08).
3860.8 *7*	5/2$^+$	<0.69 ps	T$_{1/2}$: estimated mean lifetime τ<1 ps (2019Gr08).
4102.1 *7*	(7/2$^-$)	>69 ps	J$^\pi$: 7/2$^-$ proposed by 2019Gr08 based on comparisons with shell-model calculations. T$_{1/2}$: estimated mean lifetime τ>100 ps (2019Gr08).
4381.9 *10*	(5/2$^-$)		
4494.2 *8*	(7/2$^-$)	2.29 ps *49*	J$^\pi$: 7/2$^-$ proposed by 2019Gr08 based on comparisons with shell-model calculations. T$_{1/2}$: measured mean lifetime τ=3.3 ps *7* (2019Gr08).
4767.1 *10*	(9/2$^-$)		J$^\pi$: 9/2$^-$ proposed by 2019Gr08 based on comparisons with shell-model calculations.
4869.0 *8*	(5/2$^-$,7/2$^-$)		
4962.1 *12*	(9/2$^-$)		J$^\pi$: 9/2$^-$ proposed by 2019Gr08 based on comparisons with shell-model calculations.
5089.8 *11*	(11/2$^-$)		J$^\pi$: 11/2$^-$ proposed by 2019Gr08 based on comparisons with shell-model calculations.
5488.2 *10*			
5560.1 *12*	(5/2$^-$)		
6222.4 *11*	(7/2$^-$,9/2,11/2$^-$)		

† From a least-squares fit to γ-ray energies.
‡ From the Adopted Levels.
# From differential recoil-distance method (DRDM) (2019Gr08).

A general comment (not necessary to apply to dataset. 2008Wi09 and 2015Ch56 use different beam energies, which would result in different populations of excited states. You can average branching ratios, but I wouldn't average intensities relative to a single transitions.

^{208}Pb(^{36}S,^{35}Pγ) 2008Wi09,2015Ch56,2019Gr08 (continued)

$\gamma(^{35}P)$

E_γ	I_γ	E_i(level)	J^π_i	E_f	J^π_f	Comments
128 1	10 5	5089.8	(11/2$^-$)	4962.1	(9/2$^-$)	E_γ: weighted average of 128 1 (2008Wi09) and 127 1 (2015Ch56). I_γ: unweighted average of 14 2 (2008Wi09) and 5.2 6 (2015Ch56).
241 1	61 4	4102.1	(7/2$^-$)	3860.8	5/2$^+$	E_γ: From 2008Wi09 and 2015Ch56. I_γ: From 2008Wi09. Other: 32.6 9 (2015Ch56).
273 1	12.8 8	4767.1	(9/2$^-$)	4494.2	(7/2$^-$)	E_γ: From 2008Wi09 and 2015Ch56. I_γ: weighted average of 12 2 (2008Wi09) and 12.9 8 (2015Ch56).
322 1	20 7	5089.8	(11/2$^-$)	4767.1	(9/2$^-$)	E_γ: weighted average of 321 1 (2008Wi09) and 323 1 (2015Ch56). I_γ: unweighted average of 27 3 (2008Wi09) and 12.9 8 (2015Ch56).
374[†] 1	3[†] 1	4869.0	(5/2$^-$,7/2$^-$)	4494.2	(7/2$^-$)	
392 1	30 5	4494.2	(7/2$^-$)	4102.1	(7/2$^-$)	E_γ: weighted average of 391 1 (2008Wi09) and 392 1 (2015Ch56). I_γ: unweighted average of 35 3 (2008Wi09) and 24.9 11 (2015Ch56).
468 2	16.2 12	4962.1	(9/2$^-$)	4494.2	(7/2$^-$)	E_γ: unweighted average of 466 1 (2008Wi09) and 469 1 (2015Ch56). I_γ: weighted average of 14 2 (2008Wi09) and 16.8 11 (2015Ch56).
487[†] 1	<2[†]	4869.0	(5/2$^-$,7/2$^-$)	4381.9	(5/2$^-$)	I_γ: 1 1 from 2008Wi09.
633 1	8.2 22	4494.2	(7/2$^-$)	3860.8	5/2$^+$	E_γ: weighted average of 632 1 (2008Wi09) and 633 1 (2015Ch56). I_γ: unweighted average of 6 1 (2008Wi09) and 10.4 9 (2015Ch56).
664 1	32 15	4767.1	(9/2$^-$)	4102.1	(7/2$^-$)	E_γ: weighted average of 663 1 (2008Wi09) and 665 1 (2015Ch56). I_γ: unweighted average of 47 4 (2008Wi09) and 17.8 10 (2015Ch56).
767[†] 1	5[†] 1	4869.0	(5/2$^-$,7/2$^-$)	4102.1	(7/2$^-$)	
859 3	10.7 14	4962.1	(9/2$^-$)	4102.1	(7/2$^-$)	E_γ: unweighted average of 856 1 (2008Wi09) and 861 1 (2015Ch56). I_γ: weighted average of 13 2 (2008Wi09) and 9.8 12 (2015Ch56).
993[†] 1	5[†] 1	5488.2		4494.2	(7/2$^-$)	
1009[†] 1	<1[†]	4869.0	(5/2$^-$,7/2$^-$)	3860.8	5/2$^+$	
1132[†] 1	<1[†]	6222.4	(7/2$^-$,9/2,11/2$^-$)	5089.8	(11/2$^-$)	
1260[†] 1	4[†] 1	6222.4	(7/2$^-$,9/2,11/2$^-$)	4962.1	(9/2$^-$)	
^{x}1353 1	9.2 11					E_γ,I_γ: From 2015Ch56.
1387[†] 1	3[†] 1	5488.2		4102.1	(7/2$^-$)	
1458[†] 1	7[†] 2	5560.1	(5/2$^-$)	4102.1	(7/2$^-$)	
1474 1	15.9 14	3860.8	5/2$^+$	2386.7	3/2$^+$	E_γ: weighted average of 1473 1 (2008Wi09) and 1474 1 (2015Ch56). I_γ: weighted average of 15 2 (2008Wi09) and 16.4 14 (2015Ch56).
^{x}1592 1	7.7 10					E_γ,I_γ: From 2015Ch56.
1715[†] 1	4[†] 1	4102.1	(7/2$^-$)	2386.7	3/2$^+$	
1729[†] 1	4[†] 1	6222.4	(7/2$^-$,9/2,11/2$^-$)	4494.2	(7/2$^-$)	
1995 1	8 6	4381.9	(5/2$^-$)	2386.7	3/2$^+$	E_γ: weighted average of 1995 1 (2008Wi09) and 1994 1 (2015Ch56, unplaced).

Continued on next page (footnotes at end of table)

^{208}Pb(^{36}S,^{35}Pγ) 2008Wi09,2015Ch56,2019Gr08 (continued)

γ(^{35}P) (continued)

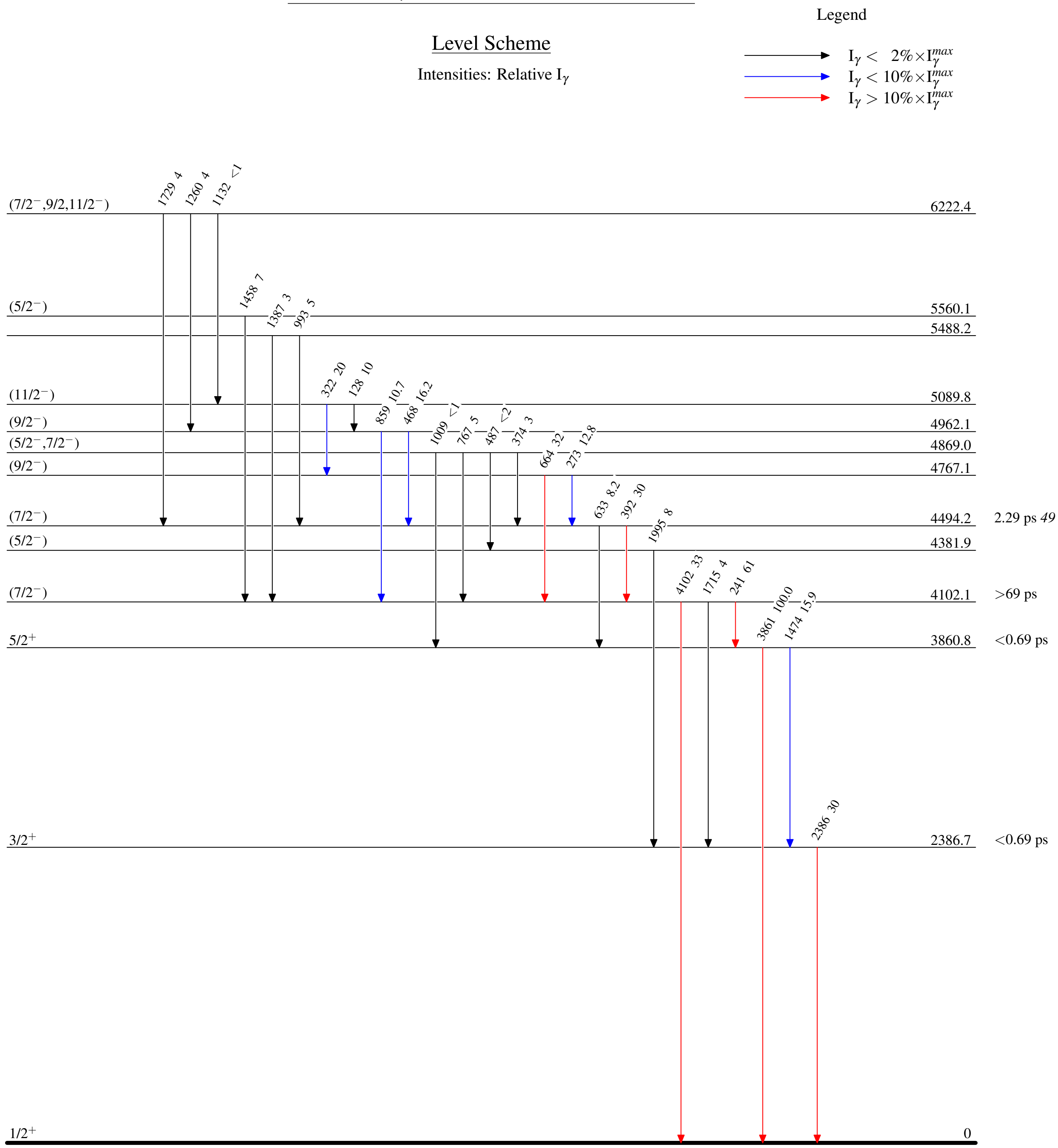
E_γ	I_γ	E_i(level)	J_i^π	E_f	J_f^π	Comments
						I_γ: unweighted average of 2 *1* (2008Wi09) and 14.2 *11* (2015Ch56).
2386 *1*	30 *4*	2386.7	3/2$^+$	0	1/2$^+$	E_γ: From 2008Wi09 and 2015Ch56. I_γ: From 2008Wi09. Other: 99.2 *28* (2015Ch56). Shell-model calculations indicate a small occupancy of the proton $1d_{3/2}$ orbit in the ground state of ^{36}S.
3861 *1*	100.0 *32*	3860.8	5/2$^+$	0	1/2$^+$	E_γ: weighted average of 3861 *1* (2008Wi09) and 3860 *1* (2015Ch56). I_γ: From 2015Ch56. Other: 100 (2008Wi09).
4102† *1*	33† *5*	4102.1	(7/2$^-$)	0	1/2$^+$	E_γ: other: 4101 (2015Ch56).

† From 2008Wi09.
^{x} γ ray not placed in level scheme.

$^{208}Pb(^{36}S,^{35}P\gamma)$ 2008Wi09,2015Ch56,2019Gr08

Level Scheme

Intensities: Relative I_γ



$^{35}_{15}P_{20}$

Adopted Levels, Gammas

Q(β^-)=167.322 *26*; S(n)=6985.84 *4*; S(p)=11586.5 *8*; Q(α)=−8322.09 *6* 2021Wa16
S(2n)=18402.99 *4*, S(2p)=22909.8 *7* (2021Wa16).
Isotope discovery (2012Th10): Cl(n,X) reaction at Physical Institute of Aarhus University (1936An01).
^{35}S decay measurements:
2014Si21, 2000Ho13, 1999Pa18, 1999Oh09.
^{35}S cross-section measurements:
^{35}Cl(n,p)^{35}S: 2024Ha18, 2023De34, 2021WaZY, 2020Ku17, 2019Ba16, 2000Ka30, 1999Gl09, 1999Gl02;
Projectile fragmentation ^{9}Be(^{40}Ar,X) and ^{181}Ta(^{40}Ar,X): 2012Zh06, 2007No13;
Spallation ^{136}Xe(p,X): 2007Na31;
^{34}S(n,γ) E=thermal: 2023KoZS, 2008FiZZ.

^{35}S Levels

Cross Reference (XREF) Flags

A	^{35}P β^- decay (47.3 s)	F	^{34}S(n,γ) E=thermal	K	^{36}S(p,d)
B	^{2}H(^{34}S,pγ)	G	^{34}S(n,n),(n,γ):resonances	L	^{37}Cl(p,^{3}He)
C	^{9}Be(^{36}S,^{35}Sγ)	H	^{34}S(pol d,p)	M	^{37}Cl(d,$\alpha\gamma$)
D	^{24}Mg(^{14}N,3pγ)	I	^{34}S(d,p)	N	^{160}Gd(^{34}S,Xγ),(^{37}Cl,Xγ)
E	^{26}Mg(^{18}O,2αnγ)	J	^{34}S(d,pγ)	O	^{208}Pb(^{36}S,^{35}Sγ)

E(level)†	J$^\pi$‡	T$_{1/2}$	XREF	Comments
0.0	3/2$^+$	87.61 d *24*	ABCDEF HIJKLMNO	%β^-=100 μ=(+)1.00 *4* (1954Bu05,2019StZV) Q=0.0483 *3* (2018Py01,2021StZZ) J$^\pi$: L=2 from 0$^+$ in ^{34}S(pol d,p), ^{34}S(d,p), ^{36}S(p,d), and ^{9}Be(^{36}S,^{35}S) and L-1/2 transfer from analyzing power. J=3/2 also from microwave spectroscopy (1949Co12,1951We11). T$_{1/2}$: unweighted average of 88 d *3* (1941Ka01), 87.1 d *12* (1943He01), 86.35 d *17* (1959Co56), 87.16 d *10* (1958Se49), 88.8 d *10* (1959Ca12), 89.0 d *5* (1961Wy01), 87.1 d *9* (1961Oz01), 87.9 d *3* (1965Fl02), 87.39 d *10* (1968Wo06), 87.5 d *4* (1969La34), and 87.38 d *3* (1999Pa18). μ: +1.00 *4* or −1.07 *4* from 1954Bu05 using the microwave absorption method, and the negative value is less likely from systematics of odd-A S isotopes. Q: from 2018Py01 using the microwave absorption method. −0.483 *3* is given in 2021StZZ. Other: 0.045 from 1954Bi40 using the microwave absorption method.
1572.369 *9*	1/2$^+$	2.29 ps *14*	ABCD F HIJKLM O	J$^\pi$: L=0 from 0$^+$ in ^{34}S(pol d,p), ^{34}S(d,p), ^{36}S(p,d), and ^{9}Be(^{36}S,^{35}S). T$_{1/2}$: lifetime τ=3.3 ps *2* in ^{208}Pb(^{36}S,^{35}Sγ) from 2022Gr07 with DRDM. Others: 3.3 ps *5* in (^{34}S,pγ) from 1973Wa10 with DSAM, >1.8 ps in ^{34}S(d,pγ) from 1970Bu18 with DSAM, and >1.6 ps in ^{34}S(d,pγ) from 1972Fr11 with DSAM.
1991.27 *4*	7/2$^-$	1.03 ns *5*	CDEF HIJKLMNO	J$^\pi$: L=3 from 0$^+$ in ^{36}S(p,d) and ^{34}S(pol d,p), and L+1/2 transfer from analyzing power. T$_{1/2}$: lifetime τ=1.48 ns *7*: weighted average of 1.7 ns *3* in ^{34}S(d,pγ) from 2024Co04 with pγ-delayed coin and 1.47 ns *7* in ^{34}S(d,pγ) from 1971Pr11 with pγ-delayed coin. Other: >4.5 ps in ^{34}S(d,pγ) from 1970Bu18 with DSAM.
2347.779 *10*	3/2$^-$	0.78 ps *12*	BCD F HIJK M O	J$^\pi$: L=1 from 0$^+$ in ^{34}S(pol d,p), ^{34}S(d,p), ^{36}S(p,d), and ^{9}Be(^{36}S,^{35}S) and L+1/2 transfer from analyzing power.

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A general comment (not necessary to apply to dataset. 2008Wi09 and 2015Ch56 use different beam energies, which would result in different populations of excited states. You can

Adopted Levels, Gammas (continued)

^{35}S Levels (continued)

A general comment (not necessary to apply to dataset. 2008Wi09 and 2015Ch56 use di

E(level)[†]	J^{π}[‡]	$T_{1/2}$	XREF	Comments
				$T_{1/2}$: lifetime τ=1.12 ps *18*: weighted average of 1.3 ps *2* in (^{34}S,pγ) from 1973Wa10 with DSAM, 0.8 ps *2* in ^{34}S(d,pγ) from 1970Bu18 with DSAM, and 1.22 ps *18* in ^{34}S(d,pγ) from 1972Fr11 with DSAM.
2717.06 *8*	5/2$^+$	69 fs *24*	CD F HIJKLM O	J^{π}: L=2 from 0$^+$ in ^{34}S(pol d,p), ^{34}S(d,p), ^{36}S(p,d), and ^{9}Be(^{36}S,^{35}S); L+1/2 transfer from J−dependence in ^{36}S(p,d) and L+1/2 transfer from analyzing power. $T_{1/2}$: lifetime τ=100 fs *35* in ^{34}S(d,pγ) from 1972Fr11 with DSAM.
2938.63 *5*	3/2$^+$		A C F HIJKLM	J^{π}: L=2 from 0$^+$ in ^{34}S(d,p), ^{36}S(p,d), ^{34}S(pol d,p), and L=(2) in ^{9}Be(^{36}S,^{35}S); L-1/2 transfer from J-dependence in ^{36}S(p,d); possibly allowed β^- feeding from 1/2$^+$ parent with log *ft*=5.0.
3421.2 *20*	5/2$^+$	<69 fs	C HIJKLM O	J^{π}: L=2 from 0$^+$ in ^{34}S(pol d,p), ^{34}S(d,p), ^{36}S(p,d), and ^{9}Be(^{36}S,^{35}S); L+1/2 transfer from J−dependence in ^{36}S(p,d) and L+1/2 transfer from analyzing power. $T_{1/2}$: lifetime τ<100 fs in ^{34}S(d,pγ) from 1972Fr11 with DSAM.
3558.080 *28*	(5/2$^+$)		F HIJK M O	J^{π}: L=(2) from 0$^+$ in ^{36}S(p,d); 1986γ to 1/2$^+$, 1572.370 and 1566.7γ to 7/2$^-$, 1991.27.
3594.62 *24*	(7/2$^+$)		CDE IJKLM O	XREF: C(3595)I(3596)K(3592)L(3598) J^{π}: 3594.5γ Q, ΔJ=2 to 3/2$^+$, g.s.; L=2+4 from 3/2$^+$ in (p,^{3}He): L=2 suggests (1/2$^+$ to 9/2$^+$) and L=4 suggests (3/2$^+$ to 13/2$^+$); L=(2) from 0$^+$ in ^{36}S(p,d) and ^{9}Be(^{36}S,^{35}S). 7/2$^+$ is suggested by shell model (2014Ay01,2021Go09). If 3594.5γ is consistent with ΔJ=0 to 3/2$^+$, the discrepancy between the in-beam γ-ray spectroscopy (^{14}N,3pγ) and transfer reactions may be resolved.
3675? *10*	(1/2$^-$,3/2$^-$)		I	J^{π}: L=(1) from 0$^+$ in ^{34}S(d,p).
3801.952 *16*	3/2$^-$	25 fs *18*	F HIJKlM	XREF: l(3811) J^{π}: L=1 from 0$^+$ in ^{34}S(d,p), ^{36}S(p,d), and ^{34}S(pol d,p) and L+1/2 transfer from analyzing power. $T_{1/2}$: lifetime τ=36 fs *26* in ^{34}S(d,pγ) from 1972Fr11 with DSAM.
3816.01 *21*	(9/2)$^-$	0.28 ps *3*	CDE H J lM O	XREF: l(3811) J^{π}: 1824.7γ M1+E2 to 7/2$^-$, 1991.27; 2061.6γ D, ΔJ=1 from (11/2$^+$), 5877.70. L=(3) from 0$^+$ in ^{9}Be(^{36}S,^{35}Sγ) is inconsistent but the fit on the parallel momentum distribution is poor. $T_{1/2}$: in (^{14}N,3pγ) from 2014Ay01 with DSAM.
3886.28	(5/2)		CD IJK M	XREF: C(3890)M(3889) J^{π}: 1894.9γ D, ΔJ=1 to 7/2$^-$, 1991.27; 3886γ to 3/2$^+$, g.s.; L=3 from 0$^+$ in ^{36}S(p,d). L=(0) from 0$^+$ in ^{9}Be(^{36}S,^{35}Sγ) is inconsistent.
4023.24 *21*	(11/2)$^-$	0.32 ps *3*	DE MNO	J^{π}: 2032.0γ E2, ΔJ=2 to 7/2$^-$, 1991.27; 207.0γ to (9/2)$^-$, 3816.01. $T_{1/2}$: in (^{14}N,3pγ) from 2014Ay01 with DSAM.
4027.2 *14*	(3/2$^+$,5/2$^+$)		C IJKLM	XREF: I(4025)K(4023) J^{π}: L=(2) from 0$^+$ in ^{36}S(p,d) and ^{9}Be(^{36}S,^{35}Sγ).
4105.6 *8*	(5/2$^+$)	<55 fs	F IJKLM	J^{π}: L=(2) from 0$^+$ in ^{36}S(p,d); 2533.1γ to 1/2$^+$, 1572.370 and 2114.3γ to 7/2$^-$, 1991.27. $T_{1/2}$: lifetime τ<80 fs in ^{34}S(d,pγ) from 1972Fr11 with DSAM.

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Adopted Levels, Gammas (continued)

^{35}S Levels (continued)

E(level)†	Jπ‡	$T_{1/2}$	XREF	Comments
4180 *3*	(1/2,3/2,5/2$^+$)		lM	XREF: l(4186) Jπ: 2608γ to 1/2$^+$, 1572.370.
4189.240 *33*	1/2$^-$	<35 fs	F HIJKlM	XREF: K(4182)l(4186)M(4187) Jπ: L=1 from 0$^+$ in ^{34}S(d,p), ^{36}S(p,d), and ^{34}S(pol d,p) and L-1/2 transfer from analyzing power. $T_{1/2}$: lifetime τ<50 fs in ^{34}S(d,pγ) from 1972Fr11 with DSAM.
4302.6 *7*	(1/2$^-$,3/2$^-$)		I KLM	XREF: L(4290) Jπ: L=(1) from 0$^+$ in ^{36}S(p,d). L=(2) from 3/2$^+$ in (p,^{3}He) suggests (1/2$^+$ to 9/2$^+$).
4477.63	(5/2$^+$)	<62 fs	F iJ LM	XREF: i(4481)L(4489) Jπ: 2905.1γ to 1/2$^+$, 1572.370 and 2486.27γ to 7/2$^-$, 1991.27. L=2 from 3/2$^+$ in (p,^{3}He) suggests (1/2$^+$ to 9/2$^+$). L=2 component of L=2,3 doublet from 0$^+$ in ^{34}S(d,p). $T_{1/2}$: lifetime τ<90 fs in ^{34}S(d,pγ) from 1972Fr11 with DSAM.
4486 *7*	7/2$^-$		Hi K	XREF: H(4482)i(4481) E(level): from ^{36}S(p,d). Jπ: L=3 from 0$^+$ in ^{36}S(p,d) and ^{34}S(pol d,p), and L+1/2 transfer from analyzing power. L=3 component of L=2,3 doublet from 0$^+$ in ^{34}S(d,p).
4575 *5*	3/2$^+$		HI KL	E(level): weighted average of 4575 *8* from ^{34}S(d,p), 4577 *10* from (p,^{3}He), and 4574 *5* from ^{36}S(p,d). Jπ: L=2 from 0$^+$ and L-1/2 transfer from J-dependence in ^{36}S(p,d). L=0+2 from 3/2$^+$ in (p,^{3}He): L=0 suggests (1/2$^+$,3/2$^+$,5/2$^+$) and L=2 suggests (1/2$^+$ to 9/2$^+$).
4615 *5*	3/2$^+$,5/2$^+$		KL	E(level): weighted average of 4614 *5* from ^{36}S(p,d) and 4617 *10* from (p,^{3}He). Jπ: L=2 from 0$^+$ in ^{36}S(p,d).
4822.79 *24*	(9/2$^+$)		DE	Jπ: 1228.1γ D, ΔJ=1 to (7/2$^+$), 3594.62; 1055.1γ D, ΔJ=1 from (11/2$^+$), 5877.70; 9/2$^+$ suggested by shell-model calculations (2021Go09).
4839 *5*	(1/2$^+$)		HI KL	E(level): weighted average of 4837 *8* from ^{34}S(d,p), 4838 *5* from ^{36}S(p,d), and 4843 *10* from (p,^{3}He). Other: 4837 from (pol d,p). Jπ: L=(0) from 0$^+$ in ^{36}S(p,d); L=2 from 3/2$^+$ in (p,^{3}He) suggests (1/2$^+$ to 9/2$^+$).
4899.82 *27*	(9/2$^+$)		DE	Jπ: 2908.3γ D, ΔJ=1 to 7/2$^-$, 1991.27; 977.9γ D+Q, ΔJ=1 from (11/2$^+$), 5877.70; 9/2$^+$ suggested by shell-model calculations (2021Go09).
4903.371 *16*	1/2$^+$		F HI K	Jπ: L=0 from 0$^+$ in ^{36}S(p,d). Other: L=1 from 0$^+$ in ^{34}S(d,p) and ^{34}S(pol d,p) and L-1/2 transfer from analyzing power. 2026Jo01 in ^{36}S(p,d) acknowledges this discrepancy and states that the J^π is confronted with strong evidence.
4963.082 *25*	5/2$^+$		C F HI KL	XREF: C(4948)K(4955) Jπ: L=2 from 0$^+$ and L+1/2 transfer from J−dependence in ^{36}S(p,d). Other: L=1 from 0$^+$ in ^{34}S(d,p) and ^{34}S(pol d,p) and L+1/2 transfer from analyzing power. 2026Jo01 in ^{36}S(p,d) acknowledges this discrepancy and states that the J^π is confronted with strong evidence.
4990 *10*	($^+$)		L	Jπ: L=0+2 from 3/2$^+$ in (p,^{3}He): L=0 suggests (1/2$^+$,3/2$^+$,5/2$^+$) and L=2 suggests (1/2$^+$ to 9/2$^+$).
5010.05 *29*	(11/2)$^-$	0.45 ps *8*	DE	Jπ: 3018.5γ E2, ΔJ=2 to 7/2$^-$, 1991.27; 986.8γ to (11/2)$^-$,

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Adopted Levels, Gammas (continued)

^{35}S Levels (continued)

E(level)†	J^{π}‡	$T_{1/2}$	XREF	Comments
				4023.24. $T_{1/2}$: in (^{14}N,3pγ) from 2014Ay01 with DSAM.
5058 *8*	7/2$^-$		HI	E(level): from ^{34}S(d,p). J^{π}: L=3 from 0$^+$ in ^{34}S(d,p) and ^{34}S(pol d,p) and L+1/2 transfer from analyzing power.
5123 *5*	1/2$^+$		HI KL	E(level): weighted average of 5126 *11* from ^{34}S(d,p), 5121 *5* from ^{36}S(p,d), and 5127 *10* from (p,^{3}He). J^{π}: L=0 from 0$^+$ in ^{36}S(p,d); L=2 from 3/2$^+$ in (p,^{3}He) suggests (1/2$^+$ to 9/2$^+$).
5286 *5*	(1/2$^-$,3/2$^-$)		K	
5343 *8*	($^-$)		C I L	E(level): weighted average of 5342 *8* from ^{34}S(d,p) and 5345 *10* from (p,^{3}He). Other: 5349 *8* from ^{9}Be(^{36}S,^{35}Sγ). J^{π}: L=3 from 3/2$^+$ in (p,^{3}He) suggests (1/2$^-$ to 11/2$^-$).
5412.28 *25*	(9/2$^+$)		DE	J^{π}: 1389.0γ D, ΔJ=1 to (11/2)$^-$, 4023.24; 1525.7γ to (5/2), 3886.28; 9/2$^+$ suggested by shell model (2014Ay01,2021Go09).
5476 *5*	3/2$^+$,5/2$^+$		I K	XREF: I(5475?) J^{π}: L=2 from 0$^+$ in ^{36}S(p,d).
5550 *10*			I L	XREF: I(5542?) J^{π}: L=(3) from 3/2$^+$ in (p,^{3}He): L=3 suggests (1/2$^-$ to 11/2$^-$).
5630 *5*	3/2$^+$,5/2$^+$		K	J^{π}: L=2 from 0$^+$ in ^{36}S(p,d).
5752.5 *8*	5/2$^-$,7/2$^-$		F I	XREF: I(5740) J^{π}: L=3 from 0$^+$ in ^{34}S(d,p).
5767 *5*	5/2$^+$		C KL	E(level): weighted average of 5766 *5* from ^{36}S(p,d) and 5771 *10* from (p,^{3}He). Other: 5775 *4* from ^{9}Be(^{36}S,^{35}Sγ). J^{π}: L=2 from 0$^+$ and L+1/2 transfer from J−dependence in ^{36}S(p,d); L=2 from 3/2$^+$ in (p,^{3}He) suggests (1/2$^+$ to 9/2$^+$).
5841.482 *31*	(3/2$^+$,5/2$^+$)		F K	J^{π}: L=(2) from 0$^+$ in ^{36}S(p,d).
5877.70 *22*	(11/2$^+$)		DE	J^{π}: 465.4γ D, ΔJ=1 to (9/2$^+$), 5412.28; 867.6γ D, ΔJ=0 to (11/2)$^-$, 5010.05; strong 2283.0γ to (7/2$^+$), 3594.62.
5890 *20*	3/2$^+$,5/2$^+$		I L	XREF: L(5915?) J^{π}: L=2 from 0$^+$ in ^{34}S(d,p).
5980 *10*			I	
6018.8 *6*			F	
6078.475 *33*	1/2$^-$,3/2$^-$		F I	J^{π}: L=1 from 0$^+$ in ^{34}S(d,p).
6123 *5*	(3/2$^+$,5/2$^+$)		KL	E(level): weighted average of 6121 *5* from ^{36}S(p,d) and 6129 *10* from (p,^{3}He). J^{π}: L=(2) from 0$^+$ in ^{36}S(p,d); L=0+2 from 3/2$^+$ in (p,^{3}He): L=0 suggests (1/2$^+$,3/2$^+$,5/2$^+$) and L=2 suggests (1/2$^+$ to 9/2$^+$).
6218 *5*	(5/2$^-$,7/2$^-$)		K	J^{π}: L=(3) from 0$^+$ in ^{36}S(p,d).
6293.93 *6*			F I	XREF: I(6292) E(level): other: 6292 *8* from ^{34}S(d,p).
6299.4 *4*	(11/2)		D	J^{π}: 887.0γ, ΔJ=1 to (9/2$^+$), 5412; 2275.9γ to (11/2)$^-$, 4023.24.
6337 *5*	1/2$^+$		I K	E(level): weighted average of 6334 *8* from ^{34}S(d,p) and 6338 *5* from ^{36}S(p,d). J^{π}: L=0 from 0$^+$ in ^{36}S(p,d).
6345 *8*	($^+$)		I L	E(level): weighted average of 6344 *8* from ^{34}S(d,p) and 6347 *10* from (p,^{3}He). J^{π}: L=(2) from 3/2$^+$ in (p,^{3}He) suggests (1/2$^+$ to 9/2$^+$).
6352.2 *5*	(13/2)$^-$	50 fs *10*	D	J^{π}: 2535.8γ E2, ΔJ=2 to (9/2)$^-$, 3816.01; 2329.5γ (M1), ΔJ=1 to (11/2)$^-$, 4023.24. $T_{1/2}$: in (^{14}N,3pγ) from 2014Ay01 with DSAM.
6354.89 *23*			F	

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Adopted Levels, Gammas (continued)

^{35}S Levels (continued)

E(level)†	Jπ‡	$T_{1/2}$	XREF	Comments
6419.9 *11*			F	
6441 *5*	(1/2+)		I K	E(level): weighted average of 6446 *8* from ^{34}S(d,p) and 6439 *5* from ^{36}S(p,d). Jπ: L=(0) from 0+ in ^{36}S(p,d).
6496 *8*			I	
6537.7 *14*			I	
6545.1 *13*	3/2+,5/2+		I K	XREF: K(6550) Jπ: L=2 from 0+ in ^{36}S(p,d).
6584 *10*			I	
6629.43 *9*			F	
6635.2 *13*	3/2+,5/2+		I K	XREF: K(6639) Jπ: L=2 from 0+ in ^{36}S(p,d).
6654 *10*	(−)		L	Jπ: L=(3) from 3/2+ in (p,^{3}He) suggests (1/2− to 11/2−).
6683 *5*	1/2+		I KL	E(level): weighted average of 6677 *8* from ^{34}S(d,p), 6682 *5* from ^{36}S(p,d), and 6696 *10* from (p,^{3}He). Jπ: L=0 from 0+ in ^{36}S(p,d); L=2 from 3/2+ in (p,^{3}He) suggests (1/2+ to 9/2+).
6761.0 *12*			F	
6790 *5*	(3/2+,5/2+)		K	Jπ: L=(2) from 0+ in ^{36}S(p,d).
6891.3 *14*	(−)		I	Jπ: L=(1,3) from 0+ in ^{34}S(d,p).
6962 *5*	(1/2−,3/2−)		K	Jπ: L=(1) from 0+ in ^{36}S(p,d).
(6986.097 *25*)	1/2+		F	Jπ: s-wave resonance capture state.
7018.89 *4*	3/2+,5/2+		G I K	XREF: I(7021)K(7019) E(level): from (n,n),(n,γ):resonances. Jπ: L=2 from 0+ in (n,n),(n,γ):resonances. Others: L=(2,3) from 0+ in ^{34}S(d,p) for 7021 *10*; L=(1) from 0+ in ^{36}S(p,d) for 7019 *5*.
7072.66 *5*		0.14 eV *4*	G	
7074.33 *6*		0.084 eV *3*	G	
7097.72 *30*	3/2+,5/2+	0.009 eV *3*	G	
7099.85 *20*	3/2+,5/2+	0.033 eV *5*	G	
7100.73 *4*	3/2−		G K	Jπ: other: L=(1) from 0+ in ^{36}S(p,d).
7143.27 *11*	(5/2−,7/2−)	0.42 eV *1*	G I KL	XREF: I(7150)K(7151)L(7151?) Jπ: L=(3) from 0+ in ^{36}S(p,d). Others: L=(0,3) from 0+ in ^{34}S(d,p); L=(4) from 3/2+ in (p,^{3}He) suggests (3/2+ to 13/2+).
7179.74 *32*	(15/2+)	3.1 ps *12*	DE	Jπ: 1302.0γ E2, ΔJ=2 to (11/2+), 5877.70; 827.8γ to (13/2)−, 6352.2. $T_{1/2}$: in (^{14}N,3pγ) from 2014Ay01 with DSAM.
7210.42 *5*	1/2−,3/2,5/2+		G i k	XREF: i(7205)k(7215) Jπ: others: L=(1,2) from 0+ in ^{34}S(d,p) for 7205 *35*; L=2 from 0+ in ^{36}S(p,d) for 7215 *5*.
7218.0 *4*		0.33 eV *2*	G i k	XREF: i(7205)k(7215)
7234.0 *5*		0.19 eV *1*	G i	XREF: i(7205)
7239.8 *6*		0.06 eV *2*	G i	XREF: i(7205)
7253.2 *6*	(3/2+,5/2+)	0.35 eV *2*	G K	XREF: K(7249) Jπ: L=(2) from 0+ in ^{36}S(p,d).
7275.93 *5*	1/2+		G k	XREF: k(7279) Jπ: other: L=(0) from 0+ in ^{36}S(p,d) for 7279 *5*.
7279.13?		0.10 eV *3*	G k	XREF: G(?)k(7279)
7289.8 *8*		0.31 eV *3*	G	
7294.19 *6*	3/2−		G	
7331.00 *5*	1/2+		G K	XREF: K(7326)

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Adopted Levels, Gammas (continued)

^{35}S Levels (continued)

E(level)†	J^{π}‡	$T_{1/2}$	XREF	Comments
				J^{π}: other: L=0 from 0^+ in ^{36}S(p,d).
7338.18 *20*	$1/2^-,3/2,5/2^+$		G	
7343.91 *20*		0.72 eV *3*	G k	XREF: k(7349)
7347.2 *5*		0.44 eV *6*	G k	XREF: k(7349)
7347.9 *5*		0.30 eV *5*	G k	XREF: k(7349)
7353.9 *5*		0.44 eV *4*	G k	XREF: k(7349)
7357.6 *5*		0.16 eV *4*	G	
7370.56 *6*	$1/2^-$		G l	XREF: l(7375)
7371.97?		0.12 eV	G l	XREF: G(?)l(7375)
7396.0 *6*	$3/2^+,5/2^+$	0.44 eV *33*	G	
7404.6 *6*		0.21 eV *5*	G	
7408.68 *11*	$3/2^-$		G	
7411.55 *6*			G	
7416.50 *4*			G	
7429.1 *6*		0.12 eV *3*	G	
7433.56 *5*			G	
7436.4 *4*	$1/2^-,3/2,5/2^+$		G	
7442.14 *5*	$1/2^+$		G K	XREF: K(7441) J^{π}: other: L=0 from 0^+ in ^{36}S(p,d).
7462.2 *6*		0.33 eV *4*	G	
7481.15 *7*	$1/2^-$		G I	E(level): weighted average of 7481.15 *6* from (n,n),(n,γ):resonances and 7482.7 *13* from ^{34}S(d,p). J^{π}: other: L=(1,2) from 0^+ in ^{34}S(d,p).
7494.5 *6*	$1/2^-,3/2^-$	0.17 eV *4*	G K	XREF: K(7490) J^{π}: L=1 from 0^+ in ^{36}S(p,d).
7542.3 *6*		0.21 eV *3*	G i	XREF: i(7570)
7604.0 *6*		0.90 eV *8*	G i	XREF: i(7570)
7609.26 *5*	$3/2^-$		G	
7648.9 *6*		0.21 eV *3*	G i	XREF: i(7640)
7655.7 *6*		0.80 eV *5*	G i	XREF: i(7640)
7663.91 *11*	$1/2^-,3/2,5/2^+$		G i	XREF: i(7640)
7678.3 *7*		0.45 eV *7*	G i	XREF: i(7640)
7731.0 *7*		0.31 eV *13*	G L	XREF: L(7712?)
7749.7 *4*	$3/2^+,5/2^+$		G K	XREF: K(7753) J^{π}: L=2 from 0^+ in ^{36}S(p,d).
7761.46 *7*	$3/2^-$		G l	XREF: l(7770)
7776.17 *11*	$1/2^-$		G l	XREF: l(7770)
7797.99 *9*	$1/2^+$		G	
7812.24 *7*	$3/2^+$		G	
7853.25 *7*	$3/2^-$		G	
7861.8 *12*		1.13 eV *19*	G	
7880.8 *12*		0.67 eV *17*	G k	XREF: k(7885) J^{π}: L=(2) from 0^+ in ^{36}S(p,d) for 7885 *5*.
7889.0 *12*		0.57 eV *25*	G k	XREF: k(7885) J^{π}: L=(2) from 0^+ in ^{36}S(p,d) for 7885 *5*.
7894.63 *7*	$3/2^-$		G	
7899.70 *30*			G	
7934.1 *15*		0.5 eV *3*	G	
7939.5 *15*		0.89 eV *4*	G	
7954.96 *11*	$3/2^-$		G	
7974.19 *11*	$3/2^-$		G K	XREF: K(7980) J^{π}: other: L=1 from 0^+ in ^{36}S(p,d).
8019.5 *2*	$3/2^+,5/2^+$		G	
8024.0 *7*	$(17/2^+)$	0.15 ps *4*	DE	J^{π}: 844.2γ (M1), ΔJ=1 to $(15/2^+)$, 7179.74. $17/2^+$ suggested by shell-model calculations (2021Go09).

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Adopted Levels, Gammas (continued)

^{35}S Levels (continued)

E(level)†	J$^{\pi}$‡	T$_{1/2}$	XREF	Comments
				T$_{1/2}$: in (^{14}N,3pγ) from 2014Ay01 with DSAM.
8041.20 *20*	3/2$^+$,5/2$^+$		G	
8076.46 *20*	3/2$^+$,5/2$^+$		G k	XREF: k(8073) J$^{\pi}$: other: L=(2) from 0$^+$ in ^{36}S(p,d) for 8073 *5*.
8077.91 *30*	3/2$^+$,5/2$^+$		G k	XREF: k(8073) J$^{\pi}$: other: L=(2) from 0$^+$ in ^{36}S(p,d) for 8073 *5*.
8093.0 *5*	1/2$^-$		G KL	XREF: K(8103)L(8103) E(level): from (n,n),(n,γ):resonances. Others: 8103 *5* from ^{36}S(p,d) and 8103 *10* from (p,^{3}He). J$^{\pi}$: other: L=1 from 0$^+$ in ^{36}S(p,d); L=(1+3) from 3/2$^+$ in (p,^{3}He).
8143.66 *20*	3/2$^+$,5/2$^+$		G	
8160 *10*			L	J$^{\pi}$: L=(1) from 3/2$^+$ in (p,^{3}He) suggests (1/2$^-$ to 7/2$^-$).
8187.17 *30*	1/2$^-$		G	
8211.25 *30*			G	
8224.2 *4*	1/2$^-$		G K	XREF: K(8221) J$^{\pi}$: other: L=1 from 0$^+$ in ^{36}S(p,d).
8228.3 *4*	3/2$^+$,5/2$^+$		G	
8243.98 *30*	3/2$^+$,5/2$^+$		G	
8256.31 *20*	3/2$^-$		G	
8262.4 *4*	5/2$^+$		G	
8273.3 *4*	(1/2$^-$)		G K	XREF: K(8270) J$^{\pi}$: other: L=1 from 0$^+$ in ^{36}S(p,d).
8298.27 *30*	3/2$^+$,5/2$^+$		G	
8333.8 *4*	3/2$^-$		G	
8335.9 *4*	3/2$^+$,5/2$^+$		G	
8391.3 *4*	3/2$^-$		G	
8393.64 *30*	3/2$^+$,5/2$^+$		G	
8406.55 *30*	3/2$^+$,5/2$^+$		G k	XREF: k(8410)
8418.8 *4*	3/2$^+$,5/2$^+$		G kL	XREF: k(8410)L(8430) J$^{\pi}$: others: L=(2) from 0$^+$ in ^{36}S(p,d); L=2 from 3/2$^+$ in (p,^{3}He) suggests (1/2$^+$ to 9/2$^+$).
8465 *7*	(3/2$^+$,5/2$^+$)		K	
8509 *5*	3/2$^+$,5/2$^+$		K	
8557 *7*	(5/2$^-$,7/2$^-$)		K	
8602 *5*	(5/2$^-$,7/2$^-$)		K	
8649 *7*	(5/2$^-$,7/2$^-$)		K	
8707 *5*	(3/2$^+$,5/2$^+$)		K	
8755.9 *11*	(17/2)	<1 ps	E	J$^{\pi}$: 1576.3γ D, ΔJ=1 to (15/2$^+$), 7179.74. T$_{1/2}$: in (^{18}O,2αnγ) from 2021Go09 with residual Doppler energy shifts observed for the 732γ and 1576γ peaks.
8761 *7*	(3/2$^+$,5/2$^+$)		K	
8822 *5*	(1/2$^-$,3/2$^-$)		K	
8876 *5*	(1/2$^-$,3/2$^-$)		K	
8947 *7*	(1/2$^-$,3/2$^-$)		K	
9032 *15*	(1/2$^-$,3/2$^-$)		K	
9065 *15*	(1/2$^-$,3/2$^-$)		K	
9126 *7*	1/2$^+$		Kl	T=5/2 XREF: l(9155) J$^{\pi}$,T: isobaric analog state of 1/2$^+$, T=5/2, g.s. of ^{35}P (2026Jo01).
9180 *5*			Kl	XREF: K(9180)l(9155) J$^{\pi}$: L=2 from 3/2$^+$ in (p,^{3}He) suggests (1/2$^+$ to 9/2$^+$).
9219 *15*	(1/2$^-$,3/2$^-$)		K	
9294 *8*	(1/2$^-$,3/2$^-$)		K	
9339 *8*	(1/2$^-$,3/2$^-$)		K	

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Adopted Levels, Gammas (continued)

^{35}S Levels (continued)

E(level)†	J^{π}‡	XREF	Comments
9395 *5*	(1/2⁻,3/2⁻)	K	
9446 *7*	(1/2⁻,3/2⁻)	K	
9510 *10*	(1/2⁻,3/2⁻)	K	
9565 *20*	(1/2⁻,3/2⁻)	K	
9615 *10*	(1/2⁻,3/2⁻)	K	
9666 *10*	(1/2⁻,3/2⁻)	K	
9716 *10*	(1/2⁻,3/2⁻)	K	
9774 *10*	(1/2⁻,3/2⁻)	K	
9849 *10*	(1/2⁻,3/2⁻)	K	
9911 *10*	(1/2⁻,3/2⁻)	K	
9989 *10*	(1/2⁻,3/2⁻)	K	
10048.6 *9*	(19/2⁺)	E	J^{π}: 2868.7γ Q, ΔJ=2 to (15/2⁺), 7179.74.
1.0049×10^{4} *10*	(1/2⁻,3/2⁻)	K	
1.0166×10^{4} *10*	(1/2⁻,3/2⁻)	K	
1.0225×10^{4} *10*	(1/2⁻,3/2⁻)	K	
1.0277×10^{4} *10*	(1/2⁻,3/2⁻)	K	
1.0350×10^{4} *10*	(1/2⁻,3/2⁻)	K	
1.0421×10^{4} *10*	(1/2⁻,3/2⁻)	K	
1.0493×10^{4} *10*	(5/2⁻,7/2⁻)	K	
1.0563×10^{4} *10*	(1/2⁻,3/2⁻)	K	
1.0663×10^{4} *10*	(1/2⁻,3/2⁻)	K	
1.0746×10^{4} *10*	(1/2⁻,3/2⁻)	K	
1.0810×10^{4} *10*	(1/2⁻,3/2⁻)	K	
1.147×10^{4} *2*	(3/2)⁺	K	T=5/2 J^{π},T: IAS of 3/2⁺, T=5/2, 2387 level in ^{35}P (2026Jo01).
12469.3 *14*	(21/2)	E	J^{π}: 2420.6γ (D), ΔJ=(1) to (19/2⁺), 10048.6.
1.288×10^{4} *10*	(5/2)⁺	K	T=5/2 J^{π},T: IAS of 5/2⁺, T=5/2, 3860 level in ^{35}P (2026Jo01).
1.368×10^{4} *10*	(5/2)⁺	K	T=5/2 J^{π},T: IAS of 5/2⁺, T=5/2, 4666 level in ^{35}P (2026Jo01).
1.422×10^{4} *10*	(5/2)⁺	K	T=5/2 J^{π},T: IAS of 5/2⁺, T=5/2, 5199 level in ^{35}P (2026Jo01).
1.470×10^{4} *10*		K	

† From a least-squares fit to γ-ray energies with uncertainties for levels connected with γ transitions; from transfer reactions or ^{34}S(n,n),(n,γ):resonances for other levels, unless otherwise noted.

‡ For high-spin states, it is assumed that spin increases as the excitation energy increases. For resonances (7018-8418 keV), J^{π} is from ^{34}S(n,n),(n,γ):resonances, unless otherwise noted; for resonances above 8418 keV, J^{π} is from ^{36}S(p,d), unless otherwise noted.

Adopted Levels, Gammas (continued)

$\gamma(^{35}S)$

E_i(level)	J^π_i	E_γ†	I_γ†	E_f	J^π_f	Mult.‡	δ‡	α#	Comments
1572.369	1/2$^+$	1572.327 *11*	100	0.0	3/2$^+$	(M1)		0.0001007 *14*	B(M1)(W.u.)=0.00248 *16* E_γ: weighted average of 1572.252 *29* from ^{35}P β^- decay, 1572.4 *3* from (^{14}N,3pγ), 1572.333 *8* from (n,γ) E=thermal, 1572.24 *17* from ^{34}S(d,pγ), and 1572 *1* from ^{208}Pb(^{36}S,^{35}Sγ). Mult.: D from (^{14}N,3pγ); $\Delta\pi$=no from level scheme.
1991.27	7/2$^-$	1991.26 *5*	100	0.0	3/2$^+$	M2+E3	+0.17 *6*	0.0001476 *22*	B(M2)(W.u.)=0.0874 *+46−48*; B(E3)(W.u.)=3.7 *+29−21* E_γ: weighted average of 1991.3 *2* from (^{14}N,3pγ), 1990.5 *4* from (^{18}O,2αnγ), 1991.27 *5* from (n,γ) E=thermal, and 1991 *1* from ^{208}Pb(^{36}S,^{35}Sγ). Mult.,δ: from ^{34}S(d,pγ). D+Q or Q+O from $\gamma(\theta)$ in ^{34}S(d,pγ); E2+M3 is ruled out by RUL; D+Q is inconsistent with level scheme from 7/2$^-$ to 3/2$^+$.
2347.779	3/2$^-$	356.66@ *9*	<0.084	1991.27	7/2$^-$	[E2]		1.15×10^{-3} *2*	E_γ,I_γ: from (n,γ) E=thermal.
		775.398 *6*	35.3 *27*	1572.369	1/2$^+$	(E1)		3.54×10^{-5} *5*	B(E1)(W.u.)=4.5×10^{-4} *+9−7* E_γ: from (n,γ) E=thermal. Other: 775.4 *4* from (^{14}N,3pγ). I_γ: weighted average of 37 *5* from (^{14}N,3pγ), 35 *4* from (n,γ) E=thermal, 37.0 *28* from ^{34}S(d,pγ), and 33.3 *27* from (d,$\alpha\gamma$). Mult.: D(+Q) from $\gamma(\theta)$ in ^{34}S(d,pγ); $\Delta\pi$=yes from level scheme; M2 ruled out by RUL.
		2347.700 *19*	100.0 *27*	0.0	3/2$^+$	(E1)		0.000880 *12*	B(E1)(W.u.)=4.6×10^{-5} *+8−6* E_γ: weighted average of 2347.8 *3* from (^{14}N,3pγ), 2347.700 *15* from (n,γ) E=thermal, and 2347 *2* from ^{208}Pb(^{36}S,^{35}Sγ). I_γ: from (d,$\alpha\gamma$). Others: 100 *8* from (^{14}N,3pγ), 100 *8* from (n,γ) E=thermal, and 100.0 *28* from ^{34}S(d,pγ). Mult.: D(+Q) from $\gamma(\theta)$ in ^{34}S(d,pγ); $\Delta\pi$=yes from level scheme; M2 component is less likely based on RUL.
2717.06	5/2$^+$	368.7 *5*	14 *8*	2347.779	3/2$^-$	[E1]		0.0002294 *33*	B(E1)(W.u.)=0.022 *+20−10* E_γ: weighted average of 368.5 *4* from (n,γ) E=thermal and 370 *1* from ^{208}Pb(^{36}S,^{35}Sγ). I_γ: unweighted average of 6.7 *23* from (n,γ) E=thermal and 22 *4* from ^{208}Pb(^{36}S,^{35}Sγ).
		725.78	<5	1991.27	7/2$^-$	[E1]		4.10×10^{-5} *6*	
		1144.67	<5	1572.369	1/2$^+$	[E2]		3.79×10^{-5} *5*	

$^{35}_{16}S_{19}$-9

$\gamma(^{35}S)$ (continued)

E_i(level)	J^π_i	E_γ†	I_γ†	E_f	J^π_f	Mult.‡	δ‡	α#	Comments
2717.06	5/2+	2716.99 *15*	100 *10*	0.0	3/2+	M1+E2	−0.39 *5*	0.000568 *9*	B(M1)(W.u.)=0.012 +*7*−*3*; B(E2)(W.u.)=0.9 +*6*−*3* E_γ: weighted average of 2717.0 *4* from (^{14}N,3pγ), 2716.99 *16* from (n,γ) E=thermal, and 2717 *2* from ^{208}Pb(^{36}S,^{35}Sγ). I_γ: from ^{208}Pb(^{36}S,^{35}Sγ). Other: 100 *17* from (n,γ) E=thermal. Mult.,δ: D+Q from $\gamma(\theta)$ in ^{34}S(d,pγ) with δ=−0.39 *5* or −15.0 *34* for J_i=5/2 (1975VaYG); −15.0 *34* is unlikely based on RUL; E1+M2 is ruled out by RUL.
2938.63	3/2+	590.85	<10	2347.779	3/2−				
		947.35	<20	1991.27	7/2−				
		1366.23	<15	1572.369	1/2+				
		2938.56 *11*	100	0.0	3/2+				E_γ: weighted average of 2938.29 *40* from ^{35}P β^- decay and 2938.58 *11* from (n,γ) E=thermal.
3421.2	5/2+	482.6	<4	2938.63	3/2+	[M1,E2]		2.7×10^{-4} *13*	
		704.1	<6	2717.06	5/2+	[M1,E2]		9.5×10^{-5} *28*	
		1073.4	<7	2347.779	3/2−	[E1]		1.836×10^{-5} *26*	
		1429.9	<30	1991.27	7/2−	[E1]		0.0002243 *31*	
		1848.8	<7	1572.369	1/2+	[E2]		0.000254 *4*	
		3421 *2*	100	0.0	3/2+	[M1,E2]		0.00090 *7*	E_γ: from ^{208}Pb(^{36}S,^{35}Sγ).
3558.080	(5/2+)	619.23 *19*	8.9 *17*	2938.63	3/2+				E_γ,I_γ: from (n,γ) E=thermal.
		1210.28 *4*	54 *6*	2347.779	3/2−				E_γ: from (n,γ) E=thermal. Other: 1211 *2* from ^{208}Pb(^{36}S,^{35}Sγ). I_γ: weighted average of 54 *6* from (n,γ) E=thermal and 54 *6* from (d,$\alpha\gamma$). Other: 18 *7* from ^{208}Pb(^{36}S,^{35}Sγ).
		1566.7 *3*	100 *6*	1991.27	7/2−				E_γ: from (n,γ) E=thermal. Other: 1567 *2* from ^{208}Pb(^{36}S,^{35}Sγ). I_γ: from (d,$\alpha\gamma$). Others: 100 *19* from (n,γ) E=thermal and 100 *10* from ^{208}Pb(^{36}S,^{35}Sγ).
		1986& *2*	39 *6*	1572.369	1/2+				E_γ,I_γ: from ^{208}Pb(^{36}S,^{35}Sγ).
		3558.1 *5*	18 *4*	0.0	3/2+				E_γ,I_γ: from (n,γ) E=thermal.
3594.62	(7/2+)	655.98	<10	2938.63	3/2+				
		877.6 *5*	9.3 *23*	2717.06	5/2+				E_γ,I_γ: from (^{14}N,3pγ). Other: Iγ<11 from ^{34}S(d,pγ).
		1246.82	<15	2347.779	3/2−				
		1603.31	<17	1991.27	7/2−				
		3594.5 *7*	100 *9*	0.0	3/2+	Q			E_γ: weighted average of 3594.7 *7* from (^{14}N,3pγ), 3594.4 *11* from (^{18}O,2αnγ), and 3593 *2* from

Adopted Levels, Gammas (continued)

$\gamma(^{35}S)$ (continued)

E_i(level)	J^π_i	E_γ†	I_γ†	E_f	J^π_f	Mult.‡	δ‡	α#	Comments
									$^{208}Pb(^{36}S,^{35}S\gamma)$. I_γ: from $(^{14}N,3p\gamma)$. Mult.: from $\gamma\gamma(\theta)$(ADO) in $(^{14}N,3p\gamma)$.
3801.952	3/2$^-$	863.28 *28*	1.3 *4*	2938.63	3/2$^+$	[E1]		2.82×10^{-5} *4*	B(E1)(W.u.)=2.4×10^{-4} *+30−12* E_γ,I_γ: from (n,γ) E=thermal. Other: Iγ<8.9 from $^{34}S(d,p\gamma)$.
		1084.79 *15*	2.2 *7*	2717.06	5/2$^+$	[E1]		1.801×10^{-5} *25*	B(E1)(W.u.)=2.0×10^{-4} *+28−11* E_γ: from (n,γ) E=thermal. I_γ: weighted average of 2.1 *4* from (n,γ) E=thermal and 11 *5* from $^{34}S(d,p\gamma)$.
		1454.09 *4*	18 *5*	2347.779	3/2$^-$	[M1,E2]		8.0×10^{-5} *12*	E_γ: from (n,γ) E=thermal. I_γ: weighted average of 18 *4* from (n,γ) E=thermal and 18 *7* from $^{34}S(d,p\gamma)$. B(M1)(W.u.)=0.024 *+32−12* if M1, B(E2)(W.u.)=4×10^1 *+6−2* if E2.
		1810.63	<8.9	1991.27	7/2$^-$	[E2]		0.0002364 *33*	
		2229.510 *16*	88 *15*	1572.369	1/2$^+$	[E1]		0.000804 *11*	B(E1)(W.u.)=9×10^{-4} *+11−4* E_γ: from (n,γ) E=thermal. I_γ: unweighted average of 110 *11* from (n,γ) E=thermal, 93 *9* from $^{34}S(d,p\gamma)$, and 61 *5* from (d,$\alpha\gamma$).
		3801.73 *4*	100 *5*	0.0	3/2$^+$	[E1]		1.60×10^{-3} *2*	B(E1)(W.u.)=2.2×10^{-4} *+27−10* E_γ: from (n,γ) E=thermal. I_γ: from (d,$\alpha\gamma$). Other: 100 *9* from $^{34}S(d,p\gamma)$, 100 *9* from (n,γ) E=thermal.
3816.01	(9/2)$^-$	1824.7 *3*	100	1991.27	7/2$^-$	M1+E2	+0.55 *9*	0.000200 *4*	B(M1)(W.u.)=0.0099 *+14−12*; B(E2)(W.u.)=3.4 *+10−9* E_γ: weighted average of 1824.6 *2* from $(^{14}N,3p\gamma)$, 1824.4 *6* from $(^{18}O,2\alpha n\gamma)$, and 1827 *1* from $^{208}Pb(^{36}S,^{35}S\gamma)$. Mult.,$\delta$: D+Q and δ from $\gamma(\theta)$ and $\gamma\gamma(\theta)$(ADO) in $(^{14}N,3p\gamma)$; E1+M2 ruled out by RUL.
3886.28	(5/2)	465	<30	3421.2	5/2$^+$				
		947.64	<12	2938.63	3/2$^+$				
		1169.20	<15	2717.06	5/2$^+$				
		1538.4 *8*	57 *33*	2347.779	3/2$^-$				E_γ: from $(^{14}N,3p\gamma)$. I_γ: unweighted average of 24 *10* from $(^{14}N,3p\gamma)$ and 89 *9* from (d,$\alpha\gamma$). Other: <18 from $^{34}S(d,p\gamma)$.
		1894.9 *4*	100 *11*	1991.27	7/2$^-$	D			E_γ,Mult.: from $(^{14}N,3p\gamma)$.

Adopted Levels, Gammas (continued)

$\gamma(^{35}S)$ (continued)

E_i(level)	J^π_i	E_γ†	I_γ†	E_f	J^π_f	Mult.‡	δ‡	α#	Comments
									I_γ: from (d,$\alpha\gamma$). Other: 100 *14* from (^{14}N,3pγ) and 100 from ^{34}S(d,pγ).
3886.28	(5/2)	2313.83	<16	1572.369	1/2$^+$				
		3886	33 *7*	0.0	3/2$^+$				I_γ: from (d,$\alpha\gamma$). Other: <16 from ^{34}S(d,pγ).
4023.24	(11/2)$^-$	207.0 *5*	0.59 *30*	3816.01	(9/2)$^-$	[M1+E2]	<0.25	0.00124 *24*	B(M1)(W.u.)=0.05 +*4*−*3* if M1. δ: calculated by evaluators from the measured half-life of the level and RUL=100 for B(E2)(W.u.).
		2032.0 *3*	100 *6*	1991.27	7/2$^-$	E2		0.000341 *5*	B(E2)(W.u.)=7.5 +*8*−*7* E_γ: weighted average of 2031.8 *3* from (^{14}N,3pγ) and 2032.4 *4* from (^{18}O,2αnγ). Other: 2032 *2* from ^{208}Pb(^{36}S,^{35}Sγ). Mult.: Q from $\gamma\gamma(\theta)$(ADO) in (^{14}N,3pγ) and M2 ruled out by RUL.
4027.2	(3/2$^+$,5/2$^+$)	469.1	<14	3558.080	(5/2$^+$)				
		606.0	<28	3421.2	5/2$^+$				
		1088.6	95 *12*	2938.63	3/2$^+$				I_γ: weighted average of 86 *22* from ^{34}S(d,pγ) and 97 *12* from (d,$\alpha\gamma$).
		1310.1	<22	2717.06	5/2$^+$				
		1679.4	97 *12*	2347.779	3/2$^-$				I_γ: from (d,$\alpha\gamma$). Other: 92 *33* from ^{34}S(d,pγ).
		2035.9	<28	1991.27	7/2$^-$				
		2454.7	100 *18*	1572.369	1/2$^+$				I_γ: from (d,$\alpha\gamma$). Other: 100 *33* from ^{34}S(d,pγ).
		4027.0	<36	0.0	3/2$^+$				
4105.6	(5/2$^+$)	547.5	<4.6	3558.080	(5/2$^+$)				
		684.4	<4.6	3421.2	5/2$^+$				
		1167.0	15 *7*	2938.63	3/2$^+$				
		1388.5	<12	2717.06	5/2$^+$				
		1757.8	<5.8	2347.779	3/2$^-$				
		2114.3	<8.1	1991.27	7/2$^-$				
		2533.1	<5.8	1572.369	1/2$^+$				
		4105.3 *8*	100 *7*	0.0	3/2$^+$				E_γ: from (n,γ) E=thermal.
4180	(1/2,3/2,5/2$^+$)	2608	22 *6*	1572.369	1/2$^+$				I_γ: from (d,$\alpha\gamma$).
		4180	100 *6*	0.0	3/2$^+$				I_γ: from (d,$\alpha\gamma$).
4189.240	1/2$^-$	631.32@ *24*	<2.8	3558.080	(5/2$^+$)				E_γ,I_γ: from (n,γ) E=thermal. Other: Iγ<7.7 from ^{34}S(d,pγ).
		1250.61 *5*	8.4 *10*	2938.63	3/2$^+$	[E1]		0.0001047 *15*	E_γ,I_γ: from (n,γ) E=thermal. Other: Iγ<9.6 from ^{34}S(d,pγ).
		1840.2 *12*	91 *10*	2347.779	3/2$^-$	[M1,E2]		0.000221 *29*	E_γ: from (n,γ) E=thermal. I_γ: weighted average of 87 *16* from (n,γ) E=thermal and 92 *10* from ^{34}S(d,pγ).

$^{35}_{16}S_{19}$-12

$\gamma(^{35}S)$ (continued)

E_i(level)	J^π_i	E_γ†	I_γ†	E_f	J^π_f	Mult.‡	α#	Comments
4189.240	1/2⁻	2616.8 *13*	9.5 *28*	1572.369	1/2⁺	[E1]	1.04×10^{-3} *2*	E_γ,I_γ: from (n,γ) E=thermal. Other: Iγ<12 from ^{34}S(d,pγ).
		4188.95 *5*	100 *10*	0.0	3/2⁺	[E1]	1.75×10^{-3} *3*	E_γ: from (n,γ) E=thermal. I_γ: other: 100 *11* from (n,γ) E=thermal.
4302.6	(1/2⁻,3/2⁻)	1954.8	100 *9*	2347.779	3/2⁻			I_γ: from (d,$\alpha\gamma$).
		4302.3	70 *9*	0.0	3/2⁺			I_γ: from (d,$\alpha\gamma$).
4477.63	(5/2⁺)	591.35	<8.2	3886.28	(5/2)		1.5×10^{-4} *6*	
		675.67	<8.2	3801.952	3/2⁻	[E1]	4.82×10^{-5} *7*	
		883.00	<8.2	3594.62	(7/2⁺)	[M1,E2]	5.4×10^{-5} *12*	
		919.54	<8.2	3558.080	(5/2⁺)	[M1,E2]	4.9×10^{-5} *10*	
		1056.4	<16	3421.2	5/2⁺	[M1,E2]	3.6×10^{-5} *6*	
		1538.96	<12	2938.63	3/2⁺	[M1,E2]	0.000106 *16*	
		1760.55 *11*	100 *12*	2717.06	5/2⁺	[M1,E2]	0.000189 *25*	E_γ: from (n,γ) E=thermal. I_γ: others: 100 *15* from (n,γ) E=thermal and 100 *11* from (d,$\alpha\gamma$).
		2129.78	<13	2347.779	3/2⁻	[E1]	0.000738 *10*	
		2486.27	<16	1991.27	7/2⁻	[E1]	0.000962 *13*	I_γ: other: 42 *11* from (d,$\alpha\gamma$).
		2905.1 *4*	90 *14*	1572.369	1/2⁺	[E2]	0.000749 *10*	E_γ: from (n,γ) E=thermal. I_γ: unweighted average of 96 *28* from (n,γ) E=thermal, 64 *12* from ^{34}S(d,pγ), and 109 *11* from (d,$\alpha\gamma$).
		4477	<20	0.0	3/2⁺	[M1,E2]	0.00126 *8*	I_γ: other: 28 *6* from (d,$\alpha\gamma$).
4822.79	(9/2⁺)	1006.8 *2*	66 *15*	3816.01	(9/2)⁻	D		E_γ: from (^{14}N,3pγ). Other: 1007.2 *11* from (^{18}O,2αnγ). I_γ: weighted average of 73 *9* from (^{14}N,3pγ) and 36 *18* from (^{18}O,2αnγ). Mult.: from $\gamma\gamma(\theta)$(ADO) in (^{14}N,3pγ) and (^{18}O,2αnγ), consistent with ΔJ=0.
		1228.1 *3*	100 *9*	3594.62	(7/2⁺)	D		E_γ,I_γ: other: 1228.3 *6* with Iγ=100 *18* from (^{18}O,2αnγ). Mult.: from (^{14}N,3pγ).
		2831.4 *9*	36 *9*	1991.27	7/2⁻			
4899.82	(9/2⁺)	1305.3 *4*	55 *18*	3594.62	(7/2⁺)			E_γ: from (^{18}O,2αnγ). Other: 1304.8 *12* from (^{14}N,3pγ).
		2908.3 *5*	100 *18*	1991.27	7/2⁻	D		Mult.: from $\gamma\gamma(\theta)$(ADO) in (^{14}N,3pγ).
4903.371	1/2⁺	1101.92 *31*	0.43 *11*	3801.952	3/2⁻			
		1964.8 *2*	4.9 *13*	2938.63	3/2⁺			
		2555.492 *14*	41 *4*	2347.779	3/2⁻			
		3330.84 *13*	100 *9*	1572.369	1/2⁺			
		4902.99 *5*	53 *5*	0.0	3/2⁺			
4963.082	5/2⁺	1161.05 *20*	0.98 *14*	3801.952	3/2⁻			
		1404.967 *24*	10.7 *11*	3558.080	(5/2⁺)			
		2615.2 *2*	19.5 *18*	2347.779	3/2⁻			
		2972.0 *4*	3.2 *11*	1991.27	7/2⁻			

A general comment (not necessary to apply to dataset. 2008Wi09 and 2015Ch56 use different beam energies, which would result in different populations of excited states. You

Adopted Levels, Gammas (continued)

$\gamma(^{35}S)$ (continued)

E_i(level)	J^π_i	E_γ†	I_γ†	E_f	J^π_f	Mult.‡	α#	Comments
4963.082	5/2$^+$	3390.62 *7*	100 *9*	1572.369	1/2$^+$			
		4963.06 *22*	47 *5*	0.0	3/2$^+$			
5010.05	(11/2)$^-$	986.8 *3*	100 *5*	4023.24	(11/2)$^-$	(M1)	3.45×10^{-5} *5*	B(M1)(W.u.)=0.036 *+8−6* E_γ: weighted average of 986.9 *3* from (^{14}N,3pγ) and 986.6 *3* from (^{18}O,2αnγ). Mult.: D from (^{18}O,2αnγ) and (^{14}N,3pγ); $\Delta\pi$=no from level scheme.
		3018.5 *7*	43 *5*	1991.27	7/2$^-$	E2	0.000798 *11*	B(E2)(W.u.)=0.22 *+5−4* Mult.: Q from $\gamma\gamma(\theta)$(ADO) and M2 ruled out by RUL in (^{14}N,3pγ).
5343	($^-$)	5343		0.0	3/2$^+$			
5412.28	(9/2$^+$)	1389.0 *4*	100 *10*	4023.24	(11/2)$^-$	D		E_γ: other: Eγ=1389.0 *8* from (^{18}O,2αnγ). Mult.: from $\gamma\gamma(\theta)$(ADO) in (^{14}N,3pγ).
		1525.7 *8*	15.4 *26*	3886.28	(5/2)			
		3420.6 *9*	28.2 *26*	1991.27	7/2$^-$			
5752.5	5/2$^-$,7/2$^-$	5752.0 *8*	100	0.0	3/2$^+$			
5767	5/2$^+$	5767		0.0	3/2$^+$			
5841.482	(3/2$^+$,5/2$^+$)	4268.65 *14*	100	1572.369	1/2$^+$			E_γ,I_γ: from (n,γ) E=thermal.
5877.70	(11/2$^+$)	465.4 *2*	38 *6*	5412.28	(9/2$^+$)	D		E_γ: weighted average of 465.3 *3* from (^{14}N,3pγ) and 465.5 *2* from (^{18}O,2αnγ). I_γ: unweighted average of 31.6 *26* from (^{14}N,3pγ) and 43.8 *32* from (^{18}O,2αnγ). Mult.: from (^{18}O,2αnγ).
		867.6 *4*	50.0 *26*	5010.05	(11/2)$^-$	D		E_γ: weighted average of 867.3 *3* from (^{14}N,3pγ) and 868.2 *4* from (^{18}O,2αnγ). I_γ: other: 62 *7* from (^{18}O,2αnγ). Mult.: from (^{18}O,2αnγ).
		977.9 *3*	43 *5*	4899.82	(9/2$^+$)	D+Q		E_γ: weighted average of 977.8 *3* from (^{14}N,3pγ) and 978.4 *10* from (^{18}O,2αnγ). I_γ: weighted average of 42.1 *26* from (^{14}N,3pγ) and 82 *22* from (^{18}O,2αnγ). Mult.: from (^{14}N,3pγ).
		1055.1 *6*	80 *7*	4822.79	(9/2$^+$)	D		E_γ: unweighted average of 1054.5 *3* from (^{14}N,3pγ) and 1055.6 *4* from (^{18}O,2αnγ). I_γ: weighted average of 79 *5* from (^{14}N,3pγ) and 122 *32* from (^{18}O,2αnγ). Mult.: from (^{18}O,2αnγ).
		1854.4 *4*	44.7 *26*	4023.24	(11/2)$^-$			E_γ,I_γ: other: 1854.5 *42* with Iγ=88 *44* from (^{18}O,2αnγ).

Adopted Levels, Gammas (continued)

$\gamma(^{35}S)$ (continued)

E_i(level)	J^π_i	E_γ†	I_γ†	E_f	J^π_f	Mult.‡	α#	Comments
5877.70	$(11/2^+)$	2061.6 *4*	100 *5*	3816.01	$(9/2)^-$	D		E_γ: weighted average of 2061.5 *3* from (^{14}N,3pγ) and 2063.3 *14* from (^{18}O,2αnγ). I_γ: other: 100 *20* from (^{18}O,2αnγ). Mult.: from (^{18}O,2αnγ).
		2283.0 *4*	56 *12*	3594.62	$(7/2^+)$			E_γ: weighted average of 2282.9 *4* from (^{14}N,3pγ) and 2283.5 *10* from (^{18}O,2αnγ). I_γ: unweighted average of 68.4 *26* from (^{14}N,3pγ) and 44 *10* from (^{18}O,2αnγ).
		3886.0 *13*	7.9 *26*	1991.27	$7/2^-$			
6018.8		6018.2 *6*	100	0.0	$3/2^+$			E_γ,I_γ: from (n,γ) E=thermal.
6078.475	$1/2^-,3/2^-$	3139.9 *5*	22 *7*	2938.63	$3/2^+$			
		6077.87 *11*	100 *12*	0.0	$3/2^+$			
6293.93		6293.2 *4*	100	0.0	$3/2^+$			E_γ,I_γ: from (n,γ) E=thermal.
6299.4	(11/2)	887.0 *7*	40 *10*	5412.28	$(9/2^+)$	D		
		2275.9 *6*	100 *20*	4023.24	$(11/2)^-$			
		2483.6 *8*	90 *20*	3816.01	$(9/2)^-$			
6352.2	$(13/2)^-$	2329.5 *9*	100 *20*	4023.24	$(11/2)^-$	(M1)	0.000392 *5*	B(M1)(W.u.)=0.023 *+6−5* Mult.: D from $\gamma\gamma(\theta)$(ADO) in (^{14}N,3pγ); $\Delta\pi$=no from level scheme.
		2535.8 *11*	50 *6*	3816.01	$(9/2)^-$	E2	0.000581 *8*	B(E2)(W.u.)=5.3 *+17−11* Mult.: Q from $\gamma\gamma(\theta)$(ADO) in (^{14}N,3pγ) and M2 ruled out by RUL.
6354.89		6355.0 *6*	100	0.0	$3/2^+$			E_γ,I_γ: from (n,γ) E=thermal.
6419.9		6419.3 *11*	100	0.0	$3/2^+$			E_γ,I_γ: from (n,γ) E=thermal.
6629.43		6628.5 *6*	100	0.0	$3/2^+$			E_γ,I_γ: from (n,γ) E=thermal.
6761.0		6760.3 *12*	100	0.0	$3/2^+$			E_γ,I_γ: from (n,γ) E=thermal.
(6986.097)	$1/2^+$	356.66@ *9*	<0.073	6629.43				
		631.32@ *24*	<0.123	6354.89				
		692.16 *5*	0.238 *31*	6293.93				
		907.608 *23*	1.00 *18*	6078.475	$1/2^-,3/2^-$			
		1144.591 *20*	0.98 *11*	5841.482	$(3/2^+,5/2^+)$			
		2022.954 *9*	19.6 *18*	4963.082	$5/2^+$			
		2082.83 *17*	28.4 *27*	4903.371	$1/2^+$			
		2508.39 *8*	0.68 *7*	4477.63	$(5/2^+)$			
		2796.74 *4*	8.8 *7*	4189.240	$1/2^-$			
		3183.95 *4*	10.9 *11*	3801.952	$3/2^-$			
		4638.14 *14*	100 *9*	2347.779	$3/2^-$			
		6985.7 *10*	0.064 *14*	0.0	$3/2^+$			
7179.74	$(15/2^+)$	827.8 *6*	2.2 *8*	6352.2	$(13/2)^-$	[E1]	3.08×10^{-5} *4*	B(E1)(W.u.)=7×10^{-6} *+6−3*
		880.2 *4*	5.9 *8*	6299.4	(11/2)			

$\gamma(^{35}S)$ (continued)

E_i(level)	J_i^π	E_γ†	I_γ†	E_f	J_f^π	Mult.‡	α#	Comments
7179.74	(15/2+)	1302.0 *3*	100 *6*	5877.70	(11/2+)	E2	5.56×10^{-5} *8*	B(E2)(W.u.)=6.6 *+41−19* E_γ: weighted average of 1302.2 *2* from (^{14}N,3pγ) and 1301.6 *3* from (^{18}O,2αnγ). Mult.: Q from $\gamma\gamma(\theta)$(ADO) in (^{14}N,3pγ) and M2 ruled out by RUL.
8024.0	(17/2+)	844.2 *6*	100	7179.74	(15/2+)	(M1)	4.66×10^{-5} *7*	B(M1)(W.u.)=0.24 *+9−5* E_γ: weighted average of 843.9 *8* from (^{14}N,3pγ) and 844.3 *6* from (^{18}O,2αnγ). Mult.: D from $\gamma\gamma$(ADO) in (^{18}O,2αnγ); $\Delta\pi$=(no) from level scheme.
8755.9	(17/2)	731.6 *15*	57 *22*	8024.0	(17/2+)			E_γ,I_γ: from (^{18}O,2αnγ).
		1576.3 *14*	100 *22*	7179.74	(15/2+)	D		E_γ,I_γ: from (^{18}O,2αnγ).
10048.6	(19/2+)	2868.7 *8*	100	7179.74	(15/2+)	Q		E_γ,I_γ: from (^{18}O,2αnγ).
12469.3	(21/2)	2420.6 *11*	100	10048.6	(19/2+)	(D)		E_γ,I_γ: from (^{18}O,2αnγ).

† For γ rays from low-spin levels (J<9/2), values are from ^{34}S(d,pγ) up to 4478 level and from (n,γ) E=thermal above that, for γ rays from high-spin level (J≥9/2), values are from ^{24}Mg(^{14}N,3pγ), unless otherwise noted. Eγ values without uncertainties are deduced from level-energy differences.

‡ Mult and δ are primarily from $\gamma(\theta)$ and $\gamma\gamma$(ADO) in ^{34}S(d,pγ), ^{24}Mg(^{14}N,3pγ), and ^{26}Mg(^{18}O,2αnγ). The magnetic or electric nature is determined based on RUL and measured $T_{1/2}$ where available, unless otherwise noted.

Total theoretical internal conversion coefficients, calculated using the BrIcc code (2008Ki07) with “Frozen Orbitals” approximation based on γ-ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

@ Multiply placed.

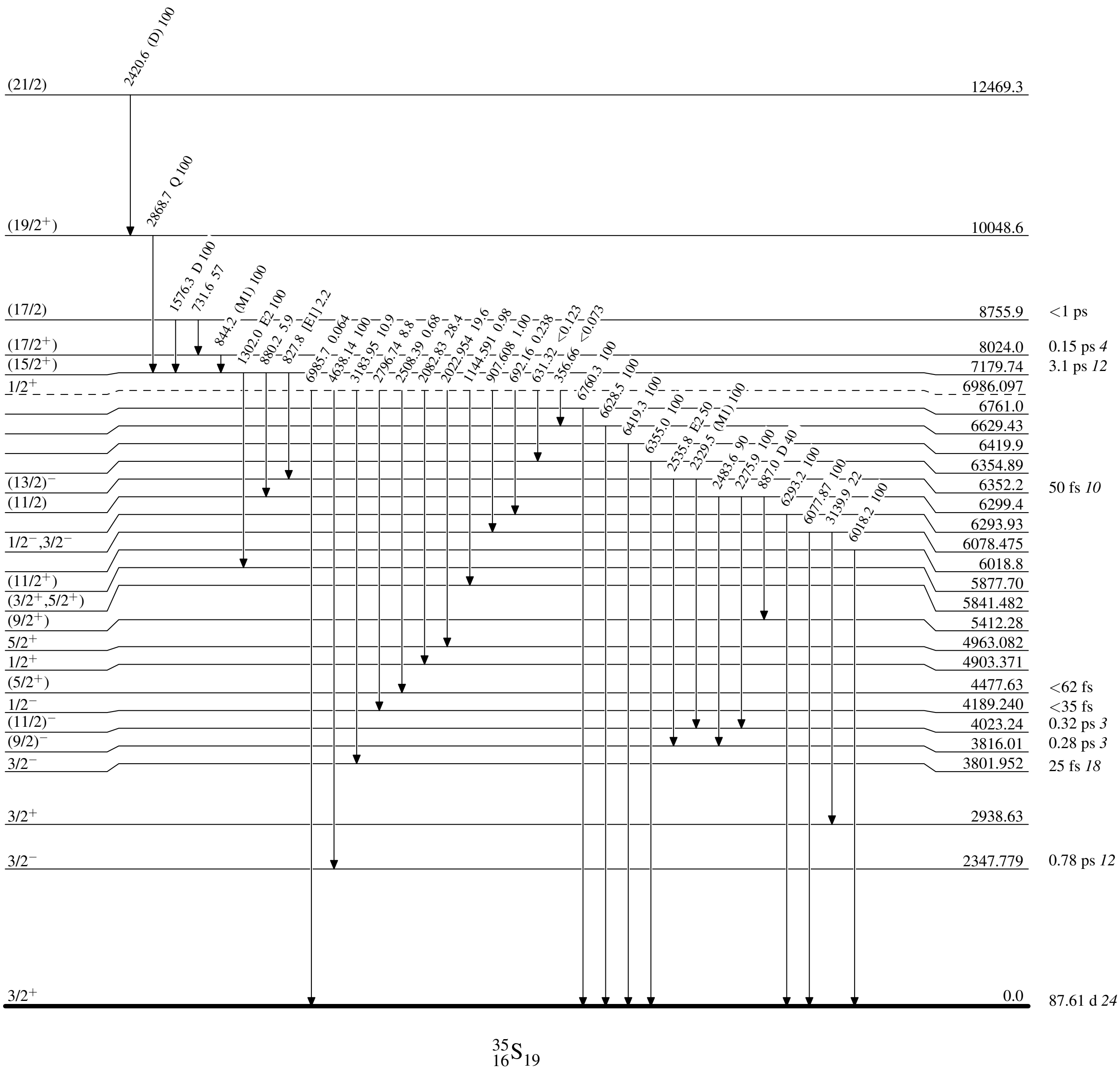
& Placement of transition in the level scheme is uncertain.

$^{35}_{16}S_{19}$-16

Adopted Levels, Gammas

Level Scheme

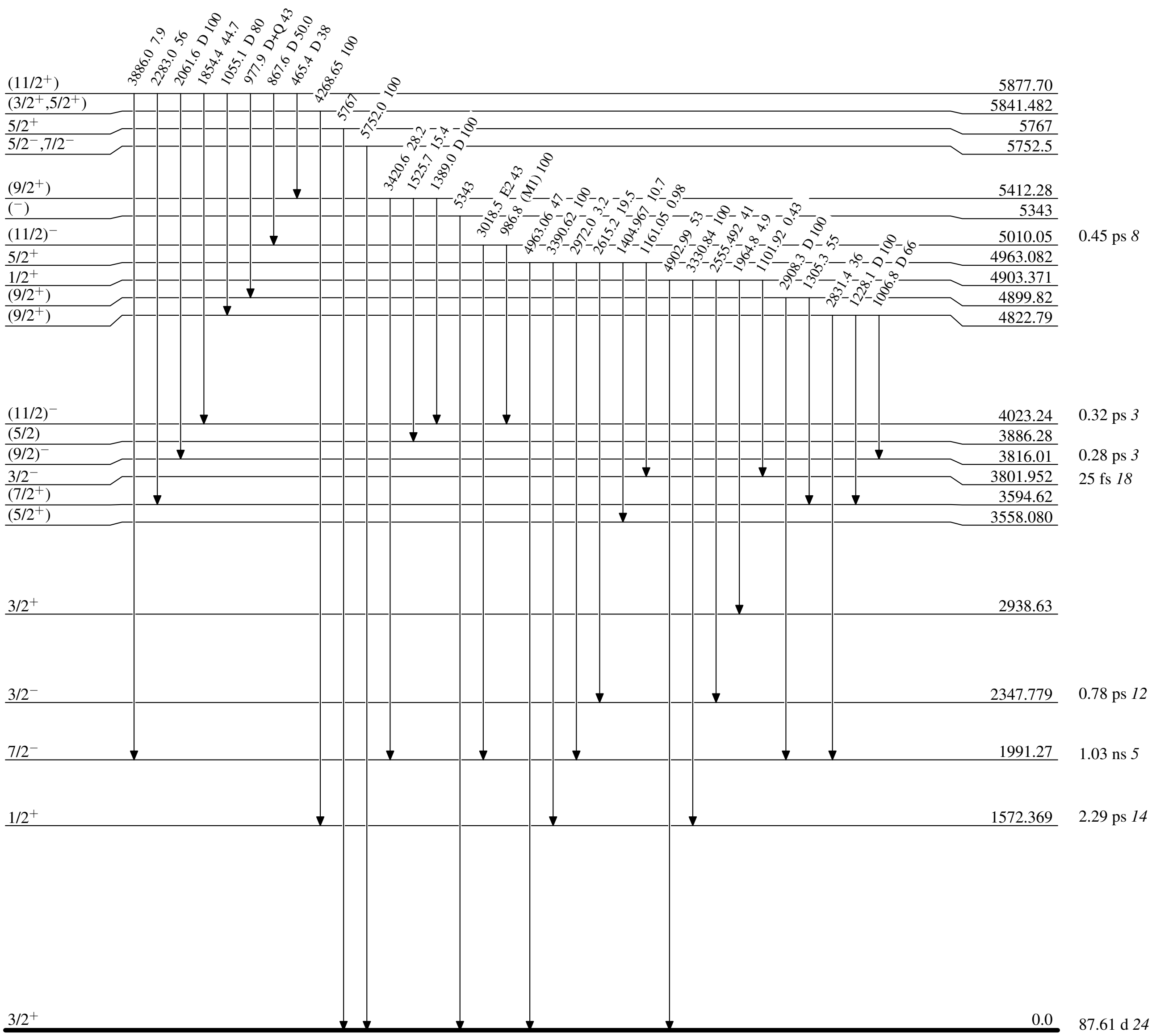
Intensities: Relative photon branching from each level



Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level

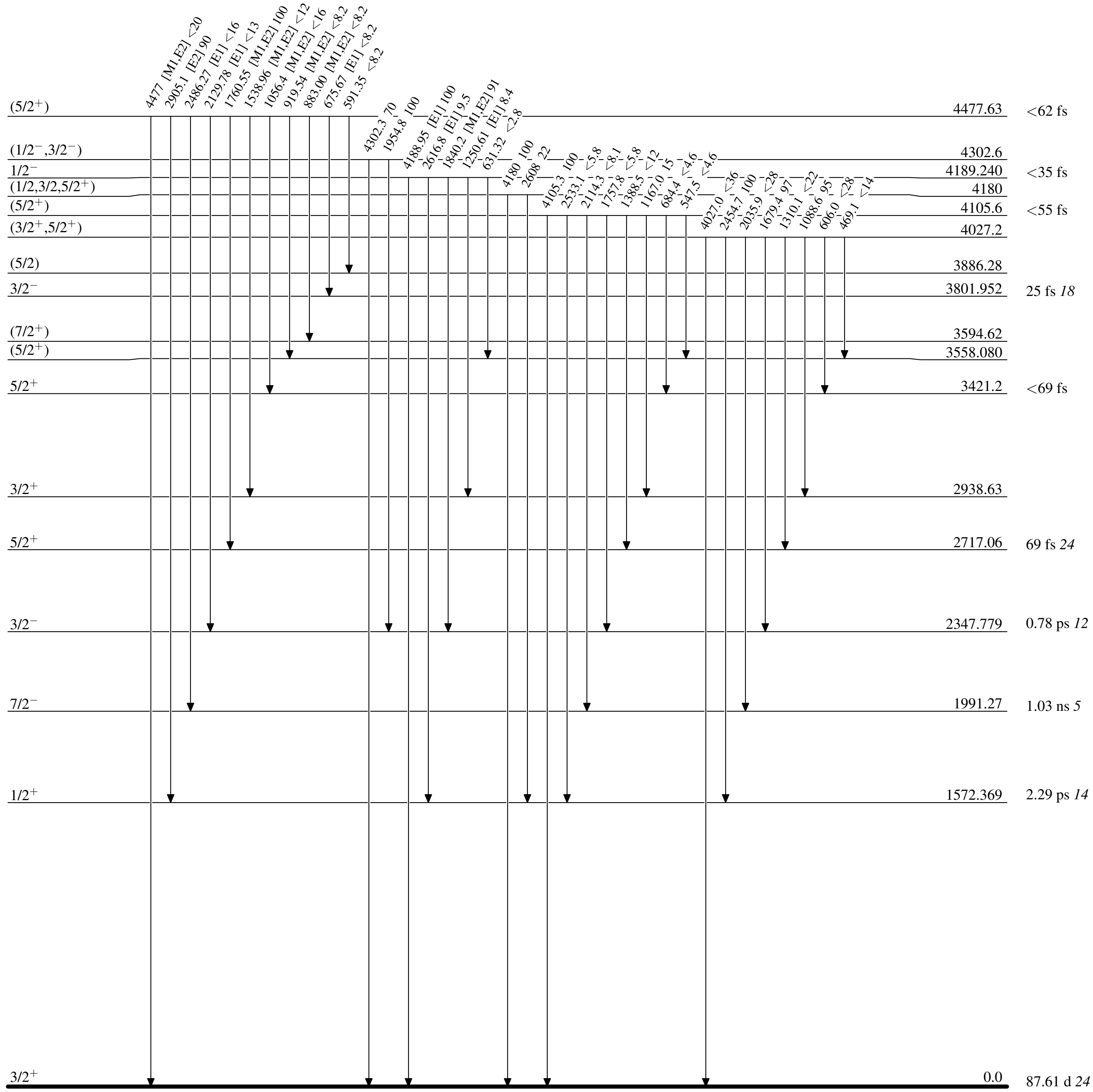


$^{35}_{16}S_{19}$

Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level



$^{35}_{16}S_{19}$

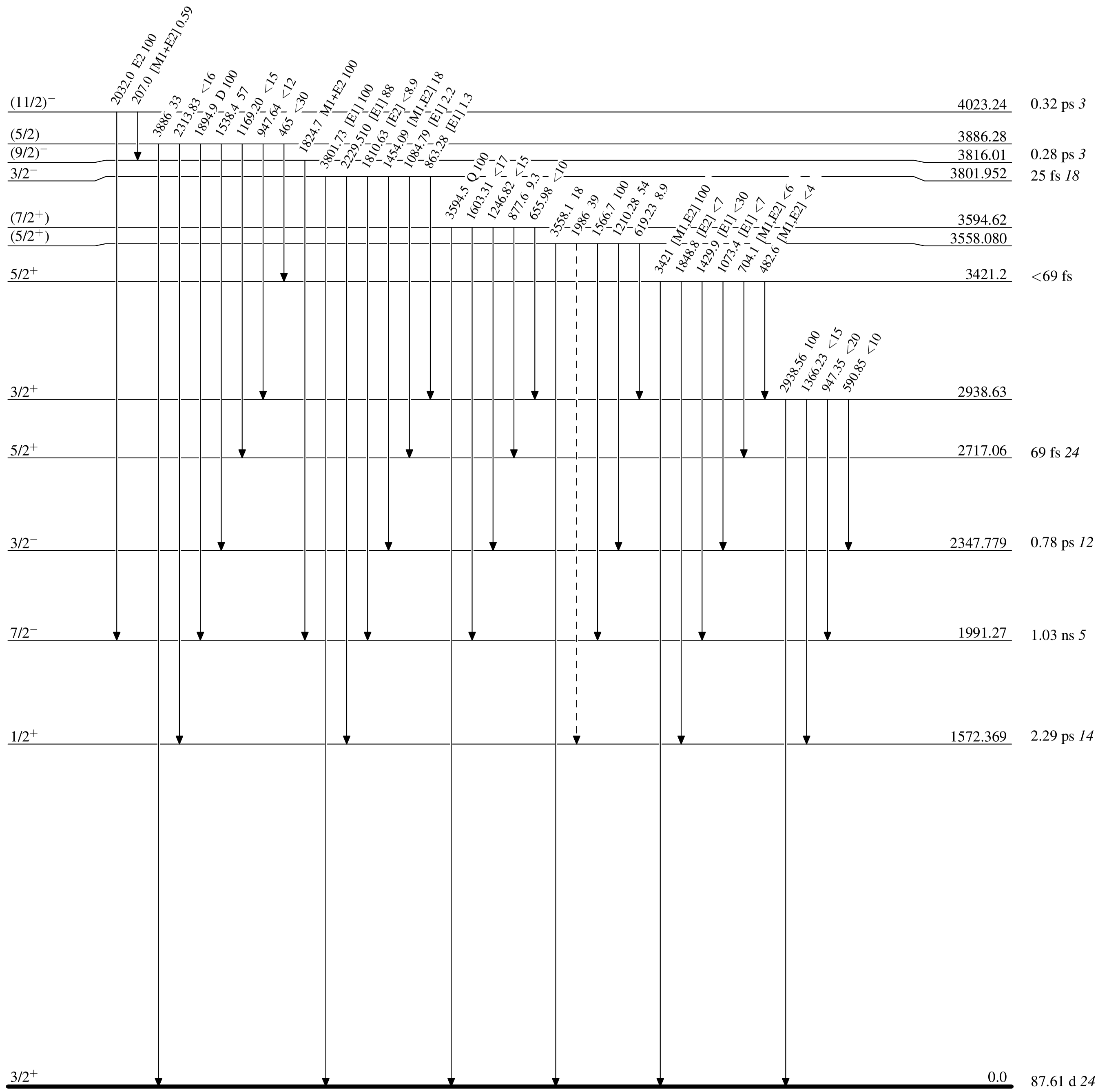
Adopted Levels, Gammas

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

γ Decay (Uncertain)

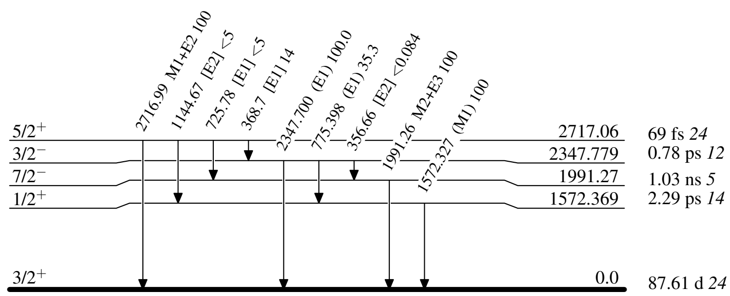


$^{35}_{16}S_{19}$

Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level



$^{35}_{16}S_{19}$

^{35}P β^- decay (47.3 s) 1986Wa22

Parent: ^{35}P: E=0; J$^\pi$=1/2$^+$; T$_{1/2}$=47.3 s *8*; Q(β^-)=3988.4 *19*; %β^- decay=100
^{35}P-J$^\pi$,T$_{1/2}$: From Adopted Levels of ^{35}P.
^{35}P-Q(β^-): From 2021Wa16.
1986Wa22: ^{35}P was produced via the ^{36}S(t,α)^{35}P reaction. 81% enriched ^{36}S targets were bombarded with 3.4-MeV tritons from the Brookhaven National Laboratory (BNL) tandem van de Graaff facility. γ rays were detected using an intrinsic coaxial Ge detector. Measured Eγ, Iγ. Deduced levels, decay branching ratios, and log *ft*. Compared with shell model calculations.
1972Ap01: ^{35}P was produced via the ^{37}Cl(t,αp)^{35}P reaction. LiCl and NaCl targets were bombarded with 16-MeV tritons from the Los Alamos tandem Van de Graaff facility. γ rays were detected using an 85 cm^3 Ge(Li) detector. β particles were detected using an end-window gas-flow proportional counter. Measured Eγ, Iγ, and Iβ. Deduced ^{35}P T$_{1/2}$, decay branching ratios, and log *ft*. Compared with shell model calculations.
1972Go31: ^{35}P was produced via the ^{18}O(^{19}F,2p)^{35}P and ^{36}S(t,α)^{35}P reactions at the second tandem of the Brookhaven National Laboratory (BNL) tandem van de Graaff facility. ^{35}P β-delayed γ rays were detected using a 60 cm^3 Ge(Li) detector. β particles were detected in coincidence with the 1572γ using a NE102 scintillator. Measured Eγ, Iγ, and Eβ. Deduced ^{35}P T$_{1/2}$, β end-point energy, mass, decay branching ratios, and log *ft*. Compared with shell model calculations.

1971Gr53: ^{35}P was produced via the ^{37}Cl(γ,2p)^{35}P reaction. 6-g NH$_4$Cl(nat) samples were irradiated with γ rays from a bremsstrahlung spectrum at the Mainz electron linear accelerator. γ rays were detected using NaI(Tl) and Ge(Li) detectors. Measured Eγ. Deduced T$_{1/2}$ of ^{35}P and levels in ^{35}S.
There is a 1-MeV gap between the highest observed level at E=2938.4 and Q(β^-) value=3988.4 *19* (2021Wa16). There may be missing weak transitions from unobserved levels in the gap. Upper limits for the unobserved β feedings:
1991: <0.08 (1986Wa22), <2.5 (1972Go31), <0.09 (1972Ap01);
2348: <0.08 (1986Wa22), <5 (1972Go31), <0.24 (1972Ap01);
2717: <0.03 (1986Wa22), <7 (1972Go31), <0.38 (1972Ap01);
3421: <0.02 (1986Wa22);
3558: <0.11 (1986Wa22);
3597: <0.02 (1986Wa22);
3675: <0.02 (1986Wa22);
3802: <0.03 (1986Wa22).

^{35}S Levels

E(level)†	J$^\pi$‡	T$_{1/2}$‡
0	3/2$^+$	87.61 d *24*
1572.290 *29*	1/2$^+$	2.29 ps *14*
2938.4 *4*	3/2$^+$	

† From a least-squares fit to γ-ray energies.
‡ From the Adopted Levels.

β^- radiations

E(decay)	E(level)	Iβ^-†‡	Log *ft*	Comments
(1050.0 *22*)	2938.4	0.47	5.0	Iβ^-: others: <8 (1972Go31), <0.45 (1972Ap01).
(2416.1 *22*)	1572.290	99.2	4.1	Iβ^-: other: 99 +*1*−*7* from measured γ/β ratio (1972Ap01), 100 (1972Go31).
(3988.4# *24*)	0	<0.73	>7.2	Iβ^-: converted by evaluators from the shell-model calculated 0.73 in 1986Wa22. Others: <9 in 1972Ap01; not given in 1972Go31.

† From Iγ intensity balance for excited levels.
‡ Absolute intensity per 100 decays.
# Existence of this branch is questionable.

^{35}P β^- decay (47.3 s) 1986Wa22 (continued)

$\gamma(^{35}S)$

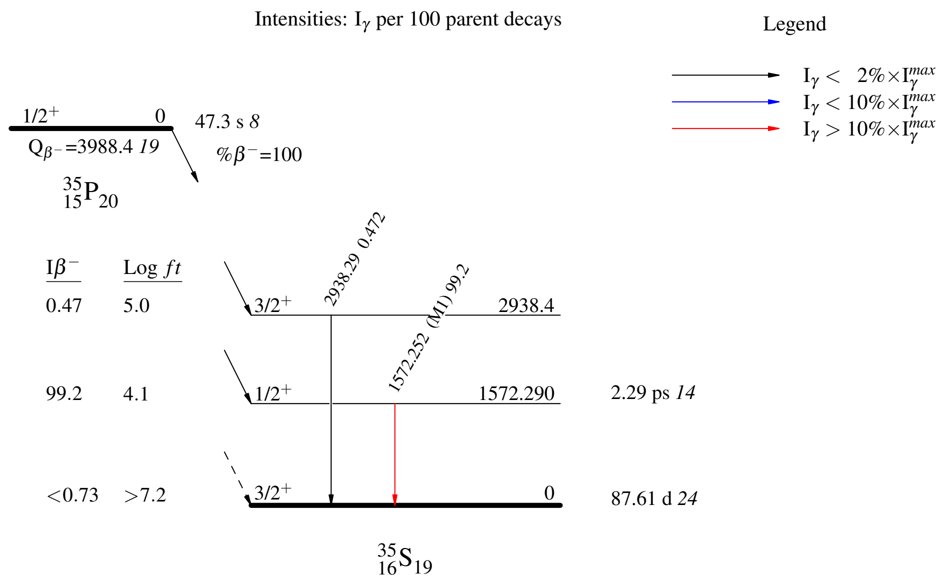
Iγ normalization: from Σ%I(γ to g.s.)=99.635 *365* deduced from 100−%Iβ^-(g.s.), where %Iβ^-(g.s.)=0.365 *365* is converted by evaluators from the shell-model calculated %Iβ^-(g.s.)=0.73 in 1986Wa22. Others: %Iβ^-(g.s.)<9 from 1972Ap01; %Iβ^-(g.s.) is not given in 1972Go31. Given that the ground state β^--feeding has not been experimentally observed, the evaluators opt to convert the shell-model calculated value to an upper limit when deducing the normalization factor.

E_γ †	I_γ †#	E_i(level)	J^π_i	E_f	J^π_f	Mult.‡	α @	Comments
1572.252 *29*	100	1572.290	1/2+	0	3/2+	(M1)	0.0001007 *14*	%Iγ=99.2 E_γ: weighted average of 1571.8 *2* (1971Gr53), 1572.2 *4* (1972Ap01), 1572.36 *24* (1972Go31), 1572.47 *25* (1972Go31), and 1572.256 *24* (1986Wa22).
2938.29 *40*	0.476 *30*	2938.4	3/2+	0	3/2+			%Iγ=0.47

† From 1986Wa22, unless otherwise noted.
‡ From the Adopted Gammas.
# For absolute intensity per 100 decays, multiply by 0.992.
@ Total theoretical internal conversion coefficients, calculated using the BrIcc code (2008Ki07) with “Frozen Orbitals” approximation based on γ-ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

^{35}P β^- decay (47.3 s) 1986Wa22

Decay Scheme



$^{2}H(^{34}S,p\gamma)$ 1973Wa10

$^{34}S(d,p)^{35}S$ in inverse kinematics.

1973Wa10: A 59.6-MeV ^{34}S beam was produced from the BNL MP-tandem Van de Graaff facility. Targets were 200-μg/cm^2 TiD prepared by evaporating titanium onto the Cu, Al, and Mg target backings in a deuterium atmosphere. γ rays were detected using a 35-cm^3 Ge(Li) detector at 0° with FWHM=2 keV at 656 keV. Measured Eγ. Deduced $T_{1/2}$ for two ^{35}S levels using Doppler Shift Attenuation lineshape analysis.

^{35}S Levels

E(level)†	J$^\pi$‡	$T_{1/2}$#	Comments
0	3/2$^+$		
1574	1/2$^+$	2.29 ps *35*	$T_{1/2}$: Lifetime=3.3 ps *5* from 1973Wa10.
2350	3/2$^-$	0.90 ps *14*	$T_{1/2}$: Lifetime=1.3 ps *2* from 1973Wa10.

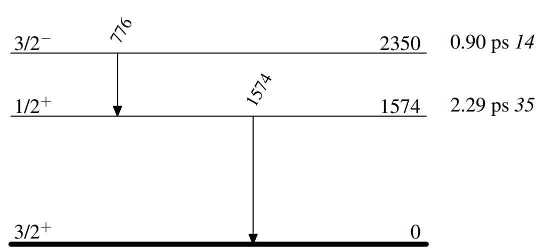
† From 1973Wa10.
‡ From the Adopted Levels.
# From Doppler Shift Attenuation Method (DSAM).

$\gamma(^{35}S)$

E$_\gamma$	E$_i$(level)	J$^\pi_i$	E$_f$	J$^\pi_f$
776	2350	3/2$^-$	1574	1/2$^+$
1574	1574	1/2$^+$	0	3/2$^+$

$^{2}H(^{34}S,p\gamma)$ 1973Wa10

Level Scheme



$^{35}_{16}S_{19}$

^{9}Be(^{36}S,^{35}Sγ) 2015MuZY,2021JoZZ

$J^{\pi}=0^{+}$ for ^{36}S ground state.

2015MuZY: a secondary beam of ^{36}S was produced via the projectile fragmentation of a 140-MeV/nucleon ^{48}Ca primary beam impinging on a 846-mg/cm^2 ^{9}Be target at the coupled cyclotron facility at NSCL, MSU. The ^{36}S nuclei were selected using the A1900 separator with an intensity of 8.1×10^{5} pps and a purity of 89.7%. The excited states of ^{35}S were populated by the one-neutron knockout reaction from the ^{36}S beam at 88 MeV/nucleon (midtarget energy) on a 100-mg/cm^2 ^{9}Be secondary target located at the reaction target position of the S800 spectrometer. The projectile-like residues were identified from their energy loss measured by an ionization chamber located at the focal plane of the S800 spectrometer and from their ToF measured between two scintillators situated at the object position and at the focal plane of the S800 spectrometer. Prompt γ rays from the deexcitation of the ^{35}S residues were detected by seven modules of the GRETINA Ge array. Measured Eγ, Iγ, (^{35}S)γ-coin, $\gamma\gamma$-coin, the inclusive knockout cross section for producing ^{35}S from ^{36}S, the fractional populations and parallel momentum distributions of 11 populated states in ^{35}S residues. Deduced levels, J, π, L-transfers, partial knockout cross sections and spectroscopic factors. Calculated single-particle cross sections (σ_{sp}) for neutron removal and parallel momentum distributions using the eikonal model.

2021JoZZ: a more careful reanalysis of 2015MuZY data.

^{35}S Levels

b_f values under comments are population fraction in the knock-out reaction, σ_f^{sp} values are for calculated single-particle cross sections, and R_s values are quenching factors from a systematics study in 2014To14. The partial cross sections to each level can be deduced from $b_f\times\sigma_{inc}^{exp}$.

The inclusive cross section σ_{inc}^{exp} for producing ^{35}S from ^{36}S is measured to be 63.7 mb *6* (2015MuZY) and 67 mb *4* (2021JoZZ).

E(level)[†]	L[‡]	C^{2}S[#]	Comments
0	2	3.33 *76*	C^{2}S: 3.2 *7* (2015MuZY). b_f=44.5% *37* (2021JoZZ), b_f=43.3% *47*, σ_f^{sp}=13.1 mb, R_s=0.66 *13* (2015MuZY).
1572 *1*	0	1.26 *25*	C^{2}S: 1.1 *3* (2015MuZY). b_f=19.9% *8* (2021JoZZ), b_f=18.6% *31*, σ_f^{sp}=16.1 mb, R_s=0.64 *13* (2015MuZY).
1990	3	0.11 *20*	C^{2}S: <0.2 (2015MuZY). b_f=1.41% *10* (2021JoZZ), b_f<2.9%, σ_f^{sp}=12.7 mb, R_s=0.69 *14* (2015MuZY).
2348 *1*	1	0.34 *1*	C^{2}S: 0.4 *1* (2015MuZY). b_f=4.33% *8* (2021JoZZ), b_f=5.3% *13*, σ_f^{sp}=13.1 mb, R_s=0.70 *14* (2015MuZY).
2717 *2*	2	0.56 *12*	C^{2}S: 0.6 *2* (2015MuZY). b_f=6.14% *7* (2021JoZZ), b_f=6.4% *16*, σ_f^{sp}=11.4 mb, R_s=0.62 *12* (2015MuZY).
2939 *2*	(2)	0.31 *6*	L: 2015MuZY reports L=0. C^{2}S: 0.4 *2* (2015MuZY). b_f=3.99% *3* (2021JoZZ), b_f=5.7% *16*, σ_f^{sp}=13.5 mb, R_s=0.61 *12* (2015MuZY).
3421 *2*	2	0.49 *11*	C^{2}S: 0.5 *1* (2015MuZY). b_f=5.08% *5* (2021JoZZ), b_f=5.1% *10*, σ_f^{sp}=11.0 mb, R_s=0.61 *12* (2015MuZY).
3595 *5*	(2)	0.21 *6*	C^{2}S: 0.3 *1* (2015MuZY). b_f=2.13% *5* (2021JoZZ), b_f=3.0% *7*, σ_f^{sp}=11.0 mb, R_s=0.60 *12* (2015MuZY).
3818 *5*	(3)	0.25 *7*	C^{2}S: 0.3 *1* (2015MuZY). b_f=0.49% *4* (2021JoZZ), b_f=4.0% *9*, σ_f^{sp}=11.8 mb, R_s=0.72 *14* (2015MuZY).
3890 *4*	(0)	0.29 *9*	C^{2}S: 0.15 *4* (2015MuZY). b_f=3.09% *6* (2021JoZZ), b_f=2.3% *5*, σ_f^{sp}=13.5 mb, R_s=0.72 *14* (2015MuZY). L: inconsistent with J^{π}=(5/2$^+$) in the Adopted Levels.
4025 *4*	(2)	0.34 *9*	C^{2}S: 0.6 *1* (2015MuZY). b_f=3.98% *5* (2021JoZZ), b_f=6.4% *10*, σ_f^{sp}=12.1 mb, R_s=0.60 *12* (2015MuZY).
4948 *5*			
5349 *8*			
5775 *4*			

† From 2021JoZZ based on measured Eγ.

Continued on next page (footnotes at end of table)

^{9}Be(^{36}S,^{35}Sγ) 2015MuZY,2021JoZZ (continued)

^{35}S Levels (continued)

‡ From 2021JoZZ. L-transfers are deduced by comparing the measured and eikonal-calculated parallel momentum distributions of residuals. Parentheses around L values are added by evaluators due to poor fits.

\# From 2021JoZZ. The spectroscopic factor $C^2S=b_f\sigma^{exp}_{inc}/R_s\sigma^{sp}_f$, where b_f is the population fraction, σ^{exp}_{inc} is the measured total inclusive cross section, R_s is the quenching factor from a systematics study 2014To14, and σ^{sp}_f is the eikonal model calculated single-particle cross section.

γ(^{35}S)

E_γ †	E_i(level)	E_f
778	2348	1572
1309	4025	2717
1542	3890	2348
1572	1572	0
1830	3818	1990
1990‡	1990	0
2036	4025	1990
2348	2348	0
2717	2717	0
2939	2939	0
3421	3421	0
3595	3595	0
4948	4948	0
5349	5349	0
5775	5775	0

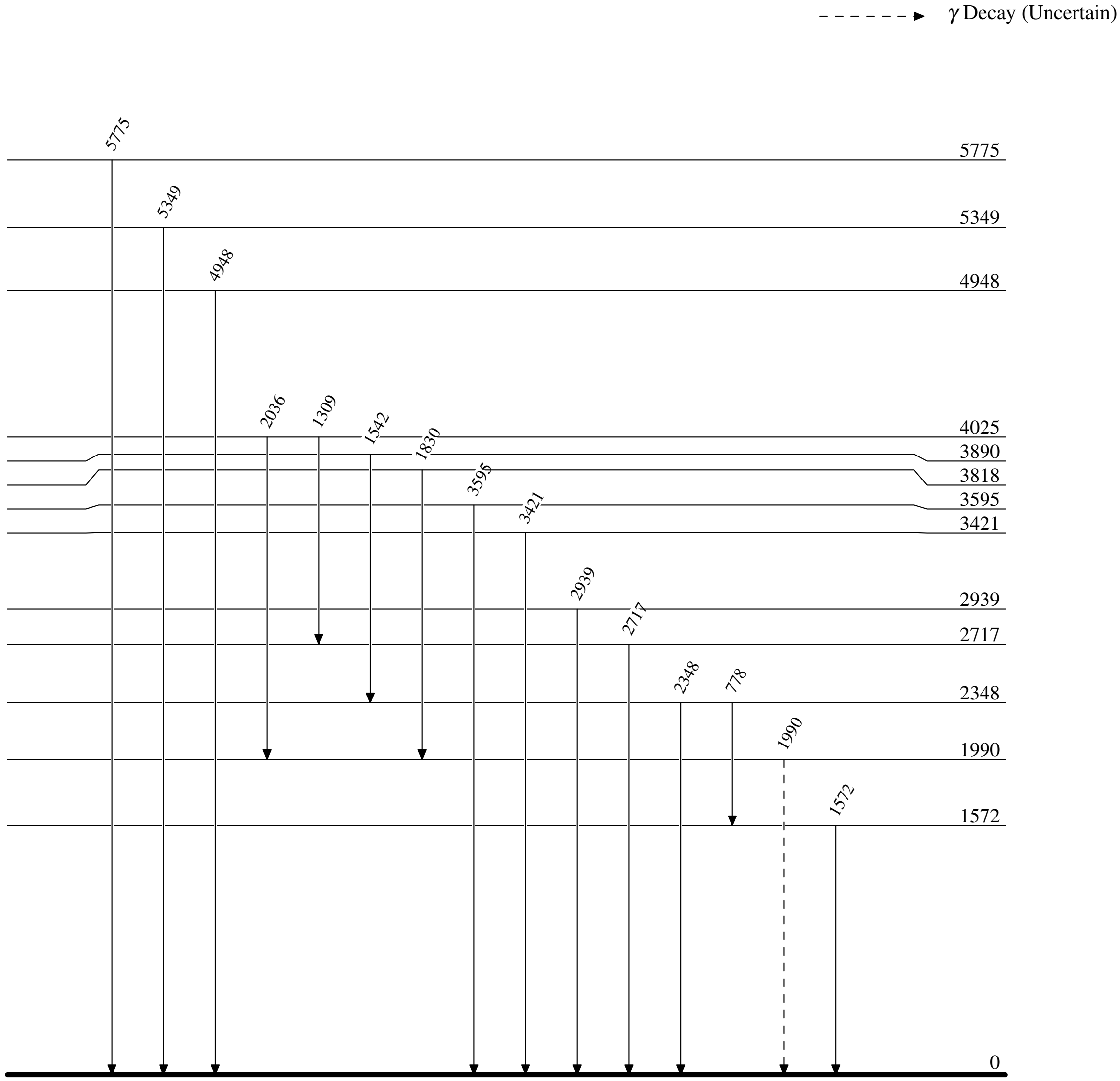
† From 2021JoZZ level scheme. γ-ray energy uncertainties are not given.

‡ Placement of transition in the level scheme is uncertain.

^{9}Be(^{36}S,^{35}Sγ) 2015MuZY,2021JoZZ

Level Scheme

Legend



$^{35}_{16}S_{19}$

^{24}Mg(^{14}N,3pγ) 2014Ay01

2014Ay01: ^{24}Mg(^{14}N,3p)^{35}S fusion-evaporation reaction. A 40-MeV, 5-pnA ^{14}N beam was delivered from the LNL XTU-Tandem accelerator and impinged on 1-mg/cm^2 thick, 99.7% isotopically enriched ^{24}Mg evaporated on an 8-mg/cm^2 gold layer. γ rays were detected using 4π-GASP array composed of 40 Compton-suppressed large volume HPGe detectors arranged in seven rings at different angles with respect to beam axis. Events were collected when at least two germanium detectors fired in coincidence. Measured Eγ, Iγ, $\gamma\gamma$-coin, $\gamma(\theta)$, $\gamma\gamma(\theta)$(ADO). Deduced levels, J, π, level lifetimes via the Doppler shift attenuation method (DSAM). Compared with large-scale shell model calculations.

^{35}S Levels

E(level)[†]	J$^\pi$[‡]	E(level)[†]	J$^\pi$[‡]	T$_{1/2}$[#]	E(level)[†]	J$^\pi$[‡]	T$_{1/2}$[#]
0.0[&]	3/2$^+$	3816.0[@] *3*	9/2$^-$	0.28 ps *3*	5412.2 *3*	9/2$^+$	
1572.5 *3*	1/2$^+$	3886.4 *4*	5/2$^+$		5877.6[&] *3*	11/2$^+$	
1991.4[@] *2*	7/2$^-$	4023.2[@] *3*	11/2$^-$	0.32 ps *3*	6299.4 *4*	11/2$^+$	
2347.9 *3*	3/2$^-$	4822.9 *3*	9/2$^+$		6352.2[@] *5*	13/2$^-$	50 fs *10*
2717.1 *3*	5/2$^+$	4899.7 *4*	9/2$^+$		7179.8[&] *3*	15/2$^+$	3.1 ps *12*
3594.7[&] *3*	7/2$^+$	5010.2 *3*	11/2$^-$	0.45 ps *8*	8023.7[&] *9*	17/2$^+$	0.15 ps *4*

† From a least-squares fit to γ-ray energies.
‡ As given in 2014Ay01, which quotes the known J^π assignments for four low-lying levels at 1572, 1991, 2347, and 2717 from 2011Ch48. 2014Ay01 assigns J^π for levels above 3 MeV based on the measured $\gamma\gamma(\theta)$(ADO) ratios. When considered in the Adopted Levels, the firm assignments here are placed within parentheses if there are no other strong arguments to support these firm assignments.
# From DSAM line-shape analyses (2014Ay01).
@ Seq.(A): ΔJ=1 sequence based on 7/2$^-$.
& Seq.(B): Sequence based on 3/2$^+$.

γ(^{35}S)

R$_{ADO}$=[Iγ(34°)+Iγ(146°)]/2Iγ(90°). Expected values are 0.8 for stretched dipole (ΔJ=1) and 1.4 for stretched quadrupole (ΔJ=2) or unstretched dipole (ΔJ=0) transitions.

E$_\gamma$	I$_\gamma$	E$_i$(level)	J$^\pi_i$	E$_f$	J$^\pi_f$	Mult.[†]	Comments
207.0 *5*	0.4 *2*	4023.2	11/2$^-$	3816.0	9/2$^-$		
465.3 *3*	1.2 *1*	5877.6	11/2$^+$	5412.2	9/2$^+$	D	R$_{ADO}$=0.71 *9*.
775.4 *4*	1.4 *2*	2347.9	3/2$^-$	1572.5	1/2$^+$	D	R$_{ADO}$=0.68 *14*.
827.8 *6*	0.3 *1*	7179.8	15/2$^+$	6352.2	13/2$^-$		
843.9 *8*	1.6 *2*	8023.7	17/2$^+$	7179.8	15/2$^+$	D	R$_{ADO}$=0.8 *3*.
867.3 *3*	1.9 *1*	5877.6	11/2$^+$	5010.2	11/2$^-$	D	R$_{ADO}$=1.36 *20*, consistent with ΔJ=0.
877.6 *5*	0.4 *1*	3594.7	7/2$^+$	2717.1	5/2$^+$		
880.2 *4*	0.8 *1*	7179.8	15/2$^+$	6299.4	11/2$^+$		
887.0 *7*	0.4 *1*	6299.4	11/2$^+$	5412.2	9/2$^+$	D	R$_{ADO}$=0.6 *3*.
977.8 *3*	1.6 *1*	5877.6	11/2$^+$	4899.7	9/2$^+$	D+Q	R$_{ADO}$=1.43 *16*.
986.9 *3*	4.4 *2*	5010.2	11/2$^-$	4023.2	11/2$^-$	D	B(E1)(W.u.)=0.00102 *+22−16*; B(M1)(W.u.)=0.036 *+8−6* R$_{ADO}$=1.30 *13*, consistent with ΔJ=0.
1006.8 *2*	0.8 *1*	4822.9	9/2$^+$	3816.0	9/2$^-$	D	R$_{ADO}$=1.37 *11*, consistent with ΔJ=0.
1054.5 *3*	3.0 *2*	5877.6	11/2$^+$	4822.9	9/2$^+$	D	R$_{ADO}$=0.72 *11*.
1228.1 *3*	1.1 *1*	4822.9	9/2$^+$	3594.7	7/2$^+$	D	R$_{ADO}$=0.78 *9*.
1302.2 *2*	13.5 *8*	7179.8	15/2$^+$	5877.6	11/2$^+$	E2	B(E2)(W.u.)=6.6 *+42−19* R$_{ADO}$=1.40 *5*, A$_2$=+0.27 *5*, A$_4$=−0.07 *5*.
1304.8 *12*	0.6 *2*	4899.7	9/2$^+$	3594.7	7/2$^+$		
1389.0 *4*	3.9 *4*	5412.2	9/2$^+$	4023.2	11/2$^-$	D	R$_{ADO}$=0.79 *11*.

Continued on next page (footnotes at end of table)

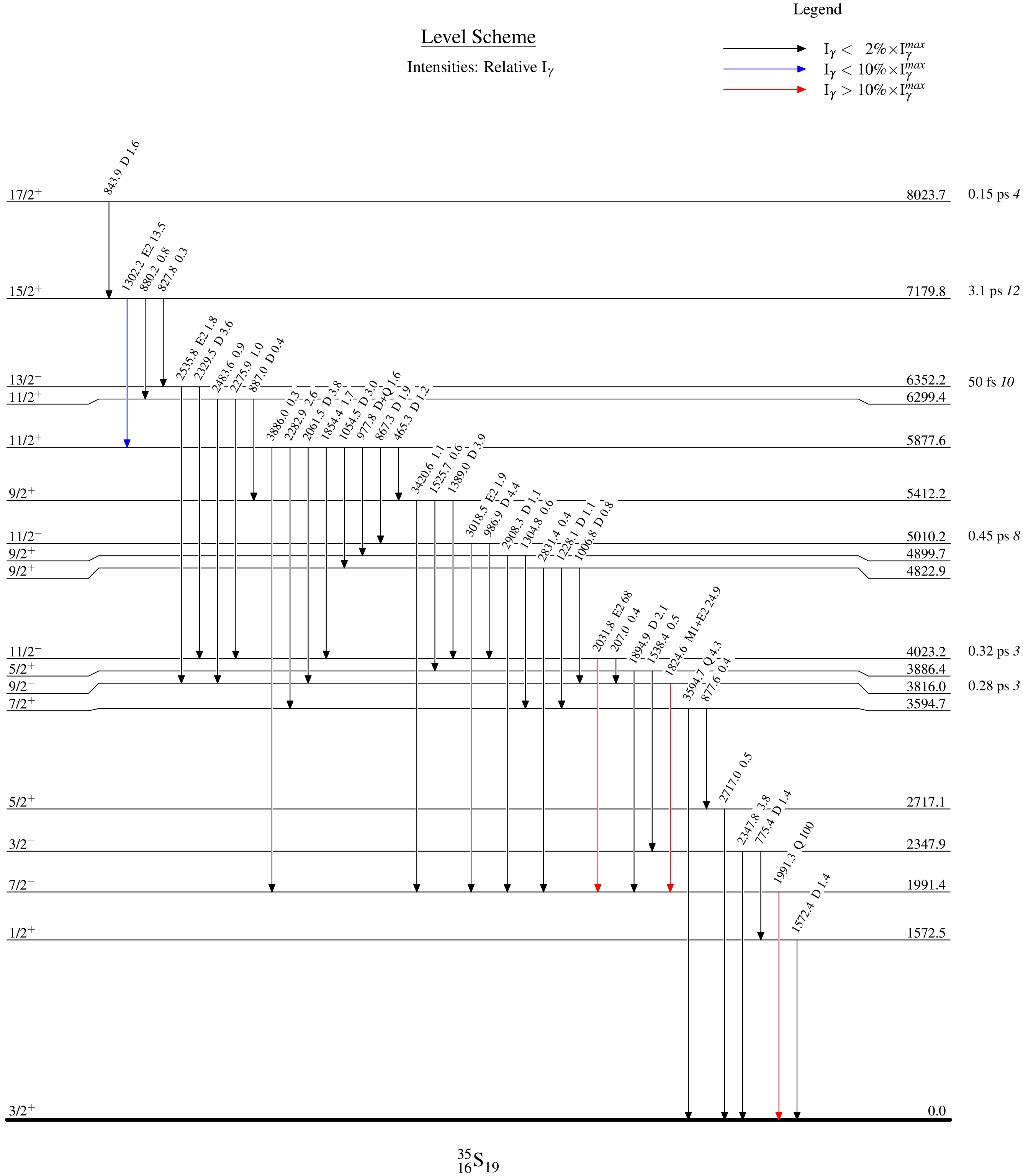
^{24}Mg(^{14}N,3pγ) 2014Ay01 (continued)

γ(^{35}S) (continued)

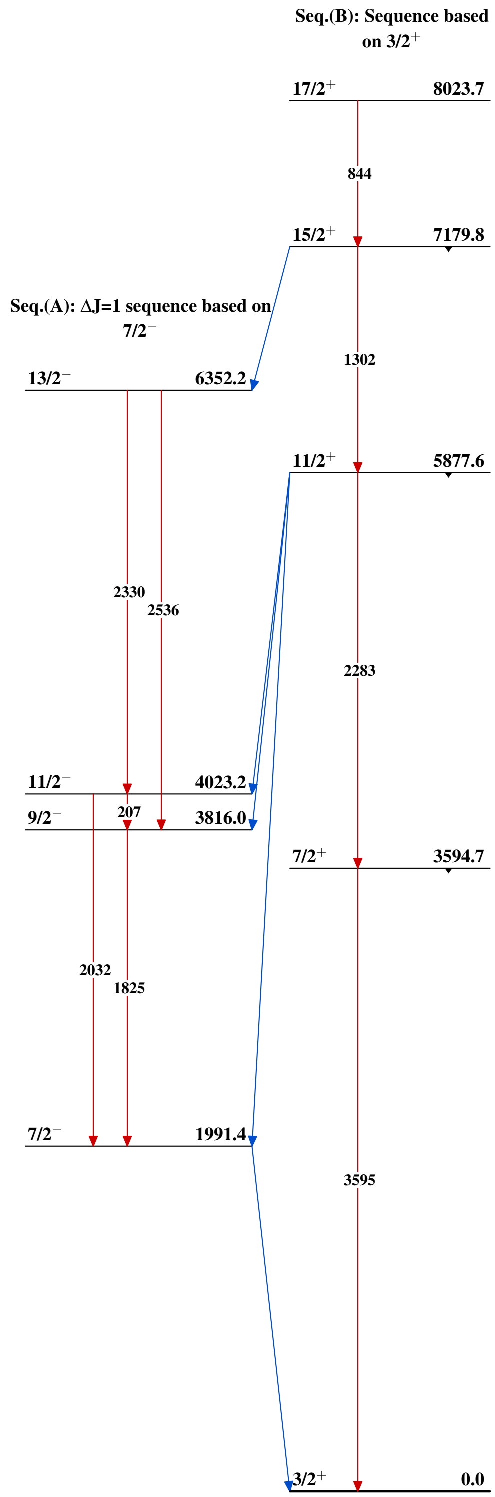
E_γ	I_γ	E_i(level)	J_i^π	E_f	J_f^π	Mult.†	δ†	Comments
1525.7 *8*	0.6 *1*	5412.2	9/2$^+$	3886.4	5/2$^+$			
1538.4 *8*	0.5 *2*	3886.4	5/2$^+$	2347.9	3/2$^-$			
1572.4 *3*	1.4 *4*	1572.5	1/2$^+$	0.0	3/2$^+$	D		R_{ADO}=0.88 *10*.
1824.6 *2*	24.9 *15*	3816.0	9/2$^-$	1991.4	7/2$^-$	M1+E2	+0.55 *9*	B(M1)(W.u.)=0.0099 *+14−12*; B(E2)(W.u.)=3.4 *+10−9* Mult.,δ: D+Q and δ from $\gamma(\theta)$ in 2014Ay01; E1+M2 ruled out by RUL. R_{ADO}=1.58 *5* indicating ΔJ=2 or 0 seems questionable considering ΔJ=1 from level scheme. A_2=+0.48 *4*, A_4=+0.08 *6*.
1854.4 *4*	1.7 *1*	5877.6	11/2$^+$	4023.2	11/2$^-$			
1894.9 *4*	2.1 *3*	3886.4	5/2$^+$	1991.4	7/2$^-$	D		R_{ADO}=0.69 *10*.
1991.3 *2*	100 *6*	1991.4	7/2$^-$	0.0	3/2$^+$	Q		R_{ADO}=1.45 *15*.
2031.8 *3*	68 *4*	4023.2	11/2$^-$	1991.4	7/2$^-$	E2		B(E2)(W.u.)=7.5 *+8−7* R_{ADO}=1.41 *10*, A_2=+0.29 *4*, A_4=−0.10 *6*.
2061.5 *3*	3.8 *2*	5877.6	11/2$^+$	3816.0	9/2$^-$	D		R_{ADO}=0.65 *9*.
2275.9 *6*	1.0 *2*	6299.4	11/2$^+$	4023.2	11/2$^-$			
2282.9 *4*	2.6 *1*	5877.6	11/2$^+$	3594.7	7/2$^+$			
2329.5 *9*	3.6 *7*	6352.2	13/2$^-$	4023.2	11/2$^-$	D		B(E1)(W.u.)=6.7×10^{-4} *+17−13*; B(M1)(W.u.)=0.023 *+6−5* R_{ADO}=0.8 *3*.
2347.8 *3*	3.8 *3*	2347.9	3/2$^-$	0.0	3/2$^+$			
2483.6 *8*	0.9 *2*	6299.4	11/2$^+$	3816.0	9/2$^-$			
2535.8 *11*	1.8 *2*	6352.2	13/2$^-$	3816.0	9/2$^-$	E2		B(E2)(W.u.)=5.3 *+17−11* R_{ADO}=1.5 *5*.
2717.0 *4*	0.5 *3*	2717.1	5/2$^+$	0.0	3/2$^+$			
2831.4 *9*	0.4 *1*	4822.9	9/2$^+$	1991.4	7/2$^-$			
2908.3 *5*	1.1 *2*	4899.7	9/2$^+$	1991.4	7/2$^-$	D		R_{ADO}=0.74 *14*.
3018.5 *7*	1.9 *2*	5010.2	11/2$^-$	1991.4	7/2$^-$	E2		B(E2)(W.u.)=0.223 *+49−39* R_{ADO}=1.35 *34*.
3420.6 *9*	1.1 *1*	5412.2	9/2$^+$	1991.4	7/2$^-$			
3594.7 *7*	4.3 *4*	3594.7	7/2$^+$	0.0	3/2$^+$	Q		R_{ADO}=1.36 *16*.
3886.0 *13*	0.3 *1*	5877.6	11/2$^+$	1991.4	7/2$^-$			

† From $\gamma(\theta)$, $\gamma\gamma(\theta)$(ADO) data, and RUL when $T_{1/2}$ is available in 2014Ay01.

^{24}Mg(^{14}N,3pγ) 2014Ay01



^{24}Mg(^{14}N,3pγ) 2014Ay01



$^{35}_{16}S_{19}$

^{26}Mg(^{18}O,2αnγ) 2021Go09

2021Go09: ^{26}Mg(^{18}O,2αn)^{35}S fusion-evaporation reaction. Exp 1 determined the beam energy for producing high-spin states of ^{35}S: an 80-MeV ^{18}O beam was delivered from the JAEA tandem accelerator facility and impinged on a 0.5-mg/cm^2 thick, self-supporting metallic enriched foil of ^{26}Mg. γ rays were detected using the GEMINI-II array of 14 Ge detectors with BGO Compton-suppressor shields. The reaction channel was selected by detecting the evaporated charged particles with a 4π silicon detector. Measured Eγ, Iγ, αγγ-coin, γγ(θ)(ADO). Exp 2 used the same ^{18}O beam energy, the same target, and the same charged-particle detectors as Exp 1. The beam was delivered from the ALTO-Tandem accelerator facility at IPN Orsay. γ rays were detected using the ORGAM array of 13 Ge detectors based on the EUROGAM array with the BGO scintillator for the anti-Compton suppressor. Compared with large-scale shell-model calculations with the SDPF-MSD4 interaction. Also see 2015GoZY, 2014GoZZ, 2013GoZW, and 2012IdZZ.

^{35}S Levels

E(level)†	Jπ‡	$T_{1/2}$	Comments
0.0#	3/2+		
1990.6# *4*	7/2−		
3594.2# *7*	(7/2+)		
3815.0 *6*	(9/2−)		
4023.1 *5*	(11/2−)		
4822.4 *6*	(9/2+)		
4899.6 *7*	(9/2+)		
5009.8 *6*	(11/2−)		
5412.4 *6*	(9/2+)		
5878.0# *6*	(11/2+)		
7179.6# *7*	(15/2+)		
8023.9# *9*	(17/2+)		
8755.7# *13*	(17/2)	<1 ps	$T_{1/2}$: estimated by 2021Go09 from residual Doppler energy shifts observed for the 732γ and 1576γ peaks.
10048.4# *10*	(19/2)		
12469.1# *15*	(21/2)		

† From a least-squares fit to γ-ray energies.
‡ As given in 2021Go09, which quotes the J^π assignments for levels below 8.5 MeV from 2014Ay01 but adds parentheses. 2021Go09 assigns J^π for levels above 8.5 MeV based on the measured γγ(θ)(ADO) ratios.
# Seq.(A): γ cascade based on the g.s.

γ(^{35}S)

$R^{\gamma 1}_{ADO}$=Iγ1(47° gated by γ_2 at all angles)/Iγ1(86° gated by γ_2 at all angles). Expected values are ≈0.8 for dipole (ΔJ=1) and ≈1.0 for stretched quadrupole (ΔJ=2) transitions.

E_γ†	I_γ†	E_i(level)	J^π_i	E_f	J^π_f	Mult.‡	Comments
465.5 *2*	14 *1*	5878.0	(11/2+)	5412.4	(9/2+)	D	R_{ADO}=0.67 *11*.
731.6 *15*	8 *3*	8755.7	(17/2)	8023.9	(17/2+)		
844.3 *6*	20 *3*	8023.9	(17/2+)	7179.6	(15/2+)	D	E_γ: 8443(6) in 2021Go09 is likely a misprint. R_{ADO}=0.51 *8*.
868.2 *4*	20 *2*	5878.0	(11/2+)	5009.8	(11/2−)	D	R_{ADO}=0.54 *21*.
978.4 *10*	26 *7*	5878.0	(11/2+)	4899.6	(9/2+)		
986.6 *3*	26 *1*	5009.8	(11/2−)	4023.1	(11/2−)		R_{ADO}=0.91 *13* consistent with ΔJ=(0).
1007.2 *11*	4 *2*	4822.4	(9/2+)	3815.0	(9/2−)	(D)	R_{ADO}=0.87 *15*.
1055.6 *4*	39 *10*	5878.0	(11/2+)	4822.4	(9/2+)	D	R_{ADO}=0.67 *18*.

Continued on next page (footnotes at end of table)

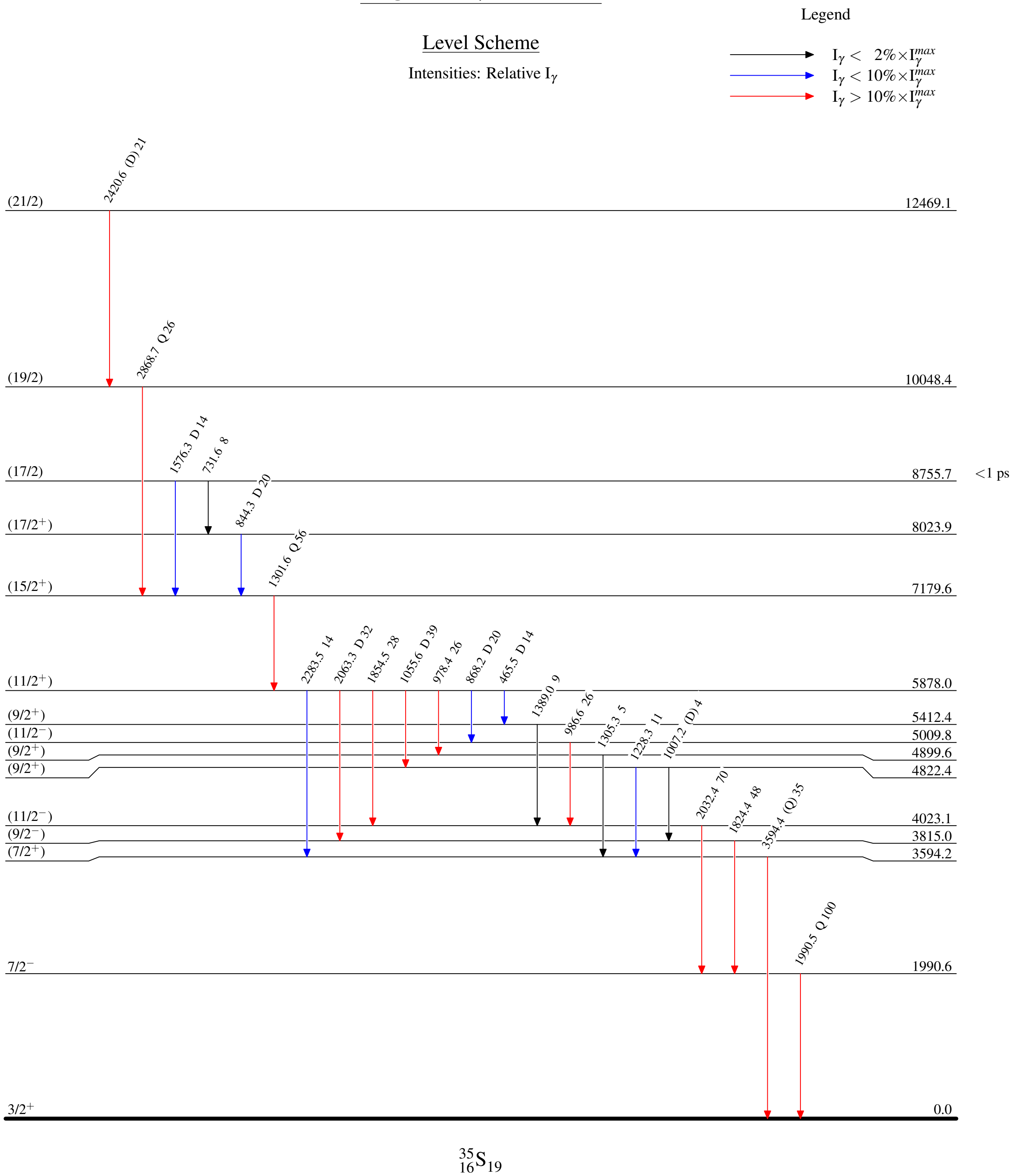
^{26}Mg(^{18}O,2αnγ) 2021Go09 (continued)

γ(^{35}S) (continued)

E_γ †	I_γ †	E_i(level)	J_i^π	E_f	J_f^π	Mult. ‡	Comments
1228.3 *6*	11 *2*	4822.4	(9/2$^+$)	3594.2	(7/2$^+$)		
1301.6 *3*	56 *11*	7179.6	(15/2$^+$)	5878.0	(11/2$^+$)	Q	R_{ADO}=1.40 *17*.
1305.3 *4*	5 *3*	4899.6	(9/2$^+$)	3594.2	(7/2$^+$)		
1389.0 *8*	9 *2*	5412.4	(9/2$^+$)	4023.1	(11/2$^-$)		
1576.3 *14*	14 *3*	8755.7	(17/2)	7179.6	(15/2$^+$)	D	R_{ADO}=0.43 *9*. 2021Go09 Table I lists this ΔJ=1 ADO ratio for the 731.6γ, which however is a (17/2) to (17/2$^+$) transition with ΔJ=0. 2021Go09 repeatedly states the 8756 level has J=(17/2) and therefore ~~Evaluators~~ reassign this ΔJ=1 ADO ratio to the 1576.3γ, corresponding to a (17/2) to (15/2$^+$) transition.
1824.4 *6*	48 *2*	3815.0	(9/2$^-$)	1990.6	7/2$^-$		
1854.5 *42*	28 *14*	5878.0	(11/2$^+$)	4023.1	(11/2$^-$)		
1990.5 *4*	100	1990.6	7/2$^-$	0.0	3/2$^+$	Q	R_{ADO}=1.30 *8*.
2032.4 *4*	70 *16*	4023.1	(11/2$^-$)	1990.6	7/2$^-$		
2063.3 *14*	32 *6*	5878.0	(11/2$^+$)	3815.0	(9/2$^-$)	D	R_{ADO}=0.58 *41*.
2283.5 *10*	14 *3*	5878.0	(11/2$^+$)	3594.2	(7/2$^+$)		
2420.6 *11*	21 *6*	12469.1	(21/2)	10048.4	(19/2)	(D)	R_{ADO}=0.50 *42*.
2868.7 *8*	26 *2*	10048.4	(19/2)	7179.6	(15/2$^+$)	Q	R_{ADO}=0.98 *13*.
3594.4 *11*	35 *2*	3594.2	(7/2$^+$)	0.0	3/2$^+$	(Q)	R_{ADO}=0.93 *20*.

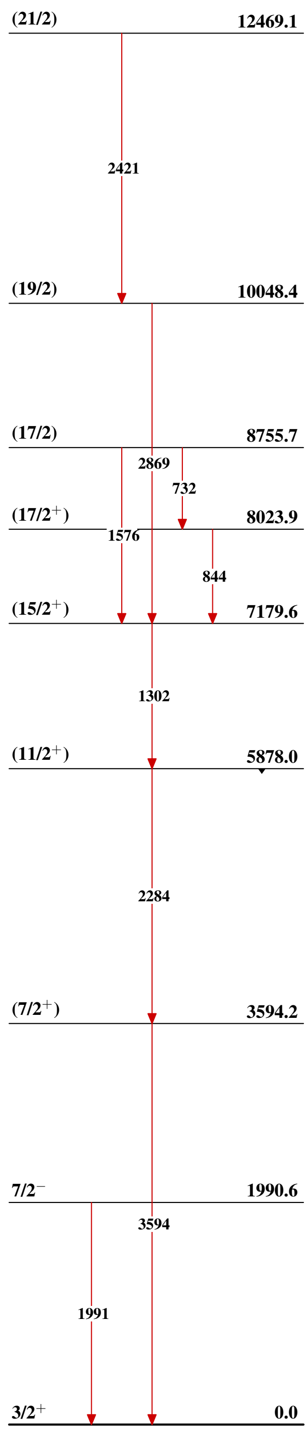
† From 2021Go09.
‡ From $\gamma\gamma(\theta)$(ADO) data in 2021Go09.

$^{26}Mg(^{18}O,2\alpha n\gamma)$ 2021Go09



^{26}Mg(^{18}O,2αnγ) 2021Go09

Seq.(A): γ cascade based on the g.s



$^{35}_{16}S_{19}$

^{34}S(n,γ) E=thermal 1985Ra15

1985Ra15: 6×10^{11} n/cm^2/s thermal neutrons at the target position were produced from the Los Alamos Omega West Reactor at a nominal 8-MW reactor power. γ rays following thermal neutron capture by a 1.09-g, 77.7% enriched ^{34}S target were detected using a 26 cm^3 coaxial Ge(Li) detector inside a 20-cm-diam by 30-cm-long NaI(Tl) annulus with FWHM=2.3 keV at 1 MeV, 5.5 keV at 6 MeV and 8.8 keV at 11 MeV. Measured σ_γ, Eγ, and Iγ for 61 γ-ray transitions. Deduced 22 levels and γ-ray branching ratios.

1972Dz13: Thermal neutrons were produced from the thermal column of the IRT reactor of the Baghdad Nuclear Research Institute. γ rays following thermal neutron capture by a 1.5-g, 97%-enriched ^{34}S target were detected using a 10-cm^3 Ge(Li) detector with FWHM=3.5 keV at 1 MeV and 8 keV at 7 MeV. Measured Eγ and Iγ for 22 γ-ray transitions. Deduced 11 levels.

1985Ke08: 5×10^{12} n/cm^2/s thermal neutrons were produced from the McMaster University Reactor. γ rays following thermal neutron capture by a natural sulfur target were detected using a 10% efficient Ge centered within a NaI(Tl) annulus with FWHM=3.2-4.4 keV at 4-7 MeV. Measured Eγ and Iγ for 13 γ-ray transitions. Deduced 9 levels.

1997Be42: Thermal neutrons were provided from the reactor BR1 of the Studiecentrum voor kernenergie in Mol, Belgium. γ rays following thermal neutron capture by a 313.6-mg, 93.26%-enriched ^{34}S and 5.933%-enriched ^{36}S target were detected using a Ge detector. Measured σ_γ, Eγ, and Iγ for 11 γ-ray transitions. Deduced 8 levels. Compared with direct-capture (DC) model calculations.

All 55 Eγ values from 2007ChZX: database of prompt gamma activation analysis (PGAA) from neutron capture are in agreement with the data below. Others: 2000Re01, 2001ZhZZ.

^{35}S Levels

E(level)†	Comments
0	
1572.373 *7*	
1991.28 *4*	
2347.782 *8*	C^{2}S=0.48 for 3/2$^-$ from 1997Be42.
2717.05 *9*	
2938.63 *5*	
3558.082 *27*	
3801.955 *15*	C^{2}S=0.08 for 3/2$^-$ from 1997Be42.
4105.6 *8*	
4189.243 *33*	C^{2}S=0.15 for 1/2$^-$ from 1997Be42.
4477.62 *7*	
4903.374 *15*	C^{2}S=0.49 for 1/2$^-$ from 1997Be42.
4963.084 *25*	C^{2}S=0.19 for 3/2$^-$ from 1997Be42.
5752.5 *8*	
5841.485 *31*	This level is introduced in S. Raman to P.M. Endt private communication (mentioned in the Adopted Levels of 2011Ch48). Populated by 1144.6γ from 6986 level and deexites by 4268.6γ to 1572 level.
6018.8 *6*	
6078.477 *32*	
6293.93 *6*	
6354.89 *23*	
6419.9 *11*	
6629.43 *9*	
6761.0 *12*	
(6986.099 *24*)	

† From a least-squares fit to γ-ray energies.

^{34}S(n,γ) E=thermal 1985Ra15 (continued)

γ(^{35}S)

Photons per 100 thermal neutron captures are reported in 1972Dz13 and 1985Ke08. γ-ray cross sections in mb are reported in 1985Ra15, multiplying these by 0.340 yields the number of photons per 100 thermal neutron captures. σ(4638)=163 mb *15*. Iγ(4638)=σ(4638)×0.340=55 *5*.

γ-ray intensities per incident neutron for all transitions and γ-ray cross sections in mb for primary transitions are reported in 1997Be42. The strongest 4638γ is used for normalization. σ(4638)=148.6 mb *55* and Iγ(4638)=σ(1997Be42)/σ(1985Ra15)×Iγ(1985Ra15)=50 *7*.

E_γ[†]	I_γ[†#]	E_i(level)	E_f	Comments
356.66[@] *9*	0.037[@] *4*	2347.782	1991.28	
356.66[@] *9*	0.037[@] *4*	(6986.099)	6629.43	
368.5 *4*	0.020 *7*	2717.05	2347.782	
619.23 *19*	0.042 *8*	3558.082	2938.63	
631.32[@] *24*	0.060[@] *9*	4189.243	3558.082	
631.32[@] *24*	0.060[@] *9*	(6986.099)	6354.89	
[x]646.9[‡] *5*	0.7 *1*			E_γ,I_γ: from 1972Dz13. 1972Dz13 assigned this γ ray as the 6986 to 6339 transition. This γ ray and its final level 6339 have not been observed in any other ^{34}S(n,γ) measurements. Evaluators consider this γ ray questionable.
[x]663.41[‡] *7*	0.09 *1*			
692.16 *5*	0.133 *17*	(6986.099)	6293.93	
775.398 *6*	17.2 *19*	2347.782	1572.373	E_γ: other: 775.50 *15* (1972Dz13). I_γ: weighted average of 19.5 *20* (1972Dz13) and 15.6 *17* (1985Ra15).
[x]803.81[‡] *9*	0.092 *14*			
863.28 *28*	0.034 *10*	3801.955	2938.63	
907.608[‡] *23*	0.56 *10*	(6986.099)	6078.477	E_γ: weighted average of 908.1 *3* (1972Dz13) and 907.607 *14* (1985Ra15). I_γ: weighted average of 0.5 *1* (1972Dz13) and 0.61 *10* (1985Ra15).
1084.79 *15*	0.054 *10*	3801.955	2717.05	
1101.92 *31*	0.033 *8*	4903.374	3801.955	
1144.591 *20*	0.55 *6*	(6986.099)	5841.485	The placement of this γ ray is changed from 2717 to 1572 (1985Ra15) to 6986 to 5841 (S. Raman to P.M. Endt private communication).
1161.05 *20*	0.055 *8*	4963.084	3801.955	
1210.28 *4*	0.252 *27*	3558.082	2347.782	
1250.61 *5*	0.214 *24*	4189.243	2938.63	
[x]1381.67[‡] *24*	0.024 *8*			
1404.967 *24*	0.60 *6*	4963.084	3558.082	
1454.09 *4*	0.45 *11*	3801.955	2347.782	
1566.7 *3*	0.47 *9*	3558.082	1991.28	
1572.333 *8*	35.0 *34*	1572.373	0	E_γ: weighted average of 1572.15 *16* (1972Dz13) and 1572.333 *7* (1985Ra15). I_γ: others: 39 *4* (1972Dz13) and 32 *4* (1997Be42).
1760.55 *11*	0.146 *21*	4477.62	2717.05	
1840.2 *12*	2.2 *4*	4189.243	2347.782	E_γ: unweighted average of 1839 *1* (1972Dz13) and 1841.426 *15* (1985Ra15). I_γ: weighted average of 3.6 *8* (1972Dz13) and 2.14 *21* (1985Ra15).
1964.8 *2*	0.37 *10*	4903.374	2938.63	
1991.27 *5*	0.54 *6*	1991.28	0	
2022.954 *9*	11 *1*	(6986.099)	4963.084	E_γ: others: 2022.80 *20* (1972Dz13) and 2023.0 *10* (1985Ke08). I_γ: weighted average of 12.0 *15* (1972Dz13), 11 *1* (1985Ke08), 11.4 *10* (1985Ra15), and 9.1 *12* (1997Be42).
2082.83 *17*	15.9 *15*	(6986.099)	4903.374	E_γ: unweighted average of 2082.65 *16* (1972Dz13), 2083.17 *7* (1985Ke08), and 2082.681 *12* (1985Ra15).

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^{34}S(n,γ) E=thermal 1985Ra15 (continued)

γ(^{35}S) (continued)

E_γ [†]	I_γ [†#]	E_i(level)	E_f	Comments
				I_γ: weighted average of 18.0 *15* (1972Dz13), 17 *2* (1985Ke08), 15.6 *17* (1985Ra15), and 12.6 *17* (1997Be42).
[x]2184.16[‡] *19*	4.7 *5*			E_γ,I_γ: from 1985Ke08. 1985Ke08 assigned this γ ray as a 4903 to 2719 transition. The final level is closest to the known 2717 level. This γ ray has not been observed in any other ^{34}S(n,γ) measurements. Including this γ ray in the level scheme would result in a significant intensity imbalance at its initial level 4903 and its final level 2717 and also lead to poor fitting in the level scheme. Evaluators therefore consider the placement of this γ ray unlikely.
2229.510 *16*	2.86 *27*	3801.955	1572.373	
2347.700 *15*	49 *4*	2347.782	0	E_γ: weighted average of 2347.50 *10* (1972Dz13), 2347.71 *3* (1985Ke08), and 2347.701 *11* (1985Ra15). I_γ: weighted average of 54 *5* (1972Dz13), 49 *4* (1985Ke08), 50 *5* (1985Ra15), and 40 *6* (1997Be42).
2508.39 *8*	0.38 *4*	(6986.099)	4477.62	
2555.492 *14*	3.14 *31*	4903.374	2347.782	E_γ: other: 2555.9 *4* (1972Dz13). I_γ: weighted average of 3.4 *7* (1972Dz13) and 3.09 *31* (1985Ra15).
2615.2 *2*	1.09 *10*	4963.084	2347.782	E_γ: other: 2614.3 *3* doublet (1972Dz13). I_γ: other: <1.3 *4* (1972Dz13).
2616.8 *13*	0.24 *7*	4189.243	1572.373	E_γ: other: 2614.3 *3* doublet (1972Dz13). I_γ: other: <1.3 *4* (1972Dz13).
2716.99 *16*	0.30 *5*	2717.05	0	
2796.74 *4*	4.9 *4*	(6986.099)	4189.243	E_γ: weighted average of 2796.83 *13* (1985Ke08) and 2796.73 *4* (1985Ra15). Other: 2796.6 *5* (1972Dz13). I_γ: weighted average of 6.1 *8* (1972Dz13), 4.3 *4* (1985Ke08), 5.4 *5* (1985Ra15), and 4.9 *7* (1997Be42).
2905.1 *4*	0.14 *4*	4477.62	1572.373	
2938.58 *11*	0.89 *9*	2938.63	0	
2972.0 *4*	0.18 *6*	4963.084	1991.28	
3139.9 *5*	0.078 *24*	6078.477	2938.63	
3183.95 *4*	6.1 *6*	(6986.099)	3801.955	E_γ: weighted average of 3184.30 *20* (1972Dz13), 3183.94 *7* (1985Ke08), and 3183.94 *4* (1985Ra15). I_γ: weighted average of 7.0 *9* (1972Dz13), 5.7 *6* (1985Ke08), 6.2 *6* (1985Ra15), and 5.7 *8* (1997Be42).
3330.84 *13*	7.6 *7*	4903.374	1572.373	E_γ: unweighted average of 3331.08 *15* (1972Dz13), 3330.64 *6* (1985Ke08), and 3330.80 *4* (1985Ra15). I_γ: weighted average of 7.8 *10* (1972Dz13), 7.7 *8* (1985Ke08), and 7.4 *7* (1985Ra15).
3390.62 *7*	5.6 *5*	4963.084	1572.373	E_γ: weighted average of 3390.86 *20* (1972Dz13), 3390.73 *7* (1985Ke08), and 3390.55 *5* (1985Ra15). I_γ: weighted average of 5.5 *8* (1972Dz13), 5.7 *6* (1985Ke08), and 5.5 *5* (1985Ra15).
3558.1 *5*	0.085 *17*	3558.082	0	
3801.73 *4*	2.59 *23*	3801.955	0	E_γ: weighted average of 3802.0 *3* (1972Dz13), 3801.49 *14* (1985Ke08), and 3801.74 *3* (1985Ra15). I_γ: weighted average of 3.6 *7* (1972Dz13), 2.3 *2* (1985Ke08), 3.09 *31* (1985Ra15), and 2.6 *4* (1997Be42).
4105.3 *8*	0.048 *17*	4105.6	0	
4188.95 *5*	2.54 *27*	4189.243	0	E_γ: weighted average of 4188.80 *13* (1985Ke08) and 4188.96 *4* (1985Ra15). Other: 4189.6 *3* (1972Dz13). I_γ: weighted average of 1.9 *4* (1972Dz13), 2.72 *27* (1985Ra15), and 2.8 *4* (1997Be42). Other: 24 *2* (1985Ke08).
4268.65 *14*	0.34 *4*	5841.485	1572.373	E_γ: other: 4268.4 *7* (assigned as 6986 to 2717 in 1972Dz13). The placement of 4268γ is changed from 6986 to 2717 (1985Ra15) to 5841 to 1572 (S. Raman to P.M. Endt, private communication, mentioned in the

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^{34}S(n,γ) E=thermal 1985Ra15 (continued)

γ(^{35}S) (continued)

E_γ†	I_γ†#	E_i(level)	E_f	Comments
				Adopted Levels of 2011Ch48). I_γ: other: 0.4 *2* (1972Dz13).
4638.14 *14*	56 *5*	(6986.099)	2347.782	E_γ: unweighted average of 4638.4 *2* (1972Dz13), 4638.10 *3* (1985Ke08), and 4637.91 *4* (1985Ra15). I_γ: weighted average of 56 *6* (1972Dz13), 59 *5* (1985Ke08), 55 *5* (1985Ra15), and 50 *7* (1997Be42).
4902.99 *5*	4.0 *4*	4903.374	0	E_γ: weighted average of 4903.07 *7* (1985Ke08) and 4902.96 *4* (1985Ra15). Other: 4903.4 *3* (1972Dz13). I_γ: weighted average of 3.2 *8* (1972Dz13), 4.3 *4* (1985Ke08), 3.7 *4* (1985Ra15), and 4.2 *6* (1997Be42).
4963.06 *22*	2.62 *27*	4963.084	0	E_γ: unweighted average of 4963.2 *5* (1972Dz13), 4963.34 *10* (1985Ke08), and 4962.63 *5* (1985Ra15). I_γ: weighted average of 2.6 *8* (1972Dz13), 2.92 *27* (1985Ra15), and 2.18 *33* (1997Be42). Other: 29 *3* (1985Ke08).
5752.0 *8*	0.018 *5*	5752.5	0	
6018.2 *6*	0.020 *7*	6018.8	0	
6077.87 *11*	0.35 *4*	6078.477	0	E_γ: other: 6078.2 *10* (1972Dz13). I_γ: weighted average of 0.3 *1* (1972Dz13) and 0.36 *4* (1985Ra15).
6293.2 *4*	0.065 *14*	6293.93	0	
6355.0 *6*	0.046 *7*	6354.89	0	
6419.3 *11*	0.016 *5*	6419.9	0	
6628.5 *6*	0.030 *6*	6629.43	0	
6760.3 *12*	0.019 *8*	6761.0	0	
6985.7 *10*	0.036 *8*	(6986.099)	0	

† From 1985Ra15, unless otherwise noted.
‡ Not observed in 2007ChZX (PGAA).
\# Intensity per 100 neutron captures.
@ Multiply placed with undivided intensity.
^{x} γ ray not placed in level scheme.

^{34}S(n,γ) E=thermal 1985Ra15

Level Scheme

Intensities: I_γ per 100 neutron captures

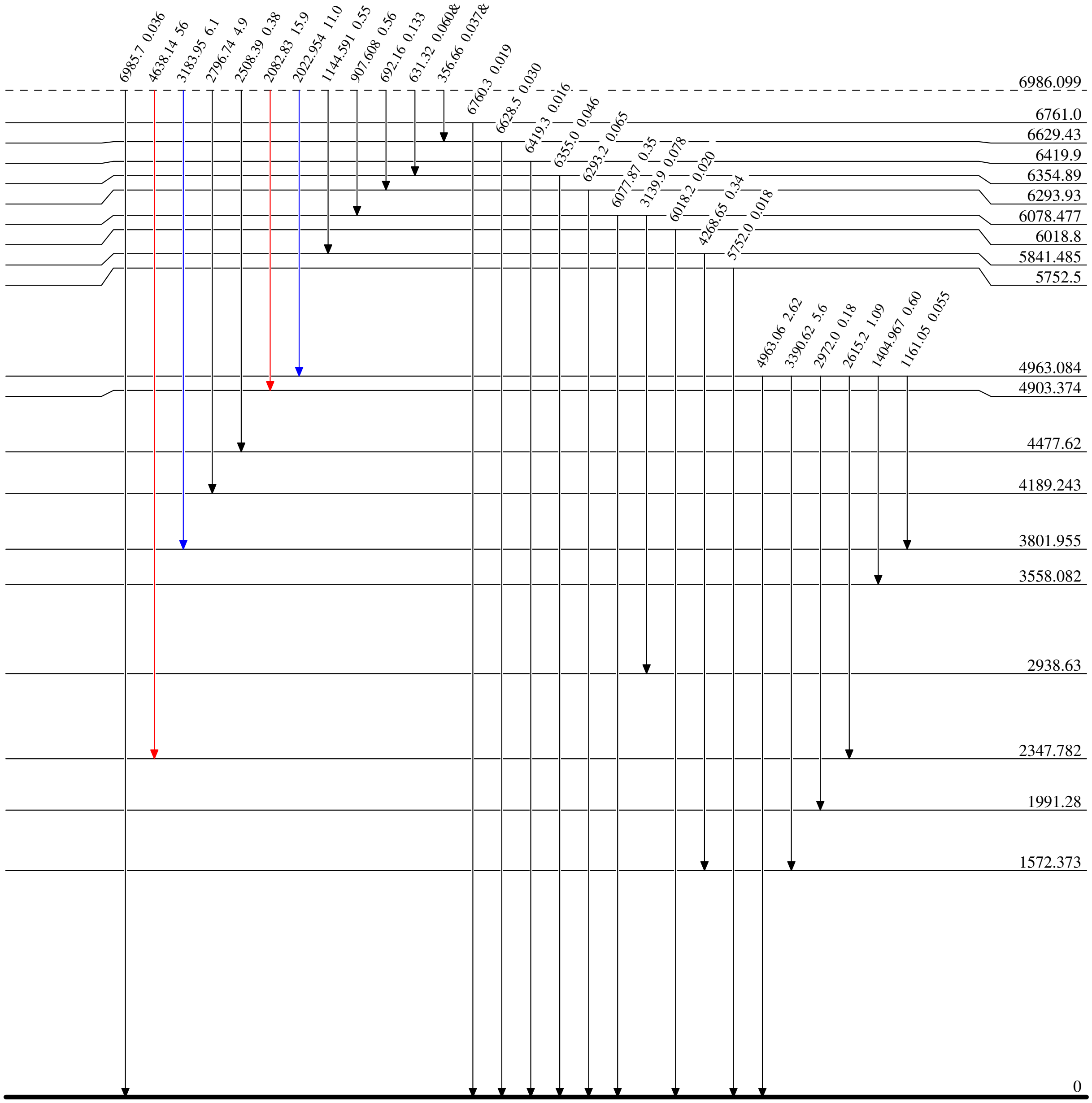
& Multiply placed: undivided intensity given

Legend

$I_\gamma < 2\% \times I_\gamma^{max}$

$I_\gamma < 10\% \times I_\gamma^{max}$

$I_\gamma > 10\% \times I_\gamma^{max}$



$^{35}_{16}S_{19}$

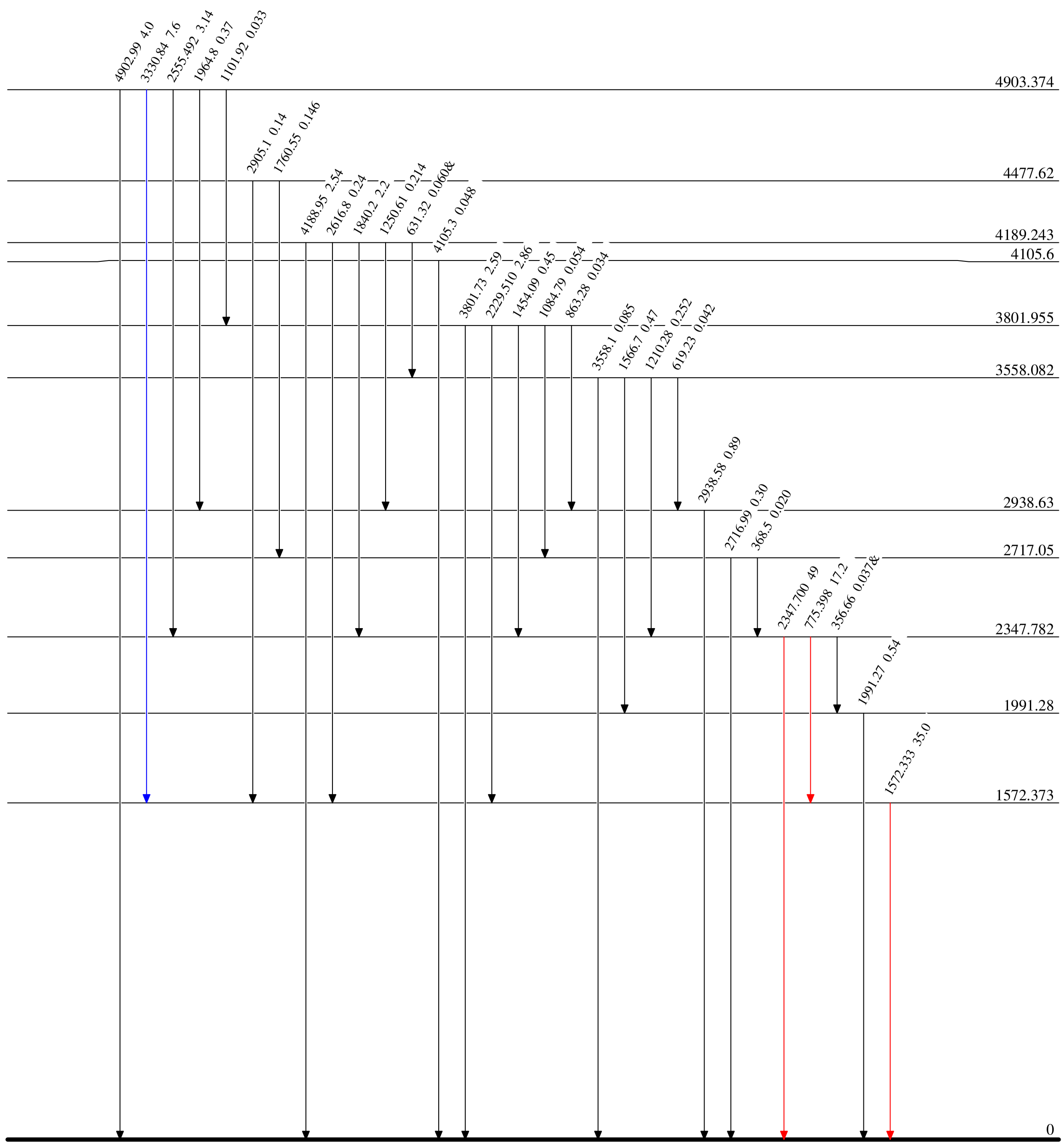
^{34}S(n,γ) E=thermal 1985Ra15

Level Scheme (continued)

Intensities: I_γ per 100 neutron captures
& Multiply placed: undivided intensity given

Legend

$I_\gamma < 2\% \times I_\gamma^{max}$
$I_\gamma < 10\% \times I_\gamma^{max}$
$I_\gamma > 10\% \times I_\gamma^{max}$



$^{35}_{16}S_{19}$

^{34}S(n,n),(n,γ):resonances 1984Ca14

1984Ca14: 30-1500-keV neutrons were produced from the Oak Ridge Electron Linear Accelerator (ORELA). The target was 94.33%-enriched ^{34}S with a 5.54% impurity of ^{32}S. Neutrons were detected using a 2-cm-thick and 7.5-cm in diameter NE110 proton-recoil detector. Prompt γ rays from neutron capture were detected using two nonhydrogenous fluorocarbon scintillators (NE-226). Measured $\sigma(E_n)$. Deduced 81 resonances, J, π, L, widths and spectroscopic factors by comparing the R-matrix predictions of the cross section with the data.

2018MuZY: Atlas of Neutron Resonances. Evaluation of neutron resonance energies, spins, width parameters, and resonance strengths for nuclei of Z=1-60.

^{35}S Levels

J^π, L, and resonance parameters E_n, $g\Gamma_n$, Γ_γ, and $g\Gamma_n\Gamma_\gamma/\Gamma$ are from 1984Ca14 due to significant discrepancies, including orders of magnitude inconsistencies, between the 1984Ca14 original data and 2018MuZY evaluation. $g\Gamma_n=(2J+1)\Gamma_n/2$.

E(level)†	J^π	$g\Gamma_n\Gamma_\gamma/\Gamma$	L	E_n(lab) (keV)	Comments
6985.54?			0	−0.305	E(level): fictitious level from analysis of resonance spectrum below the S(n) threshold. Γ_γ=0.38 eV.
7018.89 *4*		0.025 eV *1*	2	34.03 *1*	
7072.66 *5*		0.14 eV *4*		89.40 *3*	
7074.33 *6*		0.084 eV *3*		91.12 *4*	
7097.72 *30*		0.009 eV *3*	2	115.2 *3*	
7099.85 *20*		0.033 eV *5*	2	117.4 *2*	$g\Gamma_n$<1 eV.
7100.73 *4*	3/2$^-$			118.30 *1*	$g\Gamma_n$=372 eV *3*; Γ_γ=0.70 eV *4*.
7143.27 *11*		0.42 eV *1*		162.1 *1*	$g\Gamma_n$<1 eV.
7210.42 *5*			1,2	231.25 *2*	$g\Gamma_n$=62 eV *4*; Γ_γ=0.30 eV *2*.
7218.0 *4*		0.33 eV *2*		239.0 *4*	$g\Gamma_n$<4 eV.
7234.0 *5*		0.19 eV *1*		255.5 *5*	$g\Gamma_n$<4 eV.
7239.8 *6*		0.06 eV *2*		261.5 *6*	$g\Gamma_n$<2 eV.
7253.2 *6*		0.35 eV *2*		275.3 *6*	$g\Gamma_n$<1 eV.
7275.93 *5*	1/2$^+$			298.70 *3*	$g\Gamma_n$=9600 eV *150*; Γ_γ=2.82 eV *67*.
7279.13?		0.10 eV *3*		302	
7289.8 *8*		0.31 eV *3*		313.0 *8*	$g\Gamma_n$<6 eV.
7294.19 *6*	3/2$^-$			317.51 *5*	$g\Gamma_n$=2500 eV *25*; Γ_γ=0.23 eV *8*.
7331.00 *5*	1/2$^+$			355.41 *2*	$g\Gamma_n$=4810 eV *80*; Γ_γ=0.30 eV *30*.
7338.18 *20*			1,2	362.8 *2*	$g\Gamma_n$=230 eV *10*; Γ_γ=1.06 eV *7*.
7343.91 *20*		0.72 eV *3*		368.7 *2*	$g\Gamma_n$<2 eV.
7347.2 *5*		0.44 eV *6*		372.1 *5*	$g\Gamma_n$<2 eV.
7347.9 *5*		0.30 eV *5*		372.8 *5*	$g\Gamma_n$<2 eV.
7353.9 *5*		0.44 eV *4*		379.0 *5*	$g\Gamma_n$<10 eV.
7357.6 *5*		0.16 eV *4*		382.8 *5*	$g\Gamma_n$<6 eV.
7370.56 *6*	1/2$^-$			396.14 *4*	$g\Gamma_n$=6430 eV *150*; Γ_γ=3.08 eV *20*.
7371.97?		0.12 eV		397.6	
7396.0 *6*		0.44 eV *33*	2	422.3 *6*	
7404.6 *6*		0.21 eV *5*		431.2 *6*	$g\Gamma_n$<1 eV.
7408.68 *11*	3/2$^-$			435.4 *1*	$g\Gamma_n$=1570 eV *40*; Γ_γ=0.91 eV *8*.
7411.55 *6*				438.35 *5*	$g\Gamma_n$=49 eV *5*.
7416.50 *4*				443.45 *1*	$g\Gamma_n$=80 eV *8*; Γ_γ=0.51 eV *5*.
7429.1 *6*		0.12 eV *3*		456.4 *6*	$g\Gamma_n$<6 eV; Γ_γ=0.12 eV *3*.
7433.56 *5*				461.01 *2*	$g\Gamma_n$=50 eV *5*; Γ_γ=0.035 eV *20*.
7436.4 *4*			1,2	463.9 *4*	$g\Gamma_n$=296 eV *30*; Γ_γ=0.28 eV *3*.
7442.14 *5*	1/2$^+$			469.85 *2*	$g\Gamma_n$=1050 eV *35*; Γ_γ=0.37 eV *8*.
7462.2 *6*		0.33 eV *4*		490.5 *6*	$g\Gamma_n$=50 eV *10*.
7481.15 *6*	1/2$^-$			510.02 *4*	$g\Gamma_n$=375 eV *25*; Γ_γ=2.1 eV *1*.
7494.5 *6*		0.17 eV *4*		523.8 *6*	$g\Gamma_n$<4 eV.
7542.3 *6*		0.21 eV *3*		573.0 *6*	$g\Gamma_n$<1 eV.

Continued on next page (footnotes at end of table)

^{34}S(n,n),(n,γ):resonances 1984Ca14 (continued)

^{35}S Levels (continued)

E(level)†	J^π	$g\Gamma_n\Gamma_\gamma/\Gamma$	L	E_n(lab) (keV)	Comments
7604.0 *6*		0.90 eV *8*		636.5 *6*	$g\Gamma_n$<10 eV.
7609.26 *5*	3/2−			641.93 *3*	$g\Gamma_n$=2520 eV *60*; Γ_γ=0.51 eV *6*.
7648.9 *6*		0.21 eV *3*		682.7 *6*	$g\Gamma_n$<30 eV.
7655.7 *6*		0.80 eV *5*		689.7 *6*	$g\Gamma_n$<10 eV.
7663.91 *11*			1,2	698.2 *1*	$g\Gamma_n$=305 eV *50*; Γ_γ=0.80 eV *8*.
7678.3 *7*		0.45 eV *7*		713.0 *7*	$g\Gamma_n$<4 eV.
7731.0 *7*		0.31 eV *13*		767.3 *7*	$g\Gamma_n$<4 eV.
7749.7 *4*				786.5 *4*	$g\Gamma_n$=18 eV *6*; Γ_γ=2.65 eV *18*.
7761.46 *7*	3/2−			798.65 *6*	$g\Gamma_n$=25350 eV *300*; Γ_γ=1.52 eV *50*.
7776.17 *11*	1/2−			813.8 *1*	$g\Gamma_n$=860 eV *60*; Γ_γ=0.19 eV *13*.
7797.99 *9*	1/2+			836.27 *8*	$g\Gamma_n$=3600 eV *160*; Γ_γ=0.39 eV *18*.
7812.24 *7*	3/2+			850.94 *6*	$g\Gamma_n$=2650 eV *110*; Γ_γ=0.24 eV *19*.
7853.25 *7*	3/2−			893.17 *6*	$g\Gamma_n$=4950 eV *160*; Γ_γ=0.38 eV *16*.
7861.8 *12*		1.13 eV *19*		902.0 *12*	$g\Gamma_n$<3 eV.
7880.8 *12*		0.67 eV *17*		921.5 *12*	$g\Gamma_n$<20 eV.
7889.0 *12*		0.57 eV *25*		930.0 *12*	$g\Gamma_n$<20 eV.
7894.63 *7*	3/2−			935.78 *6*	$g\Gamma_n$=4100 eV *160*; Γ_γ=3.48 eV *23*.
7899.70 *30*				941.0 *3*	$g\Gamma_n$=176 eV *16*; Γ_γ=0.5 eV *3*.
7934.1 *15*		0.5 eV *3*		976.4 *15*	$g\Gamma_n$<2 eV.
7939.5 *15*		0.89 eV *4*		982.0 *15*	$g\Gamma_n$<8 eV.
7954.96 *11*	3/2−			997.9 *1*	$g\Gamma_n$=930 eV *80*; Γ_γ=3.51 eV *19*.
7974.19 *11*	3/2−			1017.7 *1*	$g\Gamma_n$=4020 eV *180*; Γ_γ=1.22 eV *17*.
8019.5 *2*			2	1064.4 *2*	$g\Gamma_n$=1032 eV *90*.
8041.20 *20*			2	1086.7 *2*	$g\Gamma_n$=2650 eV *160*.
8076.46 *20*			2	1123.0 *2*	$g\Gamma_n$=5480 eV *388*.
8077.91 *30*			2	1124.5 *3*	$g\Gamma_n$=4730 eV *320*.
8093.0 *5*	1/2−			1140.0 *5*	$g\Gamma_n$=34880 eV *830*.
8143.66 *20*			2	1192.2 *2*	$g\Gamma_n$=5510 eV *290*.
8187.17 *30*	1/2−			1237.0 *3*	$g\Gamma_n$=2945 eV *230*.
8211.25 *30*				1261.8 *3*	$g\Gamma_n$=975 eV *86*.
8224.2 *4*	1/2−			1275.1 *4*	$g\Gamma_n$=2170 eV *180*.
8228.3 *4*			2	1279.4 *4*	$g\Gamma_n$=1140 eV *100*.
8243.98 *30*			2	1295.5 *3*	$g\Gamma_n$=1980 eV *150*.
8256.31 *20*	3/2−			1308.2 *2*	$g\Gamma_n$=12475 eV *770*.
8262.4 *4*	5/2+			1314.5 *4*	$g\Gamma_n$=6240 eV *500*.
8273.3 *4*	(1/2−)			1325.7 *4*	$g\Gamma_n$=1635 eV *145*.
8298.27 *30*			2	1351.4 *3*	$g\Gamma_n$=14470 eV *650*.
8333.8 *4*	3/2−			1388.0 *4*	$g\Gamma_n$=28845 eV *1057*.
8335.9 *4*			2	1390.1 *4*	$g\Gamma_n$=1955 eV *170*.
8391.3 *4*	3/2−			1447.2 *4*	$g\Gamma_n$=27100 eV *1390*.
8393.64 *30*			2	1449.6 *3*	$g\Gamma_n$=3025 eV *260*.
8406.55 *30*			2	1462.9 *3*	$g\Gamma_n$=3885 eV *345*.
8418.8 *4*			2	1475.5 *4*	$g\Gamma_n$=3795 eV *340*.

† From $E_{c.m.}$+S(n)(^{35}S) where S(n)(^{35}S)=6985.84 *4* (2021Wa16) and $E_{c.m.}$=E_n(lab)×m(^{34}S)/[m_n+m(^{34}S)].

^{34}S(pol d,p) 1977Ab07

$J^{\pi}=0^{+}$ for ^{34}S g.s.

1977Ab07: A vector-polarized 11.8-MeV deuteron beam was produced from the University of Wisconsin Lamb-shift polarized ion source and tandem accelerator. The target was a 90% enriched ^{34}S, 610-μg/cm^2 thick, evaporated onto a 100-μg/cm^2 gold foil and covered by a 50-μg/cm^2 gold layer. The reaction products were detected using four ΔE-E counter telescopes of freon-cooled surface-barrier and Si(Li) detectors with FWHM≈60 keV. Measured angular distributions of cross sections $\sigma(E_p,\theta)$ and angular distributions of analyzing powers $iT_{11}(E_p,\theta)$. Deduced levels, J, π, L-transfers, and spectroscopic factors from finite-range, non-locality corrected DWUCK-DWBA analysis of $\sigma(E_p,\theta)$ and $iT_{11}(E_p,\theta)$. Also see 1976AbZX.

^{35}S Levels

Spectroscopic factor $C^2S=\sigma(\theta)_{exp}/\sigma(\theta)_{DWBA}/N$, where N=1.53 is a normalization factor (1977Ab07).

E(level)†	J^{π}	L@	C^2S@	Comments
0	3/2+‡	2	0.56,0.48‡	
1572	1/2+#	0	0.27,0.18#	
1992	7/2−#	3	1.16,0.73#	
2348	3/2−#	1	0.56,0.46#	
2718	5/2+#	2	0.03,0.02#	
2939	3/2+,5/2+	2	0.10,0.06	J^{π}: 1977Ab07 states that the analyzing power seems to favor 5/2+ but they do not think that the evidence is strong enough to support an unambiguous spin-parity assignment. C^2S: extracted from an incoherent superposition of direct and compound cross sections in 1977Ab07.
3421	5/2+#	2	0.04,0.03#	
3563				
3802	3/2−#	1	0.10,0.08#	
3818				
4190	1/2−‡	1	0.15,0.13‡	
4482	7/2−#	3	0.06,0.05#	
4575				
4837				
4904	1/2−‡	1	0.48,0.41‡	
4963	3/2−#	1	0.19,0.17#	
5058	7/2−#	3	0.03,0.03#	
5126				

† As given in 1977Ab07, originally from 1973EnVA rounded to the nearest integer.
‡ L-1/2 transfer from analyzing power measurements.
# L+1/2 transfer from analyzing power measurements.
@ From DWUCK-DWBA analysis of the measured $\sigma(\theta)$ in 1977Ab07. C^2S calculations use two sets of optical model potentials. Set 1: the deuteron elastic potential and the Becchetti-Greenlees proton parameters; Set 2: Adiabatic model parameters and the Becchetti-Greenlees proton parameters.

^{34}S(d,p) 2024Ku24,1971Va18,1969Mo12

$J^{\pi}=0^+$ for ^{34}S g.s.

2024Ku24: An 8-MeV/nucleon deuteron beam was produced from the 9-MV Super-FN tandem Van de Graaff accelerator at FSU. The target was carbon-backed sulfur enriched in ^{34}S. Protons were spatially dispersed by the Super-Enge Split-Pole Spectrograph and detected using a position-sensitive ionization chamber (ΔE) and a plastic scintillator (E) at the focal plane with FWHM≈45 keV. An additional ^{34}S(d,pγ)^{35}S measurement using the CeBr$_3$ Array (CeBrA) demonstrator provides information on some overlapping peaks produced from the ^{32}S contaminants in the target. Measured σ(E$_p$,θ). Deduced 23 levels, single-neutron spectroscopic factors and strengths from finite-range PTOLEMY-DWBA analysis.

1971Va18: A 10-MeV deuteron beam was produced from the Utrecht 6-MV tandem accelerator. Targets were 100 μg/cm^2 PbS on carbon plus formvar foils and 5 μg/cm^2 pure ^{34}S embedded in aluminum foil. Protons were detected using one 15-mm long and seven 30-mm long, 0.6-mm thick position-sensitive detectors (PSD) in the focal plane of the Utrecht split-pole magnetic spectrograph. Measured σ(E$_p$,θ). Deduced 17 levels, L-transfers, J, π, spectroscopic factors from DWUCK-DWBA analysis for the ground state and seven excited states of ^{35}S.

1969Mo12: A 6.495-MeV deuteron beam was produced from the ONR-CIT tandem accelerator. The target was an enriched target of 450-μg/cm^2 *25* CdS (85% ^{34}S) evaporated onto a 301-μg/cm^2 *15* gold foil and a natural target of 289-μg/cm^2 *30* Sb^{2}S^3 evaporated onto a 289-μg/cm^2 *30* gold foil. Protons were detected using an array of 16 Au-Si surface barrier detectors with FWHM=30 keV. Measured σ(E$_p$,θ). Deduced 37 levels.

1984Pi03: A 12.3-MeV deuteron beam was produced from a cyclotron. The target was CdS (61.9% in ^{36}S, 37.7% in ^{34}S), 100 μg/cm^2 on a 20 μg/cm^2 carbon backing. Protons were analyzed by a multi-angle magnetic spectrograph and detected using 700-mm long nuclear emulsion plates Ilford L4. Measured σ(E$_p$,θ). Deduced 13 levels, L-transfers, J, π, and spectroscopic factors from DWUCK-DWBA analysis for six excited states of ^{35}S.

1979So01: A 3.55-MeV deuteron beam was produced from a Van de Graaff electrostatic generator. The target was 18-μg/cm^2 GeS (80% ^{36}S, 20% ^{34}S) on a 25-μg/cm^2 carbon backing. Protons were analyzed by a magnetic spectrograph and recorded by a spark chamber with FWHM=10-15 keV. Measured σ(E$_p$,θ). Deduced levels.

1971Ko33: A 6.6-MeV deuteron beam was produced from a cyclotron. Targets were GeS films of 70-80 μg/cm^2 (98% enrichment in ^{34}S) evaporated onto a 20 μg/cm^2 carbon substrate. Protons were detected using an 800-μm-thick Si(Li). Measured σ(E$_p$,θ). Deduced levels, J, π, L-transfers, and spectroscopic factors from distorted-wave analysis for the ground state and five excited states of ^{35}S.

1971Me12: An 18-MeV deuteron beam was produced from the Yale MP tandem Van de Graaff accelerator. The target was H^{2}S gas of natural isotopic composition (95.06% ^{32}S, 4.18% ^{34}S). Protons were detected using a ΔE-E telescope of silicon surface-barrier detectors. Measured σ(E$_p$,θ). Deduced levels, J, π, L-transfers, and spectroscopic factors from local, zero-range JULIE-DWBA analysis of σ(E$_p$,θ) for the ground state and two excited states of ^{35}S.

1966Sc09: 9- and 12-MeV deuteron beams were produced from the Argonne tandem accelerator. The target was PbS. Protons were detected using silicon surface barrier detectors. Measured σ(E$_p$,θ) for the ground state and one excited state of ^{35}S.

1958En51: 6.006- and 6.542-MeV deuteron beams were produced from the MIT-ONR electrostatic generator. Targets were Sb^{2}S^3 of natural isotopic constitution (4.2% ^{34}S). Energies of charged reaction products emitted from the target at angles of 50, 90, and 130° were measured by a broad-range magnetic spectrograph and nuclear emulsions. Measured σ(E$_p$,θ). Deduced 5 levels.

Other measurements: 1976We29: ^{34}S(d,p)^{35}S angular distributions for the ground state and the first excited state of ^{35}S.
1973Co25: ^{34}S(d,p)^{35}S excitation functions for the two lowest excited states of ^{35}S.

Theoretical calculations: 1977Os07, 1974Os02.

^{35}S Levels

C^{2}S from 2024Ku24. The normalization was determined from the L=3 strengths, G$_{L=3}$, summed up through E$_x$=5.1 MeV and divided by the expected 1f$_{7/2}$ vacancy, i.e., ΣG$_{L=3}$/(2j+1), where j=7/2. This is equivalent to ΣC^{2}S$_{L=3}$=1 up to 5.1 MeV. For C^{2}S given under comments, C^{2}S=$\sigma(\theta)_{exp}/\sigma(\theta)_{DWBA}$/N/(2J$_f$+1)×(2J$_i$+1), where the normalization factor N=1.53. The C^{2}S from 1971Me12 based on N=1.58 has been renormalized by evaluators using N=1.53.

E(level)	J$^{\pi}$[&]	L[a]	C^{2}S[a]	Comments
0	3/2$^+$	2	0.54 *12*	L: 2 also from 1971Va18, 1971Ko33, and 1971Me12. C^{2}S: others: 0.43 *11* (1971Va18), 0.33 (1971Ko33), and 0.48 (1971Me12).
1571.92[#] *19*	1/2$^+$	0	0.14 *7*	L: 0 also from 1971Va18, 1971Ko33, and 1984Pi03.

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^{34}S(d,p) 2024Ku24,1971Va18,1969Mo12 (continued)

^{35}S Levels (continued)

E(level)	J$^{\pi}$[&]	L[a]	C^{2}S[a]	Comments
				C^{2}S: others: 0.17 *4* (1971Va18), 0.25 (1971Ko33), and 0.154 (1984Pi03).
1991.08[#] *16*	7/2$^-$	3	0.87 *9*	C^{2}S: others: 0.68 *17* (1971Va18), 0.38 *8* (1971Ko33), 0.65 *10* (1971Me12), and 0.841 *12* (1984Pi03).
2347.59[#] *15*	3/2$^-$	1	0.62 *7*	E(level): others: 2347 *8* (1969Mo12) and 2336 *10* (1971Va18). L: 1 also from 1971Va18 and 1984Pi03. C^{2}S: others: 0.52 *13* (1971Va18), 0.52 (1971Me12), 0.33 (1971Ko33), and 0.505 (1984Pi03).
2724 *8*	5/2$^+$	2	0.02 *1*	E(level): weighted average of 2722 *8* (1969Mo12) and 2726 *8* (1971Va18). L: other: (2,3) from 1971Va18.
2941 *10*	3/2$^+$	2	<0.02	E(level): weighted average of 2943 *10* (1969Mo12) and 2939 *10* (1971Va18).
3420 *8*	5/2$^+$	2	0.02 *1*	E(level): weighted average of 3422 *8* (1969Mo12) and 3415 *12* (1971Va18).
3560 *8*				E(level): weighted average of 3563 *8* (1969Mo12) and 3555 *9* (1971Va18).
3596[†] *8*				E(level): other: 3595 *9* (1971Va18).
3675[‡] *10*	(1/2$^-$,3/2$^-$)	(1)		J$^{\pi}$: assigned from L=(1) and also adopted in the Adopted Levels. L: from 1971Va18. C^{2}S: <2.5×10^{-3} for J^{π}=1/2$^-$ and <1.25×10^{-3} for J^{π}=3/2$^-$ (1971Va18).
3801.90[#] *30*	3/2$^-$	1	0.07 *4*	E(level): others: 3804 *8* (1969Mo12) and 3795 *10* (1971Va18). L: 1 also from 1971Va18. C^{2}S: 0.09 *3* (1971Va18).
3885[†] *10*				E(level): other: tentative 3866 *10* (1971Va18) and 3907 *10* (1971Va18).
4025 *10*				E(level): weighted average of 4025 *10* (1969Mo12) and 4025 *10* (1971Va18).
4109 *10*				E(level): weighted average of 4112 *10* (1969Mo12) and 4105 *10* (1971Va18).
4189.87[#] *26*	1/2$^-$	1	0.17 *2*	E(level): others: 4190 *8* (1969Mo12) and 4196 *12* (1971Va18). C^{2}S: other: 0.12 *3* (1971Va18).
4305 *8*				E(level): weighted average of 4302 *8* (1969Mo12) and 4312 *12* (1971Va18).
4481[†] *8*	3/2$^+$,7/2$^-$	2+3	0.07,0.05	E(level): other: 4477 doublet (2024Ku24). J$^{\pi}$: for two components, as given in 2024Ku24. L: 2,3 doublet (2024Ku24). C^{2}S: others: 0.07 *2* for 3/2$^+$ component and 0.05 *2* for 7/2$^-$ component (2024Ku24).
4575[†] *8*				
4837[†] *8*				
4903.28[#] *16*	1/2$^-$	1	0.78 *8*	E(level): other: 4903 *8* (1969Mo12). L: 1 also from 1984Pi03. C^{2}S: other: 0.776 (1984Pi03).
4963.12[#] *16*	3/2$^-$	1	0.32 *3*	E(level): other: 4965 *8* (1969Mo12). L: 1 also from 1984Pi03. C^{2}S: other: 0.218 (1984Pi03).
5058[†] *8*	7/2$^-$	3	0.08 *1*	
5126[†] *11*				
5342[†] *8*				
5475?[†] *10*				Only observed in 1969Mo12 and not included in the Adopted Levels.
5542?[†] *8*				
5740[@] *20*	5/2$^-$,7/2$^-$	3	0.03 *1*	J$^{\pi}$: assigned from L=3 and also adopted in the Adopted Levels. 2024Ku24 assigns (5/2$^-$) and states that the J^{π} assignment is purely speculative.
5890[@] *20*	3/2$^+$,5/2$^+$	2	0.11 *2*	J$^{\pi}$: assigned from L=2 and also adopted in the Adopted Levels. 2024Ku24 assigns (3/2$^+$) and states that the J^{π} assignment is purely speculative.
5980[†] *10*				
6078.6[#] *13*	1/2$^-$,3/2$^-$	1		E(level): other: 6080 *8* (1969Mo12).

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^{34}S(d,p) 2024Ku24,1971Va18,1969Mo12 (continued)

^{35}S Levels (continued)

E(level)	J$^{\pi}$[&]	L[a]	Comments
			J$^{\pi}$: assigned from L=1 and also adopted in the Adopted Levels. L: from 1984Pi03. C^{2}S=0.086 for J^{π}=1/2$^-$ and 0.042 for J^{π}=3/2$^-$.
6292[†] *8*			
6334[†] *8*			
6344[†] *8*			
6446[†] *8*			
6496[†] *8*			
6537.7[#] *14*			
6545.1[#] *13*			E(level): other: 6543 *8* (1969Mo12).
6584[†] *10*			
6635.2[#] *13*		(2,3)	E(level): others: 6634 *8* (1969Mo12) and 6625 *30* (2024Ku24).
6677[†] *8*			
6891.3[#] *14*	(1/2 to 7/2)$^{(-)}$	(1,3)	E(level): others: 6892 *10* (1969Mo12) and 6915 *30* (2024Ku24).
7021 *10*		(2,3)	E(level): weighted average of 7022 *10* (1969Mo12) and 7005 *35* (2024Ku24).
7150[@] *35*		(0,3)	
7205[@] *35*		(1,2)	
7482.7[#] *13*		(1,2)	E(level): other: 7495 *40* (2024Ku24).
7570[@] *40*		(2,3)	
7640[@] *40*		(2,3)	

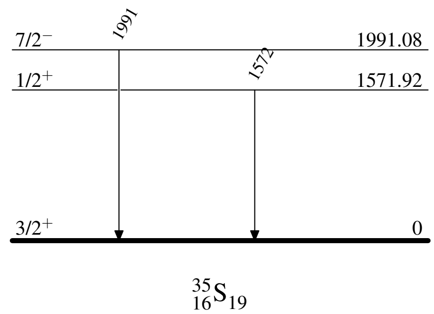
[†] From 1969Mo12.
[‡] From 1971Va18.
[#] From 1984Pi03.
[@] From 2024Ku24.
[&] Assumed for extracting C^{2}S, unless otherwise noted.
[a] From PTOLEMY-DWBA analysis of the measured $\sigma(\theta)$ in 2024Ku24, unless otherwise noted.

$\gamma(^{35}S)$

E_{γ}	E_i(level)	J_i^{π}	E_f	J_f^{π}	Comments
1572	1571.92	1/2$^+$	0	3/2$^+$	E_{γ}: from 2024Ku24.
1991	1991.08	7/2$^-$	0	3/2$^+$	E_{γ}: from 2024Ku24.

^{34}S(d,p) 2024Ku24,1971Va18,1969Mo12

Level Scheme



^{34}S(d,pγ) 1972Fr11,1975VaYG,1970Bu18

1972Fr11: 3.50- and 4.00-MeV deuteron beams were produced from the 4-MV Van de Graaff accelerator at the Nuclear Research Center of Strasbourg, France. Targets were antimony sulphide enriched to 85.6%, 550 $\mu g/cm^2$ on a 0.03 mm molybdenum layer and 150 $\mu g/cm^2$ on a 0.05 mm silver layer. Protons were detected using a surface-barrier annular silicon detector at 180°. γ rays were detected using a 40 cm^3 Ge(Li) detector at 0°, 55° and 90° with FWHM=3.2 keV at Eγ=1.332 MeV in the first experiment and a NaI crystal at 0°, 30°, 45°, 60° and 90° in the second experiment. Measured E_p, Eγ, Iγ, and pγ(θ)-coin. Deduced 12 levels, J, γ-ray branching ratios, and mixing ratios. Deduced $T_{1/2}$ for 7 levels using the Doppler-Shift Attenuation Method (DSAM).

1975VaYG,1972Va07: A 4.72-MeV deuteron beam of 175 nA was provided from the Groningen 5 MV Van de Graaff generator. Targets were 80-$\mu g/cm^2$ $Zn^{34}S$ enriched to 89% evaporated onto 10-$\mu g/cm^2$ Formvar plus 10-$\mu g/cm^2$ carbon. Protons were detected using a 2-mm annular silicon detector. γ rays were detected using a 7.6-cm×7.6-cm NaI(Tl) detector. Measured E_p, Eγ, Iγ, and pγ(θ)-coin. Deduced levels, J, γ-ray branching ratios, and mixing ratios. Also see 1971VaZG.

1970Bu18: 2-3.25 MeV deuteron beams were produced from the University of Arizona 5.5 MV Van de Graaff accelerator. Targets were 480-$\mu g/cm^2$ *120* Ag_2S made by heating 67% enriched ^{34}S powder and silver foil. Protons were detected using a 1-mm annular silicon surface barrier detector and γ rays were detected using a 7.62 cm×7.62 cm NaI(Tl) detector in the angular correlation experiment and a Ge(Li) detector in the lifetime experiment. Measured E_p, Eγ, Iγ, and pγ(θ)-coin. Deduced 3 levels, J, γ-ray branching ratios, and mixing ratios. Deduced $T_{1/2}$ for 3 levels using the Doppler-shift attenuation method (DSAM).

2024Co04: A 16-MeV deuteron beam from the John D. Fox Superconducting Linear Accelerator, FSU. Particles were momentum-analyzed by the Super-Enge Split-Pole Spectrograph and identified using a position-sensitive ionization chamber with two anode wires in the isobutane gas (ΔE) and a large plastic scintillator (E) at the focal plane. γ rays were detected using a $CeBr_3$ Array of five $CeBr_3$ detectors. Measured pγ delayed coincidence time spectra and deduced $T_{1/2}$ for the 1991 level of ^{35}S.

1971Pr11: A 4.48-MeV deuteron beam was produced from the insulated-core transformer tandem accelerator at the Aerospace Research Laboratories (ARL). The target was Ag_2S enriched to 85.61% in ^{34}S. Protons were detected using a 500 μm-thick, 50-mm^2 silicon detector and γ rays were detected using a 1-in.-thick, 1.5-in. diameter NE111 scintillator. Measured pγ-delayed coincidence time spectrum and deduced $T_{1/2}$ for the 1.99-MeV level.

1972Go31: A 3.3-MeV deuteron beam was produced from the BNL 3.5-MeV van de Graaff facility. The target was Sb_2S_3 enriched to 86% in ^{34}S. γ rays were detected using a Ge(Li) detector. Measured Eγ=1572.24 *17* deexciting the first excited state.

^{35}S Levels

E(level)[†]	J^π[‡]	$T_{1/2}$	Comments
0	$3/2^+$		
1572.33 *20*	$1/2^+$	>1.3 ps	E(level): weighted average of 1574.2 *10* (1970Bu18), 1572.3 *3* (1972Fr11), and 1572.28 *17* (1972Go31). J^π: spin=1/2 from isotropic γ(θ) in 1972Fr11 and 1972Va07. $T_{1/2}$: from τ>1.8 ps in 1970Bu18 by DSAM. Other: >1.1 ps (1972Fr11, τ>1.6 ps).
1992.9 *17*	$7/2^-$	1.03 ns *5*	E(level): unweighted average of 1994.6 *10* (1970Bu18) and 1991.2 *3* (1972Fr11). J^π: spin=5/2, 7/2 from γ(θ) in 1972Fr11, 1972Va07, and 1975VaYG. $T_{1/2}$: from τ=1.48 ns *7* from weighted average of 1.7 ns *3* (2024Co04,) and 1.47 ns *7* (1971Pr11), both from pγ(t). Other: τ>4.5 ps using DSAM (1970Bu18).
2348.8 *13*	$3/2^-$	0.71 ps *15*	E(level): unweighted average of 2350.0 *10* (1970Bu18) and 2347.5 *2* (1972Fr11). J^π: spin from γ(θ) in 1972Va07, 1975VaYG, and 1972Fr11. $T_{1/2}$: from τ=1.03 ns *21* from weighted average of 0.8 ps *2* (1970Bu18) and 1.22 ps *18* (1972Fr11), both by DSAM.
2718.0 *10*	$5/2^+$	69 fs *24*	J^π: spin=3/2, 5/2, 7/2 from γ(θ) in 1972Va07 and 1972Fr11. 2717.9γ M1+E2 to $3/2^+$. $T_{1/2}$: from τ=100 fs *35* in 1972Fr11 by DSAM.
2935 *3*	$3/2^+$		J^π: spin=3/2, 5/2 from γ(θ) in 1972Va07 and 1975VaYG.
3421.0 *10*	$5/2^+$	<69 fs	J^π: spin=3/2, 5/2, 7/2 from γ(θ) in 1972Fr11. $T_{1/2}$: from τ<100 fs in 1972Fr11 by DSAM.
3558	$(5/2^+)$		E(level): rounded to the nearest integer based on the Adopted Levels. 1972Fr11 quotes 3563 *8* from 1969Mo12 without reporting their own E(level).
3592 *5*	$(7/2^+)$		
3802.0 *10*	$3/2^-$	25 fs *18*	J^π: spin=3/2 from γ(θ) in 1972Fr11. $T_{1/2}$: from τ=36 fs *26* in 1972Fr11 by DSAM.
3815 *3*	$(9/2)^-$		
3885.0 *10*	(5/2)		

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^{34}S(d,pγ) 1972Fr11,1975VaYG,1970Bu18 (continued)

^{35}S Levels (continued)

E(level)†	J$^\pi$‡	$T_{1/2}$	Comments
4025.0 *10*	(3/2$^+$,5/2$^+$)		
4107 *2*	(5/2$^+$)	<55 fs	$T_{1/2}$: from τ<80 fs in 1972Fr11 by DSAM.
4189 *2*	1/2$^-$	<35 fs	$T_{1/2}$: from τ<50 fs in 1972Fr11 by DSAM.
4480 *2*	(5/2$^+$)	<62 fs	$T_{1/2}$: from τ<90 fs in 1972Fr11 by DSAM.

† From the average of available measurements for the lowest three excited states, and from 1972Fr11 for higher-lying states.
‡ From the Adopted Levels. Supporting arguments from this dataset are given under comments if available.

γ(^{35}S)

E_i(level)	J_i^π	E_γ†	I_γ‡	E_f	J_f^π	Mult.	δ	Comments
1572.33	1/2$^+$	1572.24 *17*	100	0	3/2$^+$			E_γ: from 1972Go31. A_2=−0.09 *5*, A_4=+0.07 *6* (1972Fr11); A_2=−0.007 *10*, A_4=+0.004 *20* (1975VaYG).
1992.9	7/2$^-$	1992.8	100	0	3/2$^+$	M2+E3	+0.17 *6*	Mult.: D+Q or Q+O from $\gamma(\theta)$ in 1970Bu18, 1972Fr11, 1972Va07, and 1975VaYG; E2+M3 is ruled out by RUL. D+Q is inconsistent with the J^π=7/2$^-$. δ: unweighted average of δ(E3/M2)=0.11 *4* (1970Bu18), +0.11 *4* (1972Fr11), and +0.28 *3* (1972Va07,1975VaYG) for J_i=7/2. Others: δ(Q/D)=−1.9 *7* (1970Bu18) and −0.43 *3* (1972Va07,1975VaYG) for J_i=5/2. A_2=+0.35 *4*, A_4=−0.39 *4* (1972Fr11); A_2=+0.16 *3*, A_4=−0.10 *4* (1975VaYG).
2348.8	3/2$^-$	776.5	27 *2*	1572.33	1/2$^+$	(E1)		Mult.,δ: D(+Q) from $\gamma(\theta)$, $\Delta\pi$=yes from level scheme. M2 component is less likely. δ<0.04 from $\gamma(\theta)$ and RUL in 1975VaYG, superseding 0<δ<+1.7 from $\gamma(\theta)$ in 1972Va07. A_2=−0.37 *3*, A_4=−0.03 *3* (1972Fr11); A_2=−0.38 *2*, A_4=+0.01 *4* (1975VaYG).
		2348.7	73 *2*	0	3/2$^+$	(E1)		Mult.,δ: D(+Q) from $\gamma(\theta)$, $\Delta\pi$=yes from level scheme. M2 component is less likely. δ<0.04 from $\gamma(\theta)$ and RUL in 1975VaYG, superseding −0.27 *12* from $\gamma(\theta)$ in 1972Va07; +0.01 *4* from $\gamma(\theta)$ in 1972Fr11 assuming pure E1 for 776.5γ. A_2=+0.33 *2*, A_4=−0.05 *2* (1972Fr11); A_2=+0.30 *2*, A_4=−0.02 *4* (1975VaYG).
2718.0	5/2$^+$	369.2	<5	2348.8	3/2$^-$			
		725.1	<5	1992.9	7/2$^-$			
		1145.7	<5	1572.33	1/2$^+$			
		2717.9	100	0	3/2$^+$	M1+E2	−0.39 *5*	Mult.,δ: D+Q from $\gamma(\theta)$ in 1975VaYG with −6.9<δ<−0.11 or +0.42<δ<+6.8 for J_i=3/2, −0.39 *5* or −15.0 *34* for J_i=5/2, 0.18 *4* or 3.8 *7* for J_i=7/2. E1+M2 is ruled out by RUL. δ=−0.39 *5* is adopted here based on J^π=5/2$^+$ in the Adopted Levels. −15.0 *34* is unlikely based on RUL. A_2=+0.47 *6*, A_4=−0.08 *7* (1972Fr11); A_2=+0.22 *2*, A_4=−0.02 *3* (1975VaYG).
2935	3/2$^+$	586	<10	2348.8	3/2$^-$			
		942	<20	1992.9	7/2$^-$			

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$^{34}S(d,p\gamma)$ 1972Fr11,1975VaYG,1970Bu18 (continued)

$\gamma(^{35}S)$ (continued)

E_i(level)	J_i^π	E_γ[†]	I_γ[‡]	E_f	J_f^π	Comments
2935	3/2+	1363	<15	1572.33	1/2+	
		2935	100	0	3/2+	A_2=−0.06 *3*, A_4=−0.01 *6* (1975VaYG).
3421.0	5/2+	486	<4	2935	3/2+	
		703.0	<6	2718.0	5/2+	
		1072.2	<7	2348.8	3/2−	
		1428.1	<30	1992.9	7/2−	
		1848.6	<7	1572.33	1/2+	
		3420.8	100	0	3/2+	A_2=−0.54 *8*, A_4=−0.11 *7* (1972Fr11).
3592	(7/2+)	657	<10	2935	3/2+	
		874	<11	2718.0	5/2+	
		1243	<15	2348.8	3/2−	
		1599	<17	1992.9	7/2−	
		2020[#@]	<17	1572.33	1/2+	
		3592	100	0	3/2+	
3802.0	3/2−	867	<4	2935	3/2+	
		1084.0	5 *2*	2718.0	5/2+	
		1453.2	8 *3*	2348.8	3/2−	
		1809.1	<4	1992.9	7/2−	
		2229.6	42 *4*	1572.33	1/2+	A_2=−0.56 *15*, A_4=0.0 *1* (1972Fr11).
		3801.8	45 *4*	0	3/2+	A_2=+0.19 *12*, A_4=0.0 *1* (1972Fr11).
3815	(9/2)−	1822	100	1992.9	7/2−	
3885.0	(5/2)	464.0	<30	3421.0	5/2+	
		950	<12	2935	3/2+	
		1167.0	<15	2718.0	5/2+	
		1536.2	<18	2348.8	3/2−	
		1892.1	100	1992.9	7/2−	
		2312.6	<16	1572.33	1/2+	
		3884.8	<16	0	3/2+	
4025.0	(3/2+,5/2+)	467	<5	3558	(5/2+)	
		604.0	<10	3421.0	5/2+	
		1090	31 *8*	2935	3/2+	
		1307.0	<8	2718.0	5/2+	
		1676.2	33 *12*	2348.8	3/2−	
		2032.0	<10	1992.9	7/2−	
		2452.6	36 *12*	1572.33	1/2+	
		4024.8	<13	0	3/2+	
4107	(5/2+)	549	<4	3558	(5/2+)	
		686	<4	3421.0	5/2+	
		1172	13 *6*	2935	3/2+	
		1389	<10	2718.0	5/2+	
		1758	<5	2348.8	3/2−	
		2114	<7	1992.9	7/2−	
		2535	<5	1572.33	1/2+	
		4107	87 *6*	0	3/2+	
4189	1/2−	387[#@]	<4	3802.0	3/2−	
		597[#@]	<4	3592	(7/2+)	
		631	<4	3558	(5/2+)	
		768[#@]	<4	3421.0	5/2+	
		1254	<5	2935	3/2+	
		1471[#@]	<5	2718.0	5/2+	
		1840	48 *5*	2348.8	3/2−	
		2196[#@]	<8	1992.9	7/2−	
		2617	<6	1572.33	1/2+	
		4189	52 *5*	0	3/2+	

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^{34}S(d,pγ) 1972Fr11,1975VaYG,1970Bu18 (continued)

γ(^{35}S) (continued)

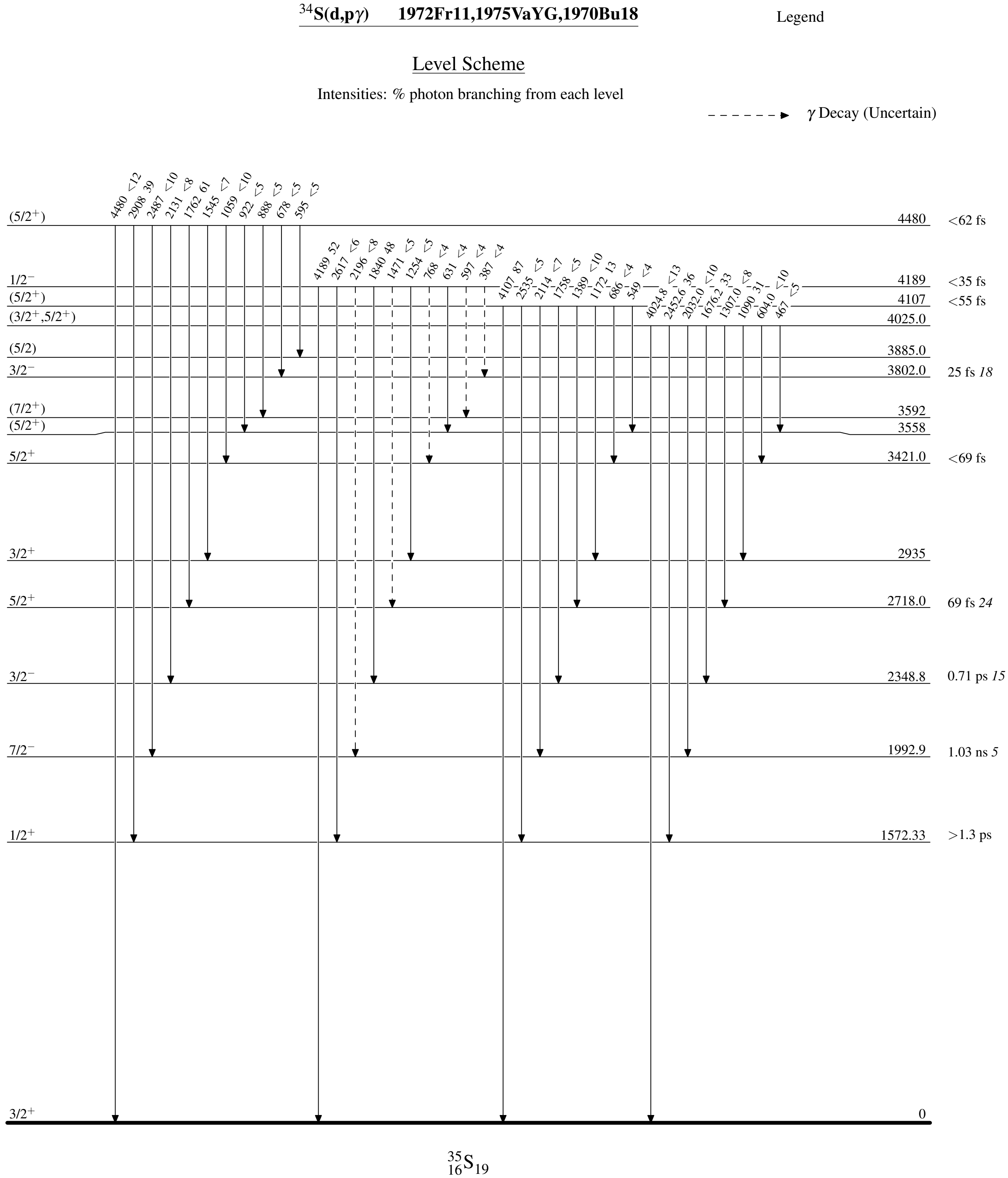
E_i(level)	J_i^π	E_γ†	I_γ‡	E_f	J_f^π
4480	(5/2$^+$)	595	<5	3885.0	(5/2)
		678	<5	3802.0	3/2$^-$
		888	<5	3592	(7/2$^+$)
		922	<5	3558	(5/2$^+$)
		1059	<10	3421.0	5/2$^+$
		1545	<7	2935	3/2$^+$
4480		1762	61 *7*	2718.0	5/2$^+$
		2131	<8	2348.8	3/2$^-$
		2487	<10	1992.9	7/2$^-$
		2908	39 *7*	1572.33	1/2$^+$
		4480	<12	0	3/2$^+$

† Deduced by evaluators from level-energy difference, unless otherwise noted.

‡ From 1972Fr11, unless otherwise noted.

\# γ transitions are considered questionable and not included in the Adopted Levels. The transition strengths estimated by evaluators exceed RUL. ^{34}S(n,γ) E=thermal observed other γ transitions from the same level but did not observe these questionable transitions.

@ Placement of transition in the level scheme is uncertain.

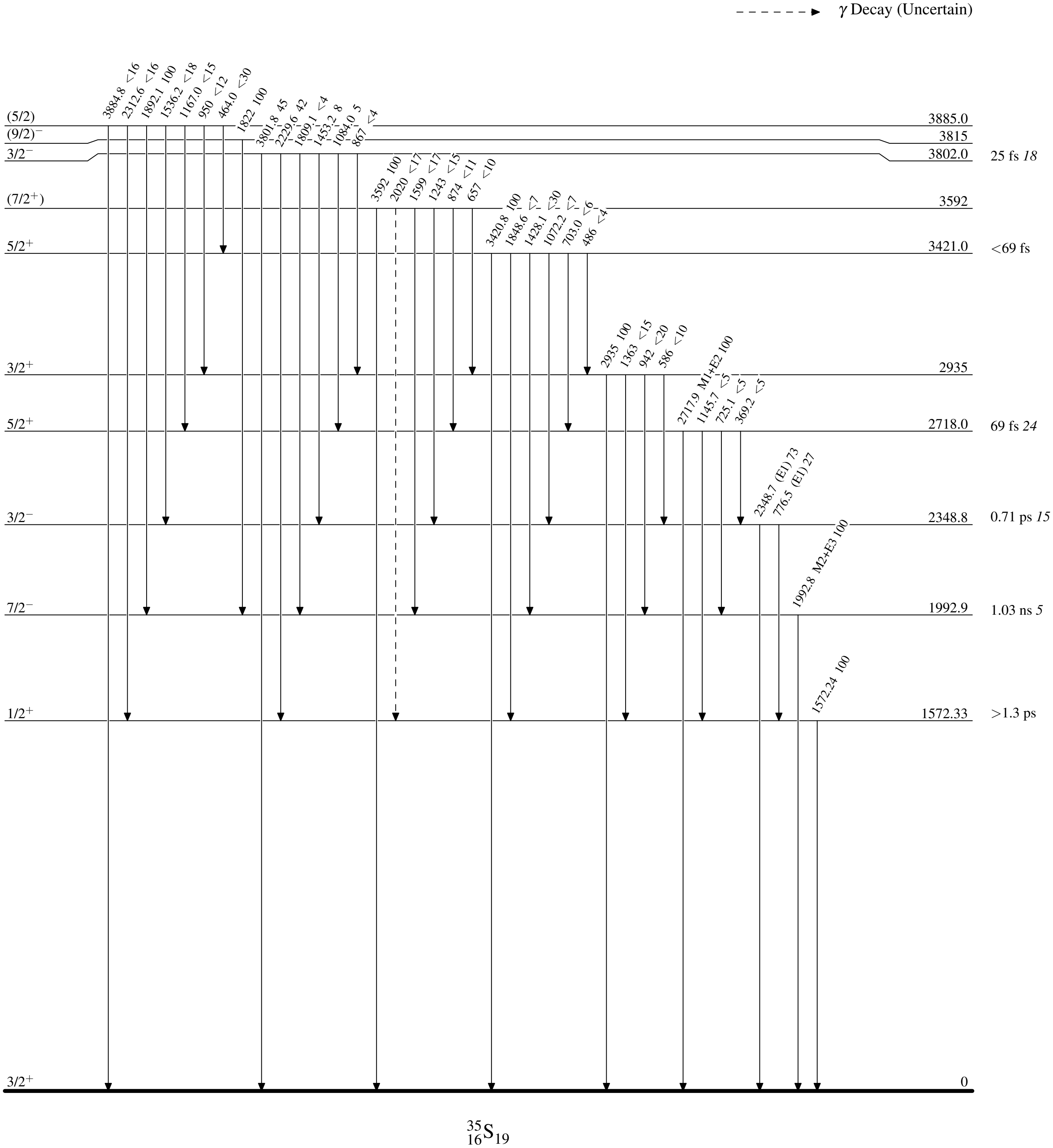


^{34}S(d,pγ) 1972Fr11,1975VaYG,1970Bu18
Legend
Level Scheme
Intensities: % photon branching from each level
γ Decay (Uncertain)
(5/2+) 4480 <62 fs
4480 <12
2908 39
2487 <10
2131 <8
1762 61
1545 <7
1059 <10
922 <5
888 <5
678 <5
595 <5
1/2− 4189 <35 fs
4189 52
2617 <6
2196 <8
1840 48
1471 <5
1254 <5
768 <4
631 <4
597 <4
387 <4
(5/2+) 4107 <55 fs
4107 87
2535 <5
2114 <7
1758 <5
1389 <10
1172 13
686 <4
549 <4
(3/2+,5/2+) 4025.0
4024.8 <13
2452.6 36
2032.0 <10
1676.2 33
1307.0 <8
1090 31
604.0 <10
467 <5
(5/2) 3885.0
3/2− 3802.0 25 fs 18
(7/2+) 3592
(5/2+) 3558
5/2+ 3421.0 <69 fs
3/2+ 2935
5/2+ 2718.0 69 fs 24
3/2− 2348.8 0.71 ps 15
7/2− 1992.9 1.03 ns 5
1/2+ 1572.33 >1.3 ps
3/2+ 0
$^{35}_{16}$S$_{19}$

$^{34}S(d,p\gamma)$ 1972Fr11,1975VaYG,1970Bu18

Level Scheme (continued)

Intensities: % photon branching from each level



^{36}S(p,d) 2026Jo01

$J^{\pi}=0^{+}$, T=2 for ^{36}S g.s.

2026Jo01: E=66 MeV dispersion-matched proton beam was produced from the Separated Sector Cyclotron facility of iThemba LABS, South Africa. The target was 1-mg/cm^2, 99.24%-enriched ^{36}S powder encapsulated between two 0.5-μm Mylar foils. The outgoing deuteron and triton reaction products were momentum-analyzed by the K=600 quadrupole-dipole-dipole magnetic spectrometer and detected by the focal-plane detectors consisting of two multiwire drift chambers (MWDC) and a downstream plastic scintillator. Measured $\sigma(E_d,\theta)$. Deduced 98 levels, L-transfers, and single-neutron spectroscopic factors for 81 levels from zero-range adiabatic distorted-wave approximation (ADWA) calculations using DWUCK4. Deduced J-dependence of L=2 angular distributions and reproduced with finite-range ADWA calculations. Comparisons with large-scale shell-model and ab initio calculations. Confirmed the N=20 shell closure in ^{36}S.

Also see Ph.D. Thesis 2021JoZZ.

^{35}S Levels

T: T=3/2 or 5/2 for ^{35}S levels populated in the ^{36}S(p,d) reaction based on T=2 for the 0$^+$ g.s. of ^{36}S. ^{35}S levels below 9 MeV are likely T=3/2 and those above 9 MeV are likely T=5/2 considering the binding, Coulomb, and proton-neutron mass differences suggest the IAS of ^{35}P ground state (T=5/2) should be around 9017 keV in ^{35}S. C^{2}S extracted for these ^{35}S levels show excellent agreement with proton-removal data from ^{35}P when scaled by expected Clebsch-Gordan ratio of 0.167 (2026Jo01).

E(level)	J$^{\pi}$ †‡	L †	C^{2}S †#	Comments
0 *5*	3/2$^+$	2	3.5 *4*	J$^{\pi}$: L-1/2 transfer from J-dependence.
1569 *5*	1/2$^+$	0	1.28 *14*	
1990 *5*	7/2$^-$	3	0.170 *19*	
2348 *5*	3/2$^-$	1	0.033 *4*	
2718 *5*	5/2$^+$	2	0.53 *6*	J$^{\pi}$: L+1/2 transfer from J−dependence.
2937 *5*	3/2$^+$	2	0.034 *4*	J$^{\pi}$: L-1/2 transfer from J-dependence.
3420 *5*	5/2$^+$	2	0.50 *6*	J$^{\pi}$: L+1/2 transfer from J−dependence.
3558 *5*	(5/2$^+$)	(2)	0.0300 *34*	
3592 *5*	(5/2$^+$)	(2)	0.0040 *5*	
3800 *5*	3/2$^-$	1	0.0200 *23*	
3883 *5*	7/2$^-$	3	0.0180 *20*	
4023 *5*	(3/2$^+$)	(2)	0.0056 *6*	
4107 *7*	(3/2$^+$)	(2)	0.00230 *26*	
4182 *5*	1/2$^-$	1	0.00100 *11*	
4300 *7*	(1/2$^-$,3/2$^-$)	(1)	3.00×10^{-4} *34*	
4486 *7*	7/2$^-$	3	0.045 *5*	
4574 *5*	3/2$^+$	2	0.080 *9*	J$^{\pi}$: L-1/2 transfer from J-dependence.
4614 *5*	5/2$^+$	2	0.110 *12*	
4838 *5*	(1/2$^+$)	(0)	0.0035 *4*	
4907 *5*	1/2$^+$	0	0.0110 *12*	
4955 *5*	5/2$^+$	2	0.250 *28*	J$^{\pi}$: L+1/2 transfer from J−dependence.
5121 *5*	1/2$^+$	0	0.0092 *10*	
5286 *5*	(1/2$^-$,3/2$^-$)	(1)	6.5×10^{-4} *7*	
5476 *5*	3/2$^+$	2	0.0090 *10*	
5630 *5*	5/2$^+$	2	0.110 *12*	
5766 *5*	5/2$^+$	2	0.76 *9*	J$^{\pi}$: L+1/2 transfer from J−dependence.
5844 *5*	(5/2$^+$)	(2)	0.080 *9*	
6121 *5*	(3/2$^+$)	(2)	0.035 *4*	
6218 *5*	(7/2$^-$)	(3)	0.00300 *34*	
6338 *5*	1/2$^+$	0	0.044 *5*	
6439 *5*	(1/2$^+$)	(0)	0.00260 *29*	
6550 *5*	3/2$^+$	2	0.0035 *4*	
6639 *5*	5/2$^+$	2	0.250 *28*	
6682 *5*	1/2$^+$	0	0.056 *6*	
6790 *5*	(5/2$^+$)	(2)	0.037 *4*	
6962 *5*	(1/2$^-$,3/2$^-$)	(1)	0.0050 *6*	

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^{36}S(p,d) 2026Jo01 (continued)

^{35}S Levels (continued)

E(level)	J$^{\pi}$†‡	L†	C^{2}S†#	Comments
7019 *5*	(1/2$^-$,3/2$^-$)	(1)	0.00150 *17*	
7102 *5*	(1/2$^-$,3/2$^-$)	(1)	0.00200 *23*	
7151 *5*	(7/2$^-$)	(3)	0.0090 *10*	
7215 *5*	3/2$^+$	2	0.0120 *14*	
7249 *5*	(3/2$^+$)	(2)	0.0120 *14*	
7279 *5*	1/2$^+$	0	0.0068 *8*	
7326 *5*	1/2$^+$	0	0.0073 *8*	
7349 *5*	(1/2$^+$)	(0)	0.0095 *11*	
7441 *5*	1/2$^+$	0	0.0094 *11*	
7490 *5*	1/2$^-$	1	0.0060 *7*	
7753 *5*	5/2$^+$	2	0.200 *23*	
7885 *5*	(5/2$^+$)	(2)	0.0300 *34*	
7980 *7*	3/2$^-$	1	0.00200 *23*	
8073 *7*	(5/2$^+$)	(2)	0.0180 *20*	
8103 *5*	1/2$^-$	1	0.00170 *19*	
8221 *5*	1/2$^-$	1	0.0040 *5*	
8270 *5*	1/2$^-$	1	0.0090 *10*	
8410 *7*	(5/2$^+$)	(2)	0.0100 *11*	
8465 *7*	(5/2$^+$)	(2)	0.0100 *11*	
8509 *5*	5/2$^+$	2	0.0120 *14*	
8557 *7*	(7/2$^-$)	(3)	0.00300 *34*	
8602 *5*	(7/2$^-$)	(3)	0.0100 *11*	
8649 *7*	(7/2$^-$)	(3)	0.0065 *7*	
8707 *5*	(5/2$^+$)	(2)	0.0220 *25*	
8761 *7*	(5/2$^+$)	(2)	0.0110 *12*	
8822 *5*	(1/2$^-$,3/2$^-$)	(1)	0.0060 *7*	
8876 *5*	(1/2$^-$,3/2$^-$)	(1)	0.00250 *28*	
8947 *7*	(1/2$^-$,3/2$^-$)	(1)	0.00100 *11*	
9032 *15*	(1/2$^-$,3/2$^-$)	(1)	0.0070 *8*	
9065 *15*	(1/2$^-$,3/2$^-$)	(1)	0.0070 *8*	
9126 *7*	1/2$^+$	0	0.35 *4*	T=5/2 J$^{\pi}$,T: isobaric analog state of 1/2$^+$, T=5/2, g.s. of ^{35}P (2026Jo01).
9180 *5*	1/2$^+$	0	0.060 *7*	
9219 *15*	(1/2$^-$,3/2$^-$)	(1)	0.00300 *34*	
9294 *8*	(1/2$^-$,3/2$^-$)	(1)	0.0100 *11*	
9339 *8*	(1/2$^-$,3/2$^-$)	(1)	0.00200 *23*	
9395 *5*	(1/2$^-$,3/2$^-$)	(1)	0.0045 *5*	
9446 *7*	(1/2$^-$,3/2$^-$)	(1)	0.00180 *20*	
9510 *10*	(1/2$^-$,3/2$^-$)	(1)	0.0035 *4*	
9565 *20*	(1/2$^-$,3/2$^-$)	(1)	0.0033 *4*	
9615 *10*	(1/2$^-$,3/2$^-$)	(1)	0.00280 *32*	
9666 *10*	(1/2$^-$,3/2$^-$)	(1)		
9716 *10*	(1/2$^-$,3/2$^-$)	(1)		
9774 *10*	(1/2$^-$,3/2$^-$)	(1)		
9849 *10*	(1/2$^-$,3/2$^-$)	(1)		
9911 *10*	(1/2$^-$,3/2$^-$)	(1)		
9989 *10*	(1/2$^-$,3/2$^-$)	(1)		
1.0049×10^4 *10*	(1/2$^-$,3/2$^-$)	(1)		
1.0166×10^4 *10*	(1/2$^-$,3/2$^-$)	(1)		
1.0225×10^4 *10*	(1/2$^-$,3/2$^-$)	(1)		
1.0277×10^4 *10*	(1/2$^-$,3/2$^-$)	(1)		
1.0350×10^4 *10*	(1/2$^-$,3/2$^-$)	(1)		
1.0421×10^4 *10*	(1/2$^-$,3/2$^-$)	(1)		
1.0493×10^4 *10*	(7/2$^-$)	(3)	0.0200 *23*	

Continued on next page (footnotes at end of table)

^{36}S(p,d) 2026Jo01 (continued)

^{35}S Levels (continued)

E(level)	J^{π}†‡	L†	C^2S†#	Comments
1.0563×10^4 *10*	(1/2$^-$,3/2$^-$)	(1)		
1.0663×10^4 *10*	(1/2$^-$,3/2$^-$)	(1)		
1.0746×10^4 *10*	(1/2$^-$,3/2$^-$)	(1)		
1.0810×10^4 *10*	(1/2$^-$,3/2$^-$)	(1)		
1.147×10^4 *2*	3/2$^+$	2	0.052 *6*	T=5/2 J^{π},T: IAS of 3/2$^+$, T=5/2, 2387 level in ^{35}P (2026Jo01).
1.288×10^4 *10*	5/2$^+$	2	0.49 *6*	T=5/2 J^{π},T: IAS of 5/2$^+$, T=5/2, 3860 level in ^{35}P (2026Jo01).
1.368×10^4 *10*	5/2$^+$	2	0.156 *18*	T=5/2 J^{π},T: IAS of 5/2$^+$, T=5/2, 4666 level in ^{35}P (2026Jo01).
1.422×10^4 *10*	5/2$^+$	2	0.247 *28*	T=5/2 J^{π},T: IAS of 5/2$^+$, T=5/2, 5199 level in ^{35}P (2026Jo01).
1.470×10^4 *10*				

† From ADWA analysis of the measured $\sigma(\theta)$ in 2026Jo01. L-transfers are not specifically given but can be inferred from the reported J^{π}.

‡ As given in 2026Jo01, which selected some J=L−1/2 or L+1/2 based on known J^{π} values (2011Ch48). When considered in the Adopted Levels, J=L−1/2 and L+1/2 are both possible unless other constraints are available.

C^2S from 2026Jo01. For levels with J^{π}=(1/2$^-$,3/2$^-$), the C^2S values are given for 3/2$^-$. C^2S is normalized to a ^{40}Ca benchmark where Woods-Saxon parameters (r_0=1.27 fm, a_0=0.70 fm) were adjusted to ensure the summed strength of the $1d_{3/2}$, $2s_{1/2}$, and $1f_{7/2}$ orbitals equals 6. A systematic uncertainty of 11.2% is associated with all spectroscopic factors due to the effective target-thickness variation (10%) and the nominal target thickness uncertainty (5%) (2026Jo01). This uncertainty is included by the evaluators.

^{37}Cl(p,^{3}He) 1975Gu15

$^{37}Cl \rightarrow ^{35}S$+1p1n from $J^{\pi}=3/2^{+}$ ^{37}Cl ground state.

1975Gu15: A 40.2-MeV proton beam was produced from the Michigan State University cyclotron. The target was 55 μg/cm^2 NaCl with 97% enriched ^{37}Cl made by evaporation of the salt onto a 30-μg/cm^2 carbon backing. The reaction products were detected using a wire-counter plastic-scintillator in the focal plane of an Enge split-pole spectrograph with FWHM=30 keV. Measured $\sigma(E(^3He),\theta)$. Deduced levels, L from zero-range DWUCK-DWBA analysis of the measured $\sigma(\theta)$.

1971Vi02: A 40-MeV proton beam was produced from the Grenoble variable energy cyclotron at intensities of 20-200 nA depending on the scattering angle with a 90-keV energy resolution. The target was natural purified chlorine gas 100 mm in diameter and 25 mm in height. The reaction particles were detected using two separate counter telescopes with each consisting of a 200-μm phosphorous-drifted silicon ΔE detector, a 2-mm lithium-drifted silicon E detector and a 3-mm lithium-drifted silicon E-reject detector with FWHM=180 keV for ^{3}He. Measured $\sigma(E(^3He),\theta)$. Deduced L for ^{35}S ground state from zero-range Nelson spin-dependent-DWBA analysis of the measured $\sigma(\theta)$.

^{35}S Levels

σ_{max} under comments are from 1975Gu15.

E(level)†	L‡	Comments
0	0+2+4	σ_{max}=26.5 μb/sr.
1575 *10*	0+2	σ_{max}=19 μb/sr.
1992 *10*	3+5	σ_{max}=1.5 μb/sr.
2717 *10*	0+2+4	σ_{max}=53.9 μb/sr.
2938 *10*	0+2+4	σ_{max}=29.2 μb/sr.
3421 *10*	0+2+4	σ_{max}=81.5 μb/sr.
3598 *10*	2+4	σ_{max}=27.8 μb/sr.
3811 *10*	3	σ_{max}=2.6 μb/sr.
4027 *10*	2	σ_{max}=2.9 μb/sr.
4114 *10*	0+2	σ_{max}=8.8 μb/sr.
4186 *10*	(2,3)	σ_{max}=5.6 μb/sr.
4290 *10*	(2)	σ_{max}=2.8 μb/sr.
4489 *10*	2	σ_{max}=2.7 μb/sr.
4577 *10*	0+2	σ_{max}=10.7 μb/sr.
4617 *10*	(1,2)	σ_{max}=13.2 μb/sr.
4843 *10*	2	σ_{max}=15.9 μb/sr.
4963 *10*	(0+2)	σ_{max}=17.3 μb/sr.
4990 *10*	0+2	σ_{max}=19.5 μb/sr.
5127 *10*	2	σ_{max}=3.5 μb/sr.
5345 *10*	3	σ_{max}=2.2 μb/sr.
5550 *10*	(3)	σ_{max}=6.2 μb/sr.
5771 *10*	2	σ_{max}=10.1 μb/sr.
5915? *10*	(2,3)	σ_{max}=3.4 μb/sr.
6129 *10*	0+2	σ_{max}=7.8 μb/sr.
6347 *10*	(2)	σ_{max}=4.8 μb/sr.
6654 *10*	(3)	σ_{max}=7.7 μb/sr.
6696 *10*	2	σ_{max}=6.7 μb/sr.
7151? *10*	(4)	σ_{max}=3.3 μb/sr.
7375? *10*		
7712? *10*	4	σ_{max}=4.6 μb/sr.
7770 *10*		σ_{max}=5.5 μb/sr.
8103 *10*	(1+3)	σ_{max}=10.6 μb/sr.
8160 *10*	(1)	σ_{max}=8.4 μb/sr.
8430 *10*	2	σ_{max}=5.7 μb/sr.
9155 *10*	2	σ_{max}=10 μb/sr.

Continued on next page (footnotes at end of table)

^{37}Cl(p,^{3}He) 1975Gu15 (continued)

^{35}S Levels (continued)

† From 1975Gu15.
‡ From DWBA analysis of the measured $\sigma(\theta)$ in 1975Gu15. L is not considered firm by the evaluators when assigning J^{π} in the Adopted Levels because the theory of (p,^{3}He) reactions is not robust.

^{37}Cl(d,$\alpha\gamma$) 1975VaYG,1972Va07

1975VaYG,1972Va07: A 4.25-MeV deuteron beam of 55 nA was provided from the Groningen 5 MV Van de Graaff generator. The target was 100 μg/cm^2 Co^{37}Cl enriched to 98% evaporated onto 10-μg/cm^2 Formvar plus 10-μg/cm^2 carbon. α particles were detected using a 60-μm annular silicon detector. γ rays were detected using a 120 cm^3 Ge(Li) detector at 90°. Measured $\sigma(E_\alpha)$, Eγ, Iγ, and $\alpha\gamma$-coin. Deduced levels, γ-ray branching ratios. Also see 1971VaZG.

1968Te06: 3.1-4.6-MeV deuteron beams of 50 nA were provided from the Groningen 5-MV Van de Graaff generator. The target was a 100-μg/cm^2 Co^{37}Cl both of natural ^{37}Cl abundance and enriched to 93% evaporated onto 10-μg/cm^2 Formvar plus 10-μg/cm^2 carbon. α particles were detected using an annular solid-state detector at 168-173°. γ rays were detected using a 3 in. by 3 in. NaI(Tl) scintillator at 55°. Measured $\sigma(E_\alpha)$, Eγ, and $\alpha\gamma$-coin. Deduced levels.

1955Pa54: ^{37}Cl(d,α)^{35}S with 3.0, 5.6, 6.0, 7.0, and 7.5-MeV deuteron beams from the MIT-ONR electrostatic generator. Targets were 80 and 300-μg/cm^2 barium chloride (75.4% ^{35}Cl, 24.6% ^{37}Cl) evaporated onto Formvar films on a gold layer. Charged reaction products emitted at 90° were magnetically analyzed by a broad-range spectrograph. Measured $\sigma(E_\alpha)$. Deduced levels.

^{35}S Levels

E(level)†	Comments
0	
1572.2 *12*	
1990.0 *10*	E(level): Other: 1992 *10* from 1955Pa54.
2348.2 *10*	E(level): Other: 2348 *10* from 1955Pa54.
2716.7 *10*	E(level): Other: 2714 *10* from 1955Pa54.
2939.2 *13*	
3423 *5*	
3560.8 *19*	
3598.4 *21*	
3803.6 *19*	
3818.1 *11*	
3889.0 *19*	
4022.2 *22*	
4027.7 *10*	E(level): Other: 4025 *10* from 1955Pa54.
4108 *3*	
4180 *3*	
4187 *3*	
4302 *4*	
4480.0 *16*	

† From 1975VaYG based on measured Eγ, but the Eγ values are not provided in 1975VaYG.

γ(^{35}S)

E$_i$(level)	E$_\gamma$†	I$_\gamma$‡	E$_f$	Comments
1572.2	1572.2	100	0	
1990.0	1989.9	100	0	
2348.2	776.0	25 *2*	1572.2	
	2348.1	75 *2*	0	
2716.7	2716.6	100	0	
2939.2	2939.1	100	0	
3423	3423	100	0	
3560.8	1212.6	35 *4*	2348.2	
	1570.8	65 *4*	1990.0	
3598.4	3598.2	100	0	I$_\gamma$: >95 in 1975VaYG.
3803.6	2231.3	38 *3*	1572.2	
	3803.4	62 *3*	0	
3818.1	1828.1	100	1990.0	
3889.0	1540.8	40 *4*	2348.2	
	1898.9	45 *5*	1990.0	

Continued on next page (footnotes at end of table)

^{37}Cl(d,$\alpha\gamma$) 1975VaYG,1972Va07 (continued)

γ(^{35}S) (continued)

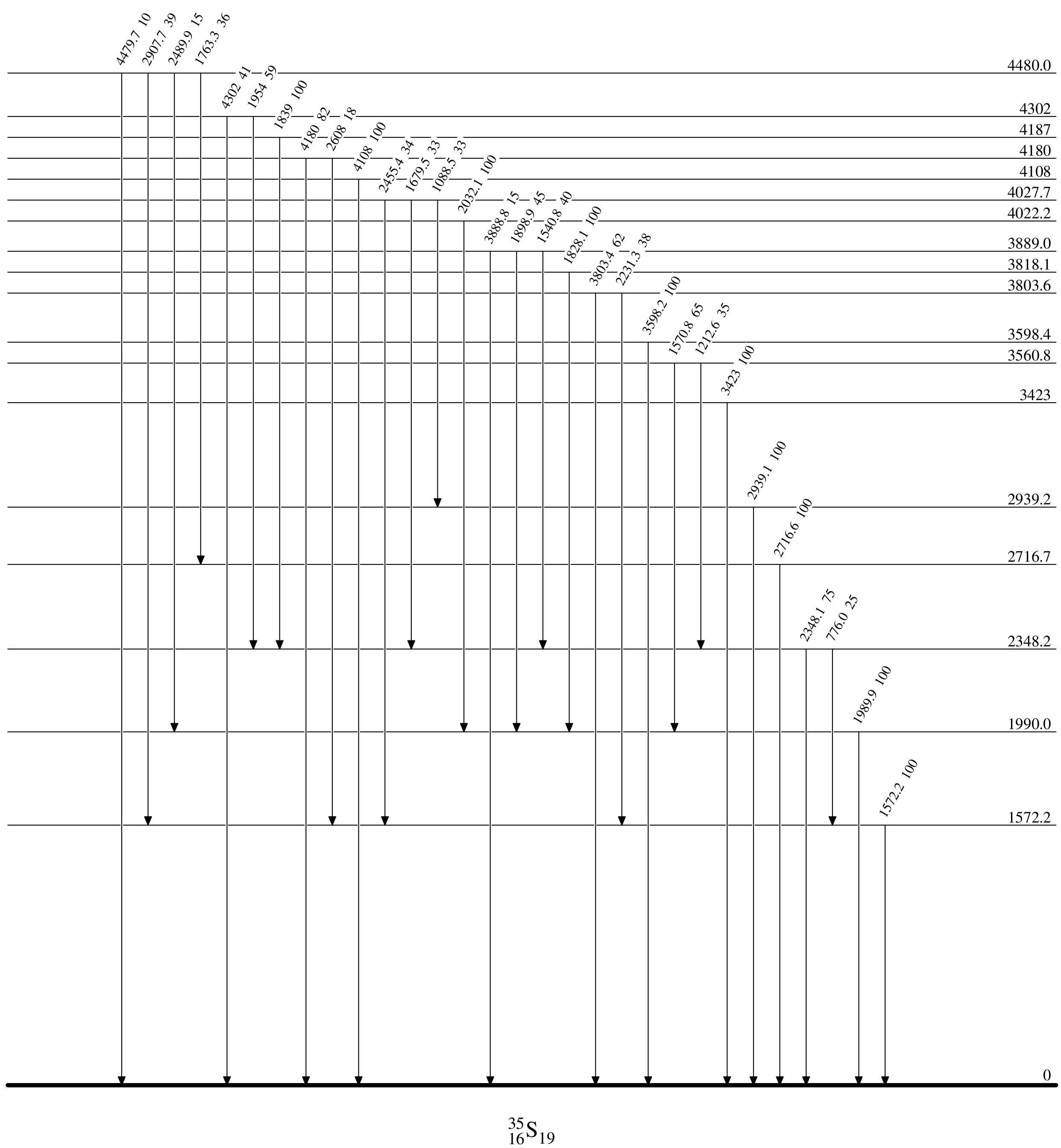
E_i(level)	E_γ †	I_γ ‡	E_f	Comments
3889.0	3888.8	15 *3*	0	
4022.2	2032.1	100	1990.0	
4027.7	1088.5	33 *4*	2939.2	
	1679.5	33 *4*	2348.2	
	2455.4	34 *6*	1572.2	
4108	4108	100	0	I_γ: >95 in 1975VaYG.
4180	2608	18 *5*	1572.2	
	4180	82 *5*	0	
4187	1839	100	2348.2	I_γ: >95 in 1975VaYG.
4302	1954	59 *5*	2348.2	
	4302	41 *5*	0	
4480.0	1763.3	36 *4*	2716.7	
	2489.9	15 *4*	1990.0	
	2907.7	39 *4*	1572.2	
	4479.7	10 *2*	0	

† Deduced by evaluators from level-energy difference in 1975VaYG.
‡ From 1975VaYG.

^{37}Cl(d,$\alpha\gamma$) 1975VaYG,1972Va07

Level Scheme

Intensities: % photon branching from each level



^{160}Gd(^{34}S,Xγ),(^{37}Cl,Xγ) 1994Fo04

1994Fo04: 159-MeV ^{34}S and 167-MeV ^{37}Cl beams were produced from the Argonne Tandem Linac Accelerator System (ATLAS). Targets were 1 mg/cm^2 98.1% enriched ^{160}Gd backed by 15 mg/cm^2 gold. γ rays were detected using the Argonne-Notre Dame BGO γ-ray facility consisting of 12 Compton-suppressed Ge detectors and a 50-element bismuth germanate (BGO) array. Measured Eγ, Iγ, $\gamma\gamma$-coin. Deduced levels. $\gamma\gamma$-coin gates were placed on known γ rays in specific A≈160 products to select individual reaction channels and identify coincident γ rays in light product partners. $\gamma\gamma$-correlation matrices produced with the fold condition $4 \leq k \leq 9$ were found to include most of the events associated with deep inelastic processes.

^{35}S Levels

E(level)†	Comments
0	
1991	
4022	J$^{\pi}$: 1994Fo04 suggests this level is a likely candidate to be the expected 9/2$^-$ yrast excitation, while (11/2)$^-$ is adopted in the Adopted Levels.

† From Eγ data in 1994Fo04.

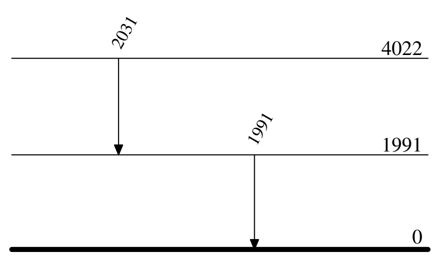
γ(^{35}S)

Eγ†	E$_i$(level)	E$_f$
1991	1991	0
2031	4022	1991

† From 1994Fo04.

^{160}Gd(^{34}S,Xγ),(^{37}Cl,Xγ) 1994Fo04

Level Scheme



$^{35}_{16}S_{19}$

^{208}Pb(^{36}S,^{35}Sγ) 2010WaZT,2022Gr07

2022Gr07: A 215-MeV ^{36}S beam was produced from the XTU-Tandem ALPI-superconducting linear accelerator complex at the INFN Legnaro National Laboratory, Italy. The target was 1 mg/cm^2 99.7% enriched ^{208}Pb with 1 mg/cm^2 Nb backing and mounted onto the Cologne differential plunger. Projectile-like fragments produced in binary grazing reactions were separated and identified by the PRISMA spectrometer. γ rays were detected using the AGATA demonstrator array of five triple cluster modules of 36-fold segmented Ge crystals covering backward angles from 135° to 175°. Measured Eγ with Doppler corrections, (^{35}S)γ-coin, and level lifetimes using the differential recoil-distance method (DRDM). Compared with shell-model calculations. Measured Eγ, Iγ, fragment-γ-coin, recoil distance. Deduced levels, lifetimes. Compared with shell-model calculations with PSDPF, SDPF-U, and FSU effective interactions.

2010WaZT,2010Wa12: A 215-MeV ^{36}S^9 beam was produced from the XTU-Tandem ALPI-superconducting linear accelerator complex at the INFN Legnaro National Laboratory, Italy. The target was 300 μg/cm^2 99.7% enriched ^{208}Pb with 20 μg/cm^2 carbon backing. Projectile-like fragments produced in binary grazing reactions were separated and identified by the PRISMA spectrometer. γ rays were detected using the CLARA array of 22 escape-suppressed Ge clover detectors covering the azimuthal angles from 98° to 180°. Measured Eγ with Doppler corrections, Iγ, and (^{35}S)γ-coin. Deduced levels. Compared with shell-model calculations.

2010WaZT states that the observed γ-ray transitions are consistent with previous published work and no attempt has been made here to construct an independent level scheme.

^{35}S Levels

E(level)†	J$^\pi$‡	T$_{1/2}$	Comments
0	3/2$^+$		2022Gr07 shell-model calculated configuration: $\pi(1d_{5/2})^6(2s_{1/2})^2\otimes\nu(1d_{5/2})^6(2s_{1/2})^2(1d_{3/2})^3$ (77%).
1572.0 *9*	1/2$^+$	2.29 ps *14*	2022Gr07 shell-model calculated configuration: $\pi(1d_{5/2})^6(2s_{1/2})^2\otimes\nu(1d_{5/2})^6(2s_{1/2})^1(1d_{3/2})^4$ (44%). T$_{1/2}$: Lifetime=3.3 ps *2* from 2022Gr07 using the recoil-distance method with decay-curve analysis.
1991.1 *9*	7/2$^-$		
2347.1 *13*	3/2$^-$		
2717.1 *14*	5/2$^+$		
3421 *2*	5/2$^+$		
3558.1 *14*	(5/2$^+$)		
3593 *2*	(7/2$^+$)		
3818.1 *14*	(9/2)$^-$		
4023.1 *22*	(11/2)$^-$		

† From a least-squares fit to γ-ray energies.
‡ From the Adopted Levels.

γ(^{35}S)

E$_\gamma$†	I$_\gamma$†	E$_i$(level)	J$^\pi_i$	E$_f$	J$^\pi_f$	Comments
370 *1*	3.2 *6*	2717.1	5/2$^+$	2347.1	3/2$^-$	
1211‡ *2*	1.7 *7*	3558.1	(5/2$^+$)	2347.1	3/2$^-$	
^{x}1227 *1*	5.2 *9*					
1567‡ *2*	9.6 *9*	3558.1	(5/2$^+$)	1991.1	7/2$^-$	
1572 *1*	100.0 *24*	1572.0	1/2$^+$	0	3/2$^+$	2022Gr07 shell-model calculated B(E2; 1/2$^+$ to 3/2$^+$) = 35.2 e^2 fm^4 (5.2 W.u.), B(M1; 1/2$^+$ to 3/2$^+$) = 0.0204 μ_N^2 (1.14×10^{-2} W.u.), and mixing ratio [δ^2 = λ(E2)/λ(M1)] = 0.30.
1827 *1*	12.9 *11*	3818.1	(9/2)$^-$	1991.1	7/2$^-$	
1986‡ *2*	3.7 *6*	3558.1	(5/2$^+$)	1572.0	1/2$^+$	
1991 *1*	37.2 *18*	1991.1	7/2$^-$	0	3/2$^+$	
2032 *2*	13.1 *12*	4023.1	(11/2)$^-$	1991.1	7/2$^-$	

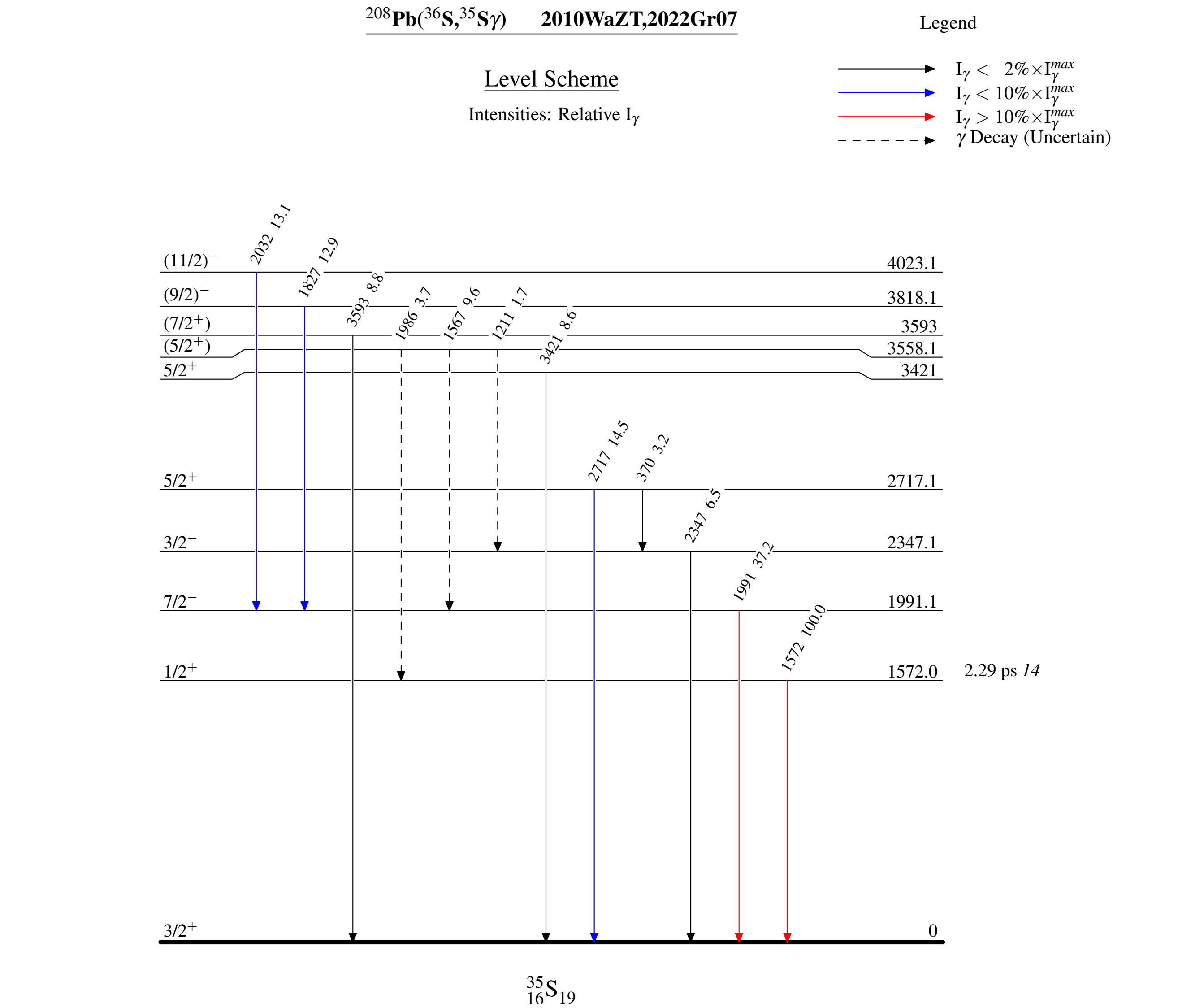
Continued on next page (footnotes at end of table)

^{208}Pb(^{36}S,^{35}Sγ) 2010WaZT,2022Gr07 (continued)

γ(^{35}S) (continued)

E_γ †	I_γ †	E_i(level)	J^π_i	E_f	J^π_f
2347 *2*	6.5 *9*	2347.1	3/2$^-$	0	3/2$^+$
2717 *2*	14.5 *15*	2717.1	5/2$^+$	0	3/2$^+$
^{x}3034 *3*	7.2 *9*				
3421 *2*	8.6 *10*	3421	5/2$^+$	0	3/2$^+$
3593 *2*	8.8 *11*	3593	(7/2$^+$)	0	3/2$^+$

† From 2010WaZT.
‡ Placement of transition in the level scheme is uncertain.
^{x} γ ray not placed in level scheme.


^{208}Pb(^{36}S,^{35}Sγ) 2010WaZT,2022Gr07
Level Scheme
Intensities: Relative I_γ
Legend
$I_\gamma < 2\% \times I_\gamma^{max}$
$I_\gamma < 10\% \times I_\gamma^{max}$
$I_\gamma > 10\% \times I_\gamma^{max}$
γ Decay (Uncertain)
(11/2)$^-$ 4023.1
(9/2)$^-$ 3818.1
(7/2$^+$) 3593
(5/2$^+$) 3558.1
5/2$^+$ 3421
5/2$^+$ 2717.1
3/2$^-$ 2347.1
7/2$^-$ 1991.1
1/2$^+$ 1572.0 2.29 ps 14
3/2$^+$ 0
2032 13.1
1827 12.9
3593 8.8
1986 3.7
1567 9.6
1211 1.7
3421 8.6
2717 14.5
370 3.2
2347 6.5
1991 37.2
1572 100.0
$^{35}_{16}S_{19}$

Adopted Levels, Gammas

Q(β^-)=−5966.2 *7*; S(n)=12644.76 *5*; S(p)=6370.81 *4*; Q(α)=−6997.90 *4* 2021Wa16
S(2n)=24152.8 *4*, S(2p)=17254.1 *11* (2021Wa16).
Isotope discovery (2012Th10): positive-ray mass spectrograph at Cambridge, UK (1919As01).

^{35}Cl Levels

Cross Reference (XREF) Flags

A	^{35}S β^- decay (87.61 d)	K	^{31}P(^{6}Li,d)	U	^{35}Cl(p,p′γ)
B	^{35}Ar ε decay (1.7755 s)	L	^{32}S(α,p)	V	^{36}Ar(n,d)
C	^{36}K εp decay (341 ms)	M	^{32}S(α,pγ)	W	^{36}Ar(d,^{3}He)
D	^{1}H(^{34}S,^{35}Clγ)	N	^{33}S(α,d)	X	^{36}Ar(pol d,^{3}He)
E	^{2}H(^{34}S,nγ)	O	^{34}S(p,γ)	Y	^{37}Cl(p,t)
F	^{12}C(^{28}Si,αpγ)	P	^{34}S(p,p),(p,p′γ):resonances	Z	^{39}K(γ,α)
G	^{16}O(^{24}Mg,αpγ)	Q	^{34}S(d,n)	Others:	
H	^{24}Mg(^{16}O,αpγ)	R	^{34}S(^{3}He,d)	AA	^{40}Ca(μ^-,$\nu\alpha$nγ)
I	^{27}Al(^{14}N,αpnγ)	S	^{35}Cl(γ,γ')	AB	Coulomb excitation
J	^{31}P(α,p),(α,n):resonances	T	^{35}Cl(n,n′γ)		

E(level)[†]	J$^\pi$[‡]	T$_{1/2}$ or Γ[&]	XREF	Comments
0.0	3/2$^+$	stable	ABC EFGHI KLM O QRSTUVWXYZ	XREF: Others: AA, AB μ=+0.821870 *2* (2013Ja17,2019StZV) Q=−0.0817 *8* (2008Py02,2021StZZ) J$^\pi$: L=2 from 0$^+$ in (d,n), (n,d), (^{3}He,d), (d,^{3}He), and (pol d,^{3}He) and L-1/2 transfer from analyzing power. Spin=3/2 from microwave spectroscopy and atomic beam methods (1948To10,1949Da14). μ: Nuclear Magnetic Resonance (2013Ja17). Q: Atomic beam magnetic resonance (2008Py02). Others: −0.08249 *2* (1972St38), −0.076 *5* (1986El09), −0.817 *8* (1993Su36), 0.0819 *11* (2000Ha64), 0.0850 *11* (2004Al08). Evaluated rms nuclear charge radius R=3.365 fm *19* (2013An02).
1219.38 *7*	1/2$^+$	0.119 ps *14*	B E H KLM O QRSTUVWXY	XREF: Others: AA, AB B(E2)↑=0.00076 *7* B(E2)↑: from Coulomb excitation. E(level): weighted average of 1219.2 *2* from ^{35}Ar ε decay, 1219.39 *7* from (p,γ), and 1219.42 *10* from (p,p′γ), 1219.0 *10* from (^{16}O,αpγ), and 1219.4 *6* from (α,pγ). J$^\pi$: L=0 from 0$^+$ in (d,n), (n,d), (^{3}He,d), (d,^{3}He), and (pol d,^{3}He). T$_{1/2}$: from lifetime τ=0.172 ps *20*: weighted average of 0.29 ps *4* from (^{34}S,nγ) (1973Wa10, DSAM), 0.21 ps *+10−8* from (α,pγ) (1969In04, DSAM), 80 fs *40* (1971Wi13, DSAM), 175 fs *20* (1972Hu11, DSAM), 150 fs *20* (1973Fa07, DSAM), 145 fs *30* (1976Me12, DSAM) from (p,γ), 0.13 ps *+3−2* from (γ,γ') (1966Ho02, resonance fluorescence), 200 fs *80* from (n,n′γ) (1989Ge09, DSAM), 0.215 ps *75* (1972Va06, DSAM), 0.21 ps *+6−4* (1972Br45, DSAM), 0.27 ps *+19−10* (1969Du08, DSAM) from (p,p′γ), 0.16 ps *5* (1969Ha17, DSAM), and 0.230 ps *25* (1977Sc36, DSAM) from Coulomb excitation.
1763.15 *4*	5/2$^+$	0.36 ps *3*	B EFGHI KLM O QRSTUVWXYZ	XREF: Others: AA, AB

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)[†]	J^{π}[‡]	$T_{1/2}$ or Γ[&]	XREF	Comments
				T=1/2 B(E2)↑=0.0106 *7* XREF: X(1785)Y(1750) B(E2)↑: from Coulomb excitation. E(level): weighted average of 1763.1 *2* from ^{35}Ar ε decay, 1763.0 *6* from (^{28}Si,αpγ), 1762.95 *17* from (^{24}Mg,αpγ), 1763.3 *2* from (^{16}O,αpγ), 1763.19 *10* from (^{14}N,αpnγ), 1763.14 *4* from (p,γ), 1763.26 *10* from (p,p′γ). J^{π}: L=2 from 0^+ in (d,n); spin=5/2 from pγ(θ) or γ(θ) in (α,pγ), (p,γ), and (p,p′γ). $T_{1/2}$: lifetime τ=0.52 ps *5*: weighted average of 0.54 ps *7* from (^{34}S,nγ) (1973Wa10, DSAM), 0.55 ps *15* from (α,pγ) (1969In04, DSAM), 0.50 ps *10* (1972Hu11, DSAM), 0.40 ps *14* (1973Fa07, DSAM), 0.53 ps *9* (1976Me12, DSAM) from (p,γ), 0.60 ps *21* from (γ,γ′) (1962Bo17, resonance fluorescence), 0.205 ps *75* (1972Va06, DSAM), 0.49 ps *+9−6* (1972Br45, DSAM), 1.15 ps *+45−63* (1969Du08, DSAM) from (p,p′γ), 0.57 ps *7* (1969Ha17, DSAM), and 0.63 ps *5* (1977Sc36, DSAM) from Coulomb excitation.
2645.67 *8*	$7/2^+$	0.159 ps *21*	B FGHI lM O Qr TU y	XREF: Others: AA, AB XREF: r(2645)y(2650) E(level): weighted average of 2645.81 *18* from (^{24}Mg,αpγ), 2645.9 *3* from (^{16}O,αpγ), 2645.45 *12* from (^{14}N,αpnγ), 2645.70 *8* from (p,γ), and 2645.74 *17* from (p,p′γ). Others: 2645.1 *10* from ^{35}Ar ε decay, 2645.8 *6* from (^{28}Si,αpγ), and 2645.8 *8* from (α,pγ). J^{π}: spin=7/2 from pγ(θ) in (α,pγ) and γ(θ) in (p,γ); parity from 882γ, M1+E2 to $5/2^+$, 1763 level in (α,pγ). $T_{1/2}$: lifetime τ=0.229 ps *30*: weighted average of 0.30 ps *9* from (α,pγ) (1969In04, DSAM), 0.255 ps *65* (1972Hu11, DSAM), 0.200 ps *30* (1976Me12, DSAM) in (p,γ), 0.27 ps *9* in (n,n′γ) (1989Ge09, DSAM), 0.185 ps *50* (1972Va06, DSAM), 0.26 ps *6* (1972Br45, DSAM), 0.35 ps *+10−7* (1969Du08, DSAM) in (p,p′γ), and 0.26 ps *9* in Coulomb excitation (1977Sc36, DSAM). Other: τ<1.3 ps in (^{28}Si,αpγ) (2007Ks01, DSAM).
2694.00 *20*	$3/2^+$	32 fs *6*	B LM O QRSTU WXy	XREF: Others: AA XREF: L(2693?)X(2676)y(2650) E(level): weighted average of 2693.8 *2* from ^{35}Ar ε decay, 2693.6 *13* from (α,pγ), 2693.9 *3* from (p,γ), 2694.50 *28* from (p,p′γ). J^{π}: spin=3/2 from γ(θ) in (p,p′γ); L=2 from 0^+ in (^{3}He,d) and (d,^{3}He). $T_{1/2}$: lifetime τ=46 fs *9*: unweighted average of 70 fs *30* in (α,pγ) from 1972Br33 with DSAM, 21 fs *3* in (α,pγ) from 1972Hu11 with DSAM, 19 fs *4* in (α,pγ) from 1973Fa07 with DSAM, 20 fs *4* in (p,γ) from 1976Me12 with DSAM, 65 fs *20* in (p,γ) from 1972Va06 with DSAM, 62 fs *16* in (p,p′γ) from 1969Du08 with DSAM, and 62 fs *8* in (n,n′γ) from 1989Ge09 with DSAM. Others: 115 fs *+95−59* in

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)†	J^{π}‡	$T_{1/2}$ or Γ&	XREF	Comments
				(p,p′γ) from 1969Du08 (using 931γ) is considered less reliable by the authors, and 11 fs *7* in (p,p′γ) from 1971Ca40 with DSAM is considered an outlier.
3002.64 *11*	5/2$^+$	22 fs *6*	B F HI LM O QRSTU WXY	E(level): weighted average of 3002.3 *3* from ^{35}Ar ε decay, 3003.1 *8* from (^{28}Si,αpγ), 3002.0 *10* from (^{16}O,αpγ), 3001.98 *22* from (^{14}N,αpnγ), 3002.6 *15* from (α,pγ), 3002.75 *8* from (p,γ), 3002.6 *5* from (p,p′γ). J^{π}: L=2 from 0$^+$ in (^{3}He,d), (d,^{3}He), and (pol d,^{3}He) with L+1/2 transfer from analyzing power. Discrepancy: L=3 from 0$^+$ in (d,n) and the angular distribution fit seems good. $T_{1/2}$: lifetime τ=32 fs *8*: unweighted average of 48 fs *9* in (α,pγ) from 1972Br33 with DSAM, 22 fs *3* in (α,pγ) from 1972Hu11 with DSAM, 14 fs *2* in (α,pγ) from 1973Fa07 with DSAM, 16 fs *4* in (p,γ) from 1976Me12 with DSAM, 72 fs *12* in (n,n′γ) from 1989Ge09 with DSAM, 18 fs *+26−15* in (p,p′γ) from 1972Va06 with DSAM, and 31 fs *13* in (p,p′γ) from 1969Du08 with DSAM.
3162.87 *9*	7/2$^−$	29.2 ps *12*	FGHI KLM O QR TU WX	T=1/2 E(level): weighted average of 3163.1 *6* from (^{28}Si,αpγ), 3163.43 *19* from (^{24}Mg,αpγ), 3162.9 *3* from (^{16}O,αpγ), 3162.64 *9* from (^{14}N,αpnγ), 3163.1 *11* from (α,pγ), 3162.99 *7* from (p,γ), 3162.75 *10* from (p,p′γ). J^{π}: spin=7/2 from pγ(θ) in (α,pγ) and γ(θ) in (p,γ) and (p,p′γ); L=3 from 0$^+$ in (d,n), (^{3}He,d), (d,^{3}He), and (pol d,^{3}He). $T_{1/2}$: lifetime τ=42.1 ps *17*: weighted average of 42 ps *2* (1974Va13, RDM), 41.7 ps *17* (1976Ke02, RDM), 45 ps *6* (1991Ja11, RDM) from ^{24}Mg(^{16}O, αpγ), 41.8 ps *18* from ^{27}Al(^{14}N, αpnγ) (1976Wa11, RDM), 60 ps *7* from (α,pγ) (1969In04, DSAM), 37 ps *4* from (p,γ), (1971Ba23, delayed coincidence), and 46 ps *6* from (p,p′γ) (1974Hu09, RDM). Others: τ=140 ps *40* (1968Az02, delayed coincidence; obtained using NaI(Tl) detectors with broad time resolution) from (p,γ).
3918.47 *12*	3/2$^+$	4.9 fs *14*	B M O RS u x	XREF: R(3908)u(3930)x(3948) E(level): weighted average of 3918.4 *5* from ^{35}Ar ε decay, 3915 *2* from (α,pγ), 3918.48 *10* from (p,γ). J^{π}: spin=3/2 determined by the feeding 5162γ(θ) from 5/2$^+$, 9081 level in (p,γ); parity=+ determined by 3918γ, M1+E2 to 3/2$^+$ ground state. L=2 from 0$^+$ in (pol d,^{3}He). $T_{1/2}$: lifetime τ=7 fs *2* from (p,γ), (1972Hu11, DSAM). Others: <22 fs from (α,pγ) (1972Br33, DSAM).
3942.9 *2*	9/2$^+$	0.18 ps *3*	F HI LM O	E(level): weighted average of 3942.0 *7* from (^{28}Si,αpγ), 3943.4 *10* from (^{16}O,αpγ), 3942.3 *10* from (^{14}N,αpnγ), 3943.4 *12* from (α,pγ), and 3942.9 *2* from (p,γ). J^{π}: spin=9/2 from pγ(θ) in (α,pγ); 2180.1γ E2 to 5/2$^+$. $T_{1/2}$: lifetime τ=0.26 ps *5*: weighted average of 0.33 ps *5* in (α,pγ) (1972Br33, DSAM), 0.39 ps *11* (1972Hu11,

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)[†]	J^{π}[‡]	$T_{1/2}$ or Γ[&]	XREF	Comments
				DSAM), 0.145 ps *50* (1973Fa07, DSAM), and 0.29 ps *5* (1976Me12, DSAM) from (p,γ). Other: τ<0.5 ps in (^{28}Si,αpγ) (2007Ks01, DSAM).
3967.6 *4*	1/2$^+$	11.8 fs *28*	B M O R	E(level): weighted average of 3967.7 *4* from ^{35}Ar ε decay, 3968 *2* from (α,pγ), and 3967.3 *6* from (p,γ). J$^\pi$: L=0 from 0$^+$ in (^{3}He,d). $T_{1/2}$: lifetime τ=17 fs *4*: weighted average of 13 fs *4* (1972Hu11, DSAM), 20 fs *5* (1973Fa07, DSAM), and 20 fs *4* (1976Me12, DSAM) in (p,γ). Other: τ<54 fs from (α,pγ) (1972Br33, DSAM).
4059.1 *4*	(3/2)$^-$	13.9 fs *21*	LM O QR	E(level): weighted average of 4056.9 *27* from (α,p), 4058 *3* from (α,pγ), 4059.2 *4* from (p,γ). J$^\pi$: L=1 from 0$^+$ in (^{3}He,d); (3/2$^-$) from FRESCO-DWBA analysis of measured $\sigma(\theta)$ of (α,p) (2019Se09). $T_{1/2}$: lifetime τ=20 fs *3*: weighted average of 31 fs *13* from (α,pγ) (1972Br33, DSAM), 22 fs *3* (1972Hu11, DSAM), 20 fs *4* (1973Fa07, DSAM), and 18 fs *3* (1976Me12, DSAM) from (p,γ).
4112.4 *16*	7/2$^+$	49 fs *11*	LM O Q	E(level): weighted average of 4111.1 *29* from (α,p), 4108 *2* from (α,pγ), and 4113.7 *10* from (p,γ). J$^\pi$: 4113.4γ E2(+M3) to 3/2$^+$. $T_{1/2}$: lifetime τ=70 fs *16* from (α,pγ) (1972Br33, DSAM).
4173.43 *23*	(5/2$^+$)	35 fs *6*	M O q	T=1/2 XREF: q(4170) E(level): weighted average of 4171 *2* from (α,pγ) and 4173.45 *19* from (p,γ). J$^\pi$: based on primary transitions from proton resonances in (p,γ): 3827.3γ from 8001.0, (7/2$^+$), 4456.4γ from 8630.1, 7/2$^-$ level, 4243.0γ from 8416.7, 1/2$^+$ level. Note that primary 3099.1γ from 7272.7, 1/2$^-$ resonance disfavors 5/2$^+$ assignment, but this γ branch is relatively weak. $T_{1/2}$: lifetime τ=51 fs *8*: weighted average of 68 fs *24* in (α,pγ) (1972Br33, DSAM), 105 fs *+35−30* (1971Wi13, DSAM), 55 fs *6* (1972Hu11, DSAM), and 34 fs *9* (1976Me12, DSAM) in (p,γ).
4177.9 *2*	3/2$^-$	27.0 fs *28*	L O qR	XREF: q(4170)R(4167) J$^\pi$: spin=3/2 from $\gamma(\theta)$ in (p,γ); L=1 from 0$^+$ in (^{3}He,d). $T_{1/2}$: lifetime τ=39 fs *4*: weighted average of 32 fs *5* (1972Hu11, DSAM), 42 fs *4* (1973Fa07, DSAM), and 47 fs *10* (1976Me12, DSAM) from (p,γ).
4347.76 *7*	9/2$^-$	1.0 ps *6*	FGHI LM O	E(level): weighted average of 4347.8 *6* from (^{28}Si,αpγ), 4348.29 *24* from (^{24}Mg,αpγ), 4347.8 *3* from (^{16}O,αpγ), 4347.43 *17* from (^{14}N,αpnγ), 4347.8 *12* from (α,pγ), and 4347.76 *6* from (p,γ). J$^\pi$: spin from $\gamma(\theta)$ in (p,γ); parity from 1184.6γ, M1+E2 to 7/2$^-$, 3163 level in (p,γ) (1974Lo17). $T_{1/2}$: lifetime τ=1.5 ps *8*: unweighted average of 1.31 ps *+30−23* from (^{28}Si,αpγ) (2007Ks01, DSAM), 2.9 ps *+14−7* (1972Br33, DSAM), and 0.33 ps *6* (1976Me12, DSAM) from (p,γ).
4624.3 *3*	3/2,5/2	40 fs *17*	B LM O	XREF: B(?) E(level): other: 4625.3 *28* from (α,p); 4618 *2* from

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)[†]	J^{π}[‡]	$T_{1/2}$ or Γ[&]	XREF	Comments
				(α,pγ) is discrepant. J^{π}: from p$\gamma(\theta)$ in (α,pγ). $T_{1/2}$: lifetime τ=58 fs *24* from (α,pγ) in 1972Br33 with DSAM.
4769.86 *23*	(5/2$^-$)	76 fs *20*	LM O	E(level): others: 4770.4 *26* from (α,p), 4770 *15* from (α,pγ). J^{π}: from primary transitions in (p,γ): 4059.2γ from 8829.3, 1/2$^-$ level, 2778.9γ from 7548.9, 7/2$^-$ level, 3231.0γ from 8001.0, (7/2$^+$) level. Other: 7/2,9/2 from p$\gamma(\theta)$ for a group at E=4770 *15* in (α,pγ) is inconsistent and disfavored by γ rays to 1/2$^+$ and 3/2$^+$. $T_{1/2}$: lifetime τ=110 fs *29*: weighted average of 270 fs *95* (1972Hu11, DSAM) and 105 fs *17* (1976Me12, DSAM) in (p,γ).
4839.10 *11*	(1/2$^+$,3/2)	9.7 fs *35*	O x	XREF: x(4839) J^{π}: from primary γ transitions in (p,γ): 2339.4γ from 7178.6, 1/2$^{(+)}$ level, 2355.4γ from 7194.6, 1/2$^-$ level, 2867.2γ from 7706.4, 5/2$^+$ level. $T_{1/2}$: lifetime τ=14 fs *5* from (p,γ) (1976Me12, DSAM).
4854.4 *19*	(1/2,3/2)	4.9 fs *14*	L O x	XREF: x(4839) E(level): other: 4859.2 *28* from (α,p). J^{π}: from primary γ transitions in (p,γ): 2340.1γ from 7194.6, 1/2$^-$ level, 3183.9γ from 8038.4, 1/2$^+$,3/2$^+$ level; decay 3634.8 γ to 1219.38, 1/2$^+$. $T_{1/2}$: lifetime τ=7 fs *2* from (p,γ) (1976Me12, DSAM).
4880.96 *15*	7/2$^{(+)}$	5.6 fs *14*	LM O	E(level): others: 4881.8 *29* from (α,p), 4880.8 *21* from (α,pγ). J^{π}: spin=7/2 from p$\gamma(\theta)$ in (α,pγ); 3417.1 primary γ from 8298.2, 3/2$^+$ resonance in (p,γ). $T_{1/2}$: lifetime τ=8 fs *2*: weighted average of 7 fs *2* (1972Hu11, DSAM), 7 fs *3* (1973Fa07, DSAM), and 10 fs *3* (1976Me12, DSAM) from (p,γ). Other: τ=0.28 ps *6* from (α,pγ) in 1972Br33 with DSAM.
5009.1 *18*	(1/2,3/2)	7.6 fs *21*	LM O R	E(level): weighted average of 5006.9 *27* from (α,p) and 5010.1 *18* from (p,γ). Others: 5015 *20* from (α,pγ) and 5010 *15* from (^{3}He,d). J^{π}: primary 2185.4γ from 7194.6, 1/2$^-$ level and 3407.4γ from 8416.7, 1/2$^+$ level. $T_{1/2}$: lifetime τ=11 fs *3* from (p,γ) (1976Me12, DSAM).
5158.6 *29*	3/2$^+$,5/2$^+$		Lm R WX	XREF: m(5175) E(level): weighted average of 5158.9 *29* from (α,p), 5163 *15* from (^{3}He,d), 5150 *20* from ^{36}Ar(d,^{3}He), and 5155 *11* from ^{36}Ar(pol d,^{3}He). J^{π}: L=2 from 0$^+$ in (pol d,^{3}He) and (d,^{3}He).
5163.34 *22*	7/2$^-$		m O	XREF: m(5175) J^{π}: spin=7/2 from the feeding 3466$\gamma(\theta)$ from 8630, 7/2$^-$ level and RUL in (p,γ); parity=$-$ from 3466γ, M1+E2 from 8630, 7/2$^-$ level.
5214.5 *20*	(5/2$^+$)[#]	<4.85 fs	LM O	XREF: M(5230) E(level): weighted average of 5209.5 *28* from (α,p),

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)[†]	J^{π}[‡]	$T_{1/2}$ or Γ[&]	XREF	Comments
				5230 *20* from (α,pγ), and 5215.8 *15* from (p,γ). J^{π}: other: 3/2,5/2 from p$\gamma(\theta)$ in (α,pγ). $T_{1/2}$: lifetime τ<7 fs from (p,γ) (1976Me12, DSAM).
5403.3 *10*	3/2^{-}	11.8 fs *28*	L O R	E(level): weighted average of 5402.0 *29* from (α,p) and 5403.5 *10* from (p,γ). Other: 5409 *12* from (^{3}He,d). J^{π}: L=1 from 0^{+} in (^{3}He,d); 2400.6γ to 5/2^{+} rules out 1/2^{-} based on RUL, since it would require a large B(M2)(W.u.) exceeding RUL. $T_{1/2}$: lifetime τ=17 fs *4* from (p,γ) (1976Me12, DSAM). The contribution of Iγ to unknown levels adds up to 36% of Iγ(4183)=100 *18* from (p,γ).
5407.07[a] *21*	11/2^{-}	0.28 ps *7*	FGHI LM	E(level): weighted average of 5407.4 *8* from (^{28}Si,αpγ), 5407.64 *27* from (^{24}Mg,αpγ), 5407.3 *3* from (^{16}O,αpγ), 5406.68 *19* from (^{14}N,αpnγ), and 5407.1 *15* from (α,pγ). J^{π}: spin=11/2 from p$\gamma(\theta)$ in (p,γ); parity from 1059γ, M1+E2 to 9/2^{-}, 4347 level. $T_{1/2}$: lifetime τ=0.4 ps *1* from (α,pγ) (1974Lo17, DSAM). Other: τ<1.1 ps from (^{28}Si,αpγ) (2007Ks01, DSAM).
5531 *5*			LM O	E(level): weighted average of 5531 *5* from (α,p) and 5535 *20* from (α,pγ).
5586.0 *20*	5/2^{+}		Lm O wx	XREF: m(5600)w(5600)x(5583) E(level): weighted average of 5586.0 *26* from (α,p) and 5586 *2* from (p,γ). J^{π}: L=2 from 0^{+} in (pol d,^{3}He) and L+1/2 transfer from analyzing power.
5598.4 *22*	3/2^{+},5/2^{+}	2.1 fs *7*	Lm O wx	XREF: m(5600)w(5600)x(5583) E(level): weighted average of 5594.6 *26* from (α,p) and 5599.7 *15* from (p,γ). J^{π}: L(d,^{3}He)=2 from 0^{+}. $T_{1/2}$: lifetime τ=3 fs *1* from (p,γ) (1976Me12, DSAM).
5633.6 *30*			Lm r w y	XREF: m(5650)r(5660)w(5600)y(5650) E(level): from (α,p).
5646 *2*	(5/2,7/2,9/2^{+})	2.8 fs *7*	Lm O r y	XREF: m(5650)r(5660)y(5650) E(level): weighted average of 5645 *4* from (α,p) and 5646 *2* from (p,γ). J^{π}: from primary transitions in (p,γ): 1903γ from 7548.9, 7/2^{-} level, 2355γ from 8001.0, (7/2^{+}) level, 3261γ from 8906.8, 5/2^{+} level. $T_{1/2}$: lifetime τ=4 fs *1* from (p,γ) in 1976Me12 with DSAM.
5655 *2*	(3/2)$^{+}$	13.9 fs *35*	Lm O r w y	T=3/2 XREF: m(5650)r(5660)w(5600)y(5650) E(level): weighted average of 5653 *4* from (α,p) and 5655 *2* from (p,γ). J^{π}: L=2 from 0^{+} in (^{3}He,d); primary 2142γ from 7797.2, 1/2^{-} level in (p,γ). Note that a weak primary 1903γ from 7548.9, 7/2^{-} level in

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)†	J^{π}‡	$T_{1/2}$ or Γ&	XREF	Comments
				(p,γ) favors 5/2$^+$. $T_{1/2}$: lifetime τ=20 fs *5* from (p,γ) (1976Me12, DSAM). The contribution of Iγ to unknown levels adds up to 18% of Iγ(2652)=100 *6* from (p,γ).
5682.2 *20*	1/2$^-$,3/2$^-$		Lm O R	XREF: m(5650) E(level): weighted average of 5679.7 *34* from (α,p), 5683 *2* from (p,γ), and 5684 *12* from (^{3}He,d). J^{π}: L=1 from 0$^+$ in (^{3}He,d). The contribution of Iγ to unknown levels adds up to 18% of Iγ(1504)=100 from (p,γ).
5723.52 *18*	5/2$^+$		L O WX	E(level): others: 5731.2 *25* from (α,p), 5720 *20* from ^{36}Ar(d,^{3}He), and 5720 *11* from ^{36}Ar(pol d,^{3}He). J^{π}: L=2 from 0$^+$ in (pol d,^{3}He) and (d,^{3}He) and L+1/2 transfer from analyzing power. The contribution of Iγ to unknown levels adds up to 100% of Iγ(3960)=100 *11* from (p,γ).
5758 *3*	(1/2,3/2$^-$)		L O R	E(level): weighted average of 5757 *4* from (α,p), 5758 *3* from (p,γ), and 5760 *12* from (^{3}He,d). J^{π}: L(^{3}He,d)=(0,1) from 0$^+$. The contribution of Iγ to unknown levels adds up to 56% of Iγ(5758)=100 *22* from (p,γ).
5806.6 *20*	(1/2,3/2,5/2)	3.5 fs *7*	L O	E(level): weighted average of 5808.1 *31* from (α,p) and 5806 *2* from (p,γ). J^{π}: primary 2890.2γ from 8696.9, 3/2$^-$ level; 5806.1γ to 3/2$^+$ g.s. $T_{1/2}$: lifetime τ=5 fs *1* from (p,γ) (1976Me12, DSAM).
5830 *4*	(5/2$^-$)#		LM	XREF: M(5850) E(level): weighted average of 5829 *4* from (α,p) and 5850 *25* from (α,pγ). J^{π}: other: spin=5/2,9/2 from pγ(θ) in (α,pγ).
5926.65 *29*	(11/2)$^-$	<0.28 ps	F HI LM	E(level): weighted average of 5927.1 *7* from (^{28}Si,αpγ), 5926.9 *5* from (^{16}O,αpγ), 5926.47 *29* from (^{14}N,αpnγ), 5928 *3* from (α,p), and 5927.1 *19* from (α,pγ). J^{π}: 1579γ M1+E2, ΔJ=1 to 4348, 9/2$^-$ in ^{12}C(^{28}Si,αpγ), also assuming spins increase with excitation energies in fusion-evaporation reactions. Spin=7/2,11/2 from pγ(θ) in (α,pγ). $T_{1/2}$: lifetime τ<0.4 ps from (^{28}Si,αpγ) (2007Ks01, DSAM).
6087.31 *24*	13/2$^-$	6.2 ps *6*	FGHI LM	E(level): weighted average of 6087.6 *8* from (^{28}Si,αpγ), 6088.05 *29* from (^{24}Mg,αpγ), 6087.2 *4* from (^{16}O,αpγ), 6086.89 *23* from (^{14}N,αpnγ), 6084.2 *29* from (α,p), and 6087.0 *18* from (α,pγ). J^{π}: spin=13/2 from pγ(θ) in (α,pγ); parity from 680γ, M1+E2 to 11/2$^-$, 5406 level in (α,pγ). $T_{1/2}$: lifetime τ=8.9 ps *8*: weighted average of 9.3 ps *8* from (α,pγ) in 1974Lo17 with RDM and 7.7 ps *14* from (^{14}N,αpnγ) in 1976Wa11.
6106.2 *21*	(3/2,5/2$^+$)	8.3 fs *21*	l O wx	XREF: l(6104)w(6140)x(6136) E(level): weighted average of 6104 *4* from (α,p)

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)[†]	J^{π}[‡]	XREF	Comments
			and 6106.2 *21* from (p,γ). J^{π}: 4886.5γ to 1/2$^+$, 4342.8γ to 5/2$^+$; primary 1675.5γ from 7781.7, (5/2$^-$) level. $T_{1/2}$: lifetime τ=12 fs *3* from (p,γ) in 1976Me12 with DSAM.
6141 *4*	5/2$^+$	L wx	XREF: w(6140)x(6136) J^{π}: L=2 from 0$^+$ in (d,^{3}He) and (pol d,^{3}He), and L+1/2 transfer from analyzing power.
6181 *3*	(1/2$^+$ to 7/2$^+$)	L nO	XREF: n(6200) E(level): other: 6180 *3* from (α,p). J^{π}: 6180γ to 3/2$^+$ g.s.; primary 1438γ from 7618.7, 5/2$^+$ level in (p,γ). Apart from the γ branch Iγ(6180)=100 observed in (p,γ), the contribution of Iγ to unknown levels adds up to 122% of Iγ(6180).
6224.4 *30*		L n	XREF: n(6200) J^{π}: L(α,d)=6 for a group at 6200 *10*.
6283 *3*	(5/2$^+$)[#]	L R	E(level): weighted average of 6283 *3* from (α,p) and 6284 *4* from (^{3}He,d). J^{π}: other: L=(2) from 0$^+$ in (^{3}He,d).
6329 *4*	(1/2,3/2$^-$)	R	J^{π}: L=(0,1) from 0$^+$ in (^{3}He,d).
6380.2 *9*	(9/2$^-$)[#]	H L R	XREF: R(6377) E(level): weighted average of 6380.8 *8* from (^{16}O,αpγ) and 6379.3 *25* from (α,p). J^{π}: other: L=(2,3) from 0$^+$ in (^{3}He,d) for a group at E=6377 *2* based on very few data points in a very narrow range of angles in the DWBA fit.
6401.2 *22*	(1/2$^-$)[#]	LM	
6428 *2*	(1/2$^+$)[#]	L R	E(level): weighted average of 6429 *3* from (α,p) and 6427 *2* from (^{3}He,d). J^{π}: other: L=(3) from 0$^+$ in (^{3}He,d) is inconsistent.
6469 *3*	(1/2$^-$,3/2$^-$)	L R	E(level): weighted average of 6475 *4* from (α,p) and 6468 *2* from (^{3}He,d). J^{π}: L=(1) from 0$^+$ in (^{3}He,d).
6492 *2*	(3/2$^+$)[#]	L O R	E(level): weighted average of 6492 *3* from (α,p), 6492 *3* from (p,γ), and 6491 *2* from (^{3}He,d). J^{π}: other: L=(2) from 0$^+$ in (^{3}He,d).
6546 *2*	(1/2$^+$)[#]	L R	E(level): weighted average of 6549 *3* from (α,p) and 6545 *2* from (^{3}He,d). J^{π}: other: L=(0,1) from 0$^+$ in (^{3}He,d), indicating (1/2,3/2$^-$).
6643 *2*	(1/2$^-$,3/2$^-$)	D R	E(level): from (^{3}He,d). J^{π}: from L=(1) from 0$^+$ in (^{3}He,d).
6659.1 *30*	(7/2$^+$)[#]	L y	XREF: y(6677)
6660.2 *8*	(11/2$^-$)	F	J^{π}: 2312.4γ to 9/2$^-$, 3497.1γ to 7/2$^-$ in (^{28}Si,αpγ), assuming spin increasing as excitation energy increases.
6675.6 *25*		D L R y	XREF: y(6677) E(level): weighted average of 6679.4 *31* from (α,p) and 6674 *2* from (^{3}He,d).
6761 *2*	3/2$^+$,5/2$^+$	D R WX	XREF: X(6745) E(level): weighted average of 6761 *2* from (^{3}He,d), 6750 *20* from ^{36}Ar(d,^{3}He), and 6745 *12* from ^{36}Ar(pol d,^{3}He). J^{π}: L(d,^{3}He)=2 from 0$^+$.
6779.0 *20*		D L R	E(level): weighted average of 6781.3 *30* from (α,p) and 6778 *2*

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)†	Jπ‡	$T_{1/2}$ or Γ&	XREF	Comments
				from (^{3}He,d).
6800 *4*			D L	XREF: D(6802) E(level): from (α,p).
6823 *2*			D R	E(level): from (^{3}He,d).
6842 *2*			D L R	E(level): weighted average of 6842 *4* from (α,p) and 6842 *2* from (^{3}He,d).
6863 *3*	$(9/2^+)$#		L	
6866.7 *6*	$(1/2,3/2^-)$		D O R	E(level): other: 6866 *2* from (^{3}He,d). Jπ: L(^{3}He,d)=(0,1) from 0^+.
6891.6 *30*	$(9/2^+)$#		L	
6948.1 *34*	$5/2^+$		L WX	XREF: W(6930) E(level): weighted average of 6948.6 *34* from (α,p) and 6930 *20* from ^{36}Ar(d,^{3}He). Other: 6953 *43* from ^{36}Ar(pol d,^{3}He). Jπ: L(pol d,^{3}He)=2 and L+1/2 from analyzing power.
6986.3 *34*	$(9/2^+)$#		L	
7066.2 *7*	$(5/2^+)$#		L O R	E(level): weighted average of 7066.2 *7* from (p,γ), and 7066 *2* from (^{3}He,d).
7103.4 *7*	$3/2^{(-)}$		L O R	XREF: L(7105?) E(level): weighted average of 7103.4 *7* from (p,γ), and 7103 *2* from (^{3}He,d). Jπ: spin=3/2 from γ(θ) in (p,γ); parity from L(^{3}He,d)=(1,3) from 0^+.
7121.7 *24*			L	
7178.6 *8*	$1/2^{(+)}$		O	T=3/2 Jπ: spin=1/2 from γ(θ) in (p,γ); parity from IAS in ^{35}S.
7179 *4*	$(7/2)^+$#		L N	E(level): weighted average of 7180 *4* from ^{32}S(α,p) and 7170 *10* from ^{33}S(α,d). Jπ: also L=6 from $3/2^+$ in ^{33}S(α,d), indicating $(7/2:17/2)^+$.
7185	$5/2^+$		O R X	XREF: R(7178)X(7181) Jπ: L(pol d,^{3}He)=2 and L+1/2 from analyzing power for a group at E=7181 *60*. Other: L(^{3}He,d)=(2) from 0^+ for a group at E=7178 *2* considered the same level.
7194.6 *7*	$1/2^-$	0.027 keV *8*	OP R	Jπ: from resonance analysis of measured σ(θ) (1971BrXT) in ^{34}S(p,p),(p,p′γ):resonances. Other: 1/2,3/2 from γ(θ) in (p,γ).
7213.7 *25*			L y	XREF: y(7250)
7225.5 *8*	5/2		O y	XREF: y(7250) Jπ: from γ(θ) in (p,γ).
7228.2 *20*	$(3/2^+)$#		L R y	XREF: R(7227)y(7250) E(level): weighted average of 7230.9 *30* from ^{32}S(α,p), 7227 *2* from ^{34}S(^{3}He,d). Jπ: other: L=(0,1) from 0^+ in (^{3}He,d) is inconsistent.
7234.0 *8*	$5/2^{(+)}$		O	Jπ: spin=5/2 from γ(θ) in (p,γ); primary 6014.1γ to $1/2^+$.
7269.2 *1*			O	
7272.7 *9*	$1/2^-$	0.014 keV *4*	OP R y	XREF: P(7274)R(7273)y(7250)

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)†	J^{π}‡	$T_{1/2}$ or Γ&	XREF	Comments
				J^{π}: from resonance analysis of measured $\sigma(\theta)$ (1971BrXT) in ^{34}S(p,p),(p,p′γ):resonances. Other: (1/2,3/2) from $\gamma(\theta)$ in (p,γ).
7348 *3*	(7/2)#		L	
7361.9 *8*	3/2$^{(+)}$		L O R	E(level): weighted average of 7362 *3* from (α,p), 7362.0 *8* from (p,γ), and 7361 *2* from (^{3}He,d). J^{π}: spin=3/2 from $\gamma(\theta)$ in (p,γ); 6141.9γ (M1+E2) to 1/2$^+$.
7396.4 *10*	7/2$^{(-)}$		O R	E(level): weighted average of 7396.0 *10* from (p,γ), and 7398 *2* from (^{3}He,d). J^{π}: spin=7/2 from $\gamma(\theta)$ in (p,γ); 4233.3γ (M1+E2) to 7/2$^-$.
7413.8 *30*			L	
7450.9 *8*	3/2		L O	E(level): weighted average of 7447 *3* from (α,p) and 7451.1 *6* from (p,γ). J^{π}: from $\gamma(\theta)$ in (p,γ).
7496			O	
7501.1 *8*	(5/2$^+$,7/2,9/2$^+$)		l O	XREF: l(7503) J^{π}: primary 3558.0γ to 9/2$^+$, 4498.2γ to 5/2$^+$.
7502.9 *7*			l O	XREF: l(7503)
7518.8 *7*	7/2		O	J^{π}: from $\gamma(\theta)$ in (p,γ).
7548.9 *6*	7/2$^-$	<0.69 fs	OP	T=3/2 XREF: P(7550) J^{π}: from resonance analysis of measured $\sigma(\theta)$ in ^{34}S(p,p),(p,p′γ):resonances. Spin=7/2 also from $\gamma(\theta)$ in (p,γ). $T_{1/2}$: lifetime τ<1 fs from (p,γ) (1973Fa07, DSAM). Other: Γ_p=4.6 eV *10* and Γ=5 eV *+4−2* from ^{34}S(p,p) elastic scattering excitation function analysis (1971BrXT). Other Γ_p, Γ_γ, Γ values/ratios are mostly deduced from (p,γ) resonance strengths, which have large discrepancies. See comments in the (p,γ) dataset for details.
7561.3 *6*	1/2,3/2		L O	E(level): other: 7566.5 *35* from (α,p). J^{π}: from $\gamma(\theta)$ in (p,γ).
7565 *20*	5/2$^+$		X	J^{π}: L=2 from 0$^+$ in (pol d,^{3}He) and L+1/2 transfer from analyzing power.
7590 *4*			L	
7600.8 *8*	5/2$^+$	<13.9 fs	O	J^{π}: spin=5/2 from $\gamma(\theta)$ in (p,γ); 7599.9γ M1+E2 to 3/2$^+$. $T_{1/2}$: lifetime τ<20 fs from (p,γ) (1971Wi13, DSAM), which is equivalent to Γ>0.033 eV.
7618.7 *5*	5/2$^+$		O	J^{π}: spin=5/2 from $\gamma(\theta)$ in (p,γ); 7617.8γ M1+E2 to 3/2$^+$.
7648.1 *30*			L	
7656.6 *8*	(1/2,3/2,5/2$^+$)		O	J^{π}: primary 3688.8γ to 1/2$^+$.
7670 *10*	(7/2 to 17/2)$^+$		N	J^{π}: L(α,d)=6 from 3/2$^+$.
7671.9 *8*	5/2$^+$		l Op	XREF: p(7686) J^{π}: spin=5/2 from $\gamma(\theta)$ in (p,γ); 4668.8γ, M1+E2 to 5/2$^+$; 5908.2γ, M1+E2 to 5/2$^+$; other: 3324.0γ to 9/2$^-$ seems to favor 5/2$^-$.
7685.5 *8*	3/2$^-$		OP	J^{π}: spin=3/2 from $\gamma(\theta)$ in (p,γ) and 3/2$^-$ from R-Matrix analysis in (p,p):resonances. Γ_p=446 eV *15* from (p,p):resonances.

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)†	Jπ‡	T1/2 or Γ&	XREF	Comments
7693.9 *8*	(1/2+,3/2,5/2)		O	Jπ: primary 7693.0γ to 3/2+, 3634.6γ to (3/2)−, 4690.9γ to 5/2+.
7706.4 *8*	5/2+		L OP	E(level): other: 7709.9 *25* from (α,p). Jπ: spin=5/2 from γ(θ) in (p,γ) and 3/2+,5/2+ from R-Matrix analysis in (p,p):resonances. Γ_p=4 eV *1* in (p,p).
7744.8 *9*	7/2−		L O	E(level): other: 7744.1 *30* from (α,p). Jπ: spin=7/2 from γ(θ) in (p,γ); 4581.6γ M1+E2 to 7/2−.
7748	(7/2 to 17/2)+		NO	XREF: N(7750) E(level): from (p,γ). Jπ: L(α,d)=6 from 3/2+.
7777.1 *9*	(5/2+)		L O	E(level): from (p,γ). Other: 7770.3 *33* from (α,p). Jπ: based on primary transitions: 4613.9γ to 7/2−, 5131.0γ to 7/2+, 6557.1γ to 1/2+.
7781.7 *11*	(5/2−)		O	Jπ: primary 3433.8γ to 9/2−; relatively weak 6561.7γ to 1/2+.
7797.2 *9*	1/2−	0.031 keV *10*	L OP	E(level): weighted average of 7799.8 *30* from (α,p), 7797.1 *9* from (p,γ), and 7796 *3* from ^{34}S(p,p),(p,p′γ):resonances. Jπ: spin=1/2 from γ(θ) in (p,γ); (1/2)− from R-Matrix analysis of in (p,p):resonances.
7837.2 *10*	3/2−	3.7 keV *4*	OP	T=3/2 E(level): weighted average of 7837.2 *10* from (p,γ), and 7837 *3* from ^{34}S(p,p),(p,p′γ):resonances. Jπ: spin=3/2 from γ(θ) in (p,γ) and 3/2− from R-Matrix analysis in (p,p):resonances. Γ: other: lifetime τ<5 fs from (p,γ) (1971Wi13, DSAM).
7839.0 *9*	(5/2+)		O	Jπ: primary 6619.0γ to 1/2+, 3895.9γ to 9/2+.
7868.7 *10*	(1/2+,3/2,5/2+)		L O	E(level): other: 7868.5 *35* from (α,p). Jπ: primary 6648.6γ to 1/2+, 2654.1γ to (5/2+).
7873.03[b] *30*	13/2+		F HI NO	XREF: N(7870) E(level): weighted average of 7873.3 *8* from (^{28}Si,αpγ), 7873.4 *4* from (^{16}O,αpγ), and 7872.78 *30* from (^{14}N,αpnγ). Jπ: 1946.35γ E1+M2, ΔJ=1 to 5927, (11/2)−; 1786.2γ D(+Q) to 13/2−; band head.
7880.8 *9*	3/2+,5/2+	0.008 keV *4*	OP	Jπ: from γ(θ) in (p,γ) and R-Matrix analysis in (p,p):resonances.
7885			O	
7889.1 *15*			O	
7899.2 *7*	(5/2)		l O	XREF: l(7901) Jπ: primary 2735.8γ to 7/2−, 3018.1γ to

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)†	Jπ‡	$T_{1/2}$ or Γ&	XREF	Comments
				$7/2^{(+)}$, 3721.1γ to $3/2^-$, 5204.9γ to $3/2^+$.
7903			l O	XREF: l(7901)
7923.3 *9*	$(3/2^+,5/2^+)$		O	Jπ: primary 5277.2γ to $7/2^+$, 6703.2γ to $1/2^+$.
7930			O	
7949			O	
7970.3 *10*	$(5/2^-)$		O	Jπ: primary 3622.3γ to $9/2^-$, weak 6750.2γ to $1/2^+$.
7978.6 *35*			L	
7987.8 *10*	$3/2^+$		O	Jπ: spin=3/2 from γ(θ) in (p,γ); 7986.8γ M1+E2 to $3/2^+$.
7995.6 *10*	(5/2)		O	Jπ: primary 2592.2γ to $3/2^-$, 7994.6γ to $3/2^+$, 2832.1γ to $7/2^-$, 5349.5γ to $7/2^+$.
8001.0 *9*	$(7/2^+)$		L O	E(level): other: 8000.7 *41* from (α,p). Jπ: from primary transitions in (p,γ): 3653.0γ to $9/2^-$, 4057.9 to $9/2^+$, 3827.3γ to $5/2^-$, 6237.4γ to $5/2^+$, 8000.0γ to $3/2^+$.
8005.0 *10*	$5/2^+$	0.011 keV *5*	OP WX	XREF: P(8004)W(8010)X(8007) Jπ: L(d,^{3}He)=2 from 0^+; spin=5/2 from γ(θ) in (p,γ).
8010 *10*	$(7/2\text{ to }17/2)^+$		N	Jπ: L(α,d)=6 from $3/2^+$.
8019.2 *35*			L	
8035.7 *10*	$1/2^-,3/2^-$	0.026 keV *7*	OP	E(level): other: 8034 *3* from ^{34}S(p,p),(p,p′γ):resonances.
8038.4 *11*	$1/2^+,3/2^+$	0.30 keV *2*	OP	E(level): weighted average of 8038.5 *11* from (p,γ), and 8038 *3* from ^{34}S(p,p),(p,p′γ):resonances. Jπ: spin=3/2 from γ(θ) in (p,γ), but $1/2^+$ from R-Matrix analysis in (p,p); 8037.5γ M1+E2 to $3/2^+$.
8075.9 *10*	(5/2)		L O	E(level): weighted average of 8075.1 *46* from (α,p) and 8075.9 *10* from (p,γ). Jπ: primary 3897.8γ to $3/2^-$, 8074.9γ to $3/2^+$; 2912.4g to $7/2^-$, 3962.0γ to $7/2^+$.
8080			O	
8096.1 *11*	(5/2)		O	Jπ: primary 2932.6γ to $7/2^-$, 5450.0γ to $7/2^+$; 3918.0γ to $3/2^-$, 4177.4γ to $3/2^+$.
8100 *10*	$(7/2\text{ to }17/2)^+$		N	Jπ: L(α,d)=6 from $3/2^+$.
8106.4 *11*	$3/2^+$		L O	E(level): other: 8103.8 *42* from (α,p). Jπ: spin=3/2 from γ(θ) in (p,γ); 8105.4γ M1+E2 to $3/2^+$.
8113.3 *11*	$(1/2^+,3/2,5/2^+)$		L O	XREF: L(8121?) Jπ: primary 2897.4γ $(5/2^+)$, 4145.4γ to $1/2^+$.
8147.2 *11*	$3/2^-$	0.56 keV *6*	OP	XREF: P(8150) Jπ: from R-Matrix analysis in (p,p); also proposed by 1976Me12 in (p,γ) based on strengths of primary γ rays. Γ: from ^{34}S(p,p),(p,p′γ):resonances in 1977Ou01.
8148 *3*	$1/2^-$	2.66 keV *27*	P	
8150 *3*	$3/2^-$	0.56 keV *6*	P	
8156.8 *11*	$(5/2^+)$		O	Jπ: primary 3978.7γ to $3/2^-$, 8155.8γ to $3/2^+$, 4213.6γ to $9/2^+$.
8171.7 *42*			L	

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)[†]	J$^{\pi}$[‡]	$T_{1/2}$ or Γ[&]	XREF	Comments
8179.4 *12*	(3/2$^-$,5/2,7/2$^+$)		O	J$^{\pi}$: primary 5016.1γ to 7/2$^-$, 5484.9γ to 3/2$^+$.
8208.2 *10*	5/2$^+$	0.033 keV *10*	OP WX	T=3/2 XREF: W(8210)X(8204) E(level): other: 8209 *3* from ^{34}S(p,p),(p,p′γ):resonances. J$^{\pi}$: L(pol d,^{3}He)=2 and L+1/2 from analyzing power.
8211 *3*	1/2$^+$	0.094 keV *15*	L OP	XREF: O(8209) E(level): from (p,p):resonances. Other: 8210.2 *46* from (α,p).
8216.3 *10*	5/2$^+$	0.014 keV *3*	l OP wx	XREF: w(8210)x(8204) J$^{\pi}$: from γ(θ) in (p,γ) and R-Matrix analysis in (p,p):resonances.
8242.4 *12*	3/2$^-$	0.140 keV *15*	l OP	XREF: l(8246.0)P(8243) J$^{\pi}$: from γ(θ) in (p,γ) and R-Matrix analysis in (p,p):resonances.
8251 *5*	(3/2$^+$,5/2$^+$)		l O	XREF: l(8246.0) J$^{\pi}$: primary 4138γ to 7/2$^+$, 7031γ to 1/2$^+$.
8269.0 *5*	5/2$^+$	0.005 keV *3*	L OP	XREF: L(8270.6) E(level): other: 8271 *3* from ^{34}S(p,p),(p,p′γ):resonances. J$^{\pi}$: 3/2$^+$,5/2$^+$ from R-Matrix analysis in (p,p):resonances; primary 5105.5γ to 7/2$^-$ rules out 3/2$^+$, since it would require Mult=M2 which is unlikely based on RUL and Γ=0.005 keV.
8277.2 *5*	(5/2)$^+$	0.006 keV *3*	OP	XREF: P(8279) J$^{\pi}$: 3/2$^+$,5/2$^+$ from R-Matrix analysis in (p,p):resonances; primary 3939.2γ to 9/2$^-$ disfavors both 3/2$^+$ and 5/2$^+$, but 3/2$^+$ is much less likely.
8282.4 *5*	(3/2$^-$,5/2)		O	J$^{\pi}$: primary 3118.9γ to 7/2$^-$, 4104.2γ to 3/2$^-$, 8281.4γ to 3/2$^+$.
8287.82 *30*	1/2$^-$	0.04 keV *1*	L OP	XREF: L(8292)P(8290) J$^{\pi}$: from γ(θ) in (p,γ) and R-Matrix analysis in (p,p):resonances.
8297.2 *9*	3/2$^-$	0.073 keV *15*	OP	XREF: P(8300)
8298.2 *5*	3/2$^+$		l O	XREF: l(8292) J$^{\pi}$: spin=3/2 from γ(θ) in (p,γ); 8297.1γ M1+E2 to 3/2$^+$.
8318.1 *3*	(5/2$^+$)		L O	XREF: L(8321.0) J$^{\pi}$: primary 4350.2γ to 1/2$^+$, 5154.8γ to 7/2$^-$, 5671.9γ to 7/2$^+$.
8319.6[a] *5*	15/2$^-$	12.5 fs *7*	F H	E(level): weighted average of 8319.4 *9* from (^{28}Si,αpγ) and 8319.7 *5* from (^{16}O,αpγ). J$^{\pi}$: 2911.9γ E2, ΔJ=2 to 11/2$^-$, 2232.7γ to 13/2$^-$; band assignment. $T_{1/2}$: lifetime τ=18 fs *1* from (^{28}Si,αpγ) (2013Bi10, DSAM). $T_{1/2}$: from (HI,xnγ).
8323.1 *13*	(3/2$^+$,5/2$^+$)		L O	XREF: L(8321.0) J$^{\pi}$: primary 4210.4γ to 7/2$^+$, 7103.0γ to 1/2$^+$.
8346 *5*			O	

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)†	J^π‡	$T_{1/2}$ or Γ&	XREF	Comments
8381.8 *8*	$5/2^+$	0.023 keV *7*	OP	XREF: P(8383) J^π: from $\gamma(\theta)$ in (p,γ) and R-Matrix analysis in (p,p):resonances.
8389.4 *16*			O	
8402.9 *5*	$5/2^-$	0.002 keV *1*	L OP	XREF: L(8398.8)P(8404) E(level): others: 8398.8 *38* from (α,p), and 8404 *3* from ^{34}S(p,p),(p,p′γ):resonances. J^π: $5/2^-$, $7/2^-$ from R-Matrix analysis in (p,p):resonances; its primary 5708.5γ to $3/2^+$, 2694 level rules out $7/2^-$ based on RUL and Γ=0.002 keV.
8405.8 *14*	$(5/2^-)$	0.001 keV *1*	OP	E(level): weighted average of 8405.7 *14* from (p,γ), and 8406 *3* from ^{34}S(p,p),(p,p′γ):resonances. J^π: $(5/2^-,7/2^-)$ from R-Matrix analysis in (p,p):resonances; primary 8404.6γ to $3/2^+$.
8409.4 *18*	$1/2^-$	0.125 keV *15*	OP	E(level): weighted average of 8409.6 *18* from ^{34}S(p,γ) and 8409 *3* from ^{34}S(p,p),(p,p′γ):resonances.
8416.7 *5*	$1/2^+$	0.026 keV *7*	OP	E(level): weighted average of 8416.7 *5* from (p,γ), and 8418 *3* from ^{34}S(p,p),(p,p′γ):resonances. J^π: $1/2^+$ from R-Matrix analysis in (p,p):resonances; but primary 4473.5γ to $9/2^+$ and 7196.5γ to $1/2^+$ in (p,γ) gives $(5/2^+)$, which may indicate two different resonances with close energies.
8430.3 *5*			L O	E(level): other: 8429.2 *40* from (α,p).
8434.8 *5*	$(3/2)^+$	0.090 keV *15*	OP	XREF: P(8435)
8464.3 *5*	$(1/2^+,3/2,5/2^+)$		O	J^π: primary 4496.4γ to $1/2^+$, 3249.6γ to $(5/2^+)$.
8484.4 *5*	$(3/2)^+$	0.012 keV *5*	l OP	E(level): weighted average of 8484.4 *5* from (p,γ), and 8486 *3* from ^{34}S(p,p),(p,p′γ):resonances. J^π: $3/2^+,5/2^+$ from R-Matrix analysis in (p,p):resonances; $3/2^+$ proposed by 2001Vo24 in (p,γ) based on proposed isobaric analog to 2940, $3/2^+$ level in ^{35}S.
8486.2 *5*	$3/2^-$	0.150 keV *15*	l OP	E(level): weighted average of 8486.1 *5* from (p,γ), and 8488 *3* from ^{34}S(p,p),(p,p′γ):resonances.
8487.5 *5*	$15/2^-$	<69 fs	F H	E(level): weighted average of 8487.8 *10* from (^{28}Si,αpγ) and 8487.4 *5* from (^{16}O,αpγ). J^π: 3080.0γ E2, ΔJ=2 to $11/2^-$, 2399.8γ D(+Q) to $13/2^-$. $T_{1/2}$: lifetime τ<0.1 ps from (^{28}Si,αpγ) (2007Ks01, DSAM).
8506.5 *5*			L O	E(level): other: 8505.1 *47* from (α,p).
8514.3 *5*	$1/2^-$	0.150 keV *15*	OP	XREF: P(8515)
8533.9 *5*			L O	E(level): weighted average of 8532.4 *43* from (α,p) and 8533.9 *5* from (p,γ).
8558.2? *13*	$(13/2^-)$	<0.14 ps	F	XREF: F(?) J^π: 2631γ D(+Q), ΔJ=1 to $11/2^-$ in

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)†	J$^{\pi}$‡	T$_{1/2}$ or Γ&	XREF	Comments
				(^{28}Si,αpγ). T$_{1/2}$: lifetime τ<0.2 ps from (^{28}Si,αpγ) (2007Ks01, DSAM).
8571.9 *5*	(5/2)$^+$	0.08 keV *1*	L OP X	T=(3/2) XREF: P(8573)X(8580) E(level): weighted average of 8571.9 *5* from (p,γ), and 8573 *3* from ^{34}S(p,p),(p,p′γ):resonances. Other: 8567.9 *37* from (α,p).
8580.5 *5*	1/2$^+$	0.75 keV *8*	OP	XREF: P(8582)
8585.8 *5*			O	
8590.1 *5*	5/2$^+$	0.003 keV *2*	OP	XREF: P(8592) E(level): weighted average of 8590.0 *5* from (p,γ), and 8592 *3* from ^{34}S(p,p),(p,p′γ):resonances. J$^{\pi}$: 3/2$^+$,5/2$^+$ from R-Matrix analysis in (p,p):resonances; primary 5426.6γ to 7/2$^-$ ruled out 3/2$^+$ based on RUL and Γ=0.003 keV.
8612.0 *5*	3/2$^+$,5/2$^+$		l O W	XREF: W(8610) J$^{\pi}$: L(d,^{3}He)=2 from 0$^+$.
8613.7 *5*	5/2$^+$	0.175 keV *20*	l OP	XREF: P(8616)
8618.2 *5*	3/2$^+$,5/2$^+$	0.002 keV *1*	l OP	XREF: P(8620) E(level): weighted average of 8618.1 *5* from (p,γ), and 8620 *3* from ^{34}S(p,p),(p,p′γ):resonances.
8630.1 *3*	7/2$^-$	0.001 keV *1*	O	J$^{\pi}$: 4282.1γ M1+E2 to 9/2$^-$, 6866.2γ D+Q to 5/2$^+$, spin=3/2,7/2 from γ(θ) in (p,γ).
8633 *3*	(3/2,5/2$^+$)	0.001 keV *1*	P	
8640	(5/2$^+$)		O	J$^{\pi}$: spin=(5/2) from γ(θ) in (p,γ); primary 4697γ to 9/2$^+$.
8641.4 *5*	3/2$^+$,5/2$^+$	0.003 keV *2*	OP	XREF: P(8643)
8654 *4*			L	
8686.2 *5*	5/2$^-$	0.001 keV *1*	OP	E(level): weighted average of 8686.2 *5* from (p,γ), and 8687 *3* from ^{34}S(p,p),(p,p′γ):resonances. J$^{\pi}$: 5/2$^-$,7/2$^-$ from R-Matrix analysis in (p,p):resonances; primary 7446.0γ to 1219, 1/2$^+$ disfavors both 5/2$^-$ and 7/2$^-$, but 7/2$^-$ is much less likely.
8688.3 *5*	1/2$^+$	0.20 keV *2*	OP	XREF: P(8689)
8691 *3*	1/2$^-$	6.4 keV *7*	P	
8696.9 *5*	3/2$^-$	0.8 keV *1*	L OP	XREF: L(8698)P(8698)
8700 *10*	(7/2 to 17/2)$^+$		N	J$^{\pi}$: L(α,d)=6 from 3/2$^+$.
8706.3 *5*			O	
8717.8 *5*			L O	E(level): weighted average of 8720.9 *37* from (α,p) and 8717.7 *5* from (p,γ).
8750.9 *5*	3/2$^-$	0.30 keV *3*	OP	E(level): weighted average of 8750.9 *5* from (p,γ), and 8750 *3* from ^{34}S(p,p),(p,p′γ):resonances.
8767.0 *5*			O	
8772.9 *5*	1/2$^-$	0.57 keV *6*	OP	XREF: P(8775)
8779.8 *5*	3/2$^-$	0.214 keV *25*	OP	XREF: P(8781) E(level): weighted average of 8779.8 *5* from (p,γ), and 8781 *3* from ^{34}S(p,p),(p,p′γ):resonances.

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)[†]	J^{π}[‡]	$T_{1/2}$ or Γ[&]	XREF	Comments
8787.2 *5*	(5/2$^+$)	0.001 keV *1*	L OP	E(level): others: 8786.3 *38* from (α,p), and 8788 *3* from ^{34}S(p,p),(p,p′γ):resonances. J$^{\pi}$: (3/2$^+$,5/2$^+$) from R-Matrix analysis in (p,p):resonances; primary 3623.7γ and 5623.8γ to 7/2$^-$.
8788.3[b] *6*	(15/2$^+$)	<0.28 ps	F H	XREF: F(?) J$^{\pi}$: 915.4γ D, ΔJ=1 to 13/2$^+$; band assignment. $T_{1/2}$: lifetime τ<0.4 ps from (^{28}Si,αpγ) (2007Ks01, DSAM).
8798.4 *5*	(3/2$^+$,5/2$^+$)	0.001 keV *1*	OP	E(level): weighted average of 8798.4 *5* from (p,γ), and 8799 *3* from ^{34}S(p,p),(p,p′γ):resonances.
8820.9 *5*	(1/2$^+$ to 7/2$^+$)		O	J$^{\pi}$: primary 5817.6γ and 7057.0γ to 5/2$^+$; 6126.3γ and 8819.7γ to 3/2$^+$, which yields (1/2$^+$,3/2±,5/2±,7/2$^+$).
8824.2 *5*	1/2$^+$	1.70 keV *17*	OP	XREF: P(8825) E(level): weighted average of 8824.2 *5* from (p,γ), and 8825 *3* from ^{34}S(p,p),(p,p′γ):resonances.
8829.3 *5*	1/2$^-$	12.2 keV *12*	OP	E(level): weighted average of 8829.3 *5* from (p,γ), and 8829 *3* from ^{34}S(p,p),(p,p′γ):resonances. J$^{\pi}$: 1/2$^-$ from R-Matrix analysis in (p,p):resonances; but primary 6183.0γ to 7/2$^+$ gives J≥3/2, which may indicate a different level.
8830 *3*	1/2$^+$	0.080 keV *15*	P	
8833.1 *5*	(5/2,7/2)		O	J$^{\pi}$: primary 3618.4γ to (5/2$^+$), 4063.0γ to (5/2$^-$); 4720.4γ to 7/2$^+$, 5669.6γ to 7/2$^-$.
8838.1 *5*	5/2$^-$,7/2$^-$	0.001 keV *1*	l OP	XREF: P(8839)
8844.4[b] *4*	17/2$^+$	5.9 ps *11*	F HI N	XREF: N(8840) E(level): weighted average of 8845.1 *9* from (^{28}Si,αpγ), 8844.6 *5* from (^{16}O,αpγ), and 8844.2 *4* from (^{14}N,αpnγ). J$^{\pi}$: 971.38γ E2, ΔJ=2 to 13/2$^+$, 524.8γ to 15/2$^-$; band assignment. $T_{1/2}$: weighted average of 6.1 ps *11* from (^{16}O,αpγ) (1991Ja11, DSAM) and 5.5 ps *14* from (^{14}N,αpnγ) (1976Wa11, DSAM).
8856.2 *5*	5/2$^+$	0.010 keV *5*	OP	E(level): weighted average of 8856.1 *5* from (p,γ), and 8858 *3* from ^{34}S(p,p),(p,p′γ):resonances. J$^{\pi}$: 3/2$^+$,5/2$^+$ from R-Matrix analysis in (p,p):resonances; primary 5692.6γ to 7/2$^-$ rules out 3/2$^+$ based on RUL and Γ=0.010 keV.
8868.6 *5*	3/2$^+$,5/2$^+$	0.027 keV *10*	OP	E(level): weighted average of 8868.6 *5* from (p,γ), and 8870 *3* from ^{34}S(p,p),(p,p′γ):resonances.
8884.1 *5*			O	
8886.1 *5*	(5/2)		L O	E(level): other: 8887.5 *41* from (α,p). J$^{\pi}$: primary 3722.6γ to 7/2$^-$, 6239.8γ to 7/2$^+$; 4707.9γ to 3/2$^-$, 4967.2γ to 3/2$^+$.

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)[†]	J^{π}[‡]	$T_{1/2}$ or Γ[&]	XREF	Comments
8893.3 *5*	(5/2$^+$,7/2$^+$)		O	J^{π}: primary 4950.0γ to 9/2$^+$, 6198.7γ to 3/2$^+$.
8904.8 *5*	(1/2,3/2,5/2$^+$)		O	J^{π}: primary 4726.6γ to 3/2$^-$, 6210.2γ to 3/2$^+$, 7684.5γ to 1/2$^+$.
8906.8 *5*	5/2$^+$	0.002 keV *1*	OP x	XREF: P(8909)x(8921) J^{π}: 3/2$^+$,5/2$^+$ from R-Matrix analysis in (p,p):resonances; primary 4963.5γ to 9/2$^+$ rules out 3/2$^+$ based on RUL and Γ=0.002 keV.
8919.8 *5*	(5/2$^-$,7/2,9/2$^+$)		O x	XREF: x(8921) J^{π}: primary 4571.7γ to 9/2$^-$, 7155.9γ to 5/2$^+$.
8933.3 *5*			O	
8953.1 *5*	3/2$^+$	0.075 keV *15*	OP	XREF: P(8955) E(level): weighted average of 8953.0 *5* from (p,γ), and 8955 *3* from ^{34}S(p,p),(p,p′γ):resonances. J^{π}: spin=3/2 from γ(θ) in (p,γ) and (3/2)$^+$ from R-Matrix analysis in (p,p):resonances.
8957.9 *5*	(1/2$^-$,3/2$^-$)	0.04 keV *1*	L OP	XREF: P(8959) E(level): others: 8957.7 *44* from (α,p) and 8959 *3* from (p,p):resonances.
8981.9 *5*	5/2$^-$,7/2$^-$	0.003 keV *2*	OP	XREF: P(8983) E(level): weighted average of 8981.9 *5* from (p,γ), and 8983 *3* from ^{34}S(p,p),(p,p′γ):resonances.
8984.2 *5*	5/2$^+$	0.025 keV *10*	OP	XREF: P(8985) J^{π}: 3/2$^+$,5/2$^+$ from R-Matrix analysis in (p,p):resonances; primary 5820.7γ to 7/2$^-$ rules out 3/2$^+$ based on RUL and Γ=0.025 keV.
8988.4 *5*			O	
8992.4 *5*			O	
8996.7 *5*	5/2$^-$,7/2$^-$	0.002 keV *1*	L OP	XREF: L(8996.9)P(8998)
9001.1 *5*			O	
9019.3 *5*	3/2$^-$	3.27 keV *30*	OP	XREF: P(9021) Γ: weighted average of 3.1 keV *3* from (p,γ) and 3.50 keV *35* from (p,p):resonances.
9024.5 *5*			l O	
9029.9 *5*	(5/2)$^+$	0.04 keV *1*	l OP	XREF: P(9031)
9033.1 *5*			O	
9038.4 *5*	1/2$^-$	0.292 keV *30*	OP	XREF: P(9040) E(level): weighted average of 9038.3 *5* from (p,γ), and 9040 *3* from ^{34}S(p,p),(p,p′γ):resonances.
9048.4 *5*	5/2$^-$,7/2$^-$	0.001 keV *1*	OP	XREF: P(9048)
9050 *3*	(5/2)$^+$	0.095 keV *15*	P	
9081.5 *4*	5/2$^+$	59 eV *10*	L OP	XREF: L(9085.7)P(9084) E(level): weighted average of 9081.4 *4* from (p,γ), and 9084 *3* from ^{34}S(p,p),(p,p′γ):resonances. J^{π}: spin=5/2 from γ(θ) in (p,γ) and (5/2)$^+$ from R-Matrix analysis in (p,p):resonances. Γ: weighted average of 65 eV *20* from (p,γ) and 57 eV *10* from (p,p):resonances.
9088.5 *5*			L OP	XREF: L(9085.7)P(9089) E(level): weighted average of 9088.5 *5* from

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)†	Jπ‡	$T_{1/2}$ or Γ&	XREF	Comments
				(p,γ), and 9089 *4* from ^{34}S(p,p),(p,p′γ):resonances.
9099.3 *5*			O	
9100.4 *5*	3/2−	0.20 keV *2*	OP	XREF: P(9101)
9107.4 *5*			L OP	XREF: L(9109.3)P(9108) E(level): weighted average of 9107.4 *5* from (p,γ), and 9108 *4* from ^{34}S(p,p),(p,p′γ):resonances.
9110.1 *5*			L OP	XREF: L(9109.3)P(9114) E(level): weighted average of 9110.0 *5* from (p,γ), and 9114 *4* from ^{34}S(p,p),(p,p′γ):resonances.
9124.0 *5*			OP	XREF: P(9120)
9127 *4*	5/2@		J P	E(level): from (p,p):resonances. Other: 9127 *9* from ^{31}P(α,p):resonances.
9135.1 *5*		2.0 keV *4*	Op	XREF: p(9136) Γ: from (p,γ).
9138.3 *5*			Op	XREF: p(9136)
9147 *4*	(7/2 to 17/2)+		N P	XREF: N(9150)P(9147) Jπ: L(α,d)=6 from 3/2+.
9155.7 *5*			O	
9157.1 *5*	5/2−		OP	XREF: P(9159) Jπ: spin=5/2 from γ(θ) in (p,γ); 5993.7γ M1+E2 to 7/2−.
9161 *4*	1/2@		J l	XREF: l(9160.0)
9163.3 *5*	(3/2−)		l OP	XREF: l(9160.0)P(9168) E(level): other: 9168 *4* from (p,p):resonances.
9184.2 *5*			Op	XREF: p(9185)
9188.8 *5*			Op	XREF: p(9185)
9194.1 *5*			L OP	E(level): others: 9194.0 *39* from (α,p), and 9196 *4* from (p,p):resonances.
9207 *4*	9/2+		P	
9222 *4*	5/2+		P X	XREF: X(9220) Jπ: L(pol d,^{3}He)=2 and L+1/2 from analyzing power for a group at E=9220 *120*.
9227 *4*	5/2−		P	
9241 *4*	7/2+		P	
9255 *4*	1/2@		J P	E(level): from (p,p):resonances. Other: 9256 *4* from ^{31}P(α,p):resonances.
9264.9 *39*			L	
9271 *4*			P	
9276 *4*			P	
9284 *4*	5/2−		P	
9294 *4*			P	
9299 *4*			P	
9316.2 *39*			L	
9325 *4*	5/2+		P	
9335 *4*	7/2+		L P	E(level): weighted average of 9336 *4* from ^{34}S(p,p),(p,p′γ):resonances and 9334 *5* from ^{32}S(α,p).
9345 *4*			P	
9349 *4*			P	
9355 *4*			P	
9376 *4*			L	

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)[†]	Jπ[‡]	$T_{1/2}$ or Γ[&]	XREF	Comments
9400 *9*	1/2[@]		J	
9450 *10*	(7/2 to 17/2)+		N	Jπ: L(α,d)=6 from 3/2+.
9456 *4*	3/2[@]		J	
9476 *4*			L	
9481.0 *18*	3/2[@]		J	
9508 *4*	(1/2−,3/2,5/2+)		L X	XREF: X(9514) Jπ: L(pol d,^{3}He)=(1,2) from 0+.
9551 *6*	5/2[@]		J	
9673 *6*	1/2[@]		J	
9712.7 *27*	1/2[@]		J L	E(level): weighted average of 9713.0 *27* from ^{31}P(α,p),(α,n):resonances and 9711.8 *44* from (α,p).
9740 *4*			L	
9751.1 *27*	7/2[@]		J	
9799 *5*			L	
9814.0 *27*	5/2(+)		J X	XREF: X(9820) Jπ: spin=5/2 from angular distributions in ^{31}P(α,p):resonances; L=(1,2) from 0+ in (pol d,^{3}He) gives (1/2−,3/2,5/2+).
9836 *5*			L	
9869.8 *27*	1/2[@]		J L	XREF: L(9870)
9900.8 *27*	3/2[@]		J	
9923 *4*	3/2−[@]		J	
9958 *6*	3/2[@]		J	
9969 *5*	(1/2,3/2)[@]		J L	XREF: J(9968) E(level): from (α,p) and (α,n):resonances.
10031 *5*	(1/2+,3/2−)[@]		J	
10058 *5*	3/2[@]		J	
10075 *5*	(1/2)[@]		J L	E(level): from (α,p) and (α,n):resonances.
10090 *5*	5/2+[@]		J	
10132 *5*	3/2−[@]		J	
10163 *5*	[@]		L	
10172 *5*	3/2−,5/2−[@]		J	
10179 *5*			L	
10181.1[a] *5*	19/2−	42.3 fs *28*	F H	E(level): weighted average of 10181.1 *10* from (^{28}Si,αpγ) and 10181.1 *5* from (^{16}O,αpγ). Jπ: 1693.5γ E2, ΔJ=2 to 15/2−; 1336.3γ D, ΔJ=1 to 17/2+; band assignment. $T_{1/2}$: lifetime τ=61 fs *4* from (^{28}Si,αpγ) (2013Bi10, DSAM).
10215 *5*	3/2[@]		J L	XREF: L(10218.0)
10221.9 *11*	(17/2−)	<0.28 ps	F H	XREF: F(?) E(level): from (^{16}O,αpγ). Jπ: 1377.8γ to 17/2+ consistent with ΔJ=0. $T_{1/2}$: lifetime τ<0.4 ps from (^{28}Si,αpγ) (2007Ks01, DSAM).
10236 *5*	3/2−[@]		J	
10278 *5*	1/2[@]		J	

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)[†]	J^{π}[‡]	$T_{1/2}$ or Γ[&]	XREF	Comments
10295 *5*			J	
10322 *5*	$5/2^+$[@]		J	
10340 *5*	(3/2,5/2)[@]		J	
10359 *5*	3/2[@]		J	
10385 *5*			J	
10395 *5*			L	
10399 *5*	$3/2^+,7/2^+$[@]		J	
10431 *5*	$5/2^+$[@]		J	
10465 *5*	$5/2^+$[@]		J L	E(level): weighted average of 10467 *5* from ^{31}P(α,p),(α,n):resonances and 10463.2 *49* from (α,p).
10486 *5*	$5/2^+$[@]		J	
10517 *6*			L	
10534 *5*	$1/2^+,5/2^+$[@]		J	
10548 *6*			L	
10584 *5*	$3/2^+$[@]		J L	E(level): weighted average of 10587 *5* from ^{31}P(α,p),(α,n):resonances and 10579.5 *54* from (α,p).
10642 *5*	$5/2^+$[@]		J L	E(level): weighted average of 10641 *5* from ^{31}P(α,p),(α,n):resonances and 10643.0 *50* from (α,p).
10678 *5*	$3/2^+,5/2^+$[@]		J	
10696 *5*	$7/2^-$[@]		J	
10733 *5*	$3/2^+$[@]		J L	E(level): weighted average of 10734 *5* from ^{31}P(α,p),(α,n):resonances and 10732.1 *50* from (α,p).
10760 *5*	$3/2^+,7/2^-$[@]		J L	E(level): weighted average of 10760 *5* from ^{31}P(α,p),(α,n):resonances and 10759.1 *55* from (α,p).
10800 *5*	$5/2^+$[@]		J	
10816 *5*	$1/2^+,3/2^-$[@]		J	
10844 *5*	$3/2^-$[@]		J	
10858.5[b] *8*	$19/2^+$	<0.28 ps	F H	XREF: F(?) J^{π}: 2069.8γ E2, ΔJ=2 to $15/2^+$, 2014.7γ D, ΔJ=1 to $17/2^+$; band assignment. $T_{1/2}$: lifetime τ<0.4 ps from (^{28}Si,αpγ) (2007Ks01, DSAM).
10870 *5*	(5/2,7/2)[@]		J	
10886 *5*			J	
10909 *5*	$3/2^+,5/2^-$[@]		J	
10928 *5*	7/2[@]		J	
10942 *5*			J	
10964 *5*	$5/2^+$[@]		J	
10973 *5*			J	
10998 *5*	$3/2^+,7/2^-$[@]		J	
11034 *5*	$3/2^+$[@]		J	
11063 *5*	$1/2^+,9/2^+$[@]		J	
11082 *5*	$5/2^+,9/2^+$[@]		J	

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)[†]	J$^{\pi}$[‡]	$T_{1/2}$ or Γ[&]	XREF	Comments
11105 *5*			J	
11123 *5*	3/2[@]		J	
11142 *5*	5/2^{+}[@]		J	
11154 *5*			J	
11169 *5*			J	
11179 *5*	3/2^{-},5/2^{+}[@]		J	
11185 *5*			J	
11198 *5*	3/2[@]		J	
11230 *5*			J	
11246 *5*	3/2[@]		J	
11265 *5*	3/2^{-}[@]		J	
11287 *5*			J	
11307 *5*			J	
11314 *5*	1/2^{-},5/2^{+}[@]		J	
11327 *5*	1/2^{-},5/2^{+}[@]		J	
11357 *5*			J	
11375 *5*			J	
11390 *5*			J	
11410 *5*			J	
11421 *5*	3/2^{-}[@]		J	
11434 *5*			J	
11459.1[b] *7*	21/2^{+}	43.0 fs *21*	F H J	E(level): weighted average of 11459.2 *12* from (^{28}Si,αpγ) and 11459.0 *7* from (^{16}O,αpγ). Other: 11460 *5* from ^{31}P(α,p),(α,n):resonances. J$^{\pi}$: 2614.5γ E2, ΔJ=2 to 17/2^{+}; band assignment. $T_{1/2}$: lifetime τ=62 fs *3* from (^{28}Si,αpγ) (2013Bi10, DSAM).
11473 *5*	1/2^{-},5/2^{-}[@]		J	
11492 *5*			J	
11505 *5*	3/2^{-},5/2^{-}[@]		J	
11525 *5*			J	
11540 *5*			J	
11551 *5*	3/2^{-},5/2^{+}[@]		J	
11565 *5*			J	
11589 *5*			J	
11609 *5*	5/2^{+},7/2^{-}[@]		J	
11627 *5*	1/2^{+},5/2^{+}[@]		J	
11638 *5*			J	
11651 *5*			J	
11684 *5*	1/2^{+},5/2^{+}[@]		J	
11773 *5*	3/2^{-},7/2^{-}[@]		J	
11783 *5*	3/2^{-}[@]		J	
1.25×10^{4} *5*			J X	T=3/2 XREF: J(12900) E(level): from (pol d,^{3}He).
12572.2[a] *6*	23/2^{-}	33 fs *4*	F H	E(level): weighted average of 12572.2 *12* from (^{28}Si,αpγ) and 12572.2 *6* from (^{16}O,αpγ). J$^{\pi}$: 2390.8γ E2, ΔJ=2 to 19/2^{-}, 1113.3γ D, ΔJ=1 to 21/2^{+}; band assignment.

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)[†]	J^{π}[‡]	$T_{1/2}$ or Γ[&]	XREF	Comments
				$T_{1/2}$: lifetime τ=48 fs *6* from (^{28}Si,αpγ) (2013Bi10, DSAM).
13900			J	T=3/2
16306.4[a] *16*	(27/2$^-$)	<9.0 fs	F	J^{π}: 3734γ to 23/2$^-$; band assignment. $T_{1/2}$: lifetime τ<13 fs from (^{28}Si,αpγ) (2013Bi10, DSAM).

† Most of Adopted Levels are reported in (p,γ) dataset with precise E(level) values, deduced based on measured γ-ray energies which however are not reported in references in (p,γ) dataset, while E(level) values from Eγ values with uncertainties in other datasets are less precise and the numbers of γ rays are much less than that of (p,γ) datasets. Therefore, a least-squares fit to available Eγ values with uncertainties would not produce E(level) values more precise and complete than those already reported in (p,γ) and the evaluators have adopted E(level) values from (p,γ) or averages of values based on Eγ data from all available datasets where applicable, as noted.

‡ For proton-resonance levels populated in (p,p):resonances, J^{π} assignments are from R-Matrix analysis, unless otherwise noted.

\# From FRESCO-DWBA analysis of measured $\sigma(\theta)$ of (α,p) (2019Se09).

@ From angular distributions in ^{31}P(α,p):resonances.

& Sources of $T_{1/2}$ values are listed under comments; Γ from ^{34}S(p,p),(p,p′γ):resonances, unless otherwise noted.

a Band(A): Band based on $f_{7/2}$ orbital.

b Band(B): Band based on 13/2$^+$.

Adopted Levels, Gammas (continued)

$\gamma(^{35}Cl)$

E_i(level)	J^π_i	E_γ[†]	I_γ[‡]	E_f	J^π_f	Mult.[#]	δ[#]	α[a]	Comments
1219.38	1/2+	1219.4 *1*	100	0.0	3/2+	M1+E2	0.093 *10*	3.66×10^{-5} *5*	B(M1)(W.u.)=0.101 *+14−11*; B(E2)(W.u.)=2.2 *+6−5* E_γ: weighted average of 1219.3 *2* from ^{35}Ar ε decay and 1219.4 *1* from (p,p′γ), and 1219.4 *6* from (α,pγ). Mult.,δ: ΔJ=1 from $\gamma(\theta)$ in (p,p′γ), E1+M2 ruled out by RUL. δ from adopted B(E2)↑=0.00076 *7* measured in Coulomb excitation and adopted lifetime τ=0.172 ps *20*.
1763.15	5/2+	543.77	<0.2	1219.38	1/2+	[E2]		0.000323 *5*	I_γ: from (p,γ). Other: <0.5 from (α,pγ).
		1763.16 *10*	100	0.0	3/2+	M1+E2	−2.85 *10*	0.0002124 *30*	B(M1)(W.u.)=0.00122 *+14−12*; B(E2)(W.u.)=12.1 *+11−9* E_γ: weighted average of 1763.0 *2* from ^{35}Ar ε decay, 1763.1 *2* from (^{24}Mg,αpγ), 1763.3 *2* from (^{16}O,αpγ), 1763.15 *10* from (^{14}N,αpnγ), and 1763.2 *1* from (p,p′γ). Other: 1763.2 *6* from (α,pγ). Mult.: from $\gamma\gamma$(ADO) in (^{16}O,αpγ) with ΔJ=1 and γ(lin pol) in (^{28}Si,αpγ). δ: weighted average of −3.0 *1* from (α,pγ), −2.66 *12* from (p,p′γ), −2.95 *45* from Coulomb excitation, −2.6 *4* from (^{16}O,αpγ), all from $\gamma(\theta)$ or p$\gamma(\theta)$. Other: −0.42 *17* from $\gamma\gamma(\theta)$ in (p,γ); 1.8 *+10−4*; deduced by the evaluators from adopted B(E2)↑=0.0106 *7* and $T_{1/2}$=0.36 ps *3*.
2645.67	7/2+	882.80 *11*	11.4 *10*	1763.15	5/2+	M1+E2	+0.21 *5*	5.29×10^{-5} *9*	B(M1)(W.u.)=0.0196 *+35−27*; B(E2)(W.u.)=4.2 *+23−18* E_γ: weighted average of 882.9 *1* from (^{24}Mg,αpγ), 882.7 *5* from (^{16}O,αpγ), 882.32 *35* from (^{14}N,αpnγ), 882.4 *10* from (α,pγ), and 882.3 *3* from (p,p′γ). I_γ: unweighted average of 8.7 *6* from (^{28}Si,αpγ), 15.2 *13* from (^{24}Mg,αpγ), 11.08 *28* from (^{16}O,αpγ), 11.1 *22* from (α,pγ), 9.5 *11* from (p,γ), and 13 *5* from (p,p′γ). Mult.: D+Q from p$\gamma(\theta)$ in (α,pγ); E1+M2 ruled out by RUL. δ: weighted average of +0.25 *5* from (α,pγ) and +0.17 *5* from (p,p′γ).
		1426.26	<1.2	1219.38	1/2+	[M3]		7.52×10^{-5} *11*	I_γ: from (α,pγ). Other: <2.2 from (p,γ).
		2645.70 *20*	100.0 *11*	0.0	3/2+	E2		0.000633 *9*	B(E2)(W.u.)=3.6 *+6−4* E_γ: weighted average of 2645.5 *4* from (^{24}Mg,αpγ), 2645.8 *5* from (^{16}O,αpγ), 2645.79 *30* from

Adopted Levels, Gammas (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ [†]	I_γ [‡]	E_f	J^π_f	Mult. [#]	δ [#]	α [a]	Comments
									(^{14}N,αpnγ), and 2645.7 *2* from (p,p′γ). Other: 2646.2 *15* from (α,pγ). I_γ: others: 100.0 *31* from (^{24}Mg,αpγ), 100.0 *22* from (^{16}O,αpγ), 100.0 *22* from (α,pγ), and 100.0 *20* from (p,p′γ). Other: 100 *6* from (^{28}Si,αpγ). Mult.: from $\gamma\gamma(\theta)$(DCO) and $\gamma\gamma$(lin pol) in (^{28}Si,αpγ); $\gamma(\theta)$ and γ(lin pol) in (p,p′γ).
2694.00	3/2$^+$	930.9 *2*	16.3 *13*	1763.15	5/2$^+$	M1+E2	+0.09 *3*	4.68×10^{-5} *7*	B(M1)(W.u.)=0.110 *+26−19*; B(E2)(W.u.)=3.9 *+33−22* E_γ: weighted average of 930.7 *2* from ^{35}Ar ε decay, 930.5 *15* from (α,pγ), and 931.3 *3* from (p,p′γ). I_γ: weighted average of 14.3 *18* from ^{35}Ar ε decay, 18.2 *26* from (α,pγ), 16.3 *13* from (p,γ), and 18.6 *25* from (p,p′γ). Mult.,δ: from $\gamma(\theta)$ and γ(lin pol) in (p,p′γ).
		1474.8 *3*	9.5 *10*	1219.38	1/2$^+$	M1+E2	−0.63 *21*	8.4×10^{-5} *4*	B(M1)(W.u.)=0.0116 *+37−29*; B(E2)(W.u.)=8.0 *+45−39* E_γ: weighted average of 1474.8 *3* from ^{35}Ar ε decay and 1475 *2* from (p,p′γ). I_γ: weighted average of 9.3 *10* from ^{35}Ar ε decay, 11.7 *26* from (α,pγ), 9.5 *13* from (p,γ), and 8.3 *25* from (p,p′γ). Mult.,δ: from $\gamma(\theta)$ and γ(lin pol) in (p,p′γ).
		2693.7 *2*	100.0 *17*	0.0	3/2$^+$	M1+E2	+0.26 *2*	0.000553 *8*	B(M1)(W.u.)=0.026 *+6−4*; B(E2)(W.u.)=0.92 *+26−19* E_γ: weighted average of 2693.7 *2* from ^{35}Ar ε decay and 2694.1 *6* from (p,p′γ). Other: 2693.2 *20* from (α,pγ). I_γ: weighted average of 100.0 *17* from ^{35}Ar ε decay, 100 *4* from (α,pγ), 100.0 *25* from (p,γ), and 100.0 *30* from (p,p′γ). Mult.: from $\gamma(\theta)$ and γ(lin pol) in (p,p′γ). δ: weighted average of +0.17 *8* from (α,pγ) and +0.26 *2* from (p,p′γ).
3002.64	5/2$^+$	308.64	<5	2694.00	3/2$^+$	[M1,E2]		0.0014 *9*	I_γ: from (α,pγ).
		356.97	<1	2645.67	7/2$^+$	[M1,E2]		9	I_γ: from (α,pγ).
		1239.47	2.2 *11*	1763.15	5/2$^+$	[M1,E2]		4.4×10^{-5} *6*	I_γ: from (p,p′γ). B(M1)(W.u.)=0.011 *+8−5* if M1, B(E2)(W.u.)=26 *+19−12* if E2.
		1783.21	3.3 *11*	1219.38	1/2$^+$	[E2]		0.0002268 *32*	B(E2)(W.u.)=6.4 *+35−22* I_γ: from (p,p′γ).
		3002.3 *4*	100.0 *10*	0.0	3/2$^+$	M1+E2	+0.08 *2*	0.000671 *9*	B(M1)(W.u.)=0.034 *+14−7*; B(E2)(W.u.)=0.09 *+7−4* E_γ: weighted average of 3002.1 *4* from ^{35}Ar ε decay,

Adopted Levels, Gammas (continued)

γ(^{35}Cl) (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[‡]	E_f	J^π_f	Mult.[#]	δ[#]	α[a]	Comments
									3002.5 *15* from (α,pγ), and 3002.5 *5* from (p,p$'\gamma$). I_γ: weighted average of 100.0 *19* from ^{35}Ar ε decay and 100.0 *10* from (p,p$'\gamma$). Mult.: from $\gamma(\theta)$ and γ(lin pol) in (p,p$'\gamma$). δ: weighted average of +0.09 *3* from (α,pγ) and +0.07 *2* from (p,p$'\gamma$). Other: −0.22 *1* from (p,γ).
3162.87	7/2$^-$	160.67 *20*	1.5 *4*	3002.64	5/2$^+$	[E1]		0.00330 *5*	B(E1)(W.u.)=6.6×10^{-5} *17* E_γ: weighted average of 161 *1* from (^{16}O,αpγ) and 160.66 *20* from (^{14}N,αpnγ). I_γ: unweighted average of 1.10 *30* from (^{28}Si,αpγ) and 1.89 *22* from (p,γ).
		517.47 *11*	12.3 *16*	2645.67	7/2$^+$	(E1)		0.0001097 *15*	B(E1)(W.u.)=1.61×10^{-5} *20* E_γ: weighted average of 517.7 *1* from (^{24}Mg,αpγ), 517.3 *5* from (^{16}O,αpγ), 517.26 *10* from (^{14}N,αpnγ), 517.4 *15* from (α,pγ), and 517.0 *8* from (p,p$'\gamma$). I_γ: unweighted average of 9.2 *5* from (^{28}Si,αpγ), 14.3 *8* from (^{24}Mg,αpγ), 11.1 *13* from (^{16}O,αpγ), 11.1 *33* from (α,pγ), 8.9 *11* from (p,γ), and 19 *8* from (p,p$'\gamma$). Mult.: D from $\gamma\gamma(\theta)$(DCO) in (^{28}Si,αpγ); $\Delta\pi$=yes from level scheme.
		1399.9 *7*	0.34 *5*	1763.15	5/2$^+$	(E1+M2)	+0.44 *12*	0.000181 *12*	B(E1)(W.u.)=1.88×10^{-8} *+31−34*; B(M2)(W.u.)=0.0085 *+43−38* I_γ: weighted average of 0.50 *20* from (^{16}O,αpγ) and 0.33 *5* from (p,γ). Mult.,δ: D+Q from $\gamma(\theta)$ in (p,γ) (1971Pr11); $\Delta\pi$=yes from level scheme.
		1943.43	<0.22	1219.38	1/2$^+$	[E3]		0.0001742 *24*	
		3162.53 *10*	100.0 *11*	0.0	3/2$^+$	M2+E3	+0.17 *2*	0.000523 *7*	B(M2)(W.u.)=0.255 *9*; B(E3)(W.u.)=4.2 *+11−9* E_γ: weighted average of 3162.7 *4* from (^{24}Mg,αpγ), 3162.6 *3* from (^{16}O,αpγ), 3162.43 *10* from (^{14}N,αpnγ), and 3162.6 *1* from (p,p$'\gamma$), and 3162.6 *15* from (α,pγ). I_γ: weighted average of 100 *5* from (^{28}Si,αpγ), 100.0 *30* from (^{24}Mg,αpγ), 100.0 *20* from (^{16}O,αpγ), 100.0 *33* from (α,pγ), 100.0 *11* from (p,γ), and 100.0 *20* from (p,p$'\gamma$). Mult.: from $\gamma(\theta)$ and γ(lin pol) in (p,γ). δ: weighted average of +0.16 *1* from (^{28}Si,αpγ), +0.18 *7* from (^{16}O,αpγ), and +0.25 *4* from (p,γ). Others: −0.14 *2* from (α,pγ) and −0.22 *5* from (p,p$'\gamma$).

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J_i^π	E_γ[†]	I_γ[‡]	E_f	J_f^π	Mult.[#]	α[a]	Comments
3918.47	3/2+	1272.78	<3.7	2645.67	7/2+	[E2]	5.56×10^{-5} 8	
		2155.1 15	24.2 28	1763.15	5/2+	[M1,E2]	0.00036 4	E_γ: from ^{35}Ar ε decay (1.7755 s). I_γ: weighted average of 29 12 from ^{35}Ar ε decay and 23.9 28 from (p,γ). B(M1)(W.u.)=0.075 +32−18 if M1, B(E2)(W.u.)=61 +26−14 if E2.
		2698.98	19 5	1219.38	1/2+	[M1,E2]	0.00060 6	I_γ: from (α,pγ). B(M1)(W.u.)=0.030 +15−9 if M1, B(E2)(W.u.)=16 +8−5 if E2.
		3918.24	100.0 19	0.0	3/2+	M1+E2	0.00108 7	I_γ: weighted average of 100 6 from ^{35}Ar ε decay, 100 5 from (α,pγ), and 100.0 19 from (p,γ). δ: −0.21 2 or +20 +10−5 from $\gamma(\theta)$ in (p,γ). B(M1)(W.u.)=0.052 +22−11 if M1, B(E2)(W.u.)=13 +6−3 if E2.
3942.9	9/2+	1297.4 15	8.0 21	2645.67	7/2+	[M1,E2]	5.2×10^{-5} 8	E_γ: from (α,pγ). I_γ: weighted average of 6.2 31 from (^{28}Si,αpγ), 8.1 21 from (α,pγ), and 8.7 22 from (p,γ). B(M1)(W.u.)=0.0042 +13−12 if M1, B(E2)(W.u.)=9.3 +29−27 if E2.
		2180.1 10	100.0 21	1763.15	5/2+	E2	0.000414 6	B(E2)(W.u.)=8.7 +17−13 E_γ: weighted average of 2180 1 from (^{16}O,αpγ) and 2180.2 15 from (α,pγ). I_γ: weighted average of 100.0 21 from (α,pγ) and 100.0 22 from (p,γ). Other: 100 11 from (^{28}Si,αpγ).
3967.6	1/2+	1273.6	2.7 6	2694.00	3/2+	[M1,E2]	4.9×10^{-5} 7	B(M1)(W.u.)=0.019 +8−6 if M1, B(E2)(W.u.)=45 +18−13 if E2.
		2204.4	≤1.3	1763.15	5/2+	[E2]	0.000426 6	
		2748.5 6	100 6	1219.38	1/2+	[M1]	0.000569 8	B(M1)(W.u.)=0.071 +23−14 E_γ: from ^{35}Ar ε decay (1.7755 s). I_γ: other: 100 7 from ^{35}Ar ε decay.
		3967.4	23 4	0.0	3/2+	[M1,E2]	0.00109 7	I_γ: weighted average of 22 5 from ^{35}Ar ε decay and 24 4 from (p,γ). B(M1)(W.u.)=0.0054 +19−13 if M1, B(E2)(W.u.)=1.31 +46−30 if E2.
4059.1	(3/2)−	1365.1	1.3 4	2694.00	3/2+	[E1]	0.0001824 26	B(E1)(W.u.)=2.1×10^{-4} +8−7
		2295.9	5.5 16	1763.15	5/2+	[E1]	0.000848 12	B(E1)(W.u.)=1.9×10^{-4} +7−6 I_γ: weighted average of 8 4 from (α,pγ) and 5.1 16 from (p,γ).
		2839.6	100.0 21	1219.38	1/2+	[E1]	1.17×10^{-3} 2	B(E1)(W.u.)=0.00179 +34−22 I_γ: weighted average of 100 4 from (α,pγ) and 100.0 21 from (p,γ).
		4058.9	<2.1	0.0	3/2+	[E1]	1.70×10^{-3} 2	

Adopted Levels, Gammas (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ†	I_γ‡	E_f	J^π_f	Mult.#	δ#	α[a]	Comments
4112.4	7/2$^+$	2349.2	89 *6*	1763.15	5/2$^+$	M1+E2	−0.16 *2*	0.000405 *6*	B(M1)(W.u.)=0.0159 *+48−31*; B(E2)(W.u.)=0.28 *+12−8* I_γ: other: 16 *5* from (α,pγ).
		4112.1	100 *6*	0.0	3/2$^+$	E2(+M3)	0.00 *10*	1.21×10^{-3} *2*	B(E2)(W.u.)=0.76 *+22−15* I_γ: other: 100 *5* from (α,pγ).
4173.43	(5/2$^+$)	1479.40	45 *14*	2694.00	3/2$^+$	[M1,E2]		9.1×10^{-5} *13*	B(M1)(W.u.)=0.048 *+20−12* if M1, B(E2)(W.u.)=82 *+34−21* if E2. B(E2)(W.u.)=82 *+34−21* upper bound exceeds RUL=100 if E2.
		1527.72	<17	2645.67	7/2$^+$	[M1,E2]		0.000106 *15*	
		2410.19	28 *9*	1763.15	5/2$^+$	[M1,E2]		0.00048 *5*	B(M1)(W.u.)=0.0068 *+33−19* if M1, B(E2)(W.u.)=4.5 *+21−13* if E2.
		2953.92	≤5.2	1219.38	1/2$^+$	[E2]		0.000771 *11*	
		4173.16	100 *17*	0.0	3/2$^+$	[M1,E2]		0.00116 *7*	B(M1)(W.u.)=0.0047 *+15−7* if M1, B(E2)(W.u.)=1.02 *+33−14* if E2.
4177.9	3/2$^-$	1175.2	<0.82	3002.64	5/2$^+$	[E1]		6.06×10^{-5} *8*	
		1483.9	13.1 *33*	2694.00	3/2$^+$	[E1]		0.000265 *4*	B(E1)(W.u.)=5.7×10^{-4} *+16−14*
		1532.2	<0.82	2645.67	7/2$^+$	[M2]		6.18×10^{-5} *9*	
		2414.7	<1.6	1763.15	5/2$^+$	[E1]		0.000921 *13*	
		2958.4	51 *8*	1219.38	1/2$^+$	(E1+M2)	+0.11 *4*	1.22×10^{-3} *2*	B(E1)(W.u.)=2.8×10^{-4} *+5−4*; B(M2)(W.u.)=1.7 *+16−11* Mult.,δ: D+Q from $\gamma(\theta)$ in (p,γ); $\Delta\pi$=yes from level scheme. B(M2)(W.u.)=1.7 *+16−11* upper bound exceeds RUL=3.
		4177.6	100 *8*	0.0	3/2$^+$	(E1+M2)	+0.06 *3*	1.74×10^{-3} *3*	B(E1)(W.u.)=1.93×10^{-4} *+28−20*; B(M2)(W.u.)=0.18 *+24−14* Mult.,δ: D+Q from $\gamma(\theta)$ in (p,γ); $\Delta\pi$=yes from level scheme.
4347.76	9/2$^-$	1184.90 *20*	100.0 *29*	3162.87	7/2$^-$	M1+E2	−0.36 *3*	3.57×10^{-5} *5*	B(M1)(W.u.)=0.007 *+7−3*; B(E2)(W.u.)=2.5 *+27−10* E_γ: weighted average of 1185.0 *2* from (^{24}Mg,αpγ), 1184.9 *3* from (^{16}O,αpγ), and 1184.80 *20* from (^{14}N,αpnγ). Other: 1184.6 *10* from (α,pγ). I_γ: weighted average of 100 *4* from (^{28}Si,αpγ), 100.0 *31* from (^{24}Mg,αpγ), 100 *4* from (^{16}O,αpγ), 100.0 *31* from (α,pγ), and 100.0 *29* from (p,γ). Mult.: from $\gamma\gamma$(DCO) and $\gamma\gamma$(lin pol) in (^{28}Si,αpγ), ΔJ=1. δ: weighted average of −0.36 *3* from (α,pγ), −0.3 *1* from (^{28}Si,αpγ), and −0.65 *30* from (^{16}O,αpγ).
		1701.88 *25*	46.8 *24*	2645.67	7/2$^+$	E1+M2	−0.018 *12*	0.000434 *6*	B(E1)(W.u.)=3.7×10^{-5} *+39−14*; B(M2)(W.u.)=0.019

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J_i^π	E_γ[†]	I_γ[‡]	E_f	J_f^π	Mult.[#]	α[a]	Comments
								+47−17 E_γ: weighted average of 1702.0 *3* from (^{24}Mg,αpγ), 1701.7 *5* from (^{16}O,αpγ), and 1701.85 *25* from (^{14}N,αpnγ). Other: 1702.1 *15* from (α,pγ). I_γ: unweighted average of 45.6 *24* from (^{28}Si,αpγ), 39.5 *24* from (^{24}Mg,αpγ), 52.0 *14* from (^{16}O,αpγ), 52 *7* from (α,pγ), and 44.9 *29* from (p,γ). Mult.: from $\gamma\gamma$(DCO) and $\gamma\gamma$(lin pol) in (^{28}Si,αpγ) with ΔJ=1 and $\gamma(\theta)$ and γ(lin pol) in (α,pγ). δ: from (p,γ). Others: +0.5 *+5−3* from (^{28}Si,αpγ), +0.5 *4* from (^{16}O,αpγ).
4347.76	$9/2^-$	2584.51	≤4.4	1763.15	$5/2^+$	[M2]	0.000329 *5*	
		4347.8 *8*	12.4 *11*	0.0	$3/2^+$	[E3]	0.001000 *14*	B(E3)(W.u.)=44 *+46−17* E_γ,I_γ: from (^{16}O,αpγ).
4624.3	3/2,5/2	4624.0	100	0.0	$3/2^+$	D+Q		E_γ: other: 4618 *2* from (α,pγ). Mult.,δ: from p$\gamma(\theta)$ with −0.08 *14* for J=3/2; +0.36 *15* for J=5/2 in (α,pγ). If J(4624=5/2), Mult would be M1+E2 based on RUL.
4769.86	$(5/2^-)$	1606.95	100 *16*	3162.87	$7/2^-$	[M1,E2]	0.000133 *19*	B(M1)(W.u.)=0.040 *+23−5* if M1, B(E2)(W.u.)=58 *+34−7* if E2.
		1767.17	54 *16*	3002.64	$5/2^+$	[E1]	0.000483 *7*	B(E1)(W.u.)=4.6×10^{-4} *+30−9*
		2075.79	<15	2694.00	$3/2^+$	[E1]	0.000702 *10*	
		3550.29	<15	1219.38	$1/2^+$	[M2]	0.000645 *9*	
		4769.51	<15	0.0	$3/2^+$	[E1]	1.95×10^{-3} *3*	
4839.10	$(1/2^+,3/2)$	1836.41	<12	3002.64	$5/2^+$			
		2145.03	<6.8	2694.00	$3/2^+$			
		2193.36[b]	<5.1	2645.67	$7/2^+$			
		3075.81	100 *9*	1763.15	$5/2^+$			
		3619.52	<12	1219.38	$1/2^+$			
		4838.74	70 *9*	0.0	$3/2^+$			
4854.4	(1/2,3/2)	3634.8	100 *7*	1219.38	$1/2^+$			
		4854.0	33 *7*	0.0	$3/2^+$			
4880.96	$7/2^{(+)}$	1878.27	15 *5*	3002.64	$5/2^+$	[M1,E2]	0.000240 *30*	B(M1)(W.u.)=0.051 *+30−16* if M1, B(E2)(W.u.)=55 *+32−17* if E2.
		2186.89	<16	2694.00	$3/2^+$	[E2]	0.000417 *6*	
		2235.21	47 *8*	2645.67	$7/2^+$	[M1,E2]	0.00040 *4*	I_γ: other: 52 *9* from (α,pγ) for a doublet. B(M1)(W.u.)=0.095 *+43−18* if M1, B(E2)(W.u.)=72 *+33−14* if E2. B(E2)(W.u.)=72 *+33−14* upper bound exceeds RUL=100 if E2.
		3117.4 *20*	100 *8*	1763.15	$5/2^+$	[M1,E2]	0.00078 *6*	E_γ: from (α,pγ). I_γ: weighted average of 100 *9* from (α,pγ) and 100 *8* from (p,γ). B(M1)(W.u.)=0.075 *+33−12* if M1, B(E2)(W.u.)=29 *+13−5* if E2.
		4880.60	<8.1	0.0	$3/2^+$	[E2]	1.46×10^{-3} *2*	

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[‡]	E_f	J^π_f	Mult.[#]	δ[#]	α[a]	Comments
5009.1	(1/2,3/2)	835.7	<25	4173.43	$(5/2^+)$				
		950.0	<10	4059.1	$(3/2)^-$				
		1846.2[b]	<50	3162.87	$7/2^-$				
		2006.4	<20	3002.64	$5/2^+$				
		2315.0	<25	2694.00	$3/2^+$				
		2363.3[b]	<5	2645.67	$7/2^+$				
		3245.8	<10	1763.15	$5/2^+$				
		3789.5	<7	1219.38	$1/2^+$				
		5008.7	100	0.0	$3/2^+$				
5163.34	$7/2^-$	2000.41	93 *12*	3162.87	$7/2^-$	D+Q	+0.44 *20*		
		2160.63	23 *9*	3002.64	$5/2^+$				
		3400.01	100 *12*	1763.15	$5/2^+$	D(+Q)	0.00 *3*		
		3943.72[b]	<23	1219.38	$1/2^+$				
		5162.93	<23	0.0	$3/2^+$				
5214.5	$(5/2^+)$	205.4	<23	5009.1	(1/2,3/2)				
		1041.1	<60	4173.43	$(5/2^+)$				
		1155.4	<10	4059.1	$(3/2)^-$				
		2051.6	<15	3162.87	$7/2^-$				
		2211.8	<25	3002.64	$5/2^+$				
		2520.4	<15	2694.00	$3/2^+$				
		2568.7	<15	2645.67	$7/2^+$				
		3451.2	<6	1763.15	$5/2^+$				
		3994.9	<4	1219.38	$1/2^+$				
		5214.1	100	0.0	$3/2^+$	[M1,E2]		0.00147 *8*	
5403.3	$3/2^-$	2400.6	46 *9*	3002.64	$5/2^+$	[E1]		0.000913 *13*	B(E1)(W.u.)=0.0011 *+6−2*
		3640.0	<18	1763.15	$5/2^+$	[E1]		1.54×10^{-3} *2*	
		4183.7	100 *18*	1219.38	$1/2^+$	[E1]		1.75×10^{-3} *2*	B(E1)(W.u.)=4.5×10^{-4} *+20−6*
		5402.9	<13	0.0	$3/2^+$	[E1]		2.14×10^{-3} *3*	
5407.07	$11/2^-$	1059.32 *20*	14.5 *22*	4347.76	$9/2^-$	M1+E2	+0.14 *3*	3.67×10^{-5} *5*	B(M1)(W.u.)=0.0082 *+30−20*; B(E2)(W.u.)=0.54 *+35−23* E_γ: weighted average of 1059.3 *2* from (^{24}Mg,αpγ), 1059.5 *2* from (^{16}O,αpγ), and 1059.17 *20* from (^{14}N,αpnγ). Other: 1059.3 *10* from (α,pγ). I_γ: unweighted average of 12.0 *12* from (^{28}Si,αpγ), 16.9 *17* from (^{24}Mg,αpγ), 19.2 *5* from (^{16}O,αpγ), and 9.9 *22* from (α,pγ). Mult.: from γγ(DCO) and γγ(lin pol) in (^{28}Si,αpγ), ΔJ=1. δ: weighted average of +0.12 *3* from (^{16}O,αpγ),

$^{35}_{17}Cl_{18}$-29

Adopted Levels, Gammas (continued)

γ(^{35}Cl) (continued)

E_i(level)	J^{π}_i	E_γ[†]	I_γ[‡]	E_f	J^{π}_f	Mult.[#]	δ[#]	α[a]	Comments
									+0.2 *1* from (^{28}Si,αpγ), and +0.25 *8* from (α,pγ).
5407.07	11/2$^-$	2243.71 *23*	100.0 *22*	3162.87	7/2$^-$	E2		0.000444 *6*	B(E2)(W.u.)=4.6 *+15−9* E_γ: weighted average of 2244.2 *3* from (^{24}Mg,αpγ), 2243.3 *2* from (^{16}O,αpγ), 2244.00 *25* from (^{14}N,αpnγ), and 2244 *3* from (α,pγ). I_γ: weighted average of 100.0 *33* from (^{28}Si,αpγ), 100.0 *32* from (^{24}Mg,αpγ), 100.0 *23* from (^{16}O,αpγ), and 100.0 *22* from (α,pγ). Mult.: from $\gamma\gamma$(DCO) and $\gamma\gamma$(lin pol) in (^{28}Si,αpγ), ΔJ=2.
5586.0	5/2$^+$	2940.2	67 *17*	2645.67	7/2$^+$				
		3822.6	100 *17*	1763.15	5/2$^+$				
		4366.3	<17	1219.38	1/2$^+$				
		5585.5	<17	0.0	3/2$^+$				
5598.4	3/2$^+$,5/2$^+$	3835.0	37 *14*	1763.15	5/2$^+$	[M1,E2]		0.00105 *7*	B(M1)(W.u.)=0.050 *+29−19* if M1, B(E2)(W.u.)=13 *+8−5* if E2.
		5597.9	100 *14*	0.0	3/2$^+$	[M1,E2]		0.00157 *8*	B(M1)(W.u.)=0.044 *+23−12* if M1, B(E2)(W.u.)=5.3 *+28−14* if E2.
5646	(5/2,7/2,9/2$^+$)	2483	100	3162.87	7/2$^-$				
		2643		3002.64	5/2$^+$				
		3883		1763.15	5/2$^+$				
		5646	<8	0.0	3/2$^+$				
5655	(3/2)$^+$	2652	100 *6*	3002.64	5/2$^+$	[M1,E2]		0.00058 *5*	B(M1)(W.u.)=0.076 *+30−14* if M1, B(E2)(W.u.)=41 *+16−7* if E2.
		3892	7.5 *25*	1763.15	5/2$^+$	[M1,E2]		0.00107 *7*	B(M1)(W.u.)=0.0018 *+10−6* if M1, B(E2)(W.u.)=0.45 *+25−15* if E2.
		4435	<7.5	1219.38	1/2$^+$	[M1,E2]		0.00125 *8*	
5682.2	1/2$^-$,3/2$^-$	1504.3	100	4177.9	3/2$^-$				
5723.52	5/2$^+$	3029.38	63 *8*	2694.00	3/2$^+$				
		3960.13	100 *11*	1763.15	5/2$^+$				
5758	(1/2,3/2$^-$)	1580	67 *22*	4177.9	3/2$^-$				
		4538	<13	1219.38	1/2$^+$				
		5758	100 *22*	0.0	3/2$^+$				
5806.6	(1/2,3/2,5/2)	5806.1	100	0.0	3/2$^+$				
5830	(5/2$^-$)	2667	100	3162.87	7/2$^-$	D+Q			I_γ: from (α,pγ). δ: −2.2 *5* or −0.6 *1* for J=5/2 from $\gamma(\theta)$ in (α,pγ).
5926.65	(11/2)$^-$	1579.09 *30*	100 *11*	4347.76	9/2$^-$	M1+E2	−0.2 *1*	0.0001074 *22*	E_γ: weighted average of 1578.9 *5* from (^{16}O,αpγ), 1579.15 *30* from (^{14}N,αpnγ), and 1579.3 *15* from (α,pγ).

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[‡]	E_f	J^π_f	Mult.[#]	δ[#]	α[a]	Comments
									Mult.,δ: D+Q and δ from $\gamma\gamma(\theta)$(DCO) in (^{28}Si,αpγ) with ΔJ=1; E1+M2 ruled out by RUL. Other: −0.8 *4* from $\gamma(\theta)$ in (α,pγ).
5926.65	(11/2)$^-$	1983.7	11.1 *22*	3942.9	9/2$^+$	[E1]		0.000639 *9*	I_γ: from (^{28}Si,αpγ).
		2763.66		3162.87	7/2$^-$				
6087.31	13/2$^-$	680.35 *10*	100.0 *21*	5407.07	11/2$^-$	M1+E2	+0.020 *15*	8.70×10^{-5} *12*	B(M1)(W.u.)=0.0107 *10* E_γ: weighted average of 680.4 *1* from (^{24}Mg,αpγ), 680.5 *5* from (^{16}O,αpγ), and 680.22 *15* from (^{14}N,αpnγ). Other: 679.9 *10* from (α,pγ). I_γ: weighted average of 100.0 *31* from (^{28}Si,αpγ), 100.0 *21* from (^{16}O,αpγ), and 100 *6* from (α,pγ). Mult.,δ: from $\gamma(\theta)$ and γ(lin pol) in (α,pγ). Other: δ=+0.1 *1* from $\gamma\gamma$(DCO) in (^{28}Si,αpγ), ΔJ=1.
		1739.4 *4*	6.9 *9*	4347.76	9/2$^-$	E2		0.0002076 *29*	B(E2)(W.u.)=0.054 *+9−8* E_γ: weighted average of 1739.6 *5* from (^{16}O,αpγ) and 1739.3 *4* from (^{14}N,αpnγ). I_γ: weighted average of 6.3 *6* from (^{28}Si,αpγ), 7.9 *9* from (^{16}O,αpγ), and 18 *6* from (α,pγ). Mult.: from $\gamma\gamma(\theta)$(DCO) and $\gamma\gamma$(lin pol) in (^{28}Si,αpγ), ΔJ=2.
6106.2	(3/2,5/2$^+$)	4342.8	10.3 *35*	1763.15	5/2$^+$				
		4886.5	62 *17*	1219.38	1/2$^+$				
		6105.6	100 *17*	0.0	3/2$^+$				
6181	(1/2$^+$ to 7/2$^+$)	6180	100	0.0	3/2$^+$				
6380.2	(9/2$^-$)	296 *1*		6087.31	13/2$^-$				I_γ: from (^{16}O,αpγ).
		971 *1*		5407.07	11/2$^-$				I_γ: from (^{16}O,αpγ).
6492	(3/2$^+$)	6491	100	0.0	3/2$^+$				
6660.2	(11/2$^-$)	2312.4		4347.76	9/2$^-$				
		3497.1		3162.87	7/2$^-$				
7066.2	(5/2$^+$)	1902.8	1.8	5163.34	7/2$^-$				
		2892.6	1.0	4173.43	(5/2$^+$)				
		2953.7	1.2	4112.4	7/2$^+$				
		3007.0	2.9	4059.1	(3/2)$^-$				
		3903.1	12	3162.87	7/2$^-$				
		4063.3	6.1	3002.64	5/2$^+$				

$^{35}_{17}Cl_{18}$-31

Adopted Levels, Gammas (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ†	I_γ‡	E_f	J^π_f	Mult.#	δ#
7066.2	$(5/2^+)$	4371.9	10	2694.00	$3/2^+$		
		4420.2	3.3	2645.67	$7/2^+$		
		5302.6	29	1763.15	$5/2^+$		
		5846.3	37	1219.38	$1/2^+$		
		7065.4	100	0.0	$3/2^+$		
7103.4	$3/2^{(-)}$	2925.4	2.3	4177.9	$3/2^-$		
		3184.8	2.3	3918.47	$3/2^+$		
		4100.5	4.7	3002.64	$5/2^+$		
		4409.1	17	2694.00	$3/2^+$	D(+Q)	0.00 *3*
		5339.8	6.3	1763.15	$5/2^+$		
		5883.5	100	1219.38	$1/2^+$	D(+Q)	0.00 *3*
		7102.6	23	0.0	$3/2^+$		
7178.6	$1/2^{(+)}$	1072.4	7.9	6106.2	$(3/2,5/2^+)$		
		1775.3	4.5	5403.3	$3/2^-$		
		2339.4	4	4839.10	$(1/2^+,3/2)$		
		3005.0	2.1	4173.43	$(5/2^+)$		
		3119.4	58	4059.1	$(3/2)^-$		
		3210.8	24	3967.6	$1/2^+$		
		3260.0	7.9	3918.47	$3/2^+$		
		4484.3	11	2694.00	$3/2^+$		
		5958.7	45	1219.38	$1/2^+$		
		7177.8	100	0.0	$3/2^+$		
7185	$5/2^+$	1970	<1.1	5214.5	$(5/2^+)$		
		2176	<1.1	5009.1	$(1/2,3/2)$		
		3011	<11	4173.43	$(5/2^+)$		
		3126	57 *13*	4059.1	$(3/2)^-$		
		4022	<6.5	3162.87	$7/2^-$		
		4182	<6.5	3002.64	$5/2^+$		
		4491	4.4 *22*	2694.00	$3/2^+$		
		4539	13 *5*	2645.67	$7/2^+$		
		5421	9 *7*	1763.15	$5/2^+$		
		5965	35 *15*	1219.38	$1/2^+$		
		7184	100 *22*	0.0	$3/2^+$		
7194.6	$1/2^-$	2185.4	1.2	5009.1	$(1/2,3/2)$		
		2340.1	3.0	4854.4	$(1/2,3/2)$		
		2355.4	2.7	4839.10	$(1/2^+,3/2)$		
		3016.6	24	4177.9	$3/2^-$		
		3135.4	4.5	4059.1	$(3/2)^-$		
		3226.8	2.7	3967.6	$1/2^+$		
		3276.0	6	3918.47	$3/2^+$		
		4500.3	1.6	2694.00	$3/2^+$		
		5431.0	0.75	1763.15	$5/2^+$		

Adopted Levels, Gammas (continued)

γ(^{35}Cl) (continued)

E_i(level)	J^π_i	E_γ†	I_γ‡	E_f	J^π_f	Mult.#	δ#	Comments
7194.6	1/2$^-$	5974.7	100	1219.38	1/2$^+$			
		7193.8	3.0	0.0	3/2$^+$			
7225.5	5/2	1119.3	3.1	6106.2	(3/2,5/2$^+$)			
		1418.9		5806.6	(1/2,3/2,5/2)			
		1543.3	3.4	5682.2	1/2$^-$,3/2$^-$			
		1571	1.4	5655	(3/2)$^+$			
		1822.2	1.5	5403.3	3/2$^-$			
		2010.9	12	5214.5	(5/2$^+$)			
		3047.5	14	4177.9	3/2$^-$			
		3166.3	3.4	4059.1	(3/2)$^-$			
		3306.9	2.5	3918.47	3/2$^+$			
		4222.6		3002.64	5/2$^+$			
		4531.2	17	2694.00	3/2$^+$	D+Q	+0.36 *4*	
		5461.9	12	1763.15	5/2$^+$	D+Q	−0.36 *7*	
		7224.7	100	0.0	3/2$^+$	D+Q	+0.36 *4*	
7234.0	5/2$^{(+)}$	1579	0.22	5655	(3/2)$^+$			
		3056.0		4177.9	3/2$^-$			
		4539.7	1.9	2694.00	3/2$^+$			
		4588.0	3.2	2645.67	7/2$^+$			
		5470.4	0.22	1763.15	5/2$^+$			
		6014.1	1.9	1219.38	1/2$^+$			
		7233.2	100	0.0	3/2$^+$	D+Q	−0.10	
7272.7	1/2$^-$	1515	1.7	5758	(1/2,3/2$^-$)			
		1869.4	0.29	5403.3	3/2$^-$			
		2263.5	2.0	5009.1	(1/2,3/2)			
		2433.5	1.5	4839.10	(1/2$^+$,3/2)			
		3099.1	0.58	4173.43	(5/2$^+$)			
		3213.4	1.3	4059.1	(3/2)$^-$			
		3304.9	1.2	3967.6	1/2$^+$			
		4578.4	1.6	2694.00	3/2$^+$			
		6052.8	35	1219.38	1/2$^+$			
		7271.9	100	0.0	3/2$^+$			
7361.9	3/2$^{(+)}$	1707	0.41	5655	(3/2)$^+$			
		2352.7	0.82	5009.1	(1/2,3/2)			
		2522.7	1.4	4839.10	(1/2$^+$,3/2)			
		3183.8	0.69	4177.9	3/2$^-$			
		3302.6	2.7	4059.1	(3/2)$^-$			
		3394.1	4.1	3967.6	1/2$^+$			
		3443.3	1.1	3918.47	3/2$^+$			
		4359.0	0.41	3002.64	5/2$^+$			
		4667.6	0.69	2694.00	3/2$^+$			
		5598.3	11	1763.15	5/2$^+$	D+Q		δ: >+2.5 or −0.01 *10* from $\gamma(\theta)$ in (p,γ).

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[‡]	E_f	J^π_f	Mult.[#]	δ[#]	α[a]	Comments
7361.9	$3/2^{(+)}$	6141.9	100	1219.38	$1/2^+$	(M1+E2)			Mult.: D+Q from $\gamma(\theta)$ in (p,γ); E1+M2 is less likely for this strong primary transition. δ: −0.05 *3* or +2.5 *3* in (p,γ).
		7361.1	14	0.0	$3/2^+$	D+Q			δ: −0.16<δ(Q/D)<−0.12 or +7.5<δ(Q/D)<+10 in (p,γ).
7396.4	$7/2^{(-)}$	3048.5	23	4347.76	$9/2^-$				
		3283.8	17	4112.4	$7/2^+$				
		3453.3	11	3942.9	$9/2^+$				
		4233.3	100	3162.87	$7/2^-$	(M1+E2)	+0.28 *25*	1.12×10^{-3} *3*	Mult.: D+Q from $\gamma(\theta)$ in (p,γ); E1+M2 is less likely for this strong primary transition.
		4393.5	30	3002.64	$5/2^+$	D(+Q)	+0.011 *18*		
		4750.4	28	2645.67	$7/2^+$				
		5632.8	4.3	1763.15	$5/2^+$				
7450.9	3/2	3532.2	6.9 *34*	3918.47	$3/2^+$				
		4448.0	13.7 *14*	3002.64	$5/2^+$				
		4756.6	11.0 *11*	2694.00	$3/2^+$				
		5687.3	5.5 *28*	1763.15	$5/2^+$				
		6230.9	100 *10*	1219.38	$1/2^+$				
7501.1	$(5/2^+,7/2,9/2^+)$	3558.0	46 *6*	3942.9	$9/2^+$				
		4498.2	23.6 *24*	3002.64	$5/2^+$				
		4855.1	100 *11*	2645.67	$7/2^+$				
		5737.5	12.7 *13*	1763.15	$5/2^+$				
7502.9		1396.7[@]	14[@]	6106.2	$(3/2,5/2^+)$				
		1904.4[@]	4[@]	5598.4	$3/2^+,5/2^+$				
		2493.7[@]	4[@]	5009.1	(1/2,3/2)				
		3324.8[@]	6[@]	4177.9	$3/2^-$				
		3443.6[@]	100[@]	4059.1	$(3/2)^-$				
		3559.8[@]	4[@]	3942.9	$9/2^+$				
		4339.7[@]	8[@]	3162.87	$7/2^-$				
		4500.0[@]	8[@]	3002.64	$5/2^+$				
		4856.9[@]	14[@]	2645.67	$7/2^+$				
		5739.2[@]	1.4[@]	1763.15	$5/2^+$				
		6282.9[@]	36[@]	1219.38	$1/2^+$				
		7502.0[@]	0.6[@]	0.0	$3/2^+$				
7518.8	7/2	1920.3	2.9 *15*	5598.4	$3/2^+,5/2^+$				
		1932.7	4.4 *22*	5586.0	$5/2^+$				
		3170.9	2.9 *15*	4347.76	$9/2^-$				
		3575.7	2.9 *15*	3942.9	$9/2^+$				

Adopted Levels, Gammas (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J_i^π	E_γ[†]	I_γ[‡]	E_f	J_f^π	Mult.[#]	δ[#]	α[a]	Comments
7518.8	7/2	4355.6	100 *10*	3162.87	$7/2^-$	D(+Q)	+0.1 *2*		
		4872.8	4.4 *22*	2645.67	$7/2^+$				
		5755.1	29.4 *30*	1763.15	$5/2^+$	D+Q	+0.098 *22*		
7548.9	$7/2^-$	1903	0.59 *7*	5646	$(5/2,7/2,9/2^+)$				
		1962.8	0.11	5586.0	$5/2^+$	[E1]		0.000624 *9*	
		2778.9	1.49 *11*	4769.86	$(5/2^-)$	[M1,E2]		0.00064 *6*	
		4385.7	100.0 *11*	3162.87	$7/2^-$	M1+E2	−0.07 *2*	1.16×10^{-3} *2*	
		4545.9	2.13 *11*	3002.64	$5/2^+$	(E1+M2)	+0.6 *4*	0.00163 *22*	Mult.,δ: D+Q and δ from $\gamma(\theta)$ in (p,γ); $\Delta\pi$=yes from level scheme.
		4902.9	0.72 *8*	2645.67	$7/2^+$	[E1]		1.99×10^{-3} *3*	
		5785.2	0.35 *4*	1763.15	$5/2^+$	[E1]		2.24×10^{-3} *3*	
		7548.0	0.287 *32*	0.0	$3/2^+$	(M2)			Mult.: Q(+O) with δ(O/Q)=0.1 *2* from $\gamma(\theta)$ in (p,γ); $\Delta\pi$=yes from level scheme; E3 component is less likely based on RUL.
7561.3	1/2,3/2	3383.2	2.6	4177.9	$3/2^-$				
		3502.0	17	4059.1	$(3/2)^-$				
		3593.5	1.4	3967.6	$1/2^+$				
		3642.6	4.6	3918.47	$3/2^+$				
		4866.9	63	2694.00	$3/2^+$				
		6341.3	100	1219.38	$1/2^+$				
		7560.4	97	0.0	$3/2^+$				
7600.8	$5/2^+$	2437.4	5.8	5163.34	$7/2^-$	[E1]		0.000934 *13*	
		2719.7	6.1	4880.96	$7/2^{(+)}$				
		2761.6	6.1	4839.10	$(1/2^+,3/2)$				
		2830.8	3.9	4769.86	$(5/2^-)$				
		2976.4	5.5	4624.3	3/2,5/2				
		3422.7	9.1	4177.9	$3/2^-$	[E1]		1.44×10^{-3} *2*	
		3427.2		4173.43	$(5/2^+)$				
		3541.5	6.1	4059.1	$(3/2)^-$	[E1]		1.50×10^{-3} *2*	
		4437.6	15	3162.87	$7/2^-$	[E1]		1.85×10^{-3} *3*	
		4597.8	24	3002.64	$5/2^+$	M1+E2	+0.50 *22*	1.25×10^{-3} *3*	Mult.: D+Q from $\gamma(\theta)$ in (p,γ); E1+M2 is less likely for this strong primary transition.
		4906.4	55	2694.00	$3/2^+$	(M1+E2)		0.00139 *8*	Mult.: D+Q from $\gamma(\theta)$ in (p,γ); $\Delta\pi$=no from level scheme.
		4954.8	5.8	2645.67	$7/2^+$	[M1,E2]		0.00140 *8*	
		5837.1	58	1763.15	$5/2^+$	M1+E2	+0.31 *7*	1.57×10^{-3} *2*	
		6380.8	3.3	1219.38	$1/2^+$	[E2]			
		7599.9	100	0.0	$3/2^+$	M1+E2	−0.19 *4*		
7618.7	$5/2^+$	1438	0.77	6181	$(1/2^+$ to $7/2^+)$				
		2404.1	2.6	5214.5	$(5/2^+)$				
		2455.3	0.9	5163.34	$7/2^-$				

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[‡]	E_f	J^π_f	Mult.[#]	δ[#]	α[a]	Comments
7618.7	5/2$^+$	2779.5	0.9	4839.10	(1/2$^+$,3/2)				
		3440.6	3.9	4177.9	3/2$^-$				
		3559.4	2.6	4059.1	(3/2)$^-$				
		4455.5	3.9	3162.87	7/2$^-$				
		5855.0	13	1763.15	5/2$^+$	M1+E2	+0.48 *11*	1.59×10^{-3} *3*	
		7617.8	100	0.0	3/2$^+$	M1+E2	+2.2 *3*		
7656.6	(1/2,3/2,5/2$^+$)	1550.4	5.3 *26*	6106.2	(3/2,5/2$^+$)				
		1974.3	1.8 *9*	5682.2	1/2$^-$,3/2$^-$				
		2002	1.8 *9*	5655	(3/2)$^+$				
		2817.4	5.3 *26*	4839.10	(1/2$^+$,3/2)				
		3478.5	1.8 *9*	4177.9	3/2$^-$				
		3597.3	56 *6*	4059.1	(3/2)$^-$				
		3688.8	3.5 *18*	3967.6	1/2$^+$				
		7655.7	100 *11*	0.0	3/2$^+$				
7671.9	5/2$^+$	2073.4	2.1	5598.4	3/2$^+$,5/2$^+$				
		2508.5	7	5163.34	7/2$^-$				
		2790.8	2.1	4880.96	7/2$^{(+)}$				
		2901.9	1.8	4769.86	(5/2$^-$)				
		3324.0	11	4347.76	9/2$^-$				
		3498.3	8.8	4173.43	(5/2$^+$)				
		4508.7	3.2	3162.87	7/2$^-$				
		4668.9	16	3002.64	5/2$^+$	M1+E2	−1.4 *5*	0.00134 *4*	
		5025.8	23	2645.67	7/2$^+$	D(+Q)	+0.3 *22*		
		5908.2	100	1763.15	5/2$^+$	M1+E2	+0.86 *14*	1.64×10^{-3} *3*	
		7671.0	1.4	0.0	3/2$^+$				
7685.5	3/2$^-$	1927		5758	(1/2,3/2$^-$)				
		2087.0	2.9	5598.4	3/2$^+$,5/2$^+$				
		2282.1	2.9	5403.3	3/2$^-$				
		2470.9	1	5214.5	(5/2$^+$)				
		2676.3	0.86	5009.1	(1/2,3/2)				
		3507.4	4.3	4177.9	3/2$^-$				
		3511.9	5.7	4173.43	(5/2$^+$)				
		3626.2	8.6	4059.1	(3/2)$^-$				
		4682.5	14	3002.64	5/2$^+$	D(+Q)	−0.01 *5*		
		5921.8	1	1763.15	5/2$^+$				
		6465.5	1.4	1219.38	1/2$^+$				
		7684.6	100	0.0	3/2$^+$	D+Q	+6.0 *7*		Mult.: D+Q with δ=+6.0 *7* from $\gamma(\theta)$ in (p,γ); $\Delta\pi$=yes from level scheme; but E1+M2 with such a large δ(M2/E1) is unlikely for this strong primary transition based on RUL, which might suggest a different level in (p,γ).

$^{35}_{17}Cl_{18}$-36

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[‡]	E_f	J^π_f	Mult.[#]	δ[#]	α[a]	Comments
7693.9	(1/2+,3/2,5/2)	2095.4	1.0 5	5598.4	3/2+,5/2+				
		3634.6	1.0 5	4059.1	(3/2)−				
		4690.9	2.1 11	3002.64	5/2+				
		7693.0	100 11	0.0	3/2+				
7706.4	5/2+	2051	4.8	5655	(3/2)+	M1+E2		0.00032 4	δ: <−2 or ≥13 from γ(θ) in (p,γ).
		2825.3	0.48	4880.96	7/2(+)				
		2867.2	1.6	4839.10	(1/2+,3/2)				
		3082.0	0.96	4624.3	3/2,5/2				
		3528.3	2.4	4177.9	3/2−				
		3593.8	2.4	4112.4	7/2+				
		3787.7	0.72	3918.47	3/2+				
		4703.4	1.2	3002.64	5/2+				
		5060.3	4.8	2645.67	7/2+	M1+E2		0.00143 8	δ: −1.01 10 or ≥13 from (p,γ).
		5942.7	0.48	1763.15	5/2+				
		6486.4	0.6	1219.38	1/2+				
		7705.5	100	0.0	3/2+	D(+Q)	+0.1 1		
7744.8	7/2−	2863.7	47	4880.96	7/2(+)				
		2974.8	4.7	4769.86	(5/2−)				
		3632.2	14	4112.4	7/2+				
		3801.7	23	3942.9	9/2+	(E1)		1.60×10^{-3} 2	Mult.: D(+Q) with δ=−0.16 17 from γ(θ) in (p,γ); Δπ=yes from level scheme; M2 component is less likely for the primary transition.
		4581.6	37	3162.87	7/2−	M1+E2	+0.18 8	1.23×10^{-3} 2	
		5098.7	100	2645.67	7/2+	(E1)		2.05×10^{-3} 3	Mult.: D(+Q) with δ=+0.1 30 from γ(θ) in (p,γ); Δπ=yes from level scheme; M2 component is less likely for the primary transition.
		5981.1	5.6	1763.15	5/2+				
		6524.8[b]	1.4	1219.38	1/2+	[E3]			This transition is considered questionable by the evaluators since it would require Mult=E3 which is unlikely for this primary transition based on RUL.
7777.1	(5/2+)	2122	1.3	5655	(3/2)+				
		2613.7	2.8	5163.34	7/2−				
		2937.9	5	4839.10	(1/2+,3/2)				
		3007.1	7.5	4769.86	(5/2−)				
		3599.0	18	4177.9	3/2−				
		3717.8	2.8	4059.1	(3/2)−				
		4613.9	5	3162.87	7/2−				
		4774.1	10	3002.64	5/2+				
		5082.7	23	2694.00	3/2+				
		5131.0	3.3	2645.67	7/2+				

$^{35}_{17}Cl_{18}$-37

Adopted Levels, Gammas (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[‡]	E_f	J^π_f	Mult.[#]	δ[#]	α[a]	Comments
7777.1	$(5/2^+)$	6013.4	100	1763.15	$5/2^+$				
		6557.1	28	1219.38	$1/2^+$				
		7776.2	45	0.0	$3/2^+$				
7781.7	$(5/2^-)$	1675.5	3.9	6106.2	$(3/2,5/2^+)$				
		2183.2		5598.4	$3/2^+,5/2^+$				
		2378.3		5403.3	$3/2^-$				
		2567.1	4.2	5214.5	$(5/2^+)$				
		3011.7	19	4769.86	$(5/2^-)$				
		3433.8	5.6	4347.76	$9/2^-$				
		3608.1	100	4173.43	$(5/2^+)$				
		3669.1	17	4112.4	$7/2^+$				
		3722.4	3.3	4059.1	$(3/2)^-$				
		3863.0	31	3918.47	$3/2^+$				
		4618.5	14	3162.87	$7/2^-$				
		5135.6	22	2645.67	$7/2^+$				
		6018.0	50	1763.15	$5/2^+$				
		6561.7	3.3	1219.38	$1/2^+$				
		7780.8	4.7	0.0	$3/2^+$				
7797.2	$1/2^-$	2039		5758	$(1/2,3/2^-)$				
		2142	1.6	5655	$(3/2)^+$				
		2393.8	2.4	5403.3	$3/2^-$				
		2788.0	2.4	5009.1	$(1/2,3/2)$				
		2942.7	1.6	4854.4	$(1/2,3/2)$				
		2958.0	1.3	4839.10	$(1/2^+,3/2)$				
		3737.9	0.36	4059.1	$(3/2)^-$				
		6577.2	11	1219.38	$1/2^+$				
		7796.3	100	0.0	$3/2^+$				
7837.2	$3/2^-$	1656[&]	1.4[&]	6181	$(1/2^+$ to $7/2^+)$				
		2238.7[&]	1.4[&]	5598.4	$3/2^+,5/2^+$				
		2622.6[&]	3.8[&]	5214.5	$(5/2^+)$				
		3659.1[&]	76[&]	4177.9	$3/2^-$	(M1+E2)	−0.05 *3*	0.000917 *13*	Mult.: D+Q from (p,γ); $\Delta\pi$=no from level scheme. B(M1)(W.u.)=1.03×10^3 *19* exceeds RUL=10. B(E2)(W.u.)=7×10^2 *+11−6* exceeds RUL=100.
		3663.6[&]	8.1[&]	4173.43	$(5/2^+)$				
		3777.9[&]	5.4[&]	4059.1	$(3/2)^-$				
		3894.1[&b]	1.4[&]	3942.9	$9/2^+$				This γ is a primary γ from the unresolved 7837+7839 doublet goes to $9/2^+$, which is unlikely from a $3/2^-$, 7837 level. It is likely from the $(5/2^+)$, 7839.0 level.
		3918.5[&]	0.54[&]	3918.47	$3/2^+$				

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[‡]	E_f	J^π_f	Mult.[#]	δ[#]	α[a]	Comments
7837.2	$3/2^-$	4834.2[&]	11[&]	3002.64	$5/2^+$				
		6073.5[&]	5.1[&]	1763.15	$5/2^+$				
		6617.2[&]	100[&]	1219.38	$1/2^+$	(E1)			Mult.: D(+Q) with δ=+0.06 *3* from (p,γ); $\Delta\pi$=yes from level scheme; M2 component is less likely for this strong primary transition. B(E1)(W.u.)=6.5 +*10*−*8* exceeds RUL=0.1.
		7836.3[&]	57[&]	0.0	$3/2^+$	(E1)			Mult.: D(+Q) with δ=+0.02 *3* from (p,γ); $\Delta\pi$=yes from level scheme; M2 component is less likely for this strong primary transition. B(E1)(W.u.)=2.24 +*47*−*44* exceeds RUL=0.1.
7839.0	$(5/2^+)$	1658[&]	1.4[&]	6181	$(1/2^+$ to $7/2^+)$				
		2240.5[&]	1.4[&]	5598.4	$3/2^+,5/2^+$				
		2624.4[&]	3.8[&]	5214.5	$(5/2^+)$				
		3660.9[&]	76[&]	4177.9	$3/2^-$				
		3665.4[&]	8.1[&]	4173.43	$(5/2^+)$				
		3779.7[&]	5.4[&]	4059.1	$(3/2)^-$				
		3895.9[&]	1.4[&]	3942.9	$9/2^+$				
		3920.3[&]	0.54[&]	3918.47	$3/2^+$				
		4836.0[&]	11[&]	3002.64	$5/2^+$				
		6075.3[&]	5.1[&]	1763.15	$5/2^+$				
		6619.0[&]	100[&]	1219.38	$1/2^+$				
		7838.1[&]	57[&]	0.0	$3/2^+$				
7868.7	$(1/2^+,3/2,5/2^+)$	2654.1	1.6	5214.5	$(5/2^+)$				
		2859.5	4.9	5009.1	$(1/2,3/2)$				
		3695.1	8.2	4173.43	$(5/2^+)$				
		6105.0	28	1763.15	$5/2^+$				
		6648.6	21	1219.38	$1/2^+$				
		7867.8	100	0.0	$3/2^+$				
7873.03	$13/2^+$	1212.8		6660.2	$(11/2^-)$				E_γ: from (^{28}Si,$\alpha p\gamma$) only.
		1786.2 *5*	9.6 *4*	6087.31	$13/2^-$	D(+Q)	−0.6 *6*		I_γ: weighted average of 8.2 *18* from (^{28}Si,$\alpha p\gamma$) and 9.7 *4* from (^{16}O,$\alpha p\gamma$). Mult.,δ: from $\gamma\gamma$(DCO) in (^{28}Si,$\alpha p\gamma$).
		1946.35 *30*	48.0 *26*	5926.65	$(11/2)^-$	E1+M2	+0.2 *1*	0.000594 *22*	E_γ: weighted average of 1946.2 *5* from (^{16}O,$\alpha p\gamma$) and 1946.40 *30* from (^{14}N,$\alpha pn\gamma$). I_γ: weighted average of 46 *5* from (^{28}Si,$\alpha p\gamma$)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[‡]	E_f	J^π_f	Mult.[#]	δ[#]	α[a]	Comments
									and 48.5 *26* from (^{16}O,αpγ). Mult.,δ: from $\gamma\gamma$(DCO) and $\gamma\gamma$(lin pol) in (^{28}Si,αpγ), ΔJ=1.
7873.03	$13/2^+$	2465.94 *30*	100.0 *20*	5407.07	$11/2^-$	E1+M2	−0.25 *10*	0.00091 *4*	E_γ: weighted average of 2466.2 *5* from (^{16}O,αpγ) and 2465.85 *30* from (^{14}N,αpnγ). I_γ: weighted average of 100 *9* from (^{28}Si,αpγ) and 100.0 *20* from (^{16}O,αpγ). Mult.,δ: from $\gamma\gamma$(DCO) and $\gamma\gamma$(lin pol) in (^{28}Si,αpγ), ΔJ=1.
7880.8	$3/2^+,5/2^+$	2226	40	5655	$(3/2)^+$				
		3041.6	5.7	4839.10	$(1/2^+,3/2)$				
		3256.3	6.7	4624.3	3/2,5/2				
		3702.7	10	4177.9	$3/2^-$				
		3768.2	4.3	4112.4	$7/2^+$				
		3821.5	6.7	4059.1	$(3/2)^-$				
		3962.1	6.7	3918.47	$3/2^+$				
		4877.8	37	3002.64	$5/2^+$				
		5186.4	83	2694.00	$3/2^+$				
		6117.1	100	1763.15	$5/2^+$				
		7879.9	33	0.0	$3/2^+$				
7899.2	(5/2)	2253	2.2	5646	$(5/2,7/2,9/2^+)$				
		2735.8	1.3	5163.34	$7/2^-$				
		3018.1	1.2	4880.96	$7/2^{(+)}$				
		3274.7	0.74	4624.3	3/2,5/2				
		3721.1	12	4177.9	$3/2^-$				
		3725.6	12	4173.43	$(5/2^+)$				
		3839.9	4.4	4059.1	$(3/2)^-$				
		4736.0	5.9	3162.87	$7/2^-$				
		5204.8	1.9	2694.00	$3/2^+$				
		6135.5	100	1763.15	$5/2^+$				
		7898.2	5.9	0.0	$3/2^+$				
7923.3	$(3/2^+,5/2^+)$	3084.1	3.4	4839.10	$(1/2^+,3/2)$				
		3153.3	4.9	4769.86	$(5/2^-)$				
		3298.8	4.6	4624.3	3/2,5/2				
		3745.2	7.3	4177.9	$3/2^-$				
		3749.7	12	4173.43	$(5/2^+)$				
		3864.0	4.2	4059.1	$(3/2)^-$				
		3955.5	3.7	3967.6	$1/2^+$				
		4004.6	2.9	3918.47	$3/2^+$				
		4920.3	2.2	3002.64	$5/2^+$				
		5228.9	27	2694.00	$3/2^+$				

$^{35}_{17}Cl_{18}$-40

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J_i^π	E_γ†	I_γ‡	E_f	J_f^π	Mult.#	Comments
7923.3	(3/2$^+$,5/2$^+$)	5277.2	2.2	2645.67	7/2$^+$		
		6159.6	3.7	1763.15	5/2$^+$		
		6703.2	66	1219.38	1/2$^+$		
		7922.3	100	0.0	3/2$^+$		
7970.3	(5/2$^-$)	2324	15	5646	(5/2,7/2,9/2$^+$)		
		2384.2	15	5586.0	5/2$^+$		
		2755.7	15	5214.5	(5/2$^+$)		
		3089.2	50	4880.96	7/2$^{(+)}$		
		3200.3	25	4769.86	(5/2$^-$)		
		3622.3	40	4347.76	9/2$^-$		
		3792.2	30	4177.9	3/2$^-$		
		3857.7	30	4112.4	7/2$^+$		
		4807.1	50	3162.87	7/2$^-$		
		4967.3	25	3002.64	5/2$^+$		
		5324.2	100	2645.67	7/2$^+$		
		6206.6	100	1763.15	5/2$^+$		
		6750.2	5	1219.38	1/2$^+$		
7987.8	3/2$^+$	1881.6	0.83	6106.2	(3/2,5/2$^+$)		
		2333	2.8	5655	(3/2)$^+$		
		3809.7	<1.3	4177.9	3/2$^-$		I_γ: total Iγ=1.3 for γ decays to 4173 and 4178 doublet final levels.
		3814.2	<1.3	4173.43	(5/2$^+$)		I_γ: total Iγ=1.3 for γ decays to 4173 and 4178 doublet final levels.
		3928.5	1.7	4059.1	(3/2)$^-$		
		6767.7	100	1219.38	1/2$^+$	M1+E2	δ: +0.21 *4* or −3.1 *4* from (p,γ).
		7986.8	60	0.0	3/2$^+$	M1+E2	δ: −0.36 *2* or −11 *3* from (p,γ).
7995.6	(5/2)	2188.9	0.77	5806.6	(1/2,3/2,5/2)		
		2397.1	1.4	5598.4	3/2$^+$,5/2$^+$		
		2592.2	3.9	5403.3	3/2$^-$		
		2832.1	1.9	5163.34	7/2$^-$		
		3817.5	5.1	4177.9	3/2$^-$		
		3936.3	1.4	4059.1	(3/2)$^-$		
		4832.4	9	3162.87	7/2$^-$		
		4992.6	2.6	3002.64	5/2$^+$		
		5301.2	0.26	2694.00	3/2$^+$		
		5349.5	0.77	2645.67	7/2$^+$		
		6231.9	1.2	1763.15	5/2$^+$		
		7994.6	100	0.0	3/2$^+$		
8001.0	(7/2$^+$)	2355	2.6	5646	(5/2,7/2,9/2$^+$)		
		2786.4	1.1	5214.5	(5/2$^+$)		
		3119.9	1	4880.96	7/2$^{(+)}$		
		3231.0	11	4769.86	(5/2$^-$)		
		3653.0	2.7	4347.76	9/2$^-$		
		3827.4	14	4173.43	(5/2$^+$)		

Adopted Levels, Gammas (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[‡]	E_f	J^π_f	Mult.[#]	δ[#]	α[a]	Comments
8001.0	(7/2+)	4057.9	4.3	3942.9	9/2+				
		4837.8	1.1	3162.87	7/2−				
		5354.9	2.3	2645.67	7/2+				
		6237.3	100	1763.15	5/2+				
		8000.0	2	0.0	3/2+				
8005.0	5/2+	2350	3.3	5655	(3/2)+				
		3123.9	1.1	4880.96	7/2(+)				
		3165.8	5.5	4839.10	(1/2+,3/2)				
		3826.9	<2.4	4177.9	3/2−				I_γ: total Iγ=2.4 for γ decays to 4173 and 4178 doublet final levels.
		3831.3	<2.4	4173.43	(5/2+)				I_γ: total Iγ=2.4 for γ decays to 4173 and 4178 doublet final levels.
		4841.8	27	3162.87	7/2−				
		5002.0	3.6	3002.64	5/2+				
		5310.6	3.1	2694.00	3/2+				
		5358.9	7.3	2645.67	7/2+				
		6241.3	27	1763.15	5/2+				
		6784.9	1.1	1219.38	1/2+				
		8004.0	100	0.0	3/2+				
8035.7	1/2−,3/2−	2229.0	10.7 *11*	5806.6	(1/2,3/2,5/2)				
		3857.6	26.8 *27*	4177.9	3/2−				
		3976.4	19.6 *20*	4059.1	(3/2)−				
		5341.3	9 *5*	2694.00	3/2+				
		6272.0	9 *5*	1763.15	5/2+				
		6815.6	100 *11*	1219.38	1/2+				
		8034.7	3.6 *18*	0.0	3/2+				
8038.4	1/2+,3/2+	3183.8	1.1	4854.4	(1/2,3/2)				
		3199.1	2.8	4839.10	(1/2+,3/2)				
		3413.9	5.3	4624.3	3/2,5/2				
		3860.3	2.8	4177.9	3/2−				
		3979.1	5.3	4059.1	(3/2)−				
		4070.6	1.4	3967.6	1/2+				
		4119.7	2.5	3918.47	3/2+				
		5035.4	7	3002.64	5/2+				
		5344.0	100	2694.00	3/2+	M1+E2	−0.20 *2*	1.44×10^{-3} *2*	Mult.,δ: D+Q with δ=−0.20 *2* or +19 *8* from (p,γ); E1+M2 and the larger δ(E2/M1) is ruled out by RUL. B(M1)(W.u.)=52.1 *+43−37* exceeds RUL=10. B(E2)(W.u.)=2.8×10^2 *+6−5* exceeds RUL=100.
		6274.7	26	1763.15	5/2+				
		6818.3	7	1219.38	1/2+				
		8037.4	14	0.0	3/2+	M1+E2	−0.13 *4*		B(M1)(W.u.)=2.20 *44*; B(E2)(W.u.)=2.2 *+16−12*

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[‡]	E_f	J^π_f	Mult.[#]	α[a]	Comments
								Mult.,δ: D+Q with δ=−0.13 *4* or +8.1 *26* from (p,γ); E1+M2 and the larger δ(E2/M1) is ruled out by RUL.
8075.9	(5/2)	2430	20	5646	(5/2,7/2,9/2$^+$)			
		2489.8	20	5586.0	5/2$^+$			
		2861.3	14	5214.5	(5/2$^+$)			
		2912.4		5163.34	7/2$^-$			
		3194.8		4880.96	7/2$^{(+)}$			
		3305.9	7.1	4769.86	(5/2$^-$)			
		3897.8	14	4177.9	3/2$^-$			
		3963.3	3.6	4112.4	7/2$^+$			
		5072.9	3.6	3002.64	5/2$^+$			
		6312.1	100	1763.15	5/2$^+$			
		8074.9		0.0	3/2$^+$			
8096.1	(5/2)	2289.4	5.1	5806.6	(1/2,3/2,5/2)			
		2450	13	5646	(5/2,7/2,9/2$^+$)			
		2497.6	13	5598.4	3/2$^+$,5/2$^+$			
		2881.5	3.3	5214.5	(5/2$^+$)			
		2932.6	5.1	5163.34	7/2$^-$			
		3256.8	7.7	4839.10	(1/2$^+$,3/2)			
		3326.1	5.1	4769.86	(5/2$^-$)			
		3918.0	4.9	4177.9	3/2$^-$			
		4036.8	5.1	4059.1	(3/2)$^-$			
		4177.4	4.6	3918.47	3/2$^+$			
		4932.9	13	3162.87	7/2$^-$			
		5401.7	13	2694.00	3/2$^+$			
		5450.0	44	2645.67	7/2$^+$			
		6332.3	21	1763.15	5/2$^+$			
		8095.1	100	0.0	3/2$^+$			
8106.4	3/2$^+$	2507.9	1.4	5598.4	3/2$^+$,5/2$^+$			
		3251.8	1.6	4854.4	(1/2,3/2)			
		3267.1	1.4	4839.10	(1/2$^+$,3/2)			
		3928.3	4.3	4177.9	3/2$^-$			
		4047.1	3.5	4059.1	(3/2)$^-$			
		4138.5	8.1	3967.6	1/2$^+$			
		5103.4	24	3002.64	5/2$^+$			
		5412.0	24	2694.00	3/2$^+$	M1+E2	0.00153 *8*	δ: +4.9 *10* or −0.05 *4*.
		6342.6	8.1	1763.15	5/2$^+$			
		6886.3	95	1219.38	1/2$^+$			
		8105.4	100	0.0	3/2$^+$	M1+E2		δ: +1.2 *1* or −0.46 *4*.
8113.3	(1/2$^+$,3/2,5/2$^+$)	1621	19	6492	(3/2$^+$)			
		2007.0	19	6106.2	(3/2,5/2$^+$)			
		2306.6	3.4	5806.6	(1/2,3/2,5/2)			

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ†	I_γ‡	E_f	J^π_f
8113.3	$(1/2^+,3/2,5/2^+)$	2458	2.8	5655	$(3/2)^+$
		2514.8		5598.4	$3/2^+,5/2^+$
		2898.7	5.6	5214.5	$(5/2^+)$
		3939.6	22	4173.43	$(5/2^+)$
		4054.0	4.1	4059.1	$(3/2)^-$
		4145.4	16	3967.6	$1/2^+$
		5110.3	5.9	3002.64	$5/2^+$
		5418.9	41	2694.00	$3/2^+$
		6349.5	16	1763.15	$5/2^+$
		6893.2	100	1219.38	$1/2^+$
		8112.3	78	0.0	$3/2^+$
8147.2	$3/2^-$	2464.9	1.6	5682.2	$1/2^-,3/2^-$
		3138.0	4.8	5009.1	(1/2,3/2)
		3969.1	13	4177.9	$3/2^-$
		4087.8	4.8	4059.1	$(3/2)^-$
		4179.3	3.2	3967.6	$1/2^+$
		4228.5	4.8	3918.47	$3/2^+$
		5452.7	4.8	2694.00	$3/2^+$
		6927.1	24	1219.38	$1/2^+$
		8146.2	100	0.0	$3/2^+$
8156.8	$(5/2^+)$	2511	7.8	5646	$(5/2,7/2,9/2^+)$
		2570.7	16	5586.0	$5/2^+$
		2942.2	12	5214.5	$(5/2^+)$
		3386.8	2.0	4769.86	$(5/2^-)$
		3978.7	9.8	4177.9	$3/2^-$
		4044.2	14	4112.4	$7/2^+$
		4097.4	2.0	4059.1	$(3/2)^-$
		4213.6	3.9	3942.9	$9/2^+$
		4993.6	9.8	3162.87	$7/2^-$
		5153.8	12	3002.64	$5/2^+$
		5510.7	100	2645.67	$7/2^+$
		6393.0	2.0	1763.15	$5/2^+$
		8155.8	5.9	0.0	$3/2^+$
8179.4	$(3/2^-,5/2,7/2^+)$	3554.9	13.2 *13*	4624.3	3/2,5/2
		5016.1	11.8 *12*	3162.87	$7/2^-$
		5176.4	5.9 *30*	3002.64	$5/2^+$
		5484.9	4.4 *22*	2694.00	$3/2^+$
		6415.6	100 *10*	1763.15	$5/2^+$
		8178.4	11.8 *12*	0.0	$3/2^+$
8208.2	$5/2^+$	2553	0.9	5655	$(3/2)^+$
		3327.1	1.0	4880.96	$7/2^{(+)}$
		3368.9	0.51	4839.10	$(1/2^+,3/2)$

$^{35}_{17}Cl_{18}$-44

Adopted Levels, Gammas (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[‡]	E_f	J^π_f	Mult.[#]	δ[#]	α[a]	Comments
8208.2	$5/2^+$	4148.8	1.9	4059.1	$(3/2)^-$				
		4289.5		3918.47	$3/2^+$				
		5205.1	1.4	3002.64	$5/2^+$				
		5513.7	0.64	2694.00	$3/2^+$				
		6444.4	18	1763.15	$5/2^+$	M1+E2	+0.33 *4*		B(M1)(W.u.)=0.75 *+35−22*; B(E2)(W.u.)=7.5 *+41−25*
		6988.1	3.9	1219.38	$1/2^+$				
		8207.2	100	0.0	$3/2^+$	M1+E2	−0.36 *4*		B(M1)(W.u.)=2.0 *+9−5*; B(E2)(W.u.)=15 *+7−4*
8216.3	$5/2^+$	2561	1.8	5655	$(3/2)^+$				
		3446.3	1.8	4769.86	$(5/2^-)$				
		4038.2	<6.7	4177.9	$3/2^-$				
		4042.6	<6.7	4173.43	$(5/2^+)$				
		5053.0	91	3162.87	$7/2^-$	(E1)		2.04×10^{-3} *3*	B(E1)(W.u.)=0.062 *+21−12* Mult.: D+Q with δ=+0.11 *2* from $\gamma(\theta)$ in (p,γ); $\Delta\pi$=yes from level scheme; but M2 component is unlikely for this strong primary transition.
		5213.2	11	3002.64	$5/2^+$				
		5521.8	6.7	2694.00	$3/2^+$				
		6452.5	3.1	1763.15	$5/2^+$				
		8215.3	100	0.0	$3/2^+$	M1+E2	+0.06 *3*		B(M1)(W.u.)=0.55 *+18−9*; B(E2)(W.u.)=0.11 *+16−8*
8242.4	$3/2^-$	4068.7	7.7	4173.43	$(5/2^+)$				
		5239.3	2.2	3002.64	$5/2^+$				
		7022.3	100	1219.38	$1/2^+$				
8251	$(3/2^+,5/2^+)$	4138	4.9	4112.4	$7/2^+$				
		5248	7.3	3002.64	$5/2^+$				
		7031	100	1219.38	$1/2^+$				
		8250	9.8	0.0	$3/2^+$				
8269.0	$5/2^+$	3499.0	4.4	4769.86	$(5/2^-)$				
		4090.8	27	4177.9	$3/2^-$				
		4350.2	6.7	3918.47	$3/2^+$				
		5105.7	16	3162.87	$7/2^-$				
		5265.9	29	3002.64	$5/2^+$				
		5574.5	13	2694.00	$3/2^+$				
		5622.9	8.9	2645.67	$7/2^+$				
		6505.2	18	1763.15	$5/2^+$				
		8268.0	100	0.0	$3/2^+$				
8277.2	$(5/2)^+$	2622	5.2	5655	$(3/2)^+$				
		3396.1	17	4880.96	$7/2^{(+)}$				
		3437.9	38	4839.10	$(1/2^+,3/2)$				
		3507.2	4.1	4769.86	$(5/2^-)$				
		3652.7	10	4624.3	3/2,5/2				
		3929.2	6.2	4347.76	$9/2^-$				
		4103.5	6.9	4173.43	$(5/2^+)$				

$^{35}_{17}Cl_{18}$-45

Adopted Levels, Gammas (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J_i^π	E_γ†	I_γ‡	E_f	J_f^π	Mult.#	Comments
8277.2	$(5/2)^+$	4217.8	5.2	4059.1	$(3/2)^-$		
		4358.4	17	3918.47	$3/2^+$		
		5274.1	35	3002.64	$5/2^+$		
		5582.7	10	2694.00	$3/2^+$		
		5631.0	10	2645.67	$7/2^+$		
		6513.4	79	1763.15	$5/2^+$		
		8276.2	100	0.0	$3/2^+$		
8282.4	$(3/2^-,5/2)$	2696.3	29	5586.0	$5/2^+$		
		3067.8	29	5214.5	$(5/2^+)$		
		3118.9	11	5163.34	$7/2^-$		
		4104.2	50	4177.9	$3/2^-$		
		4223.0	21	4059.1	$(3/2)^-$		
		5119.1	93	3162.87	$7/2^-$		
		6518.6	25	1763.15	$5/2^+$		
		8281.4	100	0.0	$3/2^+$		
8287.82	$1/2^-$	2633	4.6	5655	$(3/2)^+$		
		3278.6	6.8	5009.1	(1/2,3/2)		
		4109.7	21	4177.9	$3/2^-$		
		4228.5	4.6	4059.1	$(3/2)^-$		
		4319.9	6.8	3967.6	$1/2^+$		
		4369.06	2.3	3918.47	$3/2^+$		
		5593.3	52	2694.00	$3/2^+$		
		7067.67	100	1219.38	$1/2^+$	(E1)	B(E1)(W.u.)=0.069 *+24−14* Mult.: D+Q withδ(Q/D)=+0.21 *2* or −3.2 *2* in (p,γ); but M2 component is unlikely for this primary transition based on resonance Γ=0.04 keV.
		8286.77	30	0.0	$3/2^+$		
8298.2	$3/2^+$	3288.9	13	5009.1	(1/2,3/2)		
		3417.1	17	4880.96	$7/2^{(+)}$		
		3443.6	21	4854.4	(1/2,3/2)		
		4124.5	25	4173.43	$(5/2^+)$		
		4238.8	42	4059.1	$(3/2)^-$		
		5134.9	4.2	3162.87	$7/2^-$		This γ is questionable if J^π(8298)=$3/2^+$, since it would require Mult=M2 which is probably unlikely; however J^π(8298)=$3/2^-$ would make the strong 7078.0γ and 8297.1γ less likely.
		5295.1	4.2	3002.64	$5/2^+$		
		5603.7	42	2694.00	$3/2^+$		
		5652.0	25	2645.67	$7/2^+$		
		6534.4	33	1763.15	$5/2^+$		
		7078.1	100	1219.38	$1/2^+$	M1+E2	δ: −1.7 *1* or −0.02 *2* from $\gamma(\theta)$ in (p,γ).
		8297.1	54	0.0	$3/2^+$	M1+E2	δ: −0.15 *4* or +9.5 *36* from $\gamma(\theta)$ in (p,γ).
8318.1	$(5/2^+)$	2511.4	0.9	5806.6	(1/2,3/2,5/2)		

Adopted Levels, Gammas (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[‡]	E_f	J^π_f	Mult.[#]	δ[#]	α[a]	Comments
8318.1	(5/2$^+$)	2672	0.9	5646	(5/2,7/2,9/2$^+$)				
		2719.6	1.9	5598.4	3/2$^+$,5/2$^+$				
		3548.1	0.39	4769.86	(5/2$^-$)				
		3693.6	2.3	4624.3	3/2,5/2				
		4139.9	<3.9	4177.9	3/2$^-$				
		4144.4	<3.9	4173.43	(5/2$^+$)				
		4258.7	1.5	4059.1	(3/2)$^-$				
		4350.2	3.9	3967.6	1/2$^+$				
		5154.8	0.64	3162.87	7/2$^-$				
		5315.0	3.9	3002.64	5/2$^+$				
		5623.6	2.1	2694.00	3/2$^+$				
		5671.9	1.2	2645.67	7/2$^+$				
		6554.3	2.3	1763.15	5/2$^+$				
		7098.0	2.6	1219.38	1/2$^+$				
		8317.0	100	0.0	3/2$^+$				
8319.6	15/2$^-$	2232.7 *6*	57.2 *28*	6087.31	13/2$^-$	[M1,E2]		0.00040 *4*	E_γ,I_γ: from (^{16}O,$\alpha p\gamma$). B(M1)(W.u.)=0.0577 *41* if M1, B(E2)(W.u.)=43.8 *31* if E2.
		2911.9 *8*	100 *5*	5407.07	11/2$^-$	E2		0.000753 *11*	B(E2)(W.u.)=20.3 *12* E_γ,I_γ,Mult.: from (^{16}O,$\alpha p\gamma$). Mult=Q, ΔJ=2 from $\gamma\gamma$(DCO); M2 ruled out by RUL.
8323.1	(3/2$^+$,5/2$^+$)	4210.4	1.2	4112.4	7/2$^+$				
		5320.0	3.7	3002.64	5/2$^+$				
		5676.9	4.9	2645.67	7/2$^+$				
		6559.3	<2.4	1763.15	5/2$^+$				
		7103.0	9.8	1219.38	1/2$^+$				
		8322.0	100	0.0	3/2$^+$				
8381.8	5/2$^+$	3611.7		4769.86	(5/2$^-$)				
		3757.3		4624.3	3/2,5/2				
		4269.1	23.5 *24*	4112.4	7/2$^+$				
		4463.0	71 *7*	3918.47	3/2$^+$	M1+E2	+0.15 *3*	1.19×10^{-3} *2*	B(M1)(W.u.)=2.9 *+13−7*; B(E2)(W.u.)=13 *+8−5*
		5378.7	74 *8*	3002.64	5/2$^+$	M1+E2	+0.02 *3*	1.44×10^{-3} *2*	B(M1)(W.u.)=1.8 *+14−7*
		5687.3	2.9 *15*	2694.00	3/2$^+$	M1+E2	−0.72 *31*	0.00157 *4*	B(M1)(W.u.)=0.039 *+32−21*; B(E2)(W.u.)=2.4 *+23−18*
		5735.6	15 *8*	2645.67	7/2$^+$				
		6618.0	100 *10*	1763.15	5/2$^+$	M1+E2	+0.06 *3*		B(M1)(W.u.)=1.3 *+6−3*; B(E2)(W.u.)=0.4 *+6−3*
		8380.7	9 *5*	0.0	3/2$^+$	D+Q	1.65		
8402.9	5/2$^-$	3632.8	16 *8*	4769.86	(5/2$^-$)				
		4224.7	29 *16*	4177.9	3/2$^-$				

Adopted Levels, Gammas (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^{π}_i	E_{γ}†	I_{γ}‡	E_f	J^{π}_f
8402.9	$5/2^-$	4343.5	74 *20*	4059.1	$(3/2)^-$
		5708.4	32 *16*	2694.00	$3/2^+$
		5756.7	23 *13*	2645.67	$7/2^+$
		6639.1	48 *13*	1763.15	$5/2^+$
		8401.8	100 *26*	0.0	$3/2^+$
8405.8	$(5/2^-)$	3524.7	35 *9*	4880.96	$7/2^{(+)}$
		4232.1	3.7 *19*	4173.43	$(5/2^+)$
		4293.1	3.3 *16*	4112.4	$7/2^+$
		5402.7	35 *9*	3002.64	$5/2^+$
		6642.0	100 *26*	1763.15	$5/2^+$
		8404.7	56 *14*	0.0	$3/2^+$
8416.7	$1/2^+$	3407.4	14	5009.1	(1/2,3/2)
		4243.0	31	4173.43	$(5/2^+)$
		4473.5	8.6	3942.9	$9/2^+$
		5413.6	20	3002.64	$5/2^+$
		5770.5	23	2645.67	$7/2^+$
		6652.9	63	1763.15	$5/2^+$
		7196.5	26	1219.38	$1/2^+$
		8415.6	100	0.0	$3/2^+$
8464.3	$(1/2^+,3/2,5/2^+)$	2809	47 *24*	5655	$(3/2)^+$
		3249.6	34 *17*	5214.5	$(5/2^+)$
		3455.0	47 *24*	5009.1	(1/2,3/2)
		3609.7	19 *10*	4854.4	(1/2,3/2)
		3839.8	25 *13*	4624.3	3/2,5/2
		4286.1	31 *15*	4177.9	$3/2^-$
		4290.6	32 *17*	4173.43	$(5/2^+)$
		4496.4	31 *15*	3967.6	$1/2^+$
		4545.5	15 *8*	3918.47	$3/2^+$
		5461.2	100 *24*	3002.64	$5/2^+$
		5769.8	94 *24*	2694.00	$3/2^+$
		7244.1	53 *24*	1219.38	$1/2^+$
		8463.2	59 *18*	0.0	$3/2^+$
8484.4	$(3/2)^+$	2829	12 *6*	5655	$(3/2)^+$
		3603.2	15 *8*	4880.96	$7/2^{(+)}$
		3859.9	2.2 *11*	4624.3	3/2,5/2
		4516.5	2.6 *13*	3967.6	$1/2^+$
		4565.6	11 *6*	3918.47	$3/2^+$
		5481.3	15 *9*	3002.64	$5/2^+$
		5789.9	44 *11*	2694.00	$3/2^+$
		5838.2	6.1 *31*	2645.67	$7/2^+$
		6720.6	100 *26*	1763.15	$5/2^+$
		8483.3	9 *5*	0.0	$3/2^+$

Adopted Levels, Gammas (continued)

γ(^{35}Cl) (continued)

E$_i$(level)	J$^\pi_i$	E$_\gamma$[†]	I$_\gamma$[‡]	E$_f$	J$^\pi_f$	Mult.[#]	δ[#]	α[a]	Comments
8486.2	3/2$^-$	2728	2.4 *13*	5758	(1/2,3/2$^-$)				
		3861.7	3.9 *20*	4624.3	3/2,5/2				
		4308.0	<15	4177.9	3/2$^-$				I$_\gamma$: total Iγ=10 *5* for γ decays to 4173 and 4178 doublet levels (1976Sp08).
		4312.5	<15	4173.43	(5/2$^+$)				I$_\gamma$: total Iγ=10 *5* for γ decays to 4173 and 4178 doublet levels (1976Sp08).
		4426.8	4.1 *22*	4059.1	(3/2)$^-$				
		4518.3	15 *9*	3967.6	1/2$^+$				
		5483.1	17 *9*	3002.64	5/2$^+$				
		5791.7	41 *11*	2694.00	3/2$^+$				
		6722.4	8 *4*	1763.15	5/2$^+$				
		7266.0	14 *7*	1219.38	1/2$^+$				
		8485.1	100 *26*	0.0	3/2$^+$				
8487.5	15/2$^-$	2399.8 *8*	100.0 *34*	6087.31	13/2$^-$	(M1(+E2))	+0.2 +*3*−*2*	0.000427 *17*	Mult.,δ: D+Q and δ from $\gamma\gamma$(DCO) in (^{28}Si,αpγ); $\Delta\pi$=no from level scheme.
		3080.0 *7*	31.8 *20*	5407.07	11/2$^-$	E2		0.000825 *12*	E$_\gamma$,I$_\gamma$,Mult.: from (^{16}O,αpγ), with Mult=Q, ΔJ=2 from $\gamma\gamma$(ADO) and M2 ruled out by RUL.
8514.3	1/2$^-$	3659.7	16 *8*	4854.4	(1/2,3/2)				
		4336.1	7 *4*	4177.9	3/2$^-$				
		4454.9	100 *26*	4059.1	(3/2)$^-$				
		4546.4	85 *22*	3967.6	1/2$^+$				
		4595.5	24 *12*	3918.47	3/2$^+$				
		7294.1	37 *11*	1219.38	1/2$^+$				
		8513.2	100 *26*	0.0	3/2$^+$				
8558.2?	(13/2$^-$)	2631.4[b]	100	5926.65	(11/2)$^-$	D(+Q)	−0.2 +*2*−*3*		E$_\gamma$,Mult.,δ: from (^{28}Si,αpγ); $\Delta\pi$=no from level scheme; Mult=Q, ΔJ=1.
8571.9	(5/2)$^+$	2848.3	3.1 *16*	5723.52	5/2$^+$				
		2917	1.9 *10*	5655	(3/2)$^+$				
		2973.4	1.0 *6*	5598.4	3/2$^+$,5/2$^+$				
		3408.4	2.6 *13*	5163.34	7/2$^-$				
		3690.7	1.3 *7*	4880.96	7/2$^{(+)}$				
		4398.2	1.0 *6*	4173.43	(5/2$^+$)				
		4459.2	1.6 *9*	4112.4	7/2$^+$				
		4653.1	1.6 *9*	3918.47	3/2$^+$				
		5408.6	2.3 *12*	3162.87	7/2$^-$				
		5568.8	6.4 *33*	3002.64	5/2$^+$				
		5877.4	0.9 *4*	2694.00	3/2$^+$				
		5925.7	11 *6*	2645.67	7/2$^+$				
		7351.7	8 *4*	1219.38	1/2$^+$				
		8570.8	100 *10*	0.0	3/2$^+$				
8580.5	1/2$^+$	2088	8 *4*	6492	(3/2$^+$)				

$^{35}_{17}Cl_{18}$-49

Adopted Levels, Gammas (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^{π}_i	E_{γ}†	I_{γ}‡	E_f	J^{π}_f
8580.5	$1/2^+$	3741.2	2.4 *13*	4839.10	$(1/2^+,3/2)$
		4402.3	8 *4*	4177.9	$3/2^-$
		4521.1	3.7 *19*	4059.1	$(3/2)^-$
		4661.7	4.3 *22*	3918.47	$3/2^+$
		5886.0	10 *5*	2694.00	$3/2^+$
		7360.3	100 *11*	1219.38	$1/2^+$
		8579.4	48 *13*	0.0	$3/2^+$
8590.1	$5/2^+$	2832	22 *11*	5758	$(1/2,3/2^-)$
		2935	17 *9*	5655	$(3/2)^+$
		3820.0	73 *18*	4769.86	$(5/2^-)$
		4416.4	59 *14*	4173.43	$(5/2^+)$
		4477.4	16 *8*	4112.4	$7/2^+$
		5426.8	36 *18*	3162.87	$7/2^-$
		5587.0	100 *27*	3002.64	$5/2^+$
		5943.9	25 *13*	2645.67	$7/2^+$
		6826.2	64 *18*	1763.15	$5/2^+$
		8589.0	46 *14*	0.0	$3/2^+$
8612.0	$3/2^+,5/2^+$	6848.1	100 *10*	1763.15	$5/2^+$
		7391.8	34 *8*	1219.38	$1/2^+$
		8610.9	66 *16*	0.0	$3/2^+$
8613.7	$5/2^+$	3732.5	2.9 *14*	4880.96	$7/2^{(+)}$
		4554.3	10 *5*	4059.1	$(3/2)^-$
		4670.5	7.1	3942.9	$9/2^+$
		4694.9	13 *7*	3918.47	$3/2^+$
		5610.6	4.5 *23*	3002.64	$5/2^+$
		5919.2	8 *4*	2694.00	$3/2^+$
		5967.5	41 *11*	2645.67	$7/2^+$
		6849.8	16	1763.15	$5/2^+$
		7393.5	3.6	1219.38	$1/2^+$
		8612.6	100 *11*	0.0	$3/2^+$
8618.2	$3/2^+,5/2^+$	2894.6	6.4 *33*	5723.52	$5/2^+$
		2963	5.2 *27*	5655	$(3/2)^+$
		3763.6	4.2 *21*	4854.4	$(1/2,3/2)$
		3993.7	4.6 *24*	4624.3	$3/2,5/2$
		4650.3	9 *5*	3967.6	$1/2^+$
		4699.4	7 *4*	3918.47	$3/2^+$
		5615.1	49 *12*	3002.64	$5/2^+$
		5923.7	42 *12*	2694.00	$3/2^+$
		5972.0	5.2 *27*	2645.67	$7/2^+$
		6854.3	70 *18*	1763.15	$5/2^+$
		8617.1	100 *24*	0.0	$3/2^+$
8630.1	$7/2^-$	2906.5	0.6 *4*	5723.52	$5/2^+$

Adopted Levels, Gammas (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J_i^π	E_γ[†]	I_γ[‡]	E_f	J_f^π	Mult.[#]	δ[#]	α[a]	Comments
8630.1	7/2⁻	3466.6	17 *9*	5163.34	7/2⁻	M1+E2	+0.60 *10*	0.000884 *15*	Mult.,δ: D+Q and δ from $\gamma(\theta)$ in (p,γ); E1+M2 ruled out by RUL.
		3860.0	3.0 *15*	4769.86	(5/2⁻)				
		4282.1	26 *7*	4347.76	9/2⁻	M1+E2	−0.184 *14*	1.13×10^{-3} *2*	Mult.,δ: D+Q and δ from $\gamma(\theta)$ in (p,γ); E1+M2 ruled out by RUL.
		4456.4	8 *4*	4173.43	(5/2⁺)				
		4517.4	21 *11*	4112.4	7/2⁺	(E1+M2)	−0.06 *2*	1.86×10^{-3} *3*	Mult.,δ: D+Q and δ from $\gamma(\theta)$ in (p,γ); $\Delta\pi$=yes from level scheme.
		4686.9	21 *11*	3942.9	9/2⁺				
		5466.8	7 *4*	3162.87	7/2⁻				
		5627.0	3.0 *15*	3002.64	5/2⁺				
		5983.9	5.5 *28*	2645.67	7/2⁺				
		6866.2	100 *26*	1763.15	5/2⁺	(E1+M2)	−0.011 *3*		Mult.,δ: D+Q and δ from $\gamma(\theta)$ in (p,γ); $\Delta\pi$=yes from level scheme.
		8629.0	1.1 *7*	0.0	3/2⁺				
8640	(5/2⁺)	4697	61	3942.9	9/2⁺				
		5477	53	3162.87	7/2⁻				
		5637	16	3002.64	5/2⁺				
		5946	34	2694.00	3/2⁺				
		6876	100	1763.15	5/2⁺				
8641.4	3/2⁺,5/2⁺	2986	10 *6*	5655	(3/2)⁺				
		3055.3	9 *5*	5586.0	5/2⁺				
		3237.9	17 *9*	5403.3	3/2⁻				
		3802.1	9 *5*	4839.10	(1/2⁺,3/2)				
		3871.3	86 *22*	4769.86	(5/2⁻)				
		4016.9	7 *4*	4624.3	3/2,5/2				
		4463.2	<64	4177.9	3/2⁻				I_γ: total Iγ=50 *14* for γ decays to 4173 and 4178 doublet levels (1976Sp08).
		4467.7	<64	4173.43	(5/2⁺)				I_γ: total Iγ=50 *14* for γ decays to 4173 and 4178 doublet levels (1976Sp08).
		4582.0	32 *18*	4059.1	(3/2)⁻				
		4722.6	21 *11*	3918.47	3/2⁺				
		5946.9	19 *9*	2694.00	3/2⁺				
		8640.3	100 *25*	0.0	3/2⁺				
8686.2	5/2⁻	3100.1	2.8 *15*	5586.0	5/2⁺				
		3282.7	4.1 *21*	5403.3	3/2⁻				
		4508.0	2.6 *13*	4177.9	3/2⁻				
		4626.8	4.6 *24*	4059.1	(3/2)⁻				
		5522.9	4.9 *25*	3162.87	7/2⁻				
		5991.7	3.7 *19*	2694.00	3/2⁺				
		6922.3	6.3 *32*	1763.15	5/2⁺				
		7466.0	2.5 *13*	1219.38	1/2⁺				

γ(^{35}Cl) (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[‡]	E_f	J^π_f	Mult.[#]	α[a]	Comments
8686.2	5/2$^-$	8685.0	100 *11*	0.0	3/2$^+$			
8696.9	3/2$^-$	2890.2	3.6 *19*	5806.6	(1/2,3/2,5/2)			
		3098.4	6.0 *30*	5598.4	3/2$^+$,5/2$^+$			
		7476.7	15 *4*	1219.38	1/2$^+$			
		8695.7	100 *10*	0.0	3/2$^+$			
8750.9	3/2$^-$	2944.2	4.2 *21*	5806.6	(1/2,3/2,5/2)			
		3347.4	9 *5*	5403.3	3/2$^-$			
		3536.2	4.5 *23*	5214.5	(5/2$^+$)			
		4572.7	4.0 *21*	4177.9	3/2$^-$			
		4577.2	4.3 *23*	4173.43	(5/2$^+$)			
		6056.3	5.7 *28*	2694.00	3/2$^+$			
		6987.0	57 *15*	1763.15	5/2$^+$			
		7530.7	3.8	1219.38	1/2$^+$			
		8749.7	100 *10*	0.0	3/2$^+$			
8779.8	3/2$^-$	3181.2	16 *8*	5598.4	3/2$^+$,5/2$^+$			
		3565.1	24 *12*	5214.5	(5/2$^+$)			
		4601.6	<71	4177.9	3/2$^-$			I_γ: total Iγ=56 *15* for γ decays to 4173 and 4178 doublet levels (1976Sp08).
		4606.0	<71	4173.43	(5/2$^+$)			I_γ: total Iγ=56 *15* for γ decays to 4173 and 4178 doublet levels (1976Sp08).
		4861.0	16 *8*	3918.47	3/2$^+$			
		5776.7	100 *26*	3002.64	5/2$^+$			
		6085.2	44 *11*	2694.00	3/2$^+$			
		6133.6	48 *11*	2645.67	7/2$^+$			
		7015.9	25 *13*	1763.15	5/2$^+$			
		8778.6	41 *11*	0.0	3/2$^+$			
8787.2	(5/2$^+$)	3063.5	27 *14*	5723.52	5/2$^+$			
		3623.7	76 *20*	5163.34	7/2$^-$			
		4017.1	27 *14*	4769.86	(5/2$^-$)			
		4609.0	36 *20*	4177.9	3/2$^-$			
		5623.8	23 *12*	3162.87	7/2$^-$			
		5784.1	80 *20*	3002.64	5/2$^+$			
		7023.3	100 *24*	1763.15	5/2$^+$			
		8786.0	32 *16*	0.0	3/2$^+$			
8788.3	(15/2$^+$)	915.4 *4*	100	7873.03	13/2$^+$	(M1)	4.82×10^{-5} *7*	E_γ,Mult.: from (^{16}O,αpγ), with Mult=D, ΔJ=1 from $\gamma\gamma$(ADO); $\Delta\pi$=(no) from level scheme.
8798.4	(3/2$^+$,5/2$^+$)	4739.0	13 *7*	4059.1	(3/2)$^-$			
		5795.2	26 *7*	3002.64	5/2$^+$			
		7034.5	11 *6*	1763.15	5/2$^+$			
		7578.1	11 *6*	1219.38	1/2$^+$			
		8797.2	100 *10*	0.0	3/2$^+$			
8820.9	(1/2$^+$ to 7/2$^+$)	5817.7	17 *10*	3002.64	5/2$^+$			

$^{35}_{17}$Cl$_{18}$-52

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J_i^π	E_γ[†]	I_γ[‡]	E_f	J_f^π	Mult.[#]	α[a]	Comments
8820.9	(1/2+ to 7/2+)	6126.3	35 *10*	2694.00	3/2+			
		7057.0	40 *10*	1763.15	5/2+			
		8819.7	100 *10*	0.0	3/2+			
8824.2	1/2+	5821.0	100 *24*	3002.64	5/2+			
		6129.6	16.2 *8*	2694.00	3/2+			
		7060.3	43 *11*	1763.15	5/2+			
		7603.9	60 *16*	1219.38	1/2+			
		8823.0	51 *14*	0.0	3/2+			
8829.3	1/2−	3174	30 *15*	5655	(3/2)+			
		3990.0	3.9 *21*	4839.10	(1/2+,3/2)			
		4059.2	4.6 *24*	4769.86	(5/2−)			
		4910.5	14 *7*	3918.47	3/2+			
		5826.1	67 *18*	3002.64	5/2+			
		6134.7	39 *9*	2694.00	3/2+			
		6183.0	100 *24*	2645.67	7/2+			
		7065.4	27 *15*	1763.15	5/2+			
		7609.0	8 *4*	1219.38	1/2+			
		8828.1	12 *6*	0.0	3/2+			
8833.1	(5/2,7/2)	3618.4	30 *15*	5214.5	(5/2+)			
		4063.0	11 *6*	4769.86	(5/2−)			
		4659.3	61 *18*	4173.43	(5/2+)			
		4720.4	20 *11*	4112.4	7/2+			
		4773.7	48 *13*	4059.1	(3/2)−			
		4914.3	48 *13*	3918.47	3/2+			
		5669.7	16 *8*	3162.87	7/2−			
		5829.9	29 *15*	3002.64	5/2+			
		6138.5	20 *10*	2694.00	3/2+			
		6186.8	100 *26*	2645.67	7/2+			
		7069.2	39 *22*	1763.15	5/2+			
		8831.9	13 *7*	0.0	3/2+			
8838.1	5/2−,7/2−	3674.6		5163.34	7/2−			
		3956.9		4880.96	7/2(+)			
		4068.0	13 *7*	4769.86	(5/2−)			
		4490.0		4347.76	9/2−			
		4778.7		4059.1	(3/2)−			
		5674.7	46 *11*	3162.87	7/2−			
		7074.2	100 *10*	1763.15	5/2+			
8844.4	17/2+	524.8		8319.6	15/2−			
		971.38 *20*	100.0 *30*	7873.03	13/2+	E2	6.16×10^{-5} *9*	B(E2)(W.u.)=16.3 *+37−26* E_γ: weighted average of 971.4 *5* from (^{16}O,αpγ) and 971.38 *20* from (^{14}N,αpnγ).

$^{35}_{17}Cl_{18}$-53

Adopted Levels, Gammas (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J_i^π	E_γ[†]	I_γ[‡]	E_f	J_f^π	Comments
8844.4	$17/2^+$	2757.0		6087.31	$13/2^-$	
8856.2	$5/2^+$	3975.0	6.7 *33*	4880.96	$7/2^{(+)}$	
		4231.6	10 *5*	4624.3	3/2,5/2	
		4678.0	7 *4*	4177.9	$3/2^-$	
		4937.4	22 *6*	3918.47	$3/2^+$	
		5692.8	100 *10*	3162.87	$7/2^-$	
		6161.6	5.5 *28*	2694.00	$3/2^+$	
		6209.9		2645.67	$7/2^+$	
		7092.3	14 *8*	1763.15	$5/2^+$	
		8855.0	31 *8*	0.0	$3/2^+$	
8868.6	$3/2^+,5/2^+$	4690.4	5.3 *27*	4177.9	$3/2^-$	
		5865.4	33 *8*	3002.64	$5/2^+$	
		6174.0	25 *6*	2694.00	$3/2^+$	
		7104.7	9 *5*	1763.15	$5/2^+$	
		7648.3	33 *8*	1219.38	$1/2^+$	
		8867.4	100 *25*	0.0	$3/2^+$	
8886.1	(5/2)	3231	23 *12*	5655	$(3/2)^+$	
		3287.5	25 *13*	5598.4	$3/2^+,5/2^+$	
		3671.4	9 *4*	5214.5	$(5/2^+)$	
		3722.6	35 *16*	5163.34	$7/2^-$	
		4116.0	17 *9*	4769.86	$(5/2^-)$	
		4707.9	31 *16*	4177.9	$3/2^-$	
		4967.3	16 *8*	3918.47	$3/2^+$	
		5882.9	31 *16*	3002.64	$5/2^+$	
		6239.8	100 *27*	2645.67	$7/2^+$	
		7122.2	46 *12*	1763.15	$5/2^+$	
		8884.9	54 *16*	0.0	$3/2^+$	
8893.3	$(5/2^+,7/2^+)$	3307.1	4.3 *22*	5586.0	$5/2^+$	
		4780.6	24 *11*	4112.4	$7/2^+$	
		4950.0	11 *6*	3942.9	$9/2^+$	
		5729.9	78 *19*	3162.87	$7/2^-$	
		6198.7	100 *24*	2694.00	$3/2^+$	
		7129.4	51 *14*	1763.15	$5/2^+$	
		8892.1	2.7 *14*	0.0	$3/2^+$	
8904.8	$(1/2,3/2,5/2^+)$	2724	15 *8*	6181	$(1/2^+$ to $7/2^+)$	
		3250	12 *6*	5655	$(3/2)^+$	
		3895.5	13 *7*	5009.1	(1/2,3/2)	
		4050.2	21 *11*	4854.4	(1/2,3/2)	
		4065.5	16 *8*	4839.10	$(1/2^+,3/2)$	
		4726.6	62 *17*	4177.9	$3/2^-$	
		4845.3	20 *10*	4059.1	$(3/2)^-$	
		6210.2	52 *14*	2694.00	$3/2^+$	

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ†	I_γ‡	E_f	J^π_f	Mult.#	δ#	Comments
8904.8	(1/2,3/2,5/2$^+$)	7684.5	35 *10*	1219.38	1/2$^+$			
		8903.6	100 *24*	0.0	3/2$^+$			
8906.8	5/2$^+$	3252		5655	(3/2)$^+$			
		3261	5.4 *28*	5646	(5/2,7/2,9/2$^+$)			
		3308.2	3.2 *16*	5598.4	3/2$^+$,5/2$^+$			
		3320.6	5.8	5586.0	5/2$^+$			
		3376	3.6 *19*	5531				
		3692.1	6.4 *32*	5214.5	(5/2$^+$)			
		4136.7	8 *4*	4769.86	(5/2$^-$)			
		4963.5	19 *5*	3942.9	9/2$^+$			
		7142.9	100 *10*	1763.15	5/2$^+$			
8919.8	(5/2$^-$,7/2,9/2$^+$)	3274	10 *5*	5646	(5/2,7/2,9/2$^+$)			
		4571.7	14 *7*	4347.76	9/2$^-$			
		5756.4	100 *11*	3162.87	7/2$^-$			
		6273.5	46 *12*	2645.67	7/2$^+$			
		7155.9	7 *4*	1763.15	5/2$^+$			
8953.1	3/2$^+$	7732.8		1219.38	1/2$^+$	M1+E2	−0.34 *2*	Mult.,δ: or δ=−5.0 *4* from $\gamma(\theta)$ in (p,γ). E1+M2 ruled out by RUL.
		8951.9		0.0	3/2$^+$	M1+E2	−0.39 *5*	Mult.,δ: or δ=−8.2 *7* from $\gamma(\theta)$ in (p,γ). E1+M2 ruled out by RUL.
8984.2	5/2$^+$	3260.5	1.3 *7*	5723.52	5/2$^+$			
		4144.8	2.0 *10*	4839.10	(1/2$^+$,3/2)			
		4214.1	0.8 *5*	4769.86	(5/2$^-$)			
		4359.6	0.8 *5*	4624.3	3/2,5/2			
		4806.0	2.5 *13*	4177.9	3/2$^-$			
		4810.4	2.5 *13*	4173.43	(5/2$^+$)			
		5065.3	15 *8*	3918.47	3/2$^+$			
		5820.8	10 *5*	3162.87	7/2$^-$			
		6289.6	3.0 *15*	2694.00	3/2$^+$			
		6337.9	2.0 *10*	2645.67	7/2$^+$			
		7220.3	27 *7*	1763.15	5/2$^+$			
		8983.0	100 *10*	0.0	3/2$^+$			
9081.5	5/2$^+$	3357.8	1.7	5723.52	5/2$^+$			
		4200.3	1.7	4880.96	7/2$^{(+)}$			
		4242.1		4839.10	(1/2$^+$,3/2)			
		4311.4		4769.86	(5/2$^-$)			
		4456.9		4624.3	3/2,5/2			
		4903.2	3.3	4177.9	3/2$^-$			
		4907.7	3.3	4173.43	(5/2$^+$)			
		5162.6	15	3918.47	3/2$^+$	D(+Q)	−0.002 *9*	
		5918.1	10	3162.87	7/2$^-$			
		6386.9	3.3	2694.00	3/2$^+$			
		6435.2	1.7	2645.67	7/2$^+$			

$^{35}_{17}Cl_{18}$-55

Adopted Levels, Gammas (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[‡]	E_f	J^π_f	Mult.[#]	δ[#]	α[a]	Comments
9081.5	5/2+	7317.5	27	1763.15	5/2+	M1+E2	+0.11 *2*		B(M1)(W.u.)=1.16 *+31−25*; B(E2)(W.u.)=0.99 *+50−38*
		9080.2	100	0.0	3/2+	M1+E2	+0.089 *9*		B(M1)(W.u.)=2.26 *+49−33*; B(E2)(W.u.)=0.82 *+27−19*
9157.1	5/2−	3942.4	17 *5*	5214.5	(5/2+)				
		4386.9	16 *4*	4769.86	(5/2−)				
		4978.8	<37	4177.9	3/2−				I_γ: total Iγ=30 *7* for γ decays to 4173 and 4178 doublet final levels (1976Sp08).
		4983.3	<37	4173.43	(5/2+)				I_γ: total Iγ=30 *7* for γ decays to 4173 and 4178 doublet final levels (1976Sp08).
		5993.7	22 *6*	3162.87	7/2−	M1+E2	+0.10 *2*	1.59×10^{-3} *2*	
		6153.9	20 *5*	3002.64	5/2+				
		6462.5	3 *2*	2694.00	3/2+				
		6510.8	7 *4*	2645.67	7/2+				
		7393.1	17 *5*	1763.15	5/2+				
		9155.8	100 *9*	0.0	3/2+	(E1)			Mult.: D+Q with δ=+0.09 *1* from γ(θ) in (p,γ); Δπ=yes from level scheme; M2 component is less likely.
10181.1	19/2−	1336.3 *5*	100.0 *34*	8844.4	17/2+	(E1)		0.0001637 *23*	B(E1)(W.u.)=0.00235 *+21−18* I_γ: weighted average of 100 *17* from (^{28}Si,αpγ) and 100.0 *34* from (^{16}O,αpγ). E_γ,Mult.: from (^{16}O,αpγ), with Mult=D from γγ(ADO) with ΔJ=1; Δπ=(yes) from level scheme.
		1693.5 *5*	84 *9*	8487.5	15/2−	E2		0.0001880 *26*	B(E2)(W.u.)=44.4 *46* E_γ,I_γ,Mult.: from (^{16}O,αpγ), with Mult=Q, ΔJ=2 from γγ(ADO) and M2 ruled out by RUL.
		1861.3 *5*	83 *6*	8319.6	15/2−	E2		0.000262 *4*	B(E2)(W.u.)=27.4 *+26−24* E_γ,I_γ,Mult.: from (^{16}O,αpγ), with Mult=Q, ΔJ=2 from γγ(ADO) and M2 ruled out by RUL.
10221.9	(17/2−)	1377.8 *10*	100	8844.4	17/2+				
10858.5	19/2+	2014.7 *9*	100 *5*	8844.4	17/2+	(M1)		0.000265 *4*	E_γ,I_γ,Mult.: from (^{16}O,αpγ), with Mult=D from γγ(ADO) with ΔJ=1; Δπ=no from level scheme.
		2069.8 *10*	32.5 *22*	8788.3	(15/2+)	E2		0.000361 *5*	E_γ,I_γ,Mult.: from (^{16}O,αpγ) with Mult=Q, ΔJ=2 from γγ(ADO); M2 ruled out by RUL.
11459.1	21/2+	2614.5 *5*	100	8844.4	17/2+	E2		0.000619 *9*	B(E2)(W.u.)=15.9 *8* E_γ,Mult.: from (^{16}O,αpγ), with Mult=Q, ΔJ=2 from γγ(ADO); M2 ruled out by RUL.
12572.2	23/2−	1113.3 *5*	24.4 *18*	11459.1	21/2+	(E1)		3.60×10^{-5} *5*	B(E1)(W.u.)=0.00272 *+41−35* E_γ,I_γ,Mult.: from (^{16}O,αpγ), with Mult=D from γγ(ADO) with ΔJ=1; Δπ=yes from level scheme.

$^{35}_{17}Cl_{18}$-56

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[‡]	E_f	J^π_f	Mult.[#]	α[a]	Comments
12572.2	23/2⁻	2390.8 *5*	100 *4*	10181.1	19/2⁻	E2	0.000515 *7*	B(E2)(W.u.)=26.0 *+36−29* E_γ,I_γ,Mult.: from (^{16}O,$\alpha p\gamma$), with Mult=Q, ΔJ=2 from $\gamma\gamma$(ADO); M2 ruled out by RUL.
16306.4	(27/2⁻)	3734.0		12572.2	23/2⁻			

[†] Sources of values with uncertainties are listed under comments and those without uncertainties are deduced from level-energy differences, unless otherwise noted.

[‡] From (p,γ), unless otherwise noted.

[#] From $\gamma(\theta)$ and/or $\gamma\gamma(\theta)$ in (p,γ) for γ rays originating from low-spin levels (J<9/2) and from $\gamma\gamma(\theta)$(DCO) and γ(lin pol) in (^{28}Si,$\alpha p\gamma$) for γ rays originating from high-spin (J≥9/2) levels, unless otherwise noted. Magnetic/electric characters are determined based on γ(lin pol) or RUL with measured $T_{1/2}$, unless otherwise noted. For primary γ transitions from proton resonances where widths or resonance strengths are provided in (p,γ), a D+Q with definite δ(Q/D)>0 from $\gamma(\theta)$ is firmly assigned as M1+E2. This assignment is due to the fact that for these resonances, the proton width Γ_p is dominant over Γ_γ and thus the total width Γ is much greater than Γ_γ. The total width Γ typically falls within the same scale as the resonance strength, usually a few eV in the ^{34}S(p,γ) dataset.

[@] Primary transitions probably from the unresolved 7501+7503 doublet initial levels in 1976Me12 in (p,γ).

[&] Primary transitions probably from the unresolved 7837+7839 doublet initial levels in 1976Me12 in (p,γ).

[a] Total theoretical internal conversion coefficients, calculated using the BrIcc code (2008Ki07) with “Frozen Orbitals” approximation based on γ-ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

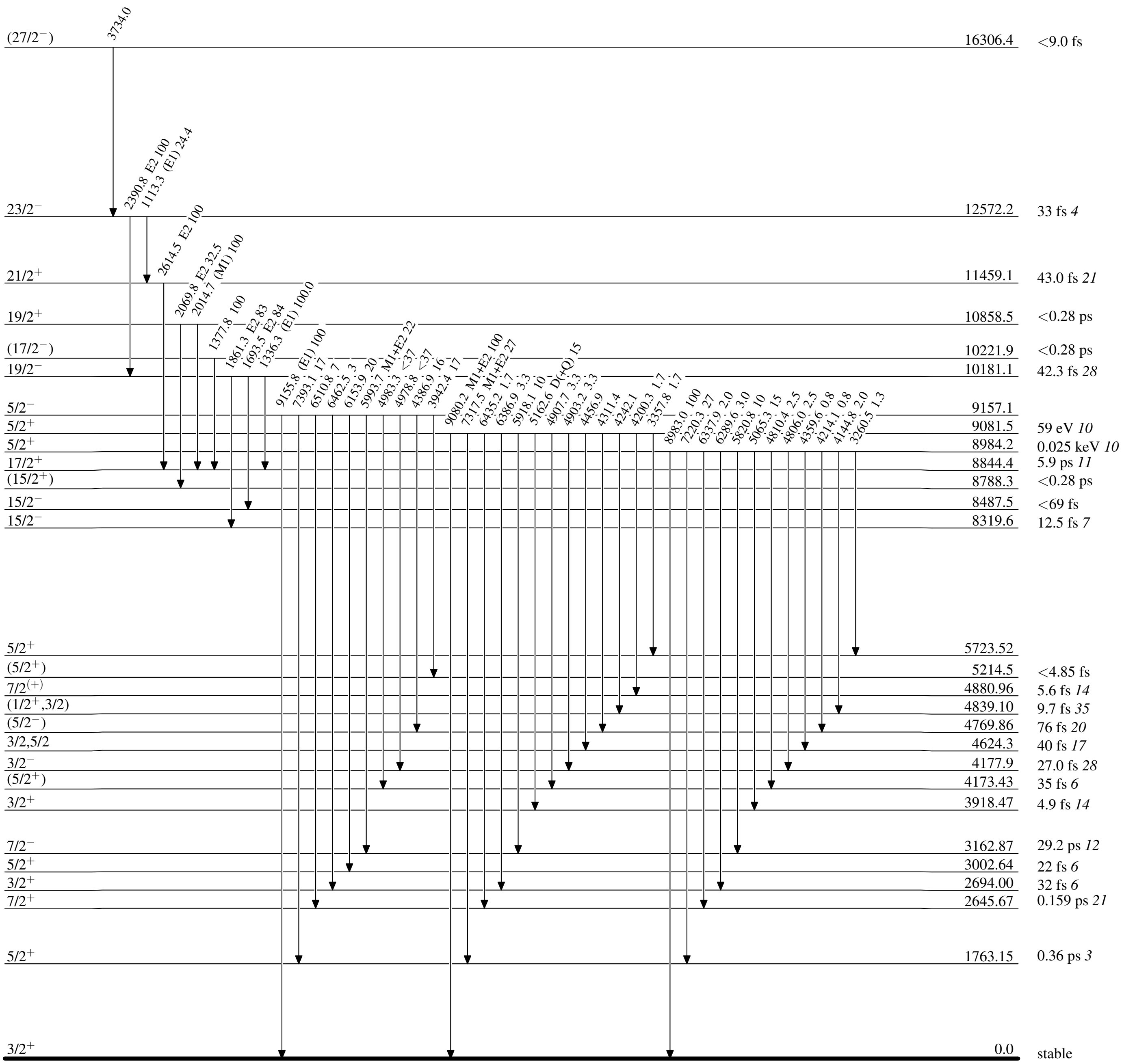
[b] Placement of transition in the level scheme is uncertain.

$^{35}_{17}Cl_{18}$-57

Adopted Levels, Gammas

Level Scheme

Intensities: Relative photon branching from each level

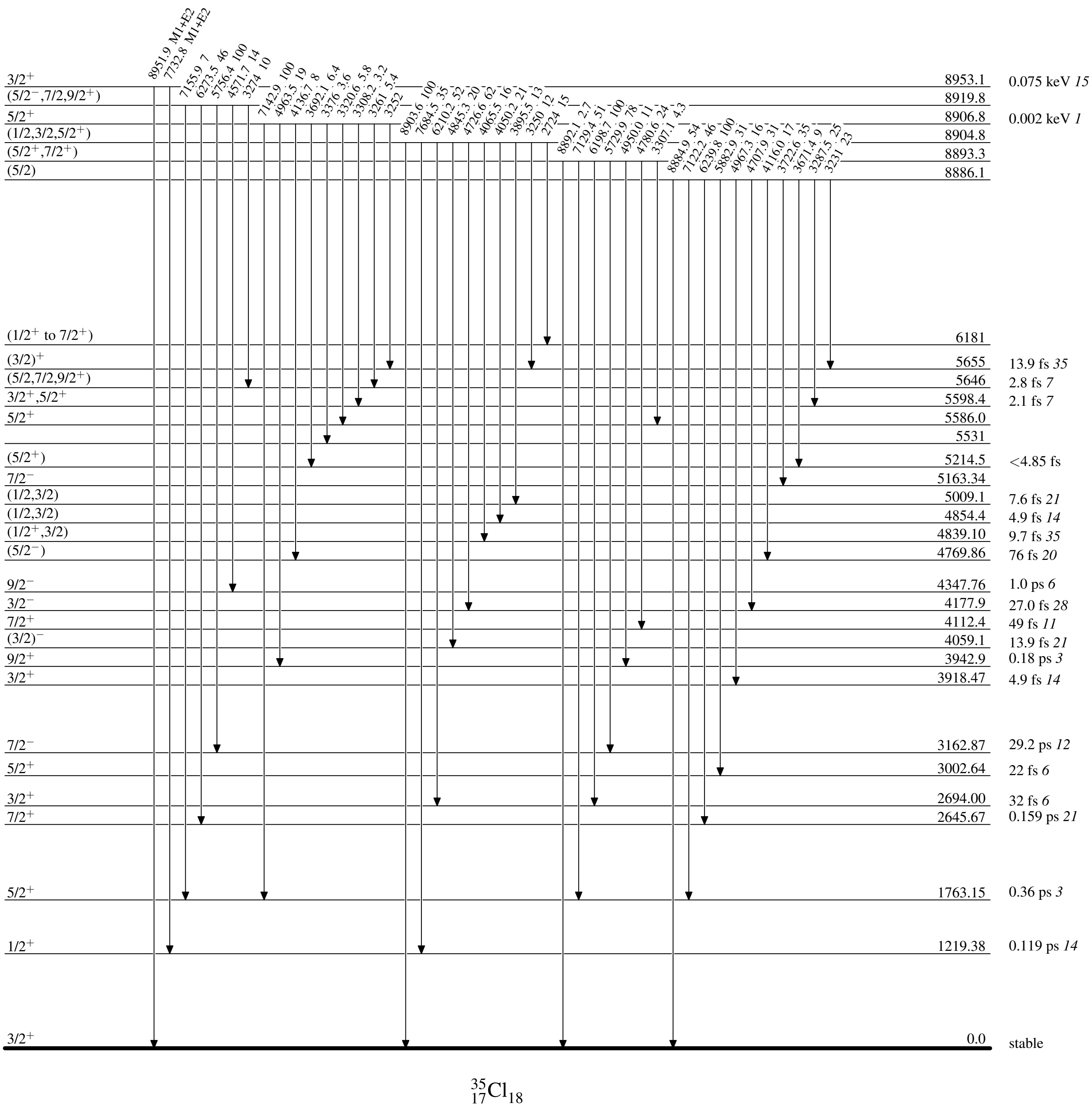


$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas

Level Scheme (continued)

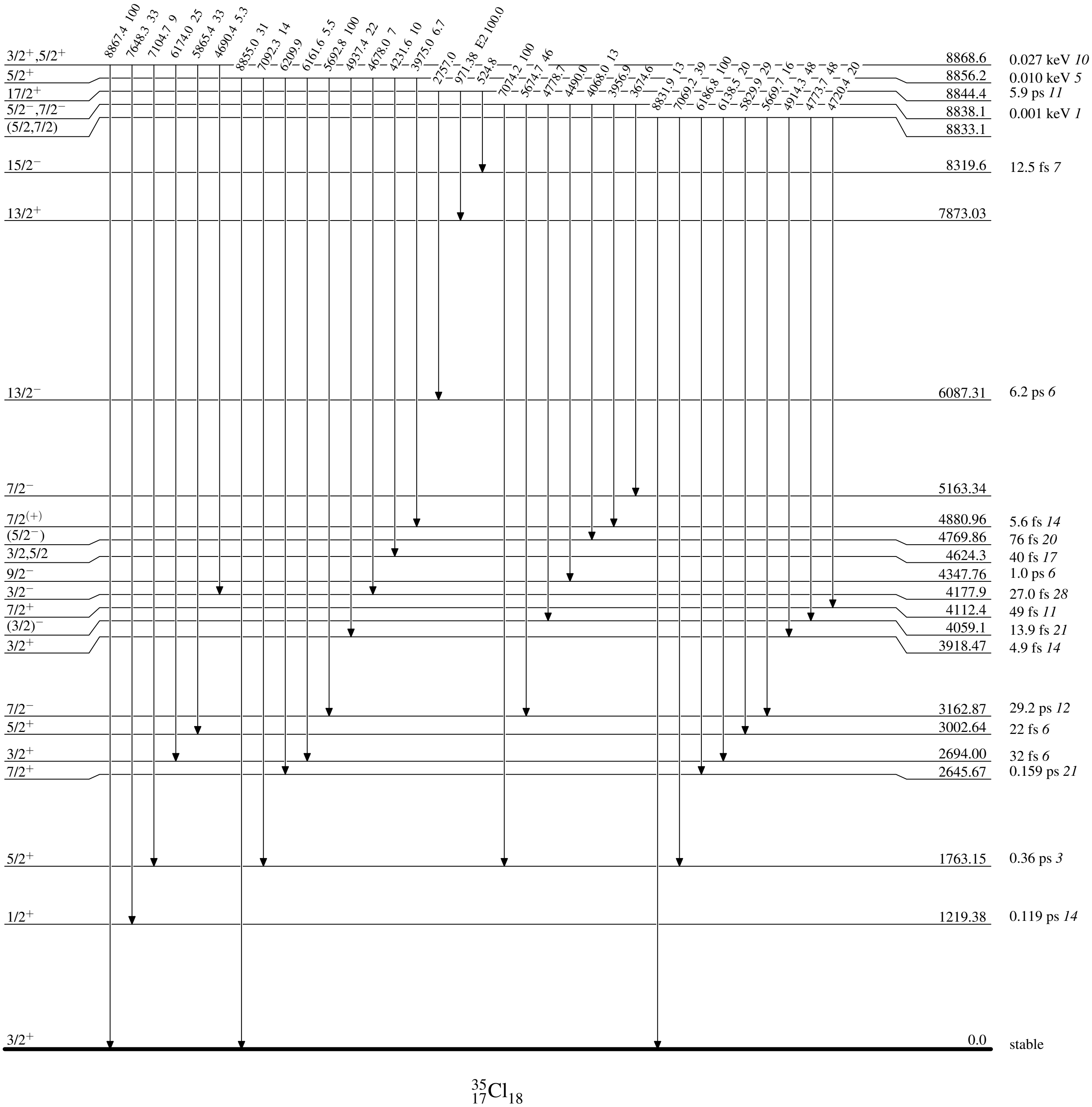
Intensities: Relative photon branching from each level



Adopted Levels, Gammas

Level Scheme (continued)

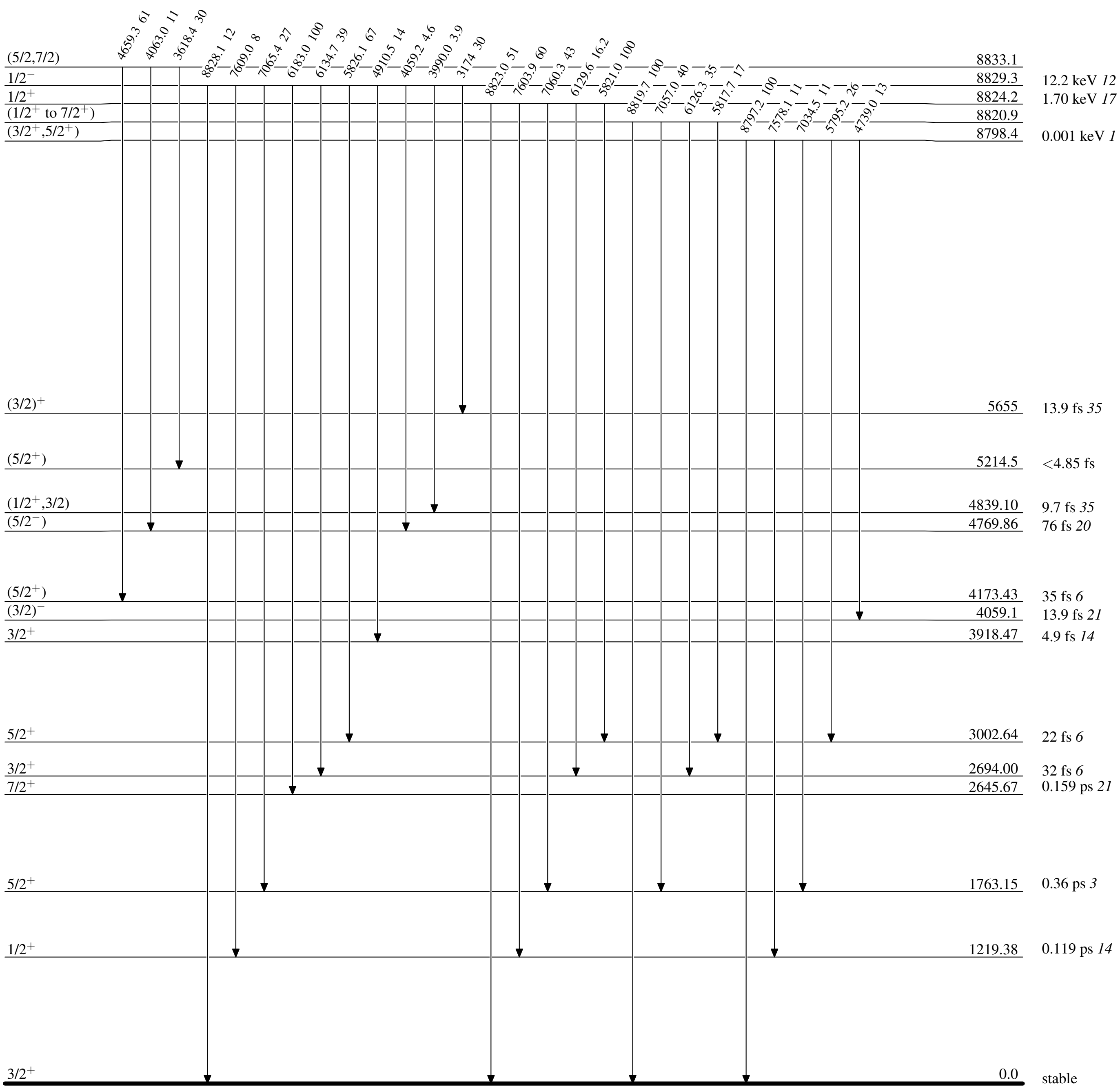
Intensities: Relative photon branching from each level



Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level

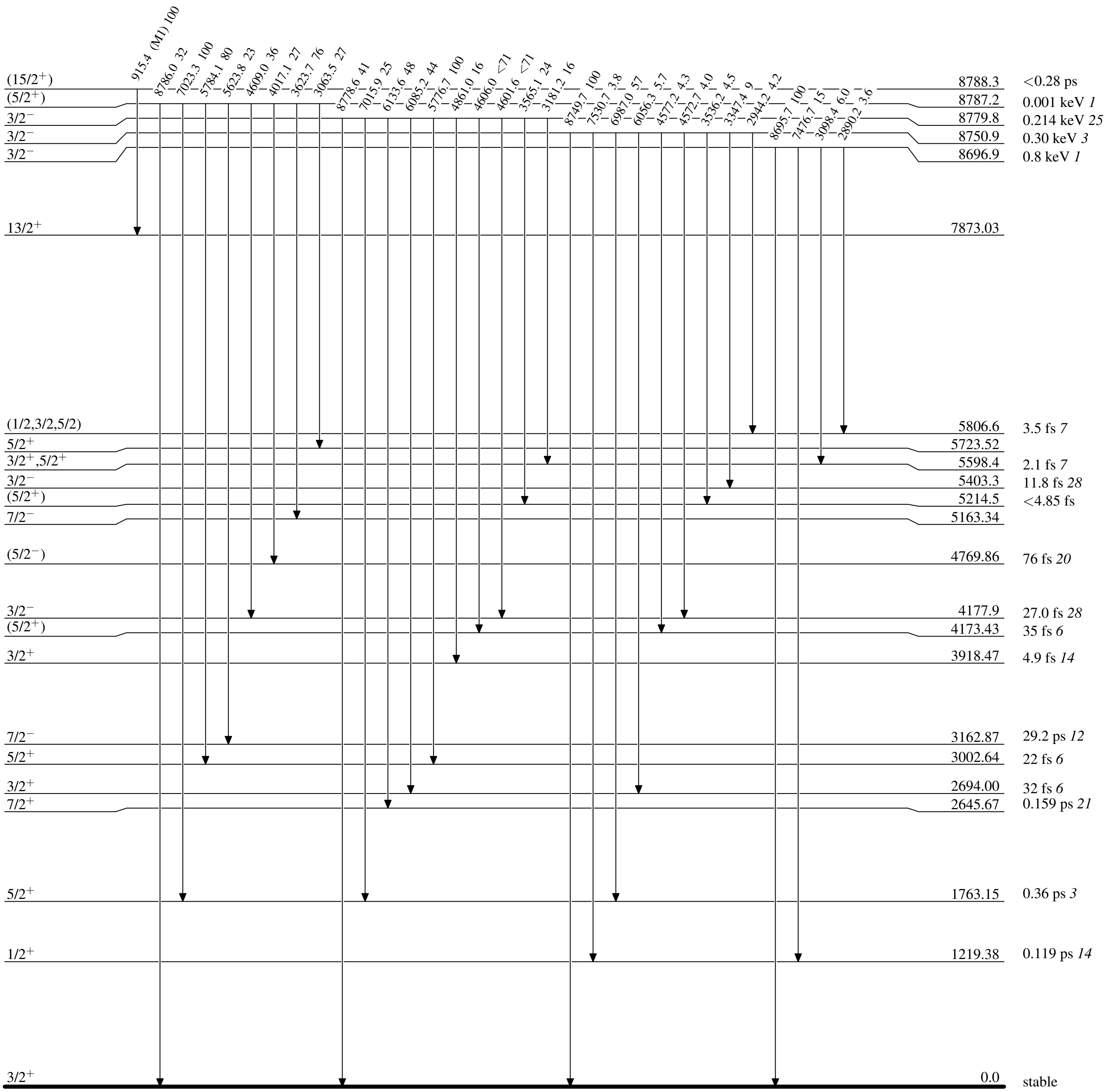


$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level

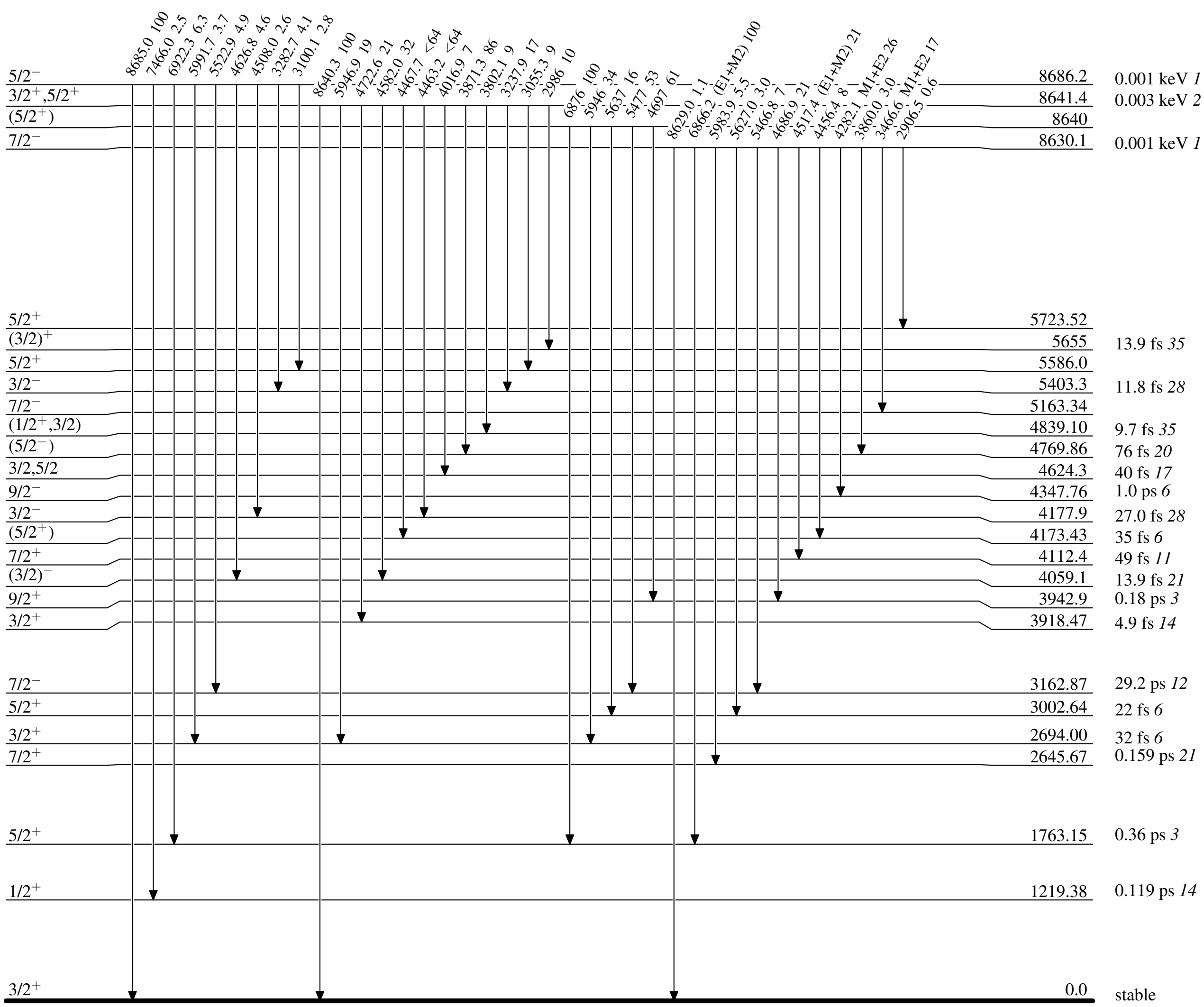


$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level

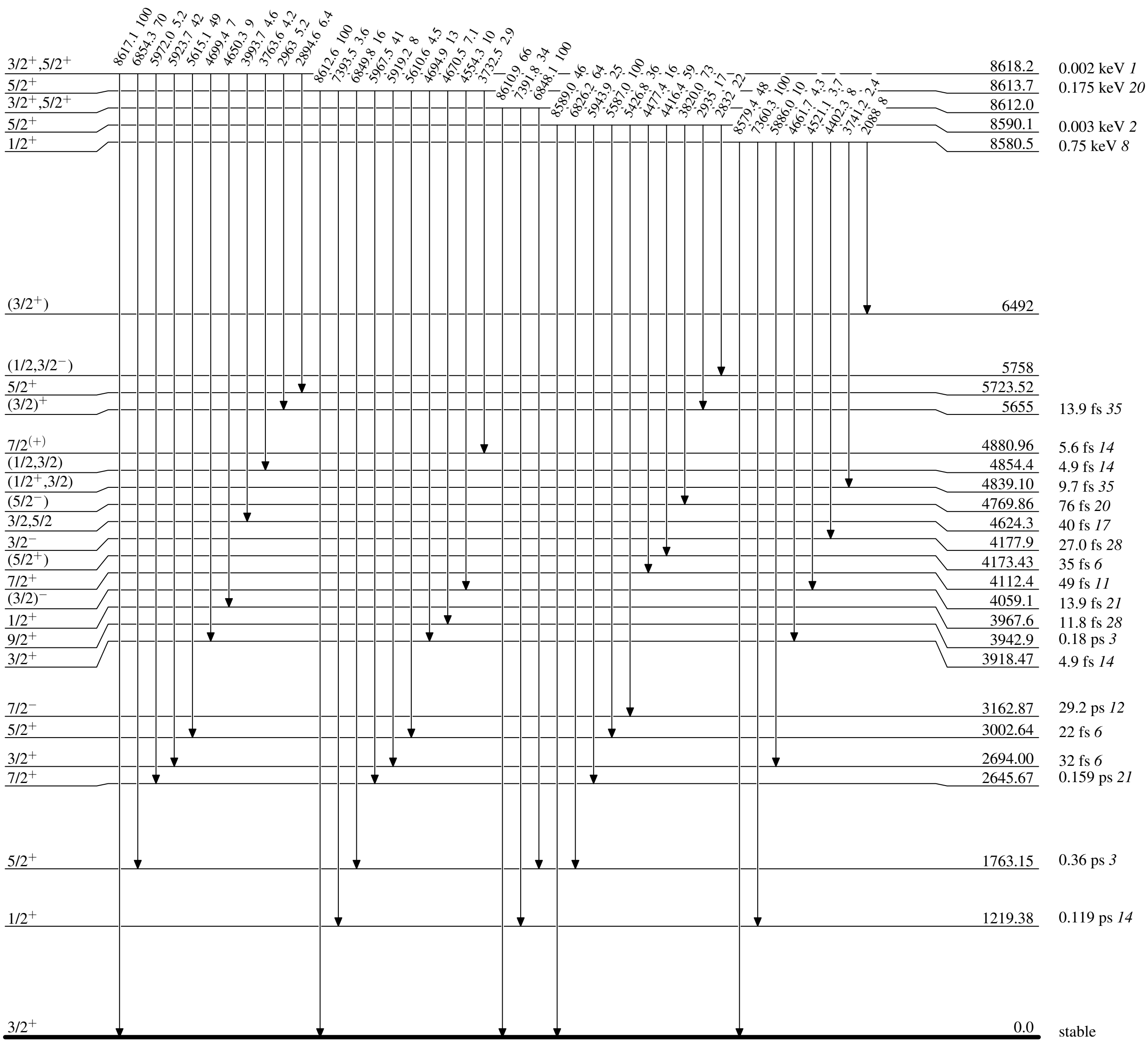


$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level



$^{35}_{17}Cl_{18}$

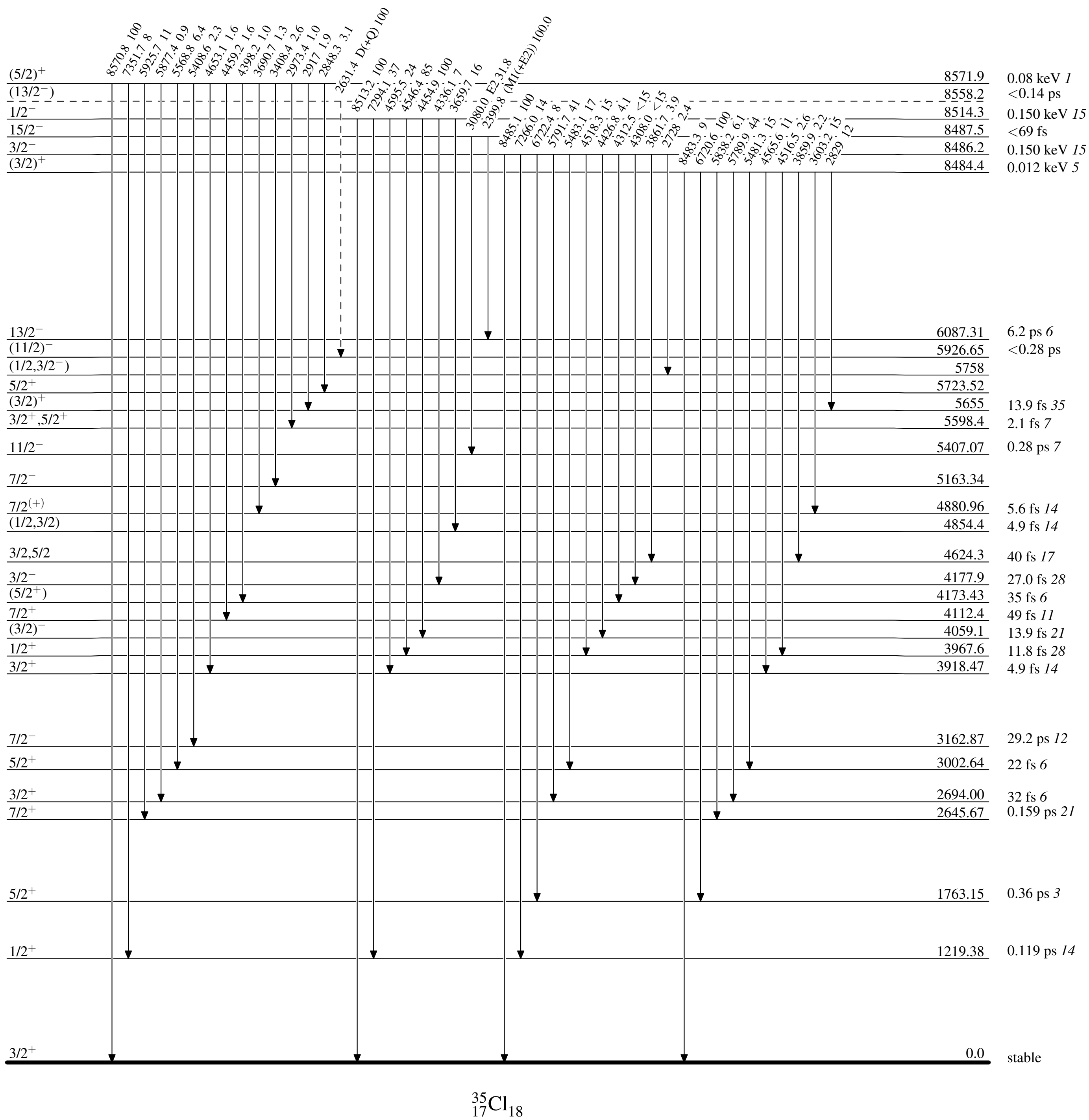
Adopted Levels, Gammas

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

γ Decay (Uncertain)

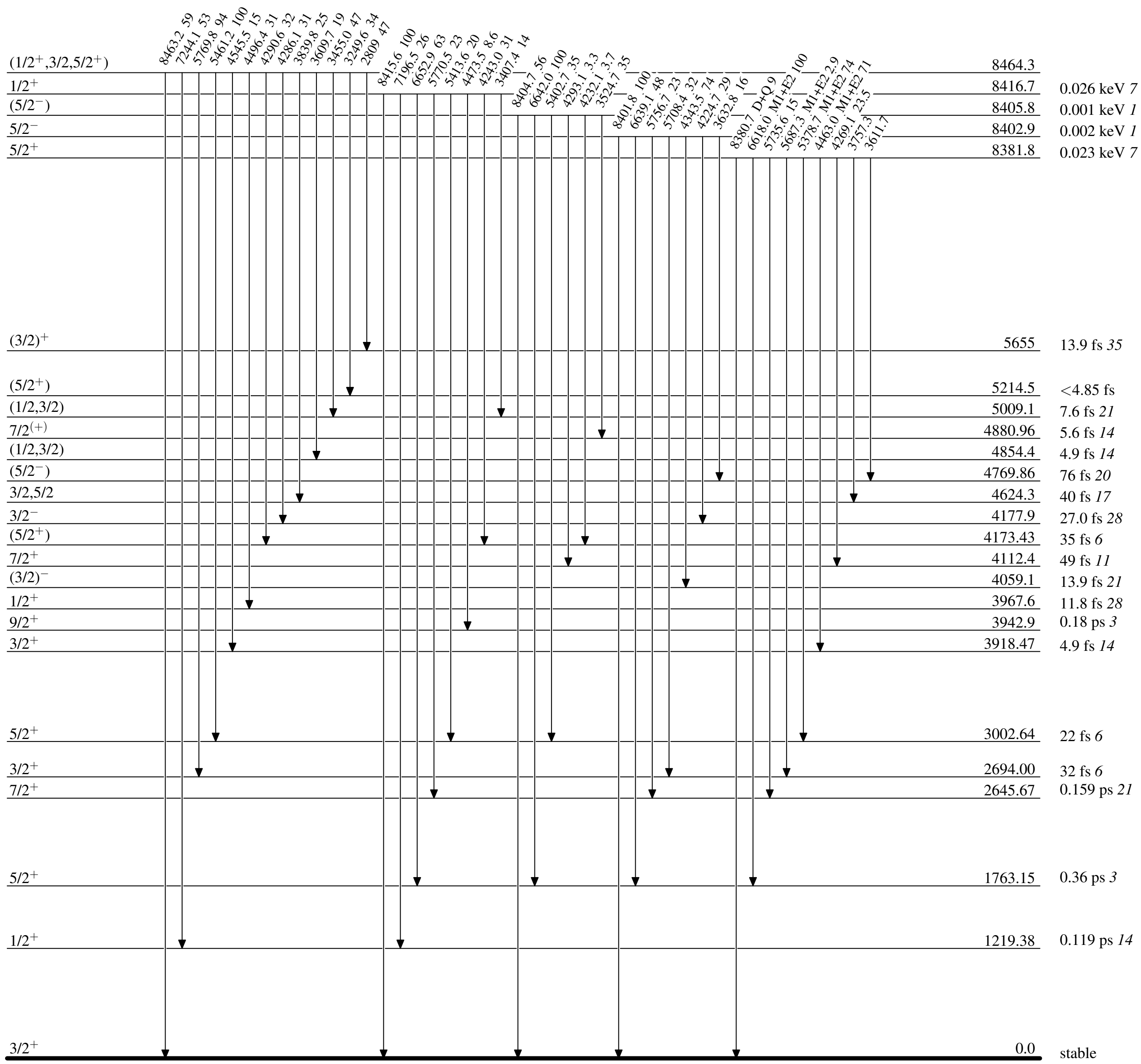


$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level

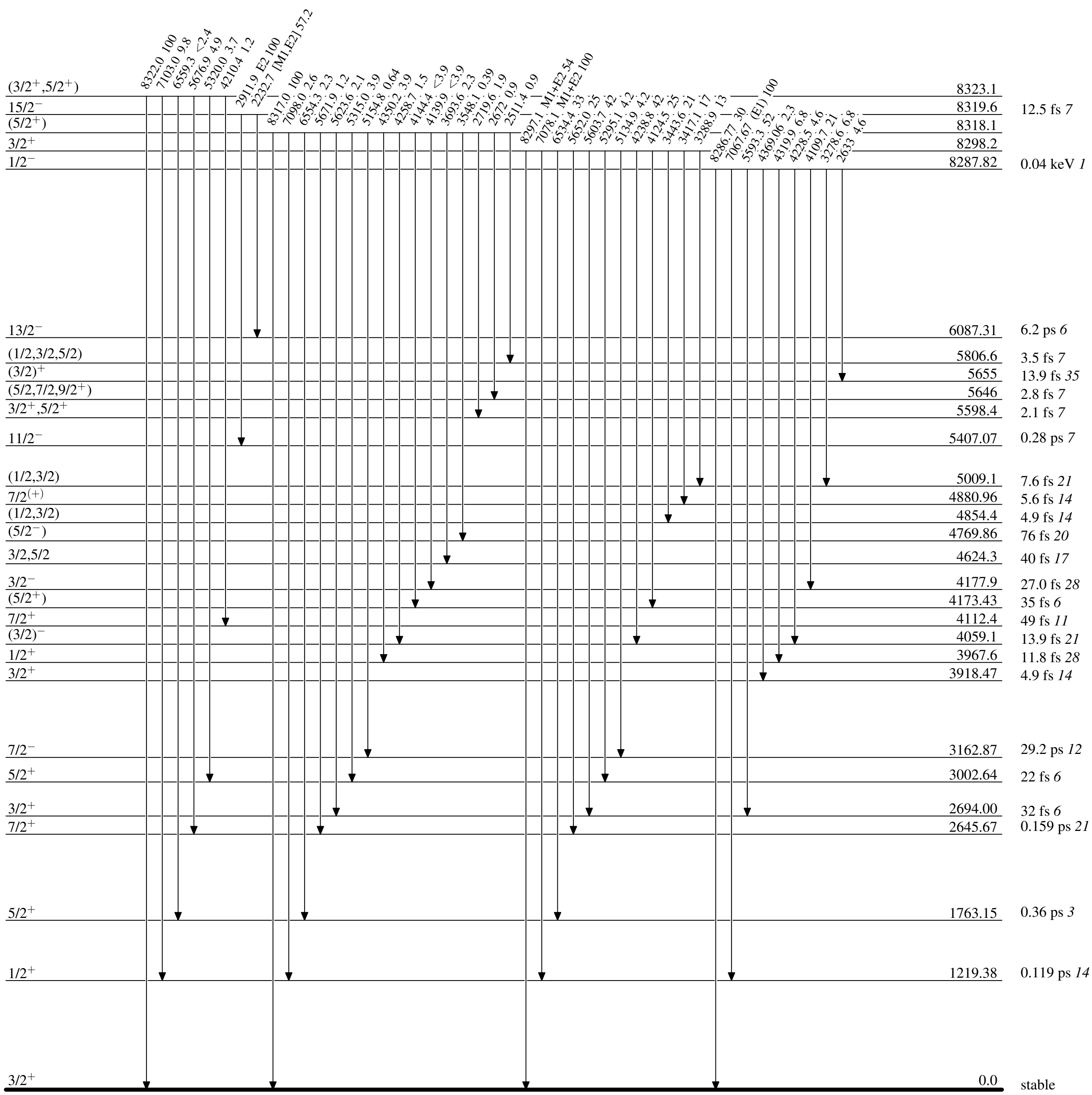


$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level

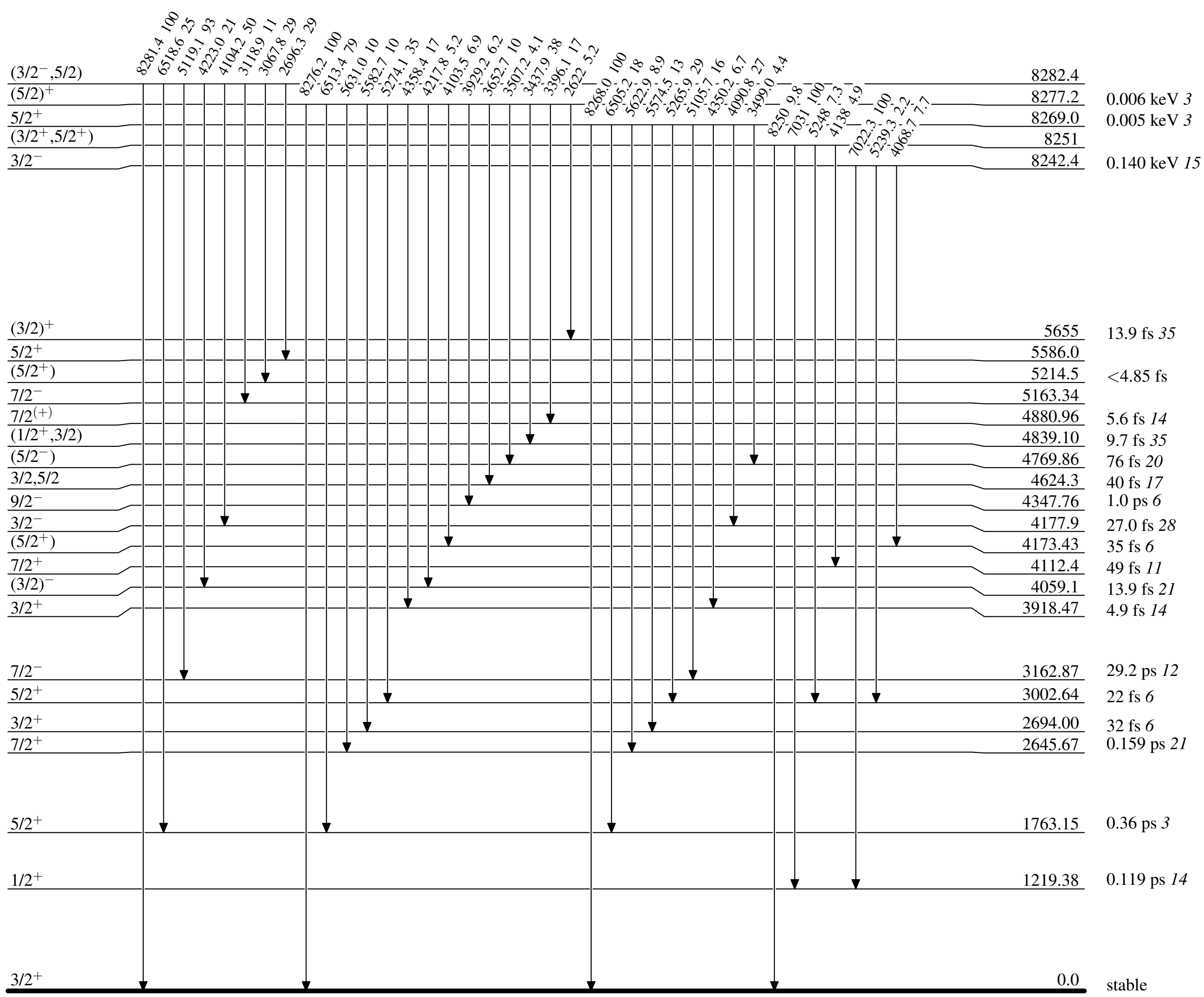


$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level

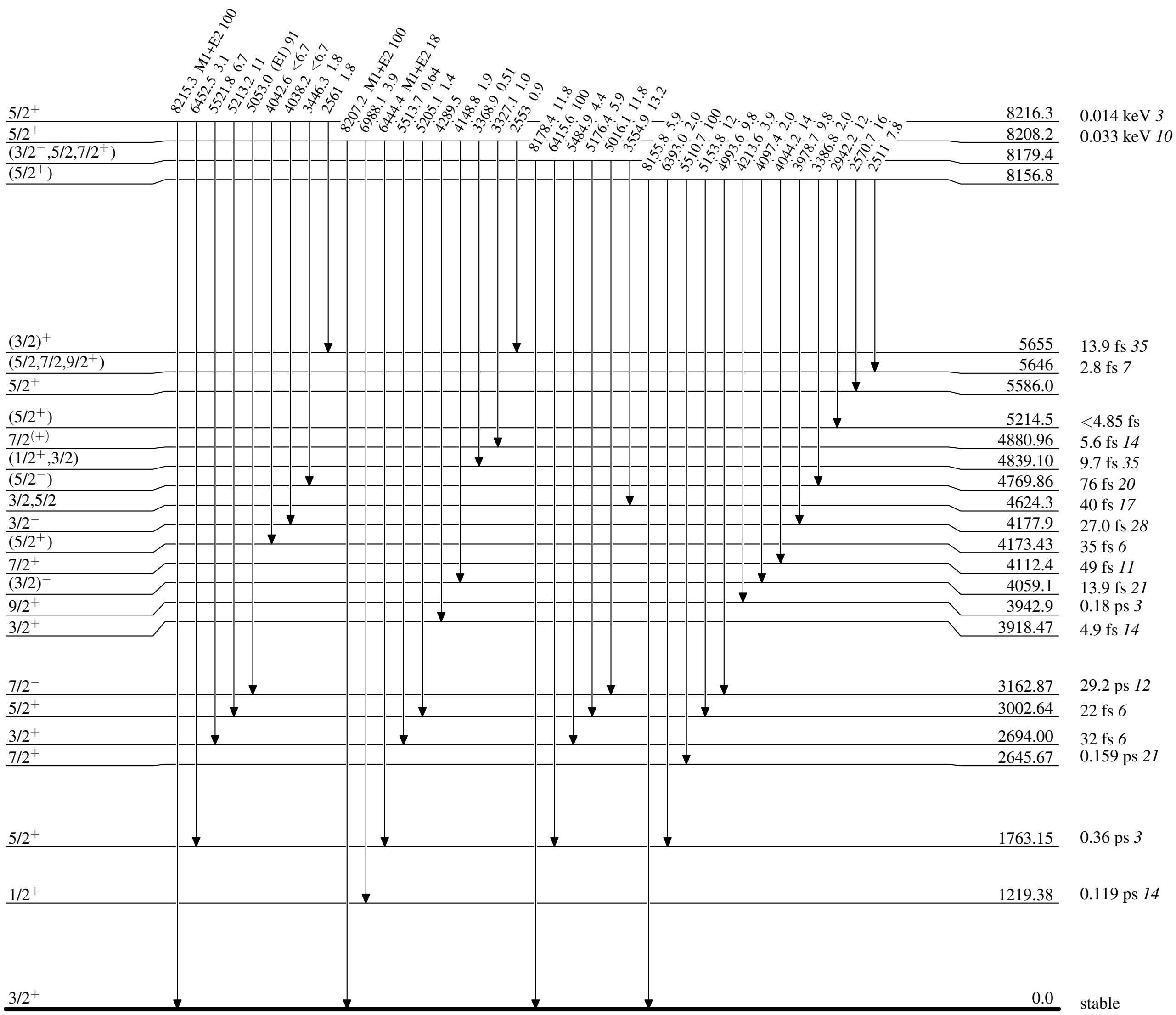


$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level

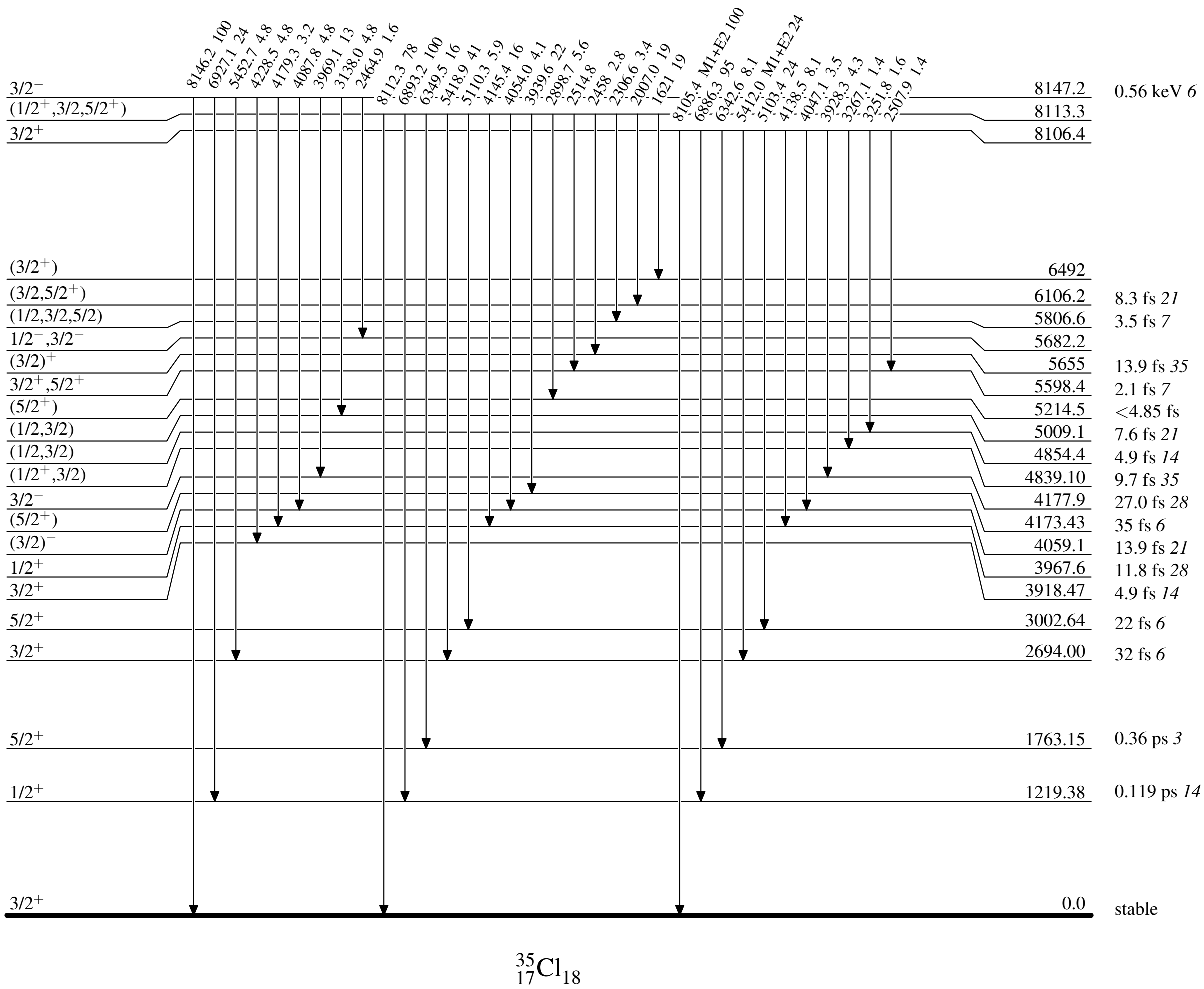


$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas

Level Scheme (continued)

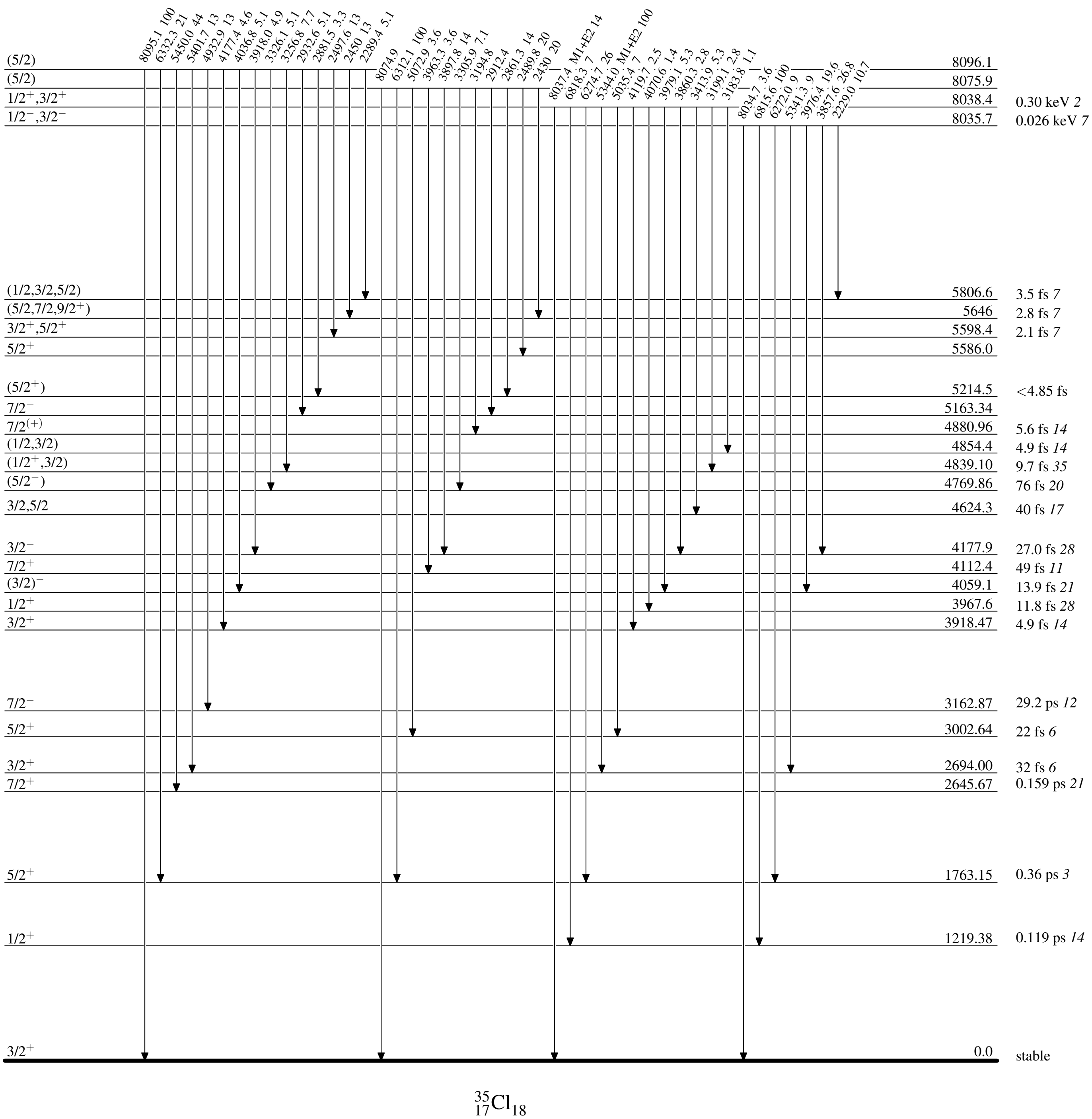
Intensities: Relative photon branching from each level



Adopted Levels, Gammas

Level Scheme (continued)

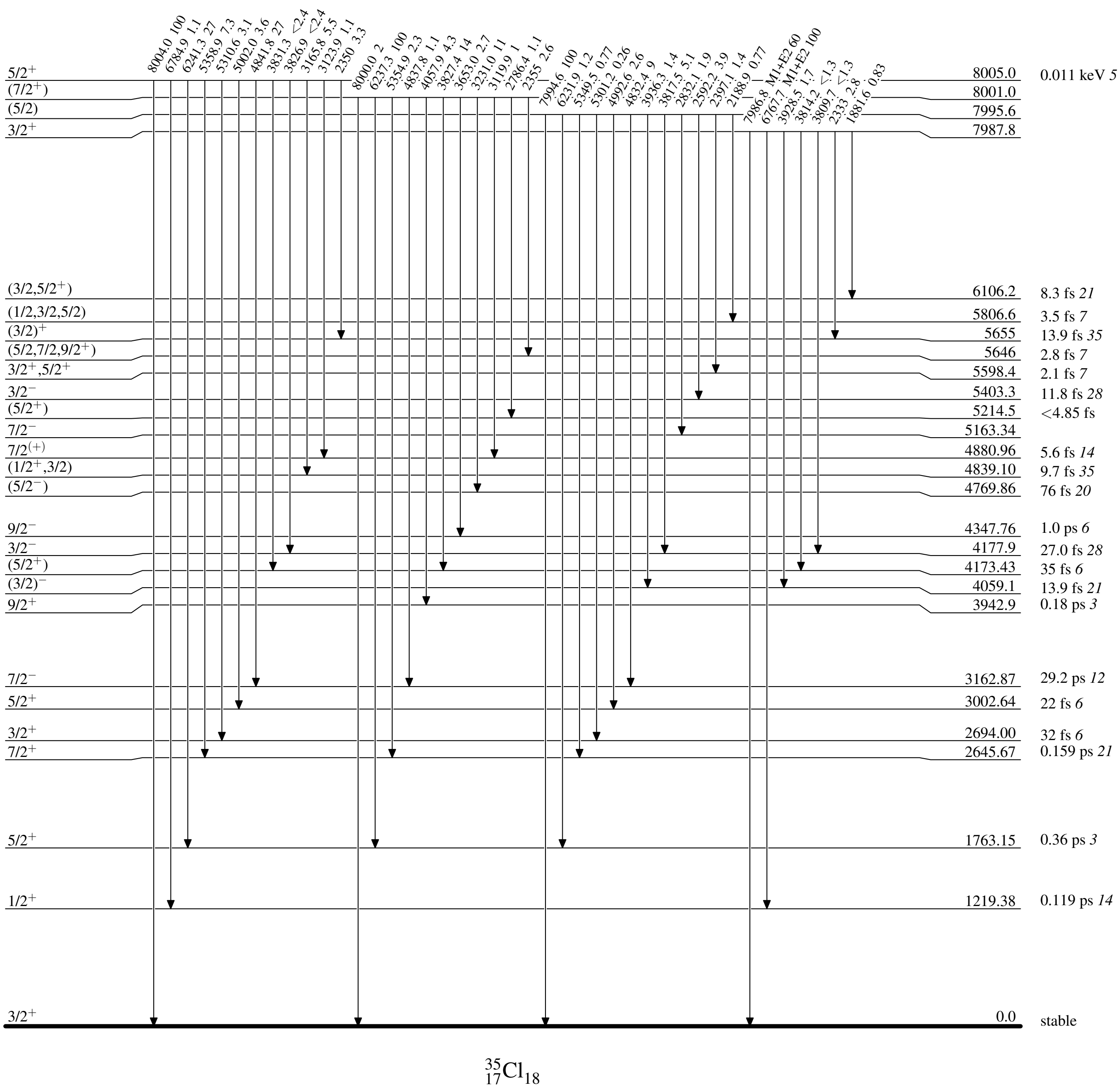
Intensities: Relative photon branching from each level



Adopted Levels, Gammas

Level Scheme (continued)

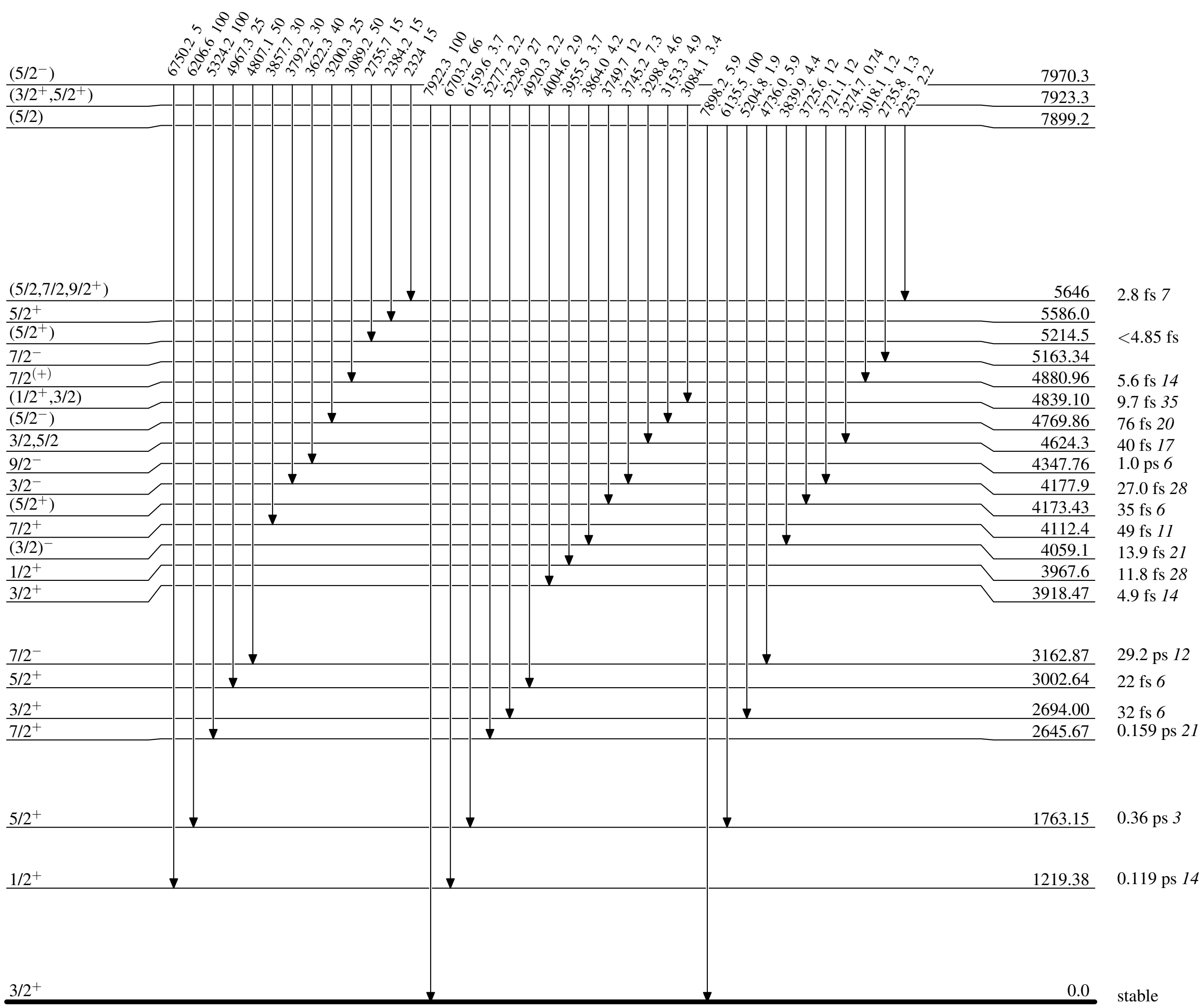
Intensities: Relative photon branching from each level



Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level

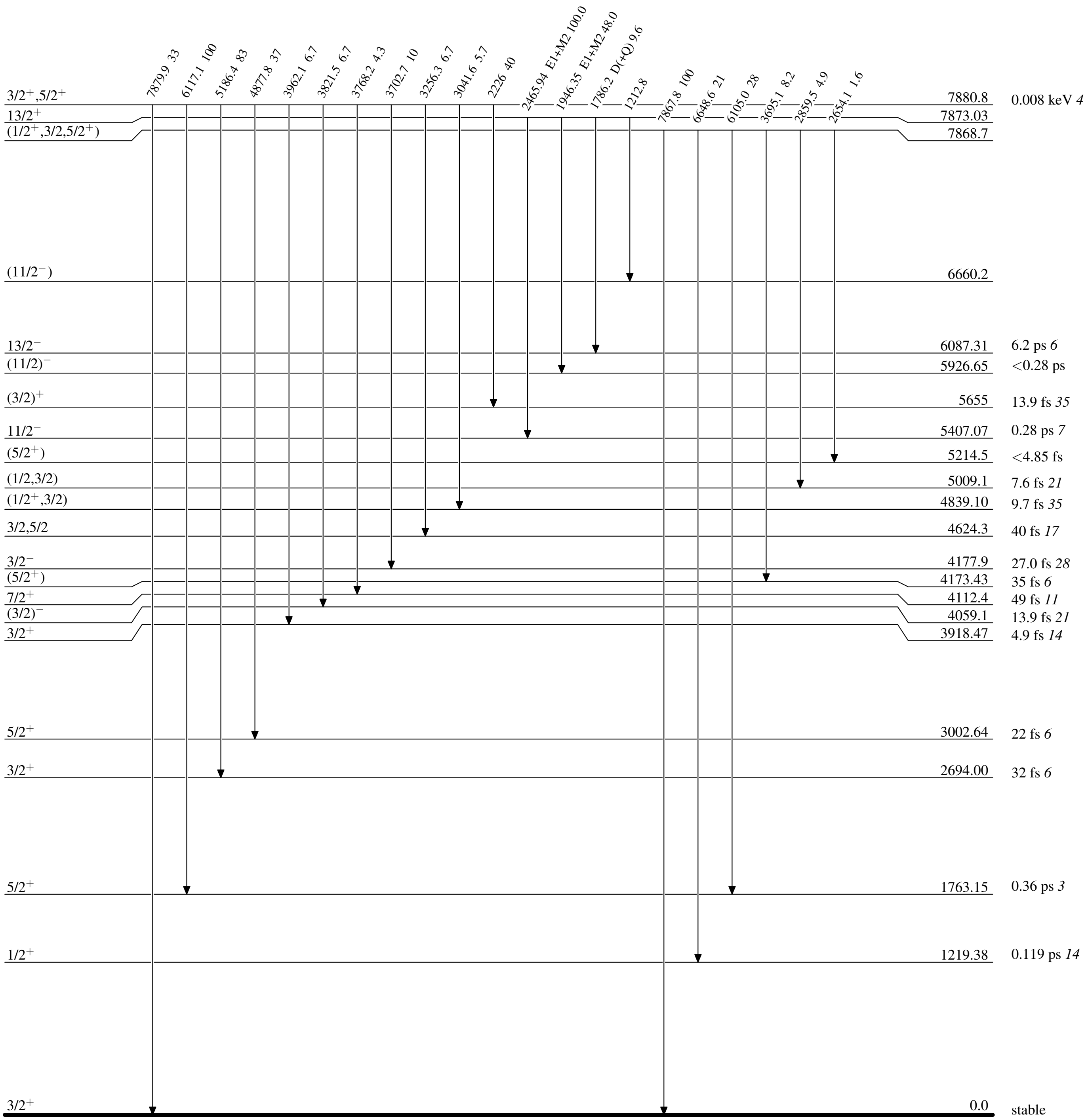


$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level

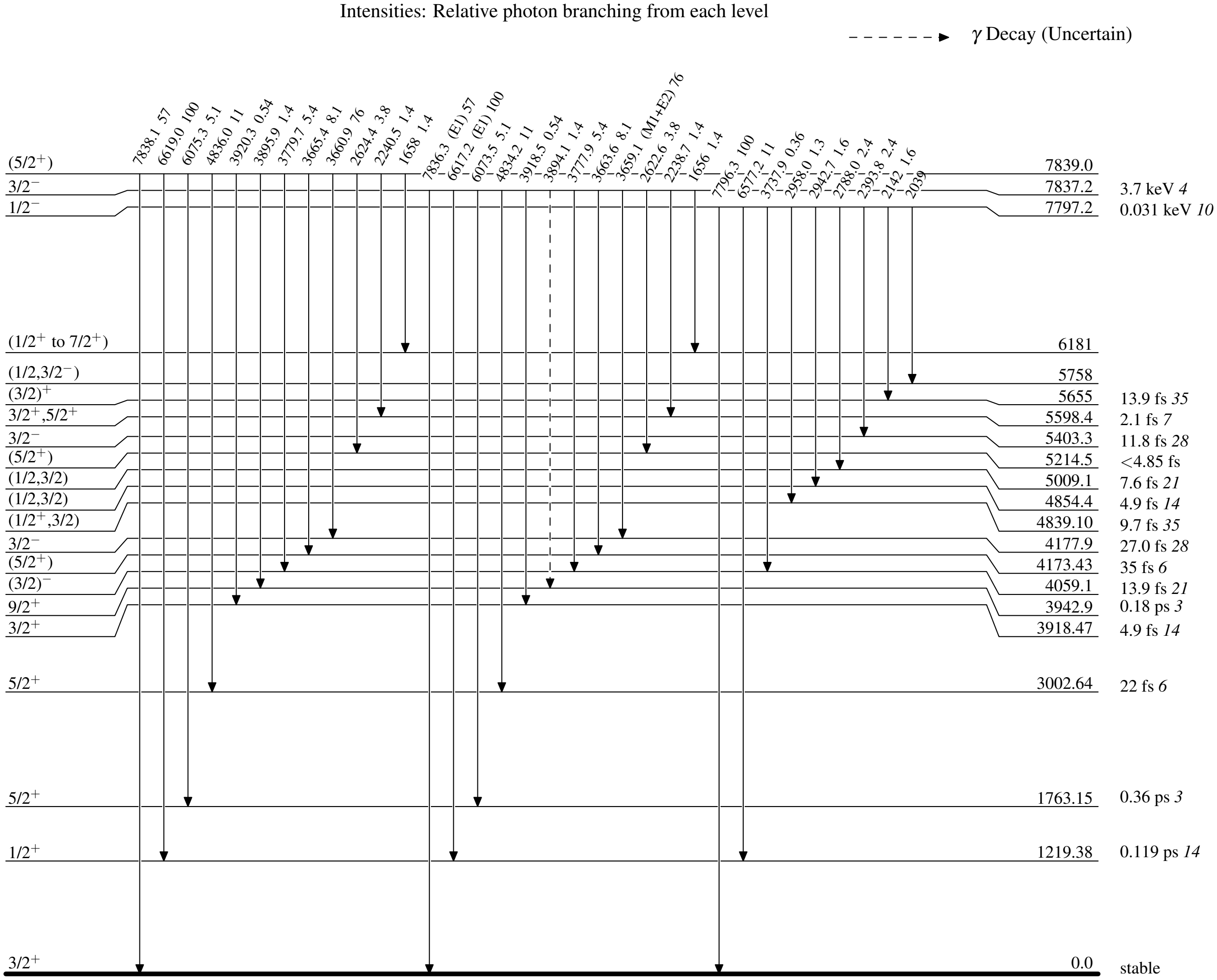


$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level



$^{35}_{17}Cl_{18}$

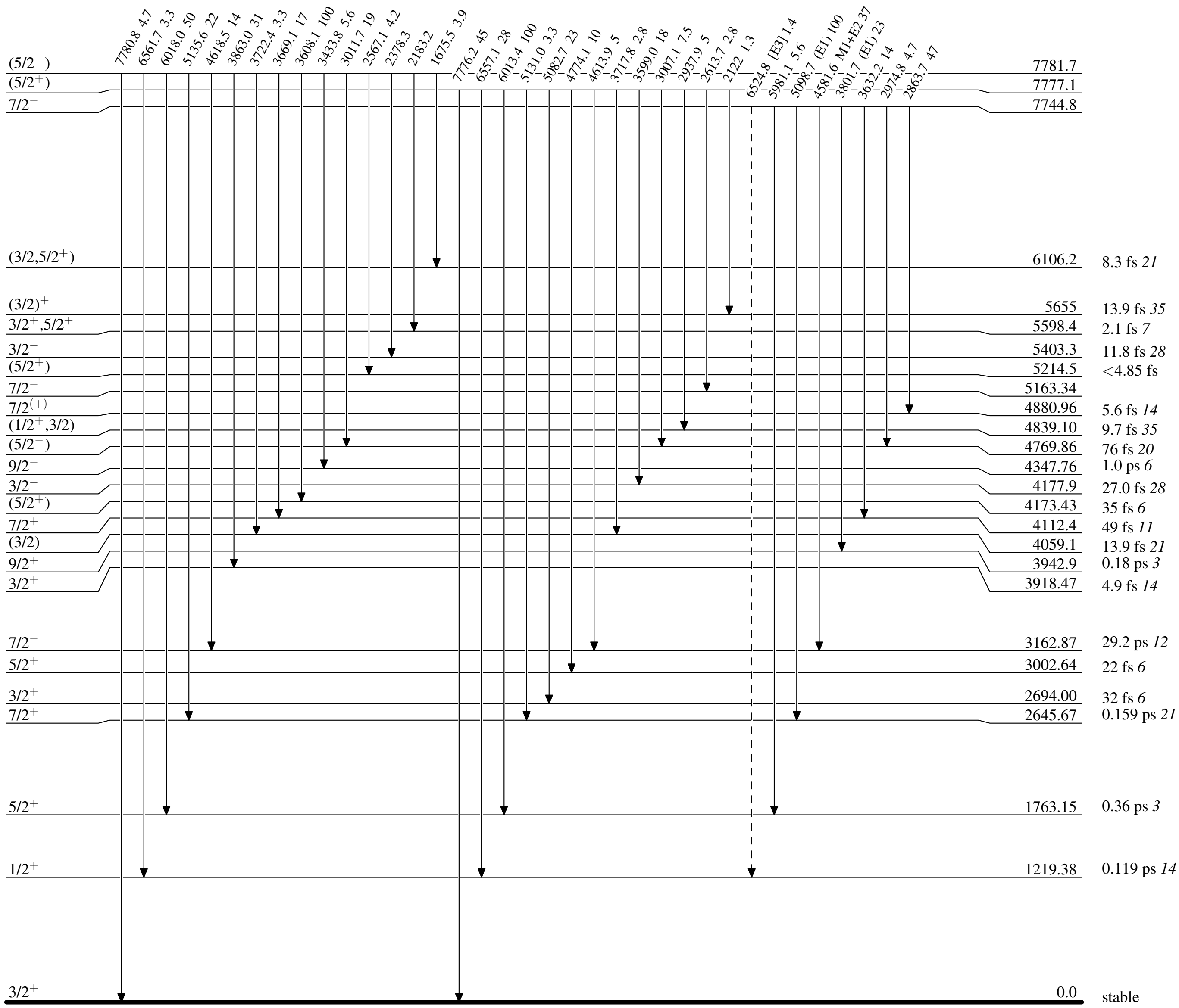
Adopted Levels, Gammas

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

γ Decay (Uncertain)

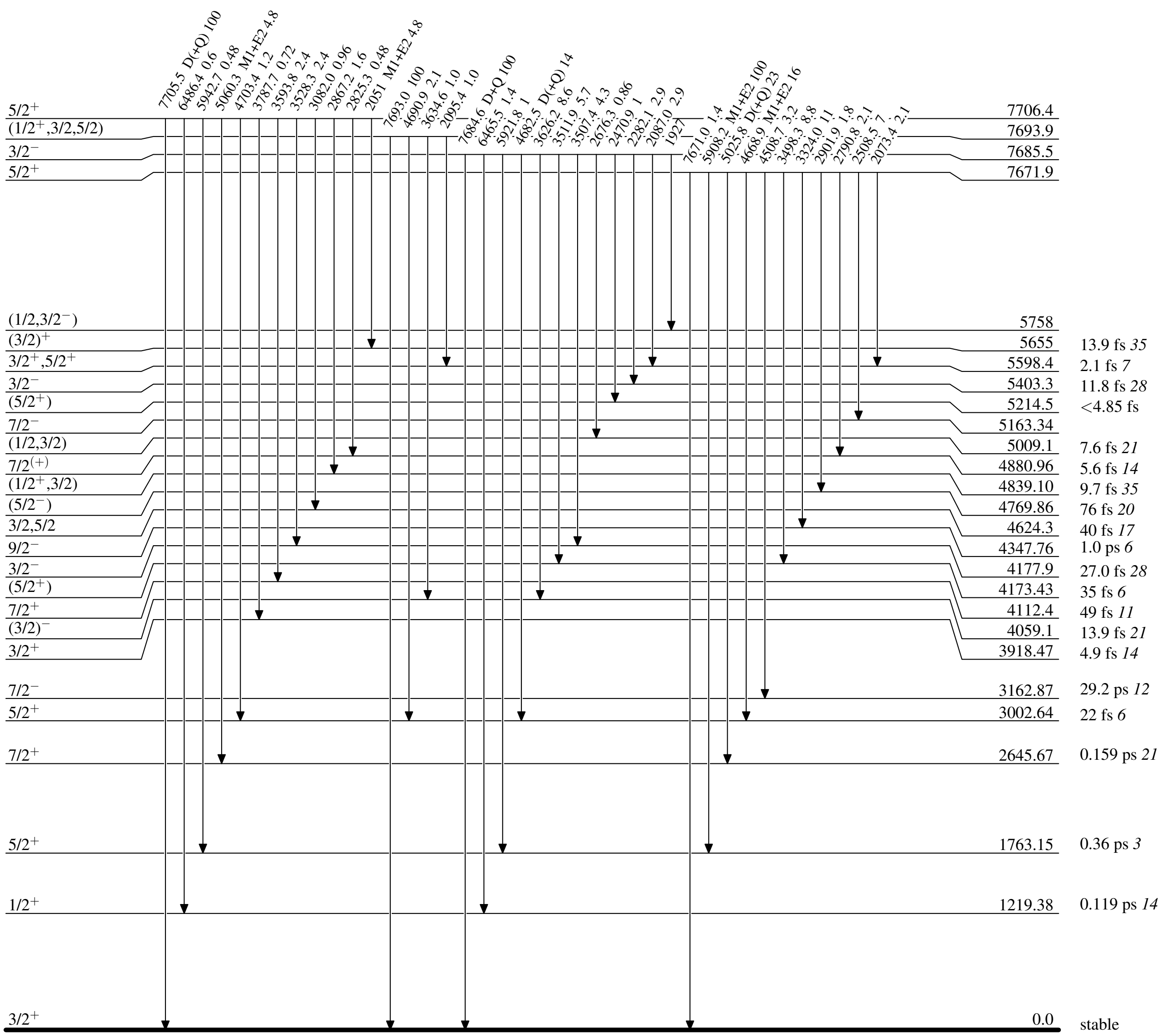


$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level

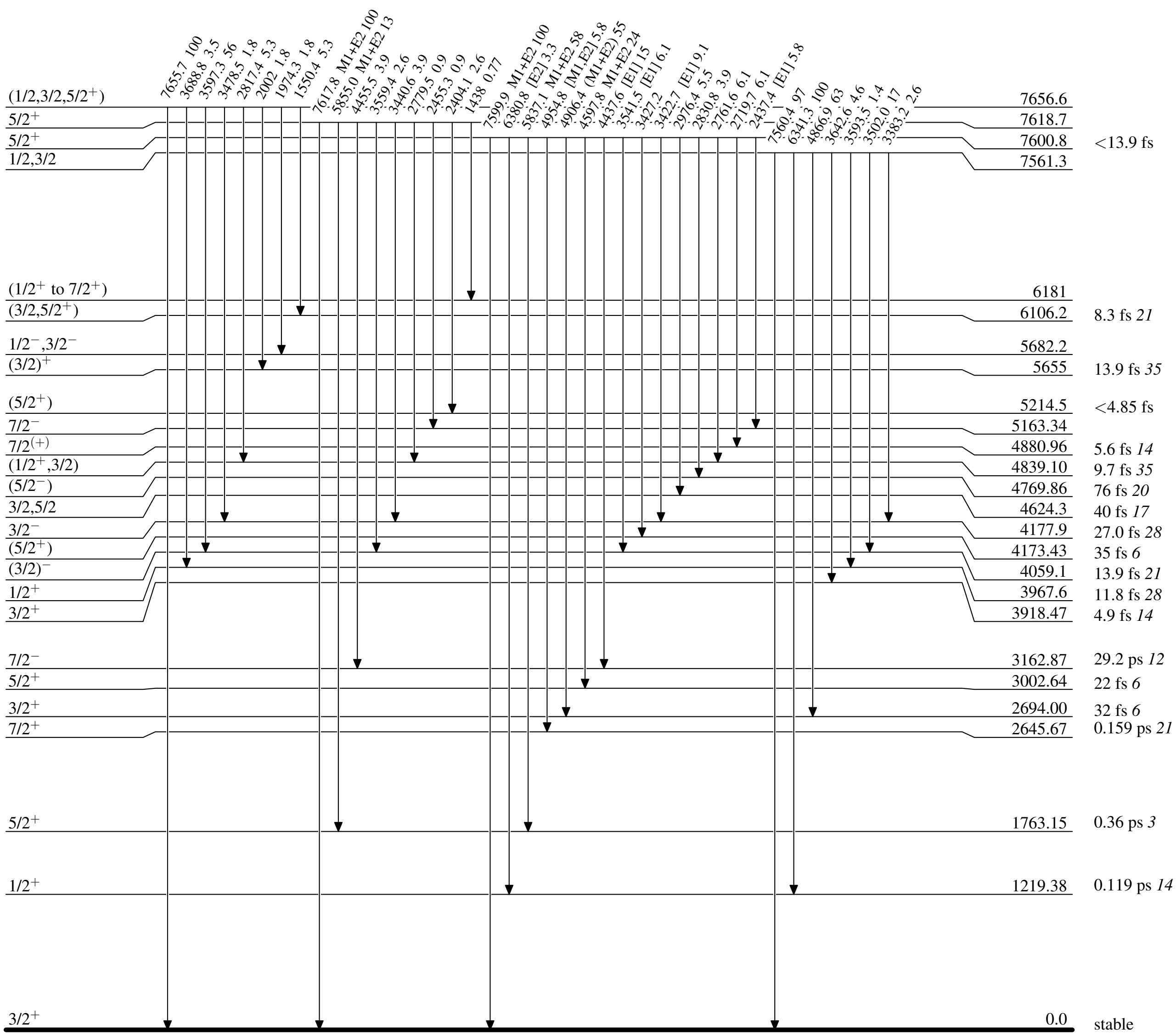


$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level

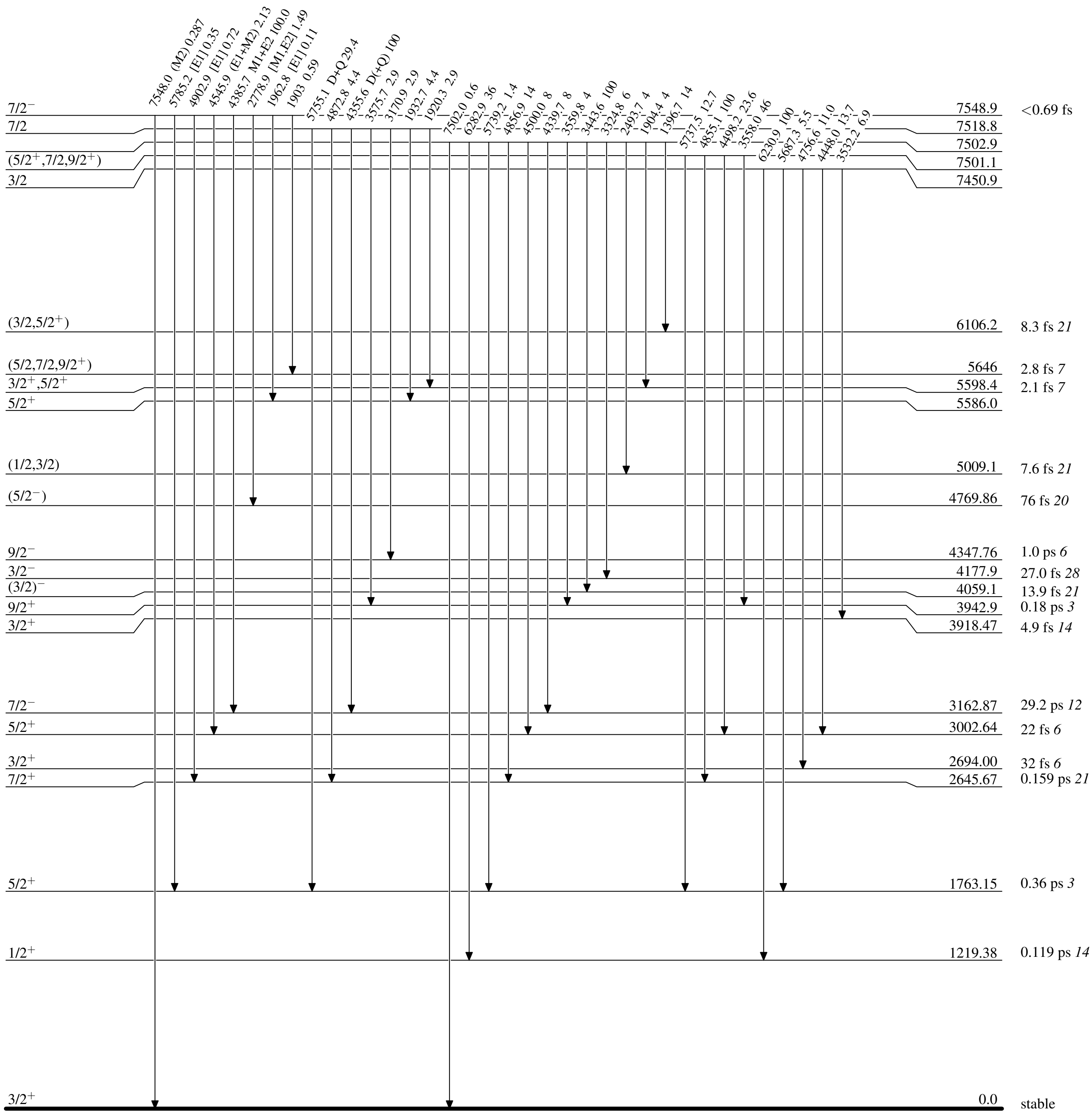


$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level

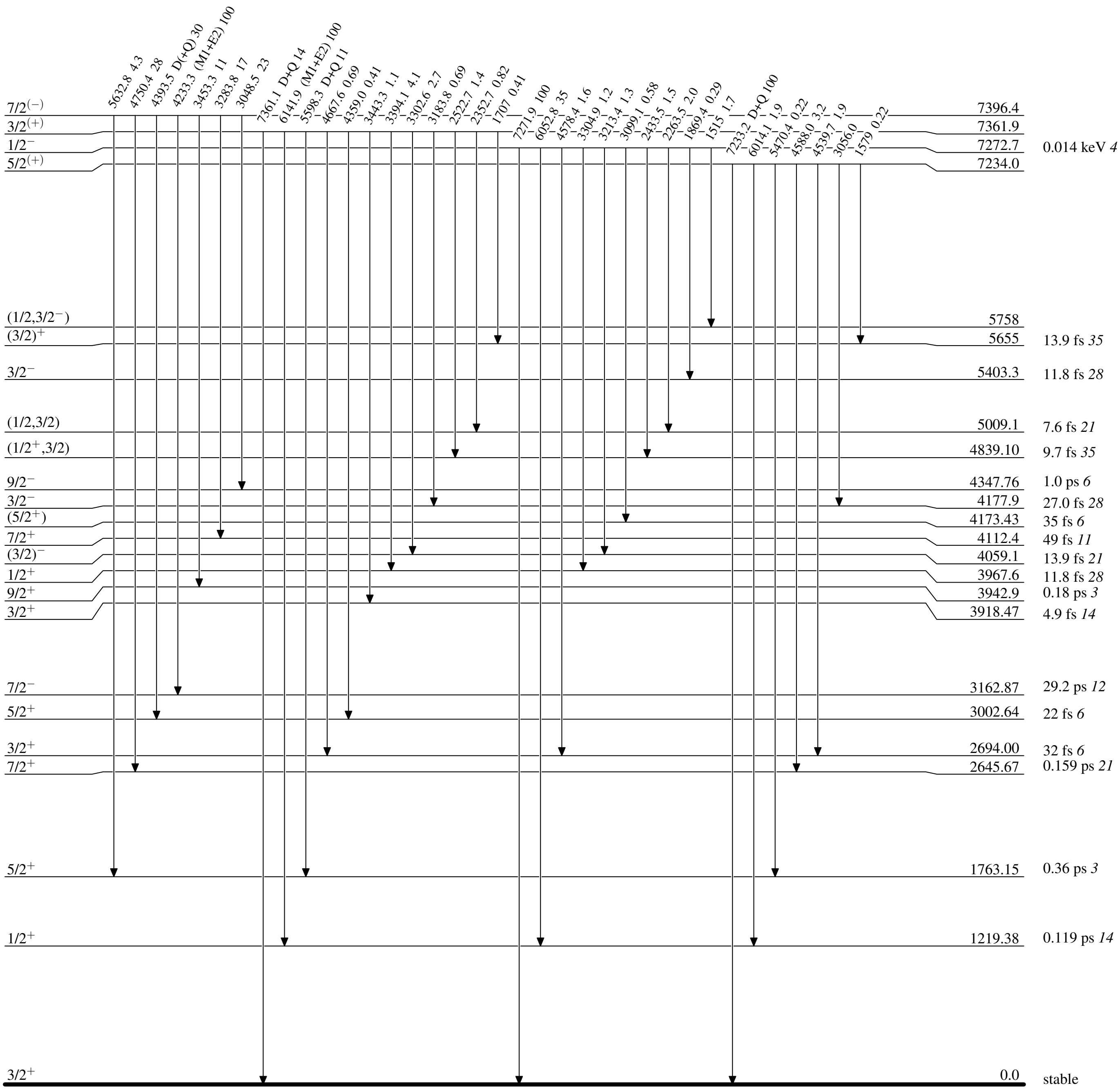


$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level

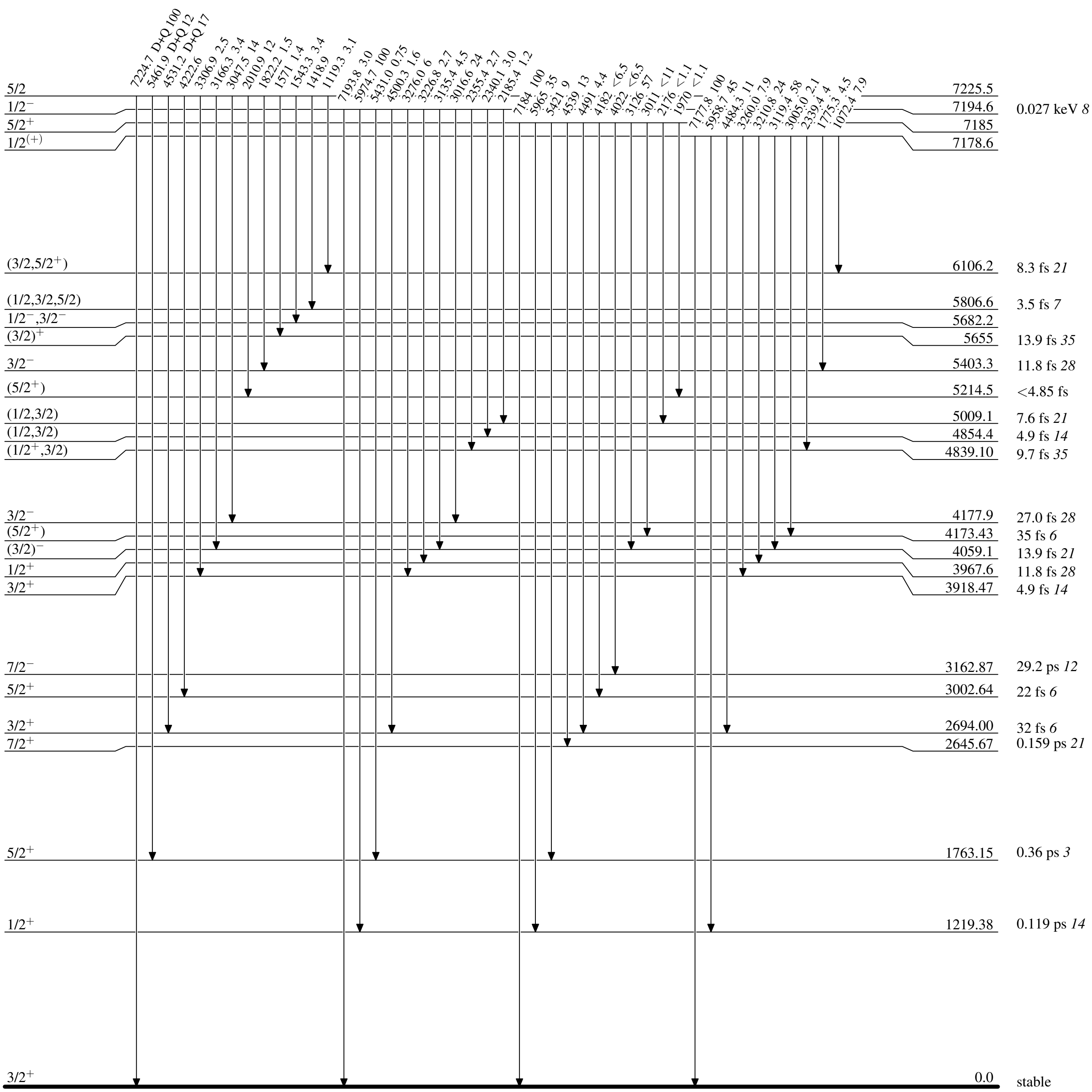


$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level

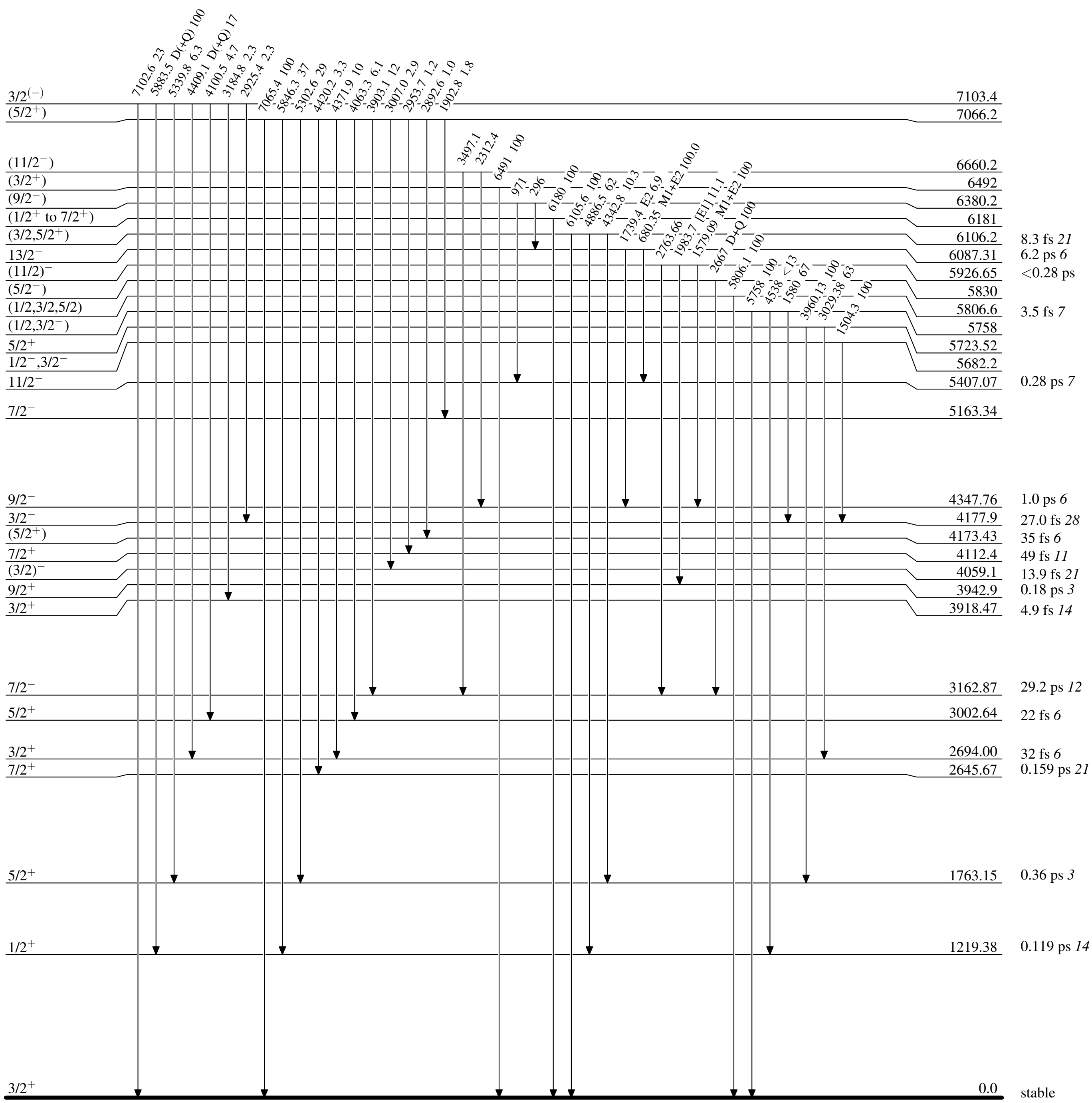


$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level

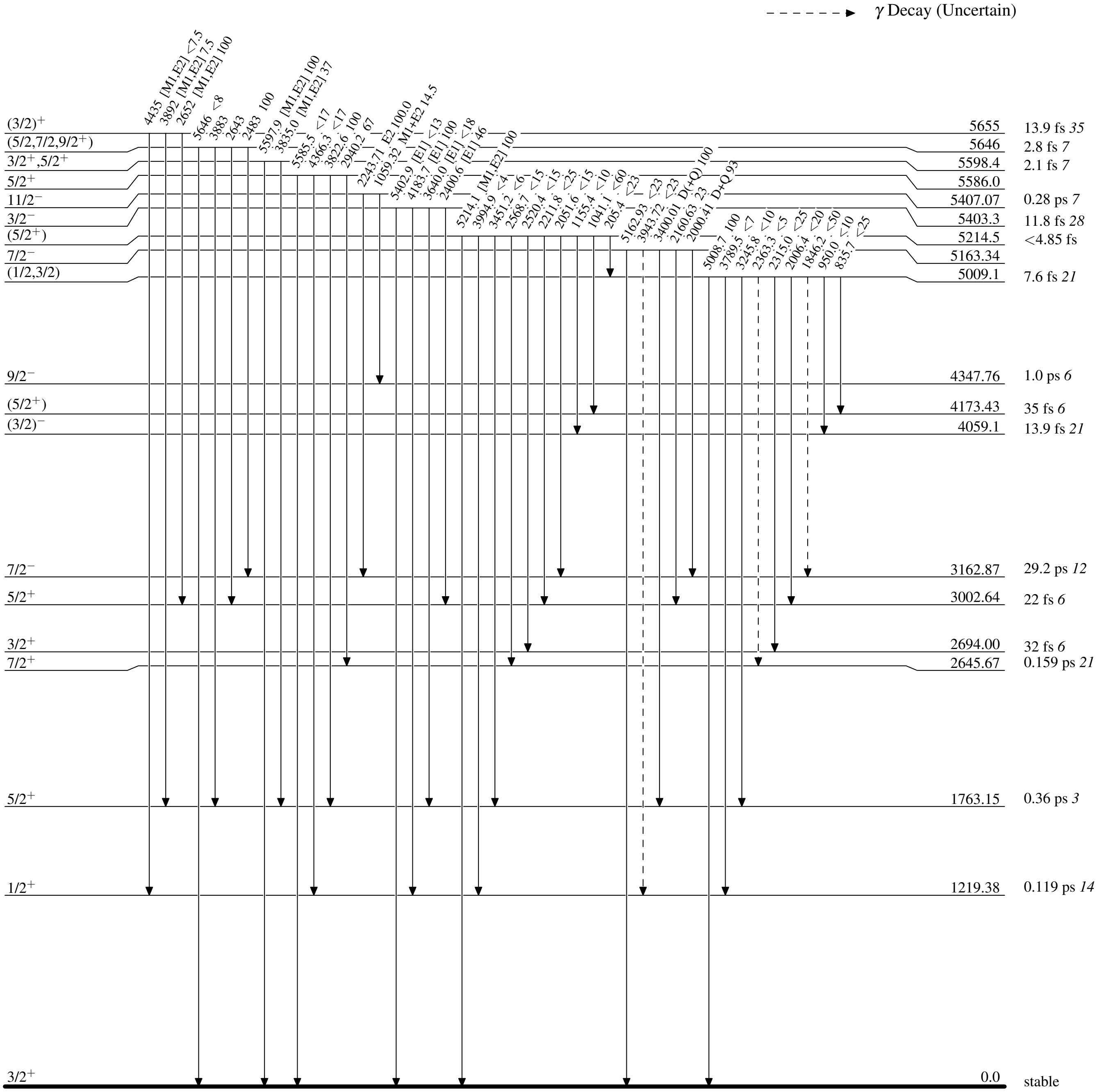


$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level



$^{35}_{17}Cl_{18}$

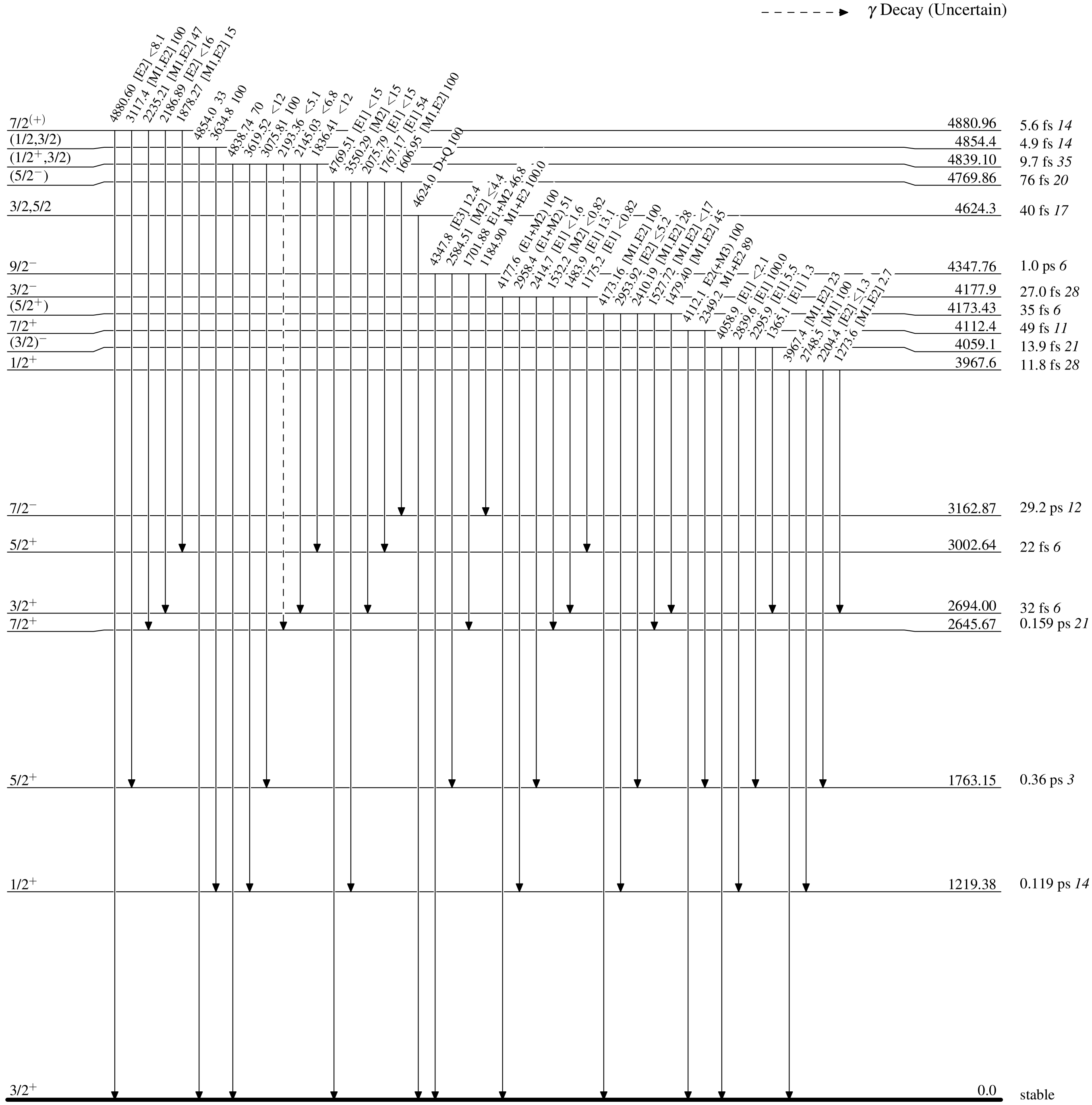
Adopted Levels, Gammas

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

γ Decay (Uncertain)

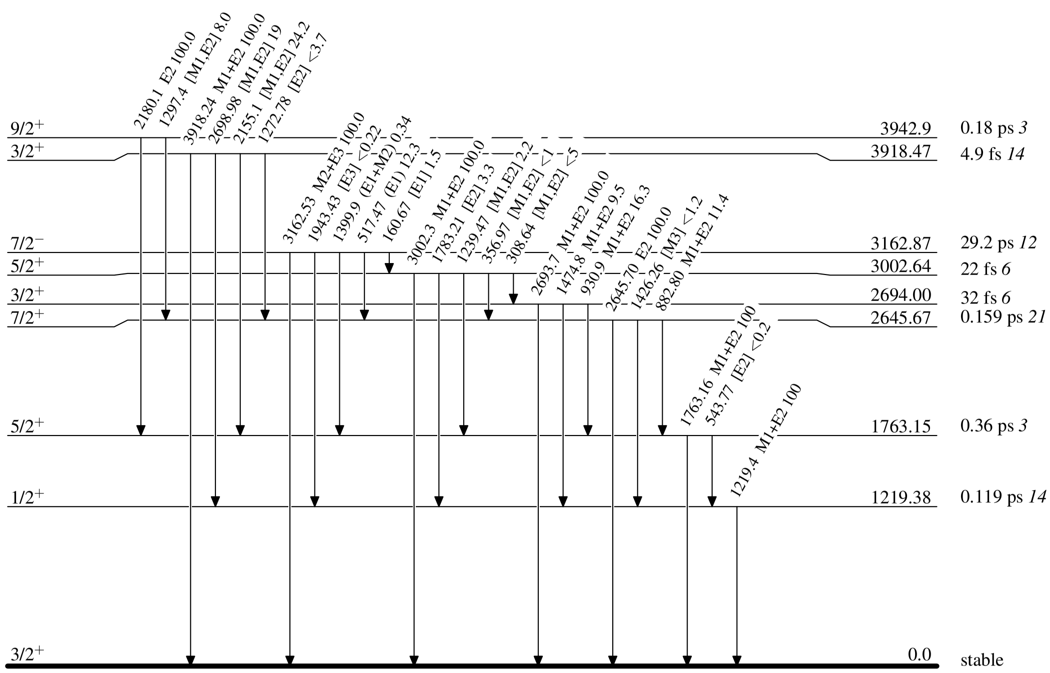


$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas

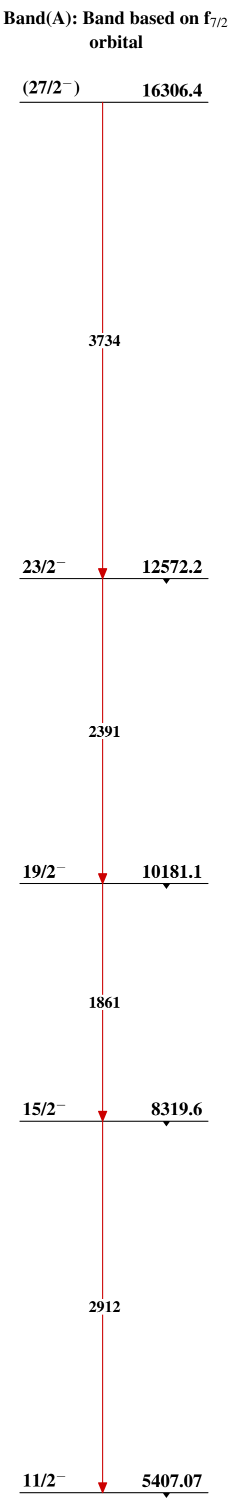
Level Scheme (continued)

Intensities: Relative photon branching from each level



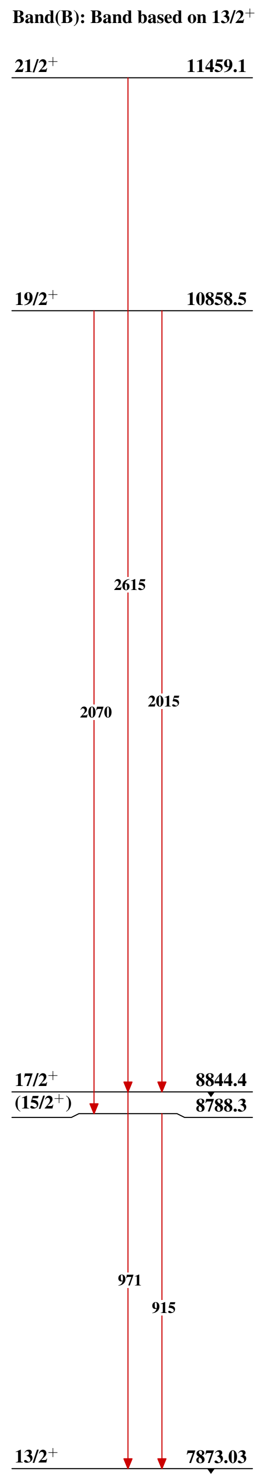
$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas



$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas (continued)



$^{35}_{17}Cl_{18}$

$^{35}S\ \beta^-$ decay (87.61 d)

Parent: ^{35}S: E=0; $J^\pi=3/2^+$; $T_{1/2}$=87.61 d *24*; $Q(\beta^-)$=167.322 *26*; %β^- decay=100
^{35}S-$Q(\beta^-)$: From 2021Wa16.
^{35}S-J^π,$T_{1/2}$: From the Adopted Levels of ^{35}S.
1983Ra04: $Q(\beta^-)$=166.74 keV *26*.
1985Ap01: $Q(\beta^-)$=167.29 keV *3*.
1985Oh06: $Q(\beta^-)$=167.4 keV *1*.
1985Al11: Upper limit for 17 keV neutrino: 0.4%.
1985Oh06, 1992Hi06, 1993Ab11, 1993Be21, 1993Mo01: Deduced evidence against 17 keV neutrino.
1994Bo33: Deduced spurious 17 keV neutrino signal origin.
1995Bo43: $Q(\beta^-)$=167.60 keV *5* and upper limit for 17 keV neutrino is 1.8×10^{-3}.
1995Mo17: Proposed 17 keV neutrino hypothesis exclusion.
Others: 2008Al39, 2000Ho13, 1999Oh09, 1996Je06, 1994Gr04, 1994Ho35, 1994Mu26, 1993Gi08, 1993Gr07, 1992Ch27, 1989Ba04, 1989Ta08, 1988Ch45, 1987Ge04, 1987Na22, 1985Ma59, 1999Pa18.
^{35}S decay leads only to the ground state of ^{35}Cl.

^{35}Cl Levels

E(level)	J^π	Comments
0	$3/2^+$	J^π: From the Adopted Levels.

β^- radiations

E(decay)	E(level)	$I\beta^-$ †	Log *ft*
(167.3 *14*)	0	100	5.023 *2*

† Absolute intensity per 100 decays.

^{35}Ar ε+β^+ decay (1.7755 s) 1980Wi13,1984Ad01

Parent: ^{35}Ar: E=0; J$^\pi$=3/2$^+$; $T_{1/2}$=1.7755 s *14*; Q(ε)=5966.2 *7*; %ε+%β^+ decay=100

^{35}Ar-J$^\pi$,$T_{1/2}$: From Adopted Levels of ^{35}Ar.

^{35}Ar-Q(ε): From 2021Wa16.

1980Wi13: ^{35}Ar was produced via the ^{35}Cl(p,n)^{35}Ar reaction with 11.0-MeV protons from the ONR-CIT tandem accelerator impinging on LiCl and $PbCl_2$ wrapped in 25-μm Ta foil. γ rays were detected using a 100-cm^3 Ge(Li) detector. Measured Eγ and Iγ for 20 γ-ray peaks. Deduced decay branching ratios for 9 levels, and log *ft*. Compared with shell-model calculations.

1984Ad01: ^{35}Ar was produced via the ^{35}Cl(p,n)^{35}Ar reaction with 13-MeV protons from the University of Washington FN tandem accelerator impinging on a thick KCl target. γ rays were detected using a 28% Ge(Li) detector. Measured Eγ and Iγ for 8 γ-ray peaks. Deduced decay branching ratios for 6 levels, and log *ft*. Compared with shell-model calculations.

1985Da04: ^{35}Ar was produced via the ^{35}Cl(p,n)^{35}Ar reaction with 9-MeV protons from the Van de Graaff accelerator at the University of Pittsburgh impinging on a NaCl target. γ rays were detected using a Ge(Li) detector with NaI anti-Compton shielding. Measured Eγ and Iγ for 4 γ-ray peaks. Deduced 3 levels.

1979Ha04: ^{35}Ar produced via the ^{51}V(p,6p11n) spallation reaction with 600-MeV protons from the CERN synchrocyclotron bombarding a 38-g/cm^2 vanadium target in the form of VC powder. γ rays were detected using a Ge(Li) detector. Measured Eγ and Iγ for the 1219γ. Deduced the decay branching ratio for the 1219-keV level.

1977Az01: ^{35}Ar was produced via the ^{35}Cl(p,n)^{35}Ar reaction with 10-13-MeV protons from the McGill synchrocyclotron impinging on CCl_4 and LiCl targets and 8-MeV protons from the University of Montreal impinging a thick LiCl pellet. Positrons were detected using an NE102 plastic scintillator coupled to an RCA 6342A photomultiplier tube. γ rays were detected using an 18% Ge(Li) detector. Measured Eγ and Iγ for 3 γ-ray peaks. Deduced decay branching ratios for 3 levels and parent ^{35}Ar $T_{1/2}$. Related: ^{35}Cl(p,n)^{35}Ar reaction threshold measurement in 1978Az01.

1971Ge04: ^{35}Ar was produced via the ^{35}Cl(p,n)^{35}Ar reaction with 10-MeV protons from the Chalk River tandem accelerator impinging on $PbCl_2$ targets. γ rays were detected using two coaxial Ge(Li) detectors. Measured Eγ and Iγ for 4 γ-ray peaks. Deduced 4 levels, decay branching ratios, log *ft*, and parent ^{35}Ar $T_{1/2}$.

1971De05: ^{35}Ar was produced via the ^{35}Cl(p,n)^{35}Ar reaction on KCl targets. γ rays were detected using a 25 cm^3 Ge(Li) detector. Measured Eγ and Iγ. Deduced 6 levels, decay branching ratios, and log *ft*.

1969Wi18,1969Fr09: ^{35}Ar was produced via the ^{35}Cl(p,n)^{35}Ar reaction with a proton beam from the Harwell tandem accelerator impinging on 3.75-mg/cm^2 KCl deposited on 0.13-mm Au backings and 160-mmHg gas CCl_4 targets. γ rays were detected using a Ge(Li) detector. Measured Eγ and Iγ for 2 γ-ray peaks. Deduced 2 levels, decay branching ratios, (p,n) reaction threshold and β end-point energy, log *ft*, and parent ^{35}Ar $T_{1/2}$.

1960Wa04: ^{35}Ar was produced at Crocker Laboratory and the Lawrence Radiation Laboratory. β spectra were measured with a uniform-field 180°-deflection single-focusing spectrometer. Deduced Coulomb-energy differences between mirror ground-state transitions.

1956Ki29: ^{35}Ar was produced via the ^{35}Cl(p,n)^{35}Ar reaction with 10-MeV protons from the BNL 60-inch cyclotron impinging on a liquid CCl_4 target. β particles were detected using a conventional thin-lens, iron-free, magnetic spectrometer. γ rays were detected using a NaI(Tl) crystal scintillation counter. Measured Eβ and total Iβ for two positron decays to ^{35}Cl excited states, Eγ and Iγ for two γ-ray peaks. Deduced two levels and decay branching ratios.

^{35}Ar β asymmetry parameter measurements: 1993Co07, 1991Na05, 1991Mi18, 1988Ga02, 1965Ca04.

^{35}Cl Levels

E(level)†	J$^\pi$‡	$T_{1/2}$‡
0	3/2$^+$	stable
1219.2 *2*	1/2$^+$	0.119 ps *14*
1763.1 *2*	5/2$^+$	0.36 ps *3*
2645.1 *10*	7/2$^+$	0.159 ps *21*
2693.9 *2*	3/2$^+$	32 fs *6*
3002.2 *4*	5/2$^+$	22 fs *6*
3918.3 *15*	3/2$^+$	4.9 fs *14*
3967.9 *6*	1/2$^+$	11.8 fs *28*
4624.5? *5*	3/2,5/2	40 fs *17*

† From a least-squares fit to γ-ray energies.
‡ From the Adopted Levels.

^{35}Ar ε+β^+ decay (1.7755 s) 1980Wi13,1984Ad01 (continued)

ε,β^+ radiations

E(decay)	E(level)	Iβ^+ [‡]	Iε[‡]	Log *ft*	I(ε+β^+)[†‡]	Comments
(1341.7[#] *13*)	4624.5?	<8.0×10^{-4}	<0.0035	>3.8	<0.0043	
(1998.3 *14*)	3967.9	0.015 *10*	0.0016 *10*	4.5 *+5−2*	0.017 *11*	
(2047.9 *19*)	3918.3	0.0096 *30*	8.3×10^{-4} *26*	4.8 *+2−1*	0.0104 *32*	
(2964.0 *13*)	3002.2	0.089 *14*	0.0011 *2*	5.0 *1*	0.090 *14*	
(3272.3 *13*)	2693.9	0.164 *9*	0.00126 *7*	5.00 *3*	0.165 *9*	
(3321.1 *16*)	2645.1	<0.00146	<2.82×10^{-5}	>8.3	<0.00149	
(4203.1 *13*)	1763.1	0.256 *13*	7.0×10^{-4} *4*	5.47 *2*	0.257 *13*	
(4747.0 *13*)	1219.2	1.22 *4*	0.00208 *7*	5.10 *2*	1.22 *4*	
(5966.2 *16*)	0	98.17 *10*	0.07309 *8*	3.759 *1*	98.24 *10*	I(ε+β^+): deduced from 100-Σ%I(ε+β^+)(excited states), where Σ%I(ε+β^+)(excited states)=1.76 *5*. Level scheme fed by β^+ decay may be complete considering the highest observed level is at 4.6 MeV. Observed Σ%I(ε)≈0.08. Evaluators assume Σ%I(ε)≤0.08 for all unobserved ε feedings to high-lying levels and include this limit in the uncertainty for Σ%I(ε+β^+)(excited states)=1.76 *10* and I(ε+β^+)(gs)=98.24 *10*.

[†] Deduced from γ-ray intensity balance, relative to Iγ(1219γ)=100, normalized to absolute %Iγ(1219γ)=1.26 *5*.
[‡] Absolute intensity per 100 decays.
[#] Existence of this branch is questionable.

γ(^{35}Cl)

Iγ normalization: absolute %Iγ(1219γ)=1.243 *30* deduced from weighted average of 1.223 *46* (1969Wi18,1969Fr09), 1.22 *20* (1971Ge04), 1.55 *15* (1977Az01), 1.34 *8* (1979Ha04), 1.23 *7* (1984Ad01), 1.228 *30* (1980Wi13).

E_γ[†]	I_γ[‡&]	E_i(level)	J_i^π	E_f	J_f^π	Mult.[#]	δ[#]	α[a]	Comments
916[b]		3918.3	3/2$^+$	3002.2	5/2$^+$				The intensity could not be determined due to being obscured by another peak (1980Wi13).
930.7 *2*	1.57 *20*	2693.9	3/2$^+$	1763.1	5/2$^+$	M1+E2	+0.09 *3*	4.69×10^{-5} *7*	%Iγ=0.0195 *25* I_γ: weighted average of 1.13 *26* (1980Wi13) and 1.66 *12* (1984Ad01).
965[b]	<0.87[@]	3967.9	1/2$^+$	3002.2	5/2$^+$				%Iγ<0.0111
1219.3 *2*	100.0 *15*	1219.2	1/2$^+$	0	3/2$^+$	M1+E2	0.093 *10*	3.66×10^{-5} *5*	%Iγ=1.24 *4* I_γ: others: 100.0 *30* from 1984Ad01, 100 *10* from 1977Az01, 100 from 1969Wi18, 1971Ge04, 1979Ha04, and 1985Da04.
1225[b]		3918.3	3/2$^+$	2693.9	3/2$^+$				The intensity could not be determined due to being obscured by another peak (1980Wi13).
1239[b]	<0.30[@]	3002.2	5/2$^+$	1763.1	5/2$^+$	[M1,E2]		4.4×10^{-5} *6*	%Iγ<0.00382
1274[b]	<0.42[@]	3967.9	1/2$^+$	2693.9	3/2$^+$	[M1,E2]		4.9×10^{-5} *7*	%Iγ<0.00535
1474.8 *3*	1.02 *11*	2693.9	3/2$^+$	1219.2	1/2$^+$	M1+E2	−0.63 *21*	8.4×10^{-5} *4*	%Iγ=0.0127 *14* I_γ: weighted average of 0.83 *13* (1980Wi13) and 1.08 *7* (1984Ad01).
1763.0 *2*	22.7 *4*	1763.1	5/2$^+$	0	3/2$^+$	M1+E2	−2.85 *10*	0.0002124 *30*	%Iγ=0.282 *9* I_γ: weighted average of 23.4 *13* (1969Wi18), 20.0 *27* (1971Ge04), 22 *6* (1977Az01), 23.08 *37* (1980Wi13), 21.85 *67* (1984Ad01), and 21.7 *9* (1985Da04).
1783[b]	<0.39[@]	3002.2	5/2$^+$	1219.2	1/2$^+$	[E2]		0.0002267 *32*	%Iγ<0.00497
1931[b]	<0.21[@]	4624.5?	3/2,5/2	2693.9	3/2$^+$				%Iγ<0.00267
2155.1 *15*	0.15 *6*	3918.3	3/2$^+$	1763.1	5/2$^+$	[M1,E2]		0.00036 *4*	%Iγ=0.0019 *8* I_γ: from 1984Ad01. Other: <0.22 (1980Wi13).
2204[b]	<0.27[@]	3967.9	1/2$^+$	1763.1	5/2$^+$	[E2]		0.000425 *6*	%Iγ<0.00344
2645[b]	<0.12[@]	2645.1	7/2$^+$	0	3/2$^+$	E2		0.000633 *9*	%Iγ<0.00153
2693.7 *2*	10.96 *18*	2693.9	3/2$^+$	0	3/2$^+$	M1+E2	+0.26 *2*	0.000553 *8*	%Iγ=0.136 *4* I_γ: weighted average of 10.2 *15* (1971Ge04), 10.96 *18* (1980Wi13) and 11.02 *34* (1984Ad01).
2699[b]	<0.33[@]	3918.3	3/2$^+$	1219.2	1/2$^+$	[M1,E2]		0.00060 *6*	%Iγ<0.00420
2748.5 *6*	0.477 *34*	3967.9	1/2$^+$	1219.2	1/2$^+$	[M1]		0.000569 *8*	%Iγ=0.0059 *5* I_γ: weighted average of 0.48 *6* (1980Wi13) and 0.476 *34* (1984Ad01).

^{35}Ar $\varepsilon+\beta^+$ decay (1.7755 s) 1980Wi13,1984Ad01 (continued)

$\gamma(^{35}Cl)$ (continued)

E_γ[†]	I_γ[‡&]	E_i(level)	J^π_i	E_f	J^π_f	Mult.[#]	δ[#]	α[a]	Comments
2861[b]	<0.099[@]	4624.5?	3/2,5/2	1763.1	$5/2^+$				%Iγ<0.00126
3002.1 *4*	7.29 *14*	3002.2	$5/2^+$	0	$3/2^+$	M1+E2	+0.08 *2*	0.000671 *9*	%Iγ=0.0906 *28* I_γ: weighted average of 9.0 *15* (1971Ge04), 7.24 *14* (1980Wi13), 7.45 *23* (1984Ad01) and 6.9 *5* (1985Da04).
3405[b]		4624.5?	3/2,5/2	1219.2	$1/2^+$				I_γ: could not be determined due to being obscured by another peak (1980Wi13).
3918	0.52 *3*	3918.3	$3/2^+$	0	$3/2^+$	M1+E2		0.00108 *7*	%Iγ=0.0065 *4*
3968	0.106 *25*	3967.9	$1/2^+$	0	$3/2^+$	[M1,E2]		0.00109 *7*	%Iγ=0.00132 *31*
4624[b]	<0.035[@]	4624.5?	3/2,5/2	0	$3/2^+$	D+Q			%Iγ<4.46×10^{-4}

† From 1984Ad01, unless otherwise noted.

‡ from 1980Wi13, unless otherwise noted.

From Adopted Gammas.

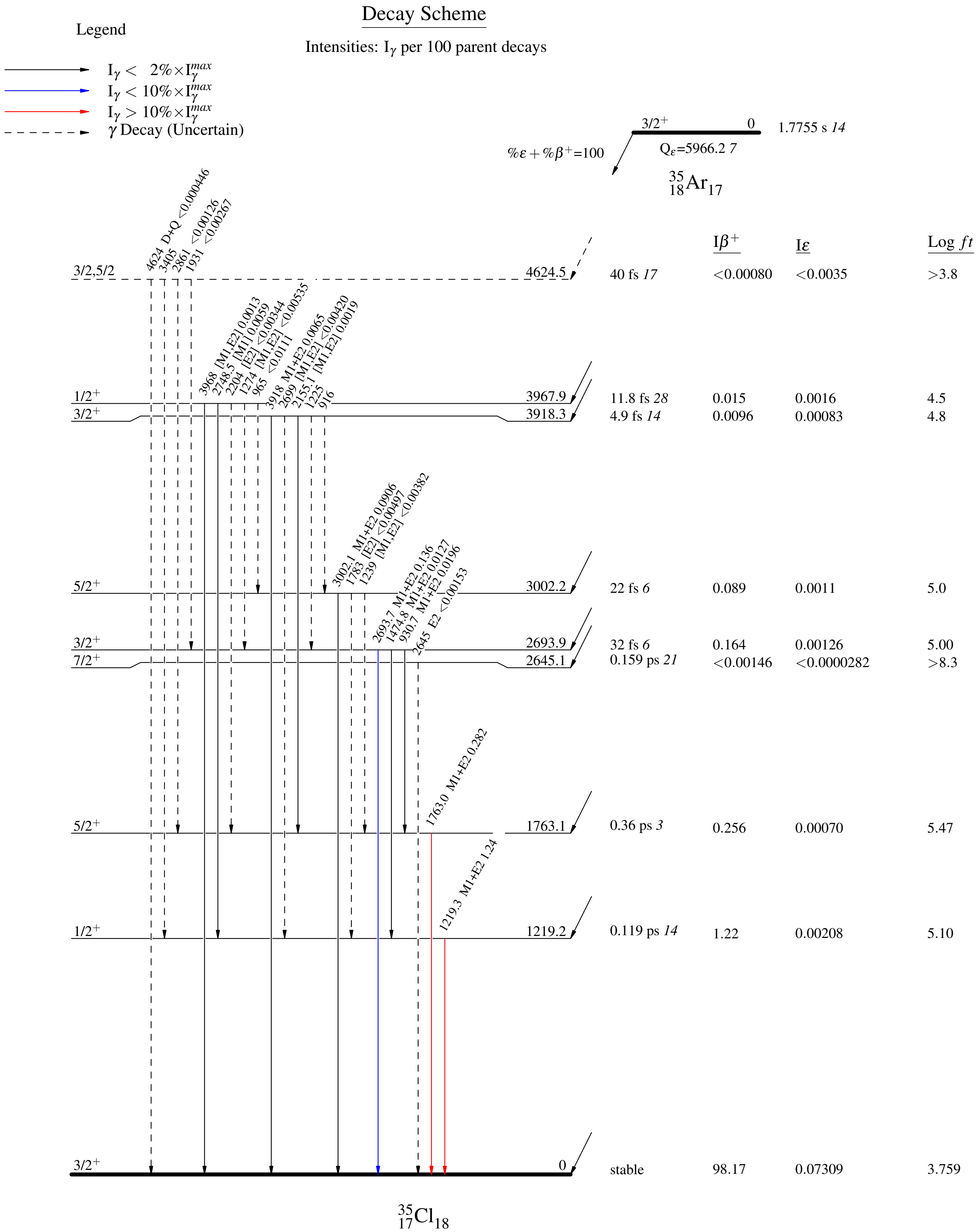
@ 1980Wi13 gives upper limits of Iγ at 98% confidence level. These peaks are not observed in the spectra of 1980Wi13 or in 1984Ad01.

& For absolute intensity per 100 decays, multiply by 0.01243 *30*.

a Total theoretical internal conversion coefficients, calculated using the BrIcc code (2008Ki07) with “Frozen Orbitals” approximation based on γ-ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

b Placement of transition in the level scheme is uncertain.

^{35}Ar ε+β^+ decay (1.7755 s) 1980Wi13,1984Ad01



^{36}K εp decay (341 ms) 1996Il02,1980Es01

Parent: ^{36}K: E=0; J$^{\pi}$=2^{+}; T$_{1/2}$=341 ms *3*; Q(εp)=4307.4 *3*; %εp decay=5.1×10^{-2}

^{36}K-J$^{\pi}$,T$_{1/2}$: From the Adopted Levels of ^{36}K (2012Ni01).

^{36}K-Q(εp): From 2021Wa16.

^{36}K-%εp decay: %ε+β^{+}p=5.1×10^{-4} is scaled by evaluators from %ε+β^{+}p=4.9×10^{-4} reported in 1996Il02. %ε+β^{+}p=4.9×10^{-4} in 1996Il02 was normalized based on the strongest 970-keV ε+β^{+}-delayed proton I$_p$=3.3×10^{-4} *9* measured by 1980Es01. 1980Es01 reported %ε+β^{+}p=4.8×10^{-4} and their I$_p$ was normalized based on the strongest 1970-keV ε+β^{+}-delayed γ ray %Iγ=79 *8* measured by 1972Mi13. Evaluators adopt a more precise %Iγ(1970)=81.8 *16* from the evaluated ^{36}K ε decay dataset (2012Ni01,1976Fr03) as the new normalization reference, and therefore, all original I$_p$ values from 1996Il02 and 1980Es01 are scaled by 1.04 *11*.

1996Il02,1997Il03: a 500-MeV proton beam was produced from the TRIUMF cyclotron and bombarded a CaO powder target. The spallation reaction products diffused out of the target and were selectively ionized at a Re surface. A=36 singly charged ions were selected by a magnetic mass analyzer and implanted into a carbon foil at 7.0×10^{5} pps. Protons and α particles were detected using a 30-μm silicon surface barrier detector with FWHM=23 keV. ^{32}S and ^{35}Cl recoils were detected using an MCP in a back-to-back geometry for timing stop signals. Measured E$_p$(>0.5 MeV) and I$_p$ for 15 proton peaks, E$_{\alpha}$(>1.5 MeV) and I$_{\alpha}$ for 19 α peaks, p(recoil)-coin, and α(recoil)-coin. Deduced levels, J, π, decay branching ratios, log *ft*, partial decay widths for ^{36}Ar, and ratios of ^{35}Cl(p,α)^{32}S to ^{35}Cl(p,γ)^{36}Ar reaction rates.

1980Es01: a 20-MeV proton beam was produced from the University of Jyvaskyla MC-20 cyclotron and bombarded a stack of four ^{36}Ar-implanted Al foils. The ^{36}Ar(p,n) reaction products were thermalized by helium, transported with NaCl-loaded helium, and deposited onto a fast-transport Al-coated mylar tape in a measurement chamber. Protons and α particles were detected using 14 and 26-μm fully depleted (FWHM=40 keV) and 100-μm partially depleted Si(Au) surface barrier detectors. A polyethene absorber foil was placed in front of Si(Au) to cause an energy shift for distinguishing p/α. Measured E$_p$(>0.5 MeV) and I$_p$ for 11 proton peaks, E$_{\alpha}$(>2.0 MeV) and I$_{\alpha}$ for 12 α peaks. Deduced levels, J, π, decay branching ratios, log *ft*, and partial decay widths for ^{36}Ar.

1980Ew01: a 600-MeV proton beam was produced from the synchrocyclotron at CERN-ISOLDE and bombarded a ScC$_2$ target. The ^{45}Sc(p,7n3p) spallation reaction products diffused out of the target and reached a tungsten surface ionization source where potassium isotopes were selectively ionized. The beam was extracted from the ion source, separated by the ISOLDE analyzing magnet, and collected by a carbon foil for proton measurements. Protons and α particles were detected using 20-μm and 100-μm silicon surface barrier detectors with FWHM=35 and 23 keV. Measured E$_p$(>0.5 MeV) and I$_p$ for 3 proton peaks, E$_{\alpha}$ and I$_{\alpha}$ for 4 α peaks. Deduced levels, J, π, and decay branching ratios for ^{36}Ar.

^{35}Cl Levels

E(level)	J$^{\pi}$†
0	3/2^{+}

† From the Adopted Levels.

Delayed Protons (^{35}Cl)

E(p)†	E(^{35}Cl)	I(p)†@	E(^{36}Ar)#	Comments
501.8 *11*	0	1.14×10^{-6} *33*	9023	E(p): other: 501 *10* (1980Es01). I(p): other: 3.6×10^{-6} *13* (1980Es01).
622.7 *14*	0	1.6×10^{-6} *5*	9147	
693.5 *8*	0	7.9×10^{-5} *22*	9220	E(p): other: 693 *5* (1980Es01), 691 *8* (1980Ew01). I(p): other: 7.8×10^{-5} *24* (1980Es01).
851.7 *10*	0	2.3×10^{-5} *7*	9383	E(p): other: 849 *5* (1980Es01). I(p): other: 2.0×10^{-5} *7* (1980Es01).
969.6 *12*	0	3.4×10^{-4} *10*	9504	E(p): other: 970 *5* (1980Es01), 959 *7* (1980Ew01). I(p): from 1980Es01 and also adopted as the normalization reference by 1996Il02.
1168.5 *14*	0	6.2×10^{-7} *7*	9709	

Continued on next page (footnotes at end of table)

^{36}K εp decay (341 ms) 1996Il02,1980Es01 (continued)

Delayed Protons (continued)

E(p)†	E(^{35}Cl)	I(p)†@	E(^{36}Ar)#	Comments
1198.5 *11*	0	2.5×10^{-6} *7*	9740	
1272.0 *20*	0	1.6×10^{-6} *5*	9815	
1334.2 *7*	0	4.5×10^{-5} *12*	9879	E(p): other: 1333 *5* (1980Es01), 1325 *7* (1980Ew01). I(p): other: 3.6×10^{-5} *11* (1980Es01).
1409.3 *22*	0	1.04×10^{-6} *33*	9957	
1874‡ *10*	0	1.1×10^{-6}‡ *5*	10435	
1992‡ *10*	0	5.0×10^{-6}‡ *20*	10556	
2048‡ *10*	0	4.9×10^{-6}‡ *19*	10613	
2458‡ *10*	0	3.0×10^{-6}‡ *13*	11035	
2640‡ *10*	0	2.1×10^{-6}‡ *10*	11222	

† From 1996Il02, unless otherwise noted. E(p) is in lab frame.

‡ From 1980Es01 and also adopted by 1996Il02.

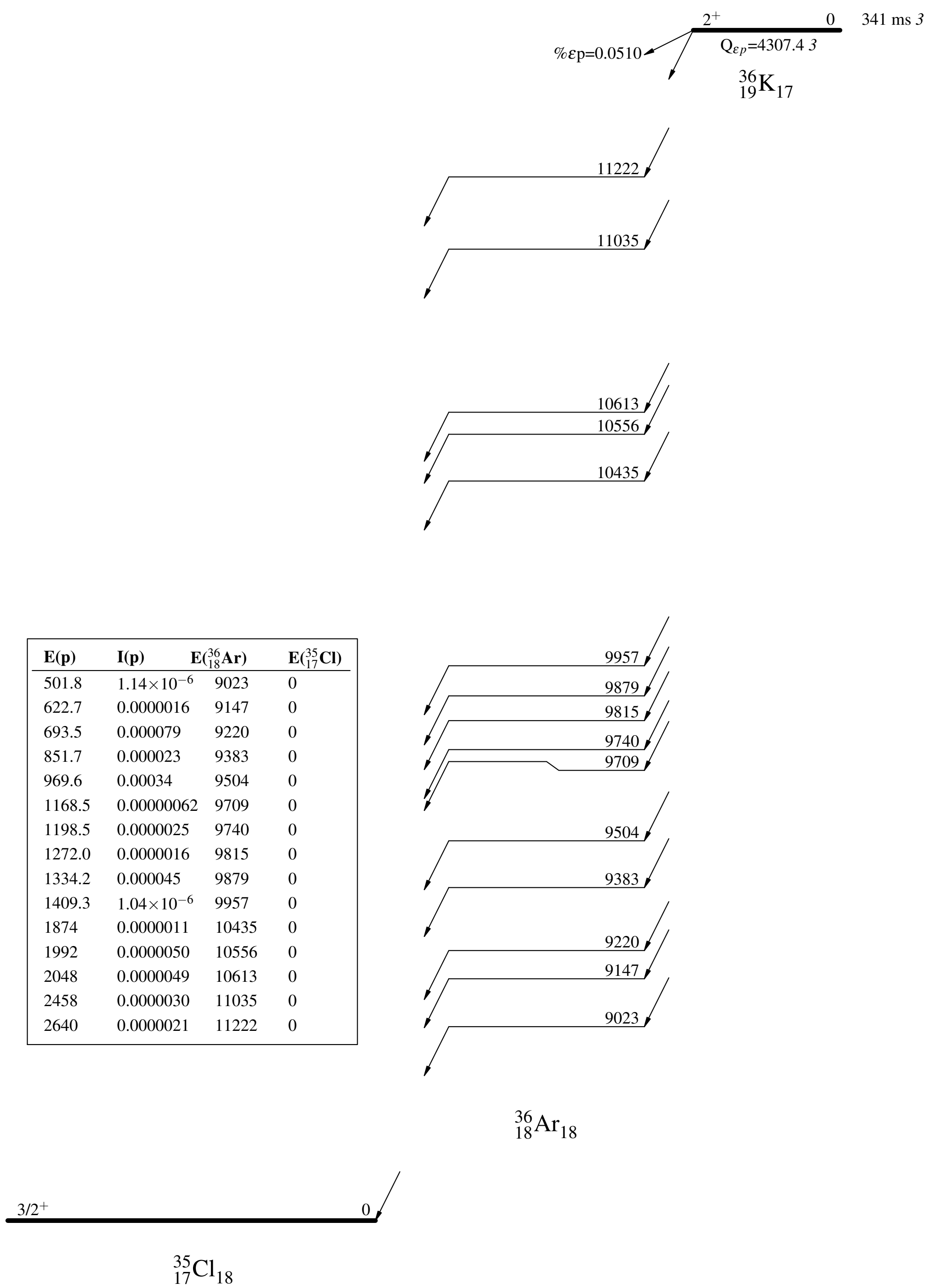
\# E(level)(^{36}Ar)=E(p)$_{lab}$×[m(p)+m(^{35}Cl)]/m(^{35}Cl)+S(p)(^{36}Ar)+E(level)(^{35}Cl), where S(p)(^{36}Ar)=8506.97 *4* (2021Wa16). Proton decay is assumed to populate only the ground state of the daughter nucleus ^{35}Cl based on the arguments proposed in 1996Il02: (A) unreasonably small log *ft* values if populating excited states; (B) no pairs of particle groups characteristic of the g.s./first excited state energy difference; (C) compatible level matching with resonance reaction studies.

@ Absolute intensity per 100 decays.

^{36}K εp decay (341 ms) 1996Il02,1980Es01

Decay Scheme

I(p) Intensities: I(p) per 100 parent decays



E(p)	I(p)	E($^{36}_{18}$Ar)	E($^{35}_{17}$Cl)
501.8	1.14×10^{-6}	9023	0
622.7	0.0000016	9147	0
693.5	0.000079	9220	0
851.7	0.000023	9383	0
969.6	0.00034	9504	0
1168.5	0.00000062	9709	0
1198.5	0.0000025	9740	0
1272.0	0.0000016	9815	0
1334.2	0.000045	9879	0
1409.3	1.04×10^{-6}	9957	0
1874	0.0000011	10435	0
1992	0.0000050	10556	0
2048	0.0000049	10613	0
2458	0.0000030	11035	0
2640	0.0000021	11222	0

^{1}H(^{34}S,^{35}Clγ) 2021Lo09

^{34}S(p,γ)^{35}Cl in inverse kinematics.

2021Lo09: ^{34}S was produced by the off-line ion source (OLIS) and accelerated by the ISAC accelerator at TRIUMF. $10^{10}-10^{11}$ pps $^{34}S^{+7}$ ions were delivered to a windowless 2-10 Torr hydrogen gas target with an effective length of 12.3 cm. Elastically scattered protons were detected by two silicon surface barrier detectors inside the gas target. Prompt γ rays from the compound recoil nuclei were detected by a BGO array in coincidence with the heavy-ion recoils separated by the DRAGON separator and identified by two MCP detectors, a DSSD, and an ionization chamber at the focal plane. Deduced resonance strengths ($\omega\gamma$) for 8 resonances and the ^{34}S(p,γ)^{35}Cl astrophysical reaction rate.

^{35}Cl Levels

Values of resonance strength $\omega\gamma$ under comments are from direct measurements in 2021Lo09.

E(level)†	Comments
6643	E_R=272 keV. $\omega\gamma$=1.22×10^{-5} eV *36*.
6674	E_R=303 keV. $\omega\gamma \le 7.2\times10^{-6}$ eV.
6761	E_R=390 keV. $\omega\gamma \le 6.9\times10^{-6}$ eV.
6778	E_R=407 keV. $\omega\gamma$=8.0×10^{-4} eV *19*.
6802	E_R=431 keV. $\omega\gamma \le 3.9\times10^{-5}$ eV.
6823	E_R=452 keV. $\omega\gamma \le 3.1\times10^{-5}$ eV.
6842	E_R=471 keV. $\omega\gamma$=9.4×10^{-5} eV *29*.
6866	E_R=496 keV. $\omega\gamma$=1.37×10^{-2} eV *22*.

† Excitation energies are deduced by evaluators from resonance energies $E_R+S_p(^{35}Cl)$, where $S_p(^{35}Cl)$=6370.81 *4* (2021Wa16).

$^{2}H(^{34}S,n\gamma)$ 1973Wa10

$^{34}S(d,n)^{35}Cl$ in inverse kinematics.

1973Wa10: A 59.6-MeV ^{34}S beam was produced from the BNL MP-tandem Van de Graaff facility. Targets were 200-μg/cm^2 TiD prepared by evaporating titanium onto the Cu, Al, and Mg target backings in a deuterium atmosphere. γ rays were detected using a 35-cm^3 Ge(Li) detector at 0° with FWHM=2 keV at 656 keV. Measured Eγ. Deduced $T_{1/2}$ for two ^{35}Cl levels using Doppler Shift Attenuation lineshape analysis.

^{35}Cl Levels

E(level)†	J^{π}‡	$T_{1/2}$#	Comments
0	3/2$^+$		
1219	1/2$^+$	201 fs *28*	$T_{1/2}$: Lifetime=0.29 ps *4* from 1973Wa10.
1762	5/2$^+$	374 fs *49*	$T_{1/2}$: Lifetime=0.54 ps *7* from 1973Wa10.

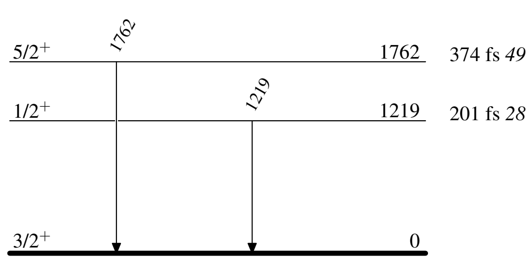
† From 1973Wa10.
‡ From the Adopted Levels.
# From Doppler Shift Attenuation Method (DSAM) in 1973Wa10.

$\gamma(^{35}Cl)$

E_{γ}	E_i(level)	J^{π}_i	E_f	J^{π}_f
1219	1219	1/2$^+$	0	3/2$^+$
1762	1762	5/2$^+$	0	3/2$^+$

$^{2}H(^{34}S,n\gamma)$ 1973Wa10

Level Scheme



$^{35}_{17}Cl_{18}$

$^{12}C(^{28}Si,\alpha p\gamma)$ 2007Ks01,2013Bi10,2020Ra14

2007Ks01: two different experiments: one with E=70 MeV ^{28}Si beam at BARC-TIFR facility and the second with E=88 MeV ^{28}Si beam at NSC facility, both using a target of about 50 μg/cm^2 ^{12}C on a ≈10.5−mg/cm^2 gold backing. γ rays were detected with almost identical (INGA) arrays of eight Compton-suppressed Clover HPGe detectors. Measured Eγ, Iγ, $\gamma\gamma$-coin, $\gamma\gamma(\theta)$(DCO), $\gamma\gamma$(lin pol), Doppler-shift attenuations. Deduced levels, J, π, $T_{1/2}$, γ-ray multipolarities and mixing ratios. Comparisons with available data and large-scale shell-model calculations.

2013Bi10: E=110 MeV ^{28}Si beam was produced from the IUAC-New Delhi accelerator facility. Target was 50 μg/cm^2 thick ^{12}C on 18 mg/cm^2 Au backing. γ rays were detected with the INGA array of thirteen Compton-suppressed Clover detectors. Measured Eγ, Iγ, $\gamma\gamma$-coin, $\gamma\gamma(\theta)$(DCO), Doppler-shift attenuations. Deduced levels, J, π, $T_{1/2}$, evidence for an SD band, γ-ray multipolarities. Comparison with large-scale shell-model calculations, two-level mixing calculation, and with structure of SD states in ^{36}Ar.

2020Ra14: E=110 MeV ^{28}Si beam was produced from the 15UD Pelletron accelerator at the Inter-University Accelerator Centre (IUAC), New Delhi. Target was 50 μg/cm^2 ^{12}C evaporated onto 18 mg/cm^2 Au backing. γ rays were detected with a multi-detector array (Indian National Gamma Array) consisting of 13 Compton-suppressed clovers. Measured Eγ, Iγ, $\gamma\gamma$-coin, $\gamma\gamma$(DCO), $\gamma\gamma$(ADO), $\gamma\gamma$(IPDCO). Deduced levels, J, π, γ-ray multipolarities. Comparisons with theoretical calculations.

Other reactions:

1982Le04: $^{28}Si(^{12}C,X\gamma)$ ^{12}C (E_{cm}=23-35 MeV) beam produced from the University of Washington FN Tandem Van de Graaff. A target of 100 μg/cm^2 ^{28}Si on a thick tantalum backing. A Ge(Li) detector. Measured σ(E), Eγ.

1985Kr21: $^{27}Al(^{12}C,X)$ E=85 MeV/nucleon. Measured particle-particle angular correlations.

^{35}Cl Levels

E(level)[†]	J^π[‡]	$T_{1/2}$[#]	Comments
0.0	3/2$^+$		
1763.0 *6*	5/2$^+$		
2645.8 *6*	7/2$^+$	<0.90 ps	
3003.1 *8*	5/2$^+$		
3163.1 *6*	7/2$^-$		
3942.0 *7*	9/2$^+$	<0.35 ps	
4347.8 *6*	9/2$^-$	0.91 ps *+21−16*	
5407.4 *8*	11/2$^-$	<0.76 ps	
5927.1 *7*	(11/2)$^-$	<0.28 ps	
6087.6 *8*	13/2$^-$		
6119.5? *13*	(11/2$^-$)		
6660.2 *8*	(11/2$^-$)		
7873.3 *8*	13/2$^+$		
8319.4[a] *9*	15/2$^-$	12.5[&] fs *7*	Tentative 8324, (13/2$^-$) level in 2007Ks01 is considered the same level here by the evaluators. $T_{1/2}$: other: <70 fs (2007Ks01). Q(transition)=0.65 *2*, β_2=0.30 (2013Bi10).
8487.8 *10*	15/2$^-$	<0.07 ps	Tentative 8481, (15/2$^-$) level in 2007Ks01 is considered the same level here by the evaluators.
8558.2? *13*	(13/2$^-$)	<0.14 ps	
8787.3? *13*	(15/2$^+$)	<0.28 ps	
8845.1[@] *9*	17/2$^+$		
10181.1[a] *10*	19/2$^-$	42.3[&] fs *28*	Tentative 10181, (19/2$^+$) level in 2007Ks01 is considered the same level here by the evaluators. $T_{1/2}$: other: <0.14 ps (2007Ks01). Q(transition)=0.82 *6*, β_2=0.37 (2013Bi10).
10223.2? *14*	(17/2$^-$)	<0.28 ps	
10862.2? *14*	19/2$^+$	<0.28 ps	
11459.2[@] *12*	21/2$^+$	43.0[&] fs *21*	Q(transition)=0.56 *2*, β_2=0.27 (2013Bi10).
12572.2[a] *12*	23/2$^-$	33[&] fs *4*	Q(transition)=0.71 *5*, β_2=0.32 (2013Bi10).
16306.4[a] *16*	(27/2$^-$)	<9.0[&] fs	Q(transition)>0.5, β_2=0.24 (2013Bi10).

Continued on next page (footnotes at end of table)

$^{12}C(^{28}Si,\alpha p\gamma)$ **2007Ks01,2013Bi10,2020Ra14 (continued)**

^{35}Cl Levels (continued)

† From a least-squares fit to γ-ray energies, assuming ΔEγ=1 keV in the fitting procedure.
‡ From the Adopted Levels, unless otherwise noted. Assignments for tentative levels not adopted in Adopted Levels are proposed by 2007Ks01.
# From DSAM in 2007Ks01, unless otherwise noted.
@ Existence of $17/2^+$ and $21/2^+$ levels provides indirect evidence in favor of SD band interpretation and parity-doublet cluster structure (2013Bi10).
& From DSAM in 2013Bi10.
a Band(A): SD band. Proposed configuration=3p−3h, $(sd)^{16}(pf)^3$. Negative-parity partner of SD band in ^{36}Ar (2013Bi10).

$\gamma(^{35}Cl)$

Δ_{IPDCO} corresponds to integrated polarization asymmetry measurement at E=70 MeV at TIFR facility (2007Ks01). The values are from e-mail reply of Dec 1, 2006 from one of the authors (M. Saha Sarkar) of 2007Ks01 to the previous evaluator B. Singh.
Δ_{IPDCO} and DCO under comments are from 2007Ks01, unless otherwise noted. Positive value of Δ_{IPDCO} indicates electric character and negative value indicates magnetic character. Expected value of DCO is ≈1.0 for a stretched transition gated by a transition of same multipolarity, ≈2.0 for dipole when gated by a stretched quadrupole (DCO(Q)) and ≈0.5 for stretched quadrupole when gated by a stretched dipole (DCO(D)) (2007Ks01). Values deviating from above values indicate a mixed transition. DCO values are also available in 2020Ra14, but expected values are opposite to those in 2007Ks01 for stretched dipole and quadrupole transitions when gated by a transition of different multipolarity, that is DCO(Q)≈0.5 for stretched dipole and DCO(D)≈2.0 for stretched quadrupole (2020Ra14).
For ADO ratios under comments from 2020Ra14, expected values are 0.6 for pure dipole transitions and 1.6 for pure quadrupole transitions or ΔJ=0 transitions (2020Ra14).

E_γ†	I_γ‡	E_i(level)	J^π_i	E_f	J^π_f	Mult.#	δ#	Comments
160@	1.1 3	3163.1	$7/2^-$	3003.1	$5/2^+$			I_γ: other: 0.6 2 at E=70 MeV.
518	9.2 5	3163.1	$7/2^-$	2645.8	$7/2^+$	D		I_γ: other: 10.0 2 at E=70 MeV. DCO(Q)=0.8 1. DCO(D)=1.5 2, R_{ADO}=1.6 4 (2020Ra14).
526		8845.1	$17/2^+$	8319.4	$15/2^-$			
680	32 1	6087.6	$13/2^-$	5407.4	$11/2^-$	M1(+E2)	+0.1 1	I_γ: other: 8.6 2 at E=70 MeV. DCO(Q)=1.3 1 at E=70 MeV. Δ_{IPDCO}=−0.09 2.
712@&	<1	6119.5?	$(11/2^-)$	5407.4	$11/2^-$			
779@&	<1	3942.0	$9/2^+$	3163.1	$7/2^-$			
883	3.2 2	2645.8	$7/2^+$	1763.0	$5/2^+$			Iγ=4.6 2 at E=70 MeV. Δ_{IPDCO}=+0.01 5.
914&	2.4 5	8787.3?	$(15/2^+)$	7873.3	$13/2^+$	D(+Q)	−0.2 2	DCO(Q)=2.2 6 at E=88 MeV.
972	33 1	8845.1	$17/2^+$	7873.3	$13/2^+$	E2		I_γ: other: 2.0 1 at E=70 MeV. DCO(Q)=1.0 2 at E=70 MeV, 0.9 1 at E=88 MeV. Δ_{IPDCO}=+0.08 4. DCO(D)=1.8 2, DCO(Q)=0.9 1, R_{ADO}=1.6 2 (2020Ra14).
1060	7.3 7	5407.4	$11/2^-$	4347.8	$9/2^-$	M1+E2	+0.2 1	I_γ: other: 6.9 3 at E=70 MeV. DCO(D)=0.9 1. Δ_{IPDCO}=−0.10 4.
1113		12572.2	$23/2^-$	11459.2	$21/2^+$			
1185	25 1	4347.8	$9/2^-$	3163.1	$7/2^-$	M1+E2	−0.3 1	I_γ: other: 25 1 at E=70 MeV. DCO(Q)=2.0 2 at E=70 MeV, 2.8 3 at E=88 MeV. Δ_{IPDCO}=−0.08 3.

Continued on next page (footnotes at end of table)

$^{12}C(^{28}Si,\alpha p\gamma)$ 2007Ks01,2013Bi10,2020Ra14 (continued)

$\gamma(^{35}Cl)$ (continued)

E_γ †	I_γ ‡	E_i(level)	J_i^π	E_f	J_f^π	Mult.#	δ#	Comments
1213		7873.3	13/2$^+$	6660.2	(11/2$^-$)			
1296	0.4 *2*	3942.0	9/2$^+$	2645.8	7/2$^+$			I_γ: other: 0.2 *1* at E=70 MeV.
1336	6 *1*	10181.1	19/2$^-$	8845.1	17/2$^+$	D(+Q)	−0.2 *2*	DCO(Q)=2.0 *4* at E=88 MeV.
1378@&	2 *1*	10223.2?	(17/2$^-$)	8845.1	17/2$^+$			
1400	<1	3163.1	7/2$^-$	1763.0	5/2$^+$			
1579	18 *2*	5927.1	(11/2)$^-$	4347.8	9/2$^-$	(M1+E2)	−0.2 *1*	I_γ: other: 12 *1* at E=70 MeV. DCO(D)=0.9 *1* at E=70 MeV, 1.0 *1* at E=88 MeV. Δ_{IPDCO}=+0.02 *3*.
1693		10181.1	19/2$^-$	8487.8	15/2$^-$			
1702	11.4 *6*	4347.8	9/2$^-$	2645.8	7/2$^+$	E1+M2	+0.5 *+5−3*	I_γ: other: 10.8 *5* at E=70 MeV. DCO(Q)=1.0 2 from 2007Ks01 indicating ΔJ=2 or 0 seems inconsistent with ΔJ=1 from level scheme. Δ_{IPDCO}=+0.04 *4*. DCO(Q)=0.49 *6*, R_{ADO}=0.60 *6* (2020Ra14).
1740	2.0 *2*	6087.6	13/2$^-$	4347.8	9/2$^-$	E2		I_γ: other: 2.3 *2* at E=70 MeV. Δ_{IPDCO}=+0.12 *7*.
1763	>9.7	1763.0	5/2$^+$	0.0	3/2$^+$	E2+M1		I_γ: other: >15.7 at E=70 MeV. Δ_{IPDCO}=+0.13 *5*.
1786	1.8 *4*	7873.3	13/2$^+$	6087.6	13/2$^-$	D(+Q)	−0.6 *6*	I_γ: other: 0.1 *1* at E=70 MeV. DCO(Q)=1.2 *3* at E=88 MeV.
1862		10181.1	19/2$^-$	8319.4	15/2$^-$			
1946	10 *1*	7873.3	13/2$^+$	5927.1	(11/2)$^-$	E1+M2	+0.2 *1*	I_γ: other: 0.5 *1* at E=70 MeV. DCO(Q)=1.3 *2* at E=88 MeV. Δ_{IPDCO}=+0.11 *6*. DCO(Q)=0.45 *8*, R_{ADO}=0.59 *8* (2020Ra14).
1985@	2.0 *4*	5927.1	(11/2)$^-$	3942.0	9/2$^+$			I_γ: other: 1.6 *6* at E=70 MeV.
2017@&	7 *1*	10862.2?	19/2$^+$	8845.1	17/2$^+$			
2179	6.5 *7*	3942.0	9/2$^+$	1763.0	5/2$^+$	E2		I_γ: other: 11.1 *6* at E=70 MeV. Δ_{IPDCO}=+0.07 *5*.
2232		8319.4	15/2$^-$	6087.6	13/2$^-$			E_γ: from 2013Bi10 only.
2244	61 *2*	5407.4	11/2$^-$	3163.1	7/2$^-$	E2		I_γ: other: 50 *3* at E=70 MeV. DCO(Q)=1.0 *1* at E=70 MeV, 1.2 *1* at E=88 MeV. Δ_{IPDCO}=+0.04 *2*.
2312		6660.2	(11/2$^-$)	4347.8	9/2$^-$			
2391		12572.2	23/2$^-$	10181.1	19/2$^-$			
2400	<1	8487.8	15/2$^-$	6087.6	13/2$^-$	D(+Q)	+0.2 *+3−2*	E_γ: other: 2394 (2007Ks01). DCO(Q)=1.4 *4* at E=88 MeV.
2466	22 *2*	7873.3	13/2$^+$	5407.4	11/2$^-$	E1+M2	−0.25 *10*	I_γ: other: 1.2 *1* at E=70 MeV. DCO(Q)=2.4 *3*. Δ_{IPDCO}=+0.21 *6*. DCO(Q)=0.51 *8*, R_{ADO}=0.65 *9* (2020Ra14).
2614		11459.2	21/2$^+$	8845.1	17/2$^+$			
2631@&	<1	8558.2?	(13/2$^-$)	5927.1	(11/2)$^-$	D(+Q)	−0.2 *+2−3*	DCO(Q)=2.4 *7* at E=88 MeV.
2646	37 *2*	2645.8	7/2$^+$	0.0	3/2$^+$	E2		I_γ: other: 47 *2* at E=70 MeV. Δ_{IPDCO}=+0.08 *5*. DCO(D)=1.7 *2*, R_{ADO}=1.6 *2* (2020Ra14).
2757		8845.1	17/2$^+$	6087.6	13/2$^-$			
2764		5927.1	(11/2)$^-$	3163.1	7/2$^-$			E_γ: from 2013Bi10 only.

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$^{12}C(^{28}Si,\alpha p\gamma)$ **2007Ks01,2013Bi10,2020Ra14 (continued)**

$\gamma(^{35}Cl)$ (continued)

E_γ †	I_γ ‡	E_i(level)	J^π_i	E_f	J^π_f	Mult.#	δ#	Comments
2912	<1	8319.4	15/2−	5407.4	11/2−			E_γ: other: 2917 (2007Ks01, tentative). Mult.,δ: D(+Q) with δ=−0.1 +*3*−*2* if J(8319)=13/2 in 2007Ks01. DCO(Q)=1.9 *6* at E=88 MeV.
3003@	>1.1	3003.1	5/2+	0.0	3/2+			I_γ: other: >0.6 at E=70 MeV.
3080	<1	8487.8	15/2−	5407.4	11/2−			E_γ: other: 3074 (2007Ks01).
3163	100 *5*	3163.1	7/2−	0.0	3/2+	M2+E3	+0.16 *1*	I_γ: other: 100 *2* at E=70 MeV. DCO(Q)=0.9 *1* at E=70 MeV, 0.7 *1* at 88 MeV. Δ_{IPDCO}=−0.06 *2*.
3497		6660.2	(11/2−)	3163.1	7/2−			
3734		16306.4	(27/2−)	12572.2	23/2−			
4347@	<1	4347.8	9/2−	0.0	3/2+			I_γ: other: <1 at E=70 MeV.

† From 2013Bi10, unless otherwise noted.

‡ From 2007Ks01, measured at E(^{28}Si)=88 MeV at the NSC. Values are also available from E=70 MeV data of 2007Ks01 at TIFR facility, but less complete and are given under comments where available.

\# Q and D+Q with δ(Q/D) are from $\gamma\gamma(\theta)$(DCO) in 2007Ks01 and magnetic/electric characters are from $\gamma\gamma$(lin pol) in 2007Ks01 where available and/or RUL where level $T_{1/2}$ is measured, unless otherwise noted.

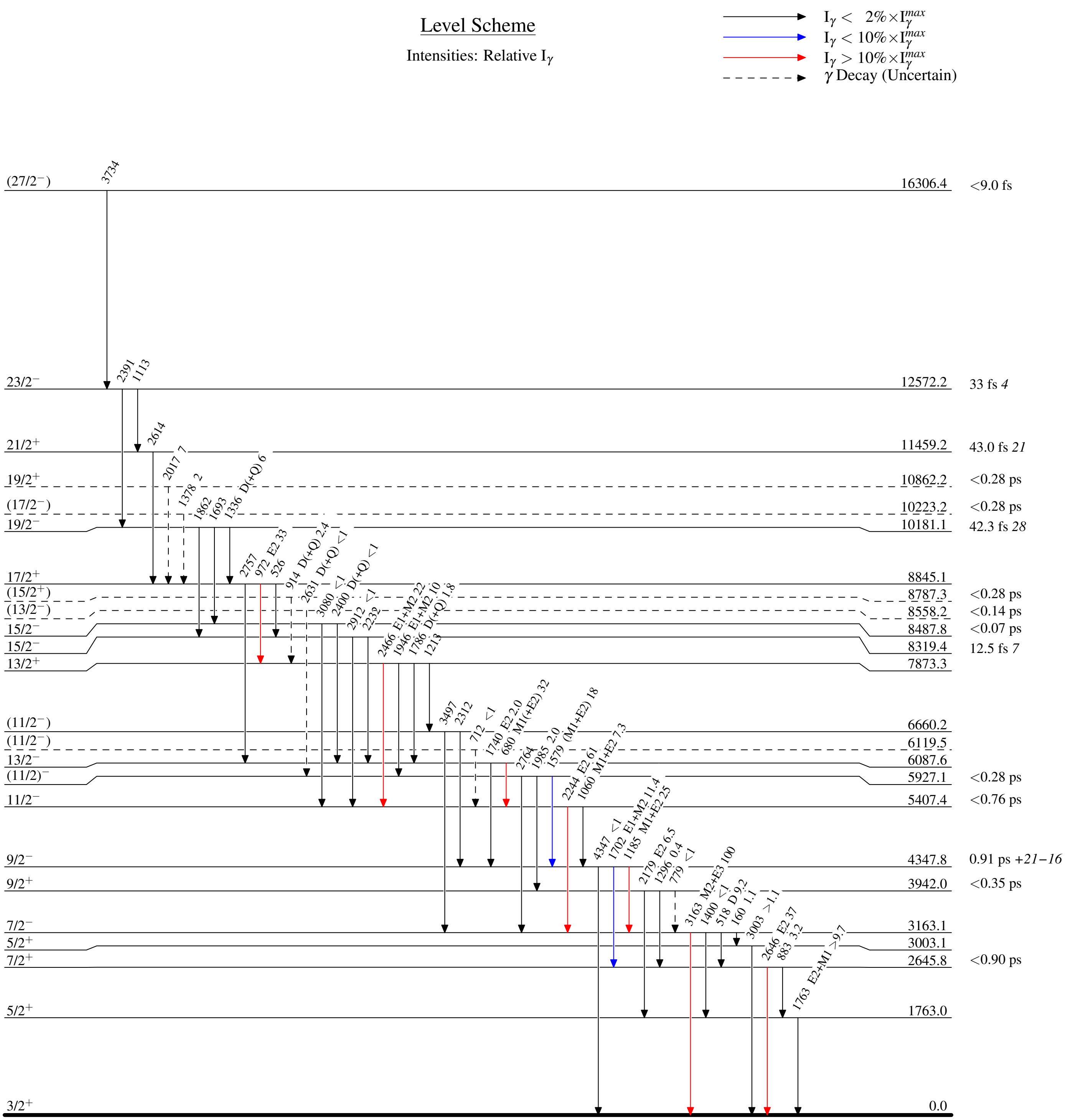
@ From 2007Ks01 only.

& Placement of transition in the level scheme is uncertain.

^{12}C(^{28}Si,αpγ) 2007Ks01,2013Bi10,2020Ra14

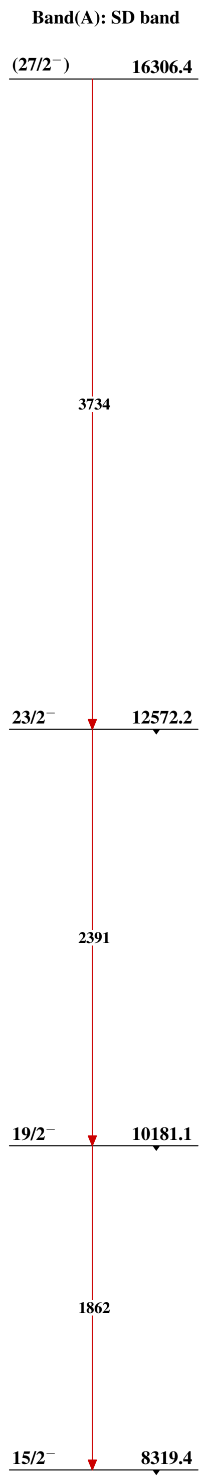
Level Scheme

Intensities: Relative I_γ



$^{35}_{17}Cl_{18}$

$^{12}C(^{28}Si,\alpha p\gamma)$ 2007Ks01,2013Bi10,2020Ra14



$^{35}_{17}Cl_{18}$

^{16}O(^{24}Mg,αpγ) 2004Ek01,2005Ek01

2004Ek01,2005Ek01: E=60 MeV ^{24}Mg beam was produced at the Legnaro National Laboratory. Target was a 0.5 mg/cm^2 enriched ^{40}Ca target with a 7 mg/cm^2 tantalum backing and with oxygen inside. γ rays were detected with the GASP array of Ge detectors and charged particles were detected with the ISIS array of 40 ΔE-E Si telescopes. Measured Eγ, Iγ, γγ-coin, γ(θ), γ(particle)-coin. Deduced levels, J, π from measured asymmetry ratios R(γ(θ)).

^{35}Cl Levels

E(level)†	Jπ‡
0.0	3/2+
1762.95 *17*	5/2+
2645.81 *18*	7/2+
3163.43 *19*	7/2−
4348.29 *24*	9/2−
5407.64 *27*	11/2−
6088.05 *29*	13/2−

† From a least-squares fit to γ-ray energies.
‡ As given in 2004Ek01 based on measured γ(θ) asymmetry and known assignments of low-lying levels. When considered in Adopted Levels, firm assignments here will be placed in parentheses if there are no strong supporting arguments.

γ(^{35}Cl)

R(γ(θ))=Asymmetry ratio measured at 35° and 81° with respect to the beam axis. Value of ≈1.2 for stretched quadrupole and ≈0.7 for stretched dipole (ΔJ=1).

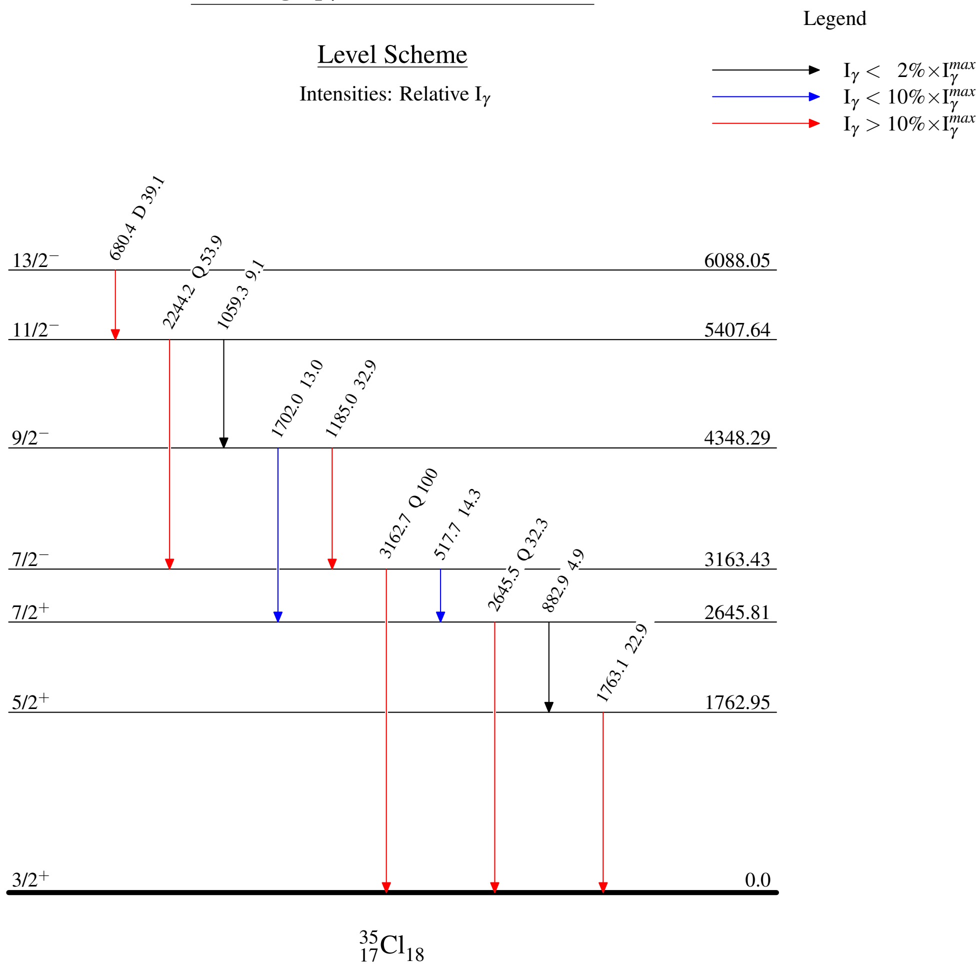
Eγ†	Iγ†	Ei(level)	J^π_i	E_f	J^π_f	Mult.‡	Comments
517.7 *1*	14.3 *8*	3163.43	7/2−	2645.81	7/2+		
680.4 *1*	39.1 *12*	6088.05	13/2−	5407.64	11/2−	D	R(γ(θ))=0.65 *3*.
882.9 *1*	4.9 *4*	2645.81	7/2+	1762.95	5/2+		
1059.3 *2*	9.1 *9*	5407.64	11/2−	4348.29	9/2−		
1185.0 *2*	32.9 *10*	4348.29	9/2−	3163.43	7/2−		
1702.0 *3*	13.0 *8*	4348.29	9/2−	2645.81	7/2+		
1763.1 *2*	22.9 *7*	1762.95	5/2+	0.0	3/2+		R(γ(θ))=1.43 *7* indicating Q seems inconsistent with ΔJ=1 from level scheme.
2244.2 *3*	53.9 *17*	5407.64	11/2−	3163.43	7/2−	Q	R(γ(θ))=1.24 *6*.
2645.5 *4*	32.3 *10*	2645.81	7/2+	0.0	3/2+	Q	R(γ(θ))=1.21 *6*.
3162.7 *4*	100 *3*	3163.43	7/2−	0.0	3/2+	Q	R(γ(θ))=1.43 *7*.

† From 2004Ek01.
‡ Deduced from γ(θ) asymmetry in 2004Ek01 by the evaluators.

$^{16}O(^{24}Mg,\alpha p\gamma)$ 2004Ek01,2005Ek01

Level Scheme

Intensities: Relative I_γ



^{24}Mg(^{16}O,αpγ) 2007De14,1976Va24

Also includes data from ^{25}Mg(^{16}O,αpnγ) in 1976Ke02 and ^{27}Al(^{16}O,2$\alpha\gamma$) in 1991Ja11.

2007De14: E=70 MeV ^{16}O beam was produced from the XTU-Tandem accelerator at Legnaro National Laboratory. Target was a 400 μg/cm^2 self-supporting target of ^{24}Mg. γ rays were detected with the GASP spectrometer of 40 Compton-suppressed HPGe detectors and a multiplicity filter of 80 BGO scintillators and charged particles were detected with the 4π array ISIS. Measured Eγ, Iγ, $\gamma\gamma$-coin, $\gamma\gamma(\theta)$(ADO), particle-γ-coin. Deduced levels, J, π, band structures, γ-ray multipolarities.

1976Va24: E=38 and 45 MeV ^{16}O beams were produced at Fysisch laboratorium in the Netherlands. Targets were 300 μg/cm^2 ^{24}Mg enriched to 99.94% evaporated onto 30 μm Au backings. γ rays were detected with a Ge(Li)-NaI(Tl) Compton-suppression spectrometer and a three-Ge(Li) Compton polarimeter. Measured Eγ, Iγ, $\gamma\gamma$-coin, $\gamma(\theta)$, γ(lin pol). Deduced levels, J, π, γ-ray multipolarities and mixing ratios.

1974Va13: E=36 MeV ^{16}O beam was produced at Fysisch laboratorium in the Netherlands. Targets were 100 and 300 μg/cm^2 Mg on 1 μm Ni backings. γ rays were detected with Ge(Li) detectors. Measured Eγ, Iγ, $\gamma\gamma$-coin, recoil distances. Deduced the half-life for the level at 3.16 MeV.

1982Ra25: E=40-51 MeV ^{16}O beams were produced at the Argonne National Laboratory. Target was 50 μg/cm^2 layer of ^{24}Mg (>99.9%) evaporated onto a 50 μg Au backing. γ rays were detected with a Ge(Li) detector. Measured γ-ray yields.

Other reactions:

1991Ja11: ^{27}Al(^{16}O,2$\alpha\gamma$) E=60 MeV ^{16}O beam was produced from the 14 UD-Pelletron at TIFR. Target was 350 μg/cm^2 Al evaporated onto a 1.8 mg/cm^2 thick Ta backing. γ rays were detected with Ge(Li) detector. Measured Eγ, recoil distances. Deduced $T_{1/2}$ for the levels of 3162 and 8844 keV.

1976Ke02: ^{25}Mg(^{16}O,αpnγ) E=43 and 50 MeV ^{16}O beams were produced at the Institut fur Kernphysik in Germany. The target was 150 μg/cm^2 enriched ^{25}Mg (99.2%) evaporated onto a Au backing. γ rays were detected with two true coaxial Ge(Li) detectors of 45 cm^3. Measured Eγ, Iγ, Doppler-shift recoil distances. Deduced $T_{1/2}$ for the level of 3162 keV.

1989De02: ^{27}Al(^{16}O,X) E=140 MeV ^{16}O from the Stony Brook LINAC facility. Measured particle-particle correlations.

1986Br26: ^{27}Al(^{16}O,X) E=65 MeV ^{16}O from the Pelletron Laboratory at the Universidade de Sao Paulo, Brazil. Measured σ.

1979Ta03: ^{27}Al(^{16}O,X) E=50, 60, and 70 MeV ^{16}O from the NBI super FN tandem. Measured particle-particle angular correlations.

2001Pi10: ^{12}C(^{32}S,X) E=19.5 MeV/nucleon ^{32}S beam from the superconducting cyclotron of the Laboratori Nazionali del Sud in Catania. Measured σ.

^{35}Cl Levels

Band assignments are from 2007De14.

E(level)[†]	J$^{\pi}$[‡]	$T_{1/2}$	Comments
0	3/2$^+$		
1219.0 *10*			
1763.3 *2*	5/2$^+$		
2645.9 *3*	7/2$^+$		
3001.9 *10*	5/2$^+$		
3162.9[#] *3*	7/2$^-$	31.0 ps *6*	$T_{1/2}$: weighted average of 29.1 ps *14* (1974Va13), 28.9 ps *12* (1976Ke02), and 31.4 ps *4* (1991Ja11), all by RDM.
3943.4 *10*	9/2$^+$		
4347.8 *3*	9/2$^-$		
5407.3[#] *3*	11/2$^-$		
5926.9 *5*	11/2$^-$		
6087.2 *4*	13/2$^-$		
6380.8 *8*			
7873.4[@] *4*	13/2$^+$		
8319.7[#] *5*	15/2$^-$		
8487.4 *5*	15/2$^-$		
8788.9[@] *6*	15/2$^+$		
8844.6[@] *5*	17/2$^+$	6.1 ps *11*	$T_{1/2}$: from RDM in 1991Ja11.

Continued on next page (footnotes at end of table)

^{24}Mg(^{16}O,αpγ) 2007De14,1976Va24 (continued)

^{35}Cl Levels (continued)

E(level)[†]	J^{π}[‡]
10181.1[#] *5*	19/2$^-$
10222.5 *11*	17/2$^-$
10859.1[@] *8*	19/2$^+$
11459.0[@] *7*	21/2$^+$
12572.2[#] *6*	23/2$^-$

[†] From a least-squares fit to γ-ray energies.
[‡] As proposed in 2007De14 based on measured $\gamma\gamma(\theta)$(ADO) and band assignments, unless otherwise noted. When considered in Adopted Levels, firm assignments here will be placed in parentheses if there are no strong supporting arguments.
[#] Band(A): Band based on 3163, 7/2$^-$ level. Configuration=$f_{7/2}$.
[@] Band(B): Band based on 7873, 13/2$^+$ level.

γ(^{35}Cl)

R_{ADO} values under comments are from $\gamma\gamma(\theta)$(ADO) in 2007De14. Expected values are ≈1.3 for ΔJ=2, quadrupole (and ΔJ=0, dipole in few cases) and ≈0.8 for ΔJ=1, dipole transitions (2007De14).

E_γ[†]	I_γ[†]	E_i(level)	J_i^π	E_f	J_f^π	Mult.[‡]	δ[#]	Comments
161[@] *1*		3162.9	7/2$^-$	3001.9	5/2$^+$			
296[@] *1*		6380.8		6087.2	13/2$^-$			
517.3 *5*	11.1 *13*	3162.9	7/2$^-$	2645.9	7/2$^+$			E_γ: weighted average of 517.4 *5* (2007De14) and 517 *1* (1976Va24).
680.5 *5*	33.0 *7*	6087.2	13/2$^-$	5407.3	11/2$^-$	D		R_{ADO}=0.84 *2*.
882.7 *5*	4.0 *1*	2645.9	7/2$^+$	1763.3	5/2$^+$	D		E_γ: weighted average of 882.9 *5* (2007De14) and 882 *1* (1976Va24). R_{ADO}=0.75 *3*.
915.4 *4*	7.2 *5*	8788.9	15/2$^+$	7873.4	13/2$^+$	D		R_{ADO}=0.69 *3*.
971[@] *1*		6380.8		5407.3	11/2$^-$			
971.4 *5*	75.3 *19*	8844.6	17/2$^+$	7873.4	13/2$^+$	Q		R_{ADO}=1.38 *3*.
1059.5 *2*	17.1 *4*	5407.3	11/2$^-$	4347.8	9/2$^-$	M1+E2	+0.12 *3*	E_γ: weighted average of 1059.6 *5* (2007De14) and 1059.5 *2* (1976Va24). I_γ: other: 7.7 *3* (1976Va24, E=38 MeV). R_{ADO}=1.05 *4*. A_2=−0.08 *6*, A_4=+0.049 *46*, POL=−0.11 *12* (1976Va24).
1113.3 *5*	4.1 *3*	12572.2	23/2$^-$	11459.0	21/2$^+$	D		R_{ADO}=0.86 *9*.
1184.9 *3*	44.4 *16*	4347.8	9/2$^-$	3162.9	7/2$^-$	M1+E2	−0.65 *30*	E_γ: weighted average of 1184.9 *5* (2007De14) and 1184.9 *3* (1976Va24). I_γ: other: 22.2 *3* (1976Va24, E=38 MeV). R_{ADO}=0.54 *2*, giving ΔJ=1. A_2=−0.663 *25*, A_4=+0.004 *25*, POL=+0.07 *7* (1976Va24).
1336.3 *5*	23.5 *8*	10181.1	19/2$^-$	8844.6	17/2$^+$	D		R_{ADO}=0.75 *2*.
1377.8 *10*	5.4 *4*	10222.5	17/2$^-$	8844.6	17/2$^+$			R_{ADO}=1.32 *12* also consistent with ΔJ=0.
1399.9 *7*	0.5 *2*	3162.9	7/2$^-$	1763.3	5/2$^+$			
1578.9 *5*	31.8 *13*	5926.9	11/2$^-$	4347.8	9/2$^-$	D		E_γ: other: 1579 *1* (1976Va24). R_{ADO}=0.48 *1*.
1693.5 *5*	19.8 *21*	10181.1	19/2$^-$	8487.4	15/2$^-$	Q		R_{ADO}=1.44 *7*.
1701.7 *5*	23.1 *6*	4347.8	9/2$^-$	2645.9	7/2$^+$	E1+M2	+0.5 *4*	E_γ: weighted average of 1701.6 *5*

Continued on next page (footnotes at end of table)

^{24}Mg(^{16}O,αpγ) 2007De14,1976Va24 (continued)

γ(^{35}Cl) (continued)

E_γ [†]	I_γ [†]	E_i(level)	J^π_i	E_f	J^π_f	Mult. [‡]	δ [#]	Comments
								(2007De14) and 1702 *1* (1976Va24). R_{ADO}=0.76 *6*, giving ΔJ=1.
1739.6 *5*	2.6 *3*	6087.2	13/2⁻	4347.8	9/2⁻	Q		E_γ: weighted average of 1739.5 *5* (2007De14) and 1740 *1* (1976Va24). R_{ADO}=1.38 *10*.
1763.3 *2*	21.2 *7*	1763.3	5/2⁺	0	3/2⁺	D+Q	−2.6 *4*	E_γ: weighted average of 1763.1 *5* (2007De14) and 1763.3 *2* (1976Va24). I_γ: other: 18.5 *3* (1976Va24, E=38 MeV). A_2=−0.368 *18*, A_4=+0.13 *2*, POL=+0.06 *20* (1976Va24). R_{ADO}=0.77 *3*, giving ΔJ=1.
1786.2 *5*	4.5 *2*	7873.4	13/2⁺	6087.2	13/2⁻			R_{ADO}=1.37 *7*, consistent with ΔJ=0 from level scheme.
1861.3 *5*	19.6 *13*	10181.1	19/2⁻	8319.7	15/2⁻	Q		R_{ADO}=1.39 *7*.
1946.2 *5*	22.4 *12*	7873.4	13/2⁺	5926.9	11/2⁻	D		R_{ADO}=0.80 *4*.
2014.7 *9*	23.1 *12*	10859.1	19/2⁺	8844.6	17/2⁺	D		R_{ADO}=0.50 *3*.
2069.8 *10*	7.5 *5*	10859.1	19/2⁺	8788.9	15/2⁺	Q		R_{ADO}=1.57 *11*.
2180 [@] *1*		3943.4	9/2⁺	1763.3	5/2⁺			
2232.7 *6*	10.3 *5*	8319.7	15/2⁻	6087.2	13/2⁻			
2244.3 *2*	89 *2*	5407.3	11/2⁻	3162.9	7/2⁻	E2(+M3)	0.000 *1*	E_γ: weighted average of 2243.9 *5* (2007De14) and 2244.3 *2* (1976Va24). I_γ: other: 30.3 *20* (1976Va24, E=38 MeV). R_{ADO}=1.26 *3*, giving ΔJ=2. A_2=+0.303 *20*, A_4=−0.119 *22*, POL=+0.26 *28* (1976Va24).
2390.8 *5*	16.8 *7*	12572.2	23/2⁻	10181.1	19/2⁻	Q		R_{ADO}=1.34 *12*.
2399.8 *8*	14.8 *5*	8487.4	15/2⁻	6087.2	13/2⁻			
2466.2 *5*	46.2 *9*	7873.4	13/2⁺	5407.3	11/2⁻	D		E_γ: other: 2466 *1* (1976Va24, unplaced). R_{ADO}=0.77 *2*.
2614.5 *5*	19.0 *8*	11459.0	21/2⁺	8844.6	17/2⁺	Q		R_{ADO}=1.33 *4*.
2645.8 *5*	36.1 *8*	2645.9	7/2⁺	0	3/2⁺	Q		E_γ: weighted average of 2645.7 *5* (2007De14) and 2646 *1* (1976Va24). R_{ADO}=1.35 *4*.
2911.9 *8*	18.0 *8*	8319.7	15/2⁻	5407.3	11/2⁻	Q		R_{ADO}=1.35 *5*.
3080.0 *7*	4.7 *3*	8487.4	15/2⁻	5407.3	11/2⁻	Q		R_{ADO}=1.20 *12*.
3162.6 *3*	100 *2*	3162.9	7/2⁻	0	3/2⁺	M2+E3	+0.18 *7*	E_γ: weighted average of 3163.1 *5* (2007De14) and 3162.4 *3* (1976Va24). I_γ: other: 100.0 *15* (1976Va24, E=38 MeV). R_{ADO}=1.30 *2*, giving ΔJ=2. A_2=+0.416 *6*, A_4=−0.001 *6*, POL=−0.57 *12* (1976Va24).
4347.8 *8*	5.5 *5*	4347.8	9/2⁻	0	3/2⁺			

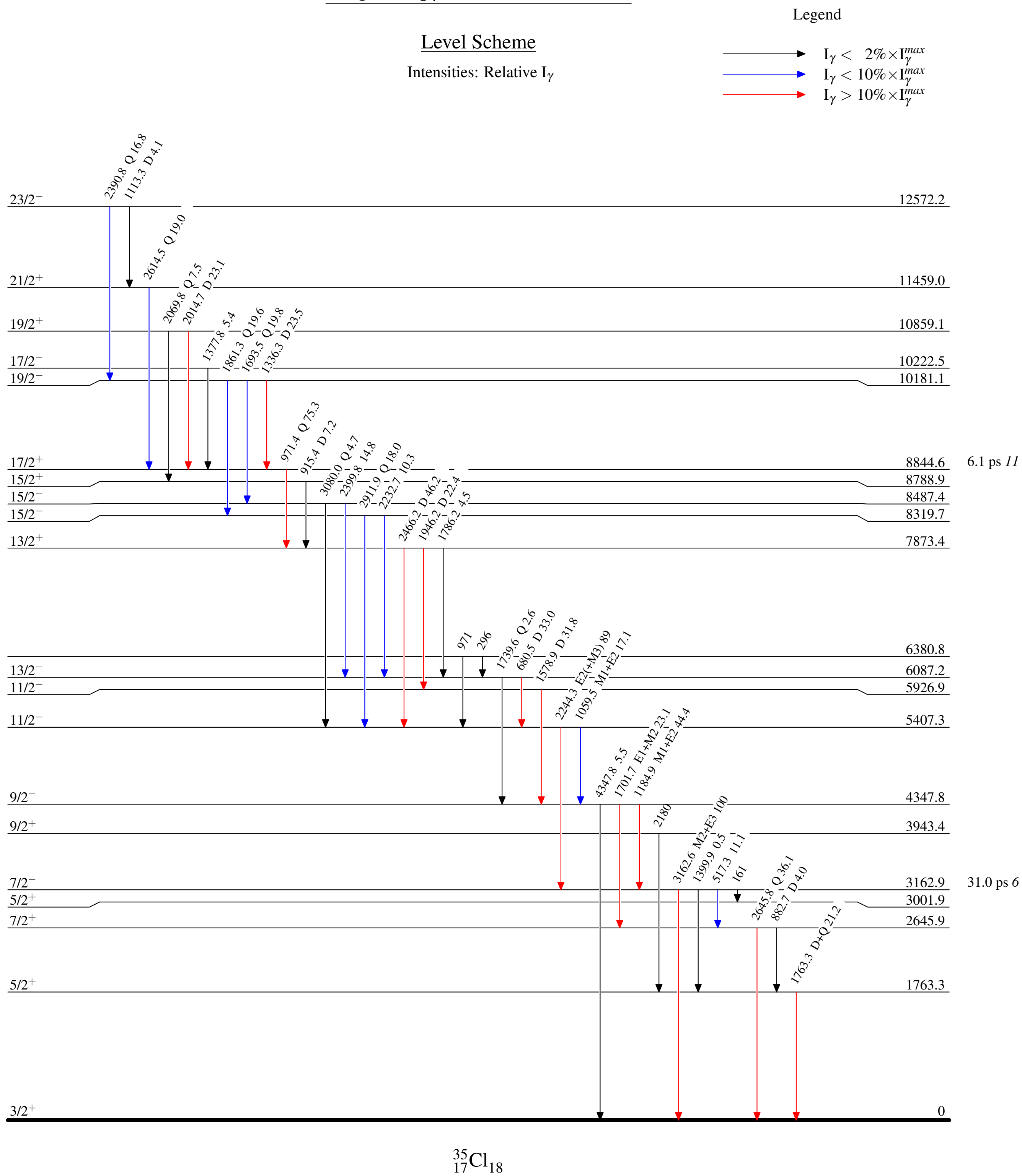
[†] From 2007De14. Note that a few Iγ values in 2007De14 have very small uncertainties (<1%) which is unrealistic for an in-beam measurement with Ge detectors. It is very likely the reported uncertainties are statistical only and the evaluators have added in quadrature a typical 2% systematic uncertainty from efficiency calibration.

[‡] D with ΔJ=1 or Q with ΔJ=2 are deduced by the evaluators based on measured γγ(θ)(ADO) in 2007De14 and others are from γ(θ) and γ(lin pol) in 1976Va24. Assignments are not explicitly listed in 2007De14.

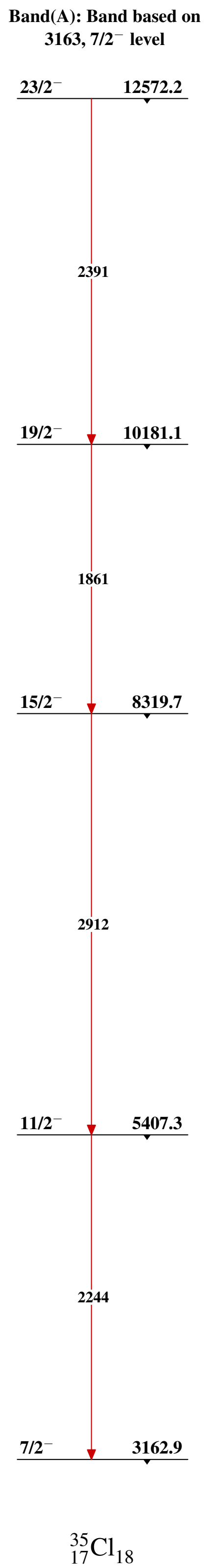
[#] From γ(θ) in 1976Va24.

[@] From 1976Va24 only.

$^{24}Mg(^{16}O,\alpha p\gamma)$ 2007De14,1976Va24

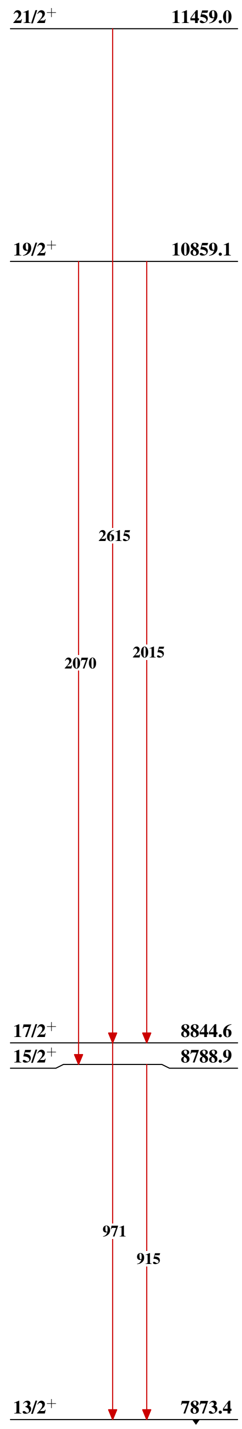


$^{24}Mg(^{16}O,\alpha p\gamma)$ 2007De14,1976Va24



^{24}Mg(^{16}O,αpγ) 2007De14,1976Va24 (continued)

Band(B): Band based on 7873, 13/2$^+$ level



$^{35}_{17}Cl_{18}$

$^{27}Al(^{14}N,\alpha pn\gamma)$ 1976Wa11

1976Wa11: E=40 MeV ^{14}N beam was produced at the Brookhaven National Laboratory. Target was ^{27}Al. γ rays were detected with Ge(Li) detectors. Measured Eγ, Iγ, γ(θ), γγ-coin, recoils distances. Deduced levels, J, π, $T_{1/2}$, γ-ray branching ratios, multipolarities.

^{35}Cl Levels

E(level)†	Jπ‡	$T_{1/2}$	Comments
0	3/2+		
1763.19 *10*	5/2+		
2645.45 *12*	7/2+		
3001.98 *22*	5/2+		
3162.64 *9*	7/2−	29.0 ps *13*	$T_{1/2}$: from τ=41.8 ps *18* (1976Wa11).
3942.3 *10*	9/2+		
4347.43 *17*	9/2−		
5406.68 *19*	11/2−		
5926.47 *29*	11/2−		
6086.89 *23*	13/2−	5.3 ps *10*	$T_{1/2}$: from τ=7.7 ps *14* (1976Wa11).
7872.78 *30*	13/2+		$T_{1/2}$: 0.5 ps<τ<50 ps (1976Wa11).
8844.2 *4*	17/2+	5.5 ps *14*	$T_{1/2}$: from τ=8 ps *2* (1976Wa11).

† From a least-squares fit to γ-ray energies.
‡ As given in 1976Wa11 based on measured γ(θ) and known assignments of low-lying levels. When considered in Adopted Levels, firm assignments here will be placed in parentheses if there are no strong supporting arguments.

γ(^{35}Cl)

A_2 and A_4 under comments are from γ(θ) in 1976Wa11, with A_4=0 if not given.

Eγ†	Iγ†	E_i(level)	J^π_i	E_f	J^π_f	Comments
160.66# *20*	2.3	3162.64	7/2−	3001.98	5/2+	A_2=−0.24 *15*.
517.26 *10*	<2.2‡	3162.64	7/2−	2645.45	7/2+	Eγ,Iγ: doublet with γ rays in ^{36}Cl (1976Wa11). A_2=+0.47 *25*.
680.22 *15*	25	6086.89	13/2−	5406.68	11/2−	A_2=−0.20 *2*.
882.32 *35*	2.3	2645.45	7/2+	1763.19	5/2+	A_2=+0.31 *40*.
971.38 *20*	8.0	8844.2	17/2+	7872.78	13/2+	A_2=+0.28 *7*, A_4=−0.18 *7*.
1059.17 *20*	4.6	5406.68	11/2−	4347.43	9/2−	A_2=−0.04 *15*. %Branching=13 *4*.
1184.80 *20*	21	4347.43	9/2−	3162.64	7/2−	A_2=−0.66 *4*, A_4=+0.08 *4*. %Branching=69 *5*.
1579.15 *30*	1.9‡	5926.47	11/2−	4347.43	9/2−	A_2=−0.6 *4*.
1701.85 *25*	9.5	4347.43	9/2−	2645.45	7/2+	A_2=−0.20 *3*. %Branching=31 *5*.
1739.3 *4*	1.7	6086.89	13/2−	4347.43	9/2−	
1763.15 *10*	32	1763.19	5/2+	0	3/2+	A_2=−0.28 *2*, A_4=+0.05 *3*.
1786#	0.63	7872.78	13/2+	6086.89	13/2−	Eγ,Iγ: inferred from the presence of 680γ-971γ coin (1976Wa11). %Branching=8 *5*.
1946.40 *30*	1.9	7872.78	13/2+	5926.47	11/2−	A_2=−0.2 *1*. %Branching=22 *5*.
2179	5.0	3942.3	9/2+	1763.19	5/2+	Eγ,Iγ: observed in γγ-coin only (1976Wa11).
2244.00 *25*	32	5406.68	11/2−	3162.64	7/2−	A_2=+0.23 *2*, A_4=−0.08 *2*. %Branching=87 *4*.
2465.85 *30*	5.9	7872.78	13/2+	5406.68	11/2−	A_2=−0.21 *6*. %Branching=70 *5*.

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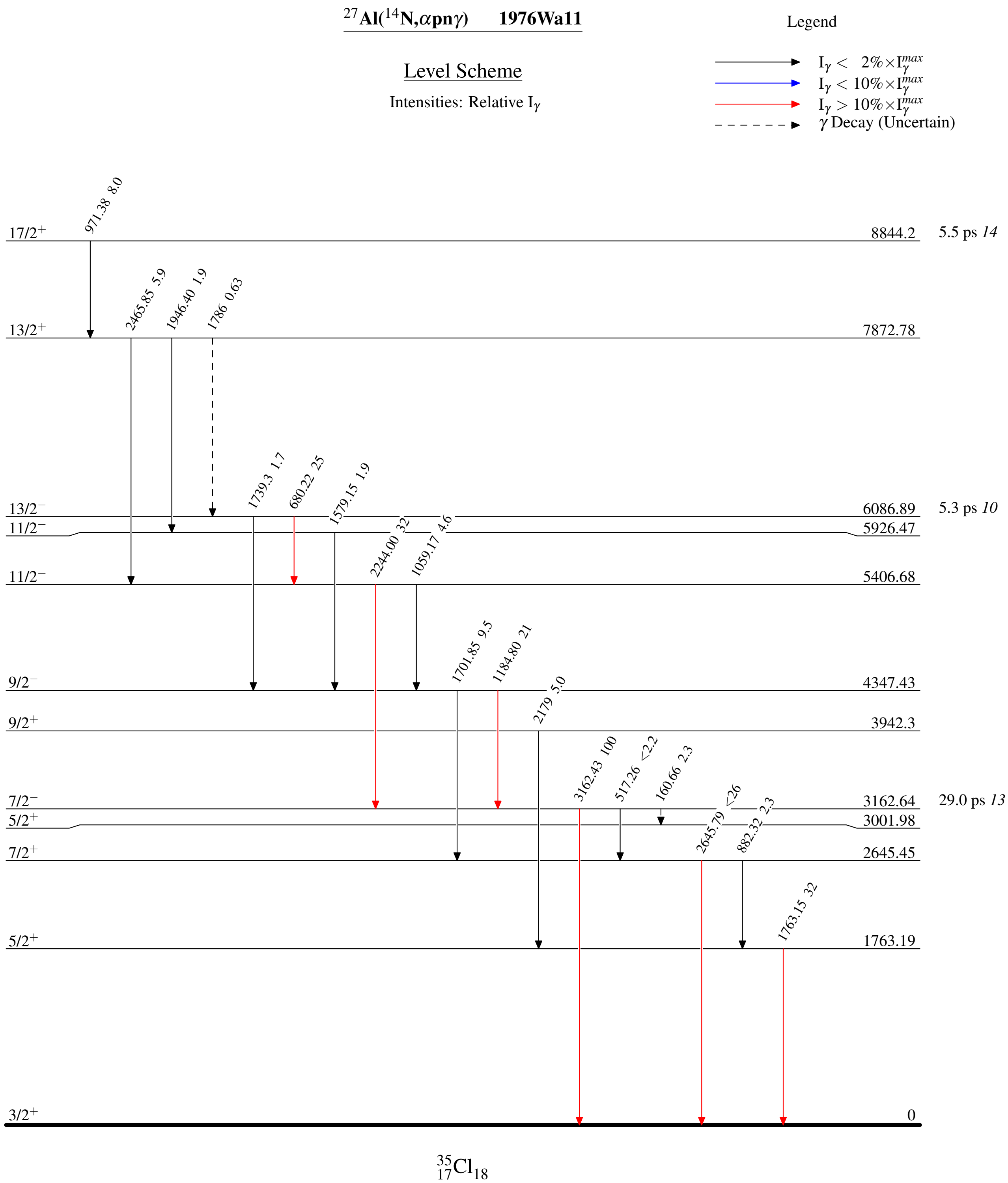
^{27}Al(^{14}N,αpnγ) 1976Wa11 (continued)

γ(^{35}Cl) (continued)

E_γ †	I_γ †	E_i(level)	J^π_i	E_f	J^π_f	Comments
2645.79 *30*	<26	2645.45	7/2$^+$	0	3/2$^+$	E_γ,I_γ: doublet with γ rays in ^{38}K (1976Wa11).
3162.43 *10*	100	3162.64	7/2$^-$	0	3/2$^+$	A_2=+0.37 *3*, A_4=−0.04 *3*.

† From 1976Wa11. Original values of relative intensities have been renormalized by the evaluators to I(3162γ)=100.
‡ Estimated from $\gamma\gamma$-coin data or known branching ratios (1976Wa11).
# Placement of transition in the level scheme is uncertain.



^{27}Al(^{14}N,αpnγ) 1976Wa11
Level Scheme
Intensities: Relative I_γ
Legend
$I_\gamma < 2\% \times I_\gamma^{max}$
$I_\gamma < 10\% \times I_\gamma^{max}$
$I_\gamma > 10\% \times I_\gamma^{max}$
γ Decay (Uncertain)
17/2+ 8844.2 5.5 ps 14
971.38 8.0
13/2+ 7872.78
2465.85 5.9
1946.40 1.9
1786 0.63
13/2− 6086.89 5.3 ps 10
11/2− 5926.47
1739.3 1.7
680.22 25
1579.15 1.9
11/2− 5406.68
2244.00 32
1059.17 4.6
9/2− 4347.43
1701.85 9.5
1184.80 21
9/2+ 3942.3
2179 5.0
7/2− 3162.64 29.0 ps 13
5/2+ 3001.98
3162.43 100
517.26 <2.2
160.66 2.3
7/2+ 2645.45
2645.79 <26
882.32 2.3
5/2+ 1763.19
1763.15 32
3/2+ 0
$^{35}_{17}Cl_{18}$

^{31}P(α,p),(α,n):resonances 1975Sc40,1971Mc23,1964Ku12

J^{π}=1/2$^+$ ^{31}P ground state.

1975Sc40: ^{31}P(α,p) with 3.25-5.55-MeV α beams produced from the 5.5-MV Van de Graaff accelerator of the University of Lowell. The target was prepared by evaporating natural ^{31}P onto carbon backings. Protons were detected using four silicon surface-barrier detectors. Measured proton yields and angular distributions. Deduced resonance levels at E_x=9.9-11.8 MeV, J, and π.

1971Mc23: ^{31}P(α,p) with 3.25-5.25-MeV α beams produced from the 5.5-MeV Van de Graaff accelerator of the Southern Universities Nuclear Institute. The target was Zn_3P_2 evaporated onto carbon backings. Protons were detected using four solid state detectors. Measured proton yields and angular distributions. Deduced resonance levels at E_x=9.9-11.7 MeV, J, and resonance strengths.

1964Ku12: ^{31}P(α,p) with 1.7-3.3-MeV α beams produced from the Utrecht Van de Graaff accelerator. A zinc phosphide target was prepared by evaporation of 40-500-mg/cm^2 Zn_3P_2 onto solid copper backings. Protons were detected using five silicon surface barrier counters at 87°, 120°, 135°, 150°, and 172°. Measured resonance yields and proton angular distributions. Deduced ^{35}Cl resonance energies, J, and resonance strengths.

1970Um01: ^{34}S(p,n), ^{31}P(α,n), ^{31}P(α,α_0) reactions with proton and α beams produced from the Florida State University Tandem Van de Graaff accelerator. Targets for the ^{34}S(p,n) experiment were a CdS target made by evaporating ^{34}S enriched CdS onto 0.12-mm gold plate and a AgS target with heating ^{34}S (67% enriched) in the presence of 0.025-mm Ag foil. Targets for the ^{31}P(α,n) experiment were prepared by evaporating natural Ca_3P_2 onto 0.075-mm Ta plate and 0.025-mm Al foil. Positrons from ^{34}Cl decay were detected using an NE102 plastic scintillator. γ rays in ^{34}S following the decay were detected using a Ge(Li) detector. Measured excitation functions. Deduced level energies and isospin T=3/2 for ^{35}Cl resonance states at 12.9 and 13.9 MeV.

^{35}Cl Levels

(2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$: from 1964Ku12 for E_x<9950 and from 1971Mc23 for E_x>9950.

E(level)[†]	J^{π}[‡]	E_{α}(lab) (keV)[&]	Comments
9127 *9*	5/2[#]	2404[#] *10*	(2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$=0.16 eV.
9161 *4*	1/2[#]	2442[#] *4*	(2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$=1.6 eV.
9256 *4*	1/2[#]	2550[#] *4*	(2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$=2.6 eV.
9400 *9*	1/2[#]	2712[#] *10*	(2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$=0.6 eV.
9456 *4*	3/2[#]	2776[#] *4*	(2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$=2.8 eV.
9481.0 *18*	3/2[#]	2804[#] *2*	(2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$=18 eV.
9551 *6*	5/2[#]	2883[#] *7*	(2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$=1.9 eV.
9673 *6*	1/2[#]	3021[#] *7*	(2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$=1.1 eV.
9713.0 *27*	1/2[#]	3066[#] *3*	(2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$=41 eV.
9751.1 *27*	7/2[#]	3109[#] *3*	(2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$=7.6 eV.
9814.0 *27*	5/2[#]	3180[#] *3*	(2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$=20 eV.
9869.8 *27*	1/2	3243 *3*	E_{α}(lab) (keV): weighted average of 3243 *3* (1964Ku12) and 3242 *5* (1975Sc40). J^{π}: 1/2 from 1964Ku12. (2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$=46 eV.
9900.8 *27*	3/2	3278 *3*	E_{α}(lab) (keV): weighted average of 3278 *3* (1964Ku12) and 3278 *5* (1975Sc40). J^{π}: 3/2 from 1964Ku12. (2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$=38 eV.
9923 *4*	3/2$^-$	3303 *4*	E_{α}(lab) (keV): weighted average of 3302 *4* (1964Ku12) and 3305 *5* (1975Sc40). J^{π}: (3/2) from 1964Ku12. (2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$=80 eV.
9958 *6*	3/2	3342 *7*	E_{α}(lab) (keV): weighted average of 3348 *5* (1971Mc23) and 3335 *5* (1975Sc40). J^{π}: 3/2 from 1971Mc23. (2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$=34 eV.
9968 *5*	(1/2,3/2)[@]	3354 *5*	(2J+1)$\Gamma_{\alpha}\Gamma_p/\Gamma$=17 eV.
10031 *5*	1/2$^+$,3/2$^-$	3425 *5*	E_{α}(lab) (keV): weighted average of 3424 *5* (1971Mc23) and 3425 *5* (1975Sc40). J^{π}: (1/2) from 1971Mc23.

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$^{31}P(\alpha,p),(\alpha,n)$:resonances 1975Sc40,1971Mc23,1964Ku12 (continued)

^{35}Cl Levels (continued)

E(level)[†]	J^{π}[‡]	E_{α}(lab) (keV)[&]	Comments
			$(2J+1)\Gamma_{\alpha}\Gamma_p/\Gamma$=124 eV.
10058 *5*	3/2[@]	3455 *5*	$(2J+1)\Gamma_{\alpha}\Gamma_p/\Gamma$=40 eV.
10075 *5*	(1/2)[@]	3475 *5*	$(2J+1)\Gamma_{\alpha}\Gamma_p/\Gamma$=52 eV.
10090 *5*	$5/2^+$	3491 *5*	E_{α}(lab) (keV): weighted average of 3491 *5* (1971Mc23) and 3490 *5* (1975Sc40). J^{π}: 5/2 from 1971Mc23. $(2J+1)\Gamma_{\alpha}\Gamma_p/\Gamma$=63 eV.
10132 *5*	$3/2^-$	3539 *5*	E_{α}(lab) (keV): weighted average of 3538 *5* (1971Mc23) and 3540 *5* (1975Sc40). J^{π}: 3/2 from 1971Mc23. $(2J+1)\Gamma_{\alpha}\Gamma_p/\Gamma$=165 eV.
10172 *5*	$3/2^-,5/2^-$	3584 *5*	E_{α}(lab) (keV): weighted average of 3587 *5* (1971Mc23) and 3580 *5* (1975Sc40). J^{π}: (5/2) from 1971Mc23. $(2J+1)\Gamma_{\alpha}\Gamma_p/\Gamma$=26 eV.
10215 *5*	3/2	3633 *5*	E_{α}(lab) (keV): weighted average of 3631 *5* (1971Mc23) and 3635 *5* (1975Sc40). J^{π}: 3/2 from 1971Mc23. $(2J+1)\Gamma_{\alpha}\Gamma_p/\Gamma$=63 eV.
10236 *5*	$3/2^-$	3656 *5*	E_{α}(lab) (keV): weighted average of 3656 *5* (1971Mc23) and 3655 *5* (1975Sc40). J^{π}: 3/2 from 1971Mc23. $(2J+1)\Gamma_{\alpha}\Gamma_p/\Gamma$=271 eV.
10278 *5*	1/2[@]	3704 *5*	$(2J+1)\Gamma_{\alpha}\Gamma_p/\Gamma$=15 eV.
10295 *5*		3723 *5*	
10322 *5*	$5/2^+,7/2^-$	3753 *5*	E_{α}(lab) (keV): weighted average of 3755 *5* (1971Mc23) and 3750 *5* (1975Sc40). J^{π}: 5/2 from 1971Mc23. $(2J+1)\Gamma_{\alpha}\Gamma_p/\Gamma$=63 eV.
10340 *5*	(3/2,5/2)[@]	3774 *5*	$(2J+1)\Gamma_{\alpha}\Gamma_p/\Gamma$=50 eV.
10359 *5*	3/2	3795 *5*	E_{α}(lab) (keV): weighted average of 3795 *5* (1971Mc23) and 3795 *5* (1975Sc40). J^{π}: (3/2) from 1971Mc23. $(2J+1)\Gamma_{\alpha}\Gamma_p/\Gamma$=155 eV.
10385 *5*		3825 *5*	
10399 *5*	$3/2^+,7/2^+$	3841 *5*	E_{α}(lab) (keV): weighted average of 3842 *5* (1971Mc23) and 3840 *5* (1975Sc40). J^{π}: (1/2,3/2) from 1971Mc23. $(2J+1)\Gamma_{\alpha}\Gamma_p/\Gamma$=61 eV.
10431 *5*	$5/2^+$	3877 *5*	E_{α}(lab) (keV): weighted average of 3876 *5* (1971Mc23) and 3877 *5* (1975Sc40). J^{π}: 5/2 from 1971Mc23. $(2J+1)\Gamma_{\alpha}\Gamma_p/\Gamma$=66 eV.
10467 *5*	$5/2^+$	3917 *5*	E_{α}(lab) (keV): weighted average of 3918 *5* (1971Mc23) and 3915 *5* (1975Sc40). J^{π}: 5/2 from 1971Mc23. $(2J+1)\Gamma_{\alpha}\Gamma_p/\Gamma$=211 eV.
10486 *5*	$5/2^+$	3939 *5*	E_{α}(lab) (keV): weighted average of 3939 *5* (1971Mc23) and 3938 *5* (1975Sc40). J^{π}: 5/2 from 1971Mc23. $(2J+1)\Gamma_{\alpha}\Gamma_p/\Gamma$=76 eV.
10534 *5*	$1/2^+,5/2^+$	3993 *5*	E_{α}(lab) (keV): weighted average of 3990 *5* (1971Mc23) and 3995 *5* (1975Sc40). J^{π}: 1/2 from 1971Mc23. $(2J+1)\Gamma_{\alpha}\Gamma_p/\Gamma$=182 eV.
10587 *5*	$3/2^+$	4053 *5*	E_{α}(lab) (keV): weighted average of 4050 *5* (1971Mc23) and 4055 *5* (1975Sc40). J^{π}: 3/2 from 1971Mc23. $(2J+1)\Gamma_{\alpha}\Gamma_p/\Gamma$=71 eV.
10641 *5*	$5/2^+$	4114 *5*	E_{α}(lab) (keV): weighted average of 4113 *5* (1971Mc23) and 4114 *5* (1975Sc40). J^{π}: 5/2 from 1971Mc23. $(2J+1)\Gamma_{\alpha}\Gamma_p/\Gamma$=272 eV.
10678 *5*	$3/2^+,5/2^+$	4156 *5*	E_{α}(lab) (keV): weighted average of 4156 *5* (1971Mc23) and 4155 *5* (1975Sc40). J^{π}: 3/2 from 1971Mc23. $(2J+1)\Gamma_{\alpha}\Gamma_p/\Gamma$=132 eV.
10696 *5*	$7/2^-$	4176 *5*	E_{α}(lab) (keV): weighted average of 4174 *5* (1971Mc23) and 4178 *5* (1975Sc40).

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^{31}P(α,p),(α,n):resonances 1975Sc40,1971Mc23,1964Ku12 (continued)

^{35}Cl Levels (continued)

E(level)†	J$^{\pi}$‡	E$_{\alpha}$(lab) (keV)&	Comments
			J$^{\pi}$: 7/2 from 1971Mc23. (2J+1)$\Gamma_{\alpha}\Gamma_{p}/\Gamma$=100 eV.
10734 *5*	3/2^{+},9/2^{+}	4219 *5*	E$_{\alpha}$(lab) (keV): weighted average of 4218 *5* (1971Mc23) and 4220 *5* (1975Sc40). J$^{\pi}$: 3/2 from 1971Mc23. (2J+1)$\Gamma_{\alpha}\Gamma_{p}/\Gamma$=269 eV.
10760 *5*	3/2^{+},7/2^{-}	4248 *5*	E$_{\alpha}$(lab) (keV): weighted average of 4246 *5* (1971Mc23) and 4250 *5* (1975Sc40).
10800 *5*	5/2^{+}	4293 *5*	E$_{\alpha}$(lab) (keV): weighted average of 4290 *5* (1971Mc23) and 4295 *5* (1975Sc40). J$^{\pi}$: 5/2 from 1971Mc23. (2J+1)$\Gamma_{\alpha}\Gamma_{p}/\Gamma$=238 eV.
10816 *5*	1/2^{+},3/2^{-}	4311 *5*	E$_{\alpha}$(lab) (keV): weighted average of 4308 *5* (1971Mc23) and 4313 *5* (1975Sc40). J$^{\pi}$: 3/2 from 1971Mc23. (2J+1)$\Gamma_{\alpha}\Gamma_{p}/\Gamma$=345 eV.
10844 *5*	3/2^{-}	4343 *5*	E$_{\alpha}$(lab) (keV): weighted average of 4342 *5* (1971Mc23) and 4343 *5* (1975Sc40). J$^{\pi}$: 3/2 from 1971Mc23. (2J+1)$\Gamma_{\alpha}\Gamma_{p}/\Gamma$=923 eV.
10870 *5*	(5/2,7/2)@	4372 *5*	(2J+1)$\Gamma_{\alpha}\Gamma_{p}/\Gamma$=137 eV.
10886 *5*		4390 *5*	
10909 *5*	3/2^{+},5/2^{-}	4416 *5*	E$_{\alpha}$(lab) (keV): weighted average of 4413 *5* (1971Mc23) and 4418 *5* (1975Sc40).
10928 *5*	7/2@	4438 *5*	(2J+1)$\Gamma_{\alpha}\Gamma_{p}/\Gamma$=231 eV.
10942 *5*		4454 *5*	
10964 *5*	5/2^{+}	4478 *5*	E$_{\alpha}$(lab) (keV): weighted average of 4478 *5* (1971Mc23) and 4477 *5* (1975Sc40). J$^{\pi}$: 5/2 from 1971Mc23. (2J+1)$\Gamma_{\alpha}\Gamma_{p}/\Gamma$=181 eV.
10973 *5*		4489 *5*	
10998 *5*	3/2^{+},7/2^{-}	4517 *5*	E$_{\alpha}$(lab) (keV): weighted average of 4517 *5* (1971Mc23) and 4517 *5* (1975Sc40). J$^{\pi}$: (5/2) from 1971Mc23. (2J+1)$\Gamma_{\alpha}\Gamma_{p}/\Gamma$=172 eV.
11034 *5*	3/2^{+}	4557 *5*	E$_{\alpha}$(lab) (keV): weighted average of 4556 *5* (1971Mc23) and 4558 *5* (1975Sc40). J$^{\pi}$: 3/2 from 1971Mc23. (2J+1)$\Gamma_{\alpha}\Gamma_{p}/\Gamma$=218 eV.
11063 *5*	1/2^{+},9/2^{+}	4590 *5*	E$_{\alpha}$(lab) (keV): weighted average of 4589 *5* (1971Mc23) and 4590 *5* (1975Sc40).
11082 *5*	5/2^{+},9/2^{+}	4612 *5*	E$_{\alpha}$(lab) (keV): weighted average of 4611 *5* (1971Mc23) and 4613 *5* (1975Sc40).
11105 *5*		4638 *5*	
11123 *5*	3/2	4658 *5*	E$_{\alpha}$(lab) (keV): weighted average of 4657 *5* (1971Mc23) and 4658 *5* (1975Sc40).
11142 *5*	5/2^{+}	4679 *5*	E$_{\alpha}$(lab) (keV): weighted average of 4678 *5* (1971Mc23) and 4680 *5* (1975Sc40).
11154 *5*		4693 *5*	
11169 *5*		4710 *5*	
11179 *5*	3/2^{-},5/2^{+}	4721 *5*	E$_{\alpha}$(lab) (keV): weighted average of 4721 *5* (1971Mc23) and 4720 *5* (1975Sc40).
11185 *5*		4728 *5*	
11198 *5*	3/2	4743 *5*	E$_{\alpha}$(lab) (keV): weighted average of 4745 *5* (1971Mc23) and 4740 *5* (1975Sc40).
11230 *5*		4779 *5*	
11246 *5*	3/2	4797 *5*	E$_{\alpha}$(lab) (keV): weighted average of 4799 *5* (1971Mc23) and 4795 *5* (1975Sc40).
11265 *5*	3/2^{-}	4818 *5*	E$_{\alpha}$(lab) (keV): weighted average of 4820 *5* (1971Mc23) and 4815 *5* (1975Sc40).
11287 *5*		4843 *5*	
11307 *5*		4866 *5*	
11314 *5*	1/2^{-},5/2^{+}	4874 *5*	E$_{\alpha}$(lab) (keV): weighted average of 4875 *5* (1971Mc23) and 4873 *5* (1975Sc40).
11327 *5*	1/2^{-},5/2^{+}	4888 *5*	E$_{\alpha}$(lab) (keV): weighted average of 4886 *5* (1971Mc23) and 4890 *5* (1975Sc40).
11357 *5*		4922 *5*	
11375 *5*		4942 *5*	
11390 *5*		4960 *5*	
11410 *5*		4982 *5*	
11421 *5*	3/2^{-}	4995 *5*	E$_{\alpha}$(lab) (keV): weighted average of 4995 *5* (1971Mc23) and 4995 *5* (1975Sc40).
11434 *5*		5009 *5*	
11460 *5*		5038 *5*	
11473 *5*	1/2^{-},5/2^{-}	5053 *5*	E$_{\alpha}$(lab) (keV): weighted average of 5056 *5* (1971Mc23) and 5050 *5* (1975Sc40).

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^{31}P(α,p),(α,n):resonances 1975Sc40,1971Mc23,1964Ku12 (continued)

^{35}Cl Levels (continued)

E(level)[†]	J^{π}[‡]	E_{α}(lab) (keV)[&]	Comments
11492 *5*		5075 *5*	
11505 *5*	3/2$^-$,5/2$^-$	5089 *5*	E_{α}(lab) (keV): weighted average of 5090 *5* (1971Mc23) and 5088 *5* (1975Sc40).
11525 *5*		5112 *5*	
11540 *5*		5129 *5*	
11551 *5*	3/2$^-$,5/2$^+$	5141 *5*	E_{α}(lab) (keV): weighted average of 5142 *5* (1971Mc23) and 5140 *5* (1975Sc40).
11565 *5*		5157 *5*	
11589 *5*		5184 *5*	
11609 *5*	5/2$^+$,7/2$^-$	5207 *5*	E_{α}(lab) (keV): weighted average of 5209 *5* (1971Mc23) and 5205 *5* (1975Sc40).
11627 *5*	1/2$^+$,5/2$^+$	5227 *5*	E_{α}(lab) (keV): weighted average of 5224 *5* (1971Mc23) and 5230 *5* (1975Sc40).
11638 *5*		5239 *5*	
11651 *5*		5254 *5*	
11684 *5*	1/2$^+$,5/2$^+$	5292[a] *5*	
11773 *5*	3/2$^-$,7/2$^-$	5392[a] *5*	
11783 *5*	3/2$^-$	5403[a] *5*	
12900[b]			T=3/2
13900[b]			T=3/2

[†] From $E_{c.m.}$+S(α)(^{35}Cl), where S(α)(^{35}Cl)=6997.91 *4* (2021Wa16). $E_{c.m.}=E_{\alpha}(lab)\times m(^{31}P)/[m_{\alpha}+m(^{31}P)]$.

[‡] From 1975Sc40 by analyzing measured angular distributions, unless otherwise noted.

[#] From 1964Ku12.

[@] From 1971Mc23.

[&] From 1971Mc23, unless otherwise noted.

[a] From 1975Sc40.

[b] From 1970Um01. Two ^{35}Cl resonances are found in ^{34}S(p,n) and ^{31}P(α,n), but not in ^{31}P(α,α_0) reactions. T=1 ^{34}S g.s. and T=1/2 protons can populate either T=1/2 or T=3/2 states in ^{35}Cl. T=1/2 ^{31}P g.s. and T=0 α particles can populate T=1/2 states in ^{35}Cl. A small isospin impurity would allow formation of T=3/2 states in ^{35}Cl. The lack of resonances in ^{31}P(α,α_0) indicates that these two resonances are most likely T=3/2. A T=3/2 resonances in ^{31}P(α,α_0) scattering is doubly isospin forbidden (both entrance and exit channels) compared with singly forbidden (entrance channel only) for ^{31}P(α,n).

^{31}P(^{6}Li,d) 1979Es05

$^{31}P+\alpha \rightarrow ^{35}Cl$ from $J^{\pi}=1/2^{+}$ ^{31}P ground state.

1979Es05,1978Es01: a 36-MeV $^{6}Li^{3+}$ beam at 150-400 nA was produced from the MP tandem accelerator at the University of Rochester and bombarded a 100-μg/cm^{2}cadmium phosphide Cd_3P_2 target. Outgoing deuterons were momentum analyzed by an Enge split-pole magnetic spectrograph and detected using Kodak NTB-50 nuclear emulsion plates with FWHM=50 keV. Measured $\sigma(E_d,\theta)$. Deduced levels, J, π, L-transfers, spectroscopic factors from finite-range, LOLA-DWBA analysis of the measured $\sigma(\theta)$.

Other: 1982Ch10 (finite-range DWBA calculations).

^{35}Cl Levels

Spectroscopic factor $g*S=\sigma(\theta)_{exp}/\sigma(\theta)_{DWBA}$, where $g=(2J_f+1)/(2J_i+1)$.

E(level)†	J^{π}‡	L#	S#
0	$3/2^{+}$	2	1.00
1220	$1/2^{+}$	0	1.49
1760	$5/2^{+}$	2	0.24
3160	$7/2^{-}$	3	0.52

† From 1979Es05.
‡ As given in 1979Es05 for extracting spectroscopic factors.
# From DWBA analysis of the measured $\sigma(\theta)$ in 1979Es05.

^{32}S(α,p) 2019Se09,1973Go16

$J^{\pi}=0^{+}$ for ^{32}S ground state.

2019Se09: a 21-MeV α beam was produced from the 10-MV FN tandem Van de Graaff accelerator at Triangle Universities Nuclear Laboratory (TUNL). Targets were 15.9-μg/cm^2 ^{nat}S in cadmium sulfide and 46.3-μg/cm^2 ^{nat}S in antimony sulfide. Reaction products were momentum-analyzed by the TUNL high-resolution Enge split-pole magnetic spectrograph and detected using a high-resolution position-sensitive focal-plane detector consisting of two position-sensitive gas avalanche counters: a gas proportionality energy loss section and a residual energy scintillator (FWHM=24 keV). Measured $\sigma(E_p,\theta)$, at θ=10° to 50°. Deduced levels, J, π from one-step finite range triton transfer FRESCO-DWBA analysis. Deduced the thermonuclear ^{34}S(p,γ)^{35}Cl reaction rate via Monte Carlo methods.

1973Go16: 19.991-, 20.025- and 20.284-MeV α beams were produced from the University of Notre Dame FN tandem Van de Graaff accelerator. Natural sulfur targets were prepared by vacuum evaporation of CdS onto 20-μg/cm^2 carbon backings. Protons were analyzed with broad-range magnetic spectrographs at 15°, 30°, 70°, 90°, and 120° and detected using nuclear track plates at the focal planes. Measured $\sigma(E_p,\theta)$. Deduced 102 levels.

1971Au07: a 28.8-MeV α beam of 0.1-0.5 μA was produced from the Naval Research Laboratory (NRL) isochronous cyclotron. The target was made by evaporating natural PbS onto a 30-μg/cm^2 carbon backing, and the effective ^{32}S thickness was 163 μg/cm^2. Protons were detected using two counter telescopes with FWHM=250 keV. Measured $\sigma(E_p,\theta)$. Deduced levels, J, π, L-transfers, and J-dependence for the g.s. and 1763-keV level from local zero-range DWUCK-DWBA analysis of the measured σ(θ).

Other: 1996So21 (cross sections).

^{35}Cl Levels

E(level)[†]	Jπ[&]	L[a]	Comments
0		2	Jπ: 3/2⁺ assumed in 1971Au07 for investigation of J-dependence of L=2 transfer.
1217.4 *26*			
1760.1 *24*		(2)	Jπ: 5/2⁺ assumed in 1971Au07 for investigation of J-dependence of L=2 transfer. L: The parentheses are added by evaluators based on the DWBA fit in 1971Au07.
2644.9 *23*			
2693?			
3001.5 *31*			
3158.2 *25*			
3945.2 *23*			E(level): weighted average of 3943.7 *23* (1973Go16) and 3947.9 *31* (2019Se09).
4056.9 *27*	(3/2⁻)[b]		E(level): 4059.12 quoted from 2011Ch48 and used for internal calibrations by 2019Se09.
4111.1 *29*	(7/2⁺)[b]		E(level): weighted average of 4110.2 *29* (1973Go16) and 4112.0 *30* (2019Se09).
4177.9 *24*	(3/2⁻)[b]		E(level): weighted average of 4177.5 *24* (1973Go16) and 4179 *4* (2019Se09).
4346.6 *24*			E(level): 4347.82 quoted from 2011Ch48 and used for internal calibrations by 2019Se09.
4625.3 *28*			E(level): weighted average of 4624.4 *31* (1973Go16) and 4626.1 *28* (2019Se09).
4770.4 *26*			E(level): weighted average of 4770.8 *26* (1973Go16) and 4769.9 *28* (2019Se09).
4859.2[‡] *28*			
4881.8 *29*			E(level): weighted average of 4883.1 *29* (1973Go16) and 4878 *5* (2019Se09).
5006.9[‡] *27*			
5158.9 *29*	(5/2⁺)[b]		E(level): weighted average of 5161.7 *33* (1973Go16) and 5156.8 *29* (2019Se09).
5209.5 *28*	(5/2⁺)		E(level): weighted average of 5206.6 *37* (1973Go16) and 5211.1 *28* (2019Se09).
5402.0 *29*			
5407.2	(11/2⁻)[b]		E(level): quoted from 2011Ch48 and used for internal calibrations by 2019Se09.
5531[‡] *5*			
5586.0 *26*			E(level): 5576 tentative level from 1973Go16.
5594.6 *26*			E(level): weighted average of 5591.8 *32* (1973Go16) and 5596.5 *26* (2019Se09).
5633.6 *30*			E(level): weighted average of 5633.1 *32* (1973Go16) and 5634 *3* (2019Se09).
5645[‡] *4*			
5653[‡] *4*			
5679.7 *34*			E(level): weighted average of 5677.7 *34* (1973Go16) and 5684 *5* (2019Se09).
5731.2[‡] *25*			
5757[‡] *4*			

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$^{32}S(\alpha,p)$ 2019Se09,1973Go16 (continued)

^{35}Cl Levels (continued)

E(level)[†]	Jπ[&]	Comments
5808.1 *31*		E(level): weighted average of 5809.2 *34* (1973Go16) and 5807.1 *31* (2019Se09).
5829[‡] *4*	(5/2⁻)[b]	E(level): 5823 tentative level from 1973Go16.
5928[‡] *3*		E(level): other: 5927.4 *35* (1973Go16).
6084.2 *29*	(13/2⁻)[b]	E(level): 6087.4 quoted from 2011Ch48 and used for internal calibrations by 2019Se09.
6104[‡] *4*		
6141 *4*		E(level): weighted average of 6140.2 *40* (1973Go16) and 6142 *4* (2019Se09).
6180[‡] *3*		
6224.4 *30*		E(level): weighted average of 6224.9 *36* (1973Go16) and 6224 *3* (2019Se09).
6283[‡] *3*	(5/2⁺)	
6379.3 *25*	(9/2⁻)	E(level): from 2019Se09. Other: 6379.0 *34* (1973Go16).
6401.2 *22*	(1/2⁻)	E(level): weighted average of 6402.4 *41* (1973Go16) and 6400.9 *22* (2019Se09).
6429[‡] *3*	(1/2⁺)	E(level): 6427 tentative level from 1973Go16.
6475[‡] *4*		
6492[‡] *3*	(3/2⁺)	E(level): other: 6491.9 *34* (1973Go16).
6549[‡] *3*	(1/2⁺)	
6659.1 *30*	(7/2⁺)	E(level): weighted average of 6656.0 *31* (1973Go16) and 6662 *3* (2019Se09).
6679.4 *31*	(1/2⁺)	E(level): weighted average of 6680.8 *31* (1973Go16) and 6677 *4* (2019Se09).
6781.3 *30*	(3/2⁻)	E(level): weighted average of 6782.8 *32* (1973Go16) and 6780 *3* (2019Se09).
6800 *4*		E(level): weighted average of 6802.1 *42* (1973Go16) and 6795 *6* (2019Se09).
6842[‡] *4*	(3/2⁺)	
6863[‡] *3*	(9/2⁺)	E(level): 6867 tentative level from 1973Go16.
6891.6 *30*	(9/2⁺)	E(level): weighted average of 6893.5 *32* (1973Go16) and 6890 *3* (2019Se09).
6948.6 *34*	(5/2⁺)	E(level): weighted average of 6947.5 *34* (1973Go16) and 6950 *4* (2019Se09).
6986.3 *34*	(9/2⁺)	E(level): weighted average of 6985.7 *34* (1973Go16) and 6987 *4* (2019Se09).
7066.2	(5/2⁺)	E(level): 7066.2 quoted from 2011Ch48 and used for internal calibrations by 2019Se09.
7105?		
7121.7[#] *24*	(5/2⁻)	E(level): weighted average of 7120.8 *34* (1973Go16) and 7122.1 *24* (2019Se09).
7180[‡] *4*	(7/2⁺)	E(level): other: 7179.0 *43* (1973Go16).
7213.7 *25*		E(level): weighted average of 7210.1 *39* (1973Go16) and 7215.2 *25* (2019Se09).
7230.9 *30*	(3/2⁺)	E(level): weighted average of 7229.5 *34* (1973Go16) and 7232 *3* (2019Se09).
7301?		
7348[‡] *3*	(7/2)	E(level): other: 7348.3 *44* (1973Go16).
7362[‡] *3*		
7413.8 *30*		E(level): weighted average of 7417.5 *44* (1973Go16) and 7412 *3* (2019Se09).
7447[‡] *3*		
7503 *4*		E(level): weighted average of 7503.0 *37* (1973Go16) and 7502 *5* (2019Se09).
7566.4 *35*		E(level): weighted average of 7567.6 *35* (1973Go16) and 7564 *5* (2019Se09).
7590 *4*		E(level): weighted average of 7587.2 *40* (1973Go16) and 7595 *6* (2019Se09).
7648.1 *30*		E(level): weighted average of 7650.1 *40* (1973Go16) and 7647 *3* (2019Se09).
7674.6[‡] *25*		E(level): 7671 tentative level from 1973Go16.
7709.9[‡] *25*		
7730?		
7744.1[#] *30*		E(level): weighted average of 7742.7 *38* (1973Go16) and 7745 *3* (2019Se09).
7770.3 *33*		E(level): weighted average of 7776.3 *45* (1973Go16) and 7768.5 *25* (2019Se09).
7799.8 *30*		E(level): weighted average of 7800.8 *35* (1973Go16) and 7799 *3* (2019Se09).
7868.5 *35*		E(level): from 1973Go16. Other: 7867 *7* (2019Se09).
7901 *4*		E(level): weighted average of 7900.6 *38* (1973Go16) and 7899 *12* (2019Se09).
7978.6 *35*		
8000.7 *41*		
8019.2 *35*		
8075.1 *46*		

Continued on next page (footnotes at end of table)

$^{32}S(\alpha$,p) 2019Se09,1973Go16 (continued)

^{35}Cl Levels (continued)

E(level)[†]	E(level)[†]	E(level)[†]	E(level)[†]
8103.8[#] *42*	8654.1 *41*	9334.0 *46*	10117?
8121?	8698?	9376.0 *41*	10162.7 *53*
8171.7 *42*	8720.9[#] *37*	9475.5 *43*	10179.1 *53*
8210.2 *46*	8786.3 *38*	9508.1[#] *41*	10218.0 *46*
8246.0 *42*	8840.3 *38*	9525?	10395.2 *54*
8270.6[#] *42*	8887.5 *41*	9711.8 *44*	10443?
8292?	8957.5 *44*	9740.4 *40*	10463.2[#] *49*
8321.0 *38*	8996.9 *40*	9799.2 *47*	10517.1 *54*
8398.8 *38*	9027.4 *38*	9836.1 *44*	10548.3 *54*
8429.2 *40*	9085.7 *38*	9870?	10579.5 *54*
8482.3 *38*	9109.3 *49*	9968.6[@] *52*	10643.0 *50*
8505.1 *47*	9160.0 *42*	9992?	10717?
8532.4 *43*	9194.0 *39*	10009?	10732.1[#] *50*
8567.9 *37*	9264.9 *39*	10075.2[@] *48*	10759.1 *55*
8616.9 *37*	9316.2 *39*	10098?	

[†] From 1973Go16, unless otherwise noted.
[‡] From 2019Se09.
[#] Doublet in 1973Go16.
[@] Triplet in 1973Go16.
[&] From 2019Se09, based on FRESCO-DWBA analysis of measured $\sigma(\theta)$, unless otherwise noted.
[a] From 1971Au07.
[b] Original firm assignments in 2019Se09 are placed in parentheses by the evaluators when DWBA fits to the measured $\sigma(\theta)$ were poor or when only a few data points were available for fitting.

$^{32}S(\alpha,p\gamma)$ 1970Ho09,1972Br33,1973Al22

$J^\pi=0^+$ for ^{32}S ground state.

1970Ho09: 10.975-, 11.960-, 12.120-MeV α beams were produced from the Chalk River MP tandem accelerator. Targets were 100-$\mu g/cm^2$ Sb_2S_3 on a 10-$\mu g/cm^2$ carbon backing. γ rays were detected using 15.2-cm-long by 12.7-cm-diam NaI(Tl) detectors. Protons were detected using an annular silicon surface barrier detector close to 180° with FWHM=30 keV for protons and FWHM=50 keV for α. Measured E_p, Eγ, Iγ, p$\gamma(\theta)$. Deduced 25 level energies from E_p, J, branching ratios, multipole mixing ratios.

1972Br33: a 10-MeV α beam was produced from the Chalk River MP tandem accelerator. Targets were 340-$\mu g/cm^2$ Sb_2S_3 on gold backings. γ rays were detected using a 44-cm^3 Ge(Li) detector inside an anticoincidence split annular NaI(Tl) detector. Measured Eγ, Iγ. Deduced 11 level energies, 16 γ branching ratios, 11 lifetimes using the Doppler Shift Attenuation Method (DSAM). 1970In01 from the same experiment deduced the lifetime using DSAM for a level at 3942 keV and the mixing ratio for a 3942→2646 γ transition.

1973Al22: a 14-MeV α beam was produced from the 80-cm cyclotron at the Research Institute for Physics in Stockholm and 12-16-MeV α beams were produced from the Uppsala tandem accelerator. Targets were prepared by depositing pulverized sulphur enriched to 99.9% in ^{32}S on lead backings. γ rays were detected using an Ortec 60-cm^3 Ge(Li) detector with FWHM=2.2 keV at 1333 keV and an additional Canberra 60-cm^3 Ge(Li) detector for $\gamma\gamma$-coincidence measurements. Measured Eγ, Iγ, $\gamma\gamma$-coin, $\gamma(\theta)$ for 4 γ rays. Deduced 12 level energies from Eγ, J for 4 levels, branching ratios, and mixing ratios.

1969In04: 6-8-MeV α beams were produced from the Chalk River MP tandem accelerator. A thick target (270 $\mu g/cm^2$) of natural sulfur on a tantalum backing and a thin target (50 $\mu g/cm^2$) of Sb_2S_3 on a gold backing were used. γ rays were detected using two Ge(Li) detectors of 15 and 40 cm^3 volume. Measured Eγ, Iγ. Deduced 6 level energies, branching ratios, 5 lifetimes using the Doppler Shift Attenuation Method (DSAM) and 1 lifetime for the 3163-keV level using the Recoil Distance Method (RDM), and transition strengths.

1974Lo17: 11.0, 12.0 and 14.7-MeV α beams were produced from the Oliver Lodge Laboratory at the University of Liverpool. Targets were natural CdS. Protons were detected using an annular ΔE-E telescope close to 180° with FWHM=80 keV. γ rays were detected using five NaI(Tl) detectors and one Ge(Li) detector. Measured Eγ, Iγ, pγ-coin, p$\gamma(\theta)$, $\gamma\gamma$-coin, γ-ray linear polarization. Deduced J and π for 3 levels, 6 γ branching ratios, 5 mixing ratios, and lifetimes for a 5408-keV level using DSAM and for a 6088-keV level using RDM.

1973Br26: a 11.2-MeV α beam was produced from the Legnaro (Padova) Van de Graaff generator. Targets were natural CdS. γ rays were detected using a Ge(Li) detector. Deduced the lifetime using the Recoil Distance Method (RDM) for a level at 3163 keV.

1973An01: 5.5- and 8.6-MeV α beams of 100-200 nA were produced from the Oxford Vertical Van de Graaff (Injector) or the Oxford EN Tandem. Targets were natural sulphur in AgS. γ rays were detected using a Ge(Li) detector at 0°. Deduced the lifetime using the Recoil Distance Method (RDM) for a level at 3163 keV.

^{35}Cl Levels

E(level)[†]	J^π[‡]	$T_{1/2}$	Comments
0	$3/2^+$		J^π: from the Adopted Levels.
1219.4 *6*	$1/2^+$	0.15 ps *+7−6*	J^π: from the Adopted Levels. $T_{1/2}$: Lifetime τ=0.21 ps *+10−8* (1969In04 DSAM).
1763.3 *6*	$5/2^+$	0.38 ps *11*	J^π: spin=5/2 from p$\gamma(\theta)$ in 1970Ho09; parity=+ from 1763.2γ M1+E2 to $3/2^+$. $T_{1/2}$: Lifetime τ=0.55 ps *15* from 1969In04 using DSAM.
2645.8 *8*	$7/2^+$	0.21 ps *6*	J^π: spin=7/2 from p$\gamma(\theta)$ in 1970Ho09; parity from 882γ, M1+E2 to $5/2^+$ in 1970Ho09. $T_{1/2}$: Lifetime τ=0.30 ps *9* from 1969In04 using DSAM.
2693.6 *13*	$3/2^+,5/2^+$	49 fs *21*	$T_{1/2}$: Lifetime τ=70 fs *30* from 1972Br33 using DSAM. Other: τ < 30 fs (1969In04, DSAM).
3002.6 *15*	$(5/2)^+$	33 fs *6*	J^π: spin=3/2,5/2 from p$\gamma(\theta)$ in 1970Ho09; parity from 3002.5γ, M1+E2 to $3/2^+$ in 1970Ho09; J=3/2 is disfavored since it would give a large δ(E2/M1)=1.2 *5* for 3002.5γ in 1970Ho09 which is less likely and also due to the absence of an expected γ to 1219.5, $1/2^+$ level if J=3/2. $T_{1/2}$: Lifetime τ=48 fs *9* from 1972Br33 using DSAM. Other: τ<50 fs (1969In04, DSAM).
3163.1 *11*	$7/2^-$	42 ps *5*	J^π: spin=7/2 from p$\gamma(\theta)$ in 1970Ho09; parity from 3162.6γ, M2+E3 to $3/2^+$ in 1970Ho09.

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^{32}S(α,pγ) 1970Ho09,1972Br33,1973Al22 (continued)

^{35}Cl Levels (continued)

E(level)†	Jπ‡	$T_{1/2}$	Comments
			$T_{1/2}$: Lifetime τ=60 ps *7* from 1969In04.
3915 *2*		<15.3 fs	E(level): from 1972Br33. $T_{1/2}$: Lifetime τ<22 fs from 1972Br33 using DSAM.
3943.4 *12*	9/2	0.229 ps *35*	$T_{1/2}$: Lifetime τ=0.33 ps *5* from 1972Br33 using DSAM.
3968 *2*		<37 fs	E(level): from 1972Br33. $T_{1/2}$: Lifetime τ<54 fs from 1972Br33 using DSAM.
4058 *3*		22 fs *9*	E(level): weighted average of 4045 *10* (1970Ho09) and 4058 *2* (1972Br33). $T_{1/2}$: Lifetime τ=31 fs *13* from 1972Br33 using DSAM.
4108 *2*	3/2,7/2	49 fs *11*	E(level): weighted average of 4110 *10* (1970Ho09) and 4108 *2* (1972Br33). $T_{1/2}$: Lifetime τ=70 fs *16* from 1972Br33 using DSAM.
4171 *2*		47 fs *17*	E(level): weighted average of 4175 *10* (1970Ho09) and 4171 *2* (1972Br33). $T_{1/2}$: Lifetime τ=68 fs *24* from 1972Br33 using DSAM.
4347.8 *12*	9/2⁻	2.0 ps *+10−5*	Jπ: spin from pγ(θ) in 1974Lo17; parity from 1184.6γ, M1+E2 to 7/2⁻ in 1974Lo17. Other: J=5/2,9/2 from pγ(θ) in 1970Ho09 and pγ(θ) in 1973Al22. $T_{1/2}$: Lifetime τ=2.9 ps *+14−7* from 1972Br33 using DSAM.
4618 *2*	3/2,5/2	40 fs *17*	E(level): from 1972Br33. $T_{1/2}$: Lifetime τ=58 fs *24* from 1972Br33 using DSAM.
4770# *15*	7/2,9/2		
4880.8 *21*	7/2	0.19 ps *4*	E(level): other: 4882 *2* from 1972Br33, 4885 *20* from 1970Ho09. $T_{1/2}$: Lifetime τ=0.28 ps *6* from 1972Br33 using DSAM.
5015# *20*			
5175# *20*			
5230# *20*	1/2,3/2,5/2		
5407.1 *15*	11/2⁻	0.28 ps *7*	E(level): other: 5415 *20* from 1970Ho09. Jπ: spin from pγ(θ) in 1974Lo17 and from γ(θ) and yield function in 1973Al22; parity from 1059.3γ, M1+E2 to 9/2⁻ in 1974Lo17. $T_{1/2}$: Lifetime τ=0.4 ps *1* from 1974Lo17 using DSAM.
5535# *20*			
5600# *20*			
5650# *20*	5/2,9/2		
5850# *25*	5/2,9/2		
5927.1 *19*	7/2,11/2		E(level): other: 5930 *25* from 1970Ho09. Jπ: spin from γ(θ) in 1973Al22.
6087.0 *18*	13/2⁻	6.5 ps *6*	Jπ: spin from pγ(θ) in 1974Lo17; parity from 679.9γ, M1+E2 to 11/2⁻ in 1974Lo17. Other: J=9/2,13/2 from γ(θ) in 1973Al22. $T_{1/2}$: Lifetime τ=9.3 ps *8* from 1974Lo17 using RDM.
6400# *25*			

† From a least-squares fit to Eγ values with uncertainties, unless otherwise noted.
‡ Spin assignments are from pγ(θ) in 1970Ho09, unless otherwise noted.
\# From 1970Ho09, derived from measured E_p.

γ(^{35}Cl)

E_i(level)	J^π_i	Eγ†	Iγ‡	E_f	J^π_f	Mult.#	δ#	Comments
1219.4	1/2⁺	1219.4 *6*	100	0	3/2⁺			Eγ: weighted average of 1219.3 *6* (1969In04) and 1219.5 *10* (1973Al22).
1763.3	5/2⁺	543.9	<0.5	1219.4	1/2⁺			
		1763.2 *6*	100	0	3/2⁺	M1+E2	3.0 *1*	Eγ: from 1969In04. Other: 1763.1 *10* (1973Al22).
2645.8	7/2⁺	882.4 *10*	11.1 *22*	1763.3	5/2⁺	M1+E2	+0.25 *5*	Iγ: from 1970Ho09. Other: 18 (1969In04), 10 (1973Al22).

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^{32}S(α,pγ) 1970Ho09,1972Br33,1973Al22 (continued)

γ(^{35}Cl) (continued)

E_i(level)	J_i^π	E_γ[†]	I_γ[‡]	E_f	J_f^π	Mult.[#]	δ[#]	Comments
2645.8	7/2+	1426.4	<1.2	1219.4	1/2+			
		2646.2 *15*	100.0 *22*	0	3/2+			E_γ: from 1969In04. Other: 2646.0 *20* (1973Al22). I_γ: from 1970Ho09. Other: 100 (1969In04), 100 (1973Al22).
2693.6	3/2+,5/2+	930.5 *15*	18.2 *26*	1763.3	5/2+			I_γ: from 1970Ho09.
		1474.2	11.7 *26*	1219.4	1/2+			I_γ: from 1970Ho09. Other: 10 (1969In04).
		2693.2 *20*	100 *4*	0	3/2+	M1+E2		I_γ: from 1970Ho09. Other: 100 (1969In04), 100 (1973Al22). δ: δ=+0.17 *8* for J=3/2; δ=−0.26 *5* for J=5/2 (1970Ho09).
3002.6	(5/2)+	309.0	<5	2693.6	3/2+,5/2+			
		356.8	<1	2645.8	7/2+			
		1239.3	<2	1763.3	5/2+			I_γ: other: <4 (1972Br33).
		1783.2	<2	1219.4	1/2+			I_γ: from 1969In04, 1972Br33.
		3002.5 *15*	100	0	3/2+	M1+E2	+0.09 *3*	E_γ: weighted average of 3003 *2* (1969In04), 3003 *2* (1972Br33), and 3002.0 *15* (1973Al22). δ: other: 1.2 *5* if J(3002.6)=3/2 (1970Ho09).
3163.1	7/2−	160.5	<0.54	3002.6	(5/2)+			
		469.5[&]	<0.54	2693.6	3/2+,5/2+			
		517.4 *15*	11.1 *33*	2645.8	7/2+			I_γ: from 1970Ho09. Other: 8.7 (1969In04), 11 (1973Al22).
		1399.8	<1.1	1763.3	5/2+			
		3162.6 *15*	100.0 *33*	0	3/2+	M2+E3	−0.14 *2*	I_γ: from 1970Ho09. Other: 100 (1969In04), 100 (1973Al22).
3915		2696	19 *5*	1219.4	1/2+			
		3915	100 *5*	0	3/2+			
3943.4	9/2	1297.4 *15*	8.1 *21*	2645.8	7/2+			I_γ: weighted average of 9.9 *22* (1970Ho09) and 6.4 *21* (1972Br33). Other: 11 (1973Al22).
		2180.2 *15*	100.0 *21*	1763.3	5/2+			I_γ: others: 100.0 *22* (1970Ho09), 100 (1973Al22).
3968		2749	100	1219.4	1/2+			
		3968	<2	0	3/2+			
4058		2295	8 *4*	1763.3	5/2+			
		2839	100 *4*	1219.4	1/2+			
		4058	<2.2	0	3/2+			
4108	3/2,7/2	2345	16 *5*	1763.3	5/2+			
		4108	100 *5*	0	3/2+			
4347.8	9/2−	1184.6 *10*	100.0 *31*	3163.1	7/2−	M1+E2	−0.36 *3*	I_γ: from 1974Lo17. Others: 100 *7* (1970Ho09) and 100 *4* (1972Br33). δ: from 1974Lo17. Other: −0.40 *+8−9* (1973Al22); +0.42 *4* (1970Ho09). A_2=−0.95 *3*, A_4=+0.08 *3* (1974Lo17). POL=−0.005 *40* (1974Lo17).
		1702.1 *15*	52 *7*	2645.8	7/2+	E1(+M2)	0.00 *3*	I_γ: unweighted average of 61 *7* (1970Ho09), 39 *4* (1972Br33), and 56.3 *31* (1974Lo17). Mult.,δ: D(+Q) and δ from pγ(θ) and electric character from γ(pol) in 1974Lo17. A_2=−0.34 *5*, A_4=+0.04 *5* (1974Lo17). POL=+0.36 *7* (1974Lo17).

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$^{32}S(\alpha,p\gamma)$ 1970Ho09,1972Br33,1973Al22 (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ †	I_γ ‡	E_f	J^π_f	Mult.#	δ#	Comments
4618	3/2,5/2	4618 *2*	100	0	$3/2^+$			E_γ,I_γ: from 1972Br33. δ: +0.08 *14* for J=3/2; −0.36 *15* for J=5/2 (1970Ho09).
4880.8	7/2	2187.1@	52@ *9*	2693.6	$3/2^+,5/2^+$			E_γ,I_γ: Final level is uncertain, either 2646 or 2694 (1970Ho09).
		2234.9@	52@ *9*	2645.8	$7/2^+$			E_γ,I_γ: Final level is uncertain, either 2646 or 2694 (1970Ho09).
		3117.4 *20*	100 *9*	1763.3	$5/2^+$			I_γ: from 1970Ho09. Other: 100 (1973Al22).
5230	1/2,3/2,5/2	5230	100	0	$3/2^+$			
5407.1	$11/2^-$	1059.3 *10*	9.9 *22*	4347.8	$9/2^-$	M1+E2	+0.25 *8*	Mult.,δ: D+Q and δ from $\gamma(\theta)$ and magnetic character from γ(pol) (1974Lo17). A_2=+0.19 *11*, A_4=−0.01 *13* (1974Lo17). POL=−0.40 *14* (1974Lo17).
		2244 *3*	100.0 *22*	3163.1	$7/2^-$	E2		Mult.: Q(+O) with δ(O/Q)=0.00 *3* from p$\gamma(\theta)$ in 1974Lo17; M2 ruled out by RUL. Other: Q(+O) with δ(O/Q)=+0.02 *9* (1973Al22). A_2=+0.44 *3*, A_4=−0.19 *3* (1974Lo17).
5850	5/2,9/2	2687	100	3163.1	$7/2^-$	D+Q		Mult.,δ: from p$\gamma(\theta)$ in 1970Ho09 with δ(Q/D)=−2.2 *5* or −0.6 *1* for J=5/2; +0.28 *4* for J=9/2 (1970Ho09).
5927.1	7/2,11/2	1579.3 *15*	100	4347.8	$9/2^-$	D+Q		Mult.,δ: from $\gamma(\theta)$ in 1973Al22, with δ(Q/D)=+1.1 *5* for J=7/2; −0.8 *4* for J=11/2.
6087.0	$13/2^-$	679.9 *10*	100 *6*	5407.1	$11/2^-$	M1+E2	+0.020 *15*	Mult.,δ: D+Q and δ from p$\gamma(\theta)$ and magnetic character from γ(pol) in 1974Lo17. Other: δ=0.00 *7* for J_i=13/2 (1973Al22). A_2=−0.22 *1*, A_4=0.005 *15* (1974Lo17). POL=−0.45 *5* (1974Lo17).
		1739.2	18 *6*	4347.8	$9/2^-$			

† Values with uncertainties are from 1973Al22 and those without uncertainties are from level-energy differences, unless otherwise noted. 1973Al22 is the only literature in this dataset that reported measured Eγ with uncertainties. 1969In04 and 1972Br33 reported E(level) with uncertainties derived from their measured Eγ but did not report Eγ. Evaluators adopt the uncertainties of E(level) from 1969In04 and 1972Br33 as the uncertainties of an Eγ if this γ transition populates the ground state and is also the dominant decay branch of that level.

‡ From 1969In04 for γ rays from level up to 3163, from 1972Br33 for γ rays from level above 3163 up to 4618 level, and from 1974Lo17 for γ rays from level above 4618, unless otherwise noted.

Mult=D+Q, Q or Q+O and δ are determined from p$\gamma(\theta)$ in 1970Ho09; magnetic or electric characters are determined based on RUL and measured level $T_{1/2}$ where available, unless otherwise noted.

@ Multiply placed with undivided intensity.

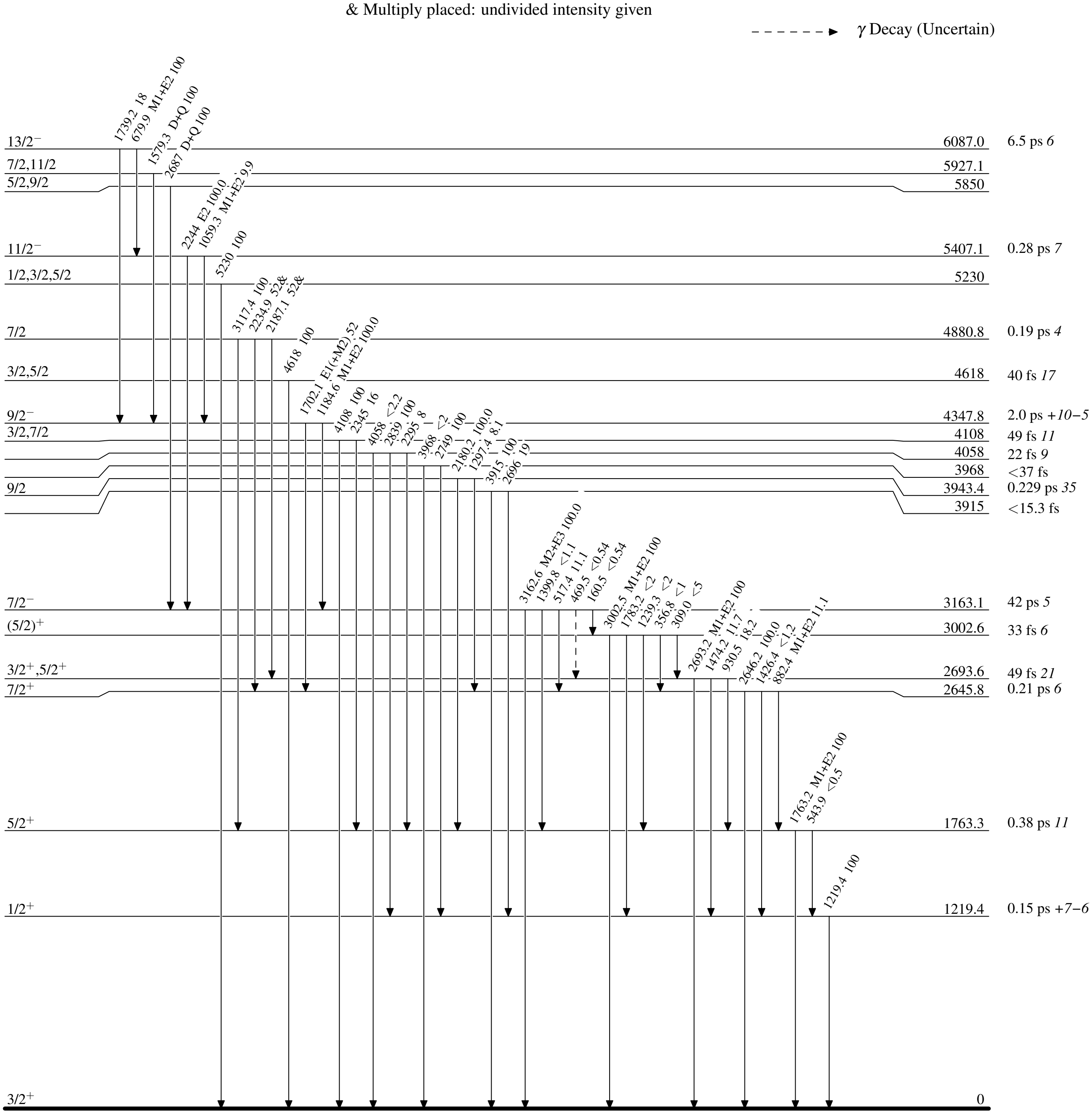
& Placement of transition in the level scheme is uncertain.

$^{32}S(\alpha,p\gamma)$ 1970Ho09,1972Br33,1973Al22

Level Scheme

Intensities: Relative photon branching from each level
& Multiply placed: undivided intensity given

Legend



$^{35}_{17}Cl_{18}$

^{33}S(α,d) 1975Na18,1977Na10

J^{π}=3/2^{+} for ^{33}S ground state.

1975Na18,1977Na10: a 40-MeV α beam was produced from the Michigan State University cyclotron. Targets were 76.8%-enriched ^{33}S of 15 μg/cm^2 sandwiched between layers of Formvar and carbon foils. Reaction products were detected in the focal plane of an Enge split-pole magnetic spectrograph with a proportional-counter plastic-scintillator combination with FWHM=40-60 keV. Measured $\sigma(E_d,\theta)$. Deduced levels, J, π, and L-transfers.

^{35}Cl Levels

E(level)†	L†	Comments
6200 *10*	6	
7170 *10*	6	
7670 *10*	6	
7750 *10*	6	
7870 *10*	4+6	L: 1977Na10 states that this is excited by a mixture of L=4 and 6 transfers. The L=6 admixture can be attributed to a $(f_{7/2})^2_{7,0}$ transfer whereas the L=4 component is associated with a mixture of $(f_{7/2})^2_{5,0}$ and $(f_{7/2},p_{3/2})_{5,0}$ transfers.
8010 *10*	6	
8100 *10*	6	
8700 *10*	6	
8840 *10*	6	
9150 *10*	6	
9450 *10*	6	

† From 1977Na10. Experimental angular distributions of L=4, 5, 6 to ^{42}Sc and ^{34}Cl levels with known J^{π} are compared with the $\sigma(E_d,\theta)$ measured in 1977Na10 to assign L-transfers.

$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10

$J^\pi=0^+$ for ^{34}S ground state.

1976Me12: 0.7-2.1-MeV proton beams were produced from the 1.1-MV Cockroft-Walton generator and a 3-MV Van de Graaff accelerator. Targets were ZnS (46% ^{34}S) and CdS (37% ^{34}S) on tantalum backings. γ rays were detected using a 10-cm by 10-cm NaI detector and a 40-cm^3 Ge(Li) detector. Measured $\sigma(E_p)$, Eγ, $\gamma(\theta)$. Deduced level energies for 33 bound states and 47 resonances, J, π, resonance strengths, γ branching ratios, mixing ratios, and lifetimes for 25 bound states using DSAM.

1976Sp08: 1.21,1.95-2.91-MeV protons were produced from the Utrecht University 3-MV Van de Graaff accelerator. Targets were Ag^2S (95% ^{34}) evaporated onto a tantalum backing. γ rays were detected using a 10 cm by 10 cm diameter NaI scintillator and a 120- or 60-cm^3 Ge(Li) detector. Measured reaction yields, Eγ. Deduced Q value, 84 resonance energies, resonance strengths, widths, J, π, and γ-branching ratios.

1976Sp09: similar setup as 1976Sp08 with 2.33-2.79-MeV protons. Measured reaction yields, Eγ, $\gamma(\theta)$. Deduced levels, J, π, and γ-branching ratios.

1972Hu10: 0.7-2.1-MeV protons were produced from the 4-MV Van de Graaff at the Centre d′Etudes Nucleaires de Bordeaux-Gradignan. Targets were 100-μg/cm^2 Ag_2S (90% ^{34}S). γ rays were detected using a 12.7-cm by 12.7-cm NaI(Tl) detector and a 60-cm^3 Ge(Li) detector. Measured $\sigma(E_p)$, Eγ, Iγ. Deduced 59 resonances, 32 bound states, resonance strengths, and γ branching ratios.

1972Hu11: similar setup as 1972Hu10. Deduced lifetimes using DSAM.

1973Fa07: 1-2-MeV protons were produced from the Helsinki 2-MV Van de Graaff accelerator. Targets were ^{34}S on tantalum backings. γ rays were detected using a 55-cm^3 Ge(Li) detector. Measured reaction yields, Eγ, Iγ. Deduced 28 resonances, resonance strengths, γ-branching ratios, and 9 lifetimes using DSAM.

2001Vo24: 1.183-2.820-MeV protons were produced from the 4-MeV electrostatic Van de Graaff accelerator (NSC KhPEI). Enriched ^{34}S target (100%). A 63 cm^3 Ge(Li) detector and a 15-cm-diam by 10-cm-long NaI(Tl) detector. Measured $\sigma(E_p)$, Eγ, Iγ. Deduced 12 resonances and resonance strengths.

2002Vo17: 2.187-MeV protons were produced from the ESU-4 accelerator (NSC KhPEI). Enriched ^{34}S target. A 63 cm^3 Ge(Li) detector. Measured Eγ, Iγ. Deduced levels, resonance strengths, widths.

1981Bi05: 1.4-2.8-MeV protons were produced from the Groningen 5-MV Van de Graaff accelerator. Targets were 4 and 1-μg/cm^2 Sb_2S_3 (90% in ^{34}S) evaporated onto 10-μg/cm^2 carbon backings. Scattered protons were detected using three silicon surface barrier detectors. γ rays were detected using a 10-cm by 10-cm NaI(Tl) detector and a Ge(Li) detector. Measured $\sigma(E_p,\theta)$, Eγ, $\gamma(\theta)$ for two γ rays. Deduced levels, J, π, widths, and resonance strengths.

1971Wi13: protons were produced from the Ontario Cancer Institute. Targets were CdS. γ rays were detected using NaI(Tl) and Ge(Li) detectors. Measured $\sigma(E_p)$,Eγ, Iγ, $\gamma(\theta)$, γ-linear polarization. Deduced 1.020-, 1.214-, 1.267-, and 1.510-MeV resonances, J^π, branchings, mixing ratios, and half-lives using DSAM.

1971Pr11: 1.211-MeV protons were produced from the 2-MeV ARL Van de Graaff at the Aerospace Research Laboratories (ARL). Targets were Ag_2S (85.6% ^{34}S). γ rays were detected using an 80-cm^3 Ge(Li) detector. Measured $\sigma(E_p)$, Eγ, Iγ, $\gamma(\theta)$, pγ-delay. Deduced resonances, levels, J, π, branchings.

1963Ha32: 0.7-1.2-MeV protons were produced from the Laboratorium Technische Fysika. Target was ^{34}S. γ rays were detected using NaI(TI) scintillation spectrometers. Measured $\sigma(E_p)$, Eγ, Iγ. Deduced levels, branchings.

1967Da12: 0.8-2.5-MeV protons were produced from the 2.5 MeV Van de Graaff generator of the U.A.R. Atomic Energy Establishment. Targets were ^{34}S on tantalum backings. γ rays were detected using two 15.2-cm-diam by 12.7-cm-thick NaI(Tl) crystals. Measured $\sigma(E_p)$, Eγ, Iγ. Deduced resonances, levels, branchings.

1960An06: 0.7-1.9-MeV protons were produced from the electrostatic generator of the Physical-Technical Institute of the Ukrainian SSR Academy of Sciences. Targets were isotopic ^{34}S (4.2%). γ rays were detected using a 2-cm-thick by 3-cm-diam CsI(Tl) crystal. Measured $\sigma(E_p)$, Eγ, $\gamma(\theta)$. Deduced levels, J, branchings.

1966Wa09: 1.214 MeV proton beam. Measured Eγ, γ-polarization. Deduced level energy and J, π for the level of 3163 and 7540 level.

1966En04: 0.3-2.1-MeV protons were produced from the Utrecht 850-keV Cockcroft-Walton generator and the 3-MV Van de Graaff generator. Targets were natural sulphur compound on tantalum backings. γ rays were detected using a cylindrical 10-cm-by-10-cm NaI crystal. Measured $\sigma(E_p)$, Eγ. Deduced the resonance strength for E_p=1214 keV.

1966Az01: 1.214-MeV protons were produced from the Ontario Cancer Institute 3-MeV Van de Graaff accelerator. Targets were CdS with enriched ^{34}S (46%). γ rays were detected using a Compton polarimeter. Measured Eγ, Iγ, $\gamma(\theta)$. Deduced levels, J, π, branchings.

1967Wa22: 1.214- and 1.512-MeV protons were produced from the Aerospace Research Laboratories. Targets were enriched CdS (40% ^{34}S) and Ag_2S (86% ^{34}S). γ rays were detected using two large volume NaI(Tl) detectors. Measured $\sigma(E_p)$, Eγ, Iγ, $\gamma(\theta)$, γ-polarization. Deduced levels, J, π, branchings.

^{34}S(p,γ) 1976Me12,1976Sp08,1972Hu10 (continued)

1967Ta08: 1.020-, 1.213- and 1.513-MeV protons were produced from the 3-MV Van de Graaff accelerator of the Ontario Cancer Institute. Targets were CdS (46% ^{34}S) on thick metal foils. γ rays were detected using a 7.6-cm-long by 7.6-cm-diam NaI(Tl) detector. Measured $\sigma(E_p)$, Eγ, Iγ, $\gamma(\theta)$, γ-polarization. Deduced levels, J, π, branchings, mixing ratios.

1977Ko35: 1.055, 1.183, 1.214, and 1.416 MeV protons were produced from the 4 MV electrostatic accelerator of the Physical-Technical Institute of the Ukrainian SSR Academy of Sciences. Target of ^{34}S. A 40 cm^3 coaxial Ge(Li) detector. Measured $\sigma(E_p)$, Eγ, $\gamma(\theta)$. Deduced levels, branchings, mixing ratios.

1977Ko34: 1.0-1.4 MeV protons. Measured Eγ, Iγ, $\gamma(\theta)$. Deduced levels, J^π, branchings, mixing ratios.

1970Ko33: 2.317-, 2.223-, and 2.336-MeV protons were produced from the 4-MV electrostatic accelerator of the Physical-Technical Institute of the Ukrainian SSR Academy of Sciences. Targets were ^{34}S on tantalum backings. γ rays were detected using a 12-cm^3 coaxial Ge(Li) detector. Measured Eγ, Iγ, $\gamma(\theta)$. Deduced 3 levels, J, and branching ratios.

1967Ko19,1967Ko20,1967Ko23: 2.079-MeV protons were produced from the 4 MV electrostatic accelerator of the Physical-Technical Institute of the Ukrainian SSR Academy of Sciences. Targets were made by implanting ^{34}S ions in a tantalum substrate. γ rays were detected using a 70-mm by 50-mm and a 70-mm by 60-mm NaI(Tl) detectors. Measured Eγ, Iγ, p$\gamma\gamma(\theta)$. Deduced levels, J, π, branchings, mixing ratios.

1968Az02: 1.214- and 1.513-MeV protons were produced from the Ontario Cancer Institute 3-MeV Van de Graaff accelerator. Targets were CdS (37% ^{34}S) evaporated onto gold backings. γ rays were detected using a 5-cm-diam by 5-cm-long NaI(Tl) crystal. Measured Eγ, $\gamma\gamma$-delay. Deduced lifetimes of the level of 3163 keV using the delayed coincidence method and DSAM.

1969Mi23: 2.079-MeV protons. ^{34}S target. Ge(Li) detector. Measured Eγ, Iγ. Deduced levels, branchings.

1971Ko32,1971Ko46: 0.8-2.6-MeV protons. Targets were made by evaporating ^{34}S ions onto tantalum backings. γ rays were detected using Ge(Li) and NaI(Tl) detectors. Measured $\sigma(E_p)$, Eγ, Iγ, and p$\gamma\gamma(\theta)$. Deduced levels, J, π, level widths, and branchings.

1971Ba23: 1.214-MeV protons were produced from the 3 MeV Dynamitron of Carleton and Ottawa Universities. Target of CdS (93% ^{34}S) on a gold backing. A 28 cm^3 Ge(Li) detector. Measured Eγ, Iγ. Deduced lifetime of the 3163 keV level using the delayed coincidence method.

1974Al04: 1.214-MeV protons. Targets were Ag_2S, CdS, and ZnS of 300-μg/cm^2. Ge(Li) detector. Measured $\sigma(E\gamma)$. Deduced resonance strengths.

1975Ke11: 1.214-MeV protons were produced from the Helsinki University 2.5-MV van de Graaff. Targets of CdS and ZnS on tantalum backings. γ rays were detected using a 120 cm^3 Ge(Li) detector with FWHM=2.9 keV at 2.6 MeV. Measured $\sigma(E\gamma)$. Deduced resonance strengths.

1979Pa16: 1.211-MeV protons were produced from the University of Melbourne 5U Pelletron accelerator. Natural sulphur target. NaI(Tl) and Ge(Li) detector. Measured $\sigma(E_p)$. Deduced resonance strength.

1988Va06: 0.4-0.7 and 1.47 MeV protons were produced from the Utrecht 3 MV Van de Graaff accelerator. Targets of Ag_2S (90% ^{34}S) of 10 and 34 μg/cm^2 on Ta backings. A 80-cm^3 coaxial Philips Ge(Li) or an γ-X intrinsic Ge detector, FWHM=1.85 and 1.75 keV at 1.33 MeV. Measured $\sigma(E_p)$, Eγ, Iγ, $\gamma(\theta)$. Deduced levels, J^π, branchings, mixing ratios.

1988Co03: 1.211-MeV protons were produced from the Geel 7-MV CN-type Van de Graaff. Targets of ^{34}S. Ge(Li) detector. Measured $\gamma\gamma$-coin. Deduced weighting function.

1996Vo20: 1.891, 1.900 and 2.070-MeV protons were produced by an electrostatic accelerator ESU-4 at the Physical and Technical Institute. Target made by implanting ^{34}S ions in a tantalum backing. A 63 cm^3 Ge(Li) detector. Measured Eγ, Iγ. Deduced levels, gamma widths.

1996Ka21: 1-3 MeV protons were produced from the electrostatic accelerator of the Institute of Physica and Technology (Kharkov). Targets were 20-μg/cm^2 Ag_2S. γ rays were detected using a 60 cm^3 Ge(Li) detector and a 150×100-mm^2 NaI(Tl) as a monitor. Measured Eγ, Iγ, $\gamma(\theta)$. Deduced J^π for 3 resonances, resonance strengths.

2000Vo21: 1.794 and 2.275 MeV protons. Measured Eγ, Iγ. Deduced gamma widths, transition strengths.

2001Ka69: 0.8-3 MeV protons produced from the ESU-4 accelerator (NSC KhPEI). Target of ^{34}S. A 15-cm-diam by 19-cm-long NaI(Tl) detector. Measured Eγ, Iγ. Deduced levels, transition strengths.

Others: 1962Du05, 1963Du08, 1964Hy01, 1966Ko22, 1966Ko15, 1967Ma48, 1969Ko26, 1969Ah02, 1969KrZW, 1972LeYW, 1974Il01, 1974Bi16, 1981Va02.

Theoretical studies: 2020Ri06 (shell model).

Standard resonance strength $\omega\gamma=(2J_r+1)/[(2J_p+1)(2J_T+1)]\times\Gamma_\gamma\Gamma_p/\Gamma$, where J_r is the spin of the resonance, J_p=1/2 is the spin of the proton, and J_T=0 is the spin of the target ^{34}S ground state. The resonance strengths S from the original papers in this dataset are mainly obtained from resonance γ-ray yields and are defined without the ($2J_p$+1) factor. Therefore, to obtain the standard resonance strength, all the resonance strengths provided in the comments below should be divided by 2.

S=21 eV *3* at E_p=1213 measured by 1966En04 has been used as a normalization standard by 1971BrXT, 1972Hu10, 1973Fa07,

^{34}S(p,γ) 1976Me12,1976Sp08,1972Hu10 (continued)

1974Bi16, and 1976Me12. 1971BrXT measured ^{34}S(p,p) elastic scattering and determined Γ_p=4.6 eV *10* and Γ=5 eV *+4−2* from a Q^2 minimization analysis of the excitation function. 1971BrXT added $\Gamma_p\Gamma_\gamma/\Gamma$=2.6 eV *4* converted from the S measured by 1966En04 to constrain their Q^2 analysis and reported a final Γ_p=5.8 eV *7*, Γ_γ=4.7 eV *17*, and Γ=10.5 eV *15*. 1974Bi16 measured ^{34}S(p,p) and determined Γ_p=7.2 eV *15* from a Q^2 minimization analysis of the excitation function and quoted S=21 eV *3* (1966En04) to deduce Γ_γ=3.8 eV *15* and Γ=11.0 eV *15*. 1974Al04 determined an absolute S_{1213}=9.7 eV *9*. 1975Ke11 determined an absolute S_{1213}=9.8 eV *10* and quoted 1974Bi16′s relative ratio of Γ_p/Γ_γ=7.2/3.9=1.9 to deduce Γ_p=3.8 eV *13*, Γ_γ=2.0 eV *13*, and Γ=5.8 eV *13*. 1976Sp09 measured the resonant absorption of the E_p=2791 resonance, which indicates that normalization using S_{1213}=9.7 eV gives a better agreement than that using S_{1213}=21 eV. 1981Bi05 determined an absolute S=4.1 eV *5* for the E_p=1892 resonance and quoted the relative ratio of S_{1213}/S_{1892} from 1978En02 to deduce S=9.5 eV *12* for E_p=1213 resonance. The S_{1213}/S_{1892}=2.3 quoted by 1981Bi05 was based on two S values: first, 1978En02 quoted S=4.2 eV for a 1892+1893 resonance doublet, which was scaled down by a factor of 21/9.7 from 1976Me12′s original S_{1892}=9.0 eV; second, 1978En02 quoted S_{1213}=9.7 eV *7* from 1976Sp08, 1974Al04, and 1975Ke11. 1979Pa16 determined S=9.0 eV *10* using the ^{27}Al(p,γ)^{28}Si resonances as normalization standards. 1996Ka21 normalized all their S to S_{1213}=9.7 eV *7* and cited 1976Sp08 but listed this value in the “this study” column. 2001Vo24 normalized all S to S_{1213}=9.7 eV *7* and cited 1976Sp08 and 1975Ke11. However, 2001Vo24 also reported a different S_{1213}=9.9 eV *7* without giving details. Although 1976Sp08 is frequently cited as the source of S_{1213}=9.7 eV *7*, they did not measure the absolute S value; instead, they calculated a weighted average of 1974Al04 and 1975Ke11.

In summary, the evaluators adopt S_{1213}=9.5 eV *9*: weighted average of all independently measured S_{1213}: 9.7 eV *9* (1974Al04), 9.8 eV *10* (1975Ke11), and 9.0 eV *10* (1979Pa16). All resonance strengths given in the comments below have been normalized to this value by the evaluators. Γ_p=4.6 eV *10* and Γ=5 eV *+4−2* from (p,p) excitation function analysis in 1971BrXT are likely the only width data that are not influenced by the questionable S_{1213}=21 eV *3*.

^{35}Cl Levels

E(level)†	J^π‡	$T_{1/2}$	Comments
0	3/2$^+$		
1219.39 *7*	1/2$^+$	105 fs *14*	E(level): weighted average of 1219.1 *5* (1972Hu10) and 1219.3 *1* (1976Me12), and 1219.44 *7* (1981Bi05). $T_{1/2}$: lifetime τ=152 fs *20*: weighted average of 80 fs *40* (1971Wi13, DSAM), 175 fs *20* (1972Hu11, DSAM), 150 fs *20* (1973Fa07, DSAM), and 145 fs *30* (1976Me12, DSAM).
1763.14 *4*	5/2$^+$	0.347 ps *63*	E(level): weighted average of 1762.8 *5* (1972Hu10), 1763.4 *7* (1976Me12), and 1763.14 *4* (1981Bi05). J^π: from $\gamma(\theta)$ in 1971Wi13. $T_{1/2}$: lifetime τ=0.50 ps *9*: weighted average of 0.50 ps *10* (1972Hu11, DSAM), 0.40 ps *14* (1973Fa07, DSAM), and 0.53 ps *9* (1976Me12, DSAM).
2645.70 *8*	(5/2,7/2)	0.146 ps *21*	E(level): weighted average of 2645.6 *5* (1972Hu10), 2644.7 *13* (1976Me12), and 2645.71 *8* (1981Bi05). J^π: from $\gamma(\theta)$ in 1971Wi13. Other: 3/2 from $\gamma(\theta)$ in 1967Ko20. $T_{1/2}$: lifetime τ=0.210 ps *30*: weighted average of 0.255 ps *65* (1972Hu11, DSAM) and 0.200 ps *30* (1976Me12, DSAM).
2693.9 *3*	(3/2,5/2)	13.9 fs *21*	E(level): weighted average of 2694.0 *3* (1972Hu10), 2694.7 *12* (1976Me12), and 2693.8 *3* (1981Bi05). J^π: from $\gamma(\theta)$ in 1971Wi13. $T_{1/2}$: lifetime τ=20 fs *3*: weighted average of 21 fs *3* (1972Hu11, DSAM), 19 fs *4* (1973Fa07, DSAM), and 20 fs *4* (1976Me12, DSAM).
3002.75 *8*	5/2	11.1 fs *21*	E(level): weighted average of 3002.7 *7* (1972Hu10), 3003.7 *8* (1976Me12), and 3002.74 *8* (1981Bi05). J^π: 5/2 from $\gamma(\theta)$ in 1967Ko20 and 1971Wi13. $T_{1/2}$: lifetime τ=16 fs *3*: weighted average of 22 fs *3* (1972Hu11, DSAM), 14 fs *2* (1973Fa07, DSAM), and 16 fs *4* (1976Me12, DSAM).
3162.99 *7*	7/2$^-$	25.7 ps *28*	T=1/2 E(level): weighted average of 3162.5 *5* (1972Hu10), 3163.9 *7* (1976Me12), and 3162.99 *6* (1981Bi05). J^π: spin from $\gamma(\theta)$ in 1966Wa09, 1967Wa22, 1967Ta08, and 1971Wi13; parity from 3163γ, M2+E3 to 3/2$^+$ ground state in 1966Wa09 and 1967Wa22.

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$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10 (continued)

^{35}Cl Levels (continued)

E(level)[†]	J^{π}[‡]	$T_{1/2}$	Comments
			$T_{1/2}$: lifetime τ=37 ps *4* (1971Ba23, delayed coincidence). Others: 140 ps *40* (1968Az02, delayed coincidence), >0.6 ps (1968Az02, DSAM), >0.2 ps (1971Wi13, DSAM), >2 ps (1972Hu11, DSAM), >0.9 ps (1973Fa07, DSAM), and >1 ps (1976Me12). 1971Ba23 states that the earlier 1968Az02 value was obtained using NaI(Tl) detectors with broad time resolution and the discrepancy between 1968Az02 and 1971Ba23 is not unreasonable.
3918.48 *10*	$3/2^+$	4.9 fs *14*	E(level): weighted average of 3917.9 *6* (1972Hu10), 3920.7 *13* (1976Me12), and 3918.48 *7* (1981Bi05). J^{π}: spin=3/2 determined by the feeding 5162$\gamma(\theta)$ from $5/2^+$, 9081 level in 1976Sp09; parity=+ determined by 3918γ, M1+E2 to $3/2^+$ ground state. $T_{1/2}$: lifetime τ=7 fs *2* (1972Hu11, DSAM). Other: <6 fs (1976Me12, DSAM).
3942.9 *2*		0.159 ps *42*	E(level): weighted average of 3942.6 *9* (1972Hu10), 3944.1 *11* (1976Me12), and 3942.9 *2* (1981Bi05). $T_{1/2}$: lifetime τ=0.23 ps *6*: weighted average of 0.39 ps *11* (1972Hu11, DSAM), 0.145 ps *50* (1973Fa07, DSAM), and 0.29 ps *5* (1976Me12, DSAM).
3967.3 *6*		11.8 fs *28*	E(level): from 1972Hu10. Other: 3979.0 *15* (1976Me12). 1976Me12 matched their 3979 level with the 3967 level observed in other work. No other work has observed a level at 3979 keV. $T_{1/2}$: lifetime τ=17 fs *4*: weighted average of 13 fs *4* (1972Hu11, DSAM), 20 fs *5* (1973Fa07, DSAM), and 20 fs *4* (1976Me12, DSAM).
4059.2 *4*		13.9 fs *21*	E(level): weighted average of 4058.4 *8* (1972Hu10), 4059.4 *4* (1976Me12), and 4059.2 *3* (1981Bi05). $T_{1/2}$: lifetime τ=20 fs *3*: weighted average of 22 fs *3* (1972Hu11, DSAM), 20 fs *4* (1973Fa07, DSAM), and 18 fs *3* (1976Me12, DSAM).
4113.7 *10*	7/2		E(level): weighted average of 4111 *3* (1972Hu10) and 4114.0 *10* (1976Me12). J^{π}: spin=7/2 from feeding 4516$\gamma(\theta)$ from 7/2, 8630 level in 1976Sp09.
4173.45 *19*		34.7 fs *63*	T=1/2 E(level): weighted average of 4173.0 *6* (1972Hu10), 4174.7 *10* (1976Me12), and 4173.45 *19* (1981Bi05). $T_{1/2}$: lifetime τ=50 fs *9*: weighted average of 105 fs *+35−30* (1971Wi13, DSAM), 55 fs *6* (1972Hu11, DSAM), and 34 fs *9* (1976Me12, DSAM).
4177.9 *2*	3/2	27.0 fs *28*	E(level): weighted average of 4177.2 *9* (1972Hu10), 4180.1 *15* (1976Me12), and 4177.9 *2* (1981Bi05). J^{π}: from $\gamma(\theta)$ in 1967Wa22. $T_{1/2}$: lifetime τ=39 fs *4*: weighted average of 32 fs *5* (1972Hu11, DSAM), 42 fs *4* (1973Fa07, DSAM), and 47 fs *10* (1976Me12, DSAM).
4347.76 *6*	5/2,9/2	0.229 ps *42*	E(level): weighted average of 4347.5 *8* (1972Hu10), 4347.2 *12* (1976Me12), and 4347.76 *6* (1981Bi05). J^{π}: spin=5/2,9/2 based on feeding 4282$\gamma(\theta)$ from 7/2, 8630 level in 1976Sp09. $T_{1/2}$: lifetime τ=0.33 ps *6* (1976Me12, DSAM). Other: >0.6 ps (1972Hu11, DSAM).
4624.4 *3*			E(level): weighted average of 4624.2 *20* (1976Me12) and 4624.4 *3* (1981Bi05). Other: 4625 *5* (1972Hu10). 15% of Iγ from this level decays to unknown final levels (1976Me12).
4769.86 *23*		76 fs *20*	E(level): weighted average of 4768.9 *12* (1972Hu10), 4766.9 *15* (1976Me12), and 4769.90 *15* (1981Bi05). $T_{1/2}$: lifetime τ=110 fs *29*: weighted average of 270 fs *95* (1972Hu11, DSAM) and 105 fs *17* (1976Me12, DSAM).
4839.10 *11*		9.7 fs *35*	E(level): weighted average of 4838 *3* (1972Hu10), 4841.7 *19* (1976Me12), and 4839.09 *11* (1981Bi05). $T_{1/2}$: lifetime τ=14 fs *5* (1976Me12, DSAM).
4854.4 *19*		4.9 fs *14*	E(level): weighted average of 4853 *2* (1972Hu10) and 4855.7 *19* (1976Me12). $T_{1/2}$: lifetime τ=7 fs *2* (1976Me12, DSAM). Other: <6 fs (1972Hu11, DSAM).
4880.96 *15*		5.6 fs *14*	E(level): weighted average of 4880.9 *1* (1972Hu10), 4885 *2* (1976Me12), and 4881.14 *20* (1981Bi05). $T_{1/2}$: lifetime τ=8 fs *2*: weighted average of 7 fs *2* (1972Hu11, DSAM), 7 fs *3* (1973Fa07, DSAM), and 10 fs *3* (1976Me12, DSAM).

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^{34}S(p,γ) 1976Me12,1976Sp08,1972Hu10 (continued)

^{35}Cl Levels (continued)

E(level)†	Jπ‡	$T_{1/2}$	E_p(lab) (keV)	Comments
5010.1 *18*		7.6 fs *21*		E(level): weighted average of 5008 *5* (1972Hu10) and 5010.4 *18* (1976Me12). $T_{1/2}$: lifetime τ=11 fs *3* (1976Me12, DSAM).
5163.34 *22*	7/2			E(level): weighted average of 5163 *5* (1972Hu10), 5166.7 *15* (1976Me12), and 5163.31 *14* (1981Bi05). Jπ: spin=3/2,7/2 from feeding 3466γ(θ) from 7/2, 8630 level in 1976Sp09. Spin=3/2 and δ(O/Q)=0.12 *1* would require a large E3 or M3 component, which is ruled out by RUL. 10% of Iγ transitions to unknown levels (1976Me12).
5215.8 *15*		<4.85 fs		E(level): weighted average of 5214 *3* (1972Hu10) and 5216.2 *15* (1976Me12). $T_{1/2}$: lifetime τ<7 fs (1976Me12, DSAM).
5403.5 *10*		11.8 fs *28*		E(level): weighted average of 5400 *5* (1972Hu10) and 5403.6 *10* (1976Me12). $T_{1/2}$: lifetime τ=17 fs *4* (1976Me12, DSAM). 20% of Iγ from this level decays (1976Me12). 50% of Iγ from this level decays to unknown levels (1972Hu10).
5520				E(level): from 1976Sp08.
5586 *2*				E(level): weighted average of 5587 *3* (1972Hu10) and 5586 *2* (1976Me12).
5599.7 *15*		2.08 fs *69*		E(level): weighted average of 5598 *3* (1972Hu10) and 5600.1 *15* (1976Me12). $T_{1/2}$: lifetime τ=3 fs *1* (1976Me12, DSAM).
5646 *2*		2.77 fs *69*		E(level): weighted average of 5645 *3* (1972Hu10) and 5646 *2* (1976Me12). $T_{1/2}$: lifetime τ=4 fs *1* (1976Me12, DSAM).
5655 *2*		13.9 fs *35*		E(level): weighted average of 5651 *3* (1972Hu10) and 5656 *2* (1976Me12). Jπ: Mirror level of ^{35}S ground state and thus $3/2^+$ (2001Vo24). Other: spin=5/2 based on the ϕeeding 2051g(θ) from the 5/2, 7706 level in 1977Ko34. $T_{1/2}$: lifetime τ=20 fs *5* (1976Me12, DSAM). Γ_γ=0.033 eV *10* (2001Vo24). 14% of Iγ from this level decays to unknown final levels (1976Me12).
5683 *2*				E(level): weighted average of 5682 *5* (1972Hu10) and 5683 *2* (1976Me12). 40% of Iγ from this level decays to unknown final levels (1976Me12).
5723.52 *18*				E(level): from 1981Bi05. 38% of Iγ from this level decays to unknown final levels (1976Sp09).
5758 *3*				E(level): weighted average of 5754 *5* (1972Hu10) and 5759 *3* (1976Me12). 25% of γ from this level decays to unknown final levels and the other 75% is attributed to two γ transitions (1976Me12). Other: 100% of γ from this level decays to unknown final levels (1972Hu10).
5806 *2*		3.47 fs *69*		E(level): weighted average of 5802 *5* (1972Hu10) and 5806 *2* (1976Me12). $T_{1/2}$: lifetime τ=5 fs *1* (1976Me12, DSAM).
6106.2 *21*		8.3 fs *21*		E(level): weighted average of 6102 *3* (1972Hu10) and 6107.2 *15* (1976Me12). $T_{1/2}$: lifetime τ=12 fs *3* (1976Me12, DSAM).
6181 *3*				E(level): from 1976Me12. 55% of Iγ from this level decays to unknown final levels (1976Me12).
6492 *3*				E(level): weighted average of 6489 *5* (1972Hu10) and 6493 *3* (1976Me12).
6866.7 *6*			510.6 *6*	E_p(lab) (keV): from 1988Va06.
7066.2 *7*			716.0 *7*	E_p(lab) (keV): weighted average of 716.0 *7* (1972Hu10) and 716.0 *10* (1976Me12). ωγ=0.09 eV *5* (1972Hu10). ωγ=0.14 eV *4* (1976Me12).
7103.4 *7*	3/2		754.3 *7*	Jπ: from γ(θ) in 1976Me12. E_p(lab) (keV): weighted average of 754.1 *7* (1972Hu10) and 754.6 *10* (1976Me12).

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^{34}S(p,γ) 1976Me12,1976Sp08,1972Hu10 (continued)

^{35}Cl Levels (continued)

E(level)†	Jπ‡	Ep(lab) (keV)	Comments
			ωγ=0.04 eV (1963Ha32). ωγ=0.23 eV *11* (1972Hu10). ωγ=0.23 eV *7* (1976Me12).
7178.6 *8*	1/2	831.8 *8*	Jπ: from γ(θ) in 1976Me12. Ep(lab) (keV): weighted average of 831.5 *8* (1972Hu10), 832.2 *15* (1976Me12), and 832 *1* (2001Vo24). ωγ=0.13 eV *2* (2001Vo24). $\Gamma_\gamma \geq 0.07$ eV (2001Vo24). ωγ=0.18 eV *9* (1972Hu10). ωγ=0.14 eV *4* (1976Me12).
7185		838	Ep(lab) (keV): from 1963Ha32. ωγ=0.056 eV (1963Ha32).
7194.6 *7*	1/2,3/2	848.2 *7*	Jπ: from γ(θ) in 1960An06. Ep(lab) (keV): weighted average of 848.2 *7* (1972Hu10) and 848.1 *10* (1976Me12). ωγ=0.17 eV (1963Ha32). ωγ=0.50 eV *15* (1972Hu10). ωγ=0.54 eV *16* (1976Me12).
7225.5 *8*	5/2	880.0 *8*	Jπ: from γ(θ) in 1976Me12. Ep(lab) (keV): weighted average of 880.5 *8* (1972Hu10) and 879.3 *10* (1976Me12). ωγ=0.073 eV (1963Ha32). ωγ=0.18 eV *9* (1972Hu10). ωγ=0.14 eV *4* (1976Me12).
7234.0 *8*	5/2+	888.8 *8*	Jπ: spin=5/2 from γ(θ) in 1960An06 and 6014γ to 1/2+, 1219 level in 1972Hu10 and 1976Me12. Ep(lab) (keV): weighted average of 889.0 *8* (1972Hu10) and 888.5 *10* (1976Me12). ωγ=0.23 eV (1963Ha32). ωγ=0.59 eV *18* (1972Hu10). ωγ=0.86 eV *26* (1976Me12).
7269.2 *1*		925.1 *1*	Ep(lab) (keV): from 1976Sp08.
7272.7 *9*	(1/2,3/2)	928.6 *9*	Jπ: from γ(θ) in 1960An06. Ep(lab) (keV): weighted average of 928.2 *9* (1972Hu10) and 929.0 *10* (1976Me12). ωγ=0.31 eV (1963Ha32). ωγ=0.63 eV *19* (1972Hu10). ωγ=1.00 eV *30* (1976Me12).
7362.0 *8*	3/2	1020.6 *8*	Jπ: from γ(θ) in 1971Wi13 and 1977Ko34. Ep(lab) (keV): weighted average of 1020.4 *8* (1972Hu10), 1020.5 *11* (1973Fa07), and 1021.0 *10* (1976Me12). ωγ=0.48 eV (1963Ha32). ωγ=1.4 eV *4* (1972Hu10). ωγ=1.13 eV (1973Fa07). ωγ=1.4 eV *4* (1976Me12).
7396.0 *10*	7/2	1055.6 *10*	Jπ: from γ(θ) in 1977Ko35. Ep(lab) (keV): weighted average of 1054.9 *10* (1972Hu10), 1055.8 *12* (1973Fa07), and 1056.9 *16* (1976Me12). ωγ=0.67 eV (1963Ha32). ωγ=0.18 eV *9* (1972Hu10). ωγ=0.18 eV (1973Fa07). ωγ=0.23 eV *7* (1976Me12).
7451.1 *6*	3/2	1112.3 *6*	Jπ: from γ(θ) in 1972LeYW. Ep(lab) (keV): weighted average of 1112.2 *10* (1972Hu10), 1112.8 *13* (1973Fa07), and 1112.2 *6* (1976Me12). ωγ=0.14 eV *7* (1972Hu10). ωγ=0.14 eV (1973Fa07). ωγ=0.18 eV *5* (1976Me12).
7496?		1158	E(level): from 1960An06.
7501.1 *8*		1163.8 *8*	Ep(lab) (keV): from 1973Fa07. Other: 1165 (1972Hu10).

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$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10 (continued)

^{35}Cl Levels (continued)

E(level)†	Jπ‡	$T_{1/2}$	E_p(lab) (keV)	Comments
				$\omega\gamma$=0.05 eV (1973Fa07). $\omega\gamma$=0.27 eV *14* for E_p=1165+1166 doublet (1972Hu10).
7502.9 *7*			1165.7 *7*	E_p(lab) (keV): from 1973Fa07. Other: 1165.4 *7* (1976Me12), 1166 (1972Hu10). $\omega\gamma$=0.27 eV *14* for E_p=1165+1166 doublet (1972Hu10). $\omega\gamma$=0.23 eV (1973Fa07). $\omega\gamma$=0.36 eV *11* (1976Me12), possible doublet.
7518.8 *7*	7/2		1182.0 *7*	J^π: from $\gamma(\theta)$ in 1977Ko35. E_p(lab) (keV): weighted average of 1183.2 *11* (1972Hu10), 1182.2 *12* (1973Fa07), and 1181.4 *7* (1976Me12). $\omega\gamma$=0.09 eV *5* (1972Hu10). $\omega\gamma$=0.14 eV (1973Fa07). $\omega\gamma$=0.18 eV *5* (1976Me12).
7548.9 *6*	$7/2^-$	<0.69 fs	1213.0 *6*	T=3/2 J^π: spin from $\gamma(\theta)$ in 1966Wa09, 1967Wa22, and 1977Ko35; parity from 4386γ, M1+E2 to $7/2^-$, 3163 level in 1966Wa09 and 1967Wa22. $T_{1/2}$: lifetime τ<1 fs (1973Fa07, DSAM). Others: τ<13 fs (1971Wi13, DSAM). E_p(lab) (keV): weighted average of 1213.7 *10* (1972Hu10), 1213.6 *6* (1973Fa07), 1212.4 *7* (1976Me12), and 1212 *1* (2001Vo24). The resonance strength of this level has been intensively studied. See the general comments at the beginning of this dataset for details. 1971Pr11 reports a 3166γ with Iγ=0.37 *13* to the 4382 final level.
7561.3 *6*	1/2,3/2		1225.8 *6*	J^π: from $\gamma(\theta)$ in 1977Ko35. E_p(lab) (keV): weighted average of 1225.6 *10* (1972Hu10), 1226.3 *6* (1973Fa07), and 1225.3 *7* (1976Me12). $\omega\gamma$=0.68 eV *20* (1972Hu10). $\omega\gamma$=0.50 eV (1973Fa07). $\omega\gamma$=0.68 eV *20* (1976Me12). $\omega\gamma$=0.7 eV *2* (2001Vo24). A_2=2 (2001Vo24).
7600.8 *8*	5/2	<13.9 fs	1266.5 *8*	J^π: spin=5/2 from $\gamma(\theta)$ in 1971Wi13 and 1976Me12. Other: spin=3/2 from 1977Ko34. $T_{1/2}$: lifetime τ<20 fs (1971Wi13, DSAM). E_p(lab) (keV): weighted average of 1266.5 *10* (1972Hu10), 1267.9 *11* (1973Fa07), and 1265.7 *8* (1976Me12). $\omega\gamma$=3.4 eV, $\Gamma_\gamma\approx$0.57 eV (1971Wi13). $\omega\gamma$=1.4 eV *4* (1972Hu10). $\omega\gamma$=1.31 eV (1973Fa07). $\omega\gamma$=1.3 eV *4* (1976Me12).
7618.7 *5*	5/2		1284.9 *5*	J^π: from $\gamma(\theta)$ in 1977Ko34. Spin=3/2,5/2 from $\gamma(\theta)$ in 1976Me12. E_p(lab) (keV): weighted average of 1285.5 *11* (1972Hu10), 1286.1 *11* (1973Fa07), and 1284.5 *5* (1976Me12). $\omega\gamma$=0.54 eV *16* (1972Hu10). $\omega\gamma$=0.50 eV (1973Fa07). $\omega\gamma$=0.77 eV *23* (1976Me12).
7656.6 *8*			1323.9 *8*	E_p(lab) (keV): weighted average of 1323.7 *12* (1972Hu10), 1323.5 *12* (1973Fa07), and 1324.2 *8* (1976Me12). $\omega\gamma$=0.23 eV *11* (1972Hu10). $\omega\gamma$=0.23 eV (1973Fa07). $\omega\gamma$=0.36 eV *11* (1976Me12).
7671.9 *8*	5/2		1339.7 *8*	J^π: spin=5/2 from $\gamma(\theta)$ in 1977Ko34. Spin=5/2,7/2 from $\gamma(\theta)$ in 1976Me12. E_p(lab) (keV): weighted average of 1339.8 *11* (1972Hu10), 1339.4 *11* (1973Fa07), and 1339.7 *8* (1976Me12). $\omega\gamma$=0.68 eV *20* (1972Hu10). $\omega\gamma$=0.63 eV (1973Fa07).

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^{34}S(p,γ) 1976Me12,1976Sp08,1972Hu10 (continued)

^{35}Cl Levels (continued)

E(level)[†]	J^{π}[‡]	$T_{1/2}$	E_p(lab) (keV)	Comments
				$\omega\gamma$=0.59 eV *18* (1976Me12).
7685.5 *8*	3/2$^-$		1353.7 *8*	J^{π}: 3/2$^-$ from $\sigma(E_p,\theta)$ in 1981Bi05. Spin=3/2 from $\gamma(\theta)$ in 1976Me12. Spin=3/2,5/2 from $\gamma(\theta)$ in 1977Ko34. E_p(lab) (keV): weighted average of 1353.7 *11* (1972Hu10), 1353.4 *11* (1973Fa07), and 1353.8 *8* (1976Me12). Γ_p=445 eV *11* (1981Bi05). $\omega\gamma$=1.13 eV *34* (1972Hu10). $\omega\gamma$=1.09 eV (1973Fa07). $\omega\gamma$=1.3 eV *4* (1976Me12).
7693.9 *8*			1362.3 *8*	E_p(lab) (keV): weighted average of 1362.4 *12* (1972Hu10) and 1362.3 *8* (1976Me12). $\omega\gamma$=0.18 eV *9* (1972Hu10). $\omega\gamma$=0.27 eV *8* (1976Me12).
7706.4 *8*	5/2$^+$		1375.2 *8*	J^{π}: 5/2$^+$ from $\sigma(E_p,\theta)$ in 1981Bi05. Spin=5/2 from $\gamma(\theta)$ in 1977Ko34. Spin=3/2,5/2 from $\gamma(\theta)$ in 1976Me12 and 5/2 is favored. E_p(lab) (keV): weighted average of 1375.7 *11* (1972Hu10), 1375.0 *11* (1973Fa07), and 1375.1 *8* (1976Me12). $\omega\gamma$=1.7 eV *5* (1972Hu10). $\omega\gamma$=1.40 eV (1973Fa07). $\omega\gamma$=2.0 eV *6* (1976Me12). Γ_p=4 eV *1* (1981Bi05).
7744.8 *9*	7/2		1414.8 *9*	J^{π}: from $\gamma(\theta)$ for J_f(2645.7)=7/2, J_f(3163)=7/2, J_f(3942.9)=9/2 (1977Ko35). E_p(lab) (keV): weighted average of 1415.6 *12* (1972Hu10), 1414.2 *12* (1973Fa07), and 1414.7 *9* (1976Me12). $\omega\gamma$=0.59 eV *18* (1972Hu10). $\omega\gamma$=0.54 eV (1973Fa07). $\omega\gamma$=0.77 eV *23* (1976Me12).
7748			1418	E(level): from 1960An06.
7777.1 *9*			1448.0 *9*	E_p(lab) (keV): weighted average of 1449.0 *11* (1972Hu10), 1447.5 *12* (1973Fa07), and 1447.5 *9* (1976Me12). $\omega\gamma$=0.68 eV *20* (1972Hu10). $\omega\gamma$=(0.63) eV (1973Fa07). $\omega\gamma$=0.68 eV *20* (1976Me12).
7781.7 *11*	5/2$^-$		1452.8 *11*	J^{π}: proposed by 1976Me12 based on strengths of primary transitions. E_p(lab) (keV): weighted average of 1453.6 *11* (1972Hu10), 1451.8 *15* (1973Fa07), and 1452.3 *13* (1976Me12). $\omega\gamma$=0.36 eV *18* (1972Hu10). $\omega\gamma$=(0.41) eV (1973Fa07). $\omega\gamma$=0.41 eV *12* (1976Me12).
7797.1 *9*	1/2		1468.6 *9*	J^{π}: from $\gamma(\theta)$ in 1988Va06, which rejects spin=3/2 from $\gamma(\theta)$ in 1976Me12. E_p(lab) (keV): weighted average of 1469.6 *12* (1972Hu10), 1468.2 *12* (1973Fa07), and 1468.2 *9* (1976Me12). $\omega\gamma$=0.68 eV *20* (1972Hu10). $\omega\gamma$=0.72 eV (1973Fa07). $\omega\gamma$=0.63 eV *19* (1976Me12).
7837.2 *10*	3/2	<3.47 fs	1509.9 *10*	T=3/2 J^{π}: spin from $\gamma(\theta)$ in 1967Wa22. $T_{1/2}$: lifetime τ<5 fs (1971Wi13, DSAM). E_p(lab) (keV): weighted average of 1509.1 *15* (1973Fa07), 1510.2 *10* (1976Me12), and 1510 *1* (2001Vo24). $\omega\gamma$=3.4 eV *6* (1967Wa22). $\omega\gamma\approx$7.5 eV, Γ_{γ}=1.9 eV (1971Wi13). $\omega\gamma$=5.0 eV *15* for E_p=1510 doublet (1972Hu10). $\omega\gamma$=0.59 eV (1973Fa07). $\omega\gamma$=2.2 eV *7* (1976Me12). $\omega\gamma$=2.1 eV *3* (2001Vo24).

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$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10 (continued)

^{35}Cl Levels (continued)

E(level)†	Jπ‡	E_p(lab) (keV)	Comments
			Γ_γ=0.55 eV *6* (2001Vo24).
7839.0 *9*		1511.8 *9*	E_p(lab) (keV): weighted average of 1512.3 *15* (1973Fa07) and 1511.6 *9* (1976Me12). $\omega\gamma$=5.0 eV *15* for E_p=1510 doublet (1972Hu10). $\omega\gamma$=3.94 eV (1973Fa07). $\omega\gamma$=2.8 eV *8* (1976Me12).
7868.7 *10*		1542.3 *10*	E_p(lab) (keV): weighted average of 1542.3 *13* (1972Hu10), 1542.3 *12* (1973Fa07), and 1542.2 *10* (1976Me12). $\omega\gamma$=0.23 eV *11* (1972Hu10). $\omega\gamma$=(0.23) eV (1973Fa07). $\omega\gamma$=0.45 eV *14* (1976Me12).
7873?		1547	E(level): from 1960An06.
7880.8 *9*	3/2,5/2	1554.8 *9*	Jπ: from $\gamma(\theta)$ in 1976Me12. E_p(lab) (keV): weighted average of 1554.9 *12* (1972Hu10), 1554.8 *11* (1973Fa07), and 1554.7 *9* (1976Me12). $\omega\gamma$=0.72 eV *22* (1972Hu10). $\omega\gamma$=0.59 eV (1973Fa07). $\omega\gamma$=1.00 eV *30* (1976Me12).
7885?		1559	E(level): from 1960An06.
7889.1 *15*		1563.3 *15*	E_p(lab) (keV): from 1973Fa07. $\omega\gamma$=(0.05) eV (1973Fa07).
7899.2 *7*	(3/2⁻,5/2)	1573.7 *7*	E_p(lab) (keV): weighted average of 1574.4 *12* (1972Hu10), 1573.2 *11* (1973Fa07), and 1573.7 *7* (1976Me12). $\omega\gamma$=0.63 eV *19* (1972Hu10). $\omega\gamma$=0.45 eV (1973Fa07). $\omega\gamma$=1.00 eV *30* (1976Me12).
7903?		1578	E(level): from 1960An06.
7923.3 *9*		1598.6 *9*	E_p(lab) (keV): weighted average of 1598.9 *13* (1972Hu10), 1598.7 *12* (1973Fa07), and 1598.4 *9* (1976Me12). $\omega\gamma$=0.36 eV *18* (1972Hu10). $\omega\gamma$=(0.32) eV (1973Fa07). $\omega\gamma$=0.45 eV *14* (1976Me12).
7930?		1605	E(level): from 1960An06.
7949?		1625	E(level): from 1960An06.
7970.3 *10*		1646.9 *10*	E_p(lab) (keV): weighted average of 1647.3 *12* (1972Hu10), 1645.9 *13* (1973Fa07), and 1647.1 *10* (1976Me12). $\omega\gamma$=0.54 eV *16* (1972Hu10). $\omega\gamma$=(0.45) eV (1973Fa07). $\omega\gamma$=0.86 eV *26* (1976Me12).
7987.8 *10*	3/2	1665.0 *10*	Jπ: from $\gamma(\theta)$ in 1976Me12. E_p(lab) (keV): weighted average of 1665.2 *12* (1972Hu10), 1663.3 *12* (1973Fa07), and 1666.1 *10* (1976Me12). $\omega\gamma$=0.68 eV *20* (1972Hu10). $\omega\gamma$=(0.59) eV (1973Fa07). $\omega\gamma$=0.54 eV *16* (1976Me12).
7995.6 *10*	5/2	1673.0 *10*	E_p(lab) (keV): weighted average of 1673.7 *11* (1972Hu10), 1672.4 *13* (1973Fa07), and 1672.7 *10* (1976Me12). $\omega\gamma$=1.2 eV *4* (1972Hu10). $\omega\gamma$=(1.04) eV (1973Fa07). $\omega\gamma$=0.95 eV *28* (1976Me12).
8001.0 *9*	(7/2,9/2⁺)	1678.6 *9*	E_p(lab) (keV): weighted average of 1679.7 *11* (1972Hu10), 1678.2 *13* (1973Fa07), and 1678.1 *9* (1976Me12). $\omega\gamma$=1.9 eV *6* (1972Hu10). $\omega\gamma$=(1.18) eV (1973Fa07). $\omega\gamma$=1.3 eV *4* (1976Me12).
8005.0 *10*	5/2⁺	1682.7 *10*	Jπ: from $\sigma(E_p,\theta)$ in 1981Bi05. Spin=5/2 from $\gamma(\theta)$ in 1981Bi05. Spin=3/2,5/2 from

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$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10 (continued)

^{35}Cl Levels (continued)

E(level)†	Jπ‡	Ep(lab) (keV)	Comments
			$\gamma(\theta)$ in 1976Me12. E_p(lab) (keV): weighted average of 1684.3 *13* (1972Hu10), 1681.9 *11* (1973Fa07), and 1682.5 *10* (1976Me12). $\omega\gamma$=(1.54) eV (1973Fa07). $\omega\gamma$=2.2 eV *7* (1976Me12). $\omega\gamma$=2.0 eV *6* (1972Hu10). Γ_p=21 eV *3* (1981Bi05).
8035.7 *10*		1714.3 *10*	E_p(lab) (keV): weighted average of 1714.0 *11* (1972Hu10), 1713.5 *14* (1973Fa07), and 1714.9 *10* (1976Me12). $\omega\gamma$=0.41 eV *20* (1972Hu10). $\omega\gamma$=(0.41) eV (1973Fa07). $\omega\gamma$=0.36 eV *11* (1976Me12).
8038.5 *11*	3/2	1717.2 *11*	J^π: from $\gamma(\theta)$ in 1976Me12. E_p(lab) (keV): weighted average of 1717.9 *11* (1972Hu10), 1716.3 *11* (1973Fa07), and 1717.3 *11* (1976Me12). $\omega\gamma$=1.6 eV *5* (1972Hu10). $\omega\gamma$=1.36 eV (1973Fa07). $\omega\gamma$=1.3 eV *4* (1976Me12).
8075.9 *10*		1755.7 *10*	E_p(lab) (keV): weighted average of 1756.0 *10* (1972Hu10), 1755.4 *11* (1973Fa07), and 1755.7 *11* (1976Me12). $\omega\gamma$=0.72 eV *22* (1972Hu10). $\omega\gamma$=(0.59) eV (1973Fa07). $\omega\gamma$=0.72 eV *22* (1976Me12).
8080?		1760	E_p(lab) (keV): from 1960An06.
8096.1 *11*	(5/2,7/2$^+$)	1776.5 *11*	E_p(lab) (keV): weighted average of 1776.7 *11* (1972Hu10), 1776.9 *11* (1973Fa07), and 1776.0 *11* (1976Me12). $\omega\gamma$=1.13 eV *34* (1972Hu10). $\omega\gamma$=1.13 eV (1973Fa07). $\omega\gamma$=1.1 eV *3* (1976Me12).
8106.4 *11*	3/2	1787.1 *11*	J^π: from $\gamma(\theta)$ in 1976Me12. E_p(lab) (keV): weighted average of 1787.7 *11* (1972Hu10), 1786.6 *11* (1973Fa07), and 1786.9 *11* (1976Me12). $\omega\gamma$=1.9 eV *6* (1972Hu10). $\omega\gamma$=1.63 eV (1973Fa07). $\omega\gamma$=1.7 eV *5* (1976Me12).
8113.3 *11*	(1/2$^+$,3/2,5/2$^+$)	1794.2 *11*	E_p(lab) (keV): weighted average of 1794.8 *13* (1972Hu10), 1793.3 *14* (1973Fa07), and 1794.2 *11* (1976Me12). $\omega\gamma$=0.54 eV *16* (1972Hu10). $\omega\gamma$=(0.50) eV (1973Fa07). $\omega\gamma$=0.72 eV *22* (1976Me12).
8147.2 *11*	3/2$^-$	1829.1 *11*	E_p(lab) (keV): weighted average of 1829.1 *13* (1972Hu10), 1828.6 *12* (1973Fa07), and 1829.5 *11* (1976Me12). $\omega\gamma$=1.09 eV *33* (1972Hu10). $\omega\gamma$=0.68 eV (1973Fa07). $\omega\gamma$=0.77 eV *23* (1976Me12).
8156.8 *11*	(5/2$^+$,7/2$^-$)	1839.0 *11*	E_p(lab) (keV): weighted average of 1839.0 *12* (1972Hu10), 1838.8 *12* (1973Fa07), and 1839.1 *11* (1976Me12). $\omega\gamma$=0.95 eV *28* (1972Hu10). $\omega\gamma$=0.95 eV (1973Fa07). $\omega\gamma$=0.86 eV *26* (1976Me12).
8179.4 *12*		1862.3 *12*	E_p(lab) (keV): weighted average of 1862.7 *13* (1972Hu10), 1862.1 *13* (1973Fa07), and 1862.1 *12* (1976Me12). $\omega\gamma$=0.36 eV *18* (1972Hu10). $\omega\gamma$=(0.32) eV (1973Fa07). $\omega\gamma$=0.41 eV *12* (1976Me12).
8208.2 *10*	5/2$^+$	1891.9 *10*	J^π: from $\sigma(E_p,\theta)$ in 1981Bi05. Spin=5/2 from $\gamma(\theta)$ in 1996Ka21 and 2001Vo24.

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$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10 (continued)

^{35}Cl Levels (continued)

E(level)†	J^{π}‡	E_p(lab) (keV)	Comments
			E_p(lab) (keV): weighted average of 1893.2 *11* (1972Hu10), 1891.3 *12* (1976Me12), 1892.0 *11* (1973Fa07), and 1891 *1* (2001Vo24). $\omega\gamma$=3.6 eV *11* (1972Hu10). $\omega\gamma$=2.71 eV (1973Fa07). $\omega\gamma$=4.1 eV *12* (1976Me12). $\omega\gamma$=4.1 eV *5*, Γ_p=39 eV *5* (1981Bi05). $\omega\gamma$=2.5 eV *7* (1996Ka21). $\omega\gamma$=4.0 eV *8* (2001Vo24), sum of $\omega\gamma$ for the resonances at E_p=1891 and 1893, Γ_γ=0.70 eV *14* (2001Vo24).
8209	1/2$^+$	1893	E(level): from 1974Bi16; a weak resonance in the tail of the 1892 resonance. J^{π}: from $\sigma(E_p,\theta)$ in 1981Bi05. Γ_p=120 eV *11* (1981Bi05).
8216.3 *10*	5/2	1900.3 *10*	J^{π}: spin=5/2 from $\gamma(\theta)$ in 2001Vo24 and 1996Ka21. E_p(lab) (keV): weighted average of 1901.8 *11* (1972Hu10), 1899.7 *12* (1973Fa07), 1899.6 *11* (1976Me12), and 1900 *1* (2001Vo24). $\omega\gamma$=2.8 eV *8* (1972Hu10). $\omega\gamma$=2.13 eV (1973Fa07). $\omega\gamma$=3.2 eV *10* (1976Me12). $\omega\gamma$=1.7 eV *5* (1996Ka21). $\omega\gamma$=3.2 eV *5* (2001Vo24). Γ_γ=0.55 eV *8* (2001Vo24).
8242.4 *12*		1927.1 *12*	E_p(lab) (keV): weighted average of 1928.2 *13* (1972Hu10), 1926.8 *13* (1973Fa07), and 1926.5 *12* (1976Me12). $\omega\gamma$=0.86 eV *26* (1972Hu10). $\omega\gamma$=0.86 eV (1973Fa07). $\omega\gamma$=1.00 eV *30* (1976Me12). $\omega\gamma$=1.0 eV *3* (2001Vo24). Γ_γ=0.25 eV *8* (2001Vo24). 20% of Iγ transitions to unknown levels (1973Fa07).
8251? *5*		1936 *5*	E_p(lab) (keV): from 1967Da12.
8268.9 *5*		1954.4 *5*	E_p(lab) (keV): weighted average of 1955.9 *13* (1972Hu10), 1952 *2* (1973Fa07), 1954.3 *12* (1976Me12), and 1954.4 *5* (1976Sp08). $\omega\gamma$=0.81 eV *24* (1972Hu10). $\omega\gamma$=(0.86) eV (1973Fa07). $\omega\gamma$=0.90 eV *27* (1976Me12). $\omega\gamma$=0.6 eV *3* (1976Sp08).
8277.2 *5*		1963.0 *5*	E_p(lab) (keV): weighted average of 1964.1 *14* (1972Hu10), 1962 *2* (1973Fa07), 1962.8 *12* (1976Me12), and 1962.9 *5* (1976Sp08). $\omega\gamma$=0.72 eV *22* (1972Hu10). $\omega\gamma$=(0.77) eV (1973Fa07). $\omega\gamma$=0.68 eV *20* (1976Me12). $\omega\gamma$=0.7 eV *4* (1976Sp08).
8282.4 *5*	(3/2$^-$,5/2)	1968.3 *5*	E_p(lab) (keV): weighted average of 1970.0 *13* (1972Hu10), 1971 *2* (1973Fa07), 1967.8 *12* (1976Me12), and 1967.9 *5* (1976Sp08). $\omega\gamma$=1.09 eV *33* (1972Hu10). $\omega\gamma$=(1.09) eV (1973Fa07). $\omega\gamma$=0.59 eV *18* (1976Me12). $\omega\gamma$=0.4 eV *2* (1976Sp08).
8287.82 *30*	1/2$^-$	1973.89 *30*	J^{π}: from $\sigma(E_p,\theta)$ in 1981Bi05, which also rejects spin=3/2 from $\gamma(\theta)$ in 1976Me12. E_p(lab) (keV): weighted average of 1975.6 *13* (1972Hu10), 1973.8 *12* (1976Me12), and 1973.80 *30* (1976Sp08). $\omega\gamma$=1.3 eV *4* (1972Hu10). $\omega\gamma$=0.72 eV *22* (1976Me12). $\omega\gamma$=0.9 eV *5* (1976Sp08). Γ_p=48 eV *6* (1981Bi05).
8297.2 *9*		1983.5 *9*	E_p(lab) (keV): weighted average of 1983.7 *5* (1976Sp08) and 1980 *2* (1973Fa07).

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^{34}S(p,γ) 1976Me12,1976Sp08,1972Hu10 (continued)

^{35}Cl Levels (continued)

E(level)†	J^{π}‡	E_p(lab) (keV)	Comments
			$\omega\gamma$=(1.36) eV (1973Fa07). $\omega\gamma$=0.6 eV *3* (1976Sp08).
8298.2 *5*	3/2	1984.6 *5*	J^{π}: from $\gamma(\theta)$ in 1976Me12. E_p(lab) (keV): weighted average of 1986.0 *13* (1972Hu10), 1983.0 *20* (1973Fa07), 1984.5 *12* (1976Me12), and 1984.5 *5* (1976Sp08). $\omega\gamma$=1.8 eV *5* (1972Hu10). $\omega\gamma$=(0.77) eV (1973Fa07). $\omega\gamma$=1.6 eV *5* (1976Me12). $\omega\gamma$=0.9 eV *5* (1976Sp08).
8318.1 *3*		2005.1 *3*	E_p(lab) (keV): weighted average of 2006.7 *12* (1972Hu10), 2004.6 *13* (1973Fa07), 2006.0 *13* (1976Me12), and 2005.0 *3* (1976Sp08). $\omega\gamma$=1.9 eV *6* (1972Hu10). $\omega\gamma$=2.08 eV (1973Fa07). $\omega\gamma$=1.9 eV *6* (1976Me12). $\omega\gamma$=2.6 eV *5* (1976Sp08).
8323.1 *13*		2010.2 *13*	E_p(lab) (keV): weighted average of 2010.5 *13* (1972Hu10) and 2009.9 *13* (1976Me12). $\omega\gamma$=0.45 eV *14* (1972Hu10). $\omega\gamma$=0.72 eV *22* (1976Me12).
8346? *5*		2034 *5*	E_p(lab) (keV): from 1967Da12.
8381.8 *8*	5/2^{+}	2070.7 *8*	J^{π}: spin=5/2 from $\gamma(\theta)$ in 2001Vo24 and 1967Ko23. E_p(lab) (keV): weighted average of 2073.5 *12* (1972Hu10), 2070.4 *5* (1976Sp08), and 2070 *1* (2001Vo24). $\omega\gamma\approx$=4 eV (1967Ko23). $\omega\gamma$=6.4 eV *19* (1972Hu10). $\omega\gamma$=6.0 eV *12* (1976Sp08). $\omega\gamma$=5.9 eV *6* (2001Vo24). Γ_{γ}=1.02eV *10* (2001Vo24).
8389.4 *16*		2078.5 *16*	E_p(lab) (keV): unweighted average of 2080.0 *14* (1972Hu10) and 2076.9 *5* (1976Sp08). $\omega\gamma$=0.27 eV *14* (1972Hu10). $\omega\gamma$=0.4 eV *2* (1976Sp08).
8402.9 *5*		2092.4 *5*	E_p(lab) (keV): from 1976Sp08. $\omega\gamma$=0.7 eV *4* (1976Sp08).
8405.7 *14*		2095.3 *14*	E_p(lab) (keV): unweighted average of 2096.6 *12* (1972Hu10) and 2093.9 *5* (1976Sp08). $\omega\gamma$=2.9 eV *9* (1972Hu10). $\omega\gamma$=2.3 eV *5* (1976Sp08).
8409.6 *18*		2099.3 *18*	E_p(lab) (keV): unweighted average of 2101.0 *14* (1972Hu10) and 2097.5 *5* (1976Sp08). $\omega\gamma$=1.6 eV *5* (1972Hu10). $\omega\gamma$=1.4 eV *3* (1976Sp08).
8416.7 *5*		2106.6 *5*	$\omega\gamma$=0.5 eV *3* (1976Sp08).
8430.3 *5*		2120.6 *5*	$\omega\gamma$=1.0 eV *2* (1976Sp08).
8434.8 *5*		2125.2 *5*	$\omega\gamma$=0.5 eV *3* (1976Sp08).
8464.3 *5*		2155.6 *5*	$\omega\gamma$=0.9 eV *5* (1976Sp08).
8484.4 *5*	3/2^{+}	2176.3 *5*	J^{π}: from 2001Vo24 based on proposed isobaric analog to 3/2^{+} in ^{35}S. E_p(lab) (keV): weighted average of 2176.4 *5* (1976Sp08) and 2176 *1* (2001Vo24). $\omega\gamma$=2.1 eV *4* (1976Sp08). $\omega\gamma$=2.1 eV *4* (2001Vo24). Γ_{γ}=0.36 eV *6* (2001Vo24).
8486.1 *5*		2178.0 *5*	$\omega\gamma$=1.9 eV *4* (1976Sp08).
8506.5 *5*		2199.1 *5*	$\omega\gamma$=0.3 eV *2* (1976Sp08).
8514.3 *5*		2207.1 *5*	$\omega\gamma$=0.5 eV *3* (1976Sp08).
8533.9 *5*		2227.3 *5*	$\omega\gamma$=0.10 eV *5* (1976Sp08).
8571.9 *5*		2266.4 *5*	$\omega\gamma$=2.8 eV *6* (1976Sp08).
8580.5 *5*		2275.3 *5*	$\omega\gamma$=1.0 eV *2* (1976Sp08).
8585.8 *5*		2280.7 *5*	$\omega\gamma$=0.10 eV *5* (1976Sp08).
8590.0 *5*		2285.0 *5*	$\omega\gamma$=0.7 eV *4* (1976Sp08).
8612.0 *5*		2307.7 *5*	$\omega\gamma$=1.0 eV *2* (1976Sp08).

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^{34}S(p,γ) 1976Me12,1976Sp08,1972Hu10 (continued)

^{35}Cl Levels (continued)

E(level)[†]	J$^{\pi}$[‡]	E$_p$(lab) (keV)	Comments
8613.7 *5*		2309.4 *5*	$\omega\gamma$=2.4 eV *5* (1976Sp08).
8618.1 *5*	(3/2$^+$,5/2$^+$)	2314.0 *5*	J$^{\pi}$: most probable spin=5/2 from $\gamma(\theta)$ in 1970Ko33. $\omega\gamma$=3.7 eV *8* (1976Sp08).
8630.1 *3*	7/2	2326.3 *3*	J$^{\pi}$: spin=3/2,7/2 from $\gamma(\theta)$ in 1976Sp09; 1976Sp09 ruled out 3/2 based on 4687γ to 9/2$^+$, 3943 level. Other: most probable spin=3/2 from $\gamma(\theta)$ in 1970Ko33. $\omega\gamma$=2.6 eV *5* (1976Sp08).
8640	5/2	2336	J$^{\pi}$: probable spin=5/2 from $\gamma(\theta)$ in 1970Ko33. E$_p$(lab) (keV): from 1970Ko33.
8641.4 *5*	3/2,5/2	2338.0 *5*	J$^{\pi}$: spin=3/2 from $\gamma(\theta)$ in 1976Sp09; spin=5/2 is also possible. $\omega\gamma$=0.9 eV *5* (1976Sp08).
8686.2 *5*		2384.1 *5*	$\omega\gamma$=2.2 eV *4* (1976Sp08).
8688.3 *5*		2386.3 *5*	$\omega\gamma$=0.7 eV *4* (1976Sp08).
8696.9 *5*		2395.1 *5*	$\omega\gamma$=0.6 eV *3* (1976Sp08).
8706.3 *5*		2404.8 *5*	$\omega\gamma$=0.2 eV *1* (1976Sp08).
8717.7 *5*		2416.5 *5*	$\omega\gamma$=0.5 eV *3* (1976Sp08).
8750.9 *5*		2450.7 *5*	$\omega\gamma$=1.7 eV *3* (1976Sp08).
8767.0 *5*		2467.3 *5*	$\omega\gamma$=0.3 eV *2* (1976Sp08).
8772.9 *5*		2473.4 *5*	$\omega\gamma$=0.6 eV *3* (1976Sp08).
8779.8 *5*		2480.5 *5*	$\omega\gamma$=1.7 eV *3* (1976Sp08).
8787.2 *5*		2488.1 *5*	$\omega\gamma$=1.2 eV *2* (1976Sp08).
8798.4 *5*		2499.6 *5*	$\omega\gamma$=1.7 eV *3* (1976Sp08).
8820.9 *5*		2522.8 *5*	$\omega\gamma$=0.7 eV *4* (1976Sp08).
8824.2 *5*		2526.2 *5*	$\omega\gamma$=3.6 eV *7* (1976Sp08).
8829.3 *5*		2531.4 *5*	$\omega\gamma$=2.1 eV *4* (1976Sp08).
8833.1 *5*		2535.3 *5*	$\omega\gamma$=0.7 eV *4* (1976Sp08).
8838.1 *5*	7/2$^-$	2540.5 *5*	J$^{\pi}$: 5/2$^-$,7/2$^-$ from $\sigma(E_p,\theta)$ in 1981Bi05. Spin=7/2 from $\gamma(\theta)$ in 1981Bi05. $\omega\gamma$=1.5 eV *3* (1976Sp08). Γ_p=4 eV *1* (1981Bi05).
8856.1 *5*		2559.0 *5*	$\omega\gamma$=4.5 eV *9* (1976Sp08).
8868.6 *5*		2571.9 *5*	$\omega\gamma$=1.0 eV *2* (1976Sp08).
8884.1 *5*		2587.9 *5*	$\omega\gamma$=0.4 eV *2* (1976Sp08).
8886.1 *5*		2589.9 *5*	$\omega\gamma$=1.4 eV *3* (1976Sp08).
8893.3 *5*	(5/2,7/2)$^+$	2597.3 *5*	J$^{\pi}$: from 2001Vo24 with no supporting arguments. E$_p$(lab) (keV): weighted average of 2597.4 *5* (1976Sp08) and 2597 *1* (2001Vo24). $\omega\gamma$=1.2 eV *2* (1976Sp08). $\omega\gamma$=1.2 eV *4* (2001Vo24). Γ_γ=0.20 eV *6* (2001Vo24).
8904.8 *5*		2609.2 *5*	$\omega\gamma$=1.0 eV *2* (1976Sp08).
8906.8 *5*	5/2$^+$	2611.2 *5*	J$^{\pi}$: from 2001Vo24 with no supporting arguments. E$_p$(lab) (keV): weighted average of 2611.3 *5* (1976Sp08) and 2611 *1* (2001Vo24). $\omega\gamma$=1.5 eV *3* (1976Sp08). $\omega\gamma$=1.5 eV *3* (2001Vo24). Γ_γ=0.25 eV *5* (2001Vo24).
8919.8 *5*		2624.6 *5*	$\omega\gamma$=1.1 eV *2* (1976Sp08).
8933.3 *5*		2638.5 *5*	$\omega\gamma$=1.0 eV *2* (1976Sp08).
8953.0 *5*	3/2	2658.8 *5*	J$^{\pi}$: from $\gamma(\theta)$ in 1976Sp09. $\omega\gamma$=2.4 eV *5* (1976Sp08).
8957.9 *5*		2663.8 *5*	$\omega\gamma$=1.2 eV *2* (1976Sp08).
8981.9 *5*		2688.6 *5*	$\omega\gamma$=2.2 eV *4* (1976Sp08).
8984.2 *5*		2690.9 *5*	$\omega\gamma$=0.6 eV *3* (1976Sp08).
8988.4 *5*		2695.3 *5*	$\omega\gamma$=0.7 eV *4* (1976Sp08).
8992.4 *5*		2699.4 *5*	$\omega\gamma$=2.3 eV *5* (1976Sp08).
8996.7 *5*		2703.8 *5*	$\omega\gamma$=1.7 eV *3* (1976Sp08).
9001.1 *5*		2708.3 *5*	$\omega\gamma$=0.6 eV *3* (1976Sp08).

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$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10 (continued)

^{35}Cl Levels (continued)

E(level)†	Jπ‡	$T_{1/2}$	E_p(lab) (keV)	Comments
9019.3 *5*		3.1 keV *3*	2727.1 *5*	$T_{1/2}$: width measured from the slope of the low-energy side of the resonance peak (1976Sp08). $\omega\gamma$=0.8 eV *4* (1976Sp08).
9024.5 *5*			2732.4 *5*	$\omega\gamma$=1.1 eV *2* (1976Sp08).
9029.9 *5*			2738.0 *5*	$\omega\gamma$=2.5 eV *5* (1976Sp08).
9033.1 *5*			2741.3 *5*	$\omega\gamma$=0.4 eV *2* (1976Sp08).
9038.3 *5*			2746.6 *5*	$\omega\gamma$=0.8 eV *4* (1976Sp08).
9048.4 *5*			2757.0 *5*	$\omega\gamma$=3.1 eV *6* (1976Sp08).
9081.4 *4*	5/2+	65 eV *20*	2791.0 *4*	Jπ: from $\sigma(E_p,\theta)$ in 1981Bi05. Spin=5/2 from $\gamma(\theta)$ in 1976Sp09. $T_{1/2}$: width Γ_p=62 eV *20*, Γ_γ=2.3 eV *4* (1976Sp09, resonant absorption). E_p(lab) (keV): weighted average of 2791.0 *4* (1976Sp08) and 2791 *1* (2001Vo24). $\omega\gamma$=15.7 eV *32* (1976Sp08). $\omega\gamma$=16.0 eV *4* (1996Ka21). $\omega\gamma$=16.0 eV *3* (2001Vo24). $\omega\gamma$=13.4 eV *22* (1976Sp09). Γ_p=59 eV *7* (1981Bi05). Γ_γ=2.7 eV *5* (2001Vo24).
9088.5 *5*			2798.3 *5*	$\omega\gamma$=0.7 eV *4* (1976Sp08).
9099.3 *5*			2809.4 *5*	$\omega\gamma$=1.2 eV *2* (1976Sp08).
9100.4 *5*			2810.6 *5*	$\omega\gamma$=2.1 eV *4* (1976Sp08).
9107.4 *5*			2817.8 *5*	$\omega\gamma$=0.5 eV *3* (1976Sp08).
9110.0 *5*			2820.5 *5*	$\omega\gamma$=0.5 eV *3* (1976Sp08).
9124.0 *5*			2834.9 *5*	$\omega\gamma$=1.2 eV *2* (1976Sp08).
9135.1 *5*		2.0 keV *4*	2846.3 *5*	$T_{1/2}$: width measured from the slope of the low-energy side of the resonance peak (1976Sp08). $\omega\gamma$=0.2 eV *1* (1976Sp08).
9138.3 *5*			2849.6 *5*	$\omega\gamma$=1.9 eV *4* (1976Sp08).
9155.7 *5*			2867.5 *5*	$\omega\gamma$=2.0 eV *4* (1976Sp08).
9157.1 *5*	5/2		2869.0 *5*	Jπ: spin=5/2 from $\gamma(\theta)$ in 2001Vo24 and 1996Ka21. $\omega\gamma$=6.6 eV *13* (1976Sp08). $\omega\gamma$=6.3 eV *1* (1996Ka21).
9163.3 *5*			2875.3 *5*	$\omega\gamma$=0.4 eV *2* (1976Sp08).
9184.2 *5*			2896.9 *5*	$\omega\gamma$=0.9 eV *5* (1976Sp08).
9188.8 *5*			2901.6 *5*	$\omega\gamma$=0.9 eV *5* (1976Sp08).
9194.1 *5*			2907.1 *5*	$\omega\gamma$=3.9 eV *8* (1976Sp08).

† For resonances above S(p)(^{35}Cl)=6370.81 *4* (2021Wa16), E(level)=$E_{c.m.}$+S(p)(^{35}Cl), where $E_{c.m.}=E_p(lab)\times m(^{34}S)/[m_p+m(^{34}S)]$. Weighted average is taken when multiple E_p(lab) measurements are available. For levels below 8.27 MeV (E_p<1.95 MeV), E_p(lab) is mainly from 1972Hu10, 1973Fa07, and 1976Me12; for levels above 8.27 MeV (E_p>1.95 MeV), E_p(lab) is mainly from 1976Sp08, unless otherwise noted.

‡ From the comparison of experimental angular distributions of γ-rays with theoretical predictions or deduced from γ-feedings, unless otherwise noted.

$\gamma(^{35}Cl)$

E_i(level)	J^π_i	E_γ†	I_γ#	E_f	J^π_f	Mult.b	δb	Comments
1219.39	1/2+	1219.37	100	0	3/2+			E_γ: other: 1220 *40* (1963Ha32). A_2=−0.03 *3*, A_4=−0.002 *50* (1971Wi13). A_2=−0.01 *2*, A_4=+0.005 *24* (1967Ta08).
1763.14	5/2+	543.75	<0.2	1219.39	1/2+			I_γ: from 1972Hu10 and 1976Me12.
		1763.09	100	0	3/2+	D+Q	−0.42 *17*	I_γ: from 1972Hu10 and 1976Me12. Mult.,δ: from $\gamma\gamma(\theta)$ in 1967Ko20 and 1967Ko23.

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^{34}S(p,γ) 1976Me12,1976Sp08,1972Hu10 (continued)

γ(^{35}Cl) (continued)

E$_i$(level)	J$_i^\pi$	E$_\gamma$†	I$_\gamma$#	E$_f$	J$_f^\pi$	Mult.b	δb	Comments
								Other: +2.8<δ<+4.0 (1971Wi13). A$_2$=−0.24 *6*, A$_4$=−0.08 *7* (1971Wi13). A$_2$=+0.25 *3*, A$_4$=−0.49 *8* (1967Ko20,1967Ko23,1970Ko33).
2645.70	(5/2,7/2)	882.55	8.7 *10*	1763.14	5/2$^+$			I$_\gamma$: weighted average of 8 *1* (1972Hu10) and 9.4 *10* (1976Me12).
		1426.28	<2	1219.39	1/2$^+$			
		2645.59	91.3 *10*	0	3/2$^+$			I$_\gamma$: weighted average of 92 *1* (1972Hu10) and 90.6 *10* (1976Me12). Mult.,δ: D+Q with δ=−0.62 *+38−36* from γ(θ) for J$_i$=3/2 (1967Ko20). A$_2$=+0.124 *17*, A$_4$=−0.230 *24* (1967Ko20,1967Ko23).
2693.9	(3/2,5/2)	930.8	13.0 *10*	1763.14	5/2$^+$			E$_\gamma$: other: 933 *7* (1970Ko33). I$_\gamma$: weighted average of 13 *2* (1972Hu10) and 13.0 *10* (1976Me12). A$_2$=−0.05 *38* (1971Wi13).
		1474.5	7.6 *10*	1219.39	1/2$^+$			E$_\gamma$: other: 1475 *7* (1970Ko33). I$_\gamma$: weighted average of 6 *2* (1972Hu10) and 8.0 *10* (1976Me12). Other: 66 *14* from 1963Ha32.
		2693.8	80 *2*	0	3/2$^+$	D+Q		I$_\gamma$: weighted average of 81 *3* (1972Hu10) and 79 *2* (1976Me12). Other: 34 *9* from 1963Ha32. δ: +0.1<δ<+0.26 or −35<δ<−7 for J=3/2, δ<−5 for J=5/2 (1971Wi13). A$_2$=+0.20 *10*, A$_4$=−0.16 *11* (1971Wi13).
3002.75	5/2	1239.59	≤1	1763.14	5/2$^+$			I$_\gamma$: from 1972Hu10. Other: Iγ<2 (1976Me12).
		1783.31	<3	1219.39	1/2$^+$			I$_\gamma$: other: Iγ≤5 (1972Hu10).
		3002.61	100	0	3/2$^+$	D+Q	−0.22 *1*	E$_\gamma$: other: 3006 *7* (1970Ko33). I$_\gamma$: from 1972Hu10 and 1976Me12. Mult.,δ: from γγ(θ) in 1967Ko20. A$_2$=+0.45 *18*, A$_4$=+0.05 *19* (1971Wi13). A$_2$=−0.025 *11*, A$_4$=−0.084 *16* (1967Ko20).
3162.99	7/2$^-$	160.24	1.7 *2*	3002.75	5/2			I$_\gamma$: from 1971Pr11. Other: <3 (1976Me12).
		517.29	8 *1*	2645.70	(5/2,7/2)			I$_\gamma$: from 1971Pr11. Others: 11 *2* (1972Hu10), 8.9 *20* (1976Me12). A$_2$=+0.31 *8*, A$_4$=−0.06 *8* (1971Wi13).
		1399.82	0.30 *4*	1763.14	5/2$^+$	E1+M2	+0.44 *12*	I$_\gamma$: from 1971Pr11. Others: ≤0.2 (1972Hu10), <0.2 (1976Me12). Mult.,δ: from γ(θ) in 1971Pr11, which states that the anomalous γ(θ) can be understood if it occurs by a mixture of M2 and E1 radiation. In view of the weakness of this transition, such a mixture is not unreasonable.
		1943.54	<0.2	1219.39	1/2$^+$			
		3162.84	90 *1*	0	3/2$^+$	M2+E3	+0.25 *4*	E$_\gamma$: other: 3163 *7* (1970Ko33). I$_\gamma$: from 1971Pr11. Others: 89 *2* (1972Hu10), 91 *2* (1976Me12). Mult.: Q+O from γ(θ) in 1966Az01, 1966Wa09, 1967Wa22, and 1967Ta08;

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^{34}S(p,γ) 1976Me12,1976Sp08,1972Hu10 (continued)

γ(^{35}Cl) (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[#]	E_f	J^π_f	Mult.[b]	δ[b]	Comments
								magnetic character from γ(pol) in 1966Wa09 and 1967Wa22. δ: from 1966Az01. Others: +0.25 *5* (1967Ta08); +0.16 *1* (1966Wa09,1967Wa22). Asymmetry ratio R=1.426 *7*, N_{90}/N_0=1.25 *11* (1971Wi13). POL=−0.75 *20*, Δπ=yes (1967Wa22). POL=−1.27 *10*, Δπ=yes (1966Wa09). 0.00<POL<0.384 (1966Az01,1967Ta08). A_2=+0.40 *8*, A_4=−0.23 *8* (1971Wi13). A_2=+0.25 *3*, A_4=+0.02 *5* (1967Ta08). A_2=+0.55 *3*, A_4=−0.008 *38* (1966Az01).
3918.48	3/2+	1272.76	<3	2645.70	(5/2,7/2)			
		2155.27	19.5 *23*	1763.14	5/2+			I_γ: weighted average of 17 *3* (1972Hu10), 25 *3* (1976Me12), and 18.0 *21* (1976Sp09).
		3918.25	81.5 *15*	0	3/2+	M1+E2		I_γ: weighted average of 83 *3* (1972Hu10), 75 *3* (1976Me12), and 82.0 *10* (1976Sp09). Mult.: D+Q from γ(θ) in 1976Sp09; δ=−0.21 *2* or +20 *+10−5* (1976Sp09). E1+M2 ruled by RUL. A_2=+0.07 *2*, A_4=−0.01 *3* (1976Sp09).
3942.9		1297.2	8 *2*	2645.70	(5/2,7/2)			I_γ: weighted average of 7 *3* (1972Hu10) and 8 *2* (1976Me12).
		2179.7	92 *2*	1763.14	5/2+			I_γ: weighted average of 93 *3* (1972Hu10) and 92 *2* (1976Me12).
		2723.4[c]	<6	1219.39	1/2+			
		3942.7[c]	<10	0	3/2+			
3967.3		1273.4	2.1 *5*	2693.9	(3/2,5/2)			I_γ: weighted average of 5 *3* (1972Hu10) and 2.0 *5* (1976Me12).
		1321.6[c]	<2	2645.70	(5/2,7/2)			
		2204.1	≤1	1763.14	5/2+			I_γ: from 1972Hu10. Other: <2 (1976Me12).
		2747.8	79 *5*	1219.39	1/2+			I_γ: weighted average of 74 *5* (1972Hu10) and 83 *5* (1976Me12).
		3967.1	19 *3*	0	3/2+			I_γ: weighted average of 21 *3* (1972Hu10) and 15 *5* (1976Me12).
4059.2		1365.3	1.2 *4*	2693.9	(3/2,5/2)			
		1413.5[c]	<1	2645.70	(5/2,7/2)			
		2296.0	4.8 *15*	1763.14	5/2+			E_γ: other: 2296 *7* (1970Ko33). I_γ: other: 3 *2* (1972Hu10).
		2839.7	94 *2*	1219.39	1/2+			E_γ: other: 2838 *7* (1970Ko33). I_γ: other: 97 *2* (1972Hu10). 45 *5* (1971Wi13).
		4059.0	<2	0	3/2+			E_γ: other: 4058 *7* (1970Ko33). I_γ: other: 20 *2* (1971Wi13).
4113.7	7/2	2350.5	47 *3*	1763.14	5/2+	D+Q	−0.16 *2*	I_γ: weighted average of 50 *20* (1972Hu10), 44 *7* (1976Me12), and 48 *3* (1976Sp09). Mult.,δ: from γ(θ) in 1976Sp09. A_2=+0.56 *3*, A_4=+0.05 *4* (1976Sp09).
		2894.2[c]	<10	1219.39	1/2+			
		4113.4	53 *3*	0	3/2+	Q(+O)	0.00 *10*	I_γ: weighted average of 50 *20* (1972Hu10) and 56 *7* (1976Me12), and 52 *3* (1976Sp09). Mult.,δ: Q(+O) from γ(θ) in 1976Sp09. A_2=+0.46 *25*, A_4=−0.23 *25* (1976Sp09).

Continued on next page (footnotes at end of table)

^{34}S(p,γ) 1976Me12,1976Sp08,1972Hu10 (continued)

γ(^{35}Cl) (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[#]	E_f	J^π_f	Mult.[b]	δ[b]	Comments
4173.45		1479.5	26 *8*	2693.9	(3/2,5/2)			I_γ: weighted average of 27 *8* (1972Hu10) and 25 *8* (1976Me12).
		1527.71	<10	2645.70	(5/2,7/2)			
		2410.22	16 *5*	1763.14	$5/2^+$			I_γ: weighted average of 18 *6* (1972Hu10) and 15 *5* (1976Me12).
		2953.93	≤3	1219.39	$1/2^+$			I_γ: from 1972Hu10. Other: <4 (1976Me12); 32 *3* (1971Wi13).
		4173.18	58 *10*	0	$3/2^+$			I_γ: weighted average of 55 *10* (1972Hu10) and 60 *10* (1976Me12). A_2=−0.01 *3*, A_4=−0.006 *40* (1967Ta08).
4177.9	3/2	1175.1	<0.5	3002.75	5/2			
		1484.0	8 *2*	2693.9	(3/2,5/2)			I_γ: weighted average of 7 *2* (1972Hu10) and 10 *3* (1976Me12).
		1532.2	<0.5	2645.70	(5/2,7/2)			
		2414.7	<1	1763.14	$5/2^+$			I_γ: other ≤1 (1972Hu10).
		2958.4	31 *5*	1219.39	$1/2^+$	D+Q	+0.11 *4*	I_γ: weighted average of 31 *5* (1972Hu10) and 30 *5* (1976Me12). Mult.,δ: from γ(θ) in 1967Wa22.
		4177.6	61 *5*	0	$3/2^+$	D+Q	+0.06 *3*	I_γ: weighted average of 62 *6* (1972Hu10) and 60 *5* (1976Me12). Mult.,δ: from γ(θ) in 1967Wa22. POL=+0.12 *31* (1967Wa22).
4347.76	5/2,9/2	1184.75	70 *2*	3162.99	$7/2^-$			I_γ: weighted average of 70 *5* (1972Hu10), 72 *5* (1976Me12), and 69 *2* (1976Sp09).
		1344.98[c]	<4	3002.75	5/2			
		1653.8[c]	<4	2693.9	(3/2,5/2)			
		1702.02	31 *2*	2645.70	(5/2,7/2)	D+Q	−0.018 *12*	I_γ: weighted average of 30 *5* (1972Hu10), 28 *5* (1976Me12), and 31 *2* (1976Sp09). Mult.,δ: D+Q from γ(θ) in 1976Sp09. A_2=−0.34 *3*, A_4=−0.00 *4* (1976Sp09).
		2584.52	≤3	1763.14	$5/2^+$			I_γ: from 1972Hu10. Other: <5 (1976Me12).
		3128.22[c]	<4	1219.39	$1/2^+$			
		4347.47	≤5	0	$3/2^+$			I_γ: from 1972Hu10. Other: <10 (1976Me12).
4624.4		4624.1	85 *10*	0	$3/2^+$			I_γ: other: 100 (1972Hu10).
4769.86		1606.83	65 *10*	3162.99	$7/2^-$			I_γ: other: 53 *10* (1972Hu10) and 47% of Iγ to unknown levels (1972Hu10).
		1767.06	35 *10*	3002.75	5/2			
		2075.9	<10	2693.9	(3/2,5/2)			
		3550.28	<10	1219.39	$1/2^+$			
		4769.51	<10	0	$3/2^+$			
4839.10		1836.30	<7	3002.75	5/2			
		2145.1	<4	2693.9	(3/2,5/2)			
		2193.33	<3	2645.70	(5/2,7/2)			
		3075.82	59 *5*	1763.14	$5/2^+$			I_γ: weighted average of 50 *20* (1972Hu10) and 60 *5* (1976Me12). Other: 48 *4* (1976Sp09).
		3619.51	<7	1219.39	$1/2^+$			
		4838.74	41 *5*	0	$3/2^+$			I_γ: weighted average of 50 *20* (1972Hu10) and 40 *5* (1976Me12). Other: 18 *3* (1976Sp09).
4854.4		3634.8	75 *5*	1219.39	$1/2^+$			
		4854.0	25 *5*	0	$3/2^+$			
4880.96		1878.16	9 *3*	3002.75	5/2			
		2187.0	<10	2693.9	(3/2,5/2)			

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^{34}S(p,γ) 1976Me12,1976Sp08,1972Hu10 (continued)

γ(^{35}Cl) (continued)

E$_i$(level)	J$^\pi_i$	E$_\gamma$[†]	I$_\gamma$[#]	E$_f$	J$^\pi_f$	Mult.[b]	δ[b]	Comments
4880.96		2235.18	29 *5*	2645.70	(5/2,7/2)			I$_\gamma$: other: 30 *10* (1972Hu10).
		3117.67	62 *5*	1763.14	5/2$^+$			I$_\gamma$: other: 70 *10* (1972Hu10).
		3661.36[c]	<4	1219.39	1/2$^+$			
		4880.60	<5	0	3/2$^+$			
5010.1		836.6	<25[a]	4173.45				
		950.9	<10[a]	4059.2				
		1847.1	<50[a]	3162.99	7/2$^-$			
		2007.3	<20[a]	3002.75	5/2			
		2316.1	<25[a]	2693.9	(3/2,5/2)			
		2364.3	<5	2645.70	(5/2,7/2)			
		3246.8	<10	1763.14	5/2$^+$			
		3790.5	<7	1219.39	1/2$^+$			
		5009.7	100	0	3/2$^+$			I$_\gamma$: from 1976Me12 and 1972Hu10.
5163.34	7/2	2000.29	40 *5*	3162.99	7/2$^-$	D+Q	+0.44 *20*	I$_\gamma$: weighted average of 40 *20* (1972Hu10), 36 *5* (1976Me12), and 54 *10* (1976Sp09). Mult.,δ: D+Q from $\gamma(\theta)$ in 1976Sp09. A$_2$=+0.52 *5*, A$_4$=−0.02 *6* (1976Sp09).
		2160.52	10 *4*	3002.75	5/2			I$_\gamma$: weighted average of 10 *5* (1972Hu10), 10 *4* (1976Me12), and 10 *5* (1976Sp09).
		3400.02	43 *5*	1763.14	5/2$^+$	D(+Q)	0.00 *3*	I$_\gamma$: weighted average of 50 *20* (1972Hu10), 44 *5* (1976Me12), and 36 *10* (1976Sp09). Mult.,δ: D(+Q) from $\gamma(\theta)$ in 1976Sp09. A$_2$=−0.23 *6*, A$_4$=−0.08 *7* (1976Sp09).
		3943.71	<10	1219.39	1/2$^+$			
		5162.93	<10	0	3/2$^+$			
5215.8		205.7	<23[a]	5010.1				
		1042.3	<60[a]	4173.45				
		1156.6	<10[a]	4059.2				
		2052.8	<15[a]	3162.99	7/2$^-$			
		2213.0	<25[a]	3002.75	5/2			
		2521.8	<15[a]	2693.9	(3/2,5/2)			
		2570.0	<15[a]	2645.70	(5/2,7/2)			
		3452.5	<6	1763.14	5/2$^+$			
		3996.2	<4	1219.39	1/2$^+$			
		5215.4	100	0	3/2$^+$			
5403.5		2400.7	25 *5*	3002.75	5/2			I$_\gamma$: other: 25 *10* (1972Hu10).
		3640.2	<10	1763.14	5/2$^+$			I$_\gamma$: other: 25 *10* from 1972Hu10.
		4183.8	55 *10*	1219.39	1/2$^+$			
		5403.1	<7	0	3/2$^+$			
5586		2940	40 *10*	2645.70	(5/2,7/2)			I$_\gamma$: weighted average of 40 *20* (1972Hu10) and 40 *10* (1976Me12).
		3823	60 *10*	1763.14	5/2$^+$			I$_\gamma$: weighted average of 60 *20* (1972Hu10) and 60 *10* (1976Me12).
		4366	<10	1219.39	1/2$^+$			
		5586	<10	0	3/2$^+$			
5599.7		3836.3	27 *10*	1763.14	5/2$^+$			I$_\gamma$: weighted average of 20 *10* (1972Hu10) and 33 *10* (1976Me12).
		5599.2	74 *10*	0	3/2$^+$			I$_\gamma$: weighted average of 80 *10* (1972Hu10) and 67 *10* (1976Me12).
5646		2483	100	3162.99	7/2$^-$			I$_\gamma$: from 1972Hu10, 1976Me12, 1971Pr11.
		2643		3002.75	5/2			I$_\gamma$: 80 from 2001Vo24 only; not observed by others.
		3883		1763.14	5/2$^+$			I$_\gamma$: 6 from 2001Vo24 only; not observed by others.

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^{34}S(p,γ) 1976Me12,1976Sp08,1972Hu10 (continued)

γ(^{35}Cl) (continued)

E_i(level)	J_i^π	E_γ[†]	I_γ[#]	E_f	J_f^π	Mult.[b]	δ[b]	Comments
5646		5646	<8	0	3/2$^+$			
5655		2652	80 *5*	3002.75	5/2			
		3892	6 *2*	1763.14	5/2$^+$			
		4435	<6	1219.39	1/2$^+$			
5683		1505	60 *10*	4177.9	3/2			
5723.52		3029.5	24 *3*	2693.9	(3/2,5/2)			I_γ: from 1976Sp09.
		3960.14	38 *4*	1763.14	5/2$^+$			I_γ: from 1976Sp09.
5758		1580	30 *10*	4177.9	3/2			
		4538	<6	1219.39	1/2$^+$			
		5758	45 *10*	0	3/2$^+$			
5806		5806	100	0	3/2$^+$			I_γ: from 1972Hu10 and 1976Me12.
6106.2		4342.8	6 *2*	1763.14	5/2$^+$			
		4886.4	36 *10*	1219.39	1/2$^+$			I_γ: other: 40 *10* (1972Hu10).
		6105.6	58 *10*	0	3/2$^+$			I_γ: other: 60 *10* (1972Hu10).
6181		6180	45 *10*	0	3/2$^+$			
6492		6491	100	0	3/2$^+$			
7066.2		1902.8	0.9	5163.34	7/2			
		2892.6	0.5	4173.45				
		2952.4	0.6	4113.7	7/2			
		3006.9	1.4	4059.2				I_γ: other: 1.0 *5* from 1972Hu10.
		3903.0	6	3162.99	7/2$^-$			I_γ: other: 6.0 *6* from 1972Hu10.
		4063.2	3	3002.75	5/2			I_γ: other: 2 *1* from 1972Hu10.
		4372.0	5	2693.9	(3/2,5/2)			I_γ: other: 6.0 *6* from 1972Hu10.
		4420.2	1.6	2645.70	(5/2,7/2)			I_γ: other: 3.0 *15* from 1972Hu10.
		5302.6	14	1763.14	5/2$^+$			I_γ: other: 16.0 *16* from 1972Hu10.
		5846.3	18	1219.39	1/2$^+$			I_γ: other: 18.0 *18* from 1972Hu10.
		7065.4	49	0	3/2$^+$			I_γ: other: 48.0 *48* from 1972Hu10.
7103.4	3/2	2925.4	1.5	4177.9	3/2			
		3184.8	1.5	3918.48	3/2$^+$			I_γ: other: 2 *1* (1972Hu10).
		4100.4	3	3002.75	5/2			I_γ: other: 3.0 *15* from 1972Hu10.
		4409.2	11	2693.9	(3/2,5/2)	D(+Q)	0.00 *3*	I_γ,Mult.,δ: from 1976Me12. Other: 13.0 *13* (1972Hu10). A_2=+0.34 *7*, A_4=+0.03 *11* (1976Me12).
		5339.8	4	1763.14	5/2$^+$			I_γ: other: 4 *2* from 1972Hu10.
		5883.5	64	1219.39	1/2$^+$	D(+Q)	0.00 *3*	I_γ,Mult.,δ: from 1976Me12. Other: 67.0 *67* (1972Hu10). A_2=−0.53 *8*, A_4=+0.01 *11* (1976Me12).
		7102.6	15	0	3/2$^+$			I_γ: other: 11.0 *11* (1972Hu10). A_2=+0.54 *5*, A_4=−0.01 *8* (1976Me12).
7178.6	1/2	1072.4	3	6106.2				I_γ: other: 4 *2* (1972Hu10).
		1775.1	1.7	5403.5				
		2339.4	1.5	4839.10				I_γ: other: 2 (2001Vo24).
		3005.0	0.8	4173.45				I_γ: other: 1 (2001Vo24).
		3119.3	22	4059.2				I_γ: others: 22 (2001Vo24), 24.0 *24* (1972Hu10). A_2=−0.02 *4*, A_4=+0.10 *66* (1976Me12). A_2=−0.01 *4*, A_4=+0.09 *24* (2001Vo24).
		3211.1	9	3967.3				I_γ: others: 9 (2001Vo24), 10.0 *10* (1972Hu10).
		3260.0	3	3918.48	3/2$^+$			I_γ: others: 3 (2001Vo24), 2 *1* (1972Hu10).
		4484.4	4	2693.9	(3/2,5/2)			I_γ: others: 4 (2001Vo24), 4 *2* (1972Hu10).
		5958.7	17	1219.39	1/2$^+$			I_γ: others: 17 (2001Vo24), 16.0 *16* (1972Hu10). A_2=−0.01 *3*, A_4=+0.06 *7* (2001Vo24). A_2=−0.01 *5*, A_4=+0.07 *8* (1976Me12).
		7177.8	38	0	3/2$^+$			I_γ: others: 38 (2001Vo24), 40.0 *40* (1972Hu10).

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$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10 (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J_i^π	E_γ[†]	I_γ[#]	E_f	J_f^π	Mult.[b]	δ[b]	Comments
								A_2=−0.01 *6*, A_4=−0.02 *7* (1976Me12). A_2=−0.02 *4*, A_4=−0.01 *5* (2001Vo24).
7185		1969	<0.5[a]	5215.8				
		2175	<0.5[a]	5010.1				
		3011	<5[a]	4173.45				
		3126	26[a] *6*	4059.2				
		4022	<3[a]	3162.99	$7/2^-$			
		4182	<3[a]	3002.75	5/2			
		4491	2[a] *1*	2693.9	(3/2,5/2)			
		4539	6[a] *2*	2645.70	(5/2,7/2)			
		5421	4[a] *3*	1763.14	$5/2^+$			
		5965	16[a] *7*	1219.39	$1/2^+$			
		7184	46[a] *10*	0	$3/2^+$			
7194.6	1/2,3/2	2184.4	0.8	5010.1				I_γ: other: 1.0 *5* (1972Hu10).
		2340.1	2	4854.4				I_γ: other: 2 *1* (1972Hu10).
		2355.4	1.8	4839.10				
		3016.6	16	4177.9	3/2			I_γ: other: 15.0 *15* (1972Hu10).
		3135.3	3	4059.2				I_γ: other: 3.0 *15* (1972Hu10).
		3227.1	1.8	3967.3				I_γ: other: 2 *1* (1972Hu10).
		3276.0	4	3918.48	$3/2^+$			I_γ: other: 5.0 *25* (1972Hu10).
		4500.4	1.1	2693.9	(3/2,5/2)			
		5431.0	0.5	1763.14	$5/2^+$			
		5974.7	67	1219.39	$1/2^+$			I_γ: other: 72.0 *72* (1972Hu10).
		7193.8	2	0	$3/2^+$			
7225.5	5/2	1119.3	1.8	6106.2				I_γ: other: 3.0 *15* (1972Hu10).
		1420		5806				I_γ: 1.0 *5* (1972Hu10).
		1543	2.0	5683				I_γ: other: 2 *1* (1972Hu10).
		1571	0.8	5655				
		1822.0	0.9	5403.5				I_γ: other: 1.0 *5* (1972Hu10).
		2009.6	7	5215.8				I_γ: other: 6.0 *30* (1972Hu10).
		3047.5	8	4177.9	3/2			I_γ: other: 7.0 *35* (1972Hu10).
		3166.2	2	4059.2				
		3306.9	1.5	3918.48	$3/2^+$			I_γ: other: 1.0 *5* (1972Hu10).
		4222.5		3002.75	5/2			I_γ: 3.0 *15* (1972Hu10).
		4531.3	10	2693.9	(3/2,5/2)	D+Q	+0.36 *4*	I_γ,Mult.,δ: from 1976Me12. Other: 10.0 *10* (1972Hu10). A_2=+0.43 *8*, A_4=−0.21 *9* (1976Me12).
		5461.9	7	1763.14	$5/2^+$	D+Q	−0.36 *7*	I_γ,Mult.,δ: from 1976Me12. Other: 8.0 *8* (1972Hu10). A_2=+0.02 *10*, A_4=−0.09 *17* (1976Me12).
		7224.7	59	0	$3/2^+$	D+Q	+0.36 *4*	I_γ,Mult.,δ: from 1976Me12. Other: 58.0 *58* (1972Hu10). A_2=+0.40 *7*, A_4=+0.04 *12* (1976Me12).
7234.0	$5/2^+$	1579	0.2	5655				
		3056.0		4177.9	3/2			I_γ: 1.0 *5* (1972Hu10).
		4539.8	1.8	2693.9	(3/2,5/2)			I_γ: other: 2 *1* (1972Hu10).
		4588.0	3	2645.70	(5/2,7/2)			I_γ: other: 2 *1* (1972Hu10).
		5470.4	0.2	1763.14	$5/2^+$			
		6014.1	1.8	1219.39	$1/2^+$			I_γ: other: 2 *1* (1972Hu10).
		7233.2	93	0	$3/2^+$	D+Q	−0.10	I_γ: other: 93.0 *93* (1972Hu10). Mult.,δ: from $\gamma(\theta)$ in 1960An06.
7272.7	(1/2,3/2)	1515	1.2	5758				I_γ: other: 1.5 *8* (1972Hu10).
		1869.2	0.2	5403.5				I_γ: other: 0.5 *3* (1972Hu10).
		2262.5	1.4	5010.1				I_γ: other: 1.5 *8* (1972Hu10).

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^{34}S(p,γ) 1976Me12,1976Sp08,1972Hu10 (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J_i^π	E_γ[†]	I_γ[#]	E_f	J_f^π	Mult.[b]	δ[b]	Comments
7272.7	(1/2,3/2)	2433.5	1.0	4839.10				I_γ: other: 1.0 *5* (1972Hu10).
		3099.1	0.4	4173.45				I_γ: other: 0.5 *3* (1972Hu10).
		3213.3	0.9	4059.2				I_γ: other: 1.0 *5* (1972Hu10).
		3305.2	0.8	3967.3				I_γ: other: 1.0 *5* from 1972Hu10.
		4578.5	1.1	2693.9	(3/2,5/2)			I_γ: other: 1.0 *5* (1972Hu10).
		6052.8	24	1219.39	1/2$^+$			I_γ: other: 23.0 *23* (1972Hu10).
		7271.9	69	0	3/2$^+$			I_γ: other: 69.0 *69* (1972Hu10).
7362.0	3/2	1707	0.3	5655				I_γ: other: 0.5 *3* (1972Hu10).
		2351.8	0.6	5010.1				I_γ: other: 0.5 *3* (1972Hu10).
		2522.8	1.0	4839.10				I_γ: other: 1.0 *5* (1972Hu10).
		3183.9	0.5	4177.9	3/2			I_γ: other: 1.0 *5* (1972Hu10).
		3302.6	2	4059.2				I_γ: others: 3.0 *15* (1972Hu10); 4 (1971Wi13).
		3394.5	3	3967.3				I_γ: others: 3.0 *15* (1972Hu10); 4 (1971Wi13).
		3443.3	0.8	3918.48	3/2$^+$			I_γ: other: 0.5 *3* (1972Hu10).
		4359.0	0.3	3002.75	5/2			I_γ: other: 0.5 *3* (1972Hu10).
		4667.8	0.5	2693.9	(3/2,5/2)			
		5598.4	8	1763.14	5/2$^+$	D(+Q)		I_γ: others: 10.0 *10* (1972Hu10); 9 (1971Wi13). Mult.,δ: δ=−0.01 *10* or δ>+2.5 (1977Ko34). A_2=−0.20 *24*, A_4=+0.92 *25* (1977Ko34).
		6142.0	73	1219.39	1/2$^+$	D+Q		I_γ: others: 70.0 *70* (1972Hu10); 72 (1971Wi13). Mult.,δ: δ=−0.05 *3* or +2.5 *3* (1977Ko34); δ=−20 or +0.22<δ<+0.26 (1971Wi13). A_2=−0.05 *3*, A_4=−0.04 *5* (1971Wi13). A_2=−0.25 *11*, A_4=+0.31 *11* (1977Ko34).
		7361.2	10	0	3/2$^+$	D+Q		I_γ: others: 10.0 *10* (1972Hu10); 11 (1971Wi13). Mult.,δ: −0.16<δ<−0.12 or +7.5<δ<+10 (1971Wi13). A_2=+0.19 *3*, A_4=−0.08 *5* (1971Wi13).
7396.0	7/2	3048.1	11	4347.76	5/2,9/2			I_γ: other: 10.0 *10* (1972Hu10).
		3282.1	8	4113.7	7/2			I_γ: other: 9.0 *9* (1972Hu10).
		3452.9	5	3942.9				I_γ: other: 8.0 *8* (1972Hu10).
		4232.7	47	3162.99	7/2$^-$	D+Q	+0.28 *25*	I_γ: other: 49.0 *49* (1972Hu10). Mult.,δ: from $\gamma(\theta)$ in 1977Ko35.
		4393.0	14	3002.75	5/2	D(+Q)	+0.011 *18*	I_γ: other: 14.0 *14* (1972Hu10). Mult.,δ: from $\gamma(\theta)$ in 1977Ko35.
		4750.0	13	2645.70	(5/2,7/2)			I_γ: other: 10.0 *10* (1972Hu10).
		5632.4	2	1763.14	5/2$^+$			
7451.1	3/2	3532.4	5.0 *25*	3918.48	3/2$^+$			I_γ: from 1972Hu10.
		4448.1	10 *1*	3002.75	5/2			I_γ: from 1972Hu10.
		4756.9	8.0 *8*	2693.9	(3/2,5/2)			I_γ: from 1972Hu10.
		5687.5	4 *2*	1763.14	5/2$^+$			I_γ: from 1972Hu10.
		6231.1	73 *7*	1219.39	1/2$^+$			I_γ: from 1972Hu10.
7501.1		3558.0	25 *3*	3942.9				I_γ: from 1972Hu10.
		4498.0	13.0 *13*	3002.75	5/2			I_γ: from 1972Hu10.
		4855.0	55 *6*	2645.70	(5/2,7/2)			I_γ: from 1972Hu10.
		5737.5	7.0 *7*	1763.14	5/2$^+$			I_γ: from 1972Hu10.
7502.9		1396.7	7	6106.2				I_γ: from 1976Me12 containing γ rays of 7501 and 7503 doublet initial levels. Other: 10.0 *10* for 7503 level only (1972Hu10).

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^{34}S(p,γ) 1976Me12,1976Sp08,1972Hu10 (continued)

γ(^{35}Cl) (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[#]	E_f	J^π_f	Mult.[b]	δ[b]	Comments
7502.9		1903.1	2	5599.7				I_γ: from 1976Me12 containing γ rays of 7501 and 7503 doublet initial levels.
		2492.7	2	5010.1				I_γ: from 1976Me12 containing γ rays of 7501 and 7503 doublet initial levels.
		3324.8	3	4177.9	3/2			I_γ: from 1976Me12 containing γ rays of 7501 and 7503 doublet initial levels.
		3443.5	50	4059.2				I_γ: from 1976Me12 containing γ rays of 7501 and 7503 doublet initial levels. Other: 70.0 *70* for 7503 level only (1972Hu10).
		3559.8	2	3942.9				I_γ: from 1976Me12 containing γ rays of 7501 and 7503 doublet initial levels.
		4339.6	4	3162.99	7/2⁻			I_γ: from 1976Me12 containing γ rays of 7501 and 7503 doublet initial levels.
		4499.8	4	3002.75	5/2			I_γ: from 1976Me12 containing γ rays of 7501 and 7503 doublet initial levels.
		4856.8	7	2645.70	(5/2,7/2)			I_γ: from 1976Me12 containing γ rays of 7501 and 7503 doublet initial levels.
		5739.3	0.7	1763.14	5/2⁺			I_γ: from 1976Me12 containing γ rays of 7501 and 7503 doublet initial levels.
		6282.9	18	1219.39	1/2⁺			I_γ: from 1976Me12 containing γ rays of 7501 and 7503 doublet initial levels. Other: 20.0 *20* for 7503 level only (1972Hu10).
		7502.0	0.3	0	3/2⁺			I_γ: from 1976Me12 containing γ rays of 7501 and 7503 doublet initial levels.
7518.8	7/2	1919.0	2 *1*	5599.7				I_γ: from 1972Hu10.
		1933	3.0 *15*	5586				I_γ: from 1972Hu10.
		3170.9	2 *1*	4347.76	5/2,9/2			I_γ: from 1972Hu10.
		3575.7	2 *1*	3942.9				I_γ: from 1972Hu10.
		4355.5	68 *7*	3162.99	7/2⁻	D(+Q)	+0.1 *2*	I_γ: from 1972Hu10. Mult.,δ: from 1977Ko35.
		4872.7	3.0 *15*	2645.70	(5/2,7/2)			I_γ: from 1972Hu10.
		5755.2	20 *2*	1763.14	5/2⁺	D+Q	+0.098 *22*	I_γ: from 1972Hu10. Mult.,δ: from γ(θ) in 1977Ko35.
7548.9	7/2⁻	1903	0.55 *6*	5646				I_γ: from 1971Pr11. Others: 0.6 (1976Me12); 0.5 *3* (1972Hu10); 1 (2001Vo24,2002Vo17).
		1963	0.1	5586				I_γ: other: 0.2 *1* (1972Hu10).
		2778.9	1.4 *1*	4769.86				I_γ: from 1971Pr11. Others: 1.3 (1976Me12); 1.5 *8* (1972Hu10).
		4385.6	94 *1*	3162.99	7/2⁻	M1+E2	−0.07 *2*	I_γ: from 1971Pr11. Others: 95 (1976Me12); 95.0 *95* (1972Hu10); ≥93 (1971Wi13). Mult.: D+Q from γ(θ) in 1967Wa22 and 1966Wa09; magnetic character from γ(pol) in 1967Wa22. δ: from 1966Wa09 and 1967Wa22. Others: δ=+0.08 *10* (1977Ko35), δ=−0.03 *3* from 2001Vo24 and 2002Vo17. N_{90}/N_0=0.72 *6* (1971Wi13). POL=+0.76 *6*, Δπ=no (1966Wa09). POL=+0.93 *10*, Δπ=no (1967Wa22). 0.021<POL<0.813 (1966Az01). A_2=+0.42 *3*, A_4=0.00 *3* (2001Vo24,2002Vo17). A_2=+0.42 *3*, A_4=−0.001 *27* (1966Az01). A_2=+0.42 *3*, A_4=+0.00 *3* (1967Ta08).
		4545.8	2.0 *1*	3002.75	5/2	D(+Q)	+0.6 *4*	I_γ: from 1971Pr11. Others: 2 (1976Me12); 1.8

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$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10 (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[#]	E_f	J^π_f	Mult.[b]	δ[b]	Comments
								9 (1972Hu10); 1.3 (1971Wi13). Mult.,δ: from $\gamma(\theta)$ in 1977Ko35.
7548.9	7/2⁻	4902.8	0.68 *7*	2645.70	(5/2,7/2)			I_γ: from 1971Pr11. Others: 0.5 (1976Me12); 0.5 *3* (1972Hu10); ≤5 (1971Wi13).
		5785.3	0.33 *4*	1763.14	5/2⁺			I_γ: from 1971Pr11. Others: 0.3 (1976Me12); 0.3 *2* (1972Hu10); ≈0.4 (1971Wi13).
		7548.0	0.27 *3*	0	3/2⁺	Q(+O)	+0.1 *2*	I_γ: from 1971Pr11. Others: 0.2 (1976Me12); 0.2 *1* (1972Hu10); ≈0.3 (1971Wi13). Mult.,δ: from $\gamma(\theta)$ in 1977Ko35.
7561.3	1/2,3/2	3383.2	0.9	4177.9	3/2			I_γ: other: 1.0 *5* (1972Hu10).
		3501.9	6	4059.2				I_γ: other: 5.0 *25* (1972Hu10).
		3593.8	0.5	3967.3				
		3642.6	1.6	3918.48	3/2⁺			I_γ: other: 1.0 *5* (1972Hu10).
		4867.0	22	2693.9	(3/2,5/2)			I_γ: other: 21.0 *21* (1972Hu10). A_2=+0.04 *3*, A_4=−0.05 *3* (1976Me12).
		6341.3	35	1219.39	1/2⁺			I_γ: other: 36.0 *36* (1972Hu10). A_2=−0.02 *2*, A_4=−0.05 *3* (1976Me12).
		7560.4	34	0	3/2⁺			I_γ: other: 36.0 *36* (1972Hu10). A_2=+0.01 *2*, A_4=−0.03 *3* (1976Me12).
7600.8	5/2	2437.4	1.9	5163.34	7/2			
		2719.7	2	4880.96				
		2761.6	2	4839.10				
		2830.8	1.3	4769.86				
		2976.3	1.8	4624.4				
		3422.7	3	4177.9	3/2			I_γ: other: 2 *1* (1972Hu10).
		3427.2		4173.45				I_γ: 2 *1* from 1972Hu10.
		3541.4	2	4059.2				I_γ: other: 1.5 *8* (1972Hu10).
		4437.5	5	3162.99	7/2⁻			I_γ: others: 4 *2* (1972Hu10); 8 (1971Wi13). Mult.,δ: Q+O with δ=−0.25 *15* for J_i=3/2 (1977Ko34). A_2=+0.19 *2*, A_4=+0.08 *2* (1977Ko34). A_2=+0.35 *32*, A_4=−0.44 *32* (1971Wi13).
		4597.7	8	3002.75	5/2	D+Q	+0.50 *22*	I_γ: others: 9.0 *9* (1972Hu10); 12 (1971Wi13). Mult.,δ: from 1971Wi13; δ=+0.85 *42* for J_i=3/2 (1977Ko34). A_2=+0.75 *7*, A_4=−0.06 *7* (1971Wi13). A_2=+0.53 *21*, A_4=−0.04 *25* (1976Me12). A_2=+0.42 *3*, A_4=−0.10 *3* (1977Ko34).
		4906.5	18	2693.9	(3/2,5/2)	D+Q	+0.32 *7*	I_γ: others: 19.0 *19* (1972Hu10); 24 (1971Wi13). δ: from 1971Wi13; other: δ=−5.3 *9* or +0.01 *5* for J_i=3/2 (1977Ko34). A_2=+0.33 *1*, A_4=−0.05 *1* (1977Ko34). A_2=+0.18 *4*, A_4=−0.02 *5* (1976Me12). A_2=+0.26 *10*, A_4=−0.05 *10* (1971Wi13).
		4954.7	1.9	2645.70	(5/2,7/2)			I_γ: other: 1.5 *8* (1972Hu10).
		5837.1	19	1763.14	5/2⁺	D+Q	+0.31 *7*	I_γ: others: 20 *2* (1971Wi13), 2 *1* (1972Hu10); 25 (1971Wi13). Mult.,δ: from 1971Wi13. δ=+0.2 *22* for J_i=3/2 (1977Ko34). A_2=+0.10 *7*, A_4=−0.10 *8* (1971Wi13). A_2=+0.12 *2*, A_4=−0.10 *3* (1976Me12). A_2=+0.30 *9*, A_4=−0.17 *9* (1977Ko34).
		6380.8	1.1	1219.39	1/2⁺			I_γ: others: 1.0 *5* (1972Hu10); ≤1.5 (1971Wi13).

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$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10 (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[#]	E_f	J^π_f	Mult.[b]	δ[b]	Comments
7600.8	5/2	7599.9	33	0	3/2+	D+Q	−0.19 *4*	I_γ: others: 31.0 *31* (1972Hu10); 31 (1971Wi13). Mult.,δ: from 1976Me12. Others: δ=+0.17 *3* (1971Wi13). A_2=−0.73 *7*, A_4=+0.04 *6* (1971Wi13). A_2=−0.62 *5*, A_4=−0.06 *5* (1976Me12). A_2=−0.56 *3*, A_4=−0.05 *2* (1977Ko34).
7618.7	5/2	1438	0.6	6181				
		2402.8	2	5215.8				I_γ: other: 3.0 *15* (1972Hu10).
		2455.3	0.7	5163.34	7/2			
		2779.5	0.7	4839.10				
		3440.6	3	4177.9	3/2			I_γ: other: 3.0 *15* (1972Hu10).
		3559.3	2	4059.2				I_γ: other: 1.0 *5* (1972Hu10).
		4455.4	3	3162.99	7/2−			I_γ: other: 3.0 *15* (1972Hu10).
		5855.0	10	1763.14	5/2+	D+Q	+0.48 *11*	I_γ: other: 10.0 *10* (1972Hu10). Mult.,δ: from $\gamma(\theta)$ in 1977Ko34. A_2=+0.38 *5*, A_4=−0.01 *6* (1976Me12). A_2=−0.11 *3*, A_4=−0.06 *3* (1977Ko34).
		7617.8	78	0	3/2+	D+Q	+2.2 *3*	I_γ: other: 80 *8* (1972Hu10). Mult.,δ: from $\gamma(\theta)$ in 1977Ko34. A_2=−0.46 *4*, A_4=+0.03 *4* (1976Me12). A_2=−0.57 *3*, A_4=+0.27 *3* (1977Ko34).
7656.6		1550.4	3.0 *15*	6106.2				I_γ: from 1972Hu10.
		1974	1.0 *5*	5683				I_γ: from 1972Hu10.
		2002	1.0 *5*	5655				I_γ: from 1972Hu10.
		2817.4	3.0 *15*	4839.10				I_γ: from 1972Hu10.
		3478.5	1.0 *5*	4177.9	3/2			I_γ: from 1972Hu10.
		3597.2	32.0 *32*	4059.2				I_γ: from 1972Hu10.
		3689.1	2.0 *10*	3967.3				I_γ: from 1972Hu10.
		7655.7	57 *6*	0	3/2+			I_γ: from 1972Hu10.
7671.9	5/2	2072.1	1.2	5599.7				
		2508.5	4	5163.34	7/2			I_γ: other: 4 *2* (1972Hu10).
		2790.8	1.2	4880.96				I_γ: other: 1.0 *5* (1972Hu10).
		2901.9	1.0	4769.86				
		3324.0	6	4347.76	5/2,9/2			I_γ: other: 6.0 *6* (1972Hu10).
		3498.3	5	4173.45				I_γ: other: 4 *2* (1972Hu10).
		4508.6	1.8	3162.99	7/2−			I_γ: other: 4 *2* (1972Hu10).
		4668.8	9	3002.75	5/2	D+Q	−1.4 *5*	I_γ: other: 8.0 *8* (1972Hu10). Mult.,δ: from $\gamma(\theta)$ in 1977Ko34. A_2=−0.35 *4*, A_4=+0.01 *5* (1976Me12). A_2=+0.52 *1*, A_4=−0.18 *1* (1977Ko34).
		5025.8	13	2645.70	(5/2,7/2)	D(+Q)	+0.3 *22*	I_γ: other: 14.0 *14* (1972Hu10). Mult.,δ: from $\gamma(\theta)$ in 1977Ko34. A_2=+0.41 *3*, A_4=−0.03 *4* (1976Me12). A_2=+0.33 *8*, A_4=−0.12 *8* (1977Ko34).
		5908.2	57	1763.14	5/2+	D+Q	+0.86 *14*	I_γ: other: 57.0 *57* (1972Hu10). Mult.,δ: from $\gamma(\theta)$ in 1977Ko34. A_2=−0.38 *4*, A_4=−0.06 *5* (1976Me12). A_2=−0.40 *9*, A_4=+0.07 *8* (1977Ko34).
		7671.0	0.8	0	3/2+			I_γ: other: 2 *1* (1972Hu10).
7685.5	3/2−	1927		5758				I_γ: 0.5 *3* (1972Hu10).
		2085.7	2	5599.7				I_γ: other: 2 *1* (1972Hu10).
		2281.9	2	5403.5				I_γ: other: 2 *1* (1972Hu10).
		2469.6	0.7	5215.8				I_γ: other: 0.5 *3* (1972Hu10).
		2675.3	0.6	5010.1				
		3507.4	3	4177.9	3/2			I_γ: other: 2 *1* (1972Hu10).

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^{34}S(p,γ) 1976Me12,1976Sp08,1972Hu10 (continued)

γ(^{35}Cl) (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[#]	E_f	J^π_f	Mult.[b]	δ[b]	Comments
7685.5	3/2$^-$	3511.9	4	4173.45				I_γ: other: 5.0 *25* (1972Hu10). A_2=+0.01 *14*, A_4=+0.23 *16* (1977Ko34).
		3626.1	6	4059.2				I_γ: other: 6.0 *6* (1972Hu10).
		4682.4	10	3002.75	5/2	D(+Q)	−0.01 *5*	I_γ: other: 9.0 *9* (1972Hu10). Mult.,δ: other: δ=−2.4 *+20−4* (1976Me12); δ=+0.01 *7* (1977Ko34). A_2=−0.04 *6*, A_4=−0.09 *7* (1976Me12). A_2=−0.06 *12*, A_4=+0.08 *13* (1977Ko34).
		5921.8	0.7	1763.14	5/2$^+$			
		6465.5	1	1219.39	1/2$^+$			I_γ: other: 1.0 *5* (1972Hu10).
		7684.6	70	0	3/2$^+$	D+Q	+6.0 *7*	I_γ,Mult.,δ: from 1976Me12. Other: Iγ=72.0 *72* (1972Hu10); δ<−3.5 for J_i=3/2 (1977Ko34). A_2=+0.25 *1*, A_4=−0.03 *1* (1976Me12). A_2=+0.18 *11*, A_4=−0.04 *11* (1977Ko34).
7693.9		2094.1	1.0 *5*	5599.7				I_γ: from 1972Hu10.
		3634.5	1.0 *5*	4059.2				I_γ: from 1972Hu10.
		4690.8	2.0 *10*	3002.75	5/2			I_γ: from 1972Hu10.
		7693.0	96 *10*	0	3/2$^+$			I_γ: from 1972Hu10.
7706.4	5/2$^+$	2051	4	5655				I_γ: other: 4 *2* (1972Hu10). Mult.,δ: D+Q with δ≤−2 or ≥13 for J_f(5655)=5/2 (1977Ko34), but J_f(5655)=(3/2)$^+$ from the Adopted Levels. A_2=+0.14 *15*, A_4=−0.39 *17* (1977Ko34).
		2825.3	0.4	4880.96				
		2867.2	1.3	4839.10				I_γ: other: 1.0 *5* (1972Hu10).
		3081.9	0.8	4624.4				
		3528.3	2	4177.9	3/2			
		3592.5	2	4113.7	7/2			I_γ: other: 1.0 *5* (1972Hu10).
		3787.7	0.6	3918.48	3/2$^+$			
		4703.3	1	3002.75	5/2			I_γ: other: 1.0 *5* (1972Hu10).
		5060.3	4	2645.70	(5/2,7/2)	D+Q	−1.01 *10*	I_γ: other: 4 *2* (1972Hu10). Mult.,δ: or ≥13 (1977Ko34). A_2=−0.03 *6*, A_4=+0.08 *8* (1976Me12). A_2=−0.18 *17*, A_4=+0.41 *18* (1977Ko34).
		5942.7	0.4	1763.14	5/2$^+$			
		6486.4	0.5	1219.39	1/2$^+$			I_γ: other: 1.0 *5* (1972Hu10).
		7705.5	83	0	3/2$^+$	D(+Q)	+0.1 *1*	I_γ: other: 88.0 *88* (1972Hu10). Mult.,δ: from γ(θ) in 1977Ko34. A_2=−0.54 *3*, A_4=−0.07 *3* (1976Me12). A_2=−0.63 *4*, A_4=+0.15 *3* (1977Ko34).
7744.8	7/2	2863.7	20	4880.96				I_γ: other: 19.0 *19* (1972Hu10).
		2974.8	2	4769.86				I_γ: other: 1.0 *5* (1972Hu10).
		3630.9	6	4113.7	7/2			I_γ: other: 5.0 *25* (1972Hu10).
		3801.7	10	3942.9		D(+Q)	−0.16 *17*	I_γ: other: 12.0 *12* (1972Hu10). Mult.,δ: from γ(θ) in 1977Ko35.
		4581.5	16	3162.99	7/2$^-$	D+Q	+0.18 *8*	Mult.,δ: from γ(θ) in 1977Ko35.
		5098.7	43	2645.70	(5/2,7/2)	D(+Q)	+0.1 *30*	Mult.,δ: from γ(θ) in 1977Ko35.
		5981.1	2.4	1763.14	5/2$^+$			
		6524.8	0.6	1219.39	1/2$^+$			
7777.1		2122	0.5	5655				
		2613.7	1.1	5163.34	7/2			
		2937.9	2	4839.10				I_γ: other: 2 *1* (1972Hu10).
		3007.1	3	4769.86				I_γ: other: 2 *1* (1972Hu10).
		3599.0	7	4177.9	3/2			I_γ: other: 7.0 *7* (1972Hu10).
		3717.7	1.1	4059.2				I_γ: other: 1.0 *5* (1972Hu10).

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$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10 (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J_i^π	E_γ[†]	I_γ[#]	E_f	J_f^π	Mult.[b]	δ[b]	Comments
7777.1		4613.8	2	3162.99	7/2$^-$			
		4774.0	4	3002.75	5/2			I_γ: other: 4 *2* (1972Hu10).
		5082.8	9	2693.9	(3/2,5/2)			I_γ: other: 7.0 *7* (1972Hu10).
		5131.0	1.3	2645.70	(5/2,7/2)			I_γ: other: 2 *1* (1972Hu10).
		6013.4	40	1763.14	5/2$^+$			I_γ: other: 46.0 *46* (1972Hu10).
		6557.1	11	1219.39	1/2$^+$			I_γ: other: 13.0 *13* (1972Hu10).
		7776.2	18	0	3/2$^+$			I_γ: other: 16.0 *16* (1972Hu10).
7781.7	5/2$^-$	1675.5	1.4	6106.2				
		2181.9		5599.7				I_γ: 1.0 *5* from 1972Hu10.
		2378.1		5403.5				I_γ: 1.0 *5* from 1972Hu10.
		2565.8	1.5	5215.8				I_γ: other: 1.0 *5* (1972Hu10).
		3011.7	7	4769.86				I_γ: other: 7.0 *7* (1972Hu10).
		3433.8	2	4347.76	5/2,9/2			I_γ: other: 2 *1* (1972Hu10).
		3608.1	36	4173.45				I_γ: other: 40.0 *40* (1972Hu10).
		3667.8	6	4113.7	7/2			I_γ: other: 3.0 *15* (1972Hu10).
		3722.3	1.2	4059.2				I_γ: other: 1.0 *5* (1972Hu10).
		3863.0	11	3918.48	3/2$^+$			I_γ: other: 10.0 *10* (1972Hu10).
		4618.4	5	3162.99	7/2$^-$			I_γ: other: 5.0 *25* (1972Hu10).
		5135.6	8	2645.70	(5/2,7/2)			I_γ: other: 9.0 *9* (1972Hu10).
		6018.0	18	1763.14	5/2$^+$			I_γ: other: 20.0 *20* (1972Hu10).
		6561.7	1.2	1219.39	1/2$^+$			
		7780.8	1.7	0	3/2$^+$			
7797.1	1/2	2039		5758				I_γ: 0.5 *3* from 1972Hu10.
		2142	1.3	5655				I_γ: other: 1.0 *5* (1972Hu10). A_2=+0.03 *8* (1988Va06).
		2393.5	2	5403.5				I_γ: other: 2 *1* (1972Hu10). A_2=+0.04 *5* (1988Va06).
		2786.9	2	5010.1				I_γ: other: 2 *1* (1972Hu10). A_2=−0.07 *5* (1988Va06).
		2942.6	1.3	4854.4				I_γ: other: 0.5 *3* (1972Hu10).
		2957.9	1.1	4839.10				
		3737.7	0.3	4059.2				
		6577.1	9	1219.39	1/2$^+$			I_γ: other: 9.0 *9* (1972Hu10). δ: δ(Q/D)=+0.21 *4* or −1.9 *2* for J=3/2 (1976Me12), which is invalidated by the J^π=1/2$^-$ assignment for the 7797 level in 1988Va06. A_2=−0.02 *6*, A_4=−0.14 *7* (1976Me12). A_2=−0.03 *3* (1988Va06).
		7796.2	83	0	3/2$^+$			I_γ: other: 85.0 *85* (1972Hu10). δ: δ(Q/D)=−0.33 *2* or −19 *+5−10* for J=3/2 (1976Me12), which is invalidated by the J^π=1/2$^-$ assignment for the 7797 level in 1988Va06. A_2=−0.06 *2*, A_4=−0.03 *2* (1976Me12). A_2=+0.006 *14* (1988Va06).
7837.2	3/2	1656	0.5	6181				I_γ: for 7837 and 7839 doublet initial levels (1976Me12). Other: 1 from 2001Vo24. .
		2237.4	0.5	5599.7				I_γ: for 7837 and 7839 doublet initial levels (1976Me12). Other: 1 (2001Vo24).
		2621.3	1.4	5215.8				I_γ: for 7837 and 7839 doublet initial levels (1976Me12). Others: 1 (2001Vo24); 1.0 *5* (1972Hu10).
		3659.1	28	4177.9	3/2	M1+E2	−0.05 *3*	I_γ: for 7837 and 7839 doublet initial levels (1976Me12). Others: 28 (2001Vo24); 35.0 *35* (1972Hu10). Mult.,δ: D+Q from $\gamma(\theta)$ in 1967Wa22; magnetic

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^{34}S(p,γ) 1976Me12,1976Sp08,1972Hu10 (continued)

γ(^{35}Cl) (continued)

E_i(level)	E_γ[†]	I_γ[#]	E_f	J^π_f	Mult.[b]	δ[b]	Comments
							character from γ(pol) in 1967Wa22. Other: δ=−0.04 *3* or +0.27 *5* from 2001Vo24. A_2=+0.17 *1*, A_4=−0.01 *2* (2001Vo24). POL=+0.66 *18*, $\Delta\pi$=no (1967Wa22).
7837.2	3663.5	3	4173.45				I_γ: for 7837 and 7839 doublet initial levels (1976Me12). Others: 3 (2001Vo24); 3.0 *15* (1972Hu10); 34 (1971Wi13). A_2=+0.17 *1*, A_4=−0.005 *13* (1967Ta08).
	3777.8	2	4059.2				I_γ: for 7837 and 7839 doublet initial levels (1976Me12). Others: 2 (2001Vo24); 2 *1* (1972Hu10); 4 (1971Wi13).
	3894.1	0.5	3942.9				I_γ: for 7837 and 7839 doublet initial levels (1976Me12). Other: 1 (2001Vo24).
	3918.5	0.2	3918.48	3/2$^+$			I_γ: for 7837 and 7839 doublet initial levels (1976Me12).
	4834.1	4	3002.75	5/2			I_γ: for 7837 and 7839 doublet initial levels (1976Me12). Others: 4 (2001Vo24); 4 *2* (1972Hu10); 4 (1971Wi13).
	6073.5	1.9	1763.14	5/2$^+$			I_γ: for 7837 and 7839 doublet initial levels (1976Me12). Others: 2 (2001Vo24); ≤1 (1971Wi13).
	6617.1	37	1219.39	1/2$^+$	D+Q	+0.06 *3*	I_γ: for 7837 and 7839 doublet initial levels (1976Me12). Others: 37 (2001Vo24); 27.0 *27* (1972Hu10); 33 (1971Wi13). Mult.,δ: from $\gamma(\theta)$ in 1967Wa22. POL=+0.17 *45* (1967Wa22). A_2=−0.46 *4*, A_4=+0.04 *6* (1967Ta08).
	7836.3	21	0	3/2$^+$	D(+Q)	+0.02 *3*	I_γ: for 7837 and 7839 doublet initial levels (1976Me12). Others: 21 (2001Vo24); 27.0 *27* (1972Hu10); 25 (1971Wi13). Mult.,δ: from $\gamma(\theta)$ in 1967Wa22. Other: 0.00 *3* from 2001Vo24. POL=−1.1 *8* (1967Wa22). A_2=+0.41 *3*, A_4=+0.012 *40* (1967Ta08). A_2=+0.40 *2*, A_4=+0.01 *4* (2001Vo24).
7839.0	1658	0.5	6181				I_γ: for 7837 and 7839 doublet initial levels (1976Me12).
	2239.2	0.5	5599.7				I_γ: for 7837 and 7839 doublet initial levels (1976Me12).
	2623.1	1.4	5215.8				I_γ: for 7837 and 7839 doublet initial levels (1976Me12).
	3660.9	28	4177.9	3/2			I_γ: for 7837 and 7839 doublet initial levels (1976Me12).
	3665.3	3	4173.45				I_γ: for 7837 and 7839 doublet initial levels (1976Me12).
	3779.6	2	4059.2				I_γ: for 7837 and 7839 doublet initial levels (1976Me12).
	3895.9	0.5	3942.9				I_γ: for 7837 and 7839 doublet initial levels (1976Me12).
	3920.3	0.2	3918.48	3/2$^+$			I_γ: for 7837 and 7839 doublet initial levels (1976Me12).
	4835.9	4	3002.75	5/2			I_γ: for 7837 and 7839 doublet initial levels (1976Me12).
	6075.3	1.9	1763.14	5/2$^+$			I_γ: for 7837 and 7839 doublet initial levels (1976Me12).
	6618.9	37	1219.39	1/2$^+$			I_γ: for 7837 and 7839 doublet initial levels (1976Me12).
	7838.1	21	0	3/2$^+$			I_γ: for 7837 and 7839 doublet initial levels (1976Me12).
7868.7	2652.8	1.0	5215.8				
	2858.5	3	5010.1				I_γ: other: 4 *2* (1972Hu10).
	3695.0	5	4173.45				I_γ: other: 4 *2* (1972Hu10).
	6105.0	17	1763.14	5/2$^+$			I_γ: other: 18.0 *18* (1972Hu10).
	6648.6	13	1219.39	1/2$^+$			I_γ: other: 12.0 *12* (1972Hu10).

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$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10 (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[#]	E_f	J^π_f	Comments
7868.7		7867.8	61	0	3/2+	I_γ: other: 62.0 *62* (1972Hu10).
7880.8	3/2,5/2	2226	12	5655		I_γ: other: 9.0 *9* (1972Hu10).
		3041.6	1.7	4839.10		I_γ: other: 3.0 *15* (1972Hu10).
		3256.2	2	4624.4		I_γ: other: 1.0 *5* (1972Hu10).
		3702.7	3	4177.9	3/2	
		3766.9	1.3	4113.7	7/2	I_γ: other: 4 *2* (1972Hu10).
		3821.4	2	4059.2		I_γ: other: 1.0 *5* (1972Hu10).
		3962.1	2	3918.48	3/2+	I_γ: other: 2 *1* (1972Hu10).
		4877.7	11	3002.75	5/2	I_γ: other: 10.0 *10* (1972Hu10).
		5186.5	25	2693.9	(3/2,5/2)	I_γ: other: 30.0 *30* (1972Hu10). A_2=+0.50 *3*, A_4=+0.09 *4* (1976Me12).
		6117.1	30	1763.14	5/2+	I_γ: other: 31.0 *31* (1972Hu10). A_2=+0.08 *3*, A_4=−0.03 *4* (1976Me12).
		7879.9	10	0	3/2+	I_γ: other: 9.0 *9* (1972Hu10). A_2=−0.06 *4*, A_4=+0.04 *5* (1976Me12).
7899.2	(3/2−,5/2)	2253	1.5	5646		
		2735.8	0.9	5163.34	7/2	
		3018.1	0.8	4880.96		
		3274.6	0.5	4624.4		
		3721.1	8	4177.9	3/2	I_γ: other: 8.0 *8* (1972Hu10).
		3725.5	8	4173.45		I_γ: other: 6.0 *6* (1972Hu10).
		3839.8	3	4059.2		I_γ: other: 2 *1* (1972Hu10).
		4735.9	4	3162.99	7/2−	I_γ: other: 4 *2* (1972Hu10).
		5204.9	1.3	2693.9	(3/2,5/2)	
		6135.5	68	1763.14	5/2+	I_γ: other: 77.0 *77* (1972Hu10).
		7898.2	4	0	3/2+	I_γ: other: 3.0 *15* (1972Hu10).
7923.3		3084.1	1.4	4839.10		
		3153.3	2	4769.86		
		3298.7	1.9	4624.4		
		3745.2	3	4177.9	3/2	I_γ: other: 2 *1* (1972Hu10).
		3749.6	5	4173.45		I_γ: other: 6.0 *6* (1972Hu10).
		3863.9	1.7	4059.2		I_γ: other: 1.0 *5* (1972Hu10).
		3955.8	1.5	3967.3		I_γ: other: 1.0 *5* (1972Hu10).
		4004.6	1.2	3918.48	3/2+	I_γ: other: 1.0 *5* (1972Hu10).
		4920.2	0.9	3002.75	5/2	
		5229.0	11	2693.9	(3/2,5/2)	I_γ: other: 12.0 *12* (1972Hu10).
		5277.2	0.9	2645.70	(5/2,7/2)	
		6159.6	1.5	1763.14	5/2+	I_γ: other: 1.0 *5* (1972Hu10).
		6703.2	27	1219.39	1/2+	I_γ: other: 28.0 *28* (1972Hu10).
		7922.3	41	0	3/2+	I_γ: other: 48.0 *48* (1972Hu10).
7970.3		2324	3	5646		I_γ: other: 3.0 *15* (1972Hu10).
		2384	3	5586		
		2754.4	3	5215.8		I_γ: other: 3.0 *15* (1972Hu10).
		3089.2	10	4880.96		I_γ: other: 9.0 *9* (1972Hu10).
		3200.3	5	4769.86		I_γ: other: 6.0 *6* (1972Hu10).
		3622.3	8	4347.76	5/2,9/2	I_γ: other: 8.0 *8* (1972Hu10).
		3792.2	6	4177.9	3/2	I_γ: other: 6.0 *6* (1972Hu10).
		3856.4	6	4113.7	7/2	I_γ: other: 2 *1* (1972Hu10).
		4807.0	10	3162.99	7/2−	I_γ: other: 13.0 *13* (1972Hu10).
		4967.2	5	3002.75	5/2	I_γ: other: 5.0 *25* (1972Hu10).
		5324.2	20	2645.70	(5/2,7/2)	I_γ: other: 22.0 *22* (1972Hu10).
		6206.6	20	1763.14	5/2+	I_γ: other: 21.0 *21* (1972Hu10).
		6750.2	1	1219.39	1/2+	
7987.8	3/2	1881.6	0.5	6106.2		I_γ: other: 1.0 *5* (1972Hu10).
		2333	1.7	5655		I_γ: other: 1.0 *5* (1972Hu10).

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^{34}S(p,γ) 1976Me12,1976Sp08,1972Hu10 (continued)

γ(^{35}Cl) (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[#]	E_f	J^π_f	Mult.[b]	δ[b]	Comments
7987.8	3/2	3809.7	<0.8	4177.9	3/2			I_γ: total Iγ=0.8 for γ decays to 4173 and 4178 doublet final levels (1976Me12).
		3814.1	<0.8	4173.45				I_γ: total Iγ=0.8 for γ decays to 4173 and 4178 doublet final levels (1976Me12).
		3928.4	1	4059.2				
		6767.7	60	1219.39	1/2$^+$	D+Q	+0.21 *4*	I_γ: other: 62.0 *62* (1972Hu10). Mult.,δ: or −3.1 *4* (1976Me12). A_2=−0.07 *2*, A_4=−0.05 *3* (1976Me12).
		7986.8	36	0	3/2$^+$	D+Q	−0.36 *2*	I_γ: other: 36.0 *36* (1972Hu10). Mult.,δ: or −11 *3* (1976Me12). A_2=−0.10 *2*, A_4=−0.06 *2* (1976Me12).
7995.6	5/2	2190	0.6	5806				
		2395.8	1.1	5599.7				
		2592.0	3	5403.5				I_γ: other: 3.0 *15* (1972Hu10).
		2832.1	1.5	5163.34	7/2			
		3817.5	4	4177.9	3/2			I_γ: other: 3.0 *15* (1972Hu10).
		3936.2	1.1	4059.2				
		4832.3	7	3162.99	7/2$^-$			I_γ: other: 6.0 *6* (1972Hu10).
		4992.5	2	3002.75	5/2			
		5301.3	0.2	2693.9	(3/2,5/2)			
		5349.5	0.6	2645.70	(5/2,7/2)			
		6231.9	0.9	1763.14	5/2$^+$			I_γ: other: 1.0 *5* (1972Hu10).
		7994.6	78	0	3/2$^+$			I_γ: other: 87.0 *87* (1972Hu10).
8001.0	(7/2,9/2$^+$)	2355	1.8	5646				I_γ: other: 3.0 *15* (1972Hu10).
		2785.1	0.8	5215.8				
		3119.9	0.7	4880.96				
		3231.0	8	4769.86				I_γ: other: 8.0 *8* (1972Hu10).
		3653.0	1.9	4347.76	5/2,9/2			
		3827.3	10	4173.45				I_γ: other: 8.0 *8* (1972Hu10).
		4057.9	3	3942.9				I_γ: other: 2 *1* (1972Hu10).
		4837.7	0.8	3162.99	7/2$^-$			
		5354.9	1.6	2645.70	(5/2,7/2)			I_γ: other: 2 *1* (1972Hu10).
		6237.3	70	1763.14	5/2$^+$			I_γ: other: 76.0 *76* (1972Hu10).
		8000.0	1.4	0	3/2$^+$			I_γ: other: 1.0 *5* (1972Hu10).
8005.0	5/2$^+$	2350	1.8	5655				
		3123.9	0.6	4880.96				
		3165.8	3	4839.10				
		3826.9	<1.3	4177.9	3/2			I_γ: total Iγ=1.3 for γ decays to 4173 and 4178 doublet final levels (1976Me12).
		3831.3	<1.3	4173.45				I_γ: total Iγ=1.3 for γ decays to 4173 and 4178 doublet final levels (1976Me12).
		4841.7	15	3162.99	7/2$^-$			I_γ: other: 17.0 *17* (1972Hu10). A_2=−0.084 *14*, A_4=−0.0041 *15* (1981Bi05).
		5001.9	2	3002.75	5/2			I_γ: other: 3.0 *15* (1972Hu10).
		5310.7	1.7	2693.9	(3/2,5/2)			
		5358.9	4	2645.70	(5/2,7/2)			
		6241.3	15	1763.14	5/2$^+$			I_γ: other: 20.0 *20* (1972Hu10). A_2=−0.11 *2*, A_4=−0.09 *3* (1976Me12).
		6784.9	0.6	1219.39	1/2$^+$			
		8004.0	55	0	3/2$^+$			I_γ: other: 60.0 *60* (1972Hu10). A_2=−0.03 *1*, A_4=−0.02 *2* (1976Me12).
8035.7		2230	6.0 *6*	5806				I_γ: from 1972Hu10.
		3857.6	15.0 *15*	4177.9	3/2			I_γ: from 1972Hu10.
		3976.3	11.0 *11*	4059.2				I_γ: from 1972Hu10.
		5341.4	5.0 *25*	2693.9	(3/2,5/2)			I_γ: from 1972Hu10.

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^{34}S(p,γ) 1976Me12,1976Sp08,1972Hu10 (continued)

γ(^{35}Cl) (continued)

E$_i$(level)	J$^\pi_i$	E$_\gamma$†	I$_\gamma$#	E$_f$	J$^\pi_f$	Mult.[b]	δ[b]	Comments
8035.7		6272.0	5.0 *25*	1763.14	5/2+			I$_\gamma$: from 1972Hu10.
		6815.6	56 *6*	1219.39	1/2+			I$_\gamma$: from 1972Hu10.
		8034.7	2.0 *10*	0	3/2+			I$_\gamma$: from 1972Hu10.
8038.5	3/2	3183.9	0.6	4854.4				
		3199.2	1.6	4839.10				
		3413.9	3	4624.4				
		3860.4	1.6	4177.9	3/2			
		3979.1	3	4059.2				I$_\gamma$: other: 4 *2* (1972Hu10).
		4071.0	0.8	3967.3				
		4119.8	1.4	3918.48	3/2+			
		5035.4	4	3002.75	5/2			I$_\gamma$: other: 3.0 *15* (1972Hu10).
		5344.2	57	2693.9	(3/2,5/2)	D+Q	−0.20 *2*	I$_\gamma$: other: 61.0 *61* (1972Hu10). Mult.,δ: or +19 *+10−5* (1976Me12). A$_2$=+0.08 *2*, A$_4$=−0.03 *2* (1976Me12).
		6274.8	15	1763.14	5/2+			I$_\gamma$: other: 17.0 *17* (1972Hu10). A$_2$=+0.24 *2*, A$_4$=+0.01 *3* (1976Me12).
		6818.4	4	1219.39	1/2+			I$_\gamma$: other: 6.0 *6* (1972Hu10).
		8037.5	8	0	3/2+	D+Q	−0.13 *4*	I$_\gamma$: other: 9.0 *9* (1972Hu10). Mult.,δ: or +8.1 *+33−18* (1976Me12). A$_2$=+0.20 *3*, A$_4$=−0.08 *3* (1976Me12).
8075.9		2430	11	5646				I$_\gamma$: other: 10.0 *10* (1972Hu10).
		2490	11	5586				I$_\gamma$: other: 11.0 *11* (1972Hu10).
		2860.0	8	5215.8				I$_\gamma$: other: 7.0 *7* (1972Hu10).
		2912.4		5163.34	7/2			I$_\gamma$: 1.0 *5* from 1972Hu10.
		3194.8		4880.96				I$_\gamma$: 2 *1* from 1972Hu10.
		3305.9	4	4769.86				I$_\gamma$: other: 4 *2* (1972Hu10).
		3897.8	8	4177.9	3/2			I$_\gamma$: other: 7.0 *7* (1972Hu10).
		3962.0	2	4113.7	7/2			I$_\gamma$: other: 2 *1* (1972Hu10).
		5072.8	2	3002.75	5/2			I$_\gamma$: other: 2 *1* (1972Hu10).
		6312.2	56	1763.14	5/2+			I$_\gamma$: other: 53.0 *53* (1972Hu10).
		8074.9		0	3/2+			I$_\gamma$: 1.0 *5* from 1972Hu10.
8096.1	(5/2,7/2+)	2290	2	5806				I$_\gamma$: other: 1.0 *5* (1972Hu10).
		2450	5	5646				I$_\gamma$: other: 5.0 *25* (1972Hu10).
		2496.3	5	5599.7				I$_\gamma$: other: 6.0 *6* (1972Hu10).
		2880.2	1.3	5215.8				I$_\gamma$: other: 1.0 *5* (1972Hu10).
		2932.6	2	5163.34	7/2			I$_\gamma$: other: 2 *1* (1972Hu10).
		3256.8	3	4839.10				I$_\gamma$: other: 3.0 *15* (1972Hu10).
		3326.1	2	4769.86				I$_\gamma$: other: 1.0 *5* (1972Hu10).
		3918.0	1.9	4177.9	3/2			
		4036.7	2	4059.2				I$_\gamma$: other: 2 *1* (1972Hu10).
		4177.4	1.8	3918.48	3/2+			I$_\gamma$: other: 3.0 *15* (1972Hu10).
		4932.7	5	3162.99	7/2−			I$_\gamma$: other: 5.0 *25* (1972Hu10).
		5401.8	5	2693.9	(3/2,5/2)			I$_\gamma$: other: 5.0 *25* (1972Hu10).
		5449.9	17	2645.70	(5/2,7/2)			I$_\gamma$: other: 17.0 *17* (1972Hu10).
		6332.4	8	1763.14	5/2+			I$_\gamma$: other: 9.0 *9* (1972Hu10).
		8095.1	39	0	3/2+			I$_\gamma$: other: 40.0 *40* (1972Hu10).
8106.4	3/2	2506.6	0.5	5599.7				I$_\gamma$: other: 0.5 *3* (1972Hu10).
		3251.8	0.6	4854.4				I$_\gamma$: other: 0.5 *3* (1972Hu10).
		3267.1	0.5	4839.10				I$_\gamma$: other: 0.5 *3* (1972Hu10).
		3928.3	1.6	4177.9	3/2			I$_\gamma$: other: 1.0 *5* (1972Hu10).
		4047.0	1.3	4059.2				I$_\gamma$: other: 0.5 *3* (1972Hu10).
		4138.8	3	3967.3				I$_\gamma$: other: 3.0 *15* (1972Hu10).
		5103.3	9	3002.75	5/2			I$_\gamma$: other: 8.0 *8* (1972Hu10).
		5412.1	9	2693.9	(3/2,5/2)	D+Q	+4.9 *10*	I$_\gamma$: other: 8.0 *8* (1972Hu10).

Continued on next page (footnotes at end of table)

^{34}S(p,γ) 1976Me12,1976Sp08,1972Hu10 (continued)

$\gamma(^{35}$Cl) (continued)

E_i(level)	J_i^π	E_γ[†]	I_γ[#]	E_f	J_f^π	Mult.[b]	δ[b]	Comments
								Mult.,δ: −0.05 *4* (1976Me12).
								A_2=+0.32 *4*, A_4=−0.08 *5* (1976Me12).
8106.4	3/2	6342.6	3	1763.14	5/2$^+$			I_γ: other: 3.0 *15* (1972Hu10).
		6886.3	35	1219.39	1/2$^+$			I_γ: other: 35.0 *35* (1972Hu10). A_2=−0.46 *2*, A_4=−0.05 *3* (1976Me12).
		8105.4	37	0	3/2$^+$	D+Q	+1.2 *1*	I_γ: other: 40.0 *40* (1972Hu10). Mult.,δ: or −0.46 *4* (1976Me12). A_2=+0.83 *3*, A_4=+0.01 *2* (1976Me12).
8113.3	(1/2$^+$,3/2,5/2$^+$)	1621	6	6492				I_γ: other: 6.0 *6* (1972Hu10).
		2007.0	6	6106.2				
		2307	1.1	5806				
		2458	0.9	5655				
		2513.5		5599.7				I_γ: 3.0 *15* (1972Hu10).
		2897.4	1.8	5215.8				
		3939.6	7	4173.45				I_γ: other: 5.0 *25* (1972Hu10).
		4053.9	1.3	4059.2				
		4145.7	5	3967.3				I_γ: other: 5.0 *25* (1972Hu10).
		5110.2	1.9	3002.75	5/2			
		5419.0	13	2693.9	(3/2,5/2)			I_γ: other: 17.0 *17* (1972Hu10).
		6349.5	5	1763.14	5/2$^+$			I_γ: other: 6.0 *6* (1972Hu10).
		6893.2	32	1219.39	1/2$^+$			I_γ: other: 32.0 *32* (1972Hu10).
		8112.3	25	0	3/2$^+$			I_γ: other: 26.0 *26* (1972Hu10).
8147.2	3/2$^-$	2464	1.0	5683				I_γ: other: 1.0 *5* (1972Hu10).
		3137.0	3	5010.1				
		3969.1	8	4177.9	3/2			
		4087.7	3	4059.2				
		4179.6	2	3967.3				
		4228.5	3	3918.48	3/2$^+$			I_γ: other: 3.0 *15* (1972Hu10).
		5452.8	3	2693.9	(3/2,5/2)			I_γ: other: 4 *2* (1972Hu10).
		6927.1	15	1219.39	1/2$^+$			I_γ: other: 20.0 *20* (1972Hu10).
		8146.2	62	0	3/2$^+$			I_γ: other: 75.0 *75* (1972Hu10).
8156.8	(5/2$^+$,7/2$^-$)	2511	4	5646				I_γ: other: 3.0 *15* (1972Hu10).
		2571	8	5586				I_γ: other: 7.0 *7* (1972Hu10).
		2940.9	6	5215.8				I_γ: other: 8.0 *8* (1972Hu10).
		3386.8	1.0	4769.86				
		3978.7	5	4177.9	3/2			I_γ: other: 8.0 *8* (1972Hu10).
		4042.9	7	4113.7	7/2			I_γ: other: 7.0 *7* (1972Hu10).
		4097.3	1	4059.2				
		4213.6	2	3942.9				
		4993.4	5	3162.99	7/2$^-$			I_γ: other: 9.0 *9* (1972Hu10).
		5153.6	6	3002.75	5/2			I_γ: other: 4 *2* (1972Hu10).
		5510.6	51	2645.70	(5/2,7/2)			I_γ: other: 50.0 *50* (1972Hu10).
		6393.0	1	1763.14	5/2$^+$			
		8155.8	3	0	3/2$^+$			I_γ: other: 4 *2* (1972Hu10).
8179.4		3554.8	9.0 *9*	4624.4				I_γ: from 1972Hu10.
		5016.0	8.0 *8*	3162.99	7/2$^-$			I_γ: from 1972Hu10.
		5176.2	4.0 *20*	3002.75	5/2			I_γ: from 1972Hu10.
		5485.0	3.0 *15*	2693.9	(3/2,5/2)			I_γ: from 1972Hu10.
		6415.6	68 *7*	1763.14	5/2$^+$			I_γ: from 1972Hu10.
		8178.4	8.0 *8*	0	3/2$^+$			I_γ: from 1972Hu10.
8208.2	5/2$^+$	2553	0.7	5655				I_γ: other: 1 (2001Vo24).
		3327.1	0.8	4880.96				I_γ: other: 1 (2001Vo24).
		3368.9	0.4	4839.10				I_γ: other: 1 (2001Vo24).
		4148.7	1.5	4059.2				I_γ: others: 2 (2001Vo24); 2 *1* (1972Hu10).

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$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10 (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[#]	E_f	J^π_f	Mult.[b]	δ[b]	Comments
8208.2	5/2+	4289.4		3918.48	3/2+			I_γ: 1.0 *5* (1972Hu10).
		5205.0	1.1	3002.75	5/2			I_γ: others: 1 (2001Vo24); 1.0 *5* (1972Hu10).
		5513.8	0.5	2693.9	(3/2,5/2)			I_γ: other: 1 (2001Vo24); 1.0 *5* (1972Hu10).
		6444.4	14	1763.14	5/2+	D+Q	+0.33 *4*	I_γ: other: 14 (2001Vo24); 16.0 *16* (1972Hu10). Mult.,δ: from 2001Vo24. A_2=+0.51 *13*, A_4=−0.21 *11*, A_6=+0.0004 (1996Ka21). A_2=+0.51 *13*, A_4=−0.21 *11* (2001Vo24).
		6988.1	3	1219.39	1/2+			I_γ: other: 3 (2001Vo24); 3.0 *15* (1972Hu10).
		8207.2	78	0	3/2+	D+Q	−0.36 *4*	I_γ: other: 78 (2001Vo24); 76.0 *76* (1972Hu10). Mult.,δ: from 2001Vo24. Other: δ(Q/D)=0.52 *42* from 1996Ka21. A_2=−0.55 *8*, A_4=+0.004 *78*, A_6=+0.006 *72* (1996Ka21). A_2=−0.55 *8*, A_4=0.00 *8* (2001Vo24).
8216.3	5/2	2561	0.8	5655				I_γ: other: 1 (2001Vo24).
		3446.3	0.8	4769.86				I_γ: other: 1 (2001Vo24).
		4038.2	<3	4177.9	3/2			I_γ: other: 3 (2001Vo24).
		4042.6	<3	4173.45				
		5052.9	41	3162.99	7/2−	D+Q	+0.11 *2*	I_γ: other: 41 (2001Vo24); 44.0 *44* (1972Hu10). Mult.,δ: from $\gamma(\theta)$ in 2001Vo24. A_2=−0.32 *7*, A_4=−0.11 *7*, A_6=+0.05 *6* (1996Ka21). A_2=−0.32 *7*, A_4=−0.11 *7* (2001Vo24).
		5213.1	5	3002.75	5/2			I_γ: other: 5 (2001Vo24); 5.0 *25* (1972Hu10).
		5521.9	3	2693.9	(3/2,5/2)			I_γ: other: 3 (2001Vo24); 4 *2* (1972Hu10).
		6452.5	1.4	1763.14	5/2+			I_γ: other: 1 (2001Vo24).
		8215.3	45	0	3/2+	D+Q	+0.06 *3*	I_γ: other: 45 (2001Vo24); 47.0 *47* (1972Hu10). Mult.,δ: from $\gamma(\theta)$ in 2001Vo24. Other: δ(Q/D)=0.59 *56* from 1996Ka21. A_2=−0.560 *8*, A_4=+0.017 *71*, A_6=+0.066 *75* (1996Ka21). A_2=−0.56 *1*, A_4=+0.02 *7* (2001Vo24).
8242.4		4068.7	7	4173.45				
		5239.2	2	3002.75	5/2			
		7022.3	91	1219.39	1/2+			
8251?		4137	4[&]	4113.7	7/2			
		5248	6[&]	3002.75	5/2			
		7031	82[&]	1219.39	1/2+			
		8250	8[&]	0	3/2+			
8268.9		3498.9	2	4769.86				I_γ: from 1976Sp08.
		4090.7	12	4177.9	3/2			
		4350.1	3	3918.48	3/2+			
		5105.5	7	3162.99	7/2−			
		5265.7	13	3002.75	5/2			
		5574.5	6	2693.9	(3/2,5/2)			
		5622.7	4	2645.70	(5/2,7/2)			
		6505.1	8	1763.14	5/2+			
		8267.9	45	0	3/2+			
8277.2		2622	1.5	5655				
		3396.1	5	4880.96				
		3437.9	11	4839.10				
		3507.2	1.2	4769.86				
		3652.6	3	4624.4				
		3929.2	1.8	4347.76	5/2,9/2			

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^{34}S(p,γ) 1976Me12,1976Sp08,1972Hu10 (continued)

γ(^{35}Cl) (continued)

E_i(level)	J_i^π	E_γ[†]	I_γ[#]	E_f	J_f^π	Mult.[b]	Comments
8277.2		4103.5	2	4173.45			
		4217.7	1.5	4059.2			
		4358.4	5	3918.48	3/2$^+$		
		5274.0	10	3002.75	5/2		
		5582.8	3	2693.9	(3/2,5/2)		
		5631.0	3	2645.70	(5/2,7/2)		
		6513.4	23	1763.14	5/2$^+$		
		8276.2	29	0	3/2$^+$		
8282.4	(3/2$^-$,5/2)	2696	8	5586			
		3066.5	8	5215.8			
		3118.9	3	5163.34	7/2		
		4104.2	14	4177.9	3/2		
		4222.9	6	4059.2			
		5119.0	26	3162.99	7/2$^-$		
		6518.6	7	1763.14	5/2$^+$		
		8281.4	28	0	3/2$^+$		
8287.82	1/2$^-$	2633	2	5655			
		3277.6	3	5010.1			
		4109.7	9	4177.9	3/2		
		4228.4	2	4059.2			
		4320.2	3	3967.3			
		4369.05	1	3918.48	3/2$^+$		
		5593.4	23	2693.9	(3/2,5/2)		A_2=−0.01 *4*, A_4=−0.02 *4* (1976Me12).
		7067.66	44	1219.39	1/2$^+$		δ: δ(Q/D)=+0.21 *2* or −3.1 *2* for J_i=3/2 (1976Me12). A_2=−0.08 *2*, A_4=−0.01 *3* (1976Me12).
		8286.77	13	0	3/2$^+$		δ: δ(Q/D)=−0.35 *4* or −11 *3* for J_i=3/2 (1976Me12). A_2=−0.11 *3*, A_4=−0.01 *4* (1976Me12).
8298.2	3/2	3287.9	3	5010.1			
		3417.1	4	4880.96			
		3443.6	5	4854.4			
		4124.5	6	4173.45			
		4238.7	10	4059.2			
		5134.8	1	3162.99	7/2$^-$		
		5295.0	1	3002.75	5/2		
		5603.8	10	2693.9	(3/2,5/2)		A_2=+0.09 *6*, A_4=−0.07 *7* (1976Me12).
		5652.0	6	2645.70	(5/2,7/2)		
		6534.4	8	1763.14	5/2$^+$		A_2=+0.43 *11*, A_4=−0.51 *12* (1976Me12).
		7078.0	24	1219.39	1/2$^+$	D+Q	Mult.,δ: −1.7 *1* or −0.02 *2* (1976Me12). A_2=−0.53 *3*, A_4=+0.11 *3* (1976Me12).
		8297.1	13	0	3/2$^+$	D+Q	Mult.,δ: −0.15 *4* or +9.5 *+48−24* (1976Me12). A_2=+0.18 *3*, A_4=−0.10 *4* (1976Me12).
8318.1		2512	0.7	5806			
		2672	0.7	5646			
		2718.3	1.5	5599.7			
		3548.1	0.3	4769.86			
		3693.5	1.8	4624.4			
		4139.9	<3	4177.9	3/2		
		4144.4	<3	4173.45			
		4258.6	1.2	4059.2			
		4350.5	3	3967.3			
		5154.7	0.5	3162.99	7/2$^-$		
		5314.9	3	3002.75	5/2		
		5623.7	1.6	2693.9	(3/2,5/2)		
		5671.9	0.9	2645.70	(5/2,7/2)		
		6554.3	1.8	1763.14	5/2$^+$		
		7097.9	2	1219.39	1/2$^+$		

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$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10 (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[#]	E_f	J^π_f	Mult.[b]	δ[b]	Comments
8318.1		8317.0	78	0	$3/2^+$			
8323.1		4209.1	1[&]	4113.7	7/2			
		5319.9	3[&]	3002.75	5/2			
		5676.9	4[&]	2645.70	(5/2,7/2)			
		6559.3	<2[&]	1763.14	$5/2^+$			
		7102.9	8[&]	1219.39	$1/2^+$			
		8322.0	82[&]	0	$3/2^+$			
8381.8	$5/2^+$	3611.7		4769.86				I_γ: 1 (2001Vo24).
		3757.2		4624.4				I_γ: 1 (2001Vo24).
		4267.8	8.0 *8*	4113.7	7/2			I_γ: from 1972Hu10. Other: 7 (2001Vo24).
		4463.0	24.0 *24*	3918.48	$3/2^+$	D+Q	+0.15 *3*	I_γ: from 1972Hu10. Other: 24 (2001Vo24). Mult.,δ: from $\gamma(\theta)$ in 2001Vo24. A_2=+0.25 *3*, A_4=−0.04 *2* (2001Vo24).
		5378.6	25.0 *25*	3002.75	5/2	D(+Q)	+0.02 *3*	I_γ: from 1972Hu10. Other: 25 (2001Vo24). Mult.,δ: from $\gamma(\theta)$ in 2001Vo24. Other: δ=−0.11 *1* from 1967Ko20, 1967Ko23. A_2=+0.60 *7*, A_4=−0.02 *7* (1967Ko20,1967Ko23). A_2=+0.51 *23*, A_4=−0.05 *24* (2001Vo24).
		5687.4	1.0 *5*	2693.9	(3/2,5/2)			I_γ: from 1972Hu10. Other: 1 (2001Vo24). A_2=+0.22 *9*, A_4=−0.02 *11* (1967Ko20,1967Ko23).
		5735.6	5.0 *25*	2645.70	(5/2,7/2)			I_γ: from 1972Hu10. Others: 5 (2001Vo24). Mult.,δ: D+Q with δ=+0.72 *+42−20* and J_f=3/2 for level 2645 (1967Ko20).
		6618.0	34.0 *34*	1763.14	$5/2^+$	D+Q	+0.06 *3*	I_γ: from 1972Hu10. Other: 34 (2001Vo24). Mult.,δ: from $\gamma(\theta)$ in 2001Vo24. Other: δ=−0.12 *10* from $\gamma(\theta)$ in 1967Ko23. A_2=+0.79 *15*, A_4=−0.08 *15* (1967Ko20,1967Ko23). A_2=+0.02 *13*, A_4=−0.08 *13* (2001Vo24).
		8380.7	3.0 *15*	0	$3/2^+$	D+Q	1.65	I_γ: from 1972Hu10. Other: 2 (2001Vo24). Mult.,δ: from $\gamma(\theta)$ in 1967Ko23. A_2=−0.83 *22*, A_4=+0.72 *33* (1967Ko20,1967Ko23).
8402.9		3632.8	4.8 *24*	4769.86				
		4224.7	9 *5*	4177.9	3/2			
		4343.4	23 *6*	4059.2				
		5708.5	10 *5*	2693.9	(3/2,5/2)			
		5756.7	7 *4*	2645.70	(5/2,7/2)			
		6639.1	15 *4*	1763.14	$5/2^+$			
		8401.8	31 *8*	0	$3/2^+$			
8405.7		3524.6	15 *4*	4880.96				
		4232.0	1.6 *8*	4173.45				
		4291.7	1.4 *7*	4113.7	7/2			
		5402.5	15 *4*	3002.75	5/2			
		6641.9	43 *11*	1763.14	$5/2^+$			
		8404.6	24 *6*	0	$3/2^+$			
8416.7		3406.4	5[&]	5010.1				
		4243.0	11[&]	4173.45				
		4473.5	3[&]	3942.9				
		5413.5	7[&]	3002.75	5/2			
		5770.5	8[&]	2645.70	(5/2,7/2)			
		6652.9	22[&]	1763.14	$5/2^+$			

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$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10 (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[#]	E_f	J^π_f	Comments
8416.7		7196.5	9[&]	1219.39	1/2+	
		8415.6	35[&]	0	3/2+	
8464.3		2809	8 *4*	5655		
		3248.3	5.8 *29*	5215.8		
		3454.0	8 *4*	5010.1		
		3609.7	3.3 *17*	4854.4		
		3839.7	4.3 *22*	4624.4		
		4286.1	5.2 *26*	4177.9	3/2	
		4290.6	5.5 *28*	4173.45		
		4496.7	5.2 *26*	3967.3		
		4545.5	2.6 *13*	3918.48	3/2+	
		5461.1	17 *4*	3002.75	5/2	
		5769.9	16 *4*	2693.9	(3/2,5/2)	
		7244.1	9 *4*	1219.39	1/2+	
		8463.2	10 *3*	0	3/2+	
8484.4	3/2+	2829	5.6 *28*	5655		I_γ: other: 6 2001Vo24.
		3603.2	7.0 *35*	4880.96		I_γ: other: 7 2001Vo24.
		3859.8	1.0 *5*	4624.4		I_γ: other: 1 2001Vo24.
		4516.8	1.2 *6*	3967.3		I_γ: other: 1 2001Vo24.
		4565.6	5.2 *26*	3918.48	3/2+	I_γ: other: 5 2001Vo24.
		5481.2	7 *4*	3002.75	5/2	I_γ: other: 7 2001Vo24.
		5790.0	20 *5*	2693.9	(3/2,5/2)	I_γ: other: 20 2001Vo24.
		5838.2	2.8 *14*	2645.70	(5/2,7/2)	I_γ: other: 3 2001Vo24.
		6720.6	46 *12*	1763.14	5/2+	I_γ: other: 46 2001Vo24.
		8483.3	4 *2*	0	3/2+	I_γ: other: 4 2001Vo24.
8486.1		2728	1.1 *6*	5758		
		3861.5	1.8 *9*	4624.4		
		4307.9	<7.2	4177.9	3/2	I_γ: total $I\gamma$=4.8 *24* for γ decays to 4173 and 4178 doublet final levels (1976Sp08).
		4312.4	<7.2	4173.45		I_γ: total $I\gamma$=4.8 *24* for γ decays to 4173 and 4178 doublet final levels (1976Sp08).
		4426.6	1.9 *10*	4059.2		
		4518.5	7 *4*	3967.3		
		5482.9	8 *4*	3002.75	5/2	
		5791.7	19 *5*	2693.9	(3/2,5/2)	
		6722.3	3.6 *18*	1763.14	5/2+	
		7265.9	6.6 *33*	1219.39	1/2+	
		8485.0	46 *12*	0	3/2+	
8514.3		3659.7	4.4 *22*	4854.4		
		4336.1	2 *1*	4177.9	3/2	
		4454.8	27 *7*	4059.2		
		4546.7	23 *6*	3967.3		
		4595.5	6.6 *33*	3918.48	3/2+	
		7294.1	10 *3*	1219.39	1/2+	
		8513.2	27 *7*	0	3/2+	
8571.9		2848.3	2.2 *11*	5723.52		
		2917	1.3 *7*	5655		
		2972.1	0.7 *4*	5599.7		
		3408.4	1.8 *9*	5163.34	7/2	
		3690.7	0.9 *5*	4880.96		
		4398.2	0.7 *4*	4173.45		
		4457.9	1.1 *6*	4113.7	7/2	
		4653.1	1.1 *6*	3918.48	3/2+	
		5408.5	1.6 *8*	3162.99	7/2−	
		5568.7	4.5 *23*	3002.75	5/2	

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^{34}S(p,γ) 1976Me12,1976Sp08,1972Hu10 (continued)

γ(^{35}Cl) (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[#]	E_f	J^π_f	Mult.[b]	δ[b]	Comments
8571.9		5877.5	0.6 *3*	2693.9	(3/2,5/2)			
		5925.7	8 *4*	2645.70	(5/2,7/2)			
		7351.7	5.4 *27*	1219.39	$1/2^+$			
		8570.8	70 *7*	0	$3/2^+$			
8580.5		2088	4.4 *22*	6492				
		3741.2	1.3 *7*	4839.10				
		4402.3	4.4 *22*	4177.9	3/2			
		4521.0	2 *1*	4059.2				
		4661.7	2.3 *12*	3918.48	$3/2^+$			
		5886.1	5.6 *28*	2693.9	(3/2,5/2)			
		7360.3	54 *6*	1219.39	$1/2^+$			
		8579.4	26 *7*	0	$3/2^+$			
8590.0		2832	4.8 *24*	5758				
		2935	3.7 *19*	5655				
		3819.9	16 *4*	4769.86				
		4416.3	13 *3*	4173.45				
		4476.0	3.6 *18*	4113.7	7/2			
		5426.6	8 *4*	3162.99	$7/2^-$			
		5586.8	22 *6*	3002.75	5/2			
		5943.8	5.5 *28*	2645.70	(5/2,7/2)			
		6826.2	14 *4*	1763.14	$5/2^+$			
		8588.9	10 *3*	0	$3/2^+$			
8612.0		6848.1	50 *5*	1763.14	$5/2^+$			
		7391.8	17 *4*	1219.39	$1/2^+$			
		8610.9	33 *8*	0	$3/2^+$			
8613.7		3732.5	1.6 *8*	4880.96				
		4554.2	5.4 *27*	4059.2				
		4670.5[c]	4[&]	3942.9				I_γ: from 1967Da12 only; not observed by others.
		4694.9	7 *4*	3918.48	$3/2^+$			
		5610.5	2.5 *13*	3002.75	5/2			
		5919.3	4.3 *22*	2693.9	(3/2,5/2)			
		5967.5	23 *6*	2645.70	(5/2,7/2)			
		6849.8[c]	9[&]	1763.14	$5/2^+$			I_γ: from 1967Da12 only; not observed by others.
		7393.5[c]	2[&]	1219.39	$1/2^+$			I_γ: from 1967Da12 only; not observed by others.
		8612.6	56 *6*	0	$3/2^+$			
8618.1	$(3/2^+,5/2^+)$	2894.5	2.1 *11*	5723.52				
		2963	1.7 *9*	5655				
		3763.5	1.4 *7*	4854.4				
		3993.5	1.5 *8*	4624.4				
		4650.5	3.0 *15*	3967.3				
		4699.3	2.3 *12*	3918.48	$3/2^+$			
		5614.9	16 *4*	3002.75	5/2			
		5923.7	14 *4*	2693.9	(3/2,5/2)			A_2=+0.147 *33*, A_4=+0.114 *52* (1970Ko33).
		5971.9	1.7 *9*	2645.70	(5/2,7/2)			
		6854.2	23 *6*	1763.14	$5/2^+$			A_2=+0.662 *23*, A_4=+0.319 *39* (1970Ko33 reported for 7855γ, 7855 is likely a misprint).
		8617.0	33 *8*	0	$3/2^+$			A_2=−0.293 *29*, A_4=−0.028 *47* (1970Ko33).
8630.1	7/2	2906.5	0.3 *2*	5723.52				
		3466.6	8 *4*	5163.34	7/2	D+Q	+0.60 *10*	Mult.,δ: D+Q from γ(θ) in 1976Sp09. A_2=+0.60 *3*, A_4=−0.08 *4* (1976Sp09).

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$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10 (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[#]	E_f	J^π_f	Mult.[b]	δ[b]	Comments
8630.1	7/2	3860.0	1.4 *7*	4769.86				
		4282.1	12 *3*	4347.76	5/2,9/2	D+Q	−0.184 *14*	Mult.,δ: D+Q from γ(θ) in 1976Sp09. A_2=+0.11 *2*, A_4=+0.02 *2* (1976Sp09).
		4456.4	3.6 *18*	4173.45				
		4516.1	10 *5*	4113.7	7/2	D+Q	−0.06 *2*	Mult.,δ: D+Q from γ(θ) in 1976Sp09. A_2=+0.43 *2*, A_4=−0.01 *2* (1976Sp09).
		4686.9	10 *5*	3942.9				
		5466.7	3.5 *18*	3162.99	$7/2^-$			
		5626.9	1.4 *7*	3002.75	5/2			
		5983.9	2.6 *13*	2645.70	(5/2,7/2)			
		6866.2	47 *12*	1763.14	$5/2^+$	D+Q	−0.011 *3*	Mult.,δ: D+Q from γ(θ) in 1976Sp09. A_2=+0.10 *33*, A_4=−0.65 *46* (1970Ko33). A_2=−0.387 *7*, A_4=+0.012 *8* (1976Sp09).
		8629.0	0.5 *3*	0	$3/2^+$			A_2=+0.398 *46*, A_4=+0.019 *78* (1970Ko33).
8640	5/2	4697	23[@]	3942.9				E_γ: other: 4689 *7* (1970Ko33).
		5477	20[@]	3162.99	$7/2^-$			E_γ: other: 5473 *7* (1970Ko33).
		5637	6[@]	3002.75	5/2			E_γ: other: 5630 *7* (1970Ko33).
		5946	13[@]	2693.9	(3/2,5/2)			E_γ: other: 5941 *7* (1970Ko33).
		6876	38[@]	1763.14	$5/2^+$			E_γ: other: 6874 *7* (1970Ko33). A_2=−0.52 *12*, A_4=−0.29 *19* (1970Ko33).
8641.4	3/2,5/2	2986	2.9 *15*	5655				
		3055	2.5 *13*	5586				
		3237.7	4.7 *24*	5403.5				
		3802.1	2.6 *13*	4839.10				
		3871.3	24 *6*	4769.86				
		4016.8	1.9 *10*	4624.4				
		4463.2	<18	4177.9	3/2			I_γ: total Iγ=14 *4* for γ decays to 4173 and 4178 doublet final levels (1976Sp08).
		4467.6	<18	4173.45				I_γ: total Iγ=14 *4* for γ decays to 4173 and 4178 doublet final levels (1976Sp08).
		4581.9	9 *5*	4059.2				
		4722.6	5.9 *30*	3918.48	$3/2^+$			
		5947.0	5.2 *26*	2693.9	(3/2,5/2)			
		8640.3	28 *7*	0	$3/2^+$			A_2=−0.12 *2*, A_4=−0.01 *2* (1976Sp09).
8686.2		3100	2.1 *11*	5586				
		3282.5	3.1 *16*	5403.5				
		4508.0	2 *1*	4177.9	3/2			
		4626.7	3.5 *18*	4059.2				
		5522.7	3.7 *19*	3162.99	$7/2^-$			
		5991.8	2.8 *14*	2693.9	(3/2,5/2)			
		6922.3	4.8 *24*	1763.14	$5/2^+$			
		7466.0	1.9 *10*	1219.39	$1/2^+$			
		8685.0	76 *8*	0	$3/2^+$			
8696.9		2891	2.9 *15*	5806				
		3097.1	4.8 *24*	5599.7				
		7476.7	12 *3*	1219.39	$1/2^+$			
		8695.7	80 *8*	0	$3/2^+$			
8750.9		2945	2.2 *11*	5806				
		3347.2	4.9 *25*	5403.5				
		3534.9	2.4 *12*	5215.8				
		4572.7	2.1 *11*	4177.9	3/2			
		4577.1	2.3 *12*	4173.45				
		6056.4	3.0 *15*	2693.9	(3/2,5/2)			
		6987.0	30 *8*	1763.14	$5/2^+$			

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$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10 (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	E_γ[†]	I_γ[#]	E_f	J^π_f	Comments
8750.9	7530.6[c]	2	1219.39	1/2+	I_γ: from 1967Da12 only; not observed by others.
	8749.7	53 *5*	0	3/2+	
8779.8	3179.9	4.3 *22*	5599.7		
	3563.8	6.6 *33*	5215.8		
	4601.6	<19	4177.9	3/2	I_γ: total $I\gamma$=15 *4* for γ decays to 4173 and 4178 doublet final levels (1976Sp08).
	4606.0	<19	4173.45		I_γ: total $I\gamma$=15 *4* for γ decays to 4173 and 4178 doublet final levels (1976Sp08).
	4861.0	4.2 *21*	3918.48	3/2+	
	5776.5	27 *7*	3002.75	5/2	
	6085.3	12 *3*	2693.9	(3/2,5/2)	
	6133.5	13 *3*	2645.70	(5/2,7/2)	
	7015.9	6.7 *34*	1763.14	5/2+	
	8778.6	11 *3*	0	3/2+	
8787.2	3063.5	6.8 *34*	5723.52		
	3623.7	19 *5*	5163.34	7/2	
	4017.1	6.7 *34*	4769.86		
	4609.0	9 *5*	4177.9	3/2	
	5623.7	5.8 *29*	3162.99	7/2−	
	5783.9	20 *5*	3002.75	5/2	
	7023.3	25 *6*	1763.14	5/2+	
	8786.0	8 *4*	0	3/2+	
8798.4	4738.9	8 *4*	4059.2		
	5795.1	16 *4*	3002.75	5/2	
	7034.5	7.0 *35*	1763.14	5/2+	
	7578.1	7.0 *35*	1219.39	1/2+	
	8797.2	62 *6*	0	3/2+	
8820.9	5817.6	9 *5*	3002.75	5/2	
	6126.4	18 *5*	2693.9	(3/2,5/2)	
	7057.0	21 *5*	1763.14	5/2+	
	8819.7	52 *5*	0	3/2+	
8824.2	5820.9	37 *9*	3002.75	5/2	
	6129.7	6.0 *3*	2693.9	(3/2,5/2)	
	7060.3	16 *4*	1763.14	5/2+	
	7603.9	22 *6*	1219.39	1/2+	
	8823.0	19 *5*	0	3/2+	
8829.3	3174	10 *5*	5655		
	3990.0	1.3 *7*	4839.10		
	4059.2	1.5 *8*	4769.86		
	4910.5	4.5 *23*	3918.48	3/2+	
	5826.0	22 *6*	3002.75	5/2	
	6134.8	13 *3*	2693.9	(3/2,5/2)	
	6183.0	33 *8*	2645.70	(5/2,7/2)	
	7065.4	9 *5*	1763.14	5/2+	
	7609.0	2.6 *13*	1219.39	1/2+	
	8828.1	3.8 *19*	0	3/2+	
8833.1	3617.1	6.9 *35*	5215.8		
	4063.0	2.6 *13*	4769.86		
	4659.3	14 *4*	4173.45		
	4719.1	4.7 *24*	4113.7	7/2	
	4773.6	11 *3*	4059.2		
	4914.3	11 *3*	3918.48	3/2+	
	5669.6	3.7 *19*	3162.99	7/2−	
	5829.8	6.7 *34*	3002.75	5/2	
	6138.6	4.5 *23*	2693.9	(3/2,5/2)	
	6186.8	23 *6*	2645.70	(5/2,7/2)	

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$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10 (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[#]	E_f	J^π_f	Comments
8833.1		7069.2	9 *5*	1763.14	5/2+	
		8831.9	2.9 *15*	0	3/2+	
8838.1	7/2−	3674.6[‡]		5163.34	7/2	
		3956.9[‡]		4880.96		
		4068.0	8 *4*	4769.86		
		4490.0[‡]		4347.76	5/2,9/2	
		4778.6[‡]		4059.2		
		5674.6	29 *7*	3162.99	7/2−	
		7074.2	63 *6*	1763.14	5/2+	A_2=−0.20 *2*, A_4=−0.00 *2* (1981Bi05).
8856.1		3974.9	3.4 *17*	4880.96		
		4231.4	5.1 *26*	4624.4		
		4677.9	3.5 *18*	4177.9	3/2	
		4937.3	11 *3*	3918.48	3/2+	
		5692.6	51 *5*	3162.99	7/2−	
		6161.6	2.8 *14*	2693.9	(3/2,5/2)	
		6209.8		2645.70	(5/2,7/2)	
		7092.2	7 *4*	1763.14	5/2+	
		8854.9	16 *4*	0	3/2+	
8868.6		4690.4	2.6 *13*	4177.9	3/2	
		5865.3	16 *4*	3002.75	5/2	
		6174.1	12 *3*	2693.9	(3/2,5/2)	
		7104.7	4.4 *22*	1763.14	5/2+	
		7648.3	16 *4*	1219.39	1/2+	
		8867.4	49 *12*	0	3/2+	
8886.1		3231	6 *3*	5655		
		3286.2	6.6 *33*	5599.7		
		3670.1	2.2 *11*	5215.8		
		3722.6	9 *4*	5163.34	7/2	
		4116.0	4.5 *23*	4769.86		
		4707.9	8 *4*	4177.9	3/2	
		4967.2	4.1 *21*	3918.48	3/2+	
		5882.8	8 *4*	3002.75	5/2	
		6239.8	26 *7*	2645.70	(5/2,7/2)	
		7122.2	12 *3*	1763.14	5/2+	
		8884.9	14 *4*	0	3/2+	
8893.3	(5/2,7/2)+	3293.4[c]		5599.7		I_γ: 1 from 2001Vo24 only; not observed by others.
		3307	1.6 *8*	5586		
		4779.3	9 *4*	4113.7	7/2	I_γ: other: 9 2001Vo24.
		4950.0	3.9 *20*	3942.9		I_γ: other: 4 2001Vo24.
		5729.8	29 *7*	3162.99	7/2−	I_γ: other: 29 2001Vo24.
		6198.8	37 *9*	2693.9	(3/2,5/2)	I_γ: other: 37 2001Vo24.
		7129.4	19 *5*	1763.14	5/2+	I_γ: other: 19 2001Vo24.
		8892.1	1.0 *5*	0	3/2+	I_γ: other: 1 2001Vo24.
8904.8		2724	4.3 *22*	6181		
		3250	3.4 *17*	5655		
		3894.5	3.9 *20*	5010.1		
		4050.2	6.2 *31*	4854.4		
		4065.5	4.5 *23*	4839.10		
		4726.6	18 *5*	4177.9	3/2	
		4845.2	5.7 *29*	4059.2		
		6210.3	15 *4*	2693.9	(3/2,5/2)	
		7684.5	10 *3*	1219.39	1/2+	
		8903.6	29 *7*	0	3/2+	
8906.8	5/2+	3252		5655		I_γ: 4 from 2001Vo24 only; not observed by others.
		3261	3.7 *19*	5646		I_γ: other: 2 2001Vo24.

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^{34}S(p,γ) 1976Me12,1976Sp08,1972Hu10 (continued)

γ(^{35}Cl) (continued)

E$_i$(level)	J$_i^\pi$	E$_\gamma$†	I$_\gamma$#	E$_f$	J$_f^\pi$	Mult.b	δb	Comments
8906.8	5/2+	3306.9	2.2 *11*	5599.7				
		3321	4	5586				I$_\gamma$: other: 4 2001Vo24.
		3387	2.5 *13*	5520				
		3690.8	4.4 *22*	5215.8				
		4136.7	5.6 *28*	4769.86				I$_\gamma$: other: 6 2001Vo24.
		4963.5	13 *3*	3942.9				I$_\gamma$: other: 15 2001Vo24.
		7142.9	69 *7*	1763.14	5/2+			I$_\gamma$: other: 69 2001Vo24.
8919.8		3274	5.6 *28*	5646				
		4571.7	8 *4*	4347.76	5/2,9/2			
		5756.3	57 *6*	3162.99	7/2−			
		6273.5	26 *7*	2645.70	(5/2,7/2)			
		7155.9	4 *2*	1763.14	5/2+			
8953.0	3/2	7732.7		1219.39	1/2+	D+Q		Mult.: from γ(θ) in 1976Sp08; δ=−0.34 *2* or δ=−5.0 *4* (1976Sp09). A$_2$=+0.12 *3*, A$_4$=−0.02 *4* (1976Sp09).
		8951.8		0	3/2+	D+Q		Mult.: from γ(θ) in 1976Sp08; δ=−0.39 *5* or δ=−8.2 *7* (1976Sp09). A$_2$=−0.19 *2*, A$_4$=+0.02 *2* (1976Sp09).
8984.2		3260.5	0.8 *4*	5723.52				
		4144.8	1.2 *6*	4839.10				
		4214.1	0.5 *3*	4769.86				
		4359.5	0.5 *3*	4624.4				
		4806.0	1.5 *8*	4177.9	3/2			
		4810.4	1.5 *8*	4173.45				
		5065.3	9 *5*	3918.48	3/2+			
		5820.7	6 *3*	3162.99	7/2−			
		6289.7	1.8 *9*	2693.9	(3/2,5/2)			
		6337.9	1.2 *6*	2645.70	(5/2,7/2)			
		7220.3	16 *4*	1763.14	5/2+			
		8983.0	60 *6*	0	3/2+			
9081.4	5/2+	3357.7	1	5723.52				I$_\gamma$: from 2001Vo24.
		4200.2	1	4880.96				I$_\gamma$: from 2001Vo24.
		4242.0‡		4839.10				
		4311.3‡		4769.86				
		4456.7‡		4624.4				
		4903.1	2	4177.9	3/2			I$_\gamma$: from 2001Vo24.
		4907.6	2	4173.45				I$_\gamma$: from 2001Vo24.
		5162.5	9	3918.48	3/2+	D(+Q)	−0.002 *9*	I$_\gamma$: from 2001Vo24. Mult.,δ: from γ(θ) in 1976Sp09. A$_2$=−0.43 *3*, A$_4$=+0.05 *3* (1976Sp09).
		5917.9	6	3162.99	7/2−			I$_\gamma$: from 2001Vo24.
		6386.9	2	2693.9	(3/2,5/2)			I$_\gamma$: from 2001Vo24.
		6435.1	1	2645.70	(5/2,7/2)			I$_\gamma$: from 2001Vo24.
		7317.4	16	1763.14	5/2+	D+Q	+0.11 *2*	I$_\gamma$: from 2001Vo24. Mult.,δ: from γ(θ) in 1976Sp09. A$_2$=+0.555 *14*, A$_4$=+0.03 *2* (1976Sp09).
		9080.1	60	0	3/2+	D+Q	+0.089 *9*	I$_\gamma$: from 2001Vo24. Mult.,δ: from γ(θ) in 1976Sp09. A$_2$=−0.243 *11*, A$_4$=+0.05 *2* (1976Sp09).
9157.1	5/2	3941.1	7.4 *19*	5215.8				
		4386.9	6.9 *17*	4769.86				
		4978.8	<16	4177.9	3/2			I$_\gamma$: total Iγ=13 *3* for γ decays to 4173 and 4178 doublet final levels (1976Sp08).
		4983.3	<16	4173.45				I$_\gamma$: total Iγ=13 *3* for γ decays to 4173 and 4178 doublet final levels (1976Sp08).

Continued on next page (footnotes at end of table)

^{34}S(p,γ) 1976Me12,1976Sp08,1972Hu10 (continued)

γ(^{35}Cl) (continued)

E_i(level)	E_γ †	I_γ #	E_f	J^π_f	Mult. b	δ b	Comments
9157.1	5993.6	9.6 *24*	3162.99	$7/2^-$	D+Q	+0.10 *2*	Mult.,δ: from $\gamma(\theta)$ in 2001Vo24. A_2=−0.15 *8*, A_4=−0.31 *9*, A_6=−0.05 *8* (1996Ka21).
	6153.8	8.4 *21*	3002.75	5/2			
	6462.6	1.3 *7*	2693.9	(3/2,5/2)			
	6510.8	2.9 *15*	2645.70	(5/2,7/2)			
	7393.1	7.5 *19*	1763.14	$5/2^+$			
	9155.8	43 *4*	0	$3/2^+$	D+Q	+0.09 *1*	Mult.,δ: from $\gamma(\theta)$ in 2001Vo24. Other: δ(Q/D)=0.47 *50* (1996Ka21). A_2=−0.29 *6*, A_4=−0.38 *8*, A_6=−0.10 *6* (1996Ka21).

† From level-energy differences, unless otherwise noted.

‡ γ branches observed in 1981Bi05. Branching ratios are not given.

For levels below 8.27 MeV (E_p<1.95 MeV), Iγ is mainly from 1976Me12 and for levels above 8.27 MeV (E_p>1.95 MeV), Iγ is mainly from 1976Sp08, unless otherwise noted.

@ From 1970Ko33.

& From 1967Da12.

a From 1963Ha32.

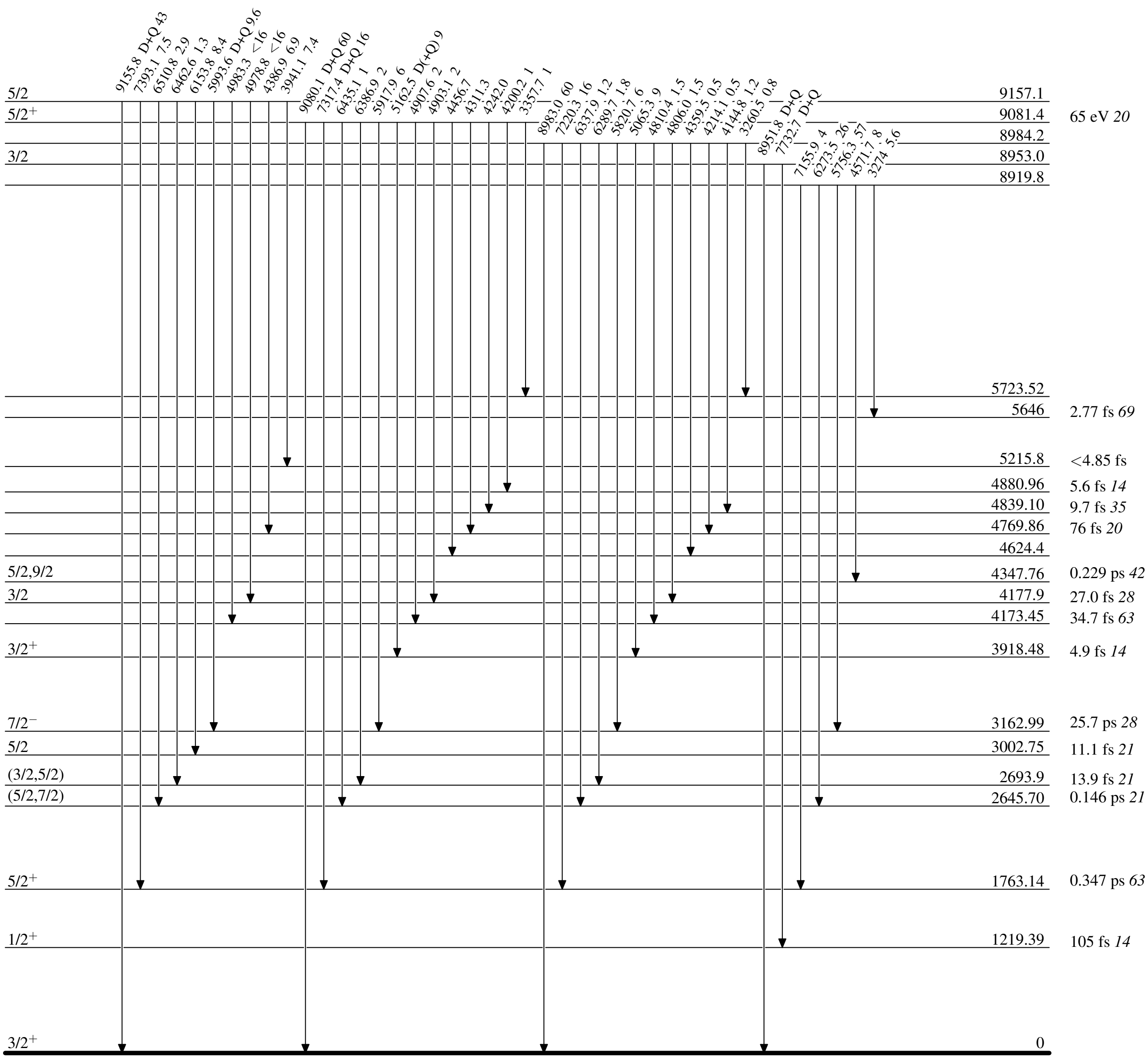
b From $\gamma(\theta)$ and/or $\gamma\gamma(\theta)$ data as given under comments.

c Placement of transition in the level scheme is uncertain.

$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme

Intensities: % photon branching from each level



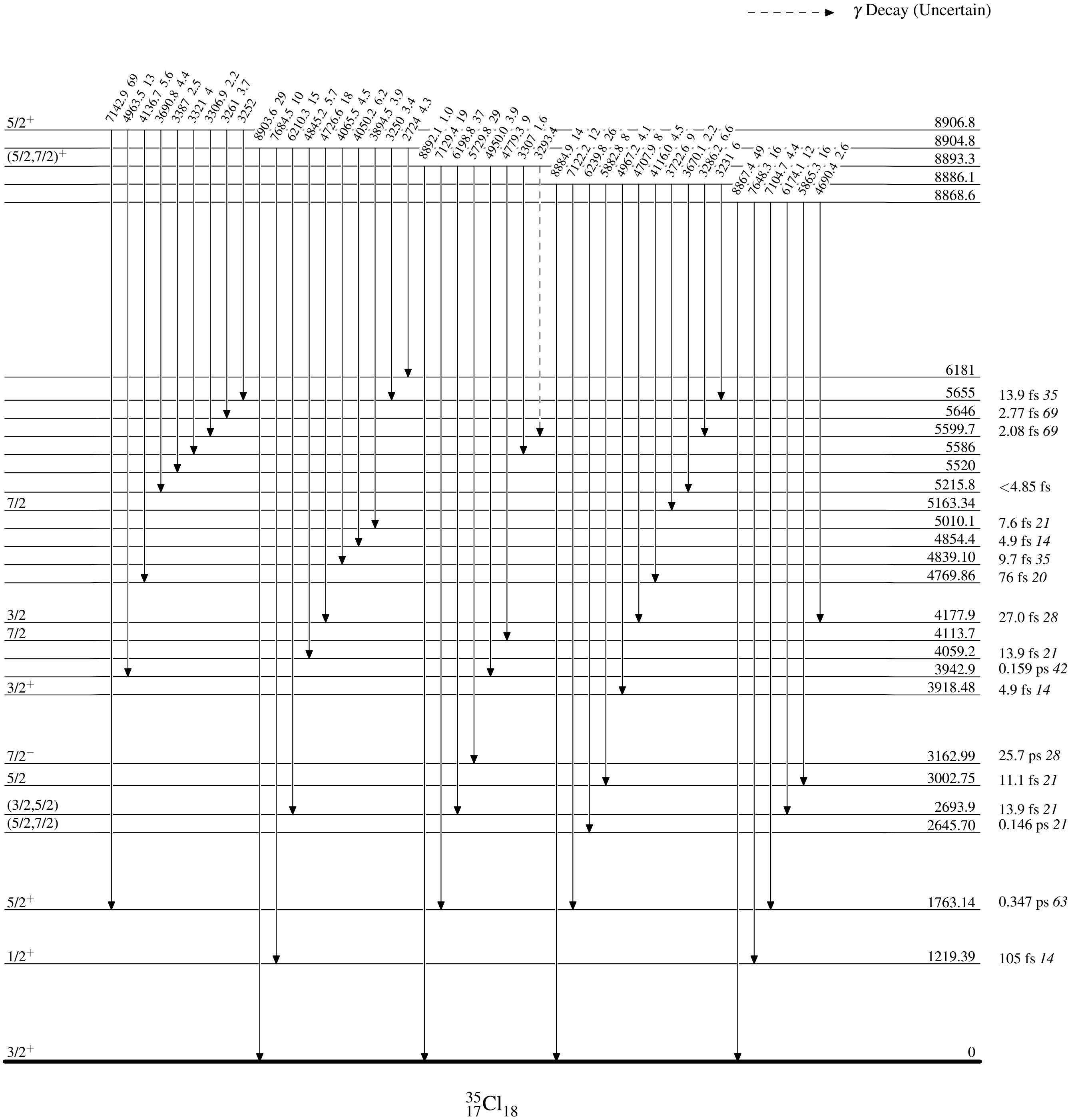
$^{35}_{17}Cl_{18}$

$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10

Legend

Level Scheme (continued)

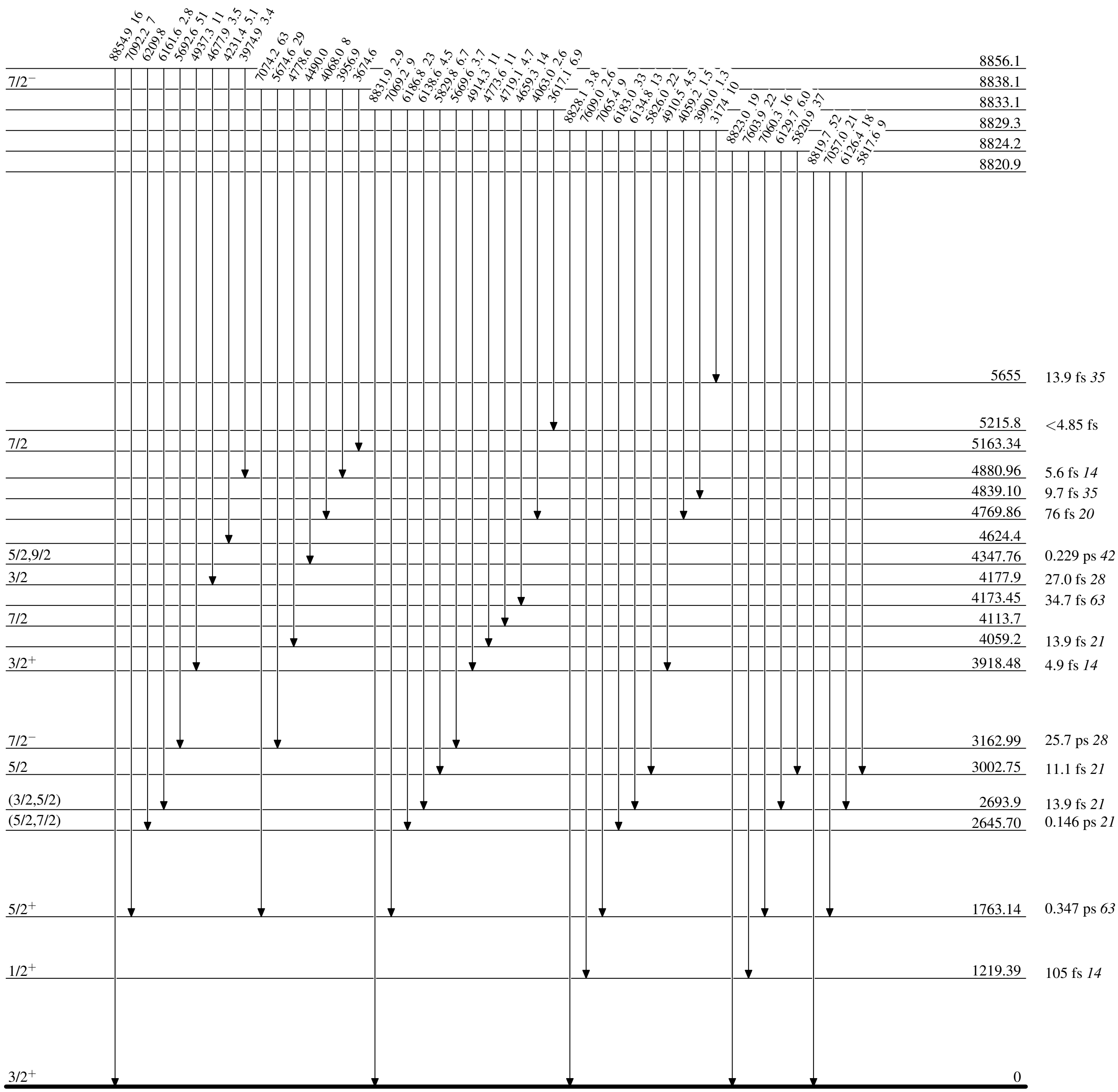
Intensities: % photon branching from each level



$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

Intensities: % photon branching from each level

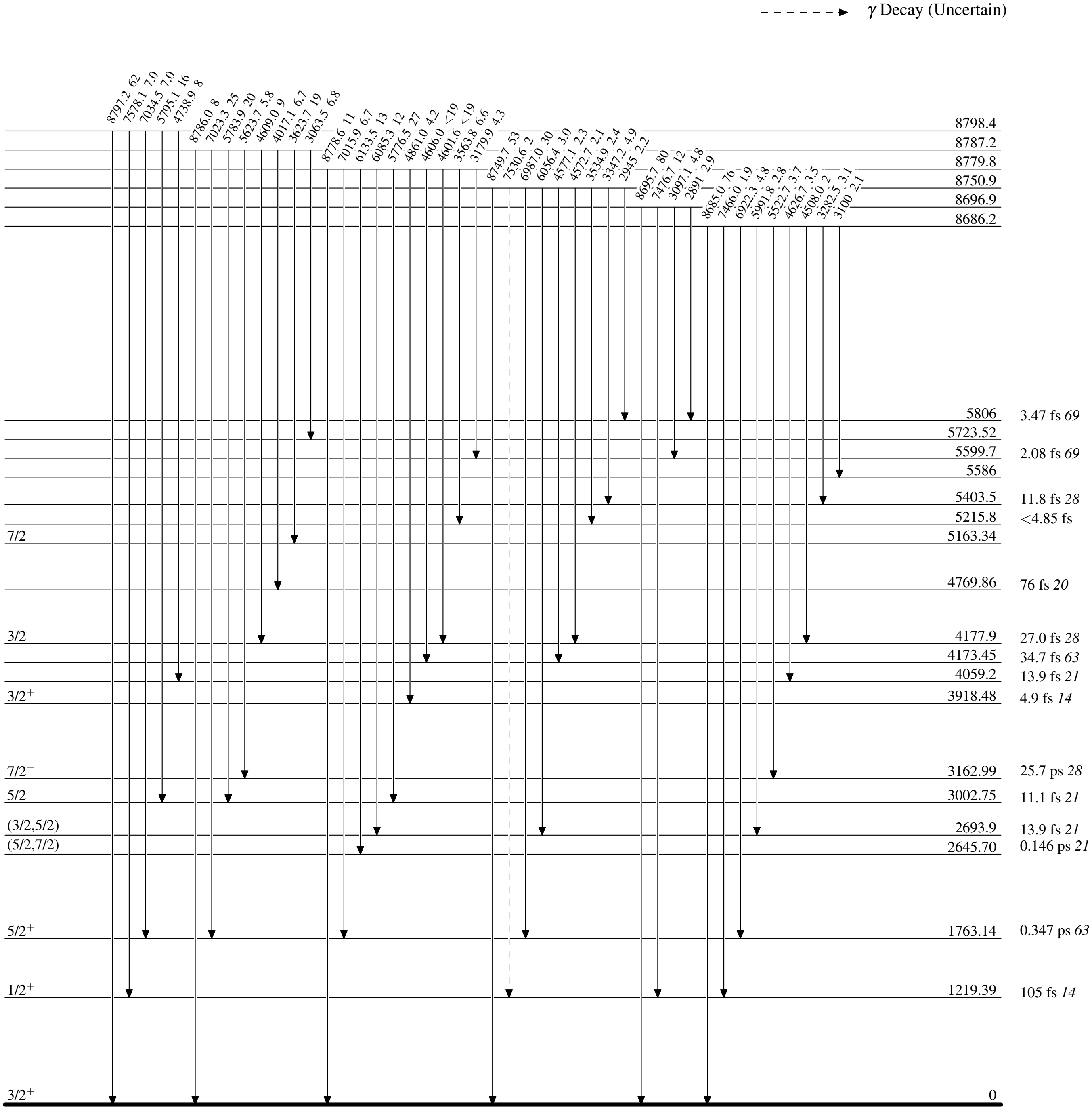


$^{35}_{17}Cl_{18}$

$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

Intensities: % photon branching from each level

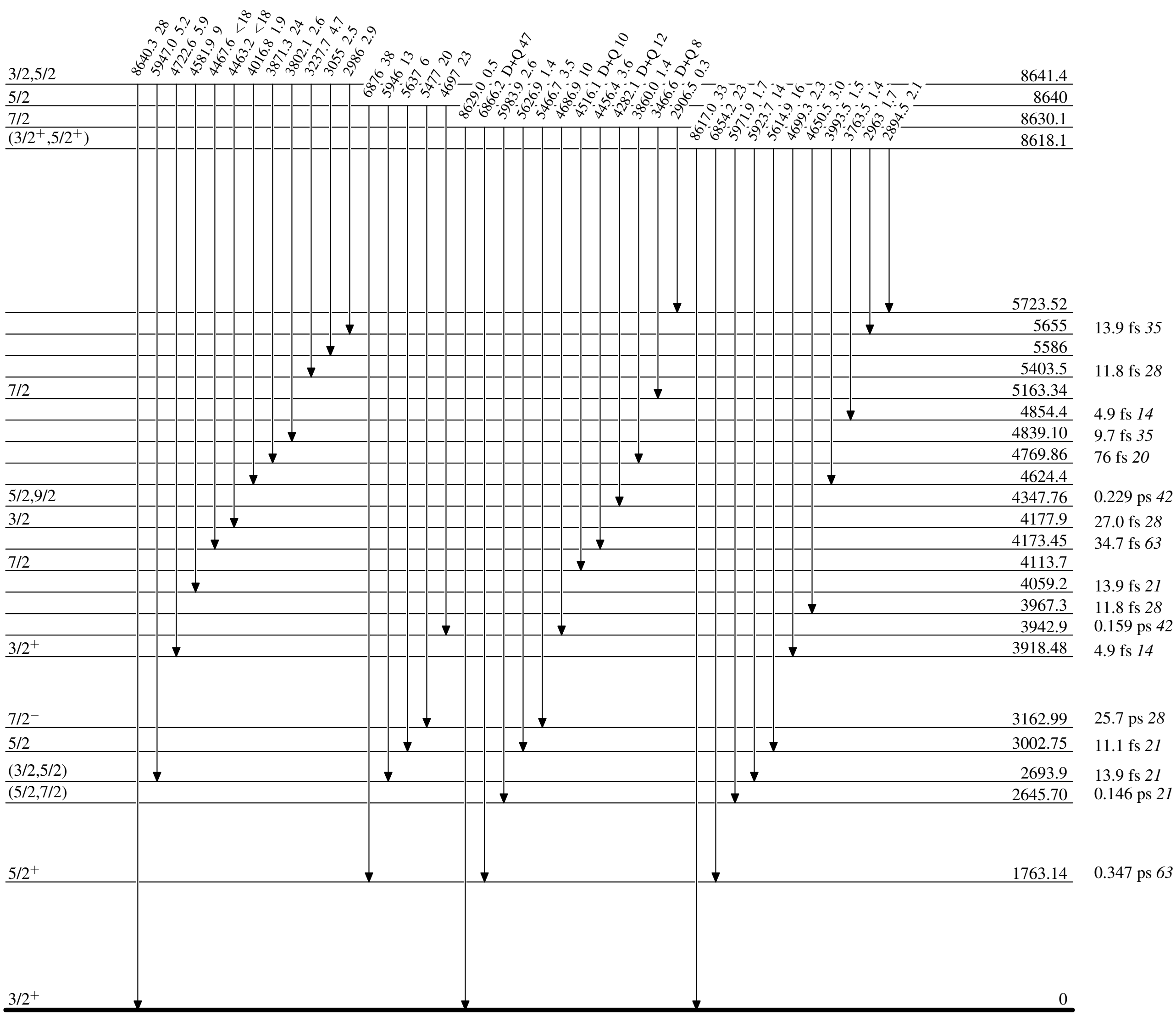


$^{35}_{17}Cl_{18}$

$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

Intensities: % photon branching from each level



$^{35}_{17}Cl_{18}$

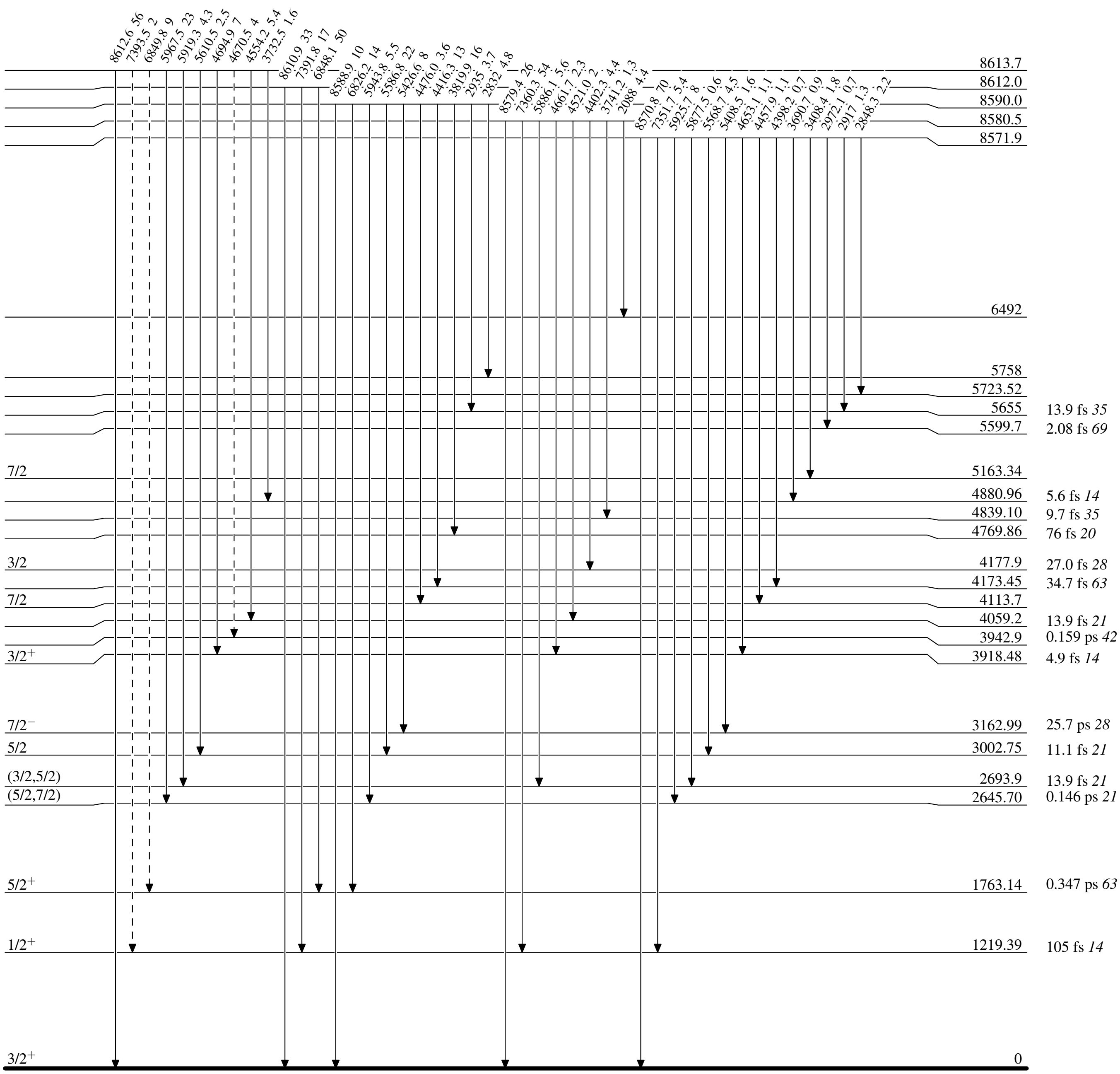
^{34}S(p,γ) **1976Me12,1976Sp08,1972Hu10**

Level Scheme (continued)

Intensities: % photon branching from each level

Legend

γ Decay (Uncertain)

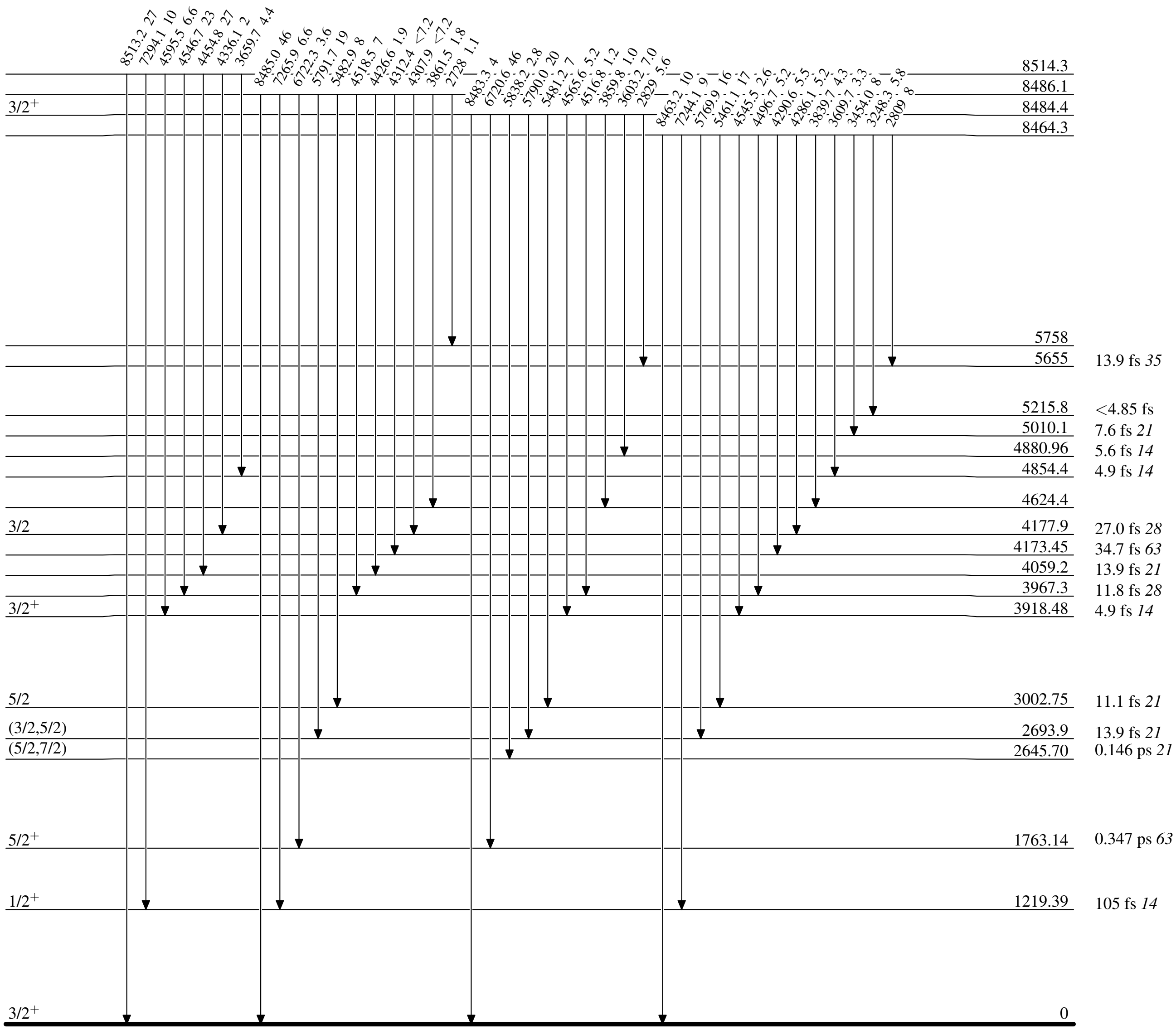


$^{35}_{17}Cl_{18}$

$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

Intensities: % photon branching from each level

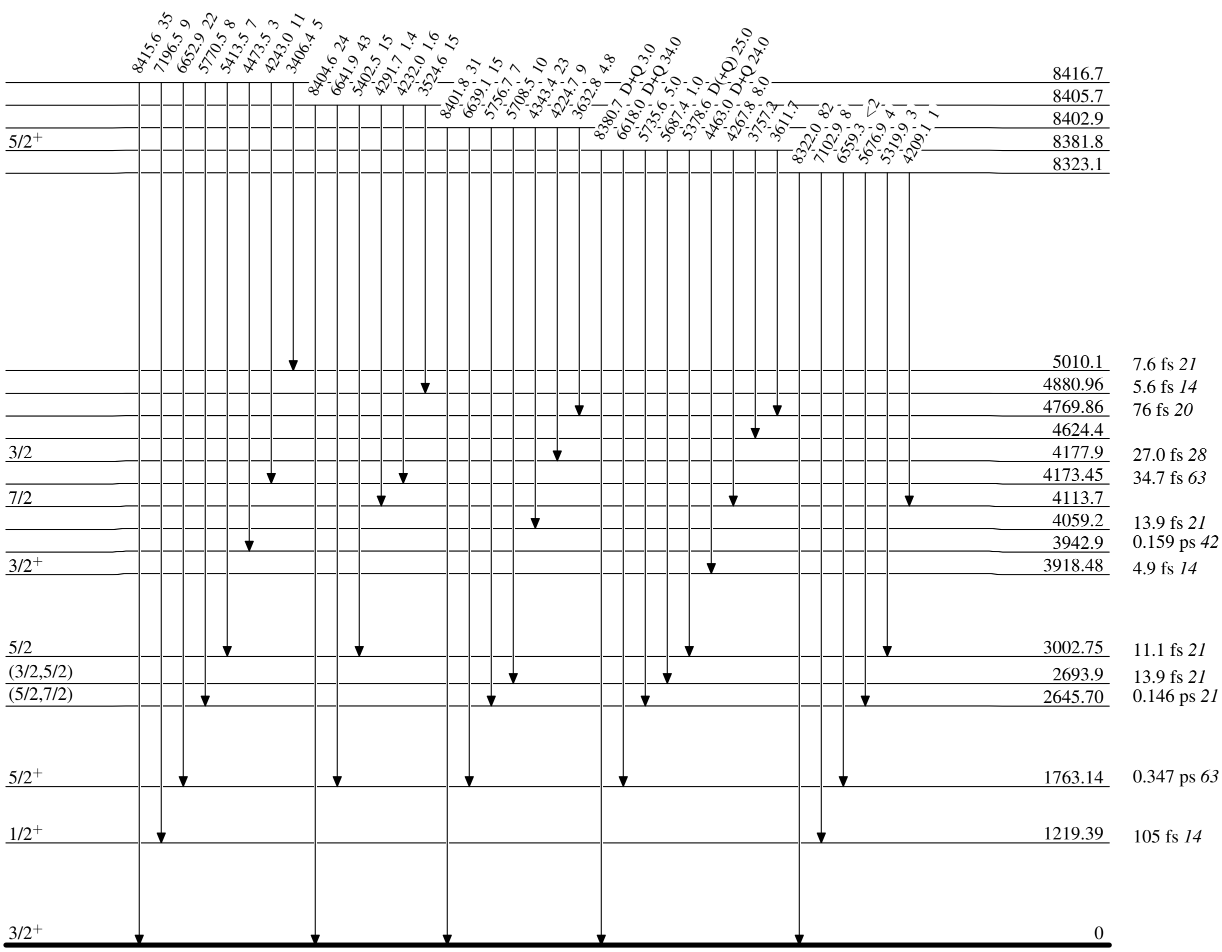


$^{35}_{17}Cl_{18}$

$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

Intensities: % photon branching from each level

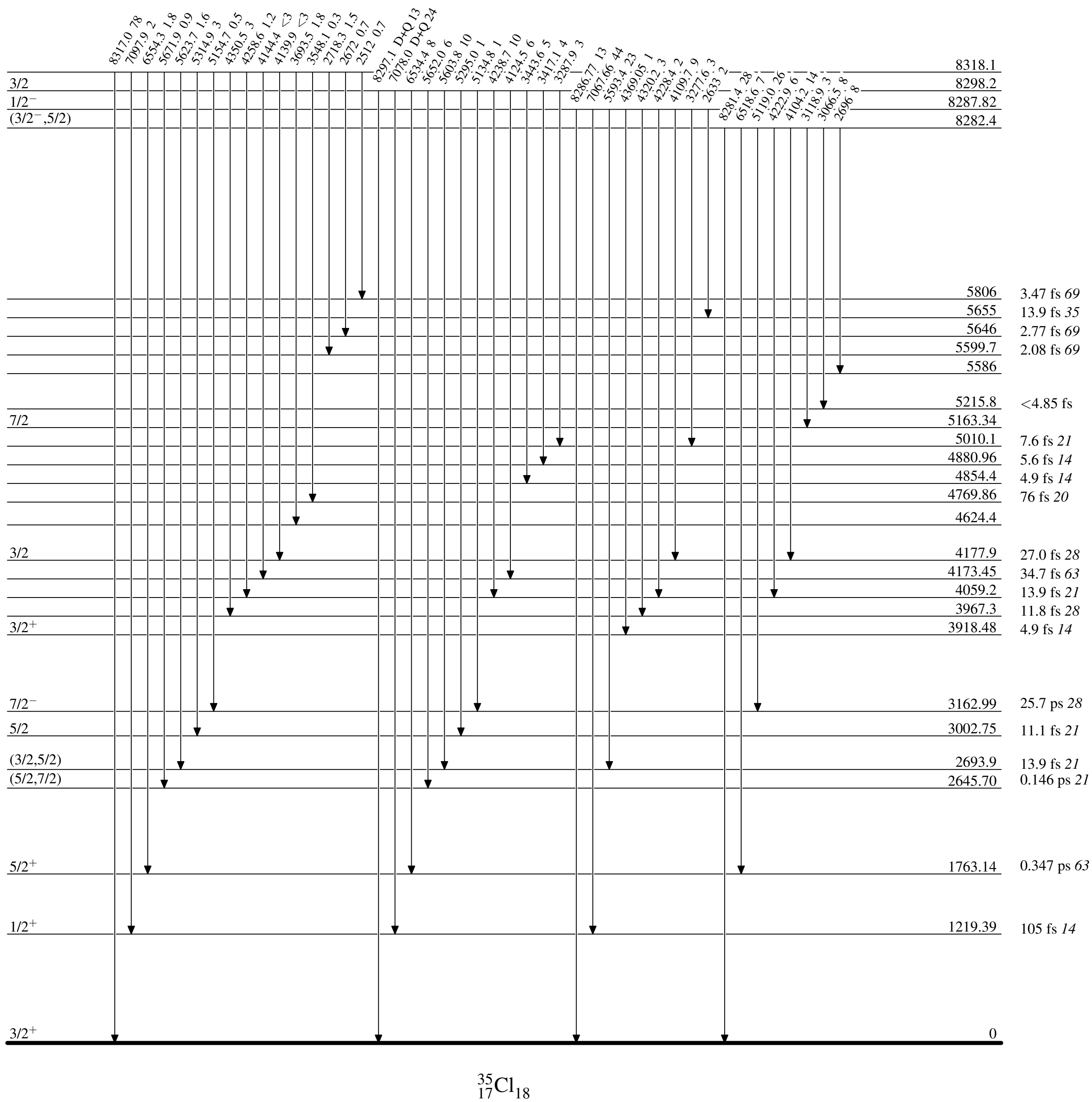


$^{35}_{17}Cl_{18}$

$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

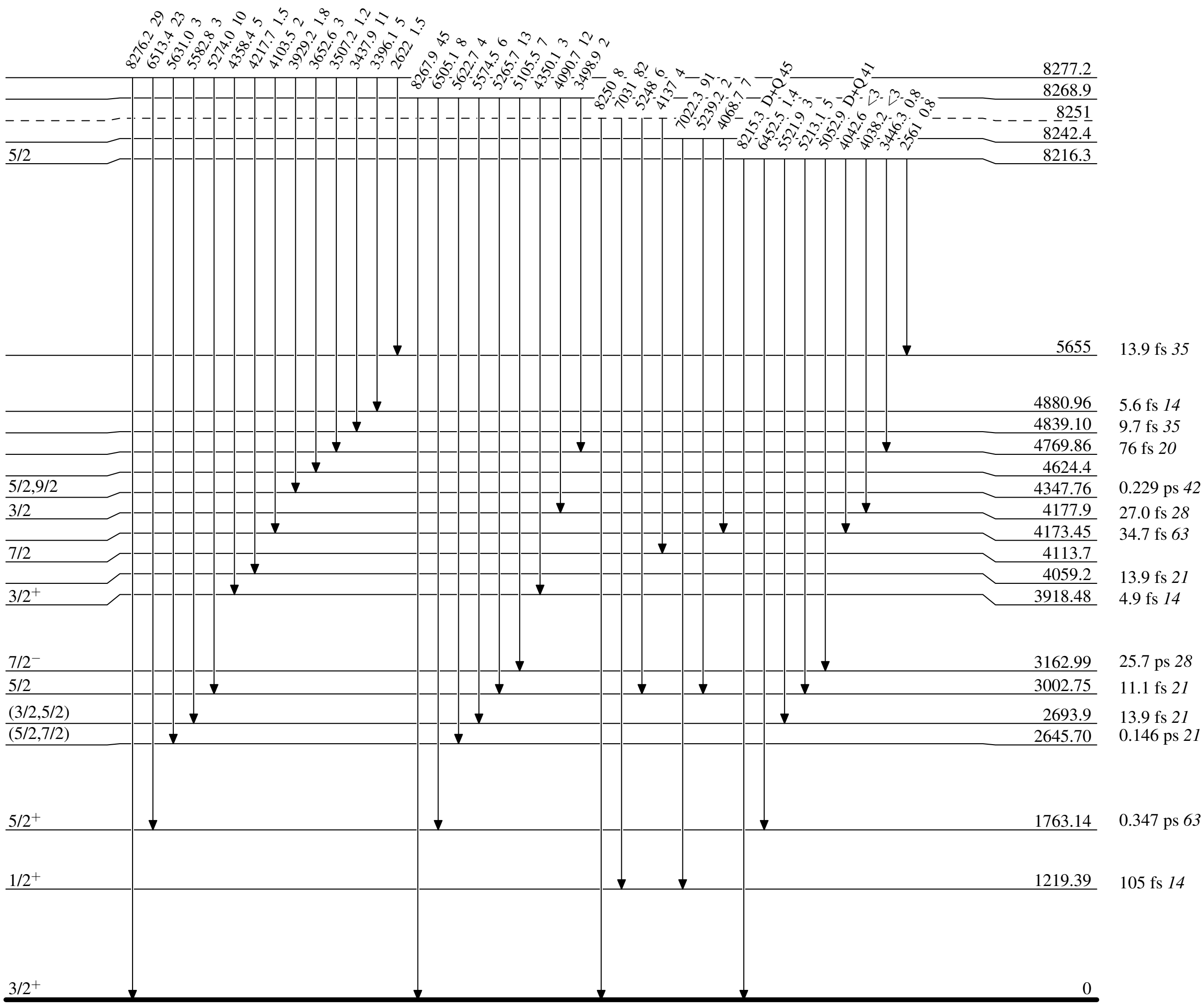
Intensities: % photon branching from each level



$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

Intensities: % photon branching from each level

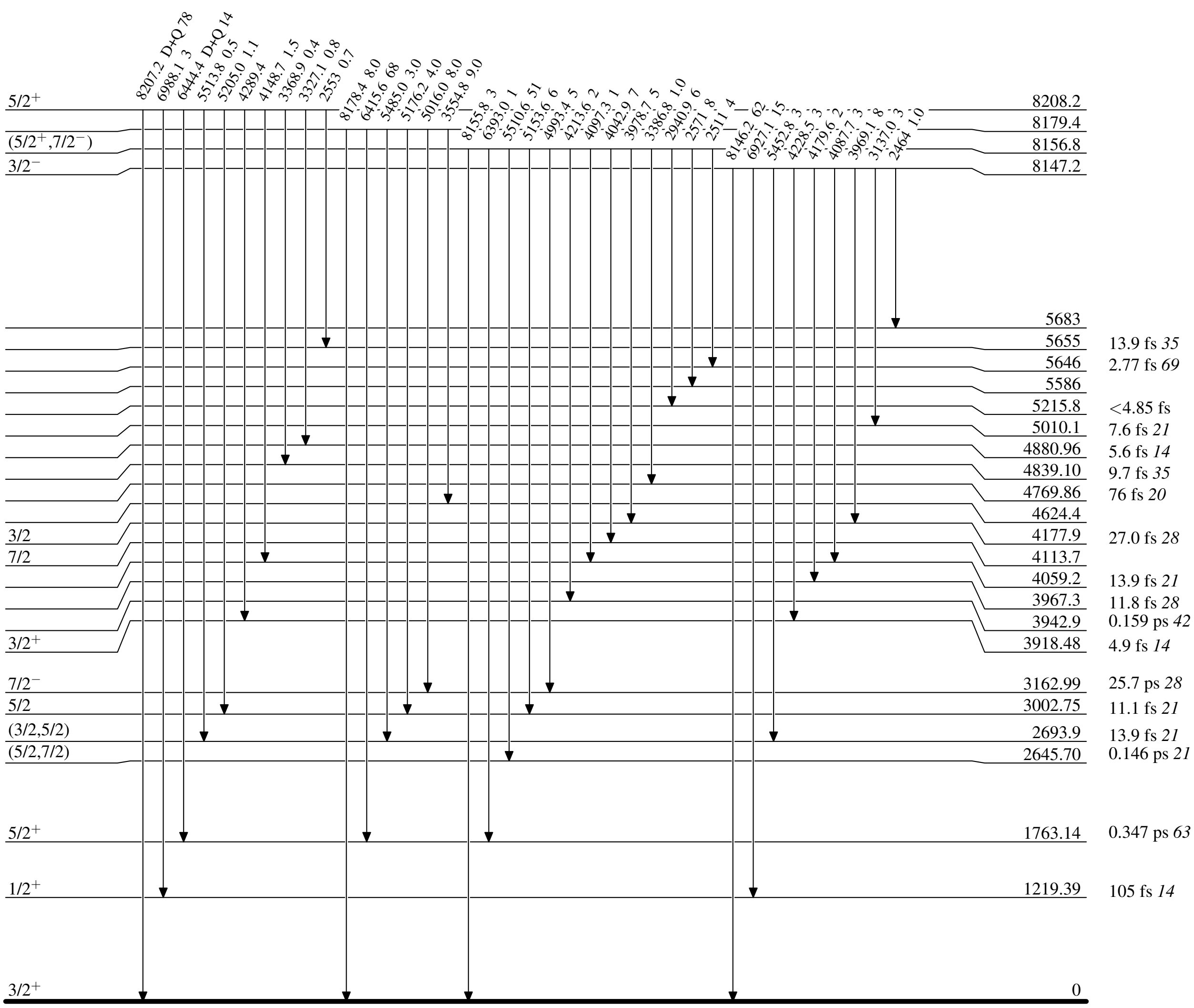


$^{35}_{17}Cl_{18}$

$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

Intensities: % photon branching from each level

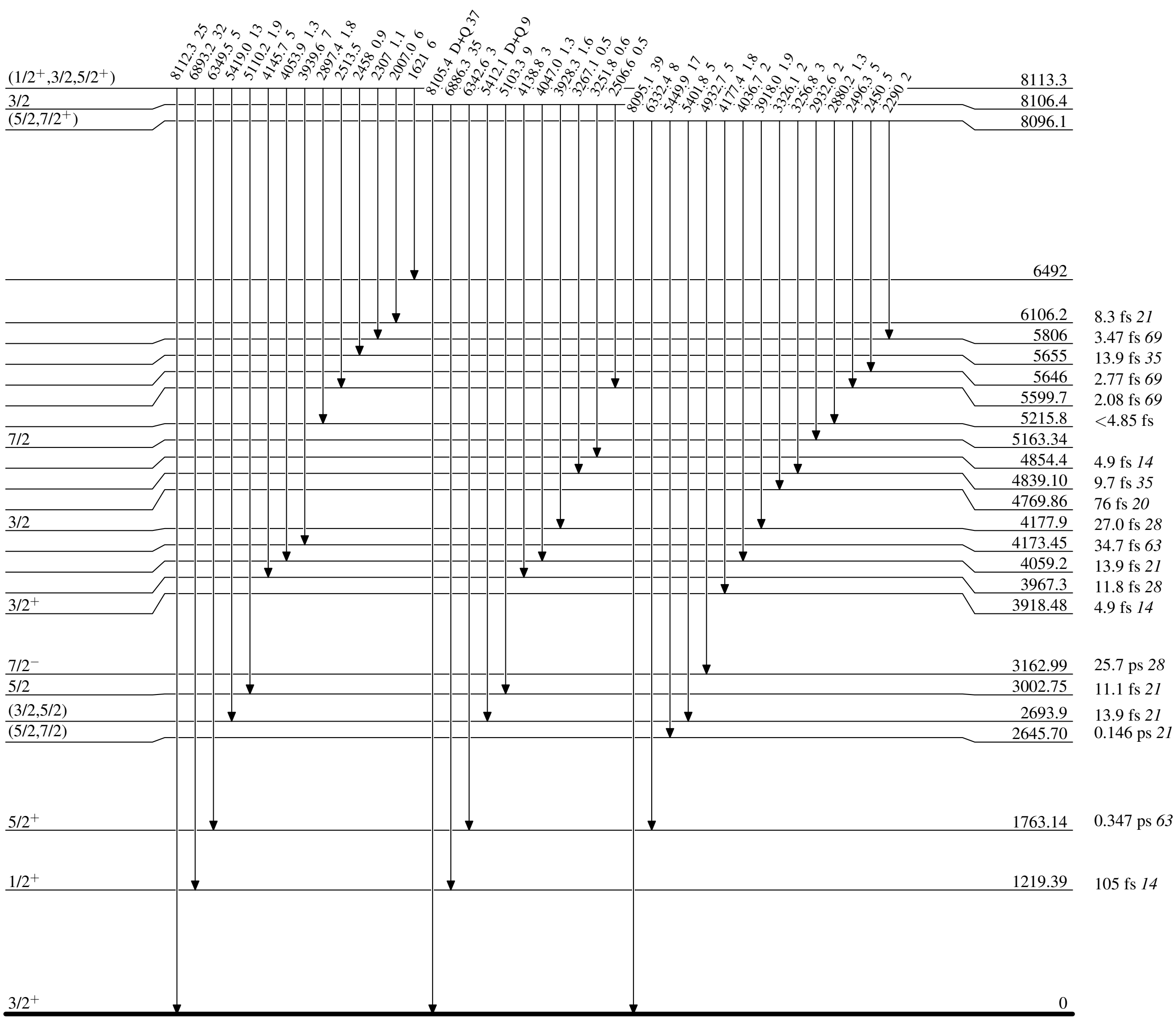


$^{35}_{17}Cl_{18}$

$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

Intensities: % photon branching from each level

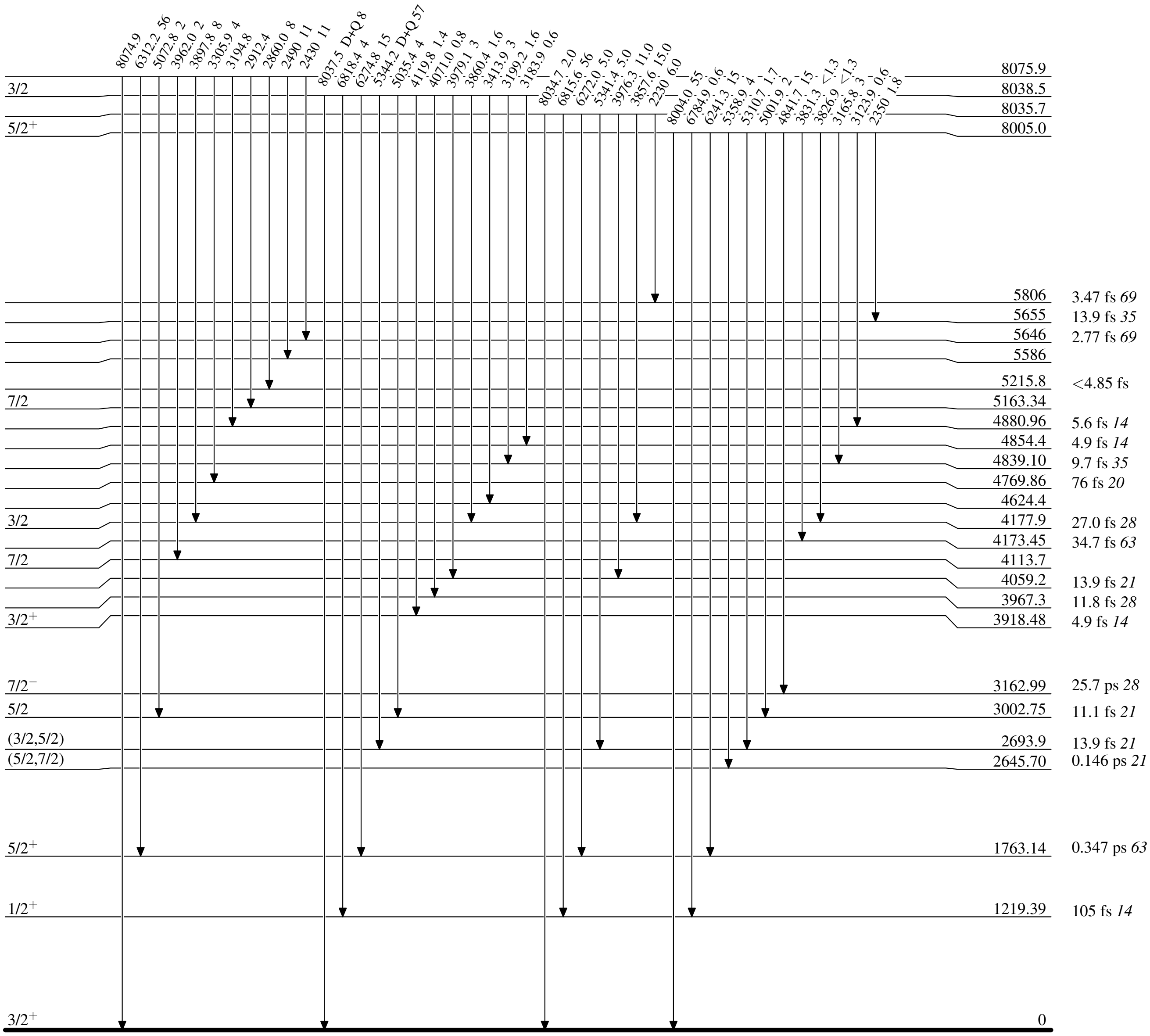


$^{35}_{17}Cl_{18}$

$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

Intensities: % photon branching from each level

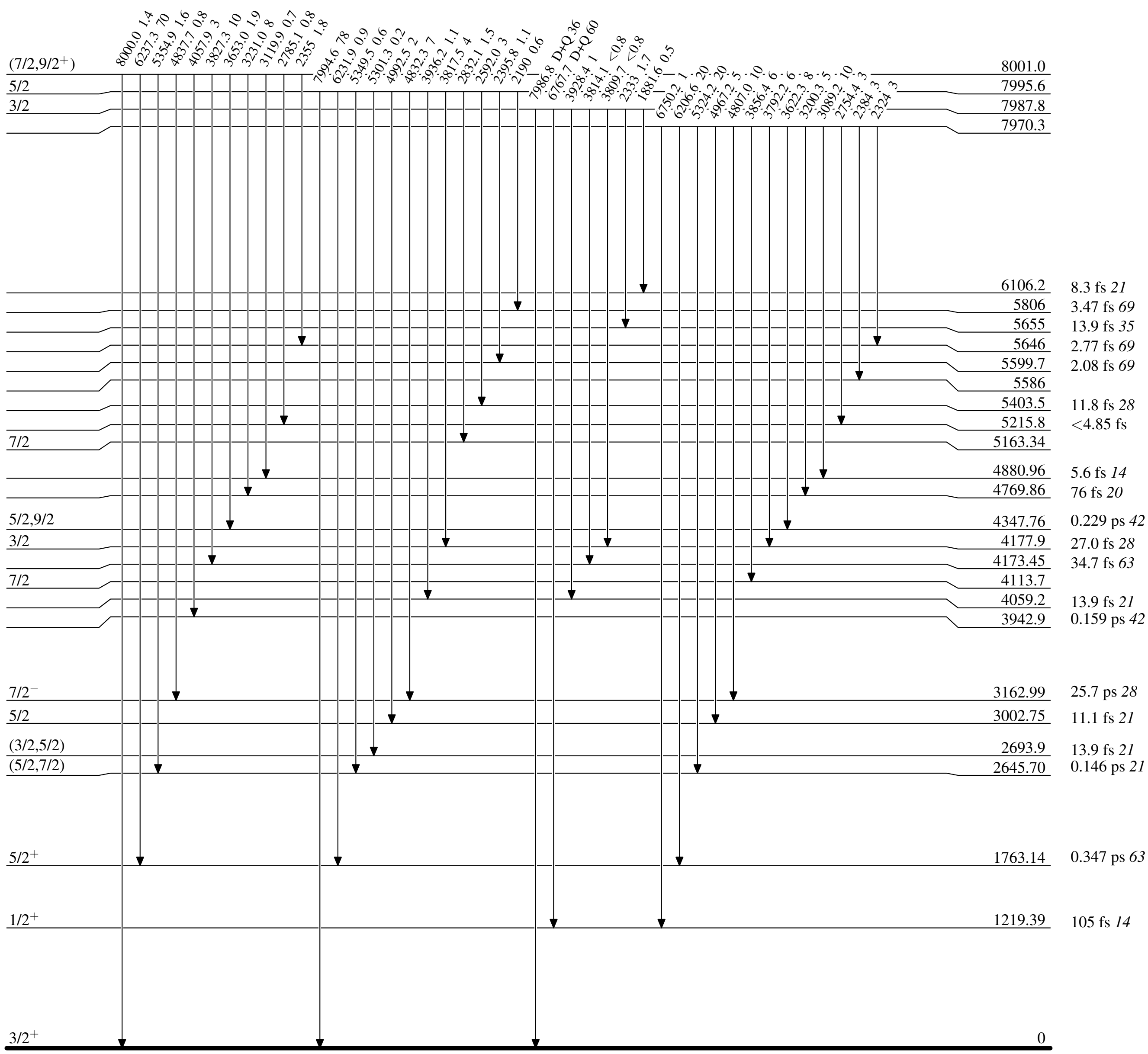


$^{35}_{17}Cl_{18}$

$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

Intensities: % photon branching from each level

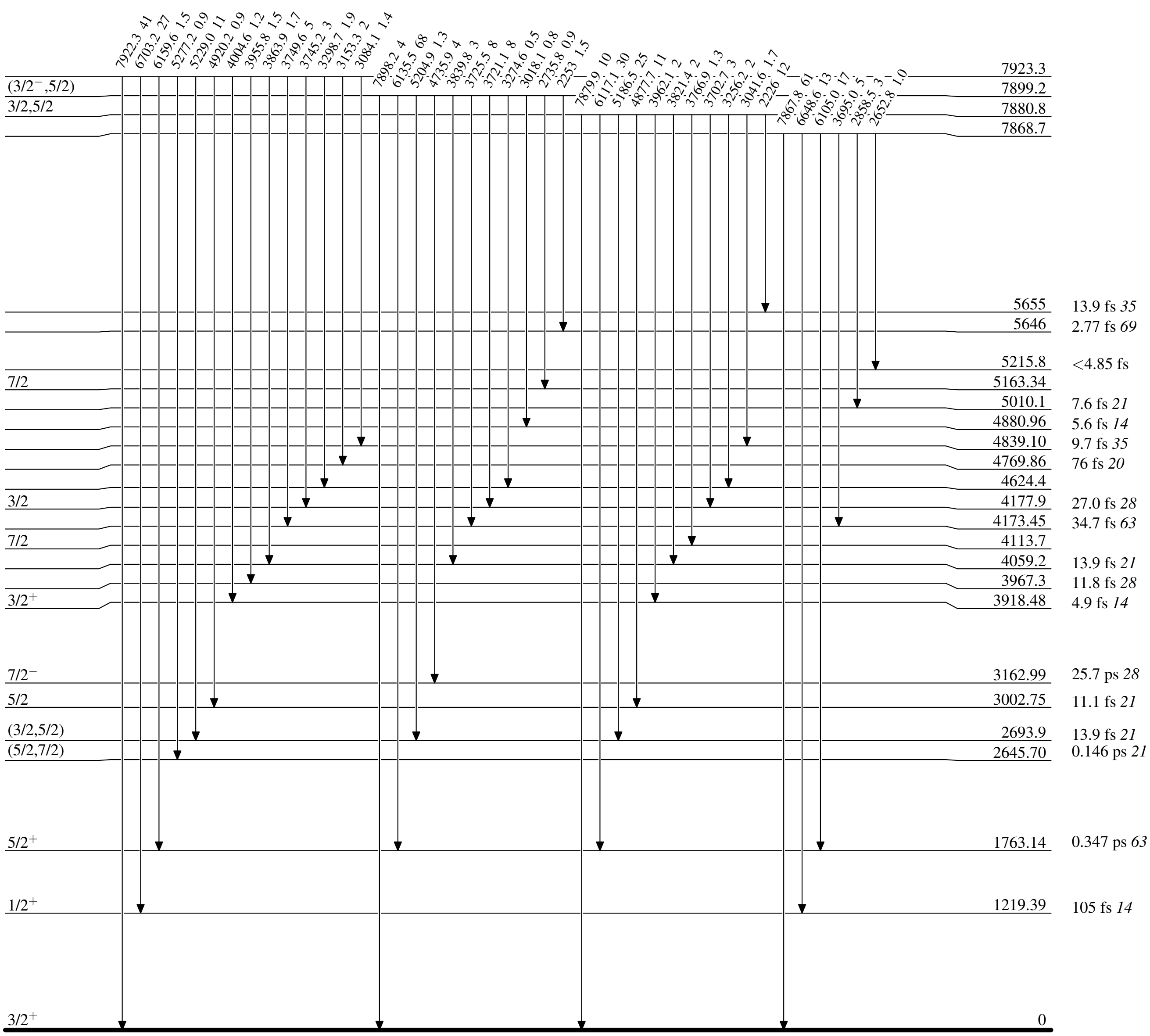


$^{35}_{17}Cl_{18}$

$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

Intensities: % photon branching from each level

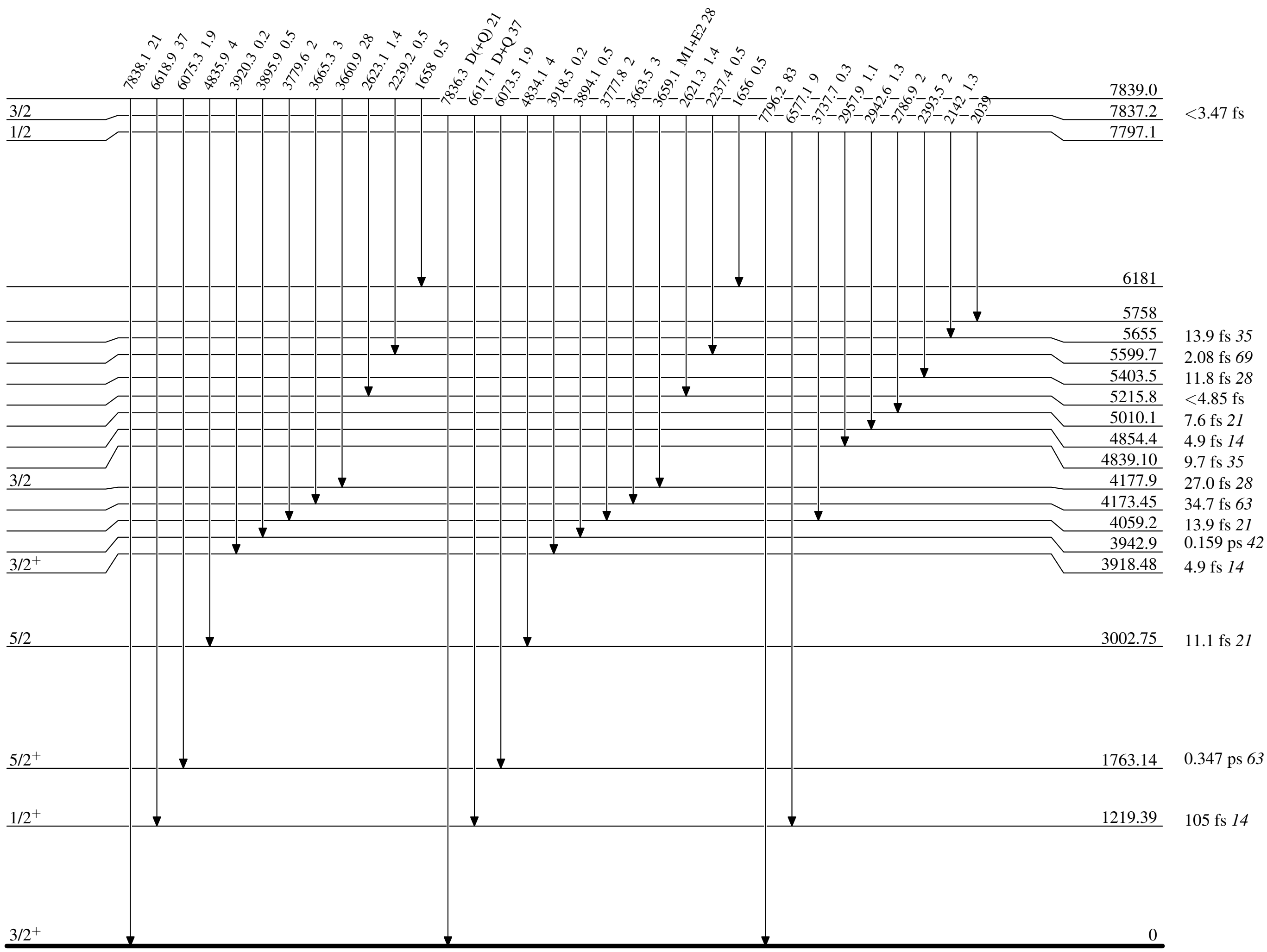


$^{35}_{17}Cl_{18}$

$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

Intensities: % photon branching from each level

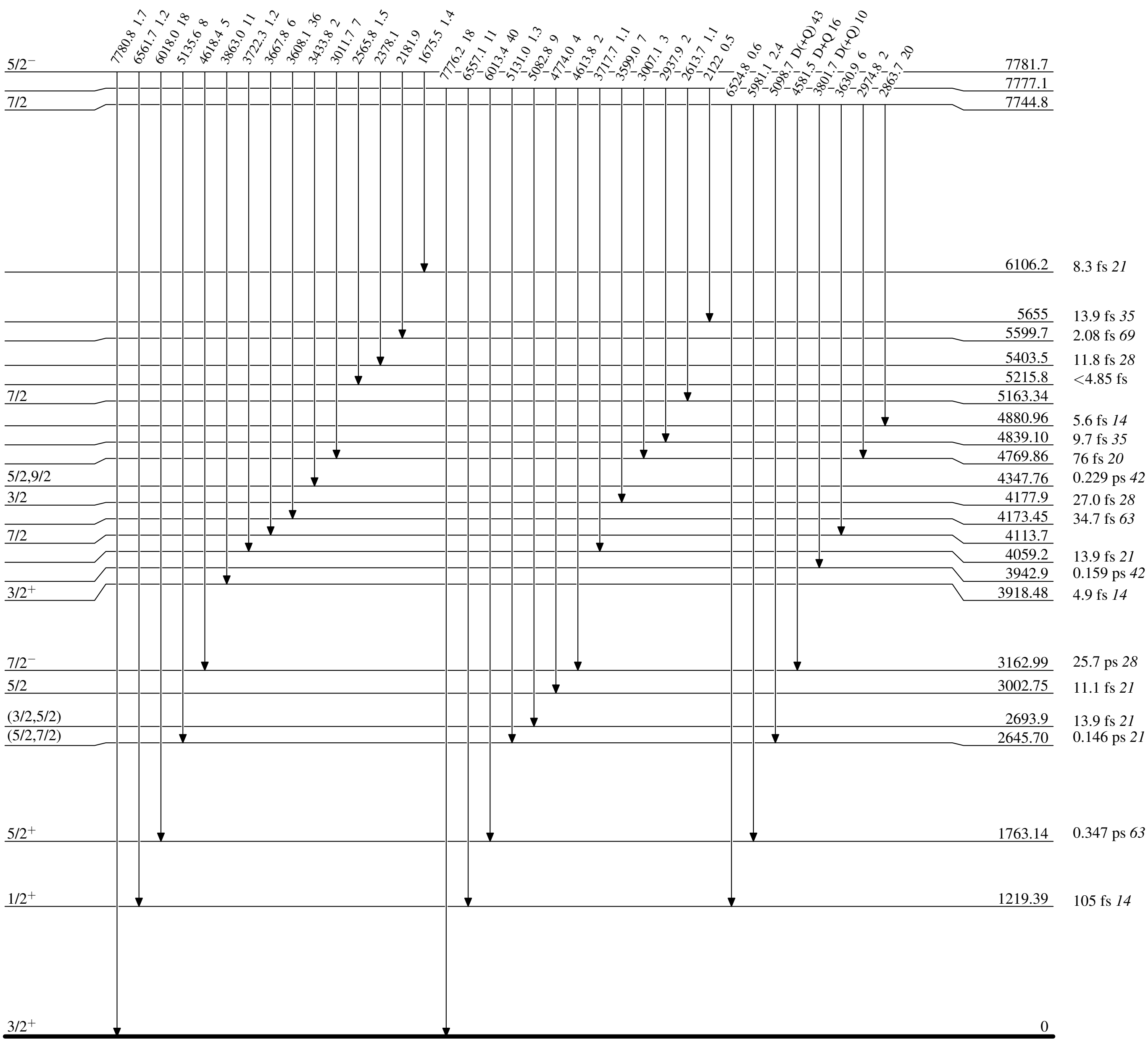


$^{35}_{17}Cl_{18}$

$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

Intensities: % photon branching from each level

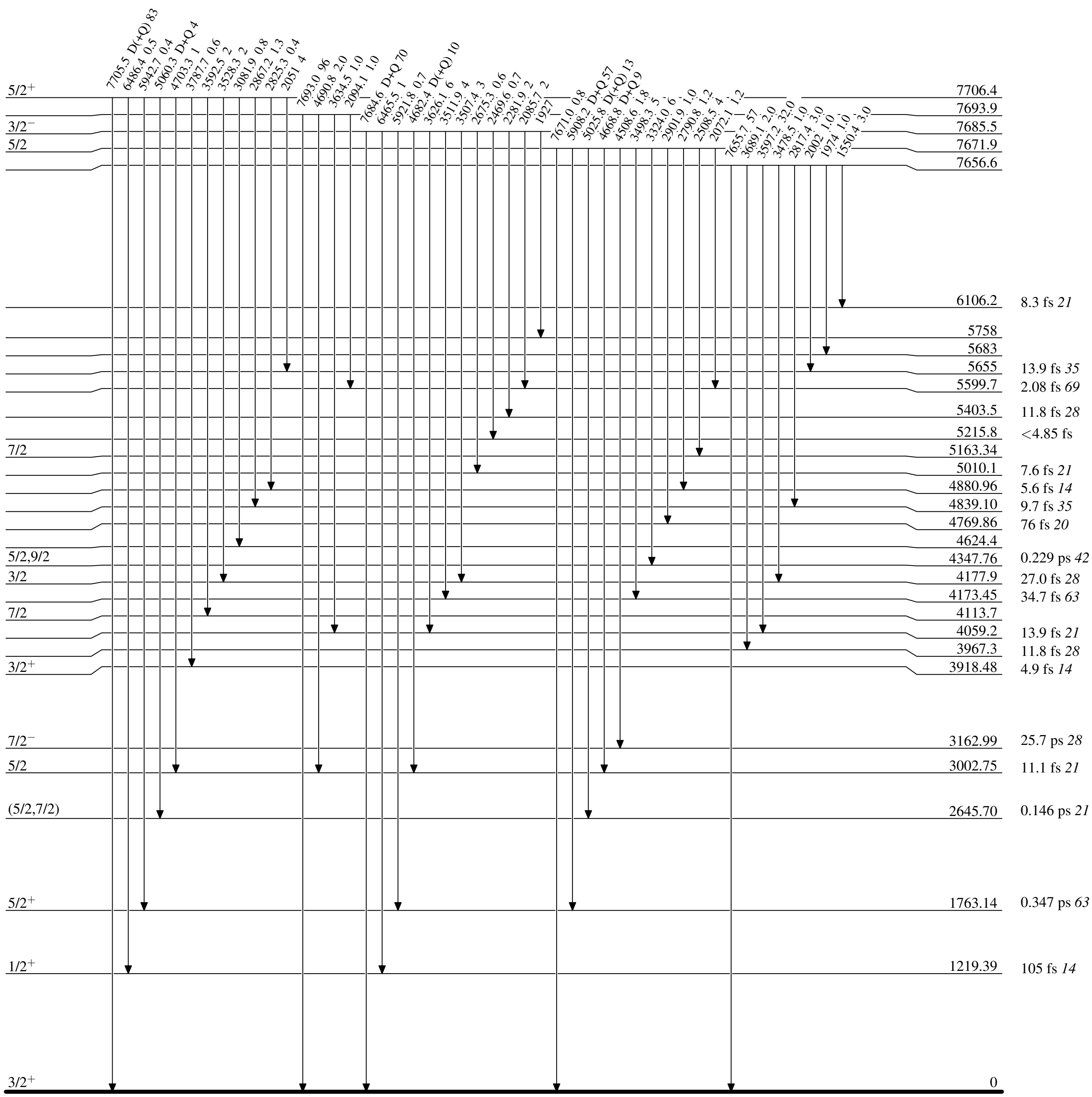


$^{35}_{17}Cl_{18}$

$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

Intensities: % photon branching from each level

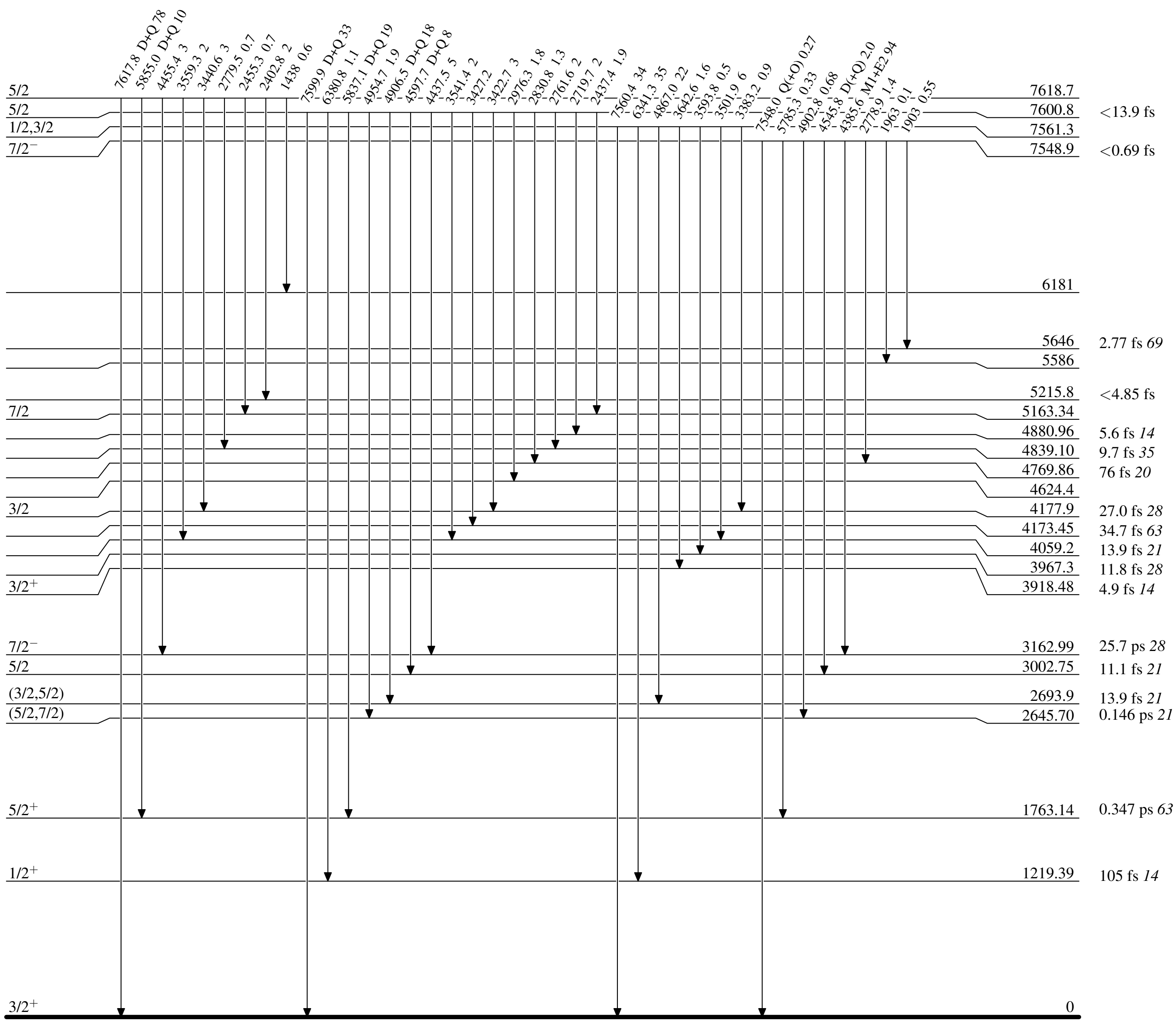


$^{35}_{17}Cl_{18}$

$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

Intensities: % photon branching from each level

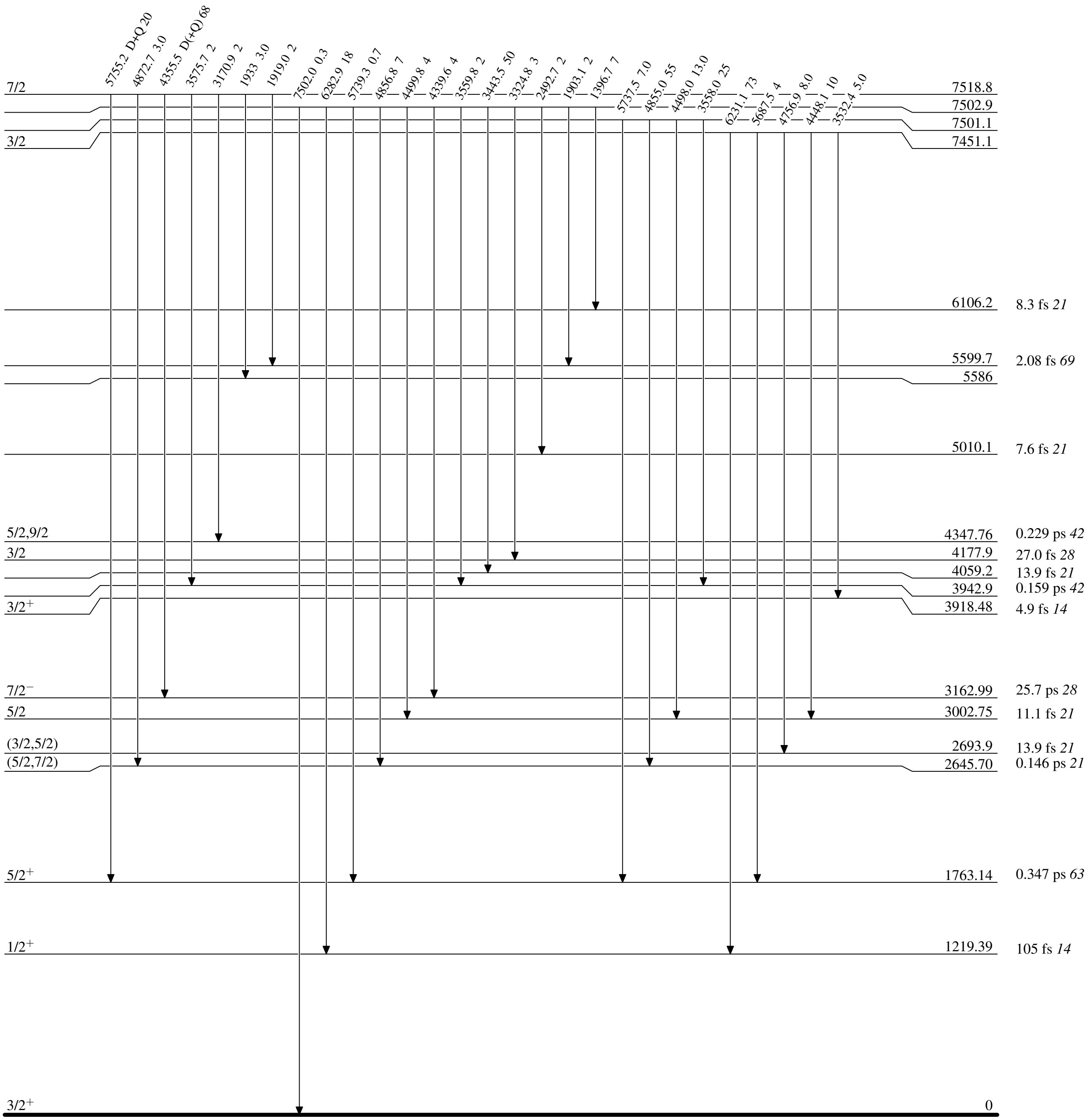


$^{35}_{17}Cl_{18}$

$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

Intensities: % photon branching from each level

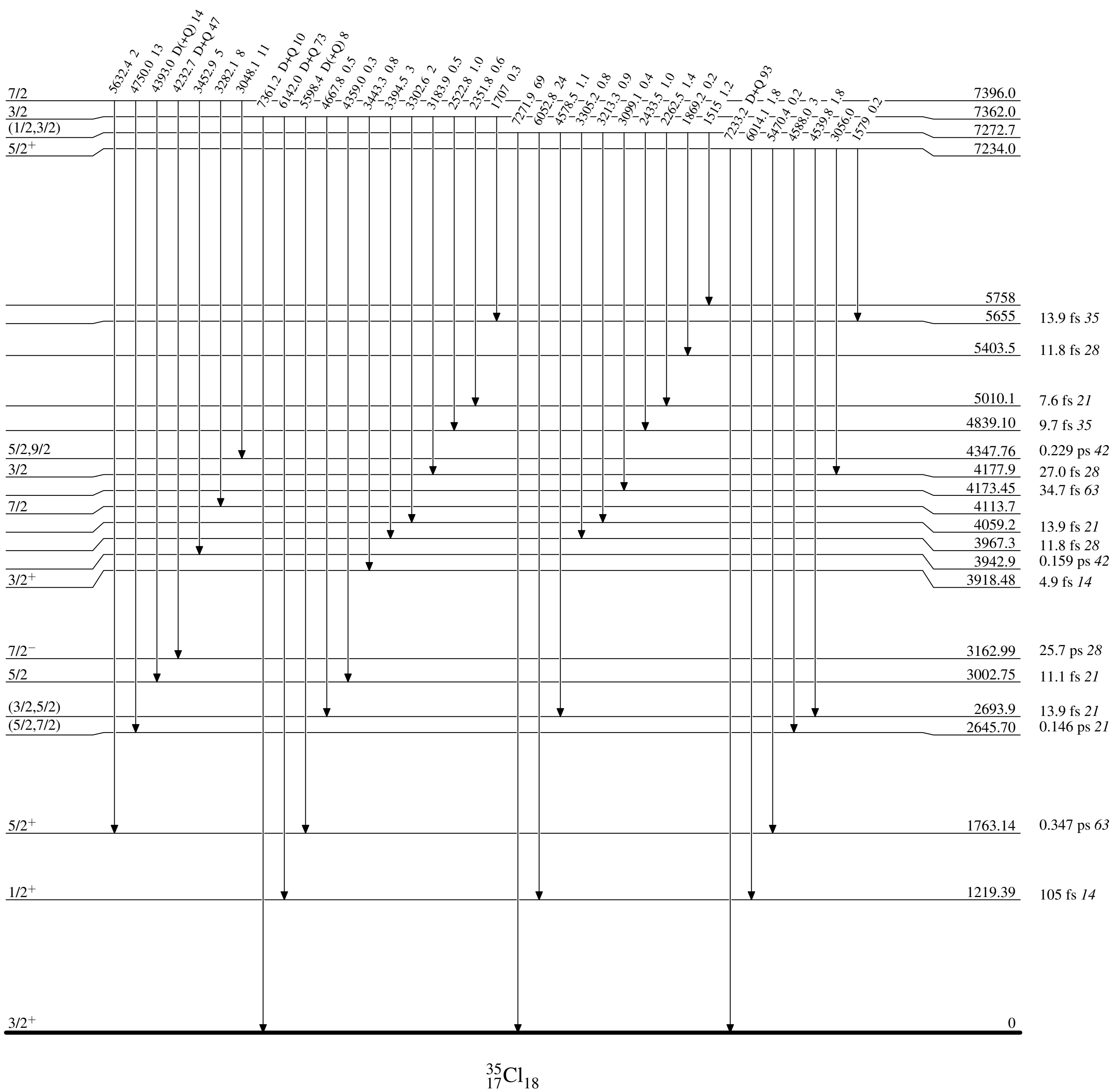


$^{35}_{17}Cl_{18}$

$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

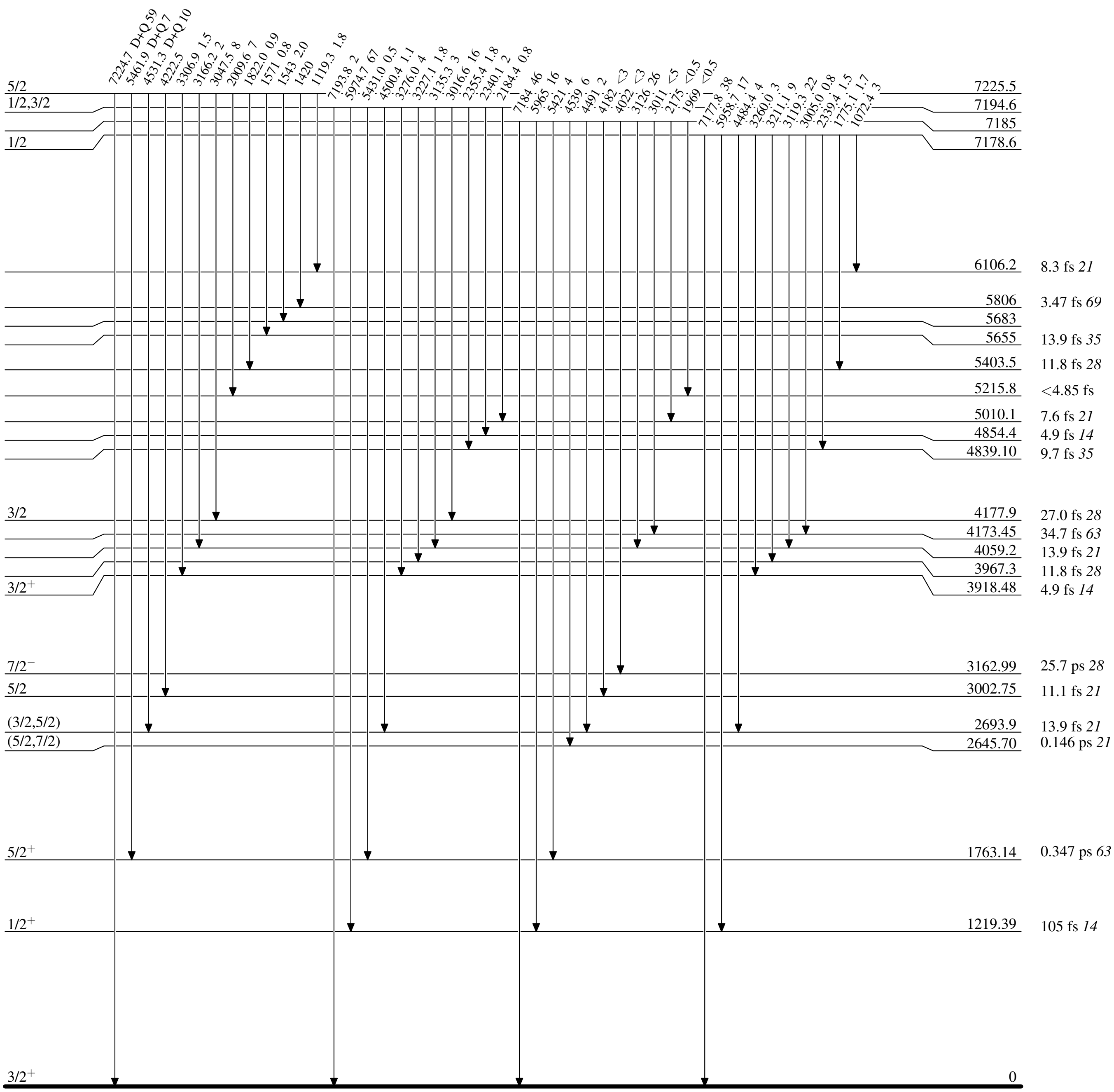
Intensities: % photon branching from each level



$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

Intensities: % photon branching from each level

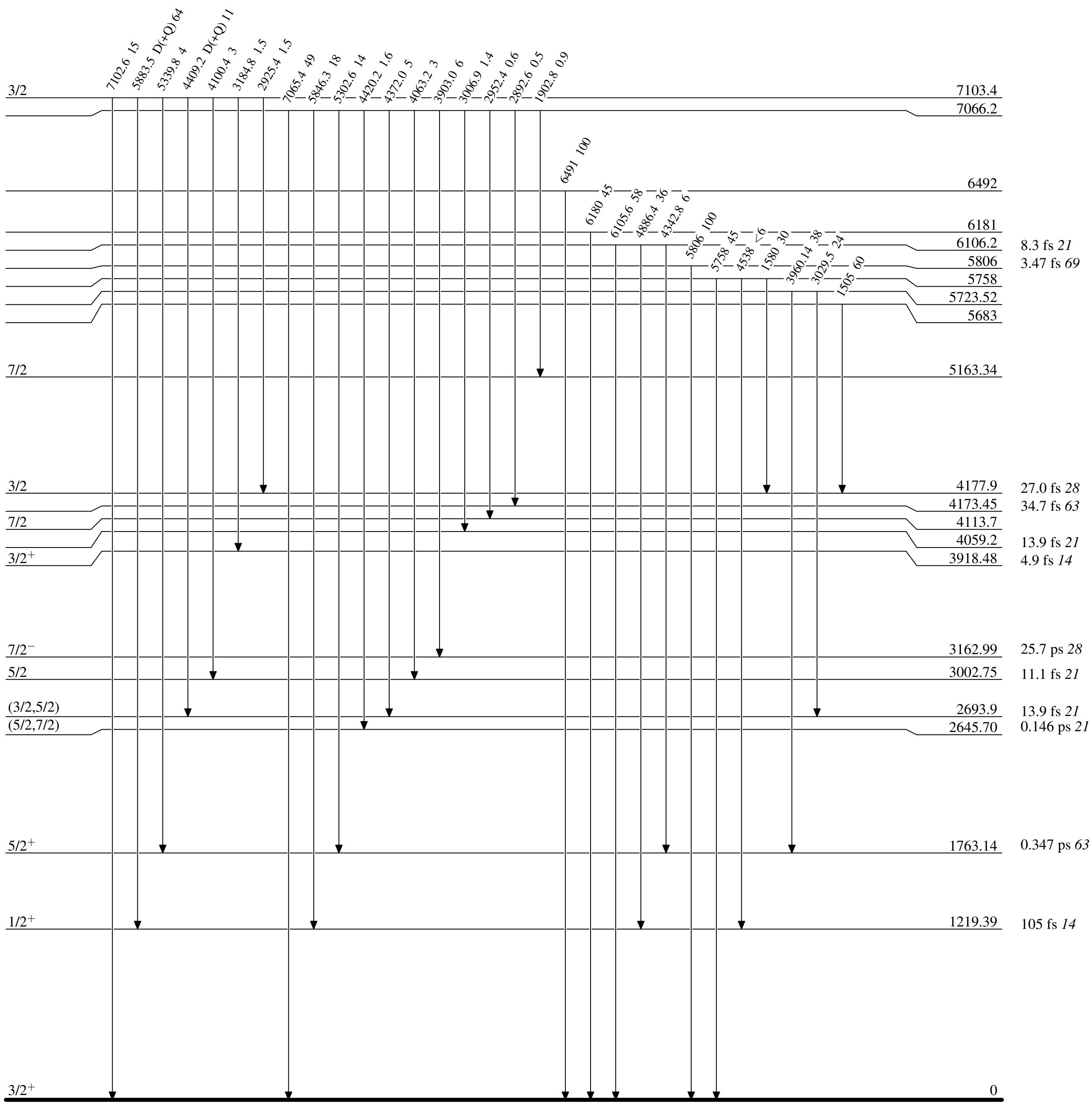


$^{35}_{17}Cl_{18}$

$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

Intensities: % photon branching from each level

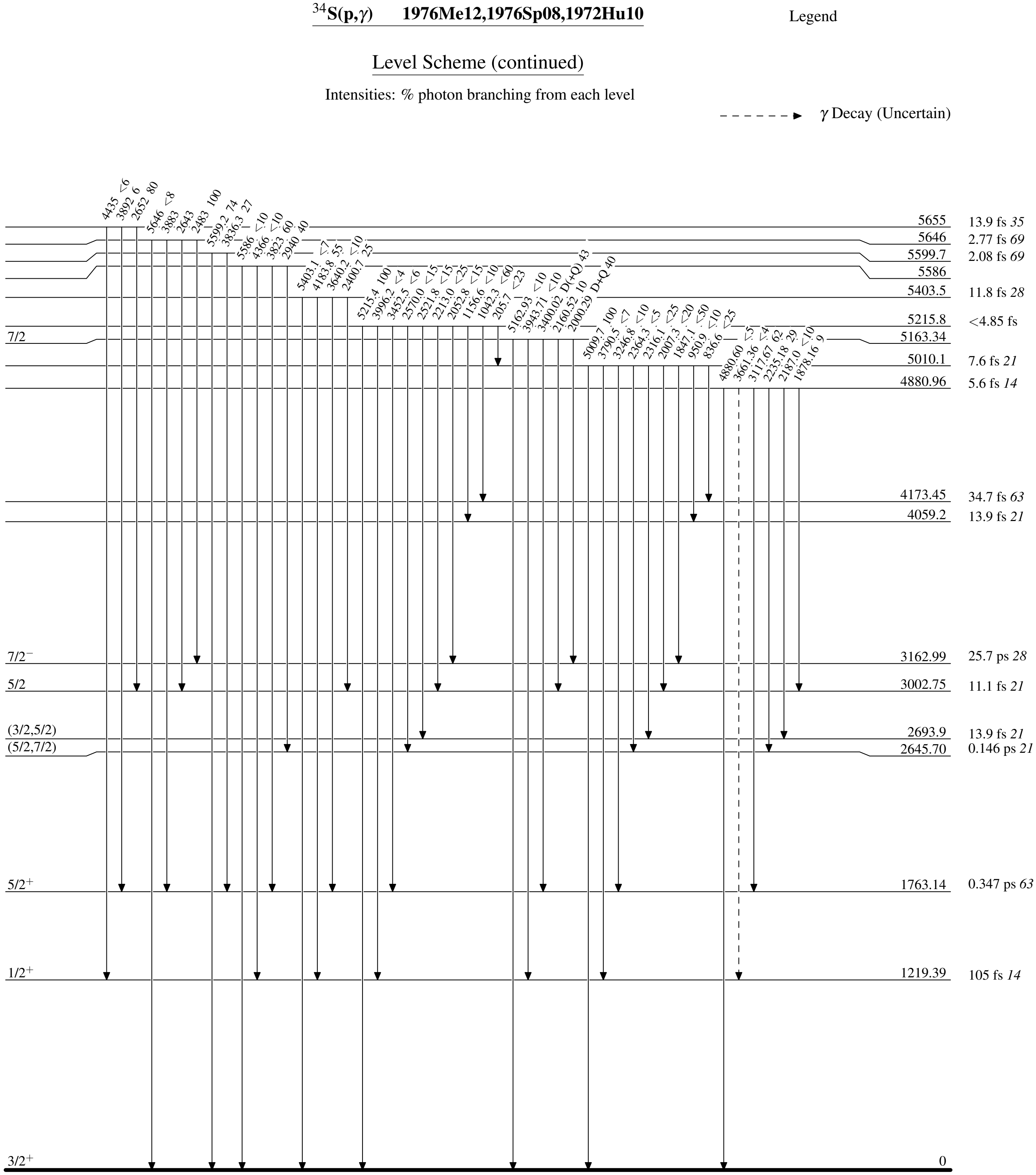


$^{35}_{17}Cl_{18}$

$^{34}S(p,\gamma)$ **1976Me12,1976Sp08,1972Hu10**

Level Scheme (continued)

Intensities: % photon branching from each level



$^{35}_{17}Cl_{18}$

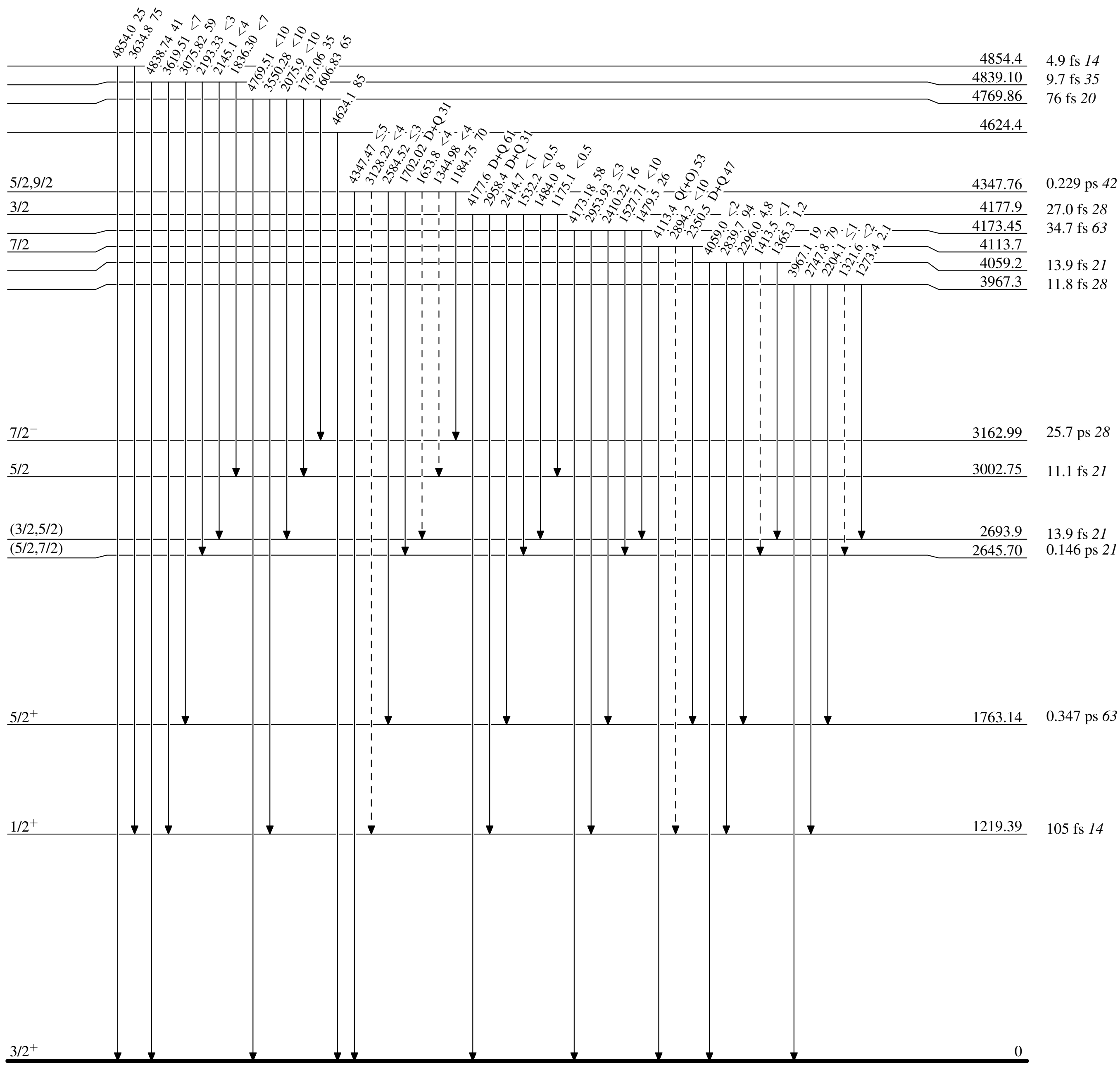
$^{34}S(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10

Legend

Level Scheme (continued)

Intensities: % photon branching from each level

γ Decay (Uncertain)

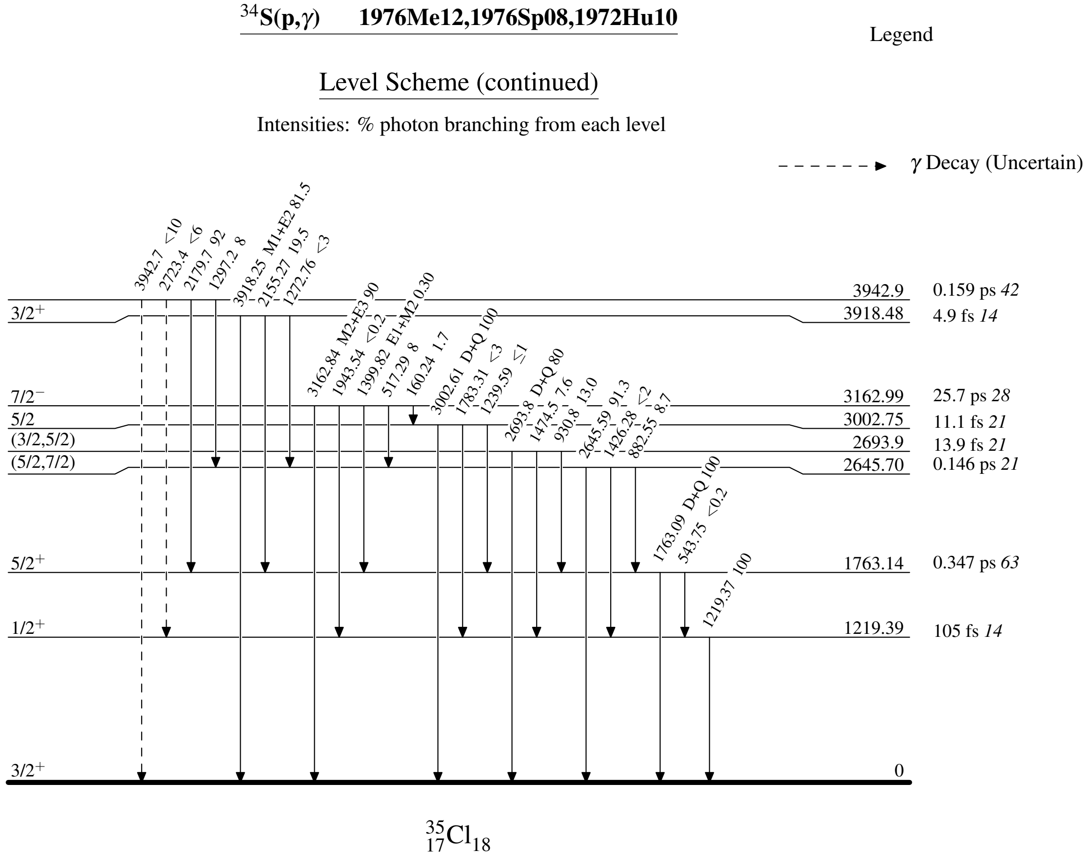


$^{35}_{17}Cl_{18}$

^{34}S(p,γ) 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

Intensities: % photon branching from each level



^{34}S(p,p),(p,p′γ):resonances 1967Ko19,1977Ou01,1981Bi05

^{34}S(p,p′γ) measurement:

1967Ko19: E=2.78-3.08−MeV proton beams were produced from the 4-MeV electrostatic accelerator of the Physical-Technical Institute of the Ukrainian SSR Academy of Sciences, in energy step of 1 keV. Target was enriched thin ^{34}S. γ rays were detected with a 70 by 60 mm NaI(Tl) crystal for detecting the 2.13 MeV γ-ray of ^{34}S decay, resolution of 11.5% for the 0.622 MeV ^{137}Cs line. Measured σ(E_p,θ) from yields of the 2.13 MeV γ-ray, pγ(θ) of the 2.13 MeV γ-ray. Deduced resonance energies, J, π, relative resonance intensities.

^{34}S(p,p) measurements:

1977Ou01: E=1.45-2.82 MeV proton beams were produced from the TUNL 3-MV Van de Graaff accelerator in energy steps of 100 eV for on-resonance and 400 eV for off-resonance, overall FWHM=350-450 eV. Target was CdS, with 90% enriched ^{34}S evaporated onto 5-10 μg/cm^2 carbon backings. Scattered protons were detected with surface barrier detectors at 90°, 105°, 135° and 160°. Measured σ(E_p,θ). Deduced resonance energies, J, π, Γ_p from R-Matrix analysis.

1981Bi05: E=1.4-2.8 MeV proton beams were produced from the Groningen 5 MV Van de Graaff accelerator. Targets were 4 and 1 μg/cm^2 Sb_2S_3 (90% in ^{34}S) evaporated onto 10 μg/cm^2 carbon backings. Scattered protons were detected with three silicon surface barrier detectors at 88°, 124° and 174°. Measured σ(E_p,θ). Deduced resonance energies, J, π, Γ_p. Also measured (p,γ).

1971BrXT: Measured σ(E_p). Deduced resonance energies, J, π, widths.

1974Bi16: E=1.2-2.1 MeV protons were produced from the Helsinki University 2.5 MV van de Graaff accelerator. Targets were prepared from natural sulphur (4.22% ^{34}S) with ^{34}S ions embedded into carbon foils. Scattered protons were detected with four silicon surface barrier detectors. Measured σ(E_p,θ). Deduced resonance energies, J, π, widths from analysis of the resonances.

Others: 1978Ce02, 1980Fa07, 1983Ch50, 1988Za04, 2019Ka53.

^{35}Cl Levels

A_2 and A_4 values under comments are from pγ(θ) of the 2.13 MeV γ-ray from ^{34}S (1967Ko19).
I(res) under comments are resonance intensities relative to 1 for E_p(lab)=2799 resonance (1967Ko19).

E(level)[†]	Jπ[‡]	Γ_p[#]	E_p(lab)(keV)[#]	Comments
7195[@]	1/2−[@]		848	Γ=27 eV *8* (1971BrXT).
7274[@]	1/2−[@]		929	Γ=14 eV *4* (1971BrXT).
7550	7/2−		1214	Γ_p=4.6 eV *10* and Γ=5 eV *+4−2* from a Q^2 minimization analysis of the (p,p) excitation function (1971BrXT). Γ=10.5 eV *15*, Γ_p=5.8 eV *7*, Γ_γ=4.7 eV *17* (1971BrXT); Γ=11.0 eV *15*, Γ_p=7.2 eV *15*, Γ_γ=3.8 eV *15* (1974Bi16), which are deduced using a questionable (p,γ) resonance strength of 21 eV *3* from 1966En04.
7686[&]	3/2−[&]		1354	Γ_p=445 eV *11* for J^π=3/2− (1981Bi05).
7706[&]	3/2+,5/2+[&]		1375	Γ_p=4 eV *1* for J^π=5/2+, 17 eV *3* for J^π=3/2+ (1981Bi05).
7796 *3*	(1/2)−	0.031 keV *10*	1467.3 *5*	
7837 *3*	3/2−	3.7 keV *4*	1510.0 *5*	Γ=3.8 keV *2*, Γ_p=3.8 keV *2* (1974Bi16).
7880 *3*	3/2+,5/2+	0.008 keV *4*	1554.0 *5*	
8004 *3*	3/2+,5/2+	0.011 keV *5*	1682.1 *5*	Γ_p=21 keV *3* for J^π=5/2+, 24 eV *3* for J^π=3/2+ (1981Bi05).
8034 *3*	1/2−,3/2−	0.026 keV *7*	1712.9 *5*	
8038 *3*	1/2+	0.30 keV *2*	1716.9 *5*	
8148 *3*	1/2−	2.66 keV *27*	1830.2 *5*	
8150 *3*	3/2−	0.56 keV *6*	1831.5 *5*	
8209 *3*	(5/2)+	0.033 keV *10*	1893.2 *5*	Γ=100 eV *20*, Γ_p=120 eV *20* (1974Bi16).
8211 *3*	1/2+	0.094 keV *15*	1894.4 *5*	Γ_p=39 eV *5* for J^π=5/2+, 44 eV *6* for J^π=3/2+, 120 eV *11* for J^π=1/2+ (1981Bi05). Γ=270 eV *50*, Γ_p=270 eV *50* (1974Bi16).
8216	5/2+	0.014 keV *3*	1900	Γ=14 eV *4*, Γ_γ=0.78 eV (1974Bi16).
8243 *3*	3/2−	0.140 keV *15*	1928.2 *5*	
8271 *3*	3/2+,5/2+	0.005 keV *3*	1956.1 *5*	
8279 *3*	3/2+,5/2+	0.006 keV *3*	1964.6 *5*	
8290 *3*	(1/2)−	0.04 keV *1*	1975.7 *5*	Γ_p=49 eV *11* for J^π=3/2−, 48 eV *6* for J^π=1/2− (1981Bi05).

Continued on next page (footnotes at end of table)

^{34}S(p,p),(p,p′γ):resonances 1967Ko19,1977Ou01,1981Bi05 (continued)

^{35}Cl Levels (continued)

E(level)†	Jπ‡	Γ_p#	E_p(lab)(keV)#	Comments
8300 *3*	3/2−	0.073 keV *15*	1986.7 *5*	
8383 *3*	3/2+,5/2+	0.023 keV *7*	2072.2 *5*	Γ=30 eV *7*, Γ_p=28 eV *6*, Γ_γ=2.4 eV (1974Bi16). Jπ: 5/2+ in 1974Bi16.
8404 *3*	5/2−,7/2−	0.002 keV *1*	2093.9 *5*	
8406 *3*	(5/2−,7/2−)	0.001 keV *1*	2095.2 *5*	
8409 *3*	1/2−	0.125 keV *15*	2098.7 *5*	
8418 *3*	1/2+	0.026 keV *7*	2107.5 *5*	
8435 *3*	(3/2)+	0.090 keV *15*	2125.9 *5*	
8486 *3*	3/2+,5/2+	0.012 keV *5*	2178.2 *5*	
8488 *3*	3/2−	0.150 keV *15*	2179.7 *5*	
8515 *3*	1/2−	0.150 keV *15*	2208.1 *5*	
8573 *3*	(5/2)+	0.08 keV *1*	2267.9 *5*	
8582 *3*	1/2+	0.75 keV *8*	2276.7 *5*	
8592 *3*	3/2+,5/2+	0.003 keV *2*	2287.4 *5*	
8616 *3*	5/2+	0.175 keV *20*	2311.6 *5*	
8620 *3*	3/2+,5/2+	0.002 keV *1*	2316.2 *5*	
8633 *3*	(3/2,5/2+)	0.001 keV *1*	2328.8 *5*	
8643 *3*	3/2+,5/2+	0.003 keV *2*	2339.9 *5*	
8687 *3*	5/2−,7/2−	0.001 keV *1*	2385.0 *5*	
8689 *3*	1/2+	0.20 keV *2*	2387.2 *5*	
8691 *3*	1/2−	6.4 keV *7*	2388.6 *5*	
8698 *3*	3/2−	0.8 keV *1*	2396.0 *5*	
8750 *3*	3/2−	0.30 keV *3*	2449.5 *5*	
8775 *3*	1/2−	0.57 keV *6*	2475.2 *5*	
8781 *3*	3/2−	0.214 keV *25*	2481.7 *5*	
8788 *3*	(3/2+,5/2+)	0.001 keV *1*	2489.1 *5*	
8799 *3*	(3/2+,5/2+)	0.001 keV *1*	2500.3 *5*	
8825 *3*	1/2+	1.70 keV *17*	2526.8 *5*	
8829 *3*	1/2−	12.2 keV *12*	2530.8 *5*	
8830 *3*	1/2+	0.080 keV *15*	2532.0 *5*	
8839 *3*	5/2−,7/2−	0.001 keV *1*	2541.7 *5*	Γ_p=4 eV *1* for Jπ=7/2−, 2 eV *1* for Jπ=5/2− (1981Bi05).
8858 *3*	3/2+,5/2+	0.010 keV *5*	2560.8 *5*	
8870 *3*	3/2+,5/2+	0.027 keV *10*	2573.2 *5*	
8909 *3*	3/2+,5/2+	0.002 keV *1*	2613.2 *5*	
8955 *3*	(3/2)+	0.075 keV *15*	2661.3 *5*	
8959 *3*	(1/2−,3/2−)	0.04 keV *1*	2665.4 *5*	
8983 *3*	5/2−,7/2−	0.003 keV *2*	2690.0 *5*	
8985 *3*	3/2+,5/2+	0.025 keV *10*	2692.1 *5*	
8998 *3*	5/2−,7/2−	0.002 keV *1*	2705.1 *5*	
9021 *3*	3/2−	3.50 keV *35*	2728.4 *5*	
9031 *3*	(5/2)+	0.04 keV *1*	2739.4 *5*	
9040 *3*	1/2−	0.292 keV *30*	2748.2 *5*	
9048 *3*	5/2−,7/2−	0.001 keV *1*	2756.7 *5*	
9050 *3*	(5/2)+	0.095 keV *15*	2758.8 *5*	
9084 *3*	(5/2)+	0.057 keV *10*	2793.2 *5*	Γ_p=59 eV *7* for Jπ=5/2+, 67 eV *8* for Jπ=3/2+ (1981Bi05).
9089 *4*			2799 *4*	I(res)=1. E_p(lab)(keV): from 1967Ko19.
9101 *3*	3/2−	0.20 keV *2*	2810.9 *5*	
9108 *4*			2818 *4*	I(res)=0.18.
9114 *4*			2825 *4*	I(res)=0.34.
9120 *4*			2831 *4*	I(res)=0.31.
9127 *4*			2838 *4*	I(res)=0.16.
9136 *4*			2847 *4*	I(res)=0.34.
9147 *4*			2859 *4*	I(res)=0.27.
9159 *4*			2871 *4*	I(res)=0.29.
9168 *4*	(3/2−)		2880 *4*	I(res)=3.70.

Continued on next page (footnotes at end of table)

^{34}S(p,p),(p,p′γ):resonances 1967Ko19,1977Ou01,1981Bi05 (continued)

^{35}Cl Levels (continued)

E(level)†	J$^{\pi}$‡	E_p(lab)(keV)#	Comments
			A_2=+0.19 *133*, A_4=+0.017 *1480* (1967Ko19).
9185 *4*		2898 *4*	I(res)=0.11.
9196 *4*		2909 *4*	
9207 *4*	9/2$^+$	2920 *4*	I(res)=1.70. A_2=+0.09 *18*, A_4=+0.37 *12* (1967Ko19).
9222 *4*		2936 *4*	I(res)=0.25.
9227 *4*	5/2$^-$	2941 *4*	I(res)=0.59. A_2=+0.59 *28*, A_4=+0.29 *32* (1967Ko19).
9241 *4*	7/2$^+$	2955 *4*	I(res)=2.32. A_2=+0.24 *40*, A_4=+0.27 *53* (1967Ko19).
9255 *4*		2970 *4*	I(res)=0.45.
9271 *4*		2986 *4*	I(res)=0.34.
9276 *4*		2991 *4*	I(res)=0.32.
9284 *4*	5/2$^-$	3000 *4*	I(res)=3.86. A_2=+0.72 *28*, A_4=−0.17 *29* (1967Ko19).
9294 *4*		3010 *4*	I(res)=0.68.
9299 *4*		3015 *4*	I(res)=0.45.
9325 *4*	5/2$^+$	3042 *4*	I(res)=1.81. A_2=+0.81 *28*, A_4=−0.48 *35* (1967Ko19).
9336 *4*	7/2$^+$	3053 *4*	I(res)=1.00. A_2=+0.17 *29*, A_4=+0.34 *40* (1967Ko19).
9345 *4*		3062 *4*	I(res)=0.50.
9349 *4*		3067 *4*	I(res)=0.50.
9355 *4*		3073 *4*	I(res)=2.23.

† From E(level)=E_p(cm)+S(p)(^{35}Cl), where S(p)=6370.81 *4* (2021Wa16) and E_p(cm) deduced from E_p(lab) by E_p(cm)=E_p(lab)*[m(^{34}S)/(m(^{34}S)+m(p))].

‡ From comparison of experimental differential cross sections with theoretical predictions by R-Matrix calculations (1977Ou01) for levels up to 9101 level and from pγ(θ) (1967Ko19) for levels after that.

From 1977Ou01 up to 9101 level and from 1967Ko19 after that, unless otherwise noted. 1977Ou01 state that although the absolute proton energies are accurate only to about 3 keV, the relative values of E_p(lab) should be reliable to within a few hundred eV, based on which the evaluators have assigned a 0.5 keV uncertainty for E_p(lab) and 3 keV uncertainty for E(level) from 1977Ou01.

@ From 1971BrXT.

& From 1981Bi05.

^{34}S(d,n) 1969Da12

$J^\pi=0^+$ for ^{34}S ground state.

1969Da12: a 4.95-MeV deuteron beam was produced from the 5.5-MeV Van de Graaff accelerator at University of Alberta and impinged on a CdS target enriched to 37% ^{34}S evaporated onto a 0.05-cm gold backing. Neutrons were detected by NE213 scintillators using the time-of-flight (tof) method with FWHM=1.0 ns. Measured $\sigma(E_n,\theta)$. Deduced levels, J, π, and L-transfers from DWUCK-DWBA analysis of the measured $\sigma(\theta)$.

1993TeZX: ^{34}S(d,n)^{35}Cl at E(d)=25 MeV at CYRJC, Tohoku University.

^{35}Cl Levels

E(level)†	L‡	Comments
0	2	
1220	0	
1762	2	
2645	(1)	L: weak stripping or possibly two unresolved levels.
2695		
3006	3	
3163	3	L: Discrepancy: L=2 from 0^+ in (^{3}He,d), (d,^{3}He), and (pol d,^{3}He).
4060		
4110		
4170		

† From 1969Da12.
‡ From DWBA analysis of the measured $\sigma(\theta)$ in 1969Da12.

^{34}S(^{3}He,d) 2017Gi09,1969Gr23

$J^\pi=0^+$ for ^{34}S g.s.

2017Gi09: a 20-MeV $^3He^{2+}$ beam was produced from the MP Van de Graaff tandem accelerator at the Maier-Leibnitz Laboratorium (MLL) and impinged on a 50-μg/cm^2 Ag_2S target 99.999% enriched in ^{34}S with a 8-μg/cm^2 carbon backing. Reaction products were momentum-separated by the Q3D magnetic spectrograph and focused on focal plane detectors consisting of two multiwire proportional counters (MWPCs) and a plastic scintillator. Deduced 22 levels, L-transfers, J, π, spectroscopic factors from FRESCO-DWBA analysis, and resonance strengths. Deduced the thermonuclear ^{34}S(p,γ)^{35}Cl reaction rate through Monte Carlo methods using the STARLIB code. Calculated the ^{34}S/^{32}S isotopic ratio using the hydrodynamical nova model SHIVA.

1969Gr23: a 11.0-MeV ^{3}He beam was produced from the MIT-ONR Van de Graaff generator. The target was 9.3-μg/cm^2 ^{34}S made by evaporating PbS (67.6%^{34}S) onto a formvar backing. Deuterons were detected on 50 μm nuclear emulsions in the MIT multiple-gap spectrograph at 23 different angles from 7.5° to 172.5°. Measured $\sigma(E_d,\theta)$. Deduced 17 levels, J, π, L-transfers, spectroscopic factors from zero-range JULIE-DWBA analysis.

1970Mo01: a 13-MeV ^{3}He beam was produced from the Argonne tandem Van de Graaff accelerator. The target was 200-μg/cm^2 86% enriched ^{34}S. Reaction products were detected using two telescopes of silicon surface-barrier detectors with FWHM≈80 keV and analyzed by a magnetic spectrograph with FHWM≈20 keV at $\theta<20°$. Measured $\sigma(E_d,\theta)$, $\theta_{c.m.}\approx5°$ to 100°. Deduced levels, J, π, L-transfers, spectroscopic factors from local, zero-range JULIE-DWBA analysis.

1994Ve04: a 25-MeV ^{3}He beam was produced from the Orsay MP Tandem Van de Graaff accelerator. The target was KCl with natural chlorine. Reaction products were momentum-analyzed by an Enge split-pole magnetic spectrograph and detected using a 50-cm long telescope of a position-sensitive proportional counter, proportional ΔE counter, and a plastic scintillator. Measured $\sigma(E_d,\theta)$. Deduced J, π, L-transfer, and spectroscopic factor for the ground state of ^{35}Cl from local, zero-range DWUCK-DWBA analysis.

^{35}Cl Levels

For levels below 6 MeV, spectroscopic factor $(2J+1)C^2S=\sigma(\theta)_{exp}/\sigma(\theta)_{DWBA}/N$, where N is a normalization factor. J is the spin of the populated ^{35}Cl level. N=4.42 is used in 1969Gr23 and 1970Mo01 from 1966Ba54. N=4.43 is used in 1994Ve04. For levels above 6 MeV, C^2S values are from 2017Gi09 without using the factor N. 2017Gi09 also states that it is not possible to distinguish between J=L+1/2 or J=L−1/2 coupling given the experimental uncertainties.

Values of resonance strengths $\omega\gamma$(min) and $\omega\gamma$(max) are calculated by 2017Gi09 using their measured C^2S and calculated Γ_p, assuming the Γ_{tot} is dominated by Γ_γ and thus $\omega\gamma$ depends only on Γ_p.

E(level)[†]	L[‡]	Comments
0	2	S: $(2J+1)C^2S$=2.88 in 1969Gr23 for $J^\pi=3/2^+$; C^2S=1.3 in 1970Mo01 for $J^\pi=3/2^+$; C^2S=1.07 in 1994Ve04 for $J^\pi=3/2^+$.
1220 *10*	0	E(level): other: 1220 *12* (1970Mo01). S: $(2J+1)C^2S$=0.26 in 1969Gr23 for $J^\pi=1/2^+$; C^2S=0.28 in 1970Mo01 for $J^\pi=1/2^+$.
1758 *10*		
2645 *10*		
2686 *10*	2	S: $(2J+1)C^2S$=0.06 in 1969Gr23 for $J^\pi=(3/2,5/2)^+$.
3002 *10*	2	E(level): weighted average of 2997 *10* (1969Gr23) and 3010 *12* (1970Mo01). S: $(2J+1)C^2S$=0.14 in 1969Gr23 for $J^\pi=(5/2)^+$; C^2S=0.04 in 1970Mo01 for $J^\pi=5/2^+$.
3158 *10*	3	E(level): weighted average of 3156 *10* (1969Gr23) and 3160 *12* (1970Mo01). S: $(2J+1)C^2S$=2.88 in 1969Gr23 for $J^\pi=7/2^-$; C^2S=0.66 in 1970Mo01 for $J^\pi=7/2^-$.
3908 *10*		
3962 *10*	0	E(level): weighted average of 3956 *10* (1969Gr23) and 3970 *12* (1970Mo01). S: $(2J+1)C^2S$=0.03 in 1969Gr23 for $J^\pi=1/2^+$; C^2S=0.06 in 1970Mo01 for $J^\pi=1/2^+$.
4053 *10*	1	E(level): weighted average of 4048 *10* (1969Gr23) and 4060 *12* (1970Mo01). S: $(2J+1)C^2S$=0.38 in 1969Gr23 for $J^\pi=(3/2)^-$; C^2S=0.15 in 1970Mo01 for $J^\pi=(3/2)^-$.
4167 *10*	1	E(level): weighted average of 4165 *10* (1969Gr23) and 4170 *12* (1970Mo01). S: $(2J+1)C^2S$=0.85 in 1969Gr23 for $J^\pi=(3/2)^-$; C^2S=0.28, 0.62 in 1970Mo01 for $J^\pi=(3/2)^-$.
5010 *15*		
5163 *15*	(2)	S: $(2J+1)C^2S$=0.11 without specifying J (1969Gr23).
5409 *12*	1	E(level): weighted average of 5407 *15* (1969Gr23) and 5410 *12* (1970Mo01). L: 1 in 1969Gr23 and L=(1,2) in 1970Mo01.

Continued on next page (footnotes at end of table)

^{34}S(^{3}He,d) 2017Gi09,1969Gr23 (continued)

^{35}Cl Levels (continued)

E(level)†	L‡	Comments
		S: (2J+1)C^{2}S=0.07 in 1969Gr23 for J^π=(3/2)$^-$.
5660 *15*	2	E(level): 5670 multiplet reported in 1970Mo01. L: 2 in 1969Gr23, possibly a component of the 5670 multiplet with L=(2) in 1970Mo01. S: (2J+1)C^{2}S=0.47 in 1969Gr23 for J^π=3/2$^+$.
5684 *12*	1	E(level): 5670 multiplet reported in 1970Mo01. L: 1 in 1969Gr23, possibly a component of the 5670 multiplet with L=(2) in 1970Mo01. S: (2J+1)C^{2}S=0.11 in 1969Gr23 for J^π=(3/2)$^-$.
5760 *12*	(0,1)	E(level): weighted average of 5760 *15* (1969Gr23) and 5760 *12* (1970Mo01). L: 1 in 1969Gr23 and L=0 in 1970Mo01. S: (2J+1)C^{2}S=0.12 in 1969Gr23 for J^π=(3/2)$^-$; C^{2}S=0.21 in 1970Mo01 for J^π=1/2$^+$.
6284 *4*	(2)	S: (2J+1)C^{2}S=0.086 *2*.
6329 *4*	(0,1)	S: (2J+1)C^{2}S=0.0030 *2* for L=0, 0.0044 *4* for L=1.
6377 *2*	(2,3)	S: (2J+1)C^{2}S=0.0012 *6* for L=2, 0.0024 *8* for L=3.
6427 *2*	(3)	S: (2J+1)C^{2}S=0.049 *2*. $\omega\gamma$(min)=6.5×10^{-24} eV *2*, $\omega\gamma$(max)=2.2×10^{-22} eV *1*.
6468 *2*	(1)	S: (2J+1)C^{2}S=0.034 *1*. $\omega\gamma$(min)=4.2×10^{-14} eV *5*, $\omega\gamma$(max)=4.7×10^{-14} eV *5*.
6491 *2*	(2)	S: (2J+1)C^{2}S=0.072 *2*. $\omega\gamma$(min)=3.7×10^{-13} eV *1*, $\omega\gamma$(max)=4.1×10^{-13} eV *1*.
6545 *2*	(0,1)	S: (2J+1)C^{2}S=0.004 *1* for L=0, 0.0028 *4* for L=1. $\omega\gamma$(min)=7.4×10^{-10} eV *13*, $\omega\gamma$(max)=4.1×10^{-9} eV *1*.
6643 *2*	(1)	S: (2J+1)C^{2}S=0.0144 *8*. $\omega\gamma$(min)=8.5×10^{-6} eV *10*, $\omega\gamma$(max)=8.9×10^{-6} eV *10*.
6674 *2*	(1,2,3)	S: (2J+1)C^{2}S=0.0020 *4* for L=1, 0.0048 *6* for L=2, 0.008 *1* for L=3. $\omega\gamma$(min)=2.4×10^{-8} eV *2*, $\omega\gamma$(max)=4.1×10^{-6} eV *8*.
6761 *2*	(0,1)	S: (2J+1)C^{2}S=0.0056 *12* for L=0, 0.0032 *4* for L=1. $\omega\gamma$(min)=2.4×10^{-4} eV *3*, $\omega\gamma$(max)=1.5×10^{-3} eV *1*.
6778 *2*	(1)	S: (2J+1)C^{2}S=0.0084 *8*. $\omega\gamma$(min)=9.4×10^{-4} eV *12*, $\omega\gamma$(max)=1.0×10^{-3} eV *1*.
6823 *2*	(1)	S: (2J+1)C^{2}S=0.006 *4*. $\omega\gamma$(min)=2.7×10^{-3} eV *3*, $\omega\gamma$(max)=2.7×10^{-3} eV *3*.
6842 *2*	(2,3)	S: (2J+1)C^{2}S=0.0216 *6* for L=2, 0.035 *2* for L=3. $\omega\gamma$(min)=3.3×10^{-5} eV *1*, $\omega\gamma$(max)=7.0×10^{-4} eV *3*.
6866 *2*	(0,2)	S: (2J+1)C^{2}S=0.0160 *2*(L=0)+0.0660 *6* (L=2). $\omega\gamma$(min)=3.0×10^{-2} eV *3*, $\omega\gamma$(max)=4.7×10^{-2} eV *5*.
7066 *2*	(1,2)	S: (2J+1)C^{2}S=0.0088 *8* for L=1, 0.012 *1* for L=2. $\omega\gamma$(min)=2.2×10^{-2} eV *1*, $\omega\gamma$(max)=2.2×10^{-2} eV *1*.
7103 *2*	(1,3)	S: (2J+1)C^{2}S=0.0184 *12* for L=1, 0.025 *2* for L=3. $\omega\gamma$(min)=6.0×10^{-2} eV *3*, $\omega\gamma$(max)=1.30 eV *7*.
7178 *2*	(2)	S: (2J+1)C^{2}S=0.032 *3*.
7194 *2*		
7227 *2*	(0,1)	S: (2J+1)C^{2}S=0.065 *15* for L=0, 0.044 *2* for L=1.
7273 *2*	(0,1)	S: (2J+1)C^{2}S=0.050 *12* for L=0, 0.030 *1* for L=1. $\omega\gamma$(min)=11 eV *2*, $\omega\gamma$(max)=34 eV *3*.
7361 *2*	(1)	S: (2J+1)C^{2}S=0.0040 *8*. $\omega\gamma$(min)=0.30 eV *4*, $\omega\gamma$(max)=3.0 eV *6*.
7398 *2*	(2,3)	S: (2J+1)C^{2}S=0.042 *2* for L=2, 0.0656 *24* for L=3. $\omega\gamma$(min)=0.18 eV *1*, $\omega\gamma$(max)=0.19 eV *1*.

† From 1969Gr23 for E(level)<6 MeV and from 2017Gi09 for E(level)>6 MeV, unless otherwise noted.

Continued on next page (footnotes at end of table)

$^{34}S(^{3}He,d)$ 2017Gi09,1969Gr23 (continued)

^{35}Cl Levels (continued)

‡ From DWBA analysis of the measured $\sigma(\theta)$ in 1969Gr23 for E(level)<6 MeV and in 2017Gi09 for E(level)>6 MeV, unless otherwise noted. Firm assignments by 2017Gi09 are placed in parentheses by the evaluators because they are based on very few data points in a very narrow range of angles in the DWBA fit which may not yield definitive and firm results.

^{35}Cl(γ,γ') 1966Ho02,1962Bo17,1991Li12

1966Ho02: a bremsstrahlung photon beam was produced by impinging 1.4-MeV electrons on a thin layer of lead embedded in a water-cooled aluminium disc at the Diamond Research Laboratory in Johannesburg. The target was C_2Cl_6. Scattered γ rays were detected using a 7.6-cm by 7.6-cm NaI crystal. Deduced the width for the 1.22-MeV first excited state using the self-absorption method of nuclear resonance fluorescence.

1962Bo17: an electron beam was produced from the MIT Van de Graaff generator, passed through a 76-μm aluminum window and a 25-μm platinum foil, and stopped in the cooling water of the electron target. γ rays were detected using a 5.1-cm by 5.1-cm NaI crystal. Lifetimes of ^{35}Cl levels are deduced from the measured intensity of the resonance fluorescence radiation which occurs when a nucleus is bombarded by bremsstrahlung.

1991Li12: a bremsstrahlung photon beam was produced by focusing a 4.1-MeV electron beam onto a radiator target of a gold disc on a water-cooled copper backing at the bremsstrahlung facility at the Stuttgart Dynamitron Accelerator. The target was a stack of discs that contain ^{13}C, ^{23}Na, ^{27}Al, ^{31}P, ^{35}Cl, ^{48}Ti. γ rays were detected using 3 Ge detectors. Deduced σ of photon scattering.

Other: 1972Sh07.

^{35}Cl Levels

$\Gamma_0\Gamma_f/\Gamma$ from 1991Li12. Γ_0, Γ_f, and Γ are the decay width to the ground state, the decay width to the final state, and the total width, respectively.

E(level)†	$T_{1/2}$	Comments
0		
1220	90 fs *+21−14*	$T_{1/2}$: from lifetime=0.13 ps *+3−2* in 1966Ho02 with resonance fluorescence. Other: lifetime=0.30 ps *10* in 1962Bo17 with resonance fluorescence assuming J=1/2.
1760	0.42 ps *15*	$T_{1/2}$: from lifetime=0.60 ps *21* in 1962Bo17 with resonance fluorescence assuming J=5/2.
2693.7		$\Gamma_0\Gamma_f/\Gamma$=12 meV *3* (1991Li12).
3002.7		$\Gamma_0\Gamma_f/\Gamma$=26 meV *11* (1991Li12).
3918.6		$\Gamma_0\Gamma_f/\Gamma$=63 meV *18* (1991Li12).

† From Eγ data.

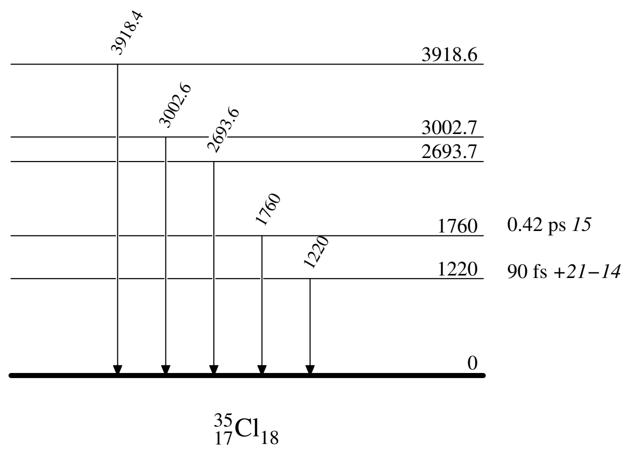
γ(^{35}Cl)

Eγ	E_i(level)	E_f	Comments
1220	1220	0	Eγ: from 1962Bo17 and 1966Ho02.
1760	1760	0	Eγ: from 1962Bo17.
2693.6†	2693.7	0	
3002.6†	3002.7	0	
3918.4†	3918.6	0	

† From 1991Li12, which quoted from 1978En02 evaluation.

^{35}Cl(γ,γ') 1966Ho02,1962Bo17,1991Li12

Level Scheme



^{35}Cl(n,n′γ) 1989Ge09,1984El12,1966Ni03

1984El12,1989Ge09: fast neutrons of diameter 26 mm were produced from the IRT-2000 reactor at the Institute for Nuclear Research and Nuclear Power of the Bulgarian Academy of Sciences in Sofia. Two ^{35}Cl targets were placed at 45° and 135° with a distance of 70 cm apart. γ rays were detected using a Ge(Li) detector. Measured Eγ. Deduced lifetimes for 4 levels using the Doppler Shift Attenuation Method (DSAM) and transition strengths.

1966Ni03: 2.5-4.2-MeV neutrons were produced via the T(p,n) reactions with protons from the university of Kentucky 6-MeV accelerator. Targets were a cylinder of CCl_4 and a 5.6-cm-diam sphere of C_2Cl_6. γ rays were detected using a 6.35-cm by 6.35-cm NaI(Tl) detector and 7.5-cc Ge(Li) detector. Measured Eγ, Iγ, $\gamma(\theta)$. Deduced levels, J.

1971Fr05: neutrons up to 10 MeV were produced from a 5-Ci Pu-Be neutron source and bombarded table salt targets. γ rays were detected using a Ge(Li) detector. Measured 2 γ rays and deduced 2 levels.

Others: 1955To32, 1977Ko19, 1983Gl07, 1992Be60 (cross sections).

^{35}Cl Levels

E(level)†	J$^\pi$#	$T_{1/2}$‡
0	3/2$^+$	
1219	1/2	139 fs *56*
1762	5/2	
2646	7/2	187 fs *62*
2694		43 fs *6*
3002		50 fs *9*
3163		

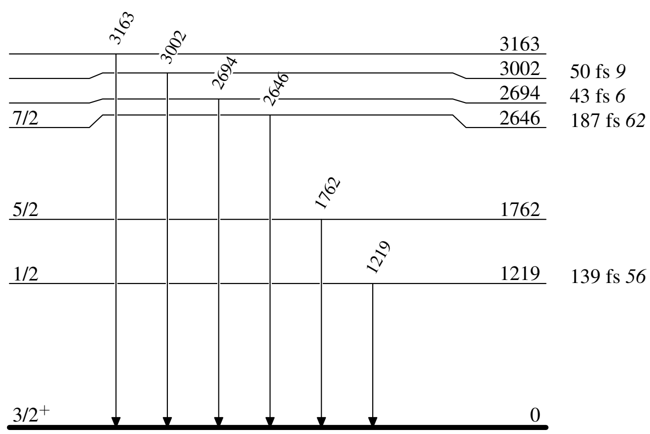
† From a least-squares fit to γ-ray energies.
‡ From 1984El12 and 1989Ge09 using DSAM.
# From $\gamma(\theta)$ in 1966Ni03, except for the ground state from the Adopted Levels.

γ(^{35}Cl)

E$_\gamma$	E$_i$(level)	J$^\pi_i$	E$_f$	J$^\pi_f$	Comments
1219	1219	1/2	0	3/2$^+$	
1762	1762	5/2	0	3/2$^+$	E$_\gamma$: from 1966Ni03. A_0=+0.942 *8*, A_2=−0.148 *23* (1966Ni03).
2646	2646	7/2	0	3/2$^+$	A_0=+1.078 *5*, A_2=+0.165 *12* (1966Ni03).
2694	2694		0	3/2$^+$	
3002	3002		0	3/2$^+$	
3163	3163		0	3/2$^+$	E$_\gamma$: from 1966Ni03.

$^{35}Cl(n,n'\gamma)$ 1989Ge09,1984El12,1966Ni03

Level Scheme



$^{35}_{17}Cl_{18}$

^{35}Cl(p,p′γ) 1969Du08,1970Ta02,1972Br45

Also include (p,p′) measurements listed below.
^{35}Cl(p,p′γ) measurements:
1969Du08: E=4.6-6.2 MeV 0.05 μA proton beam. Thin targets of $CdCl_2$ (99% ^{35}Cl), 100 μg/cm^2. A 300 μm annular surface-barrier detector on the beam axis and a movable 390 μm surface-barrier detector rotated from 20° to 148° relative to beam direction for detecting protons, FWHM=24 keV; a 7.62 by 7.62-cm-diam NaI(Tl) crystal and 15- and 35-cc Ge(Li) for detecting γ-rays. Measured σ(E_p,θ), Eγ, Iγ, pγ(θ). Deduced levels, J^π, branchings, mixing ratios, half-lives using Doppler Shift Attenuation Method (DSAM).
1970Ta02: E=4.24, 4.82, 5.08 and 5.11 MeV proton beams produced from the Universite Montreal EN tandem accelerator. Targets of 16 μg/cm^2 AgCl evaporated onto 0.013 cm gold foils, 99% enriched in ^{35}Cl. A 12.70-cm-diam by 15.24-cm-long NaI(Tl) and a 30 cm^3 Ge(Li) detector for detecting γ-rays. Measured Eγ, Iγ, γ(θ), γ(lin pol). Deduced levels, J^π, branchings, mixing ratios.
1972Br45: E=3.45-6.00 MeV protons produced from the University of Auckland folded tandem accelerator. Natural chlorine targets of AgCl, $PbCl_2$ and solid Cl (75.77% ^{35}Cl, 24.33% ^{37}Cl). A 25 cm^3 Ge(Li) detector for detecting γ-rays, FWHM=2.3 keV for Eγ=1.332 MeV. Measured Eγ. Deduced levels, half-lives using DSAM. 1972Br45 reports a few mixing ratios in their TABLE V, but it is unclear where they are taken from. They are definitely not from 1972Br45 since this work is only focused on lifetime measurements.
1972Va06: E=5.47-5.61 MeV protons from the 5.5 MV Van de Graaff accelerator of the Southern Universities Nuclear Institute. Targets of $BaCl_2$ (99% ^{35}Cl) on gold backings. A 57 cm^3 Ge(Li) detector. Measured Eγ. Deduced half-lives using DSAM.
1961St09: E=2.3-3.25 MeV. NaI(Tl) detector. Targets of AgCl. Measured σ(E_p,θ), γ(θ). Deduced levels, J for the levels of 1.22 and 1.76 MeV.
1966Er06: E=0.733-2.595 MeV protons produced from the Utrecht 3 MV Van de Graaff accelerator. $BaCl_2$ target (99.5% ^{35}Cl) on a tantalum backing. Cylindrical NaI assemblies, 10-cm by 10-cm. Measured Eγ, γ(θ). Deduced levels, J^π, mixing ratios, mainly for ^{36}Ar and for the ^{35}Cl level at E_x=1760.
1970Ta09: E=5.11 MeV proton beam. NaI(Tl) detector. Measured Eγ. Developed an analytical method to interpret the linear polarization measurement of two unresolved γ-rays: 3.00 and 3.16 MeV. Same authors as 1970Ta02.
1971Ca40: E=4.15-6.15 MeV protons from the Oxford University Tandem accelerator. Targets of 20 μg/cm^2 natural $BaCl_2$ on gold, lead, and carbon backings. A 32 cm^3 Ge(Li) detector. Measured Eγ. Deduced half-lives for the levels of 2646 and 2695 keV using DSAM.
1974Hu09: E=5.65 MeV protons from the pulsed beam facility of the University of Alberta Van de Graaff accelerator. Targets of natural KCl on tantalum backing and $PbCl_2$ on gold backing. A 24 cm^3 coaxial Ge(Li) detector. Measured Eγ. Deduced half-life for the level of 3162 keV.
Other (p,p′γ) measurement: 1985Ki07 and 1975De16 (measured 1219γ yield).
(p,p′) measurements:
1969Ta13,1971Ta02: E=3.115 and 3.345 MeV protons produced from the Universite de Montreal Model EN tandem accelerator. AgCl target (99% ^{35}Cl) of 16 μg/cm^2 on gold backing. A 12.70-cm-diam by 15.24-cm-long NaI(Tl) for detecting γ-rays. Measured Eγ, Iγ, γ(θ). Deduced levels, J^π and mixing ratio for the level of 1763 keV. Same authors as 1970Ta02.
1969An35: E=6.03 MeV proton beams on a $CdCl_2$ target. Inelastic protons detected by a Si(Li) detector. Measured σ($E_{p'}$,θ). Deduced J for a level at 3930.
Other (p,p′) measurement: 1993Bo40 (measured σ).

^{35}Cl Levels

E(level)†	J^π‡	$T_{1/2}$	Comments
0	3/2$^+$		
1219.42 *10*	1/2$^+$	0.150 ps *+42−28*	J^π: pγ(θ) in 1969Du08 is consistent with spin=1/2. $T_{1/2}$: from τ=0.216 ps *+60−40*, weighted average of 0.215 ps *75* (1972Va06), 0.21 ps *+6−4* (1972Br45), 0.27 ps *+19−10* (1969Du08).
1763.26 *10*	5/2$^+$	0.43 ps *20*	J^π: 5/2$^+$ from γ(θ) and γ(lin pol) in 1966Er06; spin=5/2 also from pγ(θ) in 1969Du08. $T_{1/2}$: from τ=0.62 ps *28*, unweighted average of 0.205 ps *75* (1972Va06), 0.49 ps *+9−6* (1972Br45), 1.15 ps *+45−63* (1969Du08).
2645.74 *17*	7/2$^+$	0.165 ps *35*	J^π: 5/2$^+$ from γ(θ) and γ(lin pol) in 1970Ta02; spin≥5/2 from pγ(θ) in 1969Du08. $T_{1/2}$: from τ=0.238 ps *50*, weighted average of 0.185 ps *50* (1972Va06) and 0.26 ps *6* (1972Br45), and 0.35 ps *+10−7* (1969Du08, using 2646γ). Other: τ=0.29 ps *+37−13*

Continued on next page (footnotes at end of table)

^{35}Cl(p,p′γ) 1969Du08,1970Ta02,1972Br45 (continued)

^{35}Cl Levels (continued)

E(level)[†]	J^π[‡]	$T_{1/2}$	Comments
			using 882γ also reported in 1969Du08, but is considered less reliable by authors; 0.083 ps *21* from 1971Ca40 is considered an outlier.
2694.50 *28*	3/2+	43 fs *11*	J^π: 3/2+ from γ(θ) and γ(lin pol) in 1970Ta02. $T_{1/2}$: from τ=63 fs *16*, weighted average of 65 fs *20* (1972Va06) and 62 fs *16* (1969Du08, using 2695γ). Other: τ=115 fs *+95−59* using 931γ also reported in 1969Du08, but is considered less reliable by authors; <120 fs (1972Br45); 11 fs *7* from 1971Ca40 is considered an outlier.
3002.6 *5*	5/2+	19 fs *9*	J^π: 5/2+ from γ(θ) and γ(lin pol) in 1970Ta02. $T_{1/2}$: from τ=27 fs *13*, weighted average of 18 fs *+26−15* (1972Va06) and 31 fs *13* (1969Du08). Other: <80 fs (1972Br45).
3162.75 *10*	7/2−	32 ps *4*	J^π: spin=7/2 from γ(θ) and π=− from γ(lin pol) in 1970Ta02. $T_{1/2}$: from τ=46 ps *6* in 1974Hu09. Other: τ>17 ps (1972Va06), >10 ps (1972Br45).
3930 *30*	3/2		E(level),J^π: from 1969An35. Spin is determined based on J-dependence of measured proton inelastic scattering cross sections and known assignments of 3163 and 3000 levels.

[†] From a least-squares fit to γ-ray energies with uncertainties, unless otherwise noted.

[‡] From Adopted Levels. Supporting arguments from this dataset are given under comments if available.

γ(^{35}Cl)

E_i(level)	J_i^π	E_γ[†]	I_γ[‡]	E_f	J_f^π	Mult.[#]	δ[#]	Comments
1219.42	1/2+	1219.4 *1*	100	0	3/2+			E_γ: weighted average of 1219.4 *1* (1972Br45) and 1219.3 *3* (1969Du08). Γ_γ=27.43×10^{-4} eV (1969Du08). A_2=−0.025 *29* for E_p=3.115 MeV; +0.007 *8* for E_p=3.345 MeV (1969Ta13). A_2=+0.031 *16*, A_4=+0.014 *20*; A_2=−0.012 *22*, A_4=−0.003 *30*; A_2=−0.009 *14*, A_4=+0.014 *6*, for different beam energies (1961St09). A_2=+0.02 *6*, A_4=+0.05 *9* (1969Du08).
1763.26	5/2+	1763.2 *1*	100	0	3/2+	M1+E2	−2.66 *12*	E_γ: weighted average of 1763.2 *1* (1972Br45), 1763.3 *3* (1969Du08). δ: weighted average of −3.0 *5* from γ(θ,pol) in 1966Er06 and −2.64 *12* from γ(θ,pol) in 1969Ta13 and 1971Ta02. Others: δ<−0.4 or δ> +0.4 from pγ(θ) in 1969Du08. A_2=−0.369 *9*, A_4=+0.213 *13*; A_2=−0.355 *14*, A_4=+0.200 *20*, for different beam energies (1969Ta13). A_2=−0.02 *2*, A_4=+0.07 *2* (1969Du08). A_2=−0.141 *13*, A_4=+0.107 *16*; A_2=−0.225 *29*, A_4=+0.083 *35*; A_2=−0.049 *12*, A_4=−0.036 *14*, for different beam energies (1961St09).
2645.74	7/2+	882.3[@] *3*	13 *5*	1763.26	5/2+	M1+E2	+0.17 *5*	I_γ: unweighted average of 8.7 *22* from 1970Ta02 and 17.6 *35* from 1969Du08. %Branching=8 *2* (1970Ta02), 15 *3* (1969Du08). A_2=−0.372 *34*, A_4=+0.034 *38*; A_2=−0.533 *31*, A_4=+0.041 *34*, for different beam energies (1970Ta02).
		1426[&a]	<1[&]	1219.42	1/2+			%Branching<1 (1970Ta02).
		2645.7 *2*	100 *2*	0	3/2+	E2		E_γ: weighted average of 2645.7 *2* (1972Br45) and 2645.6 *5* (1969Du08).

Continued on next page (footnotes at end of table)

^{35}Cl(p,p′γ) 1969Du08,1970Ta02,1972Br45 (continued)

γ(^{35}Cl) (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[‡]	E_f	J^π_f	Mult.[#]	δ[#]	Comments
								δ: δ(O/Q)=0 from γ(θ,pol) in 1970Ta02 and pγ(θ) in 1969Du08. I_γ: from 1970Ta02. Other: 100 *4* from 1969Du08. %Branching=92 *2* (1970Ta02), 85 *3* (1969Du08). A_2=+0.22 *5*, A_4=−0.13 *6* (1969Du08). A_2=+0.316 *22*, A_4=+0.024 *28*; A_2=+0.233 *11*, A_4=+0.044 *10*; A_2=+0.312 *32*, A_4=−0.117 *41*; A_2=+0.205 *18*, A_4=−0.044 *25*, for different beam energies (1970Ta02).
2694.50	3/2$^+$	931.3[@] *3*	18.6 *25*	1763.26	5/2$^+$	M1+E2	+0.09 *3*	I_γ: weighted average of 19.0 *25* (1970Ta02) and 17.7 *38* (1969Du08). %Branching=15 *2* (1970Ta02), 14 *3* (1969Du08). A_2=+0.018 *46* (1970Ta02).
		1475[@] *2*	8.3 *25*	1219.42	1/2$^+$	M1+E2	−0.63 *21*	I_γ: weighted average of 8.0 *25* (1970Ta02) and 8.9 *38* (1969Du08). %Branching=6 *2* (1970Ta02), 7 *3* (1969Du08). A_2=−0.111 *45* (1970Ta02).
		2694.1 *6*	100 *3*	0	3/2$^+$	M1+E2	+0.26 *2*	E_γ: unweighted average of 2693.5 *1* (1972Br45), 2694.6 *5* (1969Du08). I_γ: from 1970Ta02. Other: 100 *4* from 1969Du08. %Branching=79 *2* (1970Ta02), 79 *3* (1969Du08).
3002.6	5/2$^+$	1239[&]	2.2[&] *11*	1763.26	5/2$^+$			I_γ: from 1970Ta02. %Branching=2 *1* (1970Ta02).
		1783	3.3 *11*	1219.42	1/2$^+$			E_γ: from level-energy difference. I_γ: weighted average of 5.4 *32* (1970Ta02) and 3.1 *11* (1969Du08). %Branching=5 *3* (1970Ta02), 3 *1* (1969Du08).
		3002.5 *5*	100 *1*	0	3/2$^+$	M1+E2	+0.07 *2*	E_γ: weighted average of 3002.5 *5* (1972Br45) and 3002.4 *10* (1969Du08). I_γ: from 1969Du08. Other: 100 *2* from 1970Ta02. δ: others: −20<δ<−0.3 or +0.2<δ<+5 (1969Du08, pγ(θ)); −0.02<δ<0.09 (1970Ta09, γ(lin pol)). %Branching=93 *2* (1970Ta02), 97 *1* (1969Du08). A_2=−0.30 *4*(1969Du08). A_2=−0.166 *7*, A_4=+0.004 *10*; A_2=−0.104 *23*, A_4=+0.050 *40*; A_2=−0.329 *13*, A_4=−0.001 *28*; A_2=+0.342 *24*, A_4=+0.022 *43*, for different beam energies (1970Ta02).
3162.75	7/2$^-$	517.0[@] *8*	19 *8*	2645.74	7/2$^+$			I_γ: unweighted average of 11.1 *22* (1970Ta02) and 26.6 *63* (1969Du08). %Branching=10 *2* (1970Ta02), 21 *5* (1969Du08).
		1399[&a]	<1[&]	1763.26	5/2$^+$			%Branching<1 (1970Ta02).
		1943[&a]	<1[&]	1219.42	1/2$^+$			%Branching<1 (1970Ta02).

Continued on next page (footnotes at end of table)

^{35}Cl(p,p′γ) 1969Du08,1970Ta02,1972Br45 (continued)

γ(^{35}Cl) (continued)

E_i(level)	E_γ[†]	I_γ[‡]	E_f	J^π_f	Mult.[#]	δ[#]	Comments
3162.75	3162.6 *1*	100 *2*	0	3/2$^+$	M2+E3	−0.22 *5*	E_γ: weighed average of 3162.6 *1* (1972Br45) and 3162.5 *10* (1969Du08). I_γ: from 1970Ta02. Other: 100 *6* from 1969Du08. δ: weighted average of −0.14 *+8−6* (1969Du08, pγ(θ)) and −0.25 *4* (1970Ta02, γ(θ,pol)). Other: −0.31<δ<−0.21 from γ(lin pol) in 1970Ta09. %Branching=90 *2* (1970Ta02), 79 *5* (1969Du08). A_2=+0.48 *21*, A_4=−0.13 *73* (1969Du08). A_2=+0.281 *68*, A_4=+0.241 *82*; A_2=+0.453 *44*, A_4=+0.136 *63*; A_2=+0.475 *24*, A_4=+0.096 *36*, for different beam energies (1970Ta02).

† Eγ values with uncertainties for the ground-state transitions are not explicitly listed in 1972Br45 and 1969Du08 and the evaluators have taken their listed E(level) values and uncertainties (which are deduced from Eγ values by authors) as Eγ values of the corresponding ground-state transitions, considering ground-state transitions are dominant in each level.

‡ Relative photon branching ratios deduced by the evaluators based on percentage branching ratios reported in 1970Ta02 and 1969Du08, which are given under comments where available.

From γ(θ) and γ(lin pol) in 1970Ta02, unless otherwise noted.

@ From 1969Du08.

& γ reported in 1970Ta02. Energy is from level-energy difference and intensity is from 1970Ta02. The ones with Iγ reported as upper limit are considered questionable by the evaluators.

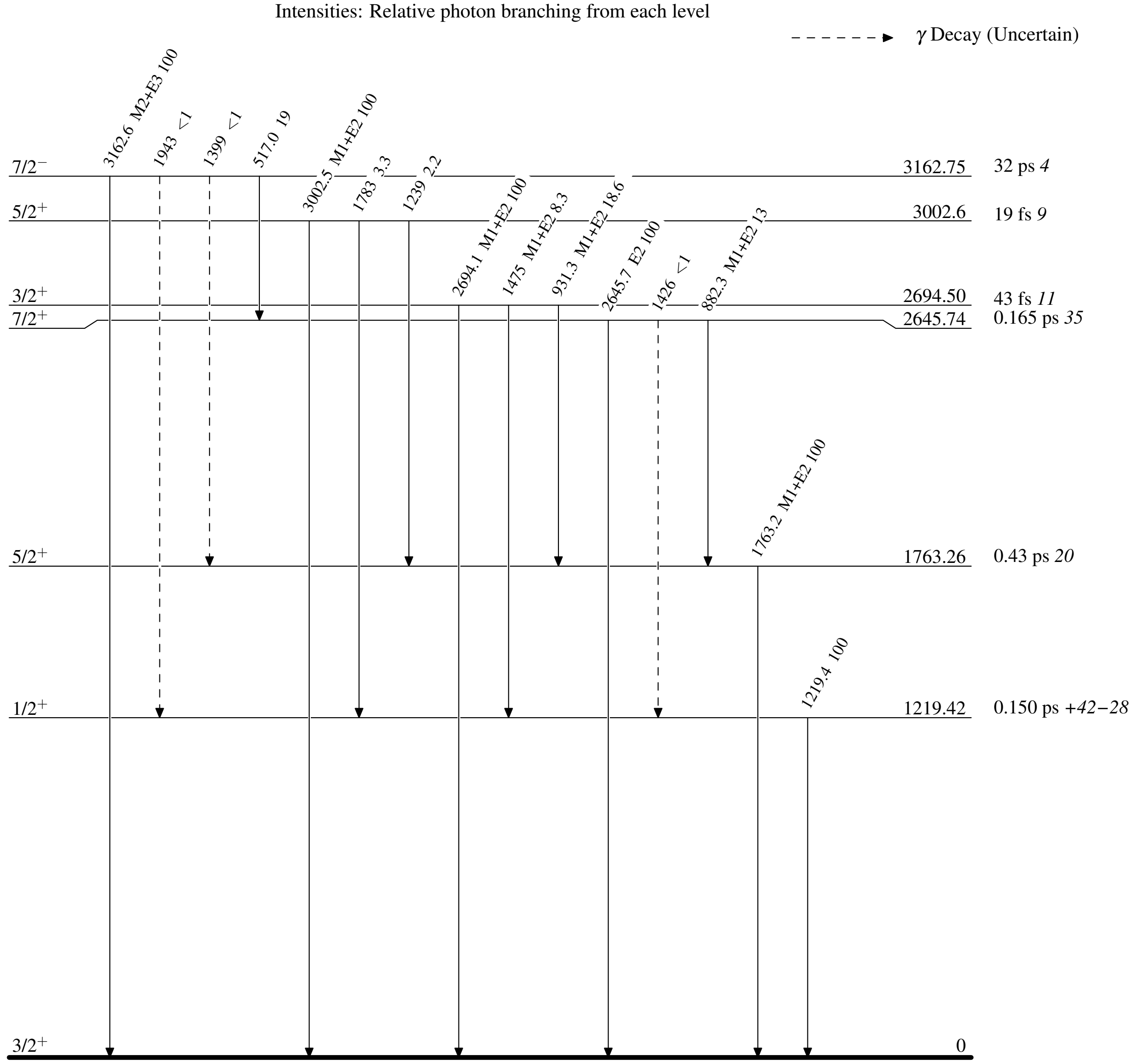
a Placement of transition in the level scheme is uncertain.

^{35}Cl(p,p′γ) 1969Du08,1970Ta02,1972Br45

Legend

Level Scheme

Intensities: Relative photon branching from each level



$^{35}_{17}Cl_{18}$

^{37}Cl(p,t) 1971Vi02,2017Ch22

$^{37}Cl \rightarrow ^{35}Cl$+2n from $J^{\pi}=3/2^{+}$ ^{37}Cl ground state.

1971Vi02: A 40-MeV proton beam was produced from the Grenoble variable energy cyclotron at intensities of 20-200 nA depending on the scattering angle with a 90-keV energy resolution. Targets were natural purified chlorine gas 100 mm in diameter and 25 mm in height and solid AgCl of isotopic ^{37}Cl. The reaction particles were detected using two separate counter telescopes consisting of a 200-μm phosphorous-drifted silicon ΔE detector, a 3-mm lithium-drifted silicon E detector, and a 3-mm lithium-drifted silicon E-reject detector with FWHM=130 keV for tritons. Measured $\sigma(E_t,\theta)$. Deduced L-transfers for 7 levels from zero-range Nelson spin-dependent-DWBA analysis of the measured $\sigma(\theta)$.

2017Ch22: a 30-MeV proton beam was produced from the Holifield Radioactive Ion Beam Facility at Oak Ridge National Laboratory. The target was ~600 μg $PbCl_2$ (~95% enriched and ~160 $\mu g/cm^2$ in ^{37}Cl) on a 20 $\mu g/cm^2$ parylene (C_8H_8) backing. Tritons were detected using the annular silicon detector array (SIDAR) consisting of 100-μm-ΔE and 1000-μm-E telescopes with FWHM=60-100 keV. Measured $\sigma(E_t,\theta)$. Deduced levels, J, π, and L-transfers from TWOFNR-DWBA analysis of the measured $\sigma(\theta)$. Provided the first spin-parity constraint of a 306-keV $^{34}S(p,\gamma)$ resonance potentially important for nova nucleosynthesis.

Other: 1985Mi06.

^{35}Cl Levels

E(level)†	L‡	Comments
0	0	L: from 2017Ch22. Other: 0+2 in 1971Vi02. Relative peak intensity=100 (1971Vi02).
1220 *40*		Relative peak intensity=0 (1971Vi02).
1750 *40*	2	Relative peak intensity=41 (1971Vi02).
2650 *40*	2	Relative peak intensity=41 (1971Vi02). Unresolved 2645 and 2694 doublet.
3000 *40*		Relative peak intensity=5 (1971Vi02).
5650 *50*	0+2	Relative peak intensity=22 (1971Vi02).
6677 *15*	(2)	E(level): from 2017Ch22. 2017Ch22 adopts L=2 and states that combinations of L=0+2 or L=1+3 are also possible. Because L=2 and L=0 are better fits than L=1 and L=3, a positive parity assignment for the 6677 level is strong.
7250 *50*		Relative peak intensity=9 (1971Vi02).

† From 1971Vi02, unless otherwise noted.

‡ From DWBA analysis of the measured $\sigma(\theta)$ in 1971Vi02, unless otherwise noted.

^{36}Ar(n,d) 1977Pa29

$J^{\pi}=0^{+}$ for ^{36}Ar ground state.

1977Pa29: 14.1-MeV neutrons were produced via the ^{3}H(d,n)^{4}He reaction at an average rate of 10^{9} neutrons per second into 4π sr from the Brown University neutron generator and impinged on a 2.85-mg/cm^{2} ^{36}Ar target in the form of a gas cell filled with 99.8% enriched ^{36}Ar. Deuterons were detected using a time-of-flight counter telescope consisting of two proportional counters and a NaI(Tl) crystal. Measured $\sigma(E_d,\theta)$. Deduced levels, L-transfers, spectroscopic factors from local zero-range DWUCK-DWBA analysis of the measured $\sigma(\theta)$.

^{35}Cl Levels

E(level)†	L‡	Comments
0	2	S: 3.83 *48* and 3.90 *47* using two sets of optical parameters.
1219#	0#	S: 1.94 *23* and 3.30 *40* using two sets of optical parameters. Other spectroscopic factors calculated using two sets of optical parameters did not show large differences.
1763#	(2)#	

† From 1977Pa29.
‡ From DWBA analysis of the measured $\sigma(\theta)$ in 1977Pa29.
# 1977Pa29 state that the first and second excited states are recorded together and not resolved, but their angular distributions could be separated due to different L-transfers, however statistical errors prevent an accurate determination of the smaller L=2 component from the second excited state.

^{36}Ar(pol d,^{3}He) 1993Ma50

$J^{\pi}=0^{+}$ for ^{36}Ar ground state.

1993Ma50: a 52-MeV polarized deuteron beam at 15 nA was produced from the Karlsruhe isochronous cyclotron at the Max-Plank-Institut fur Kernphysik. The target was 350-mbar 99.86% enriched ^{36}Ar gas. Reaction products were detected using a Si detector telescope consisting of a 250-μm ΔE-strip detector and a 1.5-mm surface-barrier E detector with FWHM=130 keV. Measured angular distributions of cross sections $\sigma(E(^{3}He),\theta)$ and angular distributions of analyzing powers $iT_{11}(E(^{3}He),\theta)$. Deduced levels, J, π, L-transfers, and spectroscopic factors from DWBA analysis of $\sigma(E(^{3}He),\theta)$ and $iT_{11}(E(^{3}He),\theta)$.

1993Ma50 states that spectroscopic factors were obtained by fitting the forward-angle maxima. The agreement with the analyzing powers is only qualitative. The determination of spins and parities of final states is mainly based on a comparison with measured angular distributions leading to final states with known spins. The comparison with DWBA predictions serves as additional check. The data were taken only in a relatively small angle interval. This is sufficient for the spin determination but not favorable for the measurement of spectroscopic factors.

^{35}Cl Levels

Spectroscopic factor $C^2S=\sigma(\theta)_{exp}/\sigma(\theta)_{DWBA}/N$, where N=2.95 is a normalization factor used by 1993Ma50 from 1966Ba54.

E(level)[†]	J^{π}[‡]	L[#]	C^2S[#]	Comments
0	$3/2^{+}$[@]	2	1.72[@]	
1213 *4*	$1/2^{+}$	0	0.78	
1785 *50*	$(5/2^{+})$[&]		0.04[a]	
2676 *7*	$(3/2^{+})$[&]		0.16[a]	
3002 *6*	$5/2^{+}$	2	1.32	
3159 *20*	$7/2^{-}$	3	0.19	
3948 *40*	$(3/2^{+},1/2^{+})$[&]	2	0.03,0.08	E(level): possible doublet of 3918 and 3968 (1990En08) suggested by Table 2 of 1993Ma50.
4839 *12*	$(5/2^{+},1/2^{+})$[&]		0.08,0.03[a]	
5155 *11*	$(5/2^{+})$	2	0.12	J^{π}: $(7/2^{-})$ and $5/2^{+}$ are given in Table 2 of 1993Ma50. Evaluators adopt $(5/2^{+})$, based on DWBA fit of the measured $\sigma(\theta)$ and $iT_{11}(\theta)$ in Fig. 3 of 1993Ma50.
5583 *13*	$5/2^{+}$	2	0.38	
5720 *11*	$5/2^{+}$	2	1.10	
6136 *8*	$5/2^{+}$	2	0.63	
6745 *12*	$(1/2^{+},5/2^{+})$[&]	(0,2)	0.09,0.28	
6953 *43*	$5/2^{+}$	2	0.49	
7181 *60*	$5/2^{+}$	2	0.20	
7565 *20*	$5/2^{+}$	2	0.30	
8007 *36*	$(1/2^{-},5/2^{+})$[&]	(1,2)	0.16,0.20	
8204 *34*	$5/2^{+}$	2	0.14	
8580 *20*	$5/2^{+}$[&]		0.22[a]	
8921 *70*	$5/2^{+}$[&]		0.18[a]	
9.22×10^{3} *12*	$5/2^{+}$[&]		0.12[a]	
9514 *20*	$(1/2^{-},5/2^{+})$[&]	(1,2)	0.18,0.30	
9.82×10^{3} *12*	$(1/2^{-},5/2^{+})$[&]	(1,2)	0.10,0.26	
1.25×10^{4} *5*				$1d_{5/2}$ is given for a range of 12.0-13.0 MeV in Fig. 4 of 1993Ma50.

[†] From 1993Ma50.
[‡] L+1/2 transfer from analyzing power measurements with $1d_{5/2}$ proton transfer assumed in DWBA calculations, unless otherwise noted.
[#] From DWBA analysis of the measured $\sigma(\theta)$ and $iT_{11}(\theta)$ in 1993Ma50. C^2S value is extracted for corresponding J^{π} assignment. Where two C^2S values are given, they correspond to respective J^{π} values.

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^{36}Ar(pol d,^{3}He) 1993Ma50 (continued)

^{35}Cl Levels (continued)

[@] L-1/2 transfer from analyzing power measurements with $1d_{3/2}$ proton transfer assumed in DWBA calculations.
[&] Assumed by 1993Ma50 for extracting spectroscopic factors.
[a] No data on $\sigma(E(^3He),\theta)$, $iT_{11}(E(^3He),\theta)$, or L-transfer is given.

^{36}Ar(d,^{3}He) **1974Do12**

$J^{\pi}=0^{+}$ for ^{36}Ar ground state.

1974Do12: a 52-MeV deuteron beam was produced from the Karlsruhe isochronous cyclotron at the Max-Plank-Institut fur Kernphysik. The target was 200-Torr 99.9% enriched ^{36}Ar gas. ^{3}He particles were detected using detector telescopes consisting of a 200-μm ΔE and a 2-mm E-surface-barrier counter with FWHM=125 keV. Measured $\sigma(E(^{3}He),\theta)$. Deduced levels, J, π, L-transfer, and spectroscopic factors from DWUCK-DWBA analysis.

Other: 1975Wa17 (theory discussions).

^{35}Cl Levels

Spectroscopic factor $C^2S=\sigma(\theta)_{exp}/\sigma(\theta)_{DWBA}/N$, where N=2.95 is a normalization factor used by 1974Do12 from 1966Ba54.

E(level)†	Jπ#	L‡	C²S‡	E(level)†	Jπ#	L‡	C²S‡	E(level)†	Jπ#	L‡	C²S‡
0	3/2+	2	2.2	3170 *20*	7/2−	3	0.22	6750 *20*	(5/2)+	2	0.79
1220 *20*	1/2+	0	1.34	5150 *20*	(5/2)+	2	0.09	6930 *20*	(5/2)+	2	0.46
1763			<0.02	5600 *20*	(5/2)+	2	0.47 *19*	8010 *20*	(5/2)+	2	0.23
2700 *20*	3/2+	2	0.39	5720 *20*	(5/2)+	2	0.73 *29*	8210 *20*	(5/2)+	2	0.28
3000 *20*	5/2+	2	1.49	6140 *20*	(5/2)+	2	0.72	8610 *20*	(5/2)+	2	0.34

† From 1974Do12.

‡ From DWBA analysis of the measured $\sigma(\theta)$ in 1974Do12.

As given in 1974Do12 for levels below 5 MeV. For levels above 5 MeV, 1974Do12 states that they tend to interpret them as $5/2^{+}$ states because the summed low-lying $C^2S(1d_{3/2})$ already exceeds the simple shell model value for ^{36}Ar and the measured $\sigma(\theta)$ resembles more closely that of the $5/2^{+}$ state at 3.0 MeV than that of the $3/2^{+}$ ground state. All C^2S values are given according to the J values.

^{39}K(γ,α) 1987Ar08

1987Ar08: E_{max}=32−MeV γ-ray beam was produced from the betatron at Moscow University Nuclear Physics Research Institute and impinged on a natural metallic potassium target of mass ≈200 g. Photons from (γ,p), (γ,n), and (γ,α) from the giant dipole resonance (GDR) in ^{39}K were detected using a 100-cm^3 Ge(Li) detector at 140°. Measured Eγ. Deduced partial photonuclear cross sections, levels, J, π.

^{35}Cl Levels

E(level)†	Jπ‡	Comments
0	3/2+	
1760	5/2+	T=1/2

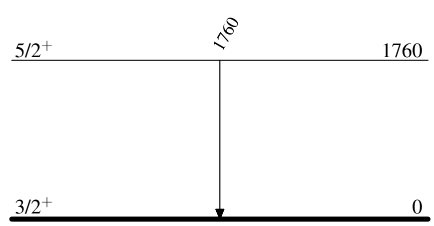
† From 1987Ar08.
‡ From Adopted Levels.

γ(^{35}Cl)

Eγ	E_i(level)	J^π_i	E_f	J^π_f
1760	1760	5/2+	0	3/2+

^{39}K(γ,α) 1987Ar08

Level Scheme



$^{35}_{17}Cl_{18}$

^{40}Ca(μ^-,$\nu\alpha$nγ) 2006Me08

2006Me08: Muons were generated from the decay of 90-MeV/c π^- beam in a 6-m, 1.2-T superconducting solenoid and were selected by a bending magnet at the M9B beamline at TRIUMF. The target was 15-g natural calcium. γ rays and x rays were detected using a 186-cm^3 HPGe detector. Charged particles caused by electrons from muon decay were tagged by a plastic scintillator in front of the HPGe detector. Measured Eγ, Iγ, E(x ray), I(x ray). Deduced levels, muon capture yields.

Muonic Lyman series for natural Calcium

μ x ray	Energy	Intensity in percent
2p-1s	783.659 *25*	83.8 *10*
3p-1s	940.63 *10*	6.2 *2*
4p-1s	995.48 *10*	2.0 *1*
5p-1s	1020.81 *10*	2.0 *1*
6p-1s	1034.62 *10*	1.8 *1*
7p-1s	1042.71 *20*	1.4 *1*
(8$^-\infty$)p-1s	1046-1063	2.8 *4*

Muonic Balmer series for natural Calcium

μ x ray	Energy	Intensity in percent
3d-2p	157.35 *13*	64.5 *9*
4d-2p	212.03 *10*	8.85 *20*
5d-2p	237.31 *10*	4.34 *20*
6d-2p	251.06 *10*	3.29 *20*
7d-2p	259.45 *10*	1.37 *20*
(8$^-\infty$)d-2p	261-277	1.4 *3*

^{35}Cl Levels

E(level)†	J$^\pi$†
0	3/2$^+$
1219.4	1/2$^+$
1763.2	5/2$^+$
2645.7	7/2$^+$
2694.0	3/2$^+$

† From Adopted Levels. Level energies are rounded-off values.

γ(^{35}Cl)

E$_\gamma$†	I$_\gamma$‡	E$_i$(level)	J$^\pi_i$	E$_f$	J$^\pi_f$
1219.4	<0.3	1219.4	1/2$^+$	0	3/2$^+$
1763.2	0.5 *3*	1763.2	5/2$^+$	0	3/2$^+$
2645.7	<0.2	2645.7	7/2$^+$	0	3/2$^+$
2693.7	<0.2	2694.0	3/2$^+$	0	3/2$^+$

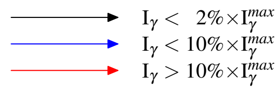
† Rounded-off values from Adopted Gammas.
‡ Percent γ-ray yield per muon capture (2006Me08).

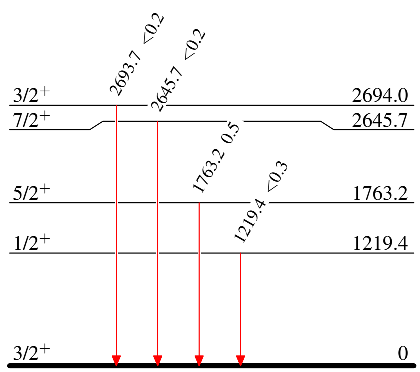
^{40}Ca(μ^-,$\nu\alpha$nγ) 2006Me08

Level Scheme

Intensities: percent γ-ray yield per muon capture

Legend





$^{35}_{17}Cl_{18}$

Coulomb excitation 1969Ha17,1977Sc36

1969Ha17: 51-61-MeV $^{35}Cl^{5,6+}$ beams of 80 nA were produced from the Chalk River MP tandem accelerator and bombarded 9.6-mg/cm^2 carbon and 5-mg/cm^2 ^{24}Mg. Deexcitation γ rays were detected using two 40-cm^3 Ge(Li) detectors at 125° and at 0° or 90°. Measured Eγ, Iγ, $\gamma(\theta)$ anisotropy, γ-ray yields. Deduced levels, mixing ratio, transition strengths, $T_{1/2}$ for 2 levels using the Doppler Shift Attenuation Method (DSAM).

1977Sc36: 53-56-MeV $^{35}Cl^{6+}$ beams of 60 nA were produced from a single MP tandem Van de Graaff accelerator at the three-stage facility of Brookhaven National Laboratory. Targets were ^{23}Na and ^{27}Al. Deexcitation γ rays were detected using a 55-cm^3 coaxial Ge(Li) detector at 0°. Measured Eγ, I$\gamma(\theta)$, γ-ray yields. Deduced levels, mixing ratios, transition strengths, $T_{1/2}$ for the levels of 1220, 1762 and 2650 keV using the Doppler Shift Attenuation Method (DSAM).

^{35}Cl Levels

Values of B(E2)↑ from 1969Ha17 and 1977Sc36 are from absolute cross-section measurements.

E(level)†	Jπ†	$T_{1/2}$	Comments
0	3/2+		
1219	1/2+	0.150 ps *20*	B(E2)↑=0.00076 *7* $T_{1/2}$: Lifetime τ=0.216 ps *28* from weighted average of τ=0.16 ps *5* (1969Ha17, DSAM) and τ=0.230 ps *25* (1977Sc36, DSAM). B(E2)↑: weighted average of 0.00081 *9* (1969Ha17) and 0.00073 *7* (1977Sc36).
1763	5/2+	0.42 ps *4*	B(E2)↑=0.0106 *7* $T_{1/2}$: Lifetime τ=0.61 ps *5* from weighted average of τ=0.57 ps *7* (1969Ha17, DSAM) and τ=0.63 ps *5* (1977Sc36, DSAM). B(E2)↑: weighted average of 0.0117 *9* (1969Ha17) and 0.0102 *5* (1977Sc36).
2646	7/2+	0.18 ps *6*	B(E2)↑=0.0029 *7* (1977Sc36) $T_{1/2}$: Lifetime τ=0.26 ps *9* (1977Sc36, DSAM).

† From the Adopted Levels. Energies are rounded values.

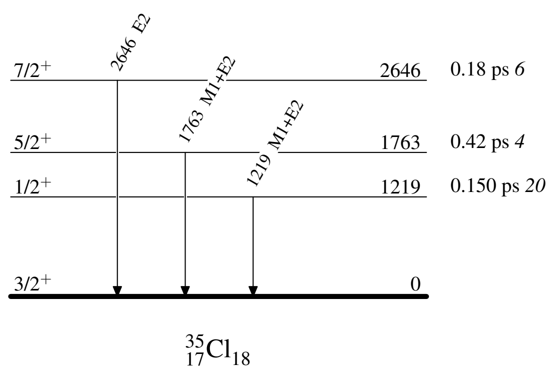
$\gamma(^{35}Cl)$

Eγ†	E_i(level)	J^π_i	E_f	J^π_f	Mult.†	δ	Comments
1219	1219	1/2+	0	3/2+	M1+E2	0.093 *10*	δ: deduced by the evaluators from adopted B(E2)↑=0.00076 *7* in this dataset and $T_{1/2}$=0.119 ps *14* from Adopted Levels. Other: 0.106 *7* deduced by 1977Sc36 based on their measured B(E2)↑ and lifetime.
1763	1763	5/2+	0	3/2+	M1+E2	−2.95 *45*	δ: from $\gamma(\theta)$ anisotropy in 1969Ha17. Other: 1.8 *+10−4*; deduced by the evaluators from adopted B(E2)↑=0.0106 *7* in this dataset and $T_{1/2}$=0.36 ps *3* from Adopted Levels; −2.7 *7* from 1977Sc36 deduced from their measured B(E2)↑ and lifetime.
2646	2646	7/2+	0	3/2+	E2		

† From Adopted Gammas. Energies are rounded values.

Coulomb excitation 1969Ha17,1977Sc36

Level Scheme



Adopted Levels, Gammas

Q(β^-)=−11874.4 *9*; S(n)=12740.3 *7*; S(p)=5896.2 *7*; Q(α)=−6429.7 *7* 2021Wa16

Q(β^-n)=−29632 *17*, from mass excesses of −1487 *17* for ^{34}K measured by 2024Dr01; −23047.3 *7* for ^{35}Ar from 2021Wa16. Value from 2021Wa16: Q(β^-n)=−29900 *200* (syst).

S(2n)=29805.6 *8*, S(2p)=11039.4 *7*, Q(ε)=5966.2 *7* (2021Wa16).

Isotope discovery (2012Th10): ^{32}S(α,n)^{35}Ar at Purdue (1940Ki12,1941Ki01,1941El04).

^{35}Ar production:

2012Zh06: ^{9}Be,^{181}Ta(^{40}Ar,X) at E(^{40}Ar)=57 MeV/nucleon at HIRFL. Measured momentum distributions and production cross sections of fragments. Observed competition between projectile fragmentation and other mechanisms. Compared with EPAX, abrasion-ablation, and HIPSE models. Studied target dependence of fragment cross sections.

2007No13: ^{181}Ta(^{40}Ar,X) at E(^{40}Ar)=100 MeV/nucleon at RIKEN. Measured fragment momentum distributions and production cross sections.

^{35}Ar decay measurements:

^{35}Ar produced via ^{35}Cl(p,n)^{35}Ar reaction (1985Da04,1984Ad01,1980Wi13,1979Ha04,1977Az01,1971Ge04, 1971De05,1969Wi18,1969Fr09,1960Wa04,1956Ki29). Measured ε+β^+-delayed γ rays, decay branching ratios, and parent ^{35}Ar $T_{1/2}$.

^{35}Ar radius measurements:

2002Oz03: C(^{35}Ar,X) at E(^{35}Ar)≈950 MeV/nucleon at RIKEN. Measured interaction cross sections. Deduced effective radii and proton skin features.

2000Ge20: ^{35}Ar produced at ISOLDE. Measured β asymmetry and hyperfine structure using β-NMR spectroscopy. Deduced mean squared charge radii and quadrupole moments.

1996Kl04,1995KlZZ: ^{35}Ar produced by ISOLDE. Measured isotope shifts and hyperfine structure using collinear fast-beam laser spectroscopy. Deduced mean square charge radii and electric quadrupole moments.

^{35}Ar mass measurement: 2011Tu09.

Theoretical calculations: 2020Ri06, 2020RiZX, 2020RiZZ.

^{35}Ar Levels

Cross Reference (XREF) Flags

A	^{35}K ε+β^+ decay (175 ms)	E	^{24}Mg(^{16}O,αnγ)	I	^{36}Ar(p,d)
B	^{36}Ca εp decay (100.9 ms)	F	^{32}S(α,n)	J	^{36}Ar(d,t)
C	^{1}H(^{36}Ar,d)	G	^{33}S(^{3}He,nγ)	K	^{36}Ar(^{3}He,α)
D	^{16}O(^{24}Mg,αnγ)	H	^{35}Cl(^{3}He,t)		

E(level)[†]	J^π	$T_{1/2}$	XREF	Comments
0.0	3/2$^+$	1.7755 s *14*	ABCDEFG IJK	%ε+%β^+=100 μ=+0.6322 *2* (2002Ma41,2019StZV) Q=−0.084 *15* (1996Kl04,2021StZZ) μ: β−NMR (2002Ma41). Others: +0.632 *2* (1965Ca04), +0.633 *7* (1996Kl04) using β-NMR. Q: β-NMR (1996Kl04). J^π: L=2 from 0$^+$ in (p,d), (d,t), (^{3}He,α), and ^{1}H(^{36}Ar,d). Allowed ε+β^+ feedings to 1/2$^+$ levels in ^{35}Cl. $T_{1/2}$: weighted average of 1.83 s *3* (1956Ki29), 1.83 s *2* (1959Al10), 1.79 s *1* (1960Ja12), 1.84 s *10* (1960Wa04), 1.76 s *3* (1963Ne05), 1.770 s *6* (1969Wi18), 1.787 s *12* (1971Ge04), 1.774 s *3* (1977Az01), and 1.7754 s *11* (2006Ia05). Evaluated rms nuclear charge radius R=3.3636 fm *42* (2013An02).
1184.08 *25*	1/2$^+$		ABC FG IJK	XREF: F(890) E(level): 1963Ne05 (α,n) observed the first excited state in ^{35}Ar at 890 *50* keV. J^π: L=0 from 0$^+$ in (p,d), (d,t), and (^{3}He,α).
1750.78 *22*	(5/2)$^+$		A DEFG IJK	XREF: F(2030)I(1700)J(1700)

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Adopted Levels, Gammas (continued)

^{35}Ar Levels (continued)

E(level)†	J^π	XREF	Comments
			E(level): 1963Ne05 (α,n) observed the second excited state in ^{35}Ar at 2030 *80* keV. J^π: L=2 from 0^+ in (^{3}He,α). Possible mirror level of $5/2^+$ at 1763 keV in ^{35}Cl.
2603.22 *28*	$7/2^{(+)}$	DE G	J^π: 2603.0γ Q, ΔJ=2 to $3/2^+$.
2638.01 *26*	$(3/2)^+$	A IJK	XREF: I(2615)K(2649) J^π: L=2 from 0^+ in (p,d) and (^{3}He,α) and spin=(3/2) from J-dependence in (p,d).
2982.79 *12*	$(5/2)^+$	A C IJK	XREF: I(2970) J^π: L=2 from 0^+ in (p,d), (d,t), and (^{3}He,α) and spin=(5/2) from J-dependence in (p,d).
3196.98‡ *26*	$7/2^-$	CDE G IJK	J^π: L=3 from 0^+ in (p,d) and (^{3}He,α); 3197.0γ Q, ΔJ=2 to $3/2^+$.
3882 *5*	$1/2^+$	K	J^π: L=0 from 0^+ in (^{3}He,α).
4001 *3*	$1/2^-,3/2^-$	K	J^π: L=1 from 0^+ in (^{3}He,α).
4065.0? *4*	$(1/2^+,3/2^+,5/2^+)$	A	XREF: A(?) J^π: possibly allowed $\varepsilon+\beta^+$ feeding from $3/2^+$ parent with log *ft*=5.6 *+4−2*.
4113 *4*		K	
4135 *4*	$1/2^-,3/2^-$	K	J^π: L=1 from 0^+ in (^{3}He,α).
4359.0 *5*	$(9/2^-)$	DE K	J^π: 1162.0γ (D), ΔJ=(1) to $7/2^-$; mirror level to 4348, $9/2^-$ level in ^{35}Cl proposed by 2007De14 in ^{24}Mg(^{16}O,αnγ).
4528.3 *4*	$(1/2^+,3/2^+,5/2^+)$	A K	XREF: K(4515) J^π: possibly allowed $\varepsilon+\beta^+$ feeding from $3/2^+$ parent with log *ft*=5.4 *+4−2*.
4725.9 *6*	$1/2^+$	A Hi K	XREF: i(4756)K(4713) J^π: L=0 from 0^+ in (^{3}He,α). Other: L=0 from 0^+ in (p,d) for a group at 4756 *28*.
4785.8 *11*	$1/2^+,3/2^+,5/2^+$	A Hi K	XREF: i(4756)K(4774) J^π: allowed $\varepsilon+\beta^+$ feeding from $3/2^+$ parent with log *ft*=5.2 *2*. Other: L=0 from 0^+ in (p,d) for a group at 4756 *28*.
5059 *11*		i K	
5116 *2*	$3/2^+,5/2^+$	Hi K	E(level): from (^{3}He,α). Other: 5102 *20* from (p,d). J^π: L=2 from 0^+ in (^{3}He,α) in 1973Be26. Discrepancy: L=3 from 0^+ in (p,d) (1968Jo04). 1973Be26 also considered L=3 but found L=2 gives a much better fit of the data than L=3 does.
5207 *3*		H K	E(level): from (^{3}He,α).
5384.2‡ *4*	$(11/2^-)$	DE HI K	XREF: I(5400) J^π: 2187.1γ Q, ΔJ=2 to $7/2^-$; band assignment.
5482 *2*	$3/2^+,5/2^+$	H K	E(level): from (^{3}He,α). J^π: L=2 from 0^+ in (^{3}He,α).
5572.67 *15*	$3/2^+$	A G	T=3/2 XREF: G(5537) J^π: isobaric analog state of $3/2^+$ ^{35}K g.s. with log *ft*=3.31 *4*. L=(0) from $3/2^+$ in (^{3}He,n). ^{35}Cl(^{3}He,t) attempted to search this T=3/2 level, but did not find it.
5594 *2*	$3/2^+,5/2^+$	C HI K	XREF: C(5570) E(level): weighted average of 5598 *20* from (p,d) and 5594 *2* from (^{3}He,α). J^π: L=2 from 0^+ in (p,d) and (^{3}He,α). Evaluators consider the 5994 level to be different from the 5572.67 T=3/2 level because (p,d) and (^{3}He,α) from T=0 targets should not populate T=3/2 levels.
5613.6 *9*	$(11/2^-)$	E	J^π: 1254.6γ to $(9/2^-)$; mirror level to 5407, $11/2^-$ level in ^{35}Cl proposed by 2007De14 in ^{24}Mg(^{16}O,αnγ).
5765.8 *5*	$(13/2^-)$	DE	J^π: 381.6γ D, ΔJ=1 to $(11/2^-)$, 1406.9γ (Q), ΔJ=(2) to $(9/2^-)$.
5915 *3*		H JK	E(level): weighted average of 5913 *5* from (d,t) and 5916 *3* from (^{3}He,α).

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Adopted Levels, Gammas (continued)

^{35}Ar Levels (continued)

E(level)†	J^{π}	XREF	Comments
5991 *3*		J	
6037 *3*	$3/2^+,5/2^+$	HIJK	XREF: I(6024) E(level): weighted average of 6037 *3* from (d,t) and 6036 *3* from (^{3}He,α). Other: 6024 *20* from (p,d). J^{π}: L=2 from 0^+ in (p,d) and (^{3}He,α).
6055? *3*		J	XREF: J(?)
6076 *3*		J	
6163 *2*		JK	E(level): weighted average of 6164 *3* from (d,t) and 6162 *2* from (^{3}He,α).
6253 *3*		Jk	XREF: k(6262)
6273 *3*		Jk	XREF: k(6262)
6302 *3*		J	
6332 *3*		J	
6345 *3*	(1/2,3/2,5/2)	A J	E(level): From (d,t). Other: 6348 *11* from ^{35}K ε decay. J^{π}: $\varepsilon+\beta^+$ feeding from $3/2^+$ parent with log *ft*=7.2 *1*.
6415 *2*		J	
6439? *4*		J	XREF: J(?)
6460 *3*		J	
6523 *3*		J	
6557 *3*		J	
6585 *3*		J	
6606 *3*		iJ	XREF: i(6620)
6616 *2*	$1/2^+$	iJK	XREF: i(6620) E(level): weighted average of 6617 *2* from (d,t) and 6615 *3* from (^{3}He,α). Other: 6620 *30* from (p,d). J^{π}: L=0 from 0^+ in (^{3}He,α) and (p,d).
6644 *3*		iJ	XREF: i(6620)
6651 *3*		iJ	XREF: i(6620)
6673 *4*	$5/2^-,7/2^-$	IJ	XREF: I(6700) E(level): weighted average of 6700 *20* from (p,d) and 6672 *3* from (d,t). J^{π}: L=3 from 0^+ in (p,d).
6823 *2*	$3/2^+,5/2^+$	I K	E(level): from (^{3}He,α). Other: 6820 *30* from (p,d). J^{π}: L=2 from 0^+ in (p,d).
6948 *2*		K	
7044 *4*	$3/2^+,5/2^+$	A I K	XREF: I(7030) E(level): weighted average of 7053 *11* from ^{35}K ε decay, 7030 *20* from (p,d), and 7043 *4* from (^{3}He,α). J^{π}: L=2 from 0^+ in (p,d).
7117 *10*		K	
7255 *11*		A	
7289 *10*		A K	E(level): weighted average of 7283 *11* from ^{35}K ε decay and 7293 *10* from (^{3}He,α).
7427 *10*		A K	E(level): weighted average of 7431 *11* from ^{35}K ε decay and 7423 *10* from (^{3}He,α).
7509 *10*	$1/2^+,3/2^+,5/2^+$	A K	E(level): weighted average of 7518 *11* from ^{35}K ε decay and 7502 *10* from (^{3}He,α). J^{π}: allowed $\varepsilon+\beta^+$ feeding from $3/2^+$ parent with log *ft*<5.0.
7840 *10*		K	
8019 *10*		K	
8109.7‡ *13*	$(15/2^-)$	E	J^{π}: 2725.7γ to $(11/2^-)$; band assignment.
8212.6 *8*	$(15/2^-)$	E	J^{π}: 2828.3γ Q, ΔJ=2 to $(11/2^-)$ and 2446.6γ to $(13/2^-)$.
8393? *20*	$1/2^+,3/2^+,5/2^+$	A	XREF: A(?) E(level): From (^{35}K ε decay). J^{π}: allowed $\varepsilon+\beta^+$ feeding from $3/2^+$ parent with log *ft*=4.6 *+3−2*.

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Adopted Levels, Gammas (continued)

^{35}Ar Levels (continued)

E(level)†	J^{π}	XREF	Comments
9906.0‡ *20*	(19/2⁻)	E	J^{π}: 1693.3γ Q, ΔJ=2 to (15/2⁻); band assignment.
12277.0‡ *32*	(23/2⁻)	E	J^{π}: 2370.9γ Q, ΔJ=2 ~~γ~~ to (19/2⁻); band assignment.

† E(level) from a least-squares fit to γ-ray energies for levels connected with γ transitions; from particle-transfer reactions or ^{35}K ε+β⁺-delayed proton decays for other levels.
‡ Band(A): Band based on $f_{7/2}$ orbital.

γ(^{35}Ar)

E_i(level)	J_i^{π}	E_{γ}†	I_{γ}†	E_f	J_f^{π}	Mult.‡	Comments
1184.08	1/2⁺	1184.1 *3*	100	0.0	3/2⁺		E_{γ}: weighted average of 1184.0 *3* from ^{35}K ε decay and 1184.3 *4* from ^{36}Ca εp decay.
1750.78	(5/2)⁺	1750.6 *3*	100	0.0	3/2⁺		E_{γ}: weighted average of 1750.5 *3* from ^{35}K ε decay, 1750.7 *4* from (^{24}Mg,αnγ), and 1750.8 *5* from (^{16}O,αnγ).
2603.22	7/2$^{(+)}$	851.9 *9*	12.3 *33*	1750.78	(5/2)⁺		E_{γ}: weighted average of 852 *1* from (^{24}Mg,αnγ) and 851.8 *9* from (^{16}O,αnγ). I_{γ}: weighted average of 10 *5* from (^{24}Mg,αnγ) and 13.3 *33* from (^{16}O,αnγ).
		2603.0 *5*	100 *10*	0.0	3/2⁺	Q	E_{γ}: weighted average of 2603.0 *5* from (^{24}Mg,αnγ) and 2602.6 *15* from (^{16}O,αnγ). I_{γ}: other: 100 *22* from (^{24}Mg,αnγ).
2638.01	(3/2)⁺	886.8# *5*	16# *6*	1750.78	(5/2)⁺		
		2638.0# *4*	100# *13*	0.0	3/2⁺		
2982.79	(5/2)⁺	1798.9# *5*	3.5# *6*	1184.08	1/2⁺		
		2982.68# *13*	100# *4*	0.0	3/2⁺		
3196.98	7/2⁻	593.7 *2*	16.4 *30*	2603.22	7/2$^{(+)}$		E_{γ}: weighted average of 593 *1* from (^{24}Mg,αnγ) and 593.7 *2* from (^{16}O,αnγ). I_{γ}: weighted average of 16 *8* from (^{24}Mg,αnγ) and 16.4 *30* from (^{16}O,αnγ).
		1446.2 *2*	100 *8*	1750.78	(5/2)⁺	D	E_{γ}: weighted average of 1446.2 *2* from (^{24}Mg,αnγ), 1446.1 *6* from (^{16}O,αnγ), and 1446.0 *6* from (^{3}He,nγ). I_{γ}: other: 100 *9* from (^{24}Mg,αnγ).
		3197.0 *7*	21 *5*	0.0	3/2⁺	Q	E_{γ}: from (^{24}Mg,αnγ). Other: 3197 *6* from (^{16}O,αnγ). I_{γ}: weighted average of 18 *5* from (^{24}Mg,αnγ) and 24 *5* from (^{16}O,αnγ).
4065.0?	(1/2⁺,3/2⁺,5/2⁺)	1426.8# *4*	100	2638.01	(3/2)⁺		
4359.0	(9/2⁻)	1162.0 *8*	65 *24*	3196.98	7/2⁻	(D)	E_{γ}: weighted average of 1162 *1* from (^{24}Mg,αnγ) and 1162.0 *8* from

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Adopted Levels, Gammas (continued)

$\gamma(^{35}Ar)$ (continued)

E_i(level)	J^π_i	E_γ†	I_γ†	E_f	J^π_f	Mult.‡	Comments
							(^{16}O,$\alpha n\gamma$). I_γ: unweighted average of 41 *11* from (^{24}Mg,$\alpha n\gamma$) and 88 *18* from (^{16}O,$\alpha n\gamma$).
4359.0	(9/2$^-$)	1756 *1*	100 *15*	2603.22	7/2$^{(+)}$		E_γ: weighted average of 1756 *1* from (^{24}Mg,$\alpha n\gamma$) and 1756.3 *14* from (^{16}O,$\alpha n\gamma$). I_γ: from (^{24}Mg,$\alpha n\gamma$). Other: 100 *53* from (^{16}O,$\alpha n\gamma$).
4528.3	(1/2$^+$,3/2$^+$,5/2$^+$)	4527.9# *7*	100	0.0	3/2$^+$		
4725.9	1/2$^+$	3542.0# *6*	100# *21*	1184.08	1/2$^+$		
		4724.5# *11*	41# *17*	0.0	3/2$^+$		
4785.8	1/2$^+$,3/2$^+$,5/2$^+$	4785.4# *11*	100	0.0	3/2$^+$		
5384.2	(11/2$^-$)	1025.2 *4*	14 *4*	4359.0	(9/2$^-$)		E_γ: weighted average of 1025 *1* from (^{24}Mg,$\alpha n\gamma$) and 1025.2 *4* from (^{16}O,$\alpha n\gamma$). I_γ: weighted average of 21 *8* from (^{24}Mg,$\alpha n\gamma$) and 12 *4* from (^{16}O,$\alpha n\gamma$).
		2187.1 *4*	100 *6*	3196.98	7/2$^-$	Q	E_γ: weighted average of 2187.4 *4* from (^{24}Mg,$\alpha n\gamma$) and 2186.8 *4* from (^{16}O,$\alpha n\gamma$). I_γ: other: 100 *13* from (^{24}Mg,$\alpha n\gamma$).
5572.67	3/2$^+$	1044.4# *4*	2.5# *8*	4528.3	(1/2$^+$,3/2$^+$,5/2$^+$)		
		1507.4# *5*	3.7# *8*	4065.0?	(1/2$^+$,3/2$^+$,5/2$^+$)		
		2589.8# *1*	100# *4*	2982.79	(5/2)$^+$		
		2934.5# *5*	6.7# *12*	2638.01	(3/2)$^+$		
		3821.7# *7*	6.7# *14*	1750.78	(5/2)$^+$		
		4387.2# *9*	6.7# *16*	1184.08	1/2$^+$		
		5572.3# *10*	11.7# *31*	0.0	3/2$^+$		
5613.6	(11/2$^-$)	1254.6 *8*	100	4359.0	(9/2$^-$)		
5765.8	(13/2$^-$)	381.6 *1*	100 *10*	5384.2	(11/2$^-$)	D	E_γ: weighted average of 381.6 *1* from (^{24}Mg,$\alpha n\gamma$) and 381.5 *3* from (^{16}O,$\alpha n\gamma$).
		1406.9 *7*	17.2 *35*	4359.0	(9/2$^-$)	(Q)	
8109.7	(15/2$^-$)	2342.6 *28*	100 *25*	5765.8	(13/2$^-$)		
		2725.7 *14*	50 *13*	5384.2	(11/2$^-$)		
8212.6	(15/2$^-$)	2446.6 *16*	21 *7*	5765.8	(13/2$^-$)		
		2828.3 *7*	100 *18*	5384.2	(11/2$^-$)	Q	
9906.0	(19/2$^-$)	1693.3 *27*	100 *20*	8212.6	(15/2$^-$)	Q	
		1796.3 *25*	67 *20*	8109.7	(15/2$^-$)	Q	
12277.0	(23/2$^-$)	2370.9 *25*	100	9906.0	(19/2$^-$)	Q	

† From (^{16}O,$\alpha n\gamma$), unless otherwise noted.

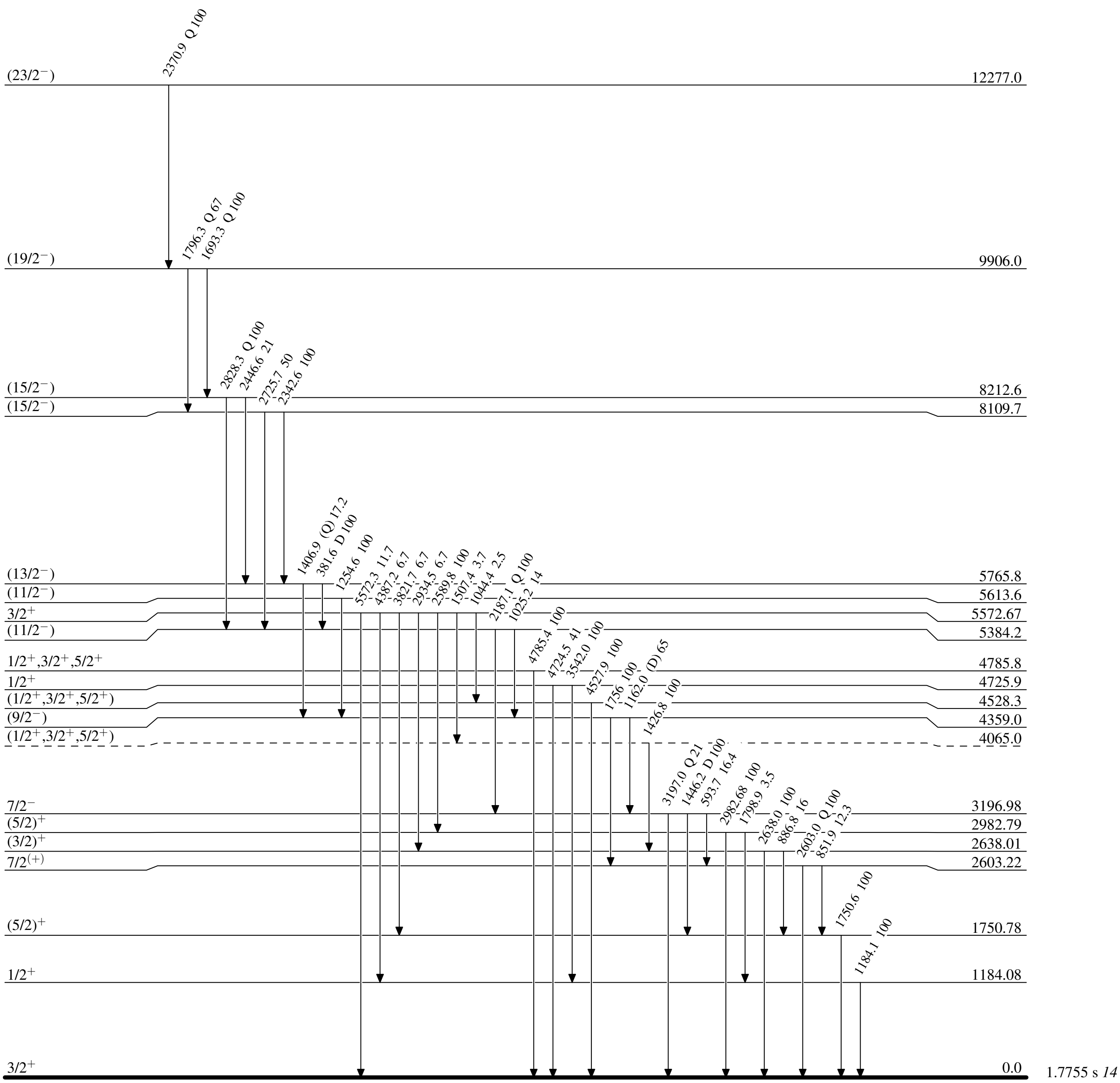
‡ From (^{16}O,$\alpha n\gamma$) and/or (^{24}Mg,$\alpha n\gamma$). Stretched quadrupole (Q, ΔJ=2) or stretched dipole (D, ΔJ=1) transitions are assigned by evaluators based on the measured $\gamma\gamma(\theta)$(ADO) and/or ratios of yields R($\gamma(\theta)$) compared with the expected values given by the authors.

From ^{35}K ε decay.

Adopted Levels, Gammas

Level Scheme

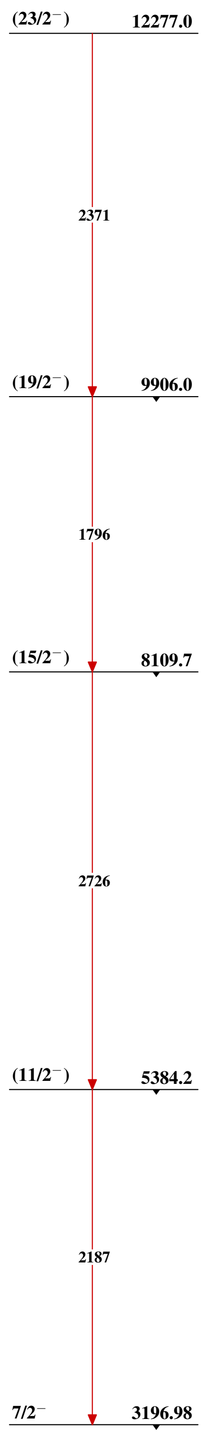
Intensities: Relative photon branching from each level



$^{35}_{18}Ar_{17}$

Adopted Levels, Gammas

Band(A): Band based on $f_{7/2}$ orbital



$^{35}_{18}Ar_{17}$

^{35}K $\varepsilon+\beta^+$ decay (175 ms) 1980Ew02

Parent: ^{35}K: E=0; $J^\pi=3/2^+$; $T_{1/2}$=175 ms *2*; $Q(\varepsilon+\beta^+)$=11874.4 *9*; %$\varepsilon+\beta^+$ decay=100

^{35}K-J^π,$T_{1/2}$: From the Adopted Levels of ^{35}K. Adopted $T_{1/2}$ from weighted average of 175 ms *2* (2018Sa54), 178 ms *8* (1998Sc19), and 190 ms *30* (1980Ew02) in this dataset.

^{35}K-$Q(\varepsilon+\beta^+)$: From 2021Wa16.

1980Ew02,1979Ca15: A 600-MeV proton beam was produced from the synchrocyclotron at CERN-ISOLDE and bombarded a ScC_2 target. The ^{45}Sc(p,8n3p) spallation reaction products diffused out of the target and reached a tungsten surface ionization source where potassium isotopes were selectively ionized. The beam was extracted from the ion source, separated by the ISOLDE analyzing magnet, and collected by a mylar foil for γ-ray measurements and then a carbon foil for proton measurements. γ rays were detected using a Ge(Li) detector. Time for positron activities were determined using a 700-μm thick silicon detector. Protons were detected using a 20-μm-700-μm thick ΔE-E telescope of silicon surface barrier detectors with FWHM=50 keV. Measured Eγ(<5 MeV), Iγ, E_p(>0.9 MeV), I_p. Deduced levels, J, π, decay branching ratios, log *ft*, parent ^{35}K $T_{1/2}$, and coefficients of the isobaric multiplet mass equation for A=35, T=3/2 quartets. Comparisons with shell-model calculations and the mirror nucleus ^{35}Cl. Also see abstracts 1978HaYH, 1979HaZY, 1979HaZT, and 1979AnZZ.

2018Sa54: A 36-MeV/nucleon ^{36}Ar primary beam was produced from the K500 cyclotron at Texas A&M University. The secondary ^{35}K beam was produced via the $^1H(^{36}Ar,^{35}K)$2n reaction of ^{36}Ar bombarding a LN_2-cooled hydrogen gas target, separated by MARS, and implanted into a 45-μm DSSD sandwiched between a 140-μm SSSD and a 1-mm Si-pad detector in a pulsed-beam mode. $\varepsilon+\beta^+$-delayed protons were detected by the implantation detector. γ rays were detected by two HPGe detectors. Measured E_p(>300 keV), I_p, Eγ, Iγ, pγ-coin, $\gamma\gamma$-coin. Deduced parent ^{35}K $T_{1/2}$.

2019ChZU: Same beam production as 2018Sa54. ^{35}K was implanted into the AstroBox2 detector filled with 800-Torr P5 gas. $\varepsilon+\beta^+$-delayed protons were detected by the implantation detector. γ rays were detected by 4 Clover Ge detectors. Measured E_p(>100 keV), I_p, Eγ, Iγ, pγ-coin, $\gamma\gamma$-coin.

1998Sc19: A polarized ^{35}K beam was produced via the fragmentation of 500-MeV/nucleon ^{40}Ca impinging on a ^9Be target at GSI, separated using ΔE-tof by FRS, momentum-selected by slits, and implanted into a KBr single crystal placed in the central region of a magnet. Positrons were detected using plastic scintillators. γ rays were detected using a Ge detector. Measured β-decay asymmetry and $\beta\gamma$-coin. Deduced polarization and *g*-factor of ^{35}K ground state from β-NMR and ^{35}K $T_{1/2}$ from $\beta\gamma$-decay time spectra.

2006Me04: A polarized ^{35}K beam was produced via the proton-pickup reaction $^{36}Ar(^9Be,^{10}Li)^{35}K$, separated by NSCL-A1900, and implanted into a KBr crystal. Positrons were detected using plastic scintillators. Deduced the magnetic dipole moment and *g*-factor of ^{35}K ground state from β-NMR.

Theoretical studies involving ^{35}K decay: shell model (1985Br29, 2003Sm02).

^{35}Ar Levels

E(level)†	J^π‡	$T_{1/2}$‡	Comments
0	$3/2^+$	1.7755 s *14*	
1184.01 *25*	$1/2^+$		
1750.72 *25*	$(5/2)^+$		
2637.99 *26*	$(3/2)^+$		
2982.79 *12*	$(5/2)^+$		
4065.0? *4*	$(1/2^+,3/2^+,5/2^+)$		
4528.2 *4*	$(1/2^+,3/2^+,5/2^+)$		
4725.9 *6*	$1/2^+$		
4785.8 *11*	$1/2^+,3/2^+,5/2^+$		
5572.66 *15*	$3/2^+$		T=3/2
6348 *11*	(1/2,3/2,5/2)		$E(p0)_{c.m.}$=452 keV *11* (2019ChZU).

Continued on next page (footnotes at end of table)

^{35}K ε+β^+ decay (175 ms) 1980Ew02 (continued)

^{35}Ar Levels (continued)

E(level)†	J$^\pi$‡	Comments
5896+x		E(level): x<5978.2 *5* from Q(ε)(^{35}K)-S(p)(^{35}Ar), where Q(ε)=11874.4 *9* and S(p)=5896.2 *7* from
7053 *11*	3/2$^+$,5/2$^+$	E(p0)$_{c.m.}$=1157 keV *11* (2019ChZU).
7255 *11*		E(p3)$_{c.m.}$=693 keV *11* (2019ChZU).
7283 *11*		E(p0)$_{c.m.}$=1387 keV *11* (2019ChZU).
7431 *11*		E(p3)$_{c.m.}$=869 keV *11* (2019ChZU).
		E(level): weighted average of E(level) of 7497 *20*, 7510 *20*, and 7527 *11*. The former two E(level) are deduced from E(p0)$_{c.m.}$=1601 *20* (1980Ew02) and E(p1)$_{c.m.}$=1467 *20* (1980Ew02), respectively, with the corresponding E(level)(^{34}Cl) (2012Ni10) and S(p)(^{35}Ar)=5896.2 *7* (2021Wa16). The 7527 *11* is from 2019ChZU with E(p3)$_{c.m.}$=965 *11*.
8393? *20*	1/2$^+$,3/2$^+$,5/2$^+$	E(level): weighted average of E(level) of 8392 *20*, 8392 *20*, and 8395 *20*, deduced from E(p0)$_{c.m.}$=2496 *20* (1980Ew02), E(p1)$_{c.m.}$=2349 *20* (1980Ew02), and E(p2)$_{c.m.}$=2038 *20* (1980Ew02), respectively, with the corresponding E(level)(^{34}Cl) (2012Ni10) and S(p)(^{35}Ar)=5896.2 *7* (2021Wa16).

† From a least-squares fit to γ-ray energies in 1980Ew02 for levels connected with γ transitions.
‡ From the Adopted Levels.

ε,β^+ radiations

E(decay)	E(level)	Iβ^+ ‡	Iε‡	Log *ft*	I($\varepsilon+\beta^+$)†‡	Comments
(3481 *20*)	8393?	0.062 *26*	4.3×10^{-4} *18*	4.6 *+3−2*	0.062 *26*	I($\varepsilon+\beta^+$): 0.062 *26* I(p0+p1+p2) from Table 3 of 1980Ew02.
(4356 *11*)	7518	>0.090	>3×10^{-4}	<5.0	>0.09	I($\varepsilon+\beta^+$): 0.15 *6* I(p0+p1) from Table 3 of 1980Ew02. Evaluators adopted a lower limit due to unreported I(p3) (2019ChZU).
(5.5×10^3# *55*)	5896+x				0.16 *16*	I($\varepsilon+\beta^+$): I(ε+β^+)=0.37 *15* for all E(p)>0.9 MeV is reported in 1980Ew02. The protons placed from the 7518 and 8393 levels in ^{35}Ar sum to I(ε+β^+)=0.21 *7*. Therefore, the remaining I(ε+β^+)=0.16 *16* is assigned by the evaluators to the unplaced protons from the 5896+x levels.
(5526 *11*)	6348	0.0025 *5*	2.9×10^{-6} *6*	7.2 *1*	2.5×10^{-3} *5*	
(6301.7 *14*)	5572.66	36.3 *24*	0.0265 *18*	3.31 *4*	36.3 *24*	
(7088.6 *18*)	4785.8	1.0 *4*	5×10^{-4} *2*	5.2 *2*	1.0 *4*	
(7148.5 *15*)	4725.9	2.1 *4*	0.0010 *2*	4.9 *1*	2.1 *4*	
(7346.2 *14*)	4528.2	0.7 *4*	3×10^{-4} *2*	5.4 *+4−2*	0.7 *4*	
(7809.4 *14*)	4065.0?	0.56 *33*	2.0×10^{-4} *12*	5.6 *+4−2*	0.56 *33*	
(8891.6 *14*)	2982.79	26.0 *22*	0.0060 *5*	4.27 *4*	26.0 *22*	
(9236.4 *14*)	2637.99	≤0.4		≥6.2	≤0.4	
(10123.7 *14*)	1750.72	11.9 *9*	0.00181 *14*	4.91 *4*	11.9 *9*	
(10690.4 *14*)	1184.01	2.2 *7*	2.8×10^{-4} *9*	5.8 *+2−1*	2.2 *7*	
(11874.4 *17*)	0	19 *4*	0.0018 *4*	5.1 *1*	19 *4*	I($\varepsilon+\beta^+$): From 1980Ew02 assuming mirror log *ft* with a small asymmetry correction.

† From γ intensity balance at each state, except for proton-emitting states. 1980Ew02 authors state that in complex decay schemes of heavy nuclides this method is known to be suspect since there is significant γ intensity that is unobserved because it lies in a

Continued on next page (footnotes at end of table)

^{35}K $\varepsilon+\beta^+$ decay (175 ms) 1980Ew02 (continued)

ε,β^+ radiations (continued)

multitude of very weak γ-ray peaks. In a nucleus as light as ^{35}K the problem is less acute. They have generated a pandemonium test in the same spirit as in 1977Ha51 and find that less than one percent of the γ intensity from ^{35}K decay should be missed for that reason.

‡ Absolute intensity per 100 decays.

Estimated for a range of levels.

$\gamma(^{35}Ar)$

Iγ normalization: From Σ%Iγ(γ to g.s.)=80.6 *40*, deduced from 100−%($\varepsilon+\beta^+$)p-%I($\varepsilon+\beta^+$)(g.s.), where %($\varepsilon+\beta^+$)p=0.37 *15* (1980Ew02) and %I($\varepsilon+\beta^+$)(g.s.)=19 *4* (1980Ew02), and the latter corresponds to log *ft*=5.07 *5*, which was deduced from the ^{35}S (g.s.) to ^{35}Cl (g.s.) mirror log *ft*=5.01 *2* with a small asymmetry correction. %($\varepsilon+\beta^+$)p=0.37 *15* for E(p)>0.9 MeV (1980Ew02). Some weak E(p)<0.9 MeV branches have also been observed (2018Sa54,2019ChZU).

Conversion coefficients are considered negligibly small.

E_γ †	I_γ †‡	E_i(level)	J_i^π	E_f	J_f^π	Comments
886.8 *5*	0.9 *3*	2637.99	(3/2)$^+$	1750.72	(5/2)$^+$	%Iγ=0.46 *+19−17*
1044.4 *4*	1.3 *4*	5572.66	3/2$^+$	4528.2	(1/2$^+$,3/2$^+$,5/2$^+$)	%Iγ=0.66 *+25−23*
1184.0 *3*	14.3 *7*	1184.01	1/2$^+$	0	3/2$^+$	%Iγ=7.2 *5*
1426.8 *4*	3.0 *5*	4065.0?	(1/2$^+$,3/2$^+$,5/2$^+$)	2637.99	(3/2)$^+$	%Iγ=1.5 *+4−3*
1507.4 *5*	1.9 *4*	5572.66	3/2$^+$	4065.0?	(1/2$^+$,3/2$^+$,5/2$^+$)	%Iγ=0.96 *+27−25*
1750.5 *3*	28 *1*	1750.72	(5/2)$^+$	0	3/2$^+$	%Iγ=14.1 *9*
1798.9 *5*	3.5 *6*	2982.79	(5/2)$^+$	1184.01	1/2$^+$	%Iγ=1.8 *4*
2589.8 *1*	52 *2*	5572.66	3/2$^+$	2982.79	(5/2)$^+$	%Iγ=26.3 *18*
2638.0 *4*	5.5 *7*	2637.99	(3/2)$^+$	0	3/2$^+$	%Iγ=2.8 *5*
^{x}2697.7 *6*						Unplaced γ ray, accounting for no more than 1.2% $\varepsilon+\beta^+$-feeding (1980Ew02). No ^{35}Ar γ rays at this energy were observed in other reaction studies.
2934.5 *5*	3.5 *6*	5572.66	3/2$^+$	2637.99	(3/2)$^+$	%Iγ=1.8 *4*
2982.68 *13*	100 *4*	2982.79	(5/2)$^+$	0	3/2$^+$	%Iγ=50.5 *27*
3542.0 *6*	2.9 *6*	4725.9	1/2$^+$	1184.01	1/2$^+$	%Iγ=1.5 *4*
3821.7 *7*	3.5 *7*	5572.66	3/2$^+$	1750.72	(5/2)$^+$	%Iγ=1.8 *5*
4387.2 *9*	3.5 *8*	5572.66	3/2$^+$	1184.01	1/2$^+$	%Iγ=1.8 *5*
4527.9 *7*	2.6 *7*	4528.2	(1/2$^+$,3/2$^+$,5/2$^+$)	0	3/2$^+$	%Iγ=1.3 *4*
4724.5 *11*	1.2 *5*	4725.9	1/2$^+$	0	3/2$^+$	%Iγ=0.61 *+30−27*
4785.4 *11*	1.9 *7*	4785.8	1/2$^+$,3/2$^+$,5/2$^+$	0	3/2$^+$	%Iγ=1.0 *4*
5572.3 *10*	6.1 *16*	5572.66	3/2$^+$	0	3/2$^+$	%Iγ=3.1 *+10−9* 1980Ew02 observed the double escape peak at 4550 keV of this γ ray. 2018Sa54 observed the photopeak at 5572 keV.

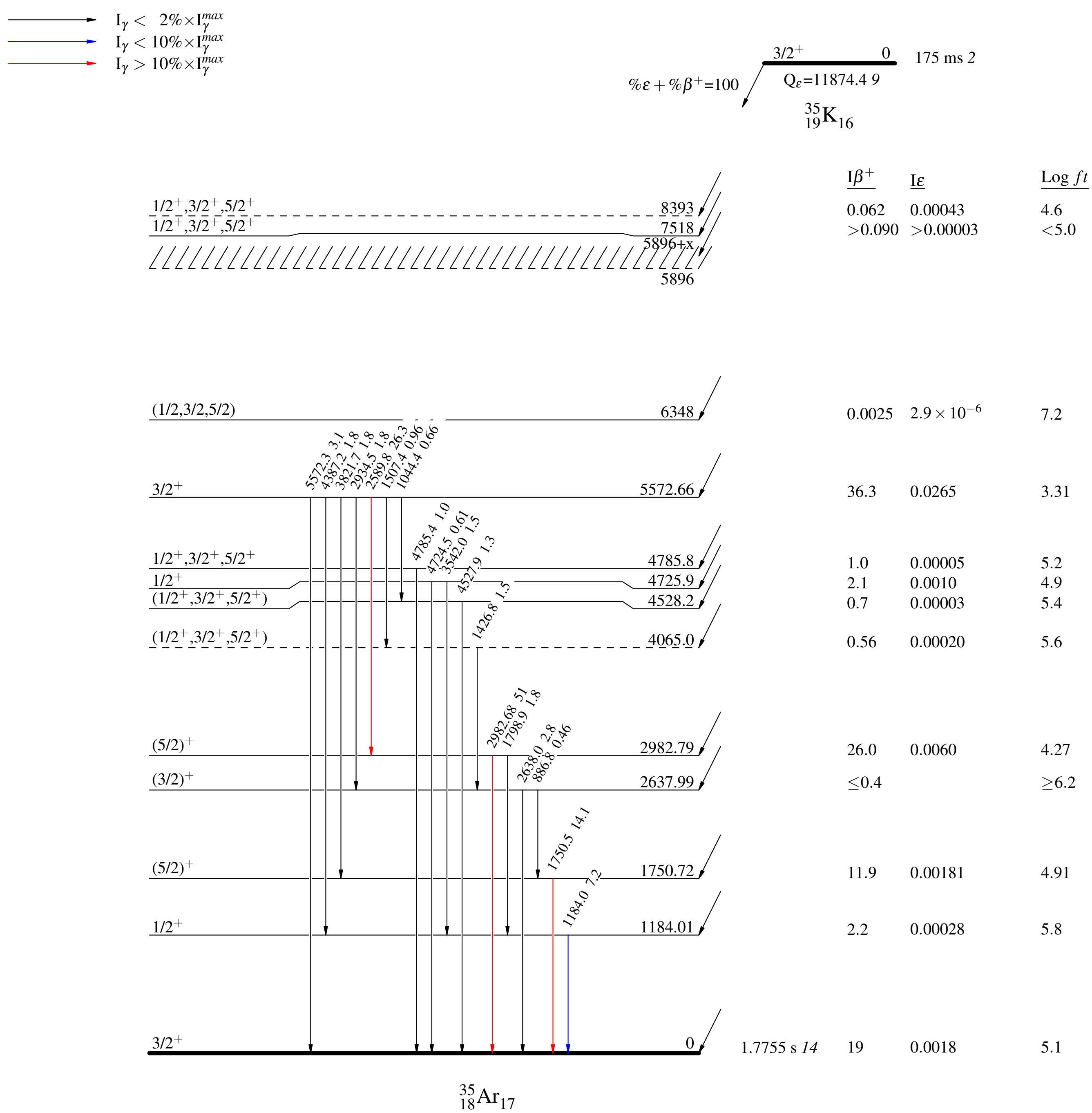
† From 1980Ew02.

‡ For absolute intensity per 100 decays, multiply by 0.505 *29*.

^{x} γ ray not placed in level scheme.

^{35}K $\varepsilon+\beta^+$ decay (175 ms) 1980Ew02

Decay Scheme



^{36}Ca εp decay (100.9 ms) 1997Tr05,2001Lo11,2015Su01

Parent: ^{36}Ca: E=0; $J^{\pi}=0^{+}$; $T_{1/2}$=100.9 ms *20*; Q(εp)=9275 *6*; %εp decay=54.1 *12*

^{36}Ca-J^{π}: From the Adopted Levels of ^{36}Ca (2012Ni01).

^{36}Ca-$T_{1/2}$: Weighted average of 102 ms *2* (1995Tr02,1997Tr05), 100.1 ms *23* (2007Do17), and 100.0 ms *24* (2015Su01). Other: 100 ms *+90−40* (1981Ay01).

^{36}Ca-Q(εp): Deduced by evaluators from ^{36}Ca mass excess of −6483 *6*; weighted average of −6483.6 *56* (2021Su04), −6480 *40* (2021La04), and −6450 *40* (2021Wa16,1977Tr03), and ^{35}Ar mass excess of −23047.3 *7* (2021Wa16). Q(εp) from 2021Wa16: 9310 *40*.

^{36}Ca-%εp decay: Unweighted average of %ε+β^{+}p=56.8 *13* (1997Tr05), 54.3 *18* (2001Lo11), 51.2 *10* (2007Do17), and 53.9 *72* (2015Su01).

1997Tr05,1995Tr02: a-300 MeV/nucleon ^{40}Ca primary beam was produced by the GSI heavy-ion synchrotron. The secondary ^{36}Ca beam was produced via the projectile fragmentation of ^{40}Ca impinging on a ^{9}Be target and was selected using ΔE-tof-Bρ by FRS at GSI, Darmstadt. A total of 2.8×10^{4} ^{36}Ca ions were implanted into a 500-μm-thick Si detector. ε+β^{+}-delayed protons were detected by the implantation detector. β particles were detected by the implantation detector and two 500-μm-thick Si counters. γ rays were detected by two Ge detectors. Measured E_p, I_p, Eγ, Iγ, βp-coin, βγ-coin, and pγ-coin. Deduced levels, decay branching ratios, log *ft*, B(F), and B(GT). Deduced parent ^{36}Ca $T_{1/2}$ from the time spectrum of proton events accumulated during the beam-off in the pulsed-beam mode. Comparisons with shell-model calculations.

2001Lo11: A 95-MeV ^{40}Ca primary beam was produced by the SISSI facility at GANIL. The secondary ^{36}Ca beam was produced via the projectile fragmentation of ^{40}Ca impinging on a ^{nat}Ni target and was selected using ΔE-tof by the LISE3 spectrometer and purified by a velocity filter. A total of 102407 ^{36}Ca ions were implanted into a 500-μm-thick Si detector. ε+β^{+}-delayed protons were detected by the implantation detector. β particles were detected by two 500-μm-thick Si counters. γ rays were detected by three Ge detectors. Measured E_p, I_p, Eγ, Iγ, βp-coin, and βγ-coin. Deduced levels, decay branching ratios, log *ft*, B(F), and B(GT). Comparisons with shell-model calculations.

2015Su01: A 69.42-MeV/nucleon ^{40}Ca primary beam was produced by the Sector Focusing Cyclotron and Separated Sector Cyclotron at the Heavy Ion Research Facility in Lanzhou (HIRFL). The secondary ^{36}Ca beam was produced via the projectile fragmentation of ^{40}Ca impinging on a ^{9}Be target and was selected using ΔE-tof-Bρ by RIBLL. A total of 22890 ^{36}Ca ions were implanted into a 525-μm-thick DSSD. ε+β^{+}-delayed protons were detected by the DSSD with a threshold of 500 keV. ε+β^{+}-delayed γ rays were detected by four Clover Ge detectors surrounding the DSSD chamber. Measured E_p, I_p, Eγ, Iγ, βp-coin, pγ-coin, and implant-decay time correlations. Deduced levels, decay branching ratios, and parent ^{36}Ca $T_{1/2}$. Also see 2016Li45.

2007Do17: A 74.5-MeV/nucleon ^{58}Ni primary beam was produced by the SISSI facility at GANIL. The secondary ^{36}Ca beam was produced via the projectile fragmentation of ^{58}Ni impinging on a ^{nat}Ni target and was selected using ΔE-tof-Bρ by the ALPHA-LISE3 separator. A total of 16991 ^{36}Ca ions were implanted into a 500-μm thick DSSD. ε+β^{+}-delayed protons were detected by the DSSD with a threshold of 60-80 keV. ε+β^{+}-delayed γ rays were detected by four Ge detectors surrounding the implantation array. A 5-mm thick lithium-drifted Si detector was used as a veto for implantation events and to detect β particles. Measured E_p, I_p, and implant-decay time correlations. Deduced levels, decay branching ratios, and parent ^{36}Ca $T_{1/2}$.

1995Ga16: 60-keV ^{36}Ca was produced by the ISOLDE general-purpose on-line isotope separator at the CERN PS/Booster and implanted into the entrance window of a gas-Si-Si ΔE-E-veto detector telescope. Measured E_p. Deduced coefficients of the isobaric multiplet mass equation for A=36, T=2 quintets. A by-product of ^{37}Ca decay study (1995Ga03).

1981Ay01,1980AyZZ: ^{36}Ca was produced via the ^{40}Ca(^{3}He,α3n) reaction using a 95-MeV ^{3}He beam from the 88-inch Cyclotron at Lawrence Berkeley Laboratory. β-delayed protons were detected using a Si surface barrier detector telescope with FWHM=55 keV and a minimum threshold of ≈1.5 MeV. Measured E_p. Deduced ^{36}Ca $T_{1/2}$ and coefficients of the isobaric multiplet mass equation for A=36, T=2 quintets.

Theoretical studies involving ^{36}Ca decay: shell model (1984Mu25,1990Br26), covariant density functional theory (2013Ni09).

^{35}Ar Levels

E(level)	J^{π}†	$T_{1/2}$†	Comments
0	$3/2^{+}$	1.7755 s *14*	
1184.3 *4*	$1/2^{+}$		E(level): From Eγ data.

† From the Adopted Levels.

^{36}Ca εp decay (100.9 ms) 1997Tr05,2001Lo11,2015Su01 (continued)

γ(^{35}Ar)

Iγ normalization: From 100/%ε+β$^+$p.

E_γ	I_γ †	E_i(level)	J^π_i	E_f	J^π_f	Comments
1184.3 *4*	1.2 *4*	1184.3	1/2$^+$	0	3/2$^+$	E_γ: weighted average of 1184.2 *4* (1997Tr05) and 1185 *1* (2001Lo11). I_γ: From 1997Tr05.

† Absolute intensity per 100 decays.

Delayed Protons (^{35}Ar)

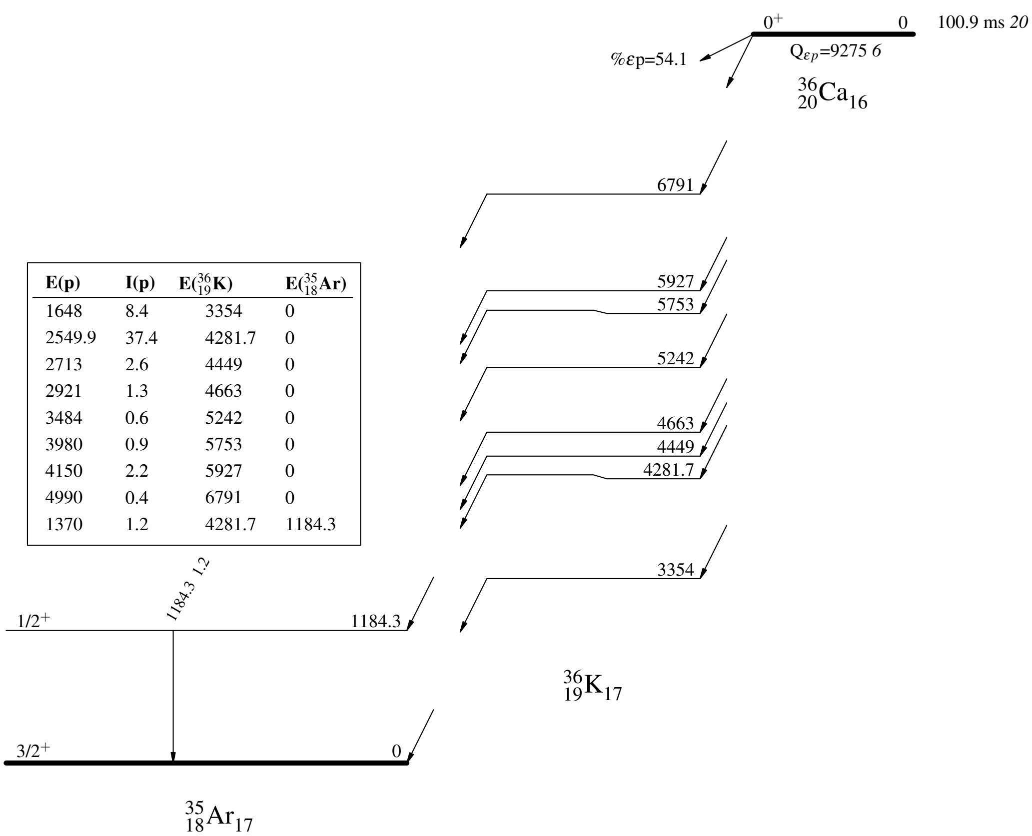
E(p)†	E(^{35}Ar)	I(p)#	E(^{36}K)‡	Comments
1.37×10^3	1184.3	1.2 *4*	4281.7	E(p),I(p): From 1997Tr05. E(^{36}K): J^π=0, T=2 isobaric analog state in ^{36}K.
1648 *18*	0	8.4 *12*	3354	E(p): weighted average of 1676 *39* (1997Tr05), 1645 *21* (2001Lo11), 1660 *18* (2007Do17), and 1624 *22* (2015Su01). I(p): unweighted average of 11.3 *6* (1997Tr05), 9.3 *8* (2001Lo11), 7.3 *8* (2007Do17), and 5.7 *16* (2015Su01).
2549.9 *22*	0	37.4 *10*	4281.7	E(p): weighted average of 2519 *21* (1981Ay01), 2550.2 *22* (1995Ga16), 2548 *37* (1997Tr05), 2551 *21* (2001Lo11), 2538 *18* (2007Do17), and 2594 *30* (2015Su01). I(p): weighted average of 37.8 *10* (1997Tr05), 32.1 *42* (2007Do17), and 34.0 *58* (2015Su01). Other: 37 *1* (2001Lo11) without separating the weaker proton branch from the same level to ^{35}Ar first excited state. E(^{36}K): J^π=0, T=2 isobaric analog state in ^{36}K.
2713 *21*	0	2.6 *9*	4449	E(p): weighted average of 2713 *31* (1997Tr05) and 2713 *21* (2001Lo11). I(p): unweighted average of 1.7 *2* (1997Tr05) and 3.5 *5* (2001Lo11).
2921 *35*	0	1.3 *2*	4663	E(p): weighted average of 2937 *35* (1997Tr05) and 2895 *44* (2001Lo11). I(p): weighted average of 1.4 *2* (1997Tr05) and 1.0 *3* (2001Lo11).
3484 *21*	0	0.6 *2*	5242	E(p),I(p): From 2001Lo11.
3.98×10^3 *7*	0	0.9 *2*	5753	E(p),I(p): From 2001Lo11.
4.15×10^3 *5*	0	2.2 *5*	5927	E(p): weighted average of 4162 *45* (1997Tr05) and 4135 *44* (2001Lo11). I(p): unweighted average of 2.7 *4* (1997Tr05) and 1.7 *3* (2001Lo11).
4.99×10^3 *7*	0	0.4 *2*	6791	E(p): weighted average of 4989 *69* (1997Tr05) and 4983 *67* (2001Lo11). I(p): weighted average of 0.7 *2* (1997Tr05) and 0.3 *1* (2001Lo11).

† In lab frame. Evaluators take a weighted average when multiple measurements of a E(p) are available. 2007Do17 and 2015Su01 reported $E_{c.m.}$. 1997Tr05 and 2001Lo11 reported the E(level) of ^{36}K proton-emitting levels. Evaluators deduced proton center-of-mass energies $E_{c.m.}$=E(level)(^{36}K)−S(p)(^{36}K)−E(level)(^{35}Ar) using their original S(p)=1666 *8*. Evaluators then deduced each $E(p)_{lab}=E_{c.m.}\times m(^{35}Ar)/[m(p)+m(^{35}Ar)]$.
‡ E(level)(^{36}K)=$E(p)_{lab}\times[m(p)+m(^{35}Ar)]/m(^{35}Ar)$+S(p)(^{36}K)+E(level)(^{35}Ar), where S(p)(^{36}K)=1658.9 *8* (2021Wa16).
# Absolute intensity per 100 decays.

^{36}Ca εp decay (100.9 ms) 1997Tr05,2001Lo11,2015Su01

Decay Scheme

γ Intensities: I_γ per 100 parent decays
I(p) Intensities: I(p) per 100 parent decays



E(p)	I(p)	E($^{36}_{19}$K)	E($^{35}_{18}$Ar)
1648	8.4	3354	0
2549.9	37.4	4281.7	0
2713	2.6	4449	0
2921	1.3	4663	0
3484	0.6	5242	0
3980	0.9	5753	0
4150	2.2	5927	0
4990	0.4	6791	0
1370	1.2	4281.7	1184.3

$^{1}H(^{36}Ar,d)$ 2010Le03,2011Le01

$J^{\pi}=0^{+}$ for ^{36}Ar ground state.

2010Le03, 2011Le01: A ^{36}Ar beam at 33 MeV/u was provided at the National Superconducting Cyclotron Laboratory, MSU. Targets were polyethylene $(CH_2)_n$. Deuterons were detected using the High-Resolution Array (HiRA) of Si and CsI(Tl) telescope detectors in coincidence with recoil residues identified in the S800 spectrometer by the focal plane ionization chamber and ToF. Measured $\sigma(E_d,\theta)$ in inverse kinematics. Deduced neutron spectroscopic factors from adiabatic distorted wave approximation (ADWA) analysis of the measured $\sigma(\theta)$ using Chapel-Hill global optical potential parameters (CH89) and JLM optical potentials and geometry for transferred neutron constrained by Hartree−Fock calculations (JLM+HF). Comparisons with shell−model calculated spectroscopic factors.

Theoretical studies involving $^{1}H(^{36}Ar,d)^{35}Ar$: 2011Nu01, 2023He15.

^{35}Ar Levels

E(level)†	J^{π}	L‡	C^2S‡	Comments
0	$3/2^{+}$	2	2.3 *2*	C^2S: others: 1.6 *1* from 2011Le01 using JLM+HF calculations; 2.10 from large basis−shell model calculations (2010Le03); 2.21 *49* from a reanalysis of the $\sigma(\theta)$ data using finite-range ADWA (2011Nu01), including theoretical uncertainties associated with optical potentials (7%) and the approximate solution of three-body problems (19%); 2.1 *+2−4* from a reanalysis of the $\sigma(\theta)$ data using ADWA within a Bayesian framework (2023He15), including theoretical uncertainties associated with optical potentials.
1180		0	1.2 *1*	
2980#				
3190#				
5570				

† From 2010Le03.
‡ From ADWA analysis of measured $\sigma(\theta)$ in 2011Le01 using CH89 optical potential parameters.
# Doublet in measured spectra.

$^{16}O(^{24}Mg,\alpha n\gamma)$ 2004Ek01,2005Ek01

2004Ek01,2005Ek01: A 60-MeV ^{24}Mg beam was produced at the Legnaro National Laboratory, Italy. The target was 0.5-mg/cm^2 enriched ^{40}Ca with a 7-mg/cm^2 tantalum backing. Oxygen was present in the target, giving rise to the fusion evaporation reactions of $^{16}O(^{24}Mg,\alpha n\gamma)^{35}Ar$ and $^{16}O(^{24}Mg,\alpha p\gamma)^{35}Cl$. γ rays were detected using the GASP array of Ge detectors and 80 BGO detectors. Charged particles were detected using the ISIS array of 40 Si ΔE-E telescopes. Neutrons were detected using a Neutron Ring replacing the six BGO elements at the most forward angles. The event trigger required one Ge detector, one BGO detector, and one neutron detector, or two Ge detectors and one BGO detector firing. Measured Eγ, Iγ, $\gamma\gamma$, $\alpha n\gamma$-coin, and $\alpha p\gamma$-coin. Deduced levels, J, π from the measured ratios of yields R($\gamma(\theta)$) and comparisons with the mirror nucleus ^{35}Cl.

^{35}Ar Levels

E(level)†	J$^\pi$‡
0.0	3/2$^+$
1750.8 *3*	5/2$^+$
2603.2 *4*	(7/2$^+$)
3197.0 *4*	7/2$^{(-)}$
4359.2 *7*	(9/2$^-$)
5384.4 *5*	11/2$^{(-)}$
5766.0 *5*	13/2$^{(-)}$

† From a least-squares fit to γ-ray energies.
‡ As given in 2004Ek01 based on known assignments of low-lying levels and mirror levels in ^{35}Cl and the measured ratios of yields R($\gamma(\theta)$). When considered in the Adopted Levels, the firm assignments here are placed within parentheses if there are no other strong arguments to support these firm assignments.

$\gamma(^{35}Ar)$

The ratios of yields R($\gamma(\theta)$) were measured at 35° and 81° with respect to the beam axis. Expected values are R($\gamma(\theta)$)≈1.2 for stretched quadrupole (ΔJ=2) and R($\gamma(\theta)$)≈0.7 for stretched dipole (ΔJ=1) transitions.

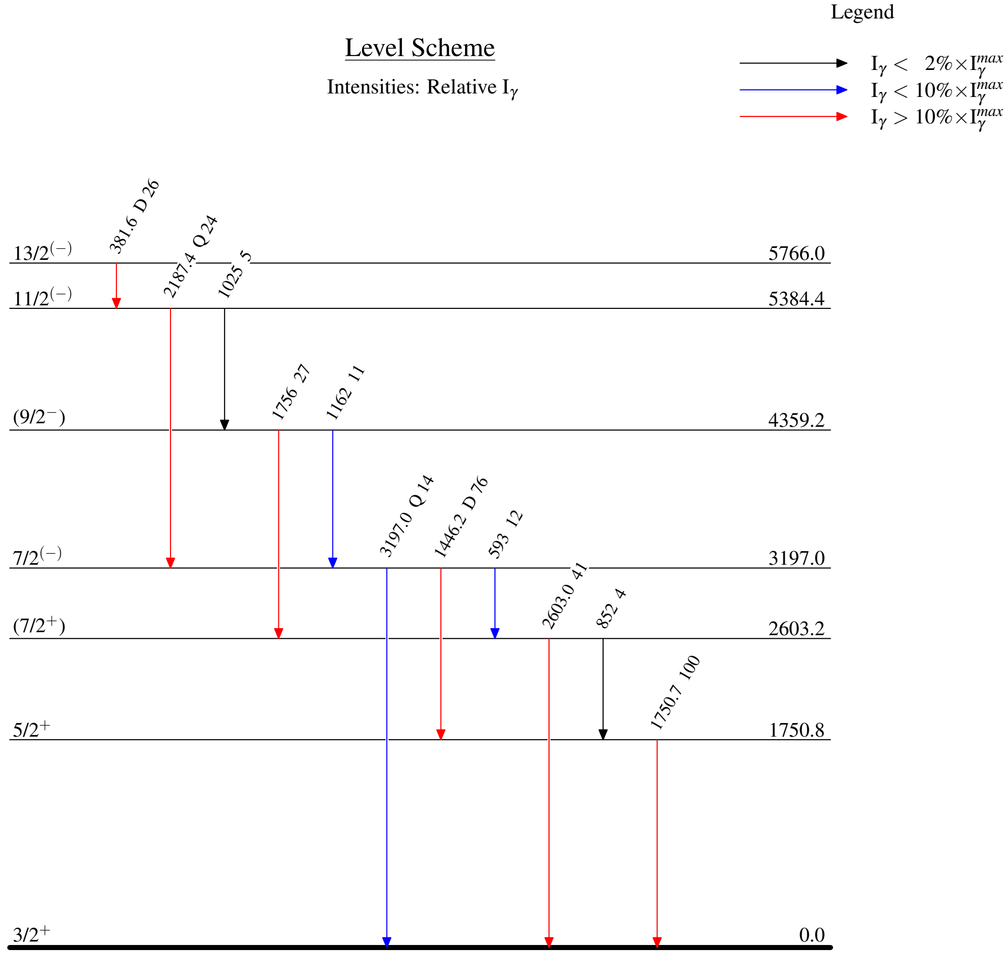
E$_\gamma$†	I$_\gamma$†	E$_i$(level)	J$^\pi_i$	E$_f$	J$^\pi_f$	Mult.#	Comments
381.6 *1*	26 *3*	5766.0	13/2$^{(-)}$	5384.4	11/2$^{(-)}$	D	R($\gamma(\theta)$)=0.69 *18*.
593 *1*	12 *6*	3197.0	7/2$^{(-)}$	2603.2	(7/2$^+$)		
852‡ *1*	4 *2*	2603.2	(7/2$^+$)	1750.8	5/2$^+$		
1025‡ *1*	5 *2*	5384.4	11/2$^{(-)}$	4359.2	(9/2$^-$)		
1162 *1*	11 *3*	4359.2	(9/2$^-$)	3197.0	7/2$^{(-)}$		
1446.2 *2*	76 *7*	3197.0	7/2$^{(-)}$	1750.8	5/2$^+$	D	R($\gamma(\theta)$)=0.71 *9*.
1750.7 *4*	100 *7*	1750.8	5/2$^+$	0.0	3/2$^+$		R($\gamma(\theta)$)=1.41 *14* indicates ΔJ=2, but ΔJ=1 from level scheme is inconsistent.
1756‡ *1*	27 *4*	4359.2	(9/2$^-$)	2603.2	(7/2$^+$)		
2187.4 *4*	24 *3*	5384.4	11/2$^{(-)}$	3197.0	7/2$^{(-)}$	Q	R($\gamma(\theta)$)=1.60 *36*.
2603.0 *5*	41 *9*	2603.2	(7/2$^+$)	0.0	3/2$^+$		R($\gamma(\theta)$)=1.01 *17*.
3197.0 *7*	14 *4*	3197.0	7/2$^{(-)}$	0.0	3/2$^+$	Q	R($\gamma(\theta)$)=1.45 *51*.

† From 2004Ek01.
‡ Tentative placements; later confirmed by $^{24}Mg(^{16}O,\alpha n\gamma)^{35}Ar$ (2007De14) and adopted by evaluators.
# Not explicitly given in 2004Ek01, but deduced by evaluators based on their measured ratios of yields R($\gamma(\theta)$) compared with expected values as stated in 2004Ek01.

$^{16}O(^{24}Mg,\alpha n\gamma)$ 2004Ek01,2005Ek01

Level Scheme

Intensities: Relative I_γ



$^{35}_{18}Ar_{17}$

^{24}Mg(^{16}O,αnγ) 2007De14,2005DeZZ

2007De14,2005DeZZ: A 70-MeV ^{16}O beam was produced by the XTU-Tandem accelerator at the Legnaro National Laboratory, Italy. The target was a 400 μg/cm^2 self-supporting target of ^{24}Mg. γ ray from fusion evaporation reactions of ^{24}Mg(^{16}O,αnγ)^{35}Ar and ^{24}Mg(^{16}O,αpγ)^{35}Cl were detected using the GASP spectrometer, which consists of an array of 40 Compton-suppressed HPGe detectors and a multiplicity filter of 80 BGO scintillators. The GASP spectrometer is operated in conjunction with the 4π charged-particle detector ISIS and a neutron ring of 6 BC501A scintillators. The events were collected when at least two Ge detectors and one BGO scintillator were fired in coincidence. Measured Eγ, Iγ, $\gamma\gamma$, αnγ-coin, αpγ-coin, $\gamma(\theta)$, $\gamma\gamma(\theta)$(ADO). Deduced levels, J, π, and transition multipolarities from γ-ray ADO ratios. Comparisons with Shell-model calculations show the multipole Coulomb interaction and the electromagnetic spin-orbit interaction contribute to the observed mirror energy differences. 2007De14 also report data on ^{35}Cl and propose the level scheme of ^{35}Ar based on mirror symmetry.

^{35}Ar Levels

E(level)†	J$^\pi$‡	E(level)†	J$^\pi$‡	E(level)†	J$^\pi$‡	E(level)†	J$^\pi$‡
0.0	3/2$^+$	3196.7# *7*	7/2$^-$	5613.2 *11*	(11/2$^-$)	8212.1 *10*	15/2$^-$
1750.8 *5*	5/2$^+$	4358.6 *8*	9/2$^-$	5765.3 *8*	13/2$^-$	9905.5# *21*	19/2$^-$
2603.0 *7*	7/2$^+$	5383.7# *7*	11/2$^-$	8109.2# *14*	15/2$^-$	12276.5# *33*	23/2$^-$

† From a least-squares fit to γ-ray energies.

‡ As given in 2007De14 based on known assignments of low-lying levels and mirror levels in ^{35}Cl, the measured $\gamma\gamma(\theta)$(ADO) ratios, and the mirror symmetry with levels with similar energies in ^{35}Cl with known assignments. When considered in the Adopted Levels, the firm assignments here are placed within parentheses if there are no other strong arguments to support these firm assignments.

Band(A): Band based on $f_{7/2}$ orbital.

γ(^{35}Ar)

R_{ADO}=[Iγ(34°)+Iγ(146°)]/2Iγ(90°). Expected values are $R_{ADO}\approx$1.3 for stretched quadrupole (ΔJ=2) and $_{ADO}\approx$0.8 for stretched dipole (ΔJ=1) transitions.

E$_\gamma$†	I$_\gamma$†	E$_i$(level)	J$^\pi_i$	E$_f$	J$^\pi_f$	Mult.‡	Comments
381.5 *3*	29 *3*	5765.3	13/2$^-$	5383.7	11/2$^-$	D	R_{ADO}=0.81 *10*.
593.7 *2*	11 *2*	3196.7	7/2$^-$	2603.0	7/2$^+$		
851.8 *9*	8 *2*	2603.0	7/2$^+$	1750.8	5/2$^+$		
1025.2 *4*	6 *2*	5383.7	11/2$^-$	4358.6	9/2$^-$		
1162.0 *8*	15 *3*	4358.6	9/2$^-$	3196.7	7/2$^-$	(D)	R_{ADO}=0.95 *25*.
1254.6 *8*	15 *5*	5613.2	(11/2$^-$)	4358.6	9/2$^-$		
1406.9 *7*	5 *1*	5765.3	13/2$^-$	4358.6	9/2$^-$	(Q)	R_{ADO}=1.3 *7*.
1446.1 *6*	67 *5*	3196.7	7/2$^-$	1750.8	5/2$^+$	D	R_{ADO}=0.87 *19*.
1693.3 *27*	15 *3*	9905.5	19/2$^-$	8212.1	15/2$^-$	Q	R_{ADO}=1.41 *23*.
1750.8 *5*	100 *9*	1750.8	5/2$^+$	0.0	3/2$^+$		R_{ADO}=1.46 *24* indicates ΔJ=2, but inconsistent with ΔJ=1 from level scheme.
1756.3 *14*	17 *9*	4358.6	9/2$^-$	2603.0	7/2$^+$		
1796.3 *25*	10 *3*	9905.5	19/2$^-$	8109.2	15/2$^-$	Q	R_{ADO}=1.8 *3*.
2186.8 *4*	49 *3*	5383.7	11/2$^-$	3196.7	7/2$^-$	Q	R_{ADO}=1.31 *15*.
2342.6 *28*	8 *2*	8109.2	15/2$^-$	5765.3	13/2$^-$		
2370.9 *25*	15 *5*	12276.5	23/2$^-$	9905.5	19/2$^-$	Q	R_{ADO}=1.8 *4*.
2446.6 *16*	6 *2*	8212.1	15/2$^-$	5765.3	13/2$^-$		
2602.6 *15*	60 *6*	2603.0	7/2$^+$	0.0	3/2$^+$	Q	R_{ADO}=1.37 *20*.
2725.7 *14*	4 *1*	8109.2	15/2$^-$	5383.7	11/2$^-$		
2828.3 *7*	28 *5*	8212.1	15/2$^-$	5383.7	11/2$^-$	Q	R_{ADO}=1.7 *6*.
3197 *6*	16 *3*	3196.7	7/2$^-$	0.0	3/2$^+$	Q	R_{ADO}=1.7 *8*.

Continued on next page (footnotes at end of table)

^{24}Mg(^{16}O,αnγ) 2007De14,2005DeZZ (continued)

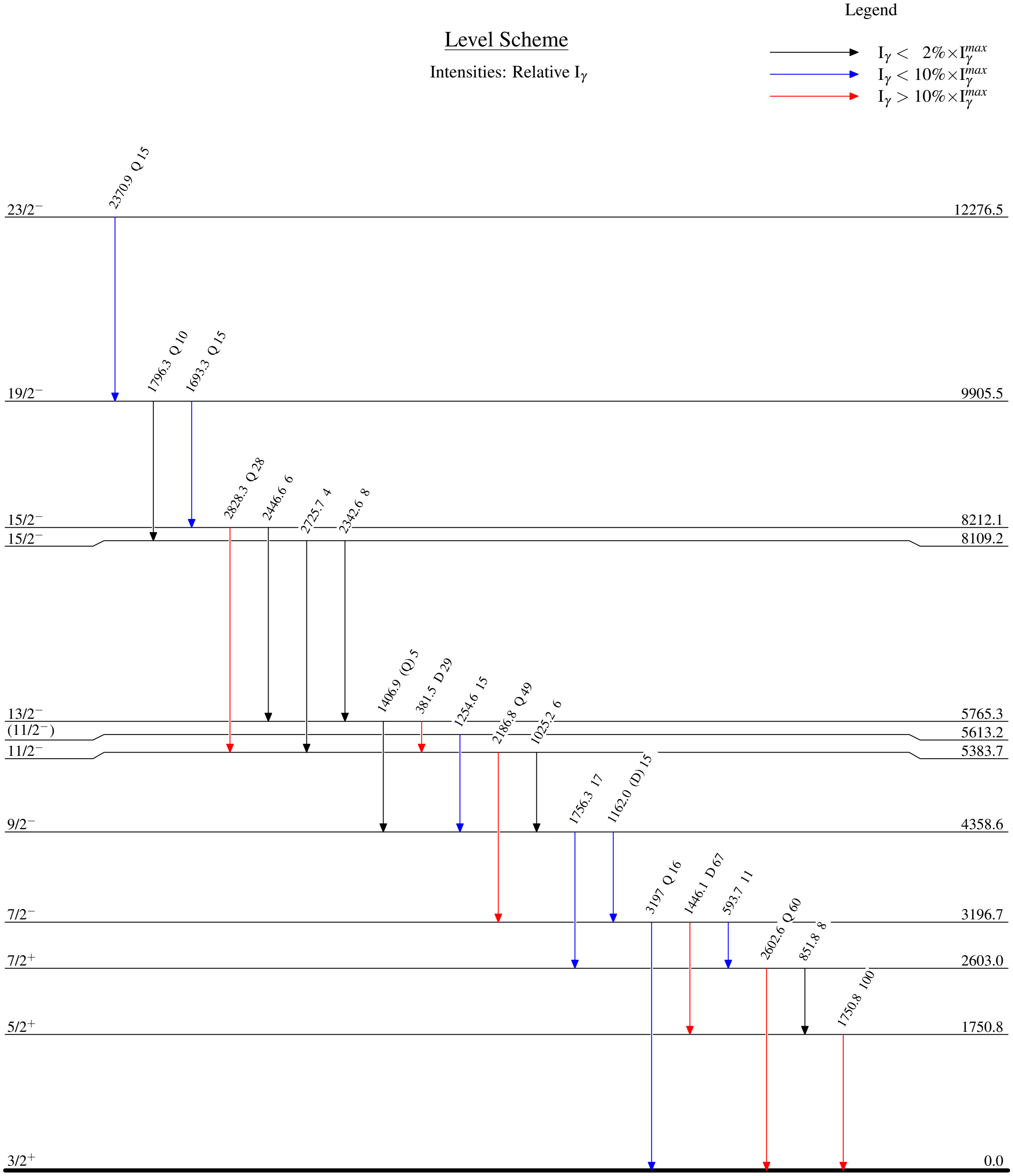
γ(^{35}Ar) (continued)

† From 2007De14.
‡ Not explicitly given in 2007De14, but deduced by evaluators based on their measured R_{ADO} ratios compared with expected values as stated in 2007De14.

^{24}Mg(^{16}O,αnγ) 2007De14,2005DeZZ

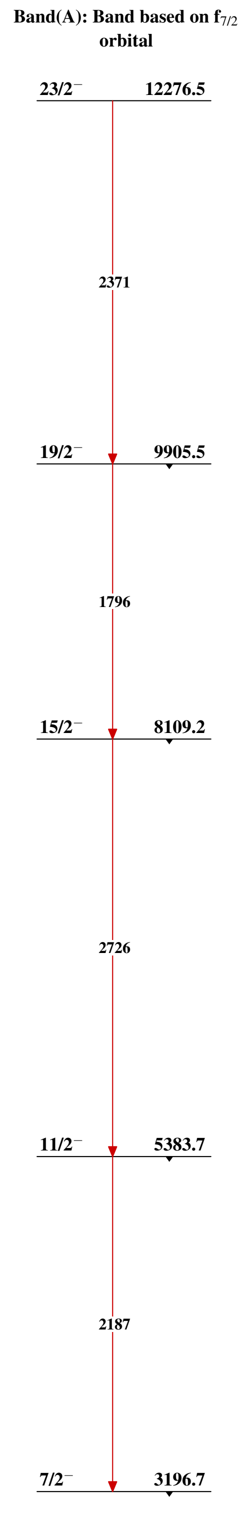
Level Scheme

Intensities: Relative I_γ



$^{35}_{18}Ar_{17}$

$^{24}Mg(^{16}O,\alpha n\gamma)$ 2007De14,2005DeZZ



$^{35}_{18}Ar_{17}$

^{32}S(α,n) 1963Ne05

1963Ne05: A 12.20-MeV α beam was produced by the Tandem Van de Graaff Accelerator at Florida State University. Targets were natural sulfur. Positrons were detected using a Ne102 plastic scintillator. Annihilation γ rays were detected using a NaI crystal. The half-life of ^{35}Ar(g.s.) was measured with a Technical Measurements Corporation 256 channel analyzer and multiscaler logic unit. Counting time per channel was determined by the sweep of a Tektronix Oscilloscope and measured using the standard frequency output of a Hewlett-Packard Electronic Counter Model 524-C. Changes in slope of the excitation curve are interpreted as excited-state thresholds due to excited levels in the product nucleus ^{35}Ar. The threshold of ^{35}Cl(p,n)^{35}Ar(g.s.) using a thick AgCl target was also measured to confirm the ^{35}Ar ground state mass.

^{35}Ar isotope discovery: ^{32}S(α,n)^{35}Ar at Purdue (1940Ki12,1941Ki01,1941El04).

^{35}Ar Levels

E(level)†	$T_{1/2}$	Comments
0	1.76 s *3*	$T_{1/2}$: from decay curve in 1963Ne05. Threshold $E(\alpha)_{lab}$=9.846 MeV *20*.
890 *50*		Threshold $E(\alpha)_{lab}$=10.69 MeV *50*.
2030 *80*		Threshold $E(\alpha)_{lab}$=11.97 MeV *80*.

† From 1963Ne05.

$^{33}S(^{3}He,n\gamma)$ 1975Da14

$^{33}S+2p\rightarrow^{35}Ar$ from $J^{\pi}=3/2^{+}$ ^{33}S ground state.

1975Da14: a $^{3}He^{+}$ beam was produced from the University of Alberta Van de Graaff accelerator. Targets were 150 $\mu g/cm^{2}$ layers of Ag_2S (59% ^{33}S) on silver backings. At $E(^{3}He)$=6.375 MeV, neutrons were detected using an NE213 liquid scintillator placed at θ_{lab}=0°, 10°, and 20°. At $E(^{3}He)$=6.660 MeV, neutrons were detected using the NE213 scintillator placed at θ_{lab}=0° and 20°. At $E(^{3}He)$=6.390 MeV, neutrons were detected using the NE213 scintillator placed at θ_{lab}=0° and neutron-coincident γ rays were detected using a 10% efficient Ge(Li) detector placed at 90°. Measured time-of-flight (TOF) spectra of neutrons, $\sigma(E_n,\theta)$, Eγ, nγ-coin. Deduced Q values, levels, L_{2p}, J, π, and isospin.

^{35}Ar Levels

E(level)†	J^{π}	L‡	Comments
0	3/2+	(0)	J^{π}: mirror 3/2+ ^{35}Cl g.s.
1184.2 *6*			
1749.8 *9*			
2600.8 *15*			J^{π}: 1975Da14 observed $L(^{3}He,n)$=(0) for a 2.60-MeV level in ^{35}Ar, implying the existence of a 3/2+ level, which possibly corresponds to the 3/2+ 2638-keV level in ^{35}Ar.
3195.8 *11*			E(level): deduced by evaluators from the unplaced 1446.0γ.
5537 *25*	(3/2+)	(0)	T=3/2 E(level): from measured neutron energy corresponding to this level. J^{π}: $(\pi 1d_{3/2})^2(\nu 1d_{3/2})^1$ configuration formed by the $J^{\pi}=0^{+}$, T=1 $(\pi 1d_{3/2})^2$ di-proton transfer from ^{3}He to ^{33}S of $^{32}S\otimes(\nu 1d_{3/2})^1$ configuration. Tentative first T=3/2 state in ^{35}Ar.

† From 1975Da14 based on measured Eγ, but the Eγ values are not provided in 1975Da14, unless otherwise noted.
‡ the observed maximum at θ=0° in $\sigma(E_n,\theta)$ implies L=0.

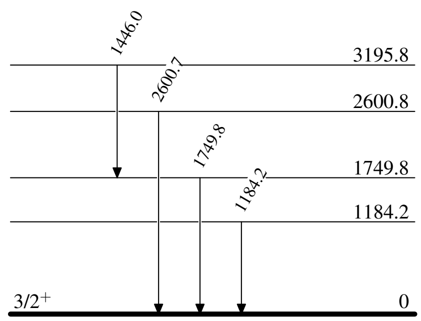
$\gamma(^{35}Ar)$

Eγ†	E_i(level)	E_f	J^{π}_f	Comments
1184.2	1184.2	0	3/2+	
1446.0 *6*	3195.8	1749.8		Unplaced γ in 1975Da14. Evaluators placed it based on the Adopted Levels from $^{16}O(^{24}Mg,\alpha n\gamma)^{35}Ar$ (2004Ek01) and $^{24}Mg(^{16}O,\alpha n\gamma)^{35}Ar$ (2007De14).
1749.8	1749.8	0	3/2+	
2600.7	2600.8	0	3/2+	

† Eγ values without uncertainties are deduced by evaluators from the level-energy differences from 1975Da14.

$^{33}S(^{3}He,n\gamma)$ 1975Da14

Level Scheme



$^{35}_{18}Ar_{17}$

^{35}Cl(^{3}He,t) 1976Be08

J^{π}=3/2^{+} for ^{35}Cl ground state.

1976Be08: ^{35}Cl(^{3}He,t)^{35}Ar is studied in an attempt to search for the member of T=3/2 isobaric quartets in ^{35}Ar. A 35-MeV ^{3}He beam from the Michigan State University cyclotron impinged on a 200 μg/cm^{2} Li-^{35}Cl target. Tritons were detected using a scintillator-proportional counter detector system at the focal plane of the Enge split-pole spectrograph. A T=3/2, 3/2^{+} level at 5537 keV *25* in ^{35}Ar was observed in ^{33}S(^{3}He,n)^{35}Ar (1975Da14) but causing an IMME breakdown. 1976Be08 measured the ^{35}K g.s. mass using ^{40}Ca(^{3}He,^{8}Li)^{35}K and predicted the T=3/2 member in ^{35}Ar to be 5579 keV *14*. 1976Be08 did not find new peaks in ^{35}Cl(^{3}He,t)^{35}Ar between 5484 keV *10* and 5591 keV *10* that were already known from ^{36}Ar(^{3}He,α)^{35}Ar (1973Be26).

^{35}Ar Levels

E(level)†
4721
4782
5116
5205
5387
5484
5591
5911
6033

† Peaks are seen in triton spectrum in 1976Be08 with no energy values reported. The authors mark those peaks with numbers corresponding to those listed in TABLE I of 1973Be26 (not the same authors) in ^{36}Ar(^{3}He,α). The energy values quoted here are taken by the evaluators from that table.

^{36}Ar(p,d) 1968Jo04,1968Ko11

$J^{\pi}=0^{+}$ for ^{36}Ar ground state.

1968Jo04: a 27.5-MeV *1* proton beam was produced by the 1.3-m FFAG cyclotron at the Nuclear Physics Laboratory at the University of Colorado. Targets were natural argon gas (99.6% ^{40}Ar) and 99.0% enriched ^{36}Ar gas. Deuterons were detected using a telescope of a 0.228-mm transmission surface barrier detector and a 0.37-cm lithium-drifted stopping counter with FWHM=120 keV. Measured $\sigma(E_d,\theta)$. Deduced levels, L-transfers, and spectroscopic factors from finite-range DWUCK-DWBA analysis of the measured $\sigma(\theta)$.

1968Ko11: a 33.6-MeV proton beam was produced by the 64-in. sector-focusing cyclotron at Michigan State University. The target was >99% isotopically enriched ^{36}Ar gas. Deuterons were detected using a telescope of a 279-μm silicon surface barrier detector and a 3-mm lithium-drifted silicon counter with FWHM=130 keV. Measured $\sigma(E_d,\theta)$. Deduced levels, J, π, L-transfers, and spectroscopic factors from zero-range Macefield-DWBA analysis of the measured $\sigma(\theta)$.

^{35}Ar Levels

Spectroscopic factor $C^2S=\sigma(\theta)_{exp}/\sigma(\theta)_{DWBA}/N$, where N=1.65*3/2 is a normalization factor adopted by 1968Jo04. The DWBA calculation in 1968Ko11 assumed neutron-well radius parameter r_{0n}=1.15 fm for $d_{5/2}$ pickup and 1.35 fm for $d_{3/2}$ pickup, respectively, to reproduce the forward-angle J-dependence observed in the L=2 deuteron angular distributions. 1968Ko11 demonstrates that a larger radius results in a larger DWBA-calculated cross section and this gives rise to problems in extracting absolute spectroscopic factors, and therefore, the evaluators put the C^2S from 1968Ko11 in comments. 1968Ko11 states that there is some evidence that the radius parameter corresponding to the best DWBA fit to the data might also result in the most trustworthy value for the spectroscopic factor.

E(level)†	J^{π}#	L‡	C^2S‡	Comments
0	$3/2^+$	2	2.92	C^2S: others: 3.03 for 1.15 fm and 1.76 for 1.35 fm (better fit) (1968Ko11).
1180 *20*	$1/2^+$	0	2.50	E(level): also from 1968Ko11. C^2S: others: 1.29 for 1.15 fm and 1.05 for 1.35 fm (1968Ko11).
1700 *30*	$(5/2^+)$		0.1	C^2S: from 1968Ko11 for radius parameter=1.15 fm.
2615 *20*	$(3/2)^+$	2		E(level): weighted average of 2630 *20* (1968Jo04) and 2600 *20* (1968Ko11). J^{π}: spin=(3/2) based on the observed J-dependence of the deuteron angular distributions (1968Ko11). L: from 1968Ko11. Other: 1 from 1968Jo04 is inconsistent. 1973Be26 determined L(n)=2 from ^{36}Ar(^{3}He,α)^{35}Ar and states that L=1 is probably in error as this level corresponds well with the known $3/2^+$ state at 2695 keV in ^{35}Cl and there are no nearby negative-parity states in ^{35}Cl. C^2S: 0.42 for 1.15 fm and 0.28 for 1.35 fm (better fit) for L=2 from 1968Ko11. Other: 0.12 for L=1 from 1968Jo04.
2970 *20*	$(5/2)^+$	2	2.47	E(level): weighted average of 2990 *20* (1968Jo04) and 2950 *20* (1968Ko11). J^{π}: spin=(5/2) based on the observed J-dependence of the deuteron angular distributions (1968Ko11). C^2S: assuming $5/2^+$ (1968Jo04). Other: 3.10 assuming $3/2^+$ (1968Jo04). 2.31 for 1.15 fm (better fit) and 1.53 for 1.35 fm from 1968Ko11.
3200 *20*	$7/2^-$	3	0.63	E(level): weighted average of 3210 *20* (1968Jo04) and 3190 *20* (1968Ko11). C^2S: others: 0.64 for 1.15 fm and 0.37 for 1.35 fm (better fit) (1968Ko11).
4756 *28*	$(1/2^+)$	0	0.172	E(level): weighted average of 4770 *20* (1968Jo04) and 4700 *40* (1968Ko11). C^2S: others: 0.05 for 1.15 fm and 0.04 for 1.35 fm (1968Ko11).
5102 *20*	$7/2^-$	3	0.46	E(level): weighted average of 5110 *20* (1968Jo04) and 5070 *40* (1968Ko11). L: From 1968Jo04. L is not reported in 1968Ko11. 1973Be26 determined L=2 from ^{36}Ar(^{3}He,α)^{35}Ar and states that L=2 gives a much better account of the data than L=3 does.
5400 *50*				E(level): From 1968Ko11 only.
5598 *20*	$3/2^+,5/2^+$	2	2.93,2.37	E(level): weighted average of 5610 *20* (1968Jo04) and 5570 *30* (1968Ko11). C^2S: others: 1.77 for 1.15 fm (better fit) and 1.25 for 1.35 fm, both for $J^{\pi}=(3/2,5/2)^+$ (1968Ko11).
6024 *20*	$3/2,5/2^+$	2	1.58,1.31	E(level): weighted average of 6030 *20* (1968Jo04) and 6010 *30* (1968Ko11).

Continued on next page (footnotes at end of table)

^{36}Ar(p,d) 1968Jo04,1968Ko11 (continued)

^{35}Ar Levels (continued)

E(level)†	J$^{\pi}$#	L‡	C^{2}S‡	Comments
				C^{2}S: others: 1.18 for 1.15 fm and 0.83 for 1.35 fm (better fit), both for J^{π}=(3/2,5/2)$^+$ (1968Ko11).
6620 *30*	(1/2$^+$)	0		E(level),L: From 1968Ko11 only. C^{2}S: 0.24 for 1.15 fm and 0.19 for 1.35 fm (1968Ko11).
6700 *20*	7/2$^-$	3	2.60	L,C^{2}S: From 1968Jo04 only.
6820 *30*		2		E(level),L: From 1968Ko11 only. C^{2}S: 0.72 for 1.15 fm (better fit) and 0.51 for 1.35 fm, both for J^{π}=(3/2,5/2)$^+$ (1968Ko11).
7030 *20*	(3/2,5/2)$^+$	2	2.20,1.79	

† From 1968Jo04, unless otherwise noted.
‡ From DWBA analysis of measured $\sigma(\theta)$. Values of L-transfers are from both 1968Jo04 and 1968Ko11 and values of C^{2}S are from 1968Jo04, unless otherwise noted. Where two C^{2}S values are given, they correspond to respective J^{π} values as given.
# Assumed in 1968Jo04 and/or 1968Ko11 for extracting C^{2}S, unless otherwise noted.

^{36}Ar(d,t) 1970Wh04,2015Fr01

$J^\pi=0^+$ for ^{36}Ar ground state.

1970Wh04: A 21.0-MeV deuteron beam was produced by the Yale MP tandem Van de Graaff accelerator. The target was a ^{36}Ar gas cell. Tritons were detected using a 140-μm-530-μm thick ΔE-E telescope of silicon surface barrier detectors with FWHM=65-70 keV. Measured $\sigma(E_t,\theta)$. Deduced levels, L-transfers, and spectroscopic factors from JULIE-DWBA analysis of the measured $\sigma(\theta)$. Comparisons with shell-model calculations.

2015Fr01: A 22-MeV deuteron beam was produced by the MP tandem Van de Graaff accelerator at the Maier-Leibnitz Laboratorium (MLL) in Garching, Germany. Targets were fabricated by implanting 25-70-keV 3-6 μg/cm^2 of ^{36}Ar ions into 30 μg/cm^2 natural abundance carbon foils. Reaction products were momentum analyzed by a Q3D magnetic spectrograph. Tritons were detected using a multiwire gas-filled proportional counter backed by a scintillator at the focal plane. Measured E_t at θ_{lab}=15°, 20°, 25° with FWHM≈9 keV and at 54° with FWHM≈16 keV. Deduced levels, proton resonance energies, level densities. Comparisons with shell-model calculations.

^{35}Ar Levels

Spectroscopic factor $C^2S=\sigma(\theta)_{exp}/\sigma(\theta)_{DWBA}/N$, where N=3.33 is a normalization factor adopted by 1970Wh04 from 1966Ba54.

E(level)†	J^π‡	L#	C^2S#	Comments
0	3/2$^+$	2	3.4	
1180 *10*	1/2$^+$	0	1.4	
1700	(5/2$^+$)	(2)	<0.2	
2635 *20*	(3/2$^+$)	(2)	0.5	C^2S: 1970Wh04 authors state that there is a large uncertainty in the spectroscopic strength. 1970Wh04 also gives S=0.11 or 0.032 assuming L=1.
2986 *20*	5/2$^+$	2	2.6	
3200 *20*	7/2$^-$	(3)	0.33,0.11	C^2S: assuming r_{0n}=1.25 F and V_n≈60 MeV, respectively. 1970Wh04 authors state that there is a large uncertainty in the spectroscopic strength.
5913 *5*				
5991 *3*				
6037 *3*				May be a doublet (2015Fr01).
6055? *3*				Tentative (2015Fr01).
6076 *3*				
6164 *3*				
6253 *3*				
6273 *3*				
6302 *3*				
6332 *3*				
6345 *3*				
6415 *2*				
6439? *4*				Tentative (2015Fr01).
6460 *3*				
6523 *3*				
6557 *3*				
6585 *3*				
6606 *3*				
6617 *2*				
6644 *3*				
6651 *3*				
6672 *3*				

† From 1970Wh04 for low-lying states up to 3200 keV and from 2015Fr01 for others.
‡ As given in 1970Wh04, used for extracting C^2S.
# From DWBA analysis of the measured $\sigma(\theta)$ in 1970Wh04.

^{36}Ar(^{3}He,α) 1973Be26,1998VoZY

J^π=0$^+$ for ^{36}Ar ground state.

1973Be26: an 18-MeV ^{3}He beam was produced by the University of Pennsylvania Tandem Van de Graaff accelerator. The target was pure argon gas enriched to 99.8% in ^{36}Ar. α particles were momentum analyzed in a multi-angle spectrograph and detected using Ilford K-1 nuclear emulsions with FWHM=35 keV. Measured $\sigma(E_\alpha,\theta)$. Deduced levels, J, π, L-transfers, and spectroscopic factors from local zero-range DWUCK-DWBA analysis of the measured $\sigma(\theta)$. Comparisons with shell-model calculations and the mirror nucleus ^{35}Cl. Also see 1972MiZO.

1998VoZY: a 25-MeV ^{3}He beam at 0.5-μA intensity was provided from the FN tandem accelerator at the Nuclear Structure Laboratory of the University of Notre Dame. The targets were made by implanting 75-keV ^{36}Ar into 40 μg/cm^2 carbon foils from the University of Toledo ion source. The reaction products were momentum analyzed by the Notre Dame broad-range magnetic spectrograph. The particles were subsequently detected at the focal surface of the spectrograph by a position sensitive proportional gas detector backed with a plastic scintillator. Measured E_α. Deduced levels.

^{35}Ar Levels

Spectroscopic factor $C^2S=(2j+1)\times\sigma(\theta)_{exp}/\sigma(\theta)_{DWBA}/N$, where the isospin Clebsch-Gordan coefficient C^2 is 1/2 in this case, j is the total angular momentum of the transferred neutron, and the normalization factor N=16.8. 1973Be26 states that the overall normalization for the (^{3}He,α) reaction is not well determined and therefore resort to empirical means to determine N. N=15.5 deduced from shell-model calculated total S=3.52 for all four 1/2$^+$ states and the 1973Be26 measured NS=54.6. N=18.1 deduced from the ^{35}Cl(^{3}He,d)^{36}Ar(g.s.) S=4.73 (1970Mo10) and the 1973Be26 measured ^{36}Ar(^{3}He,α)^{35}Ar(g.s.) NS=85.4. 1973Be26 adopted the average N=16.8.

E(level)	J^π‡	L†	C^2S†	Comments
0	3/2$^+$	2	2.545	
1175 *2*	1/2$^+$	0	1.19	E(level): weighted average of 1179 *10* (1973Be26) and 1175 *2* (1998VoZY).
1747 *2*	5/2$^+$	2	0.025	E(level): weighted average of 1738 *10* (1973Be26) and 1747 *2* (1998VoZY).
2649 *2*	3/2$^+$	2	0.57	E(level): weighted average of 2637 *10* (1973Be26) and 2649 *2* (1998VoZY).
2983 *2*	5/2$^+$	2	1.39	E(level): weighted average of 2982 *10* (1973Be26) and 2983 *2* (1998VoZY).
3197 *2*	7/2$^-$	3	0.39	E(level): weighted average of 3193 *10* (1973Be26) and 3197 *2* (1998VoZY).
3882 *5*	1/2$^+$	0	0.02	E(level): weighted average of 3884 *10* (1973Be26) and 3881 *5* (1998VoZY).
4001 *3*	(3/2)$^-$	1	0.065	E(level): weighted average of 4012 *10* (1973Be26) and 4000 *3* (1998VoZY).
4113 *4*				E(level): weighted average of 4110 *10* (1973Be26) and 4113 *4* (1998VoZY).
4135 *4*	(3/2)$^-$	1	0.025	E(level): weighted average of 4142 *10* (1973Be26) and 4134 *4* (1998VoZY).
4350 *6*				E(level): weighted average of 4350 *10* (1973Be26) and 4350 *6* (1998VoZY).
4515 *5*				E(level): weighted average of 4530 *10* (1973Be26) and 4514 *3* (1998VoZY).
4713 *6*	1/2$^+$	0	0.05	E(level): weighted average of 4721 *10* (1973Be26) and 4710 *6* (1998VoZY).
4774 *6*				E(level): weighted average of 4782 *10* (1973Be26) and 4771 *6* (1998VoZY).
5059 *11*				E(level): unweighted average of 5048 *10* (1973Be26) and 5069 *4* (1998VoZY).
5116 *2*	(3/2,5/2)$^+$	2	0.25,0.145#	E(level): weighted average of 5116 *10* (1973Be26) and 5116 *2* (1998VoZY).
5207 *3*				E(level): weighted average of 5205 *10* (1973Be26) and 5207 *3* (1998VoZY).
5389 *8*				E(level): weighted average of 5387 *10* (1973Be26) and 5391 *8* (1998VoZY).
5482 *2*	(3/2,5/2)$^+$	2	0.77,0.445#	E(level): weighted average of 5484 *10* (1973Be26) and 5482 *2* (1998VoZY).
5594 *2*	(3/2,5/2)$^+$	2	1.98,1.14#	E(level): weighted average of 5591 *10* (1973Be26) and 5594 *2* (1998VoZY).
5916 *3*				E(level): weighted average of 5911 *10* (1973Be26) and 5916 *3* (1998VoZY).
6036 *3*	(3/2,5/2)$^+$	2	1.3,0.755#	E(level): weighted average of 6033 *10* (1973Be26) and 6036 *3* (1998VoZY).
6162 *2*				E(level): weighted average of 6153 *10* (1973Be26) and 6162 *2* (1998VoZY).
6262 *10*				E(level): weighted average of 6258 *10* (1973Be26) and 6267 *12* (1998VoZY).
6615 *3*	1/2$^+$	0	0.36	E(level): weighted average of 6631 *10* (1973Be26) and 6614 *2* (1998VoZY). Probable doublet in 1973Be26.
6823 *2*				E(level): weighted average of 6827 *10* (1973Be26) and 6823 *2* (1998VoZY).
6948 *2*				E(level): weighted average of 6959 *10* (1973Be26) and 6948 *2* (1998VoZY).
7043 *4*				E(level): weighted average of 7055 *10* (1973Be26) and 7042 *3* (1998VoZY).
7117† *10*				
7293† *10*				

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^{36}Ar(^{3}He,α) 1973Be26,1998VoZY (continued)

^{35}Ar Levels (continued)

E(level)
7423† *10*
7502† *10*
7840† *10*
8019† *10*

† From 1973Be26. L and C^2S values are extracted from DWBA analysis of measured $\sigma(\theta)$. Quoted values of C^2S are converted from the S values in 1973Be26 with C^2=1/2.

‡ As given in 1973Be26, used for extracting C^2S.

1973Be26 states that the differences for j=3/2 and 5/2 are small in the DWBA-calculated L=2 shapes. It is not possible to differentiate between the two allowed j values for L=2 transitions. Both C^2S values are given for each level with two spin values. Assuming that all four levels have spins of 3/2 would lead to a summed L=2 C^2S that exceeds the simple shell-model sum rule limit of 8 for combined $1d_{3/2}$ and $1d_{5/2}$ pickup, which suggests that all four of these levels probably have $5/2^+$.

Adopted Levels

Q(β^-)=−1.595×10^4 *11*; S(n)=17757 *17*; S(p)=83.6 *5*; Q(α)=−6563 *3* 2021Wa16

Q(β^-),S(n): Deduced by evaluators using mass excesses of 4777 *105* for ^{35}Ca measured by 2023La09, and −1487 *17* for ^{34}K measured by 2024Dr01; −11172.9 *5* for ^{35}K from 2021Wa16. Values from 2021Wa16: Q(β^-)=−16360 *200* (syst), S(n)=18020 *200* (syst).

S(2n)=34860 *200* (syst), S(2p)=4747.5 *6*, Q(ε)=11874.4 *9*, Q(εp)=5978.2 *5* (2021Wa16).

Isotope discovery (2012Th10): ^{40}Ca(^{3}He,^{8}Li)^{35}K at Michigan State (1976Be08).

^{35}K decay measurements:

1980Ew02,1979Ca15: ^{35}K produced via ^{45}Sc(p,8n3p) spallation at CERN. Measured $T_{1/2}$ and $\varepsilon+\beta^+$-delayed protons and γ rays.

2018Sa54,2019ChZU: ^{35}K produced via ^{1}H(^{36}Ar,^{35}K)2n at Texas A&M. Measured $T_{1/2}$ and $\varepsilon+\beta^+$-delayed protons and γ rays.

1998Sc19: Polarized ^{35}K produced via fragmentation of ^{40}Ca on ^{9}Be target at GSI. Measured $T_{1/2}$ and g-factor of ^{35}K ground state from β-NMR.

2006Me04: Polarized ^{35}K produced via ^{36}Ar(^{9}Be,^{10}Li)^{35}K at NSCL, MSU. Measured g-factor of ^{35}K ground state from β-NMR.

^{35}K mass measurements: 2023Zh10, 2007Ya08, 1976Be08.

Theoretical calculations (binding energies, moments, radii, levels, J, π, mass, $T_{1/2}$, reaction rates, etc.): 2025Sh19, 2023Bo17, 2023Fo05, 2022Zo01, 2020Ma25, 2016Si02, 2016Me17, 2003Sm02.

^{35}K Levels

Cross Reference (XREF) Flags

A	^{35}Ca $\varepsilon+\beta^+$ decay (25.7 ms)
B	^{9}Be(^{36}Ca,^{35}K)
C	^{40}Ca(^{3}He,^{8}Li)

E(level)†	J$^\pi$‡	$T_{1/2}$	XREF	Comments
0.0	3/2$^+$	175 ms *2*	ABC	%ε+%β^+=100; %εp=0.37 *15* μ=(+)0.392 *7* (2006Me04,2019StZV) J$^\pi$: L(^{36}Ca,^{35}K)=2 from 0$^+$ and allowed $\varepsilon+\beta^+$ feeding to 4725.9, 1/2$^+$ level in ^{35}Ar. Possible mirror level of 3/2$^+$ ^{35}S g.s. $T_{1/2}$: weighted average of 175 ms *2* (2018Sa54), 178 ms *8* (1998Sc19), and 190 ms *30* (1980Ew02). μ: From β-NMR spectroscopy (2006Me04). Other: 0.36 *3* (1998Sc19, β-NMR spectroscopy). The positive sign is based on the mirror ^{35}S g.s. %εp: from 1980Ew02 for E(p)>0.9 MeV. E(p)<0.9 MeV has also been observed (2018Sa54,2019ChZU).
1553 *5*	(1/2)$^+$		A C	E(level): other: 1560 *40* from (^{3}He,^{8}Li). J$^\pi$: allowed $\varepsilon+\beta^+$ feeding from 3/2$^+$ parent; possible mirror level of 1572, 1/2$^+$ level in ^{35}S (1976Be08).
2690 *50*			C	E(level): from (^{3}He,^{8}Li) only. Possible mirror level of 2717, 5/2$^+$ level in ^{35}S (1976Be08).
3781 *26*	1/2$^+$,3/2$^+$		A	
4018 *37*	1/2$^+$,3/2$^+$		A	
4788 *49*	1/2$^+$,3/2$^+$		A	
4982 *13*	1/2$^+$,3/2$^+$		A	
5249 *73*	1/2$^+$,3/2$^+$		A	
5533 *49*	1/2$^+$,3/2$^+$		A	
5710 *49*	1/2$^+$,3/2$^+$		A	
5865 *38*	1/2$^+$,3/2$^+$		A	
6089 *62*	1/2$^+$,3/2$^+$		A	
6335 *73*	1/2$^+$,3/2$^+$		A	
9168 *23*	1/2$^+$		A	T=5/2 J$^\pi$,T: isobaric analog state of 1/2$^+$ ^{35}Ca g.s. (1985Ay01,1999Tr04).

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Adopted Levels (continued)

^{35}K Levels (continued)

† From ^{35}Ca $\varepsilon+\beta^+$ decay, unless otherwise noted.
‡ Allowed $\varepsilon+\beta^+$ feeding from $1/2^+$ ^{35}Ca parent, unless otherwise noted.

^{35}Ca ε+β^+ decay (25.7 ms) 1999Tr04,1985Ay01

Parent: ^{35}Ca: E=0; J$^\pi$=1/2$^+$; T$_{1/2}$=25.7 ms *2*; Q(ε+β^+)=1.595×10^4 *11*; %ε+β^+ decay=100

^{35}Ca-J$^\pi$,T$_{1/2}$: From the Adopted Levels of ^{35}Ca.

^{35}Ca-Q(ε+β^+): 15950 *105* deduced by evaluators from mass excesses of 4777 *105* for ^{35}Ca measured by 2023La09 and −11172.9 *5* for ^{35}K from 2021Wa16. Q(ε) from 2021Wa16: 16360 *200* (syst).

1999Tr04, 1998Le45: A secondary ^{35}Ca beam at 0.3 ions/s and 98% purity was produced via the projectile fragmentation of a 95-MeV/nucleon ^{40}Ca^{20+} primary beam impinging on a rotating ^{nat}Ni target, selected using ΔE-ToF by the GANIL LISE3 spectrometer, and implanted into a 500-μm thick silicon detector sandwiched between two 500-μm thick silicon detectors for detecting β^+ particles. 3.5×10^4 ^{35}Ca ions were stopped at a depth of 300 μm with FWHM=70 μm (setting 1) and 2.5×10^4 ^{35}Ca ions were stopped at a depth of 450 μm (setting 2). ε+β^+-delayed protons were detected by the implantation detector. γ rays were detected by three Ge detectors and two NaI detectors. Measured E$_p$, I$_p$, Eγ, Iγ, E$_{2p}$, I$_{2p}$, βp-coin, pγ-coin. Built the decay scheme consisting of 1p-emitting states in ^{35}K, a 2p-emitting state (T=5/2 IAS) in ^{35}K, 1p daughter states in ^{34}Ar, and a 2p daughter state in ^{33}Cl. Deduced decay branching ratios, B(GT) and B(F), and parent ^{35}Ca T$_{1/2}$ from implant-decay correlations.

1985Ay01: ^{35}Ca isotope discovery. ^{35}Ca was produced by bombarding a natural calcium target using a 135-MeV ^{3}He beam from the 88-inch Cyclotron at Lawrence Berkeley Laboratory. Recoiling products were slowed down, transported, and collected on a slowly rotating catcher wheel. The ε+β^+-delayed protons were detected using Si detector telescopes. Measured E$_p$, I$_p$, pp-coin. Built the decay scheme consisting of a 2p-emitting state (T=5/2 IAS) in ^{35}K, sequential 2p intermediate states in ^{34}Ar, and 2p daughter states in ^{33}Cl. Deduced ^{35}Ca T$_{1/2}$ and ^{35}Ca mass using the known members of A=35, T=5/2 sextuplets IMME.

Theoretical studies involving ^{35}Ca decay: 2003Sm02, 1991De26, 1990Br26.

The decay scheme is considered relatively complete as all observed individual I(1p) and I(2p) add up to almost 100% in 1999Tr04.

^{35}K Levels

E(level)†	J$^\pi$#	T$_{1/2}$#	Comments
0	3/2$^+$	175 ms *2*	
1553 *5*	(1/2)$^+$		E(p0)$_{lab}$=1427 *5*, proton line 1 in 1999Tr04.
3781 *26*	1/2$^+$,3/2$^+$		E(p0)$_{lab}$=3592 *25*, proton line 5 in 1999Tr04.
4018 *37*	1/2$^+$,3/2$^+$		E(p0)$_{lab}$=3822 *36*, proton line 6 in 1999Tr04.
4520‡			E(p1)$_{lab}$=1909-2647, proton group 2 in 1999Tr04, corresponding to E(level)=4140-4900.
4788 *49*	1/2$^+$,3/2$^+$		E(p0)$_{lab}$=4570 *48*, proton line 9 in 1999Tr04.
4982 *13*	1/2$^+$,3/2$^+$		E(p0)$_{lab}$=4754 *38*, proton line 10 in 1999Tr04, corresponding to E(level)=4977 *39*. E(p1)$_{lab}$=2727 *13*, proton line 3 in 1999Tr04, corresponding to E(level)=4982 *13*. E(level): weighted average of the two E(level) values of 4977 *39* (p0) and 4982 *13* (p1).
5249 *73*	1/2$^+$,3/2$^+$		E(p0)$_{lab}$=5018 *71*, proton line 11 in 1999Tr04.
5493‡			E(p1)$_{lab}$=2947-3500, proton group 4 in 1999Tr04, corresponding to E(level)=5208-5778.
5533 *49*	1/2$^+$,3/2$^+$		E(p0)$_{lab}$=5294 *48*, proton line 12 in 1999Tr04.
5710 *49*	1/2$^+$,3/2$^+$		E(p0)$_{lab}$=5466 *48*, proton line 13 in 1999Tr04.
5716‡			E(p2)$_{lab}$=1909-2647, proton group 2 in 1999Tr04, corresponding to E(level)=5336-6096.
5865 *38*	1/2$^+$,3/2$^+$		E(p0)$_{lab}$=5616 *37*, proton line 14 in 1999Tr04.
6089 *62*	1/2$^+$,3/2$^+$		E(p0)$_{lab}$=5834 *60*, proton line 15 in 1999Tr04.
6302‡			E(p3)$_{lab}$=1909-2647, proton group 2 in 1999Tr04, corresponding to E(level)=5922-6681.
6335 *73*	1/2$^+$,3/2$^+$		E(p1)$_{lab}$=4041 *71*, proton line 7 in 1999Tr04.
6585‡			E(p0)$_{lab}$=5983-6649, proton group 16 in 1999Tr04, corresponding to E(level)=6243-6928.
7813‡			E(p0)$_{lab}$=7131-7887, proton group 18 in 1999Tr04, corresponding to E(level)=7424-8203.
9168 *23*	1/2$^+$		T=5/2 E(level): weighted average of the three E(level) values of 9144 *92* (p0), 9157 *23* (p1), and 9186 *27* (2p0). J$^\pi$,T: isobaric analog state of 1/2$^+$ ^{35}Ca g.s. (1985Ay01,1999Tr04). E(p0)$_{lab}$=8802 *89*, proton line 19 in 1999Tr04, corresponding to E(level)=9144 *92*. E(p1)$_{lab}$=6783 *22*, proton line 17 in 1999Tr04, corresponding to E(level)=9157 *23*. E(2p0)$_{lab}$=4305 *26*, proton line 8 in 1999Tr04, corresponding to E(level)=9186 *27* with S(2p)(^{35}K)=4747.5 *6* (2021Wa16) and adding a +7−keV correction for the difference in the recoil effect between 1p and 2p emissions (1999Tr04).

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^{35}Ca ε+β^+ decay (25.7 ms) 1999Tr04,1985Ay01 (continued)

^{35}K Levels (continued)

E(level)†	Jπ#	$T_{1/2}$#	Comments
			Others: $E(2p0)_{lab}$=4089 *30*, $E(2p0)_{c.m.}$=4311 *40* (1985Ay01), corresponding to E(level)=9059 *41* in ^{35}K with S(2p)(^{35}K)=4747.5 *6* (2021Wa16). 1985Ay01 observed $E(2p1)_{lab}$=3287 *30* and proposed both 2p0 and 2p1 proceed via a sequential decay mechanism with the first proton $E(p)_{lab}$=2213 keV, corresponding to an intermediate state in ^{34}Ar at 6807 keV. However, 2p1 has been ruled out in 1999Tr04 due to the nonobservation of expected pγ coincidences. 1999Tr04 also states that the observed ratio I(2p0)/I(p)=0.98 *9* agrees with the calculated branching ratio I(2p)/I(p)=1 for the IAS (1991De26).

† Evaluators deduced E(level)(^{35}K)=$E(p)_{lab}$×[m(p)+m(^{34}Ar)]/m(^{34}Ar)+S(p)(^{35}K)+E(level)(^{34}Ar), with S(p)(^{35}K)=83.6 *5* (2021Wa16), E(level)(^{34}Ar) from 2012Ni10, and $E(p)_{lab}$ from 1999Tr04. For a ^{35}K proton-emitting level with multiple proton branches, evaluators take the weighted average for E(level)(^{35}K) values deduced from each proton branch. 1999Tr04 used S(p)(^{35}K)=78 *20* from 1993Au07, which causes a small difference between the original E(level)(^{35}K) in 1999Tr04 and the deduced E(level)(^{35}K) here.

‡ Unresolved proton-emitting levels corresponding to a group of unresolved protons populating one daughter state in ^{34}Ar, which are not included in the Adopted Levels.

From the Adopted Levels.

ε,β^+ radiations

E(decay)	E(level)	Iβ$^+$ #	Iε#	Log *ft*	I(ε+β$^+$)†#	Comments
(6.78×10^3 *11*)	9168	8.4 *4*	0.0056 *9*	3.3 *1*	8.4 *4*	I(ε+β$^+$): %I(p0)=0.41 *6*, %I(p1)=3.8 *2*, %I(2p0)=4.2 *3*.
(8.14×10^3 *11*)	7813	1.1 *2*	4.0×10^{-4} *9*	4.6	1.1‡ *2*	
(9.37×10^3 *11*)	6585	1.08 *17*	2.5×10^{-4} *5*	4.9	1.08‡ *17*	
(9.62×10^3 *13*)	6335	2.9 *3*	6.1×10^{-4} *10*	4.6 *1*	2.9 *3*	
(9.65×10^3 *11*)	6302	2.0 *7*	4×10^{-4} *2*	4.7	2.0‡ *7*	
(9.86×10^3 *13*)	6089	1.39 *19*	2.7×10^{-4} *5*	4.9 *1*	1.39 *19*	
(1.009×10^4 *12*)	5865	1.42 *17*	2.6×10^{-4} *5*	5.0 *1*	1.42 *17*	
(1.023×10^4 *11*)	5716	1.0 *4*	1.7×10^{-4} *7*	5.2	1.0‡ *4*	
(1.024×10^4 *12*)	5710	0.61 *15*	1.1×10^{-4} *3*	5.4 *+2−1*	0.61 *15*	
(1.042×10^4 *12*)	5533	0.72 *18*	1.2×10^{-4} *3*	5.4 *+2−1*	0.72 *18*	
(1.046×10^4 *11*)	5493	2.2 *3*	3.6×10^{-4} *7*	4.9	2.2‡ *3*	
(1.070×10^4 *13*)	5249	3.9 *3*	5.9×10^{-4} *9*	4.7 *1*	3.9 *3*	
(1.097×10^4 *11*)	4982	10.1 *7*	0.0014 *2*	4.32 *6*	10.1 *7*	I(ε+β$^+$): %I(p0)=4.2 *4*, %I(p1)=6.0 *5*.
(1.116×10^4 *12*)	4788	2.9 *3*	3.8×10^{-4} *6*	4.9 *1*	2.9 *3*	
(1.143×10^4 *11*)	4520	5.4 *9*	7×10^{-4} *2*	4.7	5.4‡ *9*	
(1.193×10^4 *12*)	4018	3.8 *3*	4.1×10^{-4} *6*	4.9 *1*	3.8 *3*	
(1.217×10^4 *11*)	3781	3.0 *3*	3.0×10^{-4} *5*	5.1 *1*	3.0 *3*	
(1.440×10^4 *11*)	1553	48.2 *13*	0.0030 *4*	4.26 *3*	48.2 *13*	

† Deduced from the I(p) values in 1999Tr04 multiplied by 0.994. The original %ΣI(1p)=96.4 *18* and %ΣI(2p)=4.2 *3* in 1999Tr04 lead to %ΣI(1p)+%ΣI(2p)=100.6. The evaluators perform a renormalization of 100.6 to 100, which yields the factor of 0.994. The original I(p) in 1999Tr04 was determined from the number of observed proton events and the total number of implants, with simulated proton-detection efficiencies.

‡ Feeding to a group of unresolved levels in ^{35}K. The corresponding Log *ft* values are displayed without uncertainties.

Absolute intensity per 100 decays.

^{9}Be(^{36}Ca,^{35}K) 2012Sh21

$J^\pi=0^+$ for ^{36}Ca ground state.

2012Sh21: A ^{36}Ca secondary beam was produced via the projectile fragmentation of a 140-MeV/nucleon ^{40}Ca primary beam impinging on a ^{9}Be target at NSCL, MSU. The ^{36}Ca nuclei were selected by the A1900 separator to a purity of 8%. The ground states of ^{35}K and ^{35}Ca were populated by the one-proton/neutron knockout reactions, respectively, from the ^{36}Ca beam at a midtarget energy of ≈70 MeV/nucleon on a 188-mg/cm^2 ^{9}Be secondary target. Knockout residues were identified from their energy loss measured by an ionization chamber at the focal plane of the S800 spectrometer and from their ToF measured between two scintillators at the object position and at the focal plane of the S800 spectrometer. The CsI(Na) γ-ray spectrometer CAESAR was placed around the Be target position of the S800 to search for γ decay of any excited states of the residuals formed in knockout reactions. Measured the knockout cross sections for producing ^{35}K and ^{35}Ca from ^{36}Ca and the longitudinal momentum distribution of residuals. Deduced J, π, orbital angular momenta of the nucleons removed from ^{36}Ca, and spectroscopic factors. Calculated single-particle cross sections (σ_{sp}) for proton removal and longitudinal momentum distributions using eikonal models.

^{35}K Levels

E(level)	J^π	L	C^2S†	Comments
0.0	3/2$^+$	2	3.19 *16*	J^π: interpreted as knockout from proton $1d_{3/2}$ orbital. L: 2012Sh21 deduced L by comparing the measured and eikonal-calculated longitudinal momentum distributions of residuals. C^2S: from eikonal/Hartree-Fock model with R_s=0.82 *4*, uncertainty only include experimental contributions. Others: 4.37 *24* from eikonal/Strong Absorption model with R_s=1.14 *6*; 3.62 from shell model calculations. σ_{exp}=51.1 mb *26*. Calculations: σ_{sp}=16.2 mb from eikonal/Hartree-Fock model; σ_{sp}=11.7 mb from eikonal/Strong Absorption model.

† Spectroscopic factor $C^2S=\sigma_{exp}/\sigma_{sp}$.

^{40}Ca(^{3}He,^{8}Li) **1976Be08**

1976Be08: ^{35}K isotope discovery. ^{40}Ca(^{3}He,^{8}Li)^{35}K is studied using 73.7- and 75.8-MeV ^{3}He beams produced by the Michigan State University cyclotron. Targets were 370 and 190 μg/cm^{2} enriched ^{40}Ca on a 20 μg/cm^{2} natural carbon backing. ^{8}Li particles were detected using a scintillator-proportional counter detector system at the focal plane of the Enge split-pole spectrograph. Measured the cross section for producing ^{35}K g.s. and the mass excess of ^{35}K. Also see 1976BeXJ and 1976BeZJ.

^{35}K Levels

E(level)	Comments
0	
1560 *40*	Assigned by 1976Be08 as the mirror state of 1/2^{+} level almost at the same energy in ^{35}S.
2690 *50*	Assigned by 1976Be08 as the mirror state of 5/2^{+} level almost at the same energy in ^{35}S.

Adopted Levels

Q(β^-)=−21910 *syst*; S(n)=17770 *syst*; S(p)=1.03×10^3 *11*; Q(α)=−8.94×10^3 *11* 2021Wa16

S(p),Q(α): Deduced by evaluators using mass excesses of 4777 *105* for ^{35}Ca measured by 2023La09, −1487 *17* for ^{34}K measured by 2024Dr01, and 11290 *16* for ^{31}Ar measured by 2024Yu13. Values from 2021Wa16: S(p)=880 *280* (syst), Q(α)=−8560 *280* (syst).

ΔQ(β^-)=450, ΔS(n)=360 (syst,2021Wa16).

S(2p)=417 *105*, Q(ε)=15950 *105*, Q(εp)=15866 *105*, from mass excesses of 4777 *105* for ^{35}Ca measured by 2023La09; −9384.3 *4* for ^{33}Ar, −11172.9 *5* for ^{35}K, and −18378.29 *8* for ^{34}Ar from 2021Wa16. Values from 2021Wa16: S(2p)=00 *200* (syst), Q(ε)=16360 *200* (syst), Q(εp)=16280 *200* (syst).

S(2n)=41980 *450* (syst) (2021Wa16).

Isotope discovery (2011Am01): ^{40}Ca(^{3}He,α4n)^{35}Ca at Berkeley (1985Ay01).

^{35}Ca production:

1986La17,1986AnZV: Ni(^{40}Ca,X) at GANIL.

^{35}Ca decay measurements:

1999Tr04,1998Le45: Ni(^{40}Ca,X) at GANIL. Measured $T_{1/2}$ and ε+β^+-delayed 1p/2p emissions.

^{35}Ca mass measurements: 2023La09, 1985Ay01.

Theoretical calculations: 2003Sm02, 1998Co30, 1997Co19, 1991De26, 1990Br26.

^{35}Ca Levels

Cross Reference (XREF) Flags

A ^{1}H(^{37}Ca,t)
B ^{9}Be(^{36}Ca,^{35}Ca)
C ^{9}Be(^{37}Ca,X)

E(level)	J$^\pi$	$T_{1/2}$	XREF	Comments
0.0	1/2$^+$	25.7 ms *2*	AB	%ε+%β^+=100; %εp=95.8 *3*; %ε2p=4.2 *3* J$^\pi$: L(^{36}Ca,^{35}Ca)=0 from 0$^+$. $T_{1/2}$: From implant-decay correlation in 1999Tr04. Other: 50 ms *30* estimated by comparison with the ^{22}Al yield from 1985Ay01. %εp and %ε2p are derived from the renormalization of %ΣI(1p)+%ΣI(2p)=100.6 in 1999Tr04 to 100. The original decay branching ratios in 1999Tr04: %ΣI(1p)=96.4 *18* and %ΣI(2p)=4.2 *3*.
2.09×10^3 *10*	3/2$^+$		A C	E(level): weighted average of 2.24×10^3 *33* from ^{1}H(^{37}Ca,t) and 2.08×10^3 *10* from (^{37}Ca,X). J$^\pi$: L=0 from 3/2$^+$ in ^{1}H(^{37}Ca,t).

^{1}H(^{37}Ca,t) 2023La09

J^{π}=3/2^{+} for ^{37}Ca ground state.

2023La09: A 50-MeV/nucleon ^{37}Ca secondary beam was produced via the fragmentation of a 95-MeV/nucleon $^{40}Ca^{20+}$ primary beam impinging on a ^{9}Be target, selected by the LISE3 spectrometer at GANIL, and then impinged on a liquid hydrogen cryogenic target (CRYPTA). Beam ions before target were tracked using two low-pressure multiwire proportional chambers (CATS). Heavy residuals after target were detected by a zero-degree detection system consisting of an ionization chamber, a set of two XY drift chambers, and a plastic scintillator. Tritons from 2n-transfer were detected using a set of six MUST2 DSSD-CsI telescopes in coincidence with the heavy residues of Ca or Ar. Measured $\sigma(E_t,\theta)$ in inverse kinematics. Deduced levels, J, π, L-transfers from FRESCO-DWBA analysis of measured $\sigma(\theta)$.

First measurement of ^{35}Ca mass excess: 4777 keV *105*.

First observation of excited states in ^{35}Ca.

Evidence of the magicity of N=16 close to the proton drip line.

^{35}Ca Levels

E(level)	J^{π}	L	Comments
0.0	(1/2)$^{+}$	2	J^{π}: mirror level: 1/2^{+} ^{35}P g.s. and shell model calculations. L: removal of one neutron from the $2s_{1/2}$ orbital and the other from the $1d_{3/2}$ orbital, leaving a single neutron in the $2s_{1/2}$ orbital.
2.24×10^{3} *33*	3/2^{+}	0	J^{π}: mirror level: 3/2^{+} ^{35}P first excited state at 2386.9 *11* and shell model calculations. L: removal of two neutrons from the $2s_{1/2}$ orbital, leaving a single neutron in the $1d_{3/2}$ orbital.

^{9}Be(^{36}Ca,^{35}Ca) 2012Sh21

$J^{\pi}=0^{+}$ for ^{36}Ca ground state.

2012Sh21: A ^{36}Ca secondary beam was produced via the projectile fragmentation of a 140-MeV/nucleon ^{40}Ca primary beam impinging on a ^{9}Be target at NSCL, MSU. The ^{36}Ca nuclei were selected by the A1900 separator to a purity of 8%. The ground states of ^{35}K and ^{35}Ca were populated by the one-proton/neutron knockout reactions, respectively, from the ^{36}Ca beam at a midtarget energy of ≈70 MeV/nucleon on a 188-mg/cm^2 ^{9}Be secondary target. Knockout residues were identified from their energy loss measured by an ionization chamber at the focal plane of the S800 spectrometer and from their ToF measured between two scintillators at the object position and at the focal plane of the S800 spectrometer. The CsI(Na) γ-ray spectrometer CAESAR was placed around the Be target position of the S800 to search for γ decay of any excited states of the residuals formed in knockout reactions. Measured knockout cross sections for producing ^{35}K and ^{35}Ca from ^{36}Ca and the longitudinal momentum distribution of residuals. Deduced J, π, orbital angular momenta of the nucleons removed from ^{36}Ca, and spectroscopic factors. Calculated single-particle cross sections (σ_{sp}) for neutron removal and longitudinal momentum distributions using eikonal models.

^{35}Ca Levels

E(level)	J$^{\pi}$	L	C^{2}S†	Comments
0.0	1/2^{+}	0	0.45 *8*	J$^{\pi}$: from L(n)=0 from 0^{+}, interpreted as knockout from neutron 2s$_{1/2}$ orbital. L: 2012Sh21 deduced L by comparing the measured and eikonal-calculated longitudinal momentum distributions of residuals. C^{2}S: from eikonal/Hartree-Fock model with R$_s$=0.24 *2*, uncertainty only include experimental contributions. Others: 0.49 *8* from eikonal/Strong Absorption model with R$_s$=0.26 *2* and also from transfer-to-continuum model; R$_s$=0.26 *2*; 1.80 from shell model. σ_{exp}=5.03 mb *46*. Calculated values: σ_{sp}=11.1 mb from eikonal/Hartree-Fock model; σ_{sp}=10.2 mb from eikonal/Strong Absorption model; σ_{sp}=10.3 mb from transfer-to-continuum model.

† Spectroscopic factor C^{2}S=σ_{exp}/σ_{sp}.

^{9}Be(^{37}Ca,X) 2024Dr01

$J^{\pi}=3/2^{+}$ for ^{37}Ca ground state.

2024Dr01: a 72-MeV/nucleon ^{37}Ca secondary beam was produced via the fragmentation of a 140-MeV/nucleon $^{40}Ca^{20+}$ primary beam impinging on a 0.5-mm-thick Be target. Experimental setup includes the CAESium-iodide scintillator ARray (CAESAR) for detecting γ rays, a DSSD-CsI(Tl) ΔE-E Ring Telescope for detecting protons, a Scintillating-Fiber Array (SFA) and the S800 spectrograph for detecting heavy residuals. Measured E_p, Eγ, and E(residual). Deduced the first excited state in ^{35}Ca via the 2p+^{33}Ar exit channel from total decay-energy spectra using the invariant-mass method. Comparisons with shell-model calculations with the USDC Hamiltonian.

^{35}Ca Levels

E(level)	J$^{\pi}$	Comments
2.08×10^{3} *10*	(3/2^{+})	E(level): From total decay energy of 2p+^{33}Ar E_T=1667 keV *20*. J$^{\pi}$: shell-model calculations.

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